Extras from Handwritten Notes

6371 points not in the condensed books or Papers 1–4, arranged by book and chapter

Every point here was found in your handwritten notes but is **not** in the 572-page condensed books (Books 2–17) or in the Paper 1–4 answer books. Where your notes disagreed with the books or papers, the book/paper version was kept and the point is not listed here. The grey tag after each point gives the notes file and page it came from.

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BP Basic Principles

OTHER TOPICS IN THIS BOOK

Biomaterials — Ideal Biomaterial and Collagen

- An ideal biomaterial should promote cellular interaction and tissue development, in addition to being biocompatible and having proper mechanical and physical properties. [Biomaterials p1]
- Collagen is the most abundant and ubiquitous material in the body. [Biomaterials p1]
- Collagen can be easily purified from both animal and human tissue. [Biomaterials p1]
- Collagen produces minimal inflammatory and antigenic responses. [Biomaterials p1]
- Intermolecular cross-linking of collagen reduces its degradation rate. [Biomaterials p1]
- Collagen can be processed into a wide variety of structures, such as sponges, fibres and films. [Biomaterials p1]
- The main limitation of collagen as a scaffold is its high degradation rate. [Biomaterials p1]

Alginate

- Alginate is a polysaccharide isolated from seaweed. [Biomaterials p2]

- Alginate has gentle gelling properties in the presence of divalent cations such as calcium (Ca2+). [Biomaterials p2]
- Alginate is a copolymer of D-mannuronate and L-guluronate. [Biomaterials p2]
- Alginate has been used as an injectable cell-delivery vehicle and as a cell-immobilisation matrix. [Biomaterials p2]
- The limitation of alginate is that its mechanical properties are not controllable. [Biomaterials p2]

Alginate and Collagen in Bioprinting

- Alginate and collagen are the materials used as bio-inks in bioprinting; they can be printed into the desired scaffold shape and serve as the backbone for the new tissue or organ. [Biomaterials p2]

Acellular Tissue Matrices

- Acellular tissue matrices are collagen-rich matrices. [Biomaterials p2]
- Acellular tissue matrices degrade after implantation and are replaced by extracellular matrix (ECM) proteins. [Biomaterials p2]
- Acellular tissue matrices show no evidence of immunogenic rejection. [Biomaterials p2]

Synthetic Polymers

- PGA, PLA and PLGA are polyesters of naturally occurring alpha-hydroxy acids. [Biomaterials p3]
- These synthetic polymers are thermoplastics, so they are easily formed into 3D scaffolds. [Biomaterials p3]
- The degradation products of PGA, PLA and PLGA are non-toxic and are eliminated from the body as carbon dioxide (CO2) and water (H2O). [Biomaterials p3]

Adverse reactions to intravascular iodinated contrast media

- Adverse reactions to intravascular iodinated contrast are classified into four groups: idiosyncratic anaphylactoid reactions, non-idiosyncratic reactions, delayed contrast reactions and contrast-induced nephropathy. [Contrast 2 p1]
- Moderate contrast reactions occur in 0.5–2% of patients. [Contrast 2 p2]
- Signs of a moderate reaction can include tachycardia or bradycardia, hypertension or hypotension, dyspnoea and pulmonary oedema, in addition to bronchospasm, laryngeal oedema and skin reactions. [Contrast 2 p2]
- A moderate reaction is treated with close observation and hydrocortisone 100 mg to 500 mg IV or IM, plus a beta-agonist for bronchospasm. [Contrast 2 p2]
- In a severe reaction, epinephrine is the drug of choice at 0.01 mg/kg in 1:10,000 dilution IV; if there is no IV access, the same dose is given as 1:1000 dilution IM. [Contrast 2 p2]
- Hypotension in a severe contrast reaction is treated with fluids and may need dopamine, with careful monitoring in an ICU setting. [Contrast 2 p2]

Delayed reactions and preventive strategies

- Delayed contrast reactions can occur from 3 hours to 7 days after contrast administration and usually resolve spontaneously. [Contrast 2 p3]
- Suggested drugs for preventing contrast reactions include H2 antagonists (with H1 antihistamines) and ephedrine, as well as corticosteroids. [Contrast 2 p3]

Ionic vs non-ionic contrast agents

- Ionic agents dissociate into an anion and a cation, whereas non-ionic agents do not dissociate in blood (the notes also write "insoluble" for non-ionic agents). [Contrast 2 p4]
- Ionic agents are less expensive (their one advantage), and non-ionic agents are more expensive (their one disadvantage). [Contrast 2 p4]
- Ionic agents have more viscosity and non-ionic agents less viscosity. [Contrast 2 p4]

Contrast agents in urology: classification

- Diatrizoate is sold as Hypaque. [Contrast 2 p5]

- Iothalamate (Conray) and ioxitalamate are ionic high-osmolality iodinated contrast agents. [Contrast 2 p5]
- Iobitridol, iopromide and ioversol are non-ionic low-osmolality agents, along with iohexol and iopamidol. [Contrast 2 p5]
- Iohexol is sold as Omnipaque (margin annotation on p 9). [Contrast 2 p5]
- Iotrolan is a non-ionic iso-osmolal contrast agent. [Contrast 2 p5]

Risk factors for CIN

- Hypertension is a patient-related risk factor for CIN. [Contrast 2 p6]
- Advancing age is given as a CIN risk factor with a threshold of 75 years. [Contrast 2 p6]
- Reduced cardiac output and hypotension are patient-related risk factors for CIN, alongside class IV heart failure. [Contrast 2 p6]
- Hypoalbuminaemia is a patient-related risk factor for CIN. [Contrast 2 p6]
- Diuretic use is a patient-related risk factor for CIN. [Contrast 2 p6]
- Increased contrast viscosity is a non-patient-related risk factor for CIN. [Contrast 2 p6]
- There is actually no standard criterion for diagnosing CIN. [Contrast 2 p6]

Pathophysiology of CIN

- When contrast enters the renal vasculature, it causes adenosine release from the macula densa, endothelin release and prostaglandin dysregulation, which together produce sustained intrarenal vasoconstriction lasting hours. [Contrast 2 p7]
- The vasoconstriction prolongs contrast transit and so increases the exposure of the tubules to contrast. [Contrast 2 p7]
- Oxidative stress and inflammation contribute to the cellular injury and death that leads to acute kidney injury in CIN. [Contrast 2 p7]

Preventive strategies / precautions for CIN

- Modifiable risk factors should be corrected before contrast, including correcting anaemia and hypoalbuminaemia. [Contrast 2 p8]
- Ascorbic acid and statins are other agents that have been used to prevent CIN. [Contrast 2 p8]
- Patients with GFR below 30 mL/min should be discussed with a nephrologist before the procedure. [Contrast 2 p8]
- After the contrast procedure, serum creatinine should be checked and hydration continued (margin annotation on p 12, which also times steps 1–5 as pre-procedure and non-ionic lowest-dose contrast as during the procedure). [Contrast 2 p8]

Haemostatic Agents — Classification

- Crosseal and Vivostat are fibrin-based glues or sealants, in addition to Tisseel (Tisseel VH) and Evicel. [Hemostatic agents p1]
- Surgiflo is a gelatin-based flowable agent; Floseal and Surgiflo are the injector-type gelatin agents. [Hemostatic agents p1]
- Oxycel and Tabotamp are cellulose-based haemostats, in addition to Surgicel. [Hemostatic agents p1]
- Avitene is the brand of microfibrillar collagen haemostat. [Hemostatic agents p1]
- Advaseal is a PEG-polymer tissue sealant, in addition to Coseal. [Hemostatic agents p1]
- Gelatin-, cellulose- and collagen-based agents give topical haemostasis only and have no tissue-sealant effect. [Hemostatic agents p1]
- Glutaraldehyde-based (BioGlue) and PEG-polymer agents are tissue sealants and are not useful for active bleeding. [Hemostatic agents p1]

Key Points: Fibrin-based Sealants

- The active ingredients of fibrin sealants are fibrinogen, calcium chloride, aprotinin (except Evicel) and thrombin. [Hemostatic agents p2]
- Tisseel takes 20 minutes to set up. [Hemostatic agents p2]
- Fibrin sealants have sealant and adhesive qualities in a dry field. [Hemostatic agents p2]

Key Points: Gelatin-based Agents

- In gelatin–thrombin agents, blood percolates through the gelatin matrix and the granules swell; the patient's fibrinogen and the agent's thrombin then form the clot. [Hemostatic agents p2]
- Gelatin-based agents are reabsorbed within 6–8 weeks. [Hemostatic agents p2]

Key Points: Collagen-based Agents

- Collagen-based agents come as a pre-loaded system. [Hemostatic agents p2]

Key Points: Glutaraldehyde-based Agents (BioGlue)

- In BioGlue, glutaraldehyde cross-links bovine serum albumin and tissue proteins. [Hemostatic agents p3]
- BioGlue needs a dry field to work. [Hemostatic agents p3]

3D Laparoscopy

- Besides depth perception, the 3D view allows correct measurement of the dimensions of anatomical spaces. [3D Lap p1]
- Depth perception is the visual ability to judge the relative distance of objects. [3D Lap p1]
- The process by which the brain combines two views into a single 3D image through retinal disparity is called stereopsis. [3D Lap p1]
- Stereoscopy is the technique of creating or enhancing the illusion of depth in an image. [3D Lap p1]

3D Laparoscopy: advantages

- 3D laparoscopy retains tactile feedback. [3D Lap p2]
- 3D laparoscopy has a low maintenance cost. [3D Lap p2]
- Conventional laparoscopic instruments can be used with 3D laparoscopy. [3D Lap p2]

MRI: Principle

- Protons (hydrogen nuclei) in the body are positively charged and also have magnetic spin, and MRI uses this property of protons to produce images. [MRI p1]
- In the neutral state, protons in the body spin in a haphazard fashion. [MRI p1]
- The MRI magnet produces a magnetic field that aligns the body's protons in a straight line in the direction of the field. [MRI p1]
- A radiofrequency pulse applied perpendicular to the magnetic field pushes the aligned protons to a higher energy level. [MRI p1]
- When the radiofrequency pulse is removed (relaxation), the protons lose energy, and the machine processes this released energy into the image. [MRI p1]

MRI: Advantages, Disadvantages and Types

- MRI gives much anatomical detail even without contrast. [MRI p2]
- MRI can image in multiple planes without moving the patient. [MRI p2]
- MRI contrast agents carry a considerably smaller risk of allergic reactions (compared with CT contrast). [MRI p2]
- The MRI radiofrequency transmitter can cause burns if mishandled. [MRI p2]
- Types of MRI are conventional (T1- and T2-weighted), diffusion-weighted, dynamic contrast-enhanced, and MR spectroscopy or chemical shift MRI. [MRI p2]

Nanotechnology in Urology

- Nanomedicine is the application of nanotechnology to medicine: the monitoring, repair, construction and control of human biological systems at the molecular level using engineered nanodevices and nanostructures. [Nano T B p1]
- Limitations of current imaging systems that nanomedicine may overcome are diagnosing malignancy at an early stage, detecting micrometastasis, and detecting recurrence at a curable stage. [Nano T B p1]

Diagnostic Applications: Imaging, PSA, Genetic Disease

- Magnetic nanoparticles, because of their small size, extravasate into the interstitial space, are transported to lymph nodes and are taken up by macrophages there; they change the magnetic properties of tissue, which MRI detects (lymphotropic nanoparticle enhanced MRI, LNMRI), for example in prostate cancer nodal metastases. [Nano T B p2]
- Agents used for LNMRI are intravenous iron oxide or superparamagnetic iron oxide with gold nanoparticles. [Nano T B p2]
- A gold nanoparticle based PSA assay is used to detect PSA recurrence after radical prostatectomy and is 300 times more sensitive than conventional assays. [Nano T B p2]
- A nanoparticle DNA based assay detects polycystic kidney disease single nucleotide polymorphisms (written "polypeptides", PKD-SNPs) using only 0.02-0.05 ml of blood. [Nano T B p2]

Diagnosis of Infections and Therapeutic Applications

- Nanosensors are used to diagnose UTI, and nanoparticle-based specific E. coli antibodies have been developed for improved UTI detection. [Nano T B p3]
- Liposomal doxorubicin formulations are an example of nanoparticle chemotherapy delivery exploiting the enhanced permeability and retention effect. [Nano T B p3]
- In prostate cancer, drug delivery has used gelatin nanoparticles carrying paclitaxel, or transferrin-conjugated paclitaxel. [Nano T B p3]
- Nanocarriers have been used to apply thermal energy (cryotherapy and HIFU) to prostate cancer cells, with encouraging animal results. [Nano T B p3]

Bladder Reconstruction, OAB/IC, ED, Artificial Kidney

- 3D porous polylactide-co-glycolide and polyether urethane scaffolds have been used to grow bladder muscle, avoiding the complications of intestinal segments. [Nano T B p4]
- Weekly intravesical empty liposomes (80 mg in 40 cc water) have been shown to be superior to oral pentosan polysulfate for OAB/IC. [Nano T B p4]
- Nanoparticles encapsulating tadalafil have been applied as a gel on the glans for ED. [Nano T B p4]
- Liposomal PGE1 formulations have been developed for intraurethral use in ED. [Nano T B p4]
- The human nephron filter (HNF) is a portable artificial kidney worn by the patient, made of two membranes in a cartridge: the G-membrane mimics the glomerulus and produces an ultrafiltrate containing low molecular weight solutes, and the T-membrane performs tubular reabsorption. [Nano T B p4]

HNF Advantages, Biodegradable Stents, Emerging Tools, Limitations

- The human nephron filter can provide a GFR of 30 ml/min, allows patient mobility and improves quality of life. [Nano T B p5]
- A poly-epsilon-caprolactone / polylactide-co-glycolide ureteric stent with nanofibres and micropores, studied in a porcine model, degraded gradually over 10 weeks. [Nano T B p5]
- Other emerging nano tools are nanotweezers for microsurgical procedures, nano-urobots for use during cystoscopy and ureteroscopy, and the nano laser (a laser with nanoscale dimensions). [Nano T B p5]
- Ethical, socioeconomic, political and environmental concerns must be fully addressed before nanotechnology reaches clinical practice. [Nano T B p5]
- National and international regulations are needed to prevent misuse of nanotechnology, such as "genetic colonialism" and terrorist activities. [Nano T B p5]

Role of nuclear medicine in urology: radionuclide agents — Basic Principles (Book 1)

- Iodine-131 or iodine-123 orthoiodohippurate (OIH) is a tubular-secreted renal agent. [Nuclear Medicine p1]
- Tc-99m ethylenedicysteine (EC) is a tubular-secreted renal agent. [Nuclear Medicine p1]
- Tc-99m glucoheptonate (GHA) is a cortical-binding renal agent, like DMSA. [Nuclear Medicine p1]

Excretory mechanisms — Basic Principles (Book 1)

- DMSA undergoes some glomerular filtration, and about 60% is fixed in the tubules (tubular fixation). [Nuclear Medicine p2]

MAG3 — Basic Principles (Book 1)

- MAG3 has limited availability. [Nuclear Medicine p3]
- MAG3 shows high hepatobiliary uptake. [Nuclear Medicine p3]

Tc-99m ethylenedicysteine (EC) — Basic Principles (Book 1)

- EC gives less radiation than DMSA and is less time-consuming than DMSA. [Nuclear Medicine p3]
- EC gives reliable measurement of differential renal function and can be used as a substitute for MAG3. [Nuclear Medicine p3]
- EC has lower sensitivity than DMSA for detecting cortical lesions. [Nuclear Medicine p3]

DMSA — Basic Principles (Book 1)

- DMSA is also indicated to evaluate renal space-occupying lesions. [Nuclear Medicine p4]
- In acute pyelonephritis DMSA may show diffusely decreased uptake. [Nuclear Medicine p4]
- DMSA gives a higher overall radiation dose and is time-consuming. [Nuclear Medicine p4]

DTPA and diuretic renogram — Basic Principles (Book 1)

- DTPA is the most commonly used renal agent. [Nuclear Medicine p5]
- The maximal diuretic response after furosemide occurs within 15–20 minutes. [Nuclear Medicine p5]
- The rationale of the diuretic renogram is to increase the sensitivity of the dynamic renal study. [Nuclear Medicine p5]

O'Reilly curves — Basic Principles (Book 1)

- An equivocal type IIIb curve (slow response to diuretic) should be repeated with the F−15 protocol. [Nuclear Medicine p6]
- Type IV curve is "delayed compensation" (Homsy's sign): a double-peak response to the diuretic; the F−15 protocol usually shows an obstructed system. [Nuclear Medicine p6]

Washout half-time, advantages and pitfalls — Basic Principles (Book 1)

- Renography is short in duration and non-nephrotoxic. [Nuclear Medicine p7]
- Renography (DTPA) gives poor cortical imaging. [Nuclear Medicine p7]
- Low-grade obstruction can give a false-negative renogram. [Nuclear Medicine p7]

Metabolic & Physiological Changes During Pneumoperitoneum

- CO2 is also preferred for pneumoperitoneum because it is readily available. [Pneumoperitoneun p1]

Cardiac Effects

- The fall in cardiac output and cardiac index is compensated by a rise in heart rate, which increases myocardial oxygen demand; young patients tolerate this, but myocardial infarction can occur in the elderly. [Pneumoperitoneun p2]

Effect on Brain

- Raised intra-abdominal pressure compresses the IVC, reducing venous return from the brain and so reducing CSF absorption, which raises intracranial pressure. [Pneumoperitoneun p2]
- Absorbed CO2 causes cerebral vasodilatation, increasing arterial inflow to the brain, which also raises intracranial pressure. [Pneumoperitoneun p2]
- The rise in intracranial pressure can be detrimental in patients with a brain tumour (as well as head injury). [Pneumoperitoneun p2]

Effect on Lungs

- Poor ventilation from diaphragmatic splinting causes hypoxaemia particularly in obese patients (as well as COPD patients). [Pneumoperitoneun p3]

CO2 Absorption

- Absorbed CO2 combines with water to form carbonic acid, which dissociates into H+ and HCO3- (CO2 + H2O ⇌ H2CO3 ⇌ H+ + HCO3-), so more H+ in blood produces acidosis. [Pneumoperitoneun p4]
- Uncompensated acidosis (when respiration is compromised, as in obesity or COPD) decreases myocardial contractility and causes vasodilatation. [Pneumoperitoneun p4]
- The direct sympathetic effect of CO2 (peripheral vasoconstriction and raised SVR) increases myocardial workload and oxygen demand, which can cause myocardial infarction in the elderly with poor cardiac reserve. [Pneumoperitoneun p4]

Robotics: da Vinci System Components

- At the surgeon console the surgeon rests the head on the viewport; this interrupts the head sensors and activates the masters, so the instruments move only while the surgeon is looking into the viewer. [Robot p1]
- The surgeon operates with two masters positioned directly under the magnified 3D display. [Robot p1]
- Foot switches at the console control the clutch, camera, focus and cautery. [Robot p1]
- The vision cart also carries the flat-screen monitor, the insufflator and a xenon light source. [Robot p2]
- Set-up joints on the patient-side cart (robotic tower) hold and position the camera arm and the instrument arms. [Robot p2]

Robotics: Current Status in Urology

- Robotic renal artery aneurysm repair is listed among robotic renal procedures. [Robot p3]
- Robotic ureteric procedures listed include ileal ureter replacement and megaureter repair. [Robot p3]

Robotics in Paediatric Urology

- The benefits of robotic surgery in paediatric urology remain controversial. [Robot p4]
- Rapid gastric emptying in children leads to increased small-bowel distension, which compromises access and visualisation. [Robot p4]
- Children have a higher risk of bowel or vascular injury during access because of their lax abdominal wall. [Robot p4]
- Limited surgeon experience and few training centres limit paediatric robotic surgery. [Robot p4]
- Paediatric robotic procedures that have been performed include urachal cyst excision and bladder neck reconstruction. [Robot p5]

Virtual Simulators in Urology — definition

- A simulator is a device that enables the operator to reproduce or represent test conditions or phenomena likely to occur in actual performance. [Simulation etc p1]
- Simulation can be described as a "dress rehearsal" for the finale. [Simulation etc p1]

Need for simulators

- Medicine is a safety-critical domain where real lives are at stake, so widespread use of simulators is an important adjunct to training. [Simulation etc p1]
- Limited work hours for residents are a reason simulators are needed. [Simulation etc p1]
- Financial constraints are a reason simulators are needed. [Simulation etc p1]
- Fear of litigation is a reason simulators are needed. [Simulation etc p1]
- An increasing number of complex cases is a reason simulators are needed. [Simulation etc p1]

Barriers to surgical simulation (disadvantages)

- The market for surgical simulators is small, which is a barrier to their use. [Simulation etc p2]
- Few simulators have been validated for teaching. [Simulation etc p2]
- A simulator can never substitute for real patient-based training. [Simulation etc p2]

Surgical simulators available in urology

- The UroMentor is a virtual simulator for cystoscopy and ureteroscopy. [Simulation etc p2]
- The Uro-Scopic Trainer is a simulator for ureteroscopy. [Simulation etc p2]

- The SurgicalSIM TURP simulator and the UroTrainer VR simulator are used for TURP and TURBT training. [Simulation etc p2]
- The PERC Mentor and the SimPORTAL trainer are simulators for PCNL and renal access. [Simulation etc p2]
- UroSim and a HoLEP simulator are used for laser prostate surgery training. [Simulation etc p2]
- Procedicus MIST is a simulator for laparoscopic urology. [Simulation etc p2]
- The da Vinci Skills Simulator is used for robot-assisted surgery training. [Simulation etc p2]

VR and AR in surgical navigation

- Surgical navigation is a method of guiding surgery with computer-based technology, also called computer-aided surgery or image-guided surgery. [Simulation etc p3]
- In surgical navigation the surgeon uses special instruments that are tracked by the navigation system, which matters most when the surgeon cannot see the instrument tip, as in minimally invasive surgery. [Simulation etc p3]
- iPad-assisted puncture of the pelvicalyceal system reduces radiation exposure. [Simulation etc p3]
- Electromagnetic tracking of the puncture is an example of surgical navigation. [Simulation etc p3]
- Computer-based tracking of ultrasound-guided biopsy or puncture of the kidney is an example of surgical navigation. [Simulation etc p3]
- Marker-based endoscopic tracking during laparoscopic radical prostatectomy or partial nephrectomy is used to reduce positive margins. [Simulation etc p3]
- Stereotactic radiosurgery (CyberKnife) is an example of navigation-based treatment. [Simulation etc p3]

Advantages and disadvantages of navigation

- Navigation increases surgical precision. [Simulation etc p4]
- Navigation improves ergonomics. [Simulation etc p4]
- Navigation decreases the risk of surgical errors. [Simulation etc p4]
- A disadvantage of navigation systems is their huge, bulky equipment. [Simulation etc p4]

Virtual reality

- On the computer, virtual reality is experienced mainly through two of the five senses: vision and sound. [Simulation etc p4]
- For pre-operative planning with VR, the data must be accurate. [Simulation etc p4]
- For pre-operative planning with VR, the surgical model must be built quickly. [Simulation etc p4]
- In VR-assisted robotic surgery, the VR system gives visual access to areas not originally visible, while haptic feedback gives the impression that the surgeon's hands are inside the patient. [Simulation etc p5]

Augmented reality

- Augmented reality has mainly been tested in the urological and cardiovascular domains. [Simulation etc p5]
- ARAUS stands for augmented reality assisted urological surgery. [Simulation etc p5]
- Google Glass loaded with an application demonstrating penile prosthesis placement from start to finish has been used as part of a teaching programme. [Simulation etc p6]

Difference between VR and AR

- In virtual reality, users in theory cannot tell the difference between what is real and what is virtual. [Simulation etc p6]
- Augmented reality does not need a headset. [Simulation etc p6]
- In augmented reality, users can distinguish between what is real and what is virtual. [Simulation etc p6]

Randomised Control Trial: Basic Design and Steps — Basic Principles (Book 1); compared against Paper 1–4 answer books only (Paper 1 p67–70 covers clinical trial methodology)

- The basic design of an RCT runs in this order: protocol, selection of a suitable population (the reference or target population), selection of a suitable sample (the study population), necessary

exclusions, randomisation into experimental and control groups, manipulation and follow-up, and assessment. [Statistics p1]

- Before randomisation, two groups are excluded: those who are not eligible and those who are not willing. [Statistics p1]
- The RCT protocol fixes the aims and objectives of the study, the inclusion and exclusion criteria, and the intervention to be done. [Statistics p1]
- The reference (target) population is the population to which the findings of the trial will apply if the trial succeeds. [Statistics p2]
- The study population is drawn from the reference population. It should be chosen at random and should be representative of the reference population. It is also called the experimental population. [Statistics p2]
- Randomisation is best done using random number tables. [Statistics p2]
- Manipulation is the deliberate application, withdrawal or reduction of the suspected causal factor, such as a drug or vaccine. [Statistics p2]
- Follow-up means examining the study and control groups at different time intervals. Some losses, for example from death or migration, are inevitable. [Statistics p2]
- In the final step, the trial is assessed in terms of its positive and negative results. [Statistics p3]
- An advantage of an RCT is that multiple outcome variables can be measured in a single study. [Statistics p3]
- The disadvantages of RCTs are time and cost, the danger of conflict of interest, ethical restrictions, and non-compliance of controls, which threatens the validity of the study. [Statistics p3]
- By outcome of interest, RCTs are classified as explanatory or pragmatic trials. [Statistics p3]

Concept of Blinding in RCT — Basic Principles (Book 1)

- Blinding means concealing group allocation from one or more of the people involved in a study. It is also called masking. [Statistics p4]
- Randomisation minimises the differences between treatment groups, but it does nothing to prevent differential treatment of the groups later in the trial or differential assessment of outcomes. Either can bias the estimates of treatment effect, and blinding minimises both problems. [Statistics p4]
- In a double-blind trial, the participant does not know which group they are in, so their beliefs about the treatment are less likely to influence the outcome. [Statistics p10]
- In a double-blind trial, the researcher does not know which subjects are receiving the real treatment, so they are less likely to accidentally reveal subtle clues that might influence the outcome. [Statistics p10]

Randomisation in RCTs — Basic Principles (Book 1)

- One advantage of randomisation is that it makes blinding possible. [Statistics p5]
- Randomisation also allows probability theory to be used to express how likely it is that any difference in outcome between treatment groups is due to chance alone. [Statistics p5]
- Randomising patients involves two processes: choosing a randomisation procedure, and allocation concealment. [Statistics p5]
- Simple randomisation is the easiest randomisation method. [Statistics p5]

Concept of Allocation Concealment — Basic Principles (Book 1)

- Allocation concealment means the strict precautions taken so that patients' group assignments are not revealed before they are finally allocated to their groups. The person randomising a patient does not know what the next treatment allocation will be. [Statistics p5-6]
- Methods of allocation concealment are: sequentially numbered opaque sealed envelopes (SNOSE), numbered or coded containers, a secured computer method, a pharmacy-controlled method, and a centralised service, where the researcher calls the trial office to learn the allocation sequence. [Statistics p5-6]
- A centralised service is the best method of allocation concealment. (The corpus says only "central or web-based".) [Statistics p5-6]

- The notes say that trials using allocation concealment yield up to 40% larger estimates of effect and greater heterogeneity in results. (Reviewer's note: the published finding, Schulz 1995, is the reverse. It is trials with inadequate or unclear concealment that overestimate effects by up to about 40%. Check before learning this point.) [Statistics p7]
- Example: if a healthcare provider knows which subjects will go to the intervention group, they may try to steer a particular subject who they think will benefit more. This introduces selection bias, which allocation concealment helps to eliminate. [Statistics p7]

Bias — Basic Principles (Book 1)

- Bias is a systematic error. It is defined as disproportionate weight for or against one thing, person or group compared with another, usually in a way considered unfair. [Statistics p8]
- The notes say selection bias is also called "Berksonian bias" and involves some individuals being more likely than others to be selected for a study. (Strictly, Berkson's bias is one subtype of selection bias, seen in hospital-based samples.) [Statistics p8]
- Funding bias can lead to outcomes that favour the study's financial sponsor. [Statistics p8]
- Analytical bias arises from the way the results are evaluated. [Statistics p8]
- Exclusion bias arises from the systematic exclusion of certain individuals from the study. [Statistics p8]
- Recall bias arises when participants cannot recall past events accurately. [Statistics p9]
- Observer bias is the researcher's subconscious cognitive bias in judgement. [Statistics p9]
- In positive confounding, the observed association is biased away from the null. In negative confounding, it is biased towards the null. [Statistics p9]

Kaplan–Meier Plots — Basic Principles (Book 1)

- Kaplan–Meier analysis is a non-parametric statistical method for estimating the survival function from lifetime data. [Statistics p11]
- In medical research, it is often used to measure the fraction of patients still alive at a certain time after treatment. [Statistics p11]
- It is named after Edward L. Kaplan and Paul Meier. [Statistics p11]
- A Kaplan–Meier plot is a series of declining horizontal steps. With a large enough sample, it approaches the true survival function for that population. [Statistics p11]
- To draw a Kaplan–Meier plot, at least two pieces of data are needed for each patient: their status at the last observation, and the time to the event (or to censoring). [Statistics p11]
- The length of each horizontal line along the x-axis is the survival duration for that interval. The vertical axis is the estimated probability of survival. [Statistics p11]

Kaplan-Meier plots (continued)

- Worked example: a KM plot of percent survival against years shows that patients with a "Gene B signature" have lower survival than those with a "Gene A signature" (the comparative word is left blank in the notes; the graph shows Gene B lower). [Statistics p12]
- The main advantage of KM plots is that they can include censored data, meaning that information about patients who are lost at any point in time, for any reason, can still be used in the analysis. [Statistics p12]
- KM plots give professionals an improved understanding of disease processes. [Statistics p12]
- KM plots can be used in clinical practice for counselling patients with different diseases. [Statistics p12]

Types of research studies

- Epidemiological studies are classified into four groups: descriptive, analytical, experimental and integrative studies. [Statistics p13]
- Analytical studies comprise ecological studies, case-control studies, cohort studies and cross-sectional studies. [Statistics p13]
- Experimental studies comprise RCTs and non-randomised controlled trials, plus uncontrolled trials. [Statistics p13]
- Integrative studies comprise systematic reviews and meta-analyses. [Statistics p13]

Case-control study

- A case-control study is an observational study of persons with the disease of interest (or another variable) and a suitable control group of persons without the disease. [Statistics p13]
- Because it is observational, no intervention is attempted and no attempt is made to alter the course of the disease. [Statistics p13]
- Case-control studies are retrospective analyses. [Statistics p13]
- Advantages of case-control studies: good for studying rare diseases, less time consuming, and multiple risk factors can be looked at simultaneously. [Statistics p13]
- Disadvantages of case-control studies: being retrospective, data quality is a problem; and it can be difficult to find a suitable control group. [Statistics p13]
- A case-control study proceeds from effect to cause, whereas a cohort study proceeds from cause to effect. [Statistics p13]
- A case-control study is relatively inexpensive, whereas a cohort study is expensive. [Statistics p13]
- A case-control study tests whether the suspected cause occurs more frequently in those with the disease than in those without it; a cohort study tests whether the disease occurs more frequently in exposed than in unexposed individuals. [Statistics p13]
- A case-control study is retrospective, whereas a cohort study may be retrospective or prospective. [Statistics p13]
- A case-control study involves few subjects only, whereas a cohort study involves a large number of subjects. [Statistics p13]
- A case-control study gives quick results, whereas a cohort study requires a long follow-up period. [Statistics p13]
- A case-control study is suitable for the study of rare diseases. [Statistics p13]
- A case-control study generally yields only an estimate of relative risk (the odds ratio), whereas a cohort study yields the incidence rate, relative risk (RR) and attributable risk (AR). [Statistics p13]
- A case-control study is usually the first approach to testing a hypothesis, whereas a cohort study is reserved for testing a precisely formulated hypothesis. [Statistics p13]

Cohort study

- A cohort study is an analytical study undertaken to obtain additional evidence to refute or support the existence of an association between a suspected cause and a disease. [Statistics p15]
- Other names for a cohort study are longitudinal study, incidence study and forward-looking study. [Statistics p15]

Level of evidence and grade of recommendation (EAU 2018 / Campbell)

- Level 3 evidence also includes well-designed non-experimental comparative studies (the corpus lists case-control, correlation studies, case reports and case series). [Statistics p16]
- Grades B and C can also be "extrapolated" recommendations: grade B is directly based on level II evidence or extrapolated from level 1 evidence, and grade C is directly based on level III evidence or extrapolated from level 1-2 evidence. [Statistics p16]

Nominal vs ordinal data

- Data are either qualitative (categorical), divided into nominal and ordinal, or quantitative (numerical), divided into interval and ratio. [Statistics p18]
- Nominal and ordinal data are analysed with non-parametric tests; interval and ratio data are analysed with parametric tests. [Statistics p18]
- Nominal data (for example sex or eye colour) cannot be arranged in ascending or descending order. [Statistics p18]
- Ordinal data can be counted or ordered in ascending or descending manner, for example stages of cancer (I, II, III, IV) or house numbers. [Statistics p18]
- Interval data are measured along a scale in which each portion is equidistant from the next, for example pain rated from 1 to 10 on a scale. [Statistics p18]

- Ratio data are numbers that can be compared as multiples of one another, and they can be multiplied or divided. [Statistics p18]
- A non-parametric test is a statistical test used for non-metric independent variables. [Statistics p18]
- A parametric test is a statistical test in which assumptions are made about the population parameter. [Statistics p18]

Parametric and non-parametric tests; Student t-test

- Non-parametric tests include the chi-square test, Mann-Whitney U test, Fisher's exact test and Wilcoxon test, among others. [Statistics p19]
- Parametric tests include the z-test for large samples, and for small samples the paired t-test, unpaired t-test, one-way ANOVA and two-way ANOVA. [Statistics p19]
- The t-test is any statistical hypothesis test in which the test statistic follows Student's t-distribution under the null hypothesis. [Statistics p19]
- Requirements of the t-test: a random sample, an unknown population standard deviation, and a population that follows a normal distribution. [Statistics p19]
- The one-sample t-test compares the mean of a population with a theoretical value. [Statistics p19]
- The unpaired two-sample t-test compares the means of two independent samples (for example blood pressure of patients versus blood pressure of normal persons). [Statistics p19]
- The paired t-test compares the means of two related groups of samples (for example blood pressure before and after a drug in the same patients, so n1 = n2 = n). [Statistics p19]
- To test the significance of a sample mean (whether its difference from the population mean is significant or due to chance), t = (x - mu) / (SD x square root of 1/n), where x is the sample mean, mu the population mean and n the sample size. [Statistics p19]
- For the unpaired t-test, t = (x1 - x2) / (SD x square root of (1/n1 + 1/n2)), where x1 and x2 are the sample means and n1 and n2 the sample sizes. [Statistics p19]
- For the paired t-test, t = (x1 - x2) / (SD x square root of (1/n + 1/n)). [Statistics p19]

Significance of P value

- The P value (probability value) is defined as the probability of finding the difference between two results by chance; in other words, the probability of the null hypothesis being true. [Statistics p21]
- The null hypothesis assumes that there is no significant difference between specified populations. [Statistics p21]
- Example given in the notes: when tossing a coin the probability of head or tail is 50%, so the P value is 0.5. [Statistics p21]
- Diagram: on a probability density curve the P value is the shaded tail area beyond the observed data point, in the region of very unlikely observations, with the confidence interval marked on the set of possible results. [Statistics p21]
- The smaller the P value, the smaller the probability that the event occurred by chance. [Statistics p21]
- As the confidence interval widens, a previously significant value may become insignificant. [Statistics p21]

Significance of sample size

- Sample size is the number of participants or observations included in a study and is usually represented by n. [Statistics p22]
- One formula for sample size is n = Z squared x p x (1 - p) / c squared, where Z = 1.96 for a 95% CI, c is the confidence interval and p is the prevalence in previous studies. [Statistics p22]
- Sample size influences two statistical parameters: the precision of the estimates and the power of the study to draw conclusions. [Statistics p22]
- Power is defined as the probability of finding a statistically significant result. [Statistics p22]
- An adequate sample size minimises sampling error; larger samples tend to have a smaller margin of error. [Statistics p22]
- Beyond a certain point, increasing sample size no longer reduces sampling error; this is known as the law of diminishing returns. [Statistics p22]

Tissue Engineering — Strategies and cell-based therapy

- There are three strategies to create a tissue substitute: cell-based therapy alone, unseeded biomaterials, and biomaterials seeded with cells. [Tissue Engineering p1]
- In cell-based therapy, stem cells are preferred over native (differentiated) cells. [Tissue Engineering p1]
- Fetal or neonatal amniotic fluid and placenta are a source of stem cells. [Tissue Engineering p1]
- A defining property of stem cells is the ability to easily form a clonal population. [Tissue Engineering p1]

Stem cell types

- Totipotent stem cells produce all the different cells of the organism, for example the zygote. [Tissue Engineering p2]
- Pluripotent stem cells can differentiate into all three germ layers. [Tissue Engineering p2]
- Multipotent stem cells produce only a closely related family of cells. [Tissue Engineering p2]

Biomaterials

- An ideal biomaterial should promote cellular interaction so that new functional tissue develops. [Tissue Engineering p2]
- The problem with naturally derived materials (collagen, alginate, or a collagen–alginate combination) is high degradation and uncontrollable mechanical properties. [Tissue Engineering p2]

Tissue Engineering in Urology

- Tissue-engineered ureter is still experimental and has not yet been used in humans. [Tissue Engineering p3]
- For VUR and urinary incontinence, endoscopic injection of stem cells has been used. [Tissue Engineering p3]
- Microencapsulated Leydig cells with long-term testosterone release have been used for testicular dysfunction (androgen deficiency). [Tissue Engineering p3]
- For female infertility, oocytes derived from stem cells have been implanted (experimental). [Tissue Engineering p4]
- The kidney is one of the most difficult tissues to replicate in the laboratory. [Tissue Engineering p4]
- Selective cell therapy, isolating one cell type with a specific function, is a promising renal approach: for example, cultured erythropoietin-producing cells used to treat anaemia in ESRD. [Tissue Engineering p4]
- Cells of the proximal tubule, distal tubule and glomerulus have been isolated separately in vitro, expanded and seeded onto polyglycolic acid scaffolds, showing that renal cells can be harvested and cultured. [Tissue Engineering p4]
- Embryonic renal cells harvested and expanded in vitro produced functional renal units with unidirectional urine flow. [Tissue Engineering p4]

2 Oncology, DSD, Gender & Ageing

CH 24 · BASIC PRINCIPLES OF IMMUNOLOGY AND IMMUNOTHERAPY IN UROLOGIC ONCOLOGY

Angiogenesis: definition and steps

- Angiogenesis is the physiological process by which new blood vessels form from pre-existing vessels. [Angiogenesis p1]
- Angiogenesis is distinct from vasculogenesis, which is the differentiation of precursor cells (angioblasts) into endothelial cells and the de novo formation of a primitive vascular network. [Angiogenesis p1]
- Tumour angiogenesis has 4 basic steps: injury to the tissue basement membrane, causing destruction and hypoxia; activation of endothelial cells; proliferation and stabilisation of endothelial

cells; and continued action of angiogenic factors, leading to formation of new blood vessels. [Angiogenesis p1]

Activators of Angiogenesis

- Endogenous growth factors that activate angiogenesis include FGF, angiogenin, hepatocyte growth factor, epidermal growth factor, placental growth factor, TNF-alpha and granulocyte colony-stimulating factor, in addition to VEGF, PDGF and TGF. [Angiogenesis p2]
- Small molecules that activate angiogenesis are adenosine, nicotinamide, prostaglandins E1 and E2, and butyryl glycerol. [Angiogenesis p2]
- Angiogenic factors released by tumour cells act on nearby blood vessels, so the tumour grows with its own blood supply and can then metastasise. [Angiogenesis p2]

Inhibitors of Angiogenesis

- Angiogenesis inhibitors are endogenous or exogenous (drugs). [Angiogenesis p3]
- Matrix-derived endogenous inhibitors of angiogenesis are arresten, canstatin, endorepellin, endostatin, fibronectin fragments, fibulin, thrombospondin 1 and 2, and TIMP 1, 2 and 3. [Angiogenesis p3]
- Non-matrix-derived endogenous inhibitors of angiogenesis are interferons, interleukins, platelet factor 4, angiostatin and cleaved antithrombin III. [Angiogenesis p3]

Apoptosis: definition and caspases

- Apoptosis (programmed cell death, or cellular suicide) is a process of sequential biochemical events that results in programmed cell death. [Angiogenesis p1]
- Caspases are a family of protease enzymes with essential roles in both programmed cell death and inflammation. [Angiogenesis p1]
- Once activated, executioner caspases attack several intracellular target proteins. [Angiogenesis p1]

Intrinsic Pro-apoptotic Pathway

- In the intrinsic pathway, a pro-apoptotic signal shifts the balance between BCL-2 family members in favour of the pro-apoptotic proteins; these include BAD, BIK and BOK, in addition to BAX, BAK and BID. [Angiogenesis p2]
- This shift increases the permeability of the outer mitochondrial membrane, which releases cytochrome c into the cytoplasm. [Angiogenesis p2]
- Caspase-9 activates the executioner caspase-3, which cleaves cellular proteins and activates nucleases, causing apoptosis. [Angiogenesis p2]

Extrinsic Apoptotic Pathway

- In the extrinsic pathway, the signal from a death receptor passes into the cell through its death domain, to which the cytosolic adaptor proteins TRADD and FADD and pro-caspase-8 attach to form active caspase-8. [Angiogenesis p3]

Difference between Apoptosis and Necrosis

- Apoptosis is programmed, coordinated cell death, whereas necrosis is cell degradation by hydrolytic enzymes along with tissue degradation. [Angiogenesis p4]
- Apoptosis is caused by physiological and pathological processes, whereas necrosis is caused by hypoxia and toxins. [Angiogenesis p4]
- Apoptosis causes no inflammatory reaction, whereas inflammation is always present in necrosis. [Angiogenesis p4]
- Apoptosis affects single cells, whereas necrosis kills many adjacent cells. [Angiogenesis p4]
- In apoptosis the lysosomes and other organelles stay intact, whereas in necrosis the lysosomes break down and release hydrolytic enzymes. [Angiogenesis p4]
- Apoptosis is triggered by the extrinsic and intrinsic pathways and executed by caspases, whereas necrotic cell death comes from ATP-dependent membrane damage and free-radical injury. [Angiogenesis p4]

Apoptosis and Cancer

- An intact apoptosis cascade is essential for the response to immunotherapy. [Angiogenesis p4]
- Advanced cancer cells can protect themselves from apoptosis. [Angiogenesis p4]

Immunotherapy: definition and types

- Immunotherapy is defined as treatment that stimulates or restores the ability of the immune system to fight disease. [Immunotherapy p1]
- Immunotherapies fall into two groups. Some help the immune system act directly against cancer: immune checkpoint inhibitors, adoptive cell transfer (CAR-T and CIK cells), monoclonal antibodies and cancer vaccines. Others enhance the body's own immune response: cytokines (interferons and interleukins) and BCG. [Immunotherapy p1]

Immune checkpoint inhibitors

- In prostate cancer, the CTLA-4 inhibitor ipilimumab benefited men with bone disease but not men with visceral metastases. [Immunotherapy p2]

CAR-T cell therapy

- CAR-T production steps: draw blood to collect the patient's T cells, insert the CAR gene (for example, against CD19) in the laboratory, grow millions of CAR-T cells in culture, then infuse them so they bind and kill cancer cells. Because the cells keep working inside the patient, CAR-T is called a "living drug". [Immunotherapy p3]
- CAR-T cells can be allogeneic (from a healthy donor's T cells) or autologous (from the patient's own blood). [Immunotherapy p3]
- CAR-T cells destroy cancer cells through extensive stimulated proliferation, increased cytotoxicity and increased secretion of cytokines, interleukins and growth factors. [Immunotherapy p4]
- Cytokine release syndrome after CAR-T presents with fever, fatigue, myalgia, tachycardia and capillary leak, and can progress to cardiac, hepatic and renal failure. [Immunotherapy p4]
- CAR-T neurological toxicity causes delirium, aphasia and seizures due to cerebral oedema. [Immunotherapy p4]
- Anaphylaxis is a toxicity of CAR-T therapy. [Immunotherapy p4]
- There are no long-term data on the safety of CAR-T therapy in pregnancy. [Immunotherapy p4]

CIK cell therapy

- CIK stands for cytokine-induced killer cells. [Immunotherapy p5]
- To make CIK cells, monocytes are isolated from a blood sample and incubated with interferon-γ, anti-CD3 antibody and recombinant human IL-1 and IL-2. The resulting CIK cells are amplified, quality-tested and infused into the patient, where they selectively kill infected or cancer cells. [Immunotherapy p5]
- CIK cell therapy can be autologous or allogeneic. [Immunotherapy p5]

Monoclonal antibodies

- There are four types of monoclonal antibody: murine (0% human, suffix -omab), chimeric (65% human, -ximab), humanised (more than 90% human, -zumab) and fully human (100% human, -umab). [Immunotherapy p6]
- The potential for immunogenicity falls from high with murine antibodies to low with fully human antibodies. [Immunotherapy p6]
- Common side effects of monoclonal antibodies are allergic reactions, flu-like symptoms and nausea. [Immunotherapy p6]
- Rare serious reactions to monoclonal antibodies are infusion reactions, leucopenia, skin problems and bleeding. [Immunotherapy p6]

Cancer vaccines

- Anti-idiotype vaccines are one of the types of cancer vaccine. [Immunotherapy p7]
- Oncophage (HSPPC-96, the heat shock protein–peptide complex-96 vaccine) was approved in Russia in 2008. [Immunotherapy p7]

- GVAX-Prostate is based on two prostate cancer cell lines, one androgen-sensitive and one castration-resistant. [Immunotherapy p8]
- GVAX-Prostate is given intradermally. [Immunotherapy p8]

CH 25 · MOLECULAR GENETICS AND CANCER BIOLOGY

Molecular Genetics & Cancer Biology

- To become cancer, a normal cell must acquire at least 8 key attributes: genetic instability and mutation, autonomous growth, angiogenesis, resistance to apoptosis, insensitivity to internal and external antiproliferative signals, local invasion with or without metastasis, unlimited cell-division potential, and immune-system evasion. [Cancer Biology p1]
- Homozygous gene deletion is one way a tumour suppressor gene loses its function. [Cancer Biology p1]

TP53 (Guardian of the Genome)

- TP53 is called the "guardian of the genome". [Cancer Biology p2]
- Besides DNA damage and oncogene activation, TP53 is activated by replication stress, hypoxia, viral infection and translational stress. [Cancer Biology p2]
- Among TP53's transcriptional targets, FAS/APO-1 mediates apoptosis. [Cancer Biology p2]
- Among TP53's transcriptional targets, BAI1 (brain angiogenesis inhibitor 1) acts on angiogenesis. [Cancer Biology p2]

Circulating tumour cells (biology)

- Circulating tumour cells (CTCs) are shed from tumours as an early step in blood-borne metastasis, and in most tumours they are scarce (<10/mL). [CTC p1]
- CTCs leave the primary tumour by EMT-mediated (epithelial–mesenchymal transition) active entry or by non-EMT-mediated passive entry into vessels. [CTC p1]
- In colonisation, CTCs escape immune surveillance, extravasate and invade the parenchyma of the foreign tissue to form a metastasis. [CTC p1]
- The CTC metastatic cascade runs: intravasation, transit to distant organs, extravasation and invasion, then adaptation and proliferation into a metastasis. [CTC p1]
- CTCs could be used for early tumour detection (CTC-positive versus CTC-negative). [CTC p2]
- Physical CTC detection methods use size (filtration), density (gradient centrifugation) or electrical properties (dielectrophoresis). [CTC p2]
- Biological CTC detection methods include the CellSearch system and the CTC-chip. [CTC p2]

Gene therapy in urology: basics and vectors

- Genes are specific base sequences that encode instructions for making proteins, and these proteins carry out most life functions and make up most cell structures. [Gene Therapy p1]
- Viral vector gene delivery proceeds by adhesion of the virus to the cell surface, internalisation in endosomes, release into the cytoplasm, transport along microtubules to the nucleus, release of the DNA and then protein production. [Gene Therapy p1]
- A liposome is an artificial lipid sphere (lipid bilayer) with an aqueous core holding the DNA and an outer protective layer against immune detection; it can carry DNA across the target cell membrane. [Gene Therapy p1]

Strategies to improve specificity, and role in urology

- Transductional targeting directs the vector to the cell surface of target cells for selective adhesion and gene delivery. [Gene Therapy p2]
- Transcriptional targeting controls transcription of the therapeutic gene so that it is produced selectively in the nucleus of target cells. [Gene Therapy p2]
- Targeted transgenes are directed at dysregulated metabolic pathways specific to tumours, and ideally the three approaches are combined. [Gene Therapy p2]
- The two roles of gene therapy in urology are repair and cytoreduction; cytoreduction works by increasing cytotoxic gene products or by increasing the immune response against cancer cells.

[Gene Therapy p2]

- In the kidney, an adenovirus vector has been used for transductional targeting of the G250 tumour antigen (in vitro). [Gene Therapy p2]
- In the kidney, ultrasound-directed non-viral gene therapy uses intravenous microbubbles, with ultrasound applied to increase cell membrane permeability (in vitro). [Gene Therapy p2]

Organ-specific gene therapy

- Gene therapy is not needed for testis cancer, because current treatments are excellent. [Gene Therapy p3]
- In bladder cancer, gene therapy aims to reduce recurrence or progression; systemic delivery uses non-viral vectors to target p53. [Gene Therapy p3]
- Mesenchymal stem cells transfected with VEGF have been implanted for erectile dysfunction. [Gene Therapy p3]
- Nitric oxide synthase gene therapy to the penile corpora stops the gradual loss of smooth muscle cells to fibrosis. [Gene Therapy p3]
- In prostate cancer, the non-targeted suicide-gene (enzyme–prodrug) approach uses thymidine kinase alone or combined with cytosine deaminase. [Gene Therapy p3]
- Prostate specificity can be enhanced by hybrid promoters built from the TARP promoter and/or PSMA and PSA enhancer elements. [Gene Therapy p4]

miRNA (micro RNA)

- MicroRNAs are found only in eukaryotic cells and play an important role in gene regulation (gene silencing). [micro RNA p1]
- MicroRNAs may have a role in disease diagnosis and management. [micro RNA p1]
- A microRNA is single stranded. [micro RNA p1]
- A microRNA binds to the 3' untranslated region (3' UTR) of its target mRNA. [micro RNA p1]
- In nomenclature, "mir" denotes the precursor microRNA and "miR" denotes the mature microRNA. [micro RNA p1]
- MicroRNA genes are located either in introns or within exons of non-coding genes. [micro RNA p1]
- A microRNA can act as a tumour suppressor gene or as an oncogene. [micro RNA p1]
- MicroRNAs silence genes by causing mRNA degradation and inhibiting translation. [micro RNA p1]
- MicroRNA biogenesis: the miRNA gene is transcribed by RNA polymerase II into pri-miRNA, which is processed to pre-miRNA in the nucleus and then to mature miRNA in the cytoplasm. [micro RNA p2]
- Mature miRNA joins the RNA-induced silencing complex (RISC) to form the RNA-induced silencing unit, which targets the 3' UTR of mRNA and degrades it. [micro RNA p2]
- MicroRNAs may help define cancer subtypes, for example basal versus luminal breast cancer. [micro RNA p2]
- MicroRNAs may help define the developmental origin of tumours. [micro RNA p2]
- miR-143 and miR-187 are potential urine-based markers in prostate cancer. [micro RNA p2]
- miR-187 is an independent predictor of prostate cancer on biopsy. [micro RNA p2]
- MicroRNAs can act as prognostic markers, for example in bronchogenic carcinoma. [micro RNA p2]
- A microRNA can modulate an entire gene programme, affecting hundreds of transcripts and shutting down entire deregulated pathways. [micro RNA p3]
- RNA interference therapeutics include siRNA and shRNA, which target a single transcript, and miRNA. [micro RNA p3]
- PSMA-based RNA therapy in prostate cancer is an example of single-transcript RNA interference therapy. [micro RNA p3]
- MicroRNA replacement therapy uses drug mimics of the endogenous microRNA, for example miR-16. [micro RNA p3]
- MicroRNA inhibitor therapy uses drugs that target the microRNA, for example miR-122. [micro RNA p3]
- miR-141 has a role in tumour prognosis. [micro RNA p4]
- miR-122 has a role in sepsis. [micro RNA p4]

- MicroRNAs also have roles in cardiovascular disease, inflammation, diabetes and perioperative organ protection. [micro RNA p4]

CH 59 · DIFFERENCES OF SEXUAL DEVELOPMENT

Klinefelter syndrome

- Further Klinefelter karyotype variants include 48,XXYY, 49,XXXYY and 49,XXXXY (in addition to 48,XXXY). [DSD p2]
- Klinefelter syndrome predisposes to Leydig cell tumour and Sertoli cell tumour (as well as breast cancer and extragonadal germ cell tumour). [DSD p3]
- Post-pubertal Klinefelter patients need surveillance for malignancy, with scrotal ultrasound every 4 months. [DSD p3]

Turner syndrome

- Turner mosaicism may be 45,X/46,XX as well as 45,X/46,XY. [DSD p4]
- Turner syndrome results from non-disjunction during gametogenesis or from an error in mitosis. [DSD p4]
- Turner syndrome occurs in about 1 in 2500 live births. [DSD p4]
- Turner syndrome is associated with neurophysiological and cognitive deficits. [DSD p4]
- Facial features of Turner syndrome include ptosis, hypertelorism, retrognathia and ear malformations. [DSD p4]
- Body features of Turner syndrome include webbed neck, broad chest with widely spaced nipples, lymphoedema and peripheral oedema at birth, pigmentation, hypoplastic nails, a short 4th metacarpal and cubitus valgus. [DSD p5]
- Mitral valve prolapse is a further cardiac association of Turner syndrome. [DSD p5]
- Renal anomalies in Turner syndrome include, besides horseshoe kidney, duplication, renal agenesis, malrotation and multiple renal arteries. [DSD p5]
- Turner syndrome may be diagnosed prenatally from ultrasound findings (renal, cardiac) or abnormal fetal karyotype, and most affected pregnancies end in spontaneous abortion. [DSD p5]
- Turner syndrome should be considered in any infant with lymphoedema. [DSD p5]
- Turner syndrome should be considered in any young woman with short stature and primary amenorrhoea. [DSD p6]
- Turner syndrome shows elevated gonadotropins with low oestrogen (and testosterone). [DSD p6]
- Assisted reproduction in Turner syndrome carries a high rate of miscarriage, stillbirth and malformed infants. [DSD p6]

Ovotesticular DSD (true hermaphroditism)

- Karyotype in ovotesticular DSD is 46,XX in 60%, mosaic in 33% and 46,XY in 7%. [DSD p7]
- The more common gonadal arrangement in ovotesticular DSD is one or two ovotestes rather than a separate ovary and testis. [DSD p7]
- In ovotesticular DSD the ovary is more commonly on the left side, and the testis or ovotestis more commonly on the right. [DSD p7]
- An ovotestis is accompanied by a fallopian tube in two-thirds of patients and a vas in one-third. [DSD p7]
- Ovotesticular DSD typically shows clitoromegaly or hypospadias with chordee. [DSD p7]
- The approximately 3% gonadal tumour risk in ovotesticular DSD applies to 46,XY patients; tumours are rare with 46,XX. [DSD p7]
- Patients with ovotesticular DSD raised female have a potential for fertility, unlike other forms of gonadal dysgenesis; all testicular and Wolffian tissue is removed and surveillance for gonadal tumours is still advisable. [DSD p8]
- Male rearing was historically the most common in ovotesticular DSD; all ovarian and Müllerian tissue is removed. [DSD p8]

- In ovotesticular DSD raised male, gonadectomy at puberty with androgen replacement should be considered, because of high malignancy risk and unlikely male fertility; if the gonads are kept, long-term surveillance by scrotal ultrasound is needed. [DSD p8]

Bilateral vanishing testis

- Bilateral vanishing testis must be distinguished from pure gonadal dysgenesis, in which there is no evidence of testicular function in utero. [DSD p9]
- Causes of bilateral vanishing testis include genetic mutation and teratogenesis as well as bilateral torsion. [DSD p9]

Congenital adrenal hyperplasia

- Cholesterol side-chain cleavage enzyme deficiency is one of the five enzyme defects that cause CAH. [DSD p11]
- The 21-hydroxylase gene lies on chromosome 6 within the HLA complex. [DSD p12]
- Boys with non-salt-wasting 21-hydroxylase deficiency show isosexual precocity from about 2–3 years: enlarged penis, scrotum and prostate, deepening voice, pubic hair, acne and muscle development. [DSD p12]
- Long-term consequences in these boys are short stature and infertility. [DSD p12]
- Salt-losing CAH presents in the first few weeks with progressive weight loss, dehydration and vomiting. [DSD p13]
- The 11β-hydroxylase gene lies on chromosome 8 and is not HLA-linked. [DSD p13]
- Hypertension in 11β-hydroxylase deficiency is due to raised deoxycorticosterone, and severe forms show marked virilisation. [DSD p13]
- 11β-hydroxylase deficiency is diagnosed by raised plasma 11-deoxycortisol and deoxycorticosterone (DOC). [DSD p13]
- 3β-HSD deficiency is autosomal recessive and is diagnosed by raised serum 17-OH pregnenolone (with DHEA). [DSD p14]
- Prenatal diagnosis of CAH was formerly by amniotic fluid 17-OH progesterone at 16–17 weeks; it is now by HLA genotyping or DNA analysis of chorionic villus cells at 9–11 weeks. [DSD p14]
- Prenatal treatment of CAH is dexamethasone to the mother, started no later than 9 weeks after the LMP; it crosses the placenta, suppresses fetal ACTH and prevents virilisation. [DSD p14]
- Only a female fetus needs prenatal dexamethasone; detecting the Y chromosome by PCR of free fetal DNA as early as 9 weeks avoids treating male fetuses. [DSD p15]
- The fludrocortisone dose for salt-losing CAH is 0.1–0.2 mg/day. [DSD p15]
- Prophylactic adrenalectomy is an option in salt losers and extremely virilised females when adrenal suppression is harder to maintain than adrenal crisis is to prevent; it gives better quality of life and fertility on replacement therapy. [DSD p15]

Persistent Müllerian duct syndrome and MRKH

- In PMDS the uterus and upper vagina drain into the prostatic utricle. [DSD p16]
- The MIS (AMH) gene in PMDS is on chromosome 19. [DSD p16]
- About 10% of MRKH cases are atypical, with a greater chance of urinary tract anomaly. [DSD p16]

Evaluation of the newborn with ambiguous genitalia

- A uterus may be palpable on rectal examination as an anterior midline cord-like structure. [DSD p17]
- On ultrasound, a cyst within a gonad suggests an ovotestis. [DSD p18]
- PCR characterisation of the androgen receptor is a further investigation in ambiguous genitalia. [DSD p18]
- In ambiguous genitalia with impalpable gonads, Müllerian structures, normal 17-OH progesterone and a 46,XX karyotype, consider non-adrenal masculinisation such as aromatase deficiency. [DSD p21]

OTHER TOPICS IN THIS BOOK

Next-generation sequencing

- Next-generation sequencing is also called second-generation sequencing or massively parallel sequencing. [Immunotherapy p9]
- NGS is a DNA and RNA sequencing technique that differs from first-generation (Sanger) sequencing. [Immunotherapy p9]
- DNA sequencing is the process of determining the precise order of nucleotides within a DNA molecule. [Immunotherapy p9]
- The advantages of NGS are low cost, speed and high yield. [Immunotherapy p9]
- Clinical uses of NGS include sequencing entire cancer genomes, detecting pathogens, discovering new tumour markers, and providing accurate prognostic and diagnostic data. [Immunotherapy p9]

3 Infections & Inflammation

CH 26 · INFECTIONS OF THE URINARY TRACT

Asymptomatic bacteriuria

- In institutionalised patients, asymptomatic bacteriuria is commonly caused by Pseudomonas and Proteus. [Baecteruria in diff Situations p1]
- Antimicrobial treatment may be considered for asymptomatic women whose catheter-associated bacteriuria persists 48 hours after the catheter is removed. [Baecteruria in diff Situations p1]

UTI in spinal cord injury patients: prevalence, risk factors and catheters

- About 33% of SCI patients have bacteriuria at any given time, and eventually nearly all become bacteriuric. [Baecteruria in diff Situations p2]
- Risk factors for UTI in SCI are impaired voiding, bladder overdistension, elevated intravesical pressure, obstruction, vesicoureteric reflux, instrumentation and stones. [Baecteruria in diff Situations p2]
- Further risk factors in SCI are reduced fluid intake, poor hygiene, perineal colonisation, decubitus ulcers, local trauma and the reduced immunity of chronic illness. [Baecteruria in diff Situations p2]
- With a suprapubic catheter, bacteriuria may start later than with an indwelling urethral catheter, but the eventual infection rate is the same. [Baecteruria in diff Situations p2]
- Condom drainage carries a high rate of bacteriuria (60–100%). [Baecteruria in diff Situations p2]

SCI: clinical features and organisms

- Other symptoms of UTI in SCI are flank, back or abdominal discomfort, malaise or lethargy, and malodorous urine. [Baecteruria in diff Situations p3]
- In SCI, the common urinary pathogens are enterococci, Proteus mirabilis and Pseudomonas, followed by E. coli, Klebsiella, staphylococci, Candida and Serratia. [Baecteruria in diff Situations p3]
- In SCI, the two most persistent species are E. coli and Providencia stuartii. [Baecteruria in diff Situations p3]

SCI: antibiotic choice and failure to improve

- For afebrile SCI patients with UTI, an oral fluoroquinolone is the agent of choice. [Baecteruria in diff Situations p4]
- SCI patients with fever or chills are treated with a penicillin plus an aminoglycoside, or with a third-generation cephalosporin. [Baecteruria in diff Situations p4]
- If there is no clinical improvement within 24–48 hours, re-culture the urine and adjust therapy to the previous culture report. Image the tract to exclude obstruction, stones or abscess. [Baecteruria in diff Situations p4]

SCI: follow-up

- After treatment in SCI, post-therapy cultures are usually not needed, because asymptomatic recolonisation is common and not significant. [Baecteruria in diff Situations p5]

- If a urea-splitting organism was identified, obtain a follow-up culture to confirm that it has been eradicated. [Baecteruria in diff Situations p5]

Catheter-associated bacteriuria

- With short-term catheters, 10–30% of patients with catheter-associated bacteriuria have typical UTI symptoms. [Baecteruria in diff Situations p6]
- With long-term catheters, symptomatic infection occurs at about 1 episode per 100 days of catheterisation. [Baecteruria in diff Situations p6]

CAUTI: prevention and treatment

- Adding silver oxide (as well as silver alloy) to the catheter, and hydrogen peroxide to the drainage bag, has been reported to lower the incidence of catheter-associated bacteriuria. [Baecteruria in diff Situations p7]
- In symptomatic CAUTI, antimicrobial therapy is stopped within 48 hours of resolution of the infection. [Baecteruria in diff Situations p7]

Catheter removal with probable bacteriuria

- When a catheter is to be removed and bacteriuria is likely, or the dipstick is positive, send a urine culture 24 hours before removal. [Baecteruria in diff Situations p8]
- Start empirical TMP-SMX or a fluoroquinolone just before decatheterisation and continue it for 2 days. [Baecteruria in diff Situations p8]
- Send a post-therapy culture 7–10 days later to document eradication. [Baecteruria in diff Situations p8]

Bacteriuria in the elderly: prevalence and pathogenesis

- After the age of 65 years, about 20% of women and 10% of men have bacteriuria, and prevalence rises with age. [Baecteruria in diff Situations p9]
- The pathogenesis is multifactorial and poorly understood. Probable factors are an age-related decline in cell-mediated immunity, neurogenic bladder dysfunction, more perineal soiling from incontinence, greater catheter use, and oestrogen depletion with vaginal changes in women. [Baecteruria in diff Situations p9]
- E. coli remains the commonest organism in the elderly, but Proteus, Klebsiella, Enterobacter, Serratia, Pseudomonas and enterococci become significantly more frequent. [Baecteruria in diff Situations p9]

Elderly: diagnosis and management

- In the elderly, concomitant diseases can mask or mimic UTI, so a high index of suspicion is needed. [Baecteruria in diff Situations p10]
- Urinary tract abnormalities are often present in elderly patients with UTI, so a thorough urological evaluation is needed (ultrasound, CT, urodynamics, cystoscopy). [Baecteruria in diff Situations p10]
- The goal of treating symptomatic bacteriuria in the elderly is to eliminate symptoms, not to sterilise the urine. [Baecteruria in diff Situations p10]

Elderly: prompt therapy and fluoroquinolone cautions

- Elderly patients need prompt therapy if the UTI occurs with structural urinary tract abnormalities or systemic disease such as diabetes, or is caused by a urea-splitting organism, because sepsis and its sequelae are common in the elderly. [Baecteruria in diff Situations p11]
- Fluoroquinolones work well in the elderly, but use them with caution when there is QT prolongation or uncorrected hypokalaemia or hypomagnesaemia. [Baecteruria in diff Situations p11]
- Systemic diseases such as diabetes and structural abnormalities such as obstruction should also be controlled. [Baecteruria in diff Situations p11]

Bacteriuria in pregnancy: natural history

- Of pregnant women who develop pyelonephritis, 10–20% have another episode before or just after delivery. [Baecteruria in diff Situations p12]

Pregnancy: clinical features, complications and screening

- Pyelonephritis in pregnancy may also cause nausea and vomiting. [Baecteruria in diff Situations p13]

- Stillbirth (fetal death) was a complication of bacteriuria in pregnancy mainly in the pre-antibiotic era, and the link with maternal anaemia is reported inconsistently. [Baecteruria in diff Situations p13]
- If the first-trimester screening culture shows no growth, repeat cultures are unnecessary, because these women are unlikely to develop bacteriuria later in pregnancy. [Baecteruria in diff Situations p13]

Pregnancy: prophylaxis

- Low-dose prophylaxis in pregnancy is indicated for women with a history of recurrent UTI and VUR before conception. [Baecteruria in diff Situations p14]
- Post-coital prophylaxis in pregnancy (cephalexin 250 mg or nitrofurantoin 50 mg) is given once the treatment course for the UTI is complete. [Baecteruria in diff Situations p14]
- If a follow-up culture is positive, unresolved infection needs a properly chosen alternative drug, while bacterial persistence or rapid reinfection needs prophylaxis or antimicrobial suppression. [Baecteruria in diff Situations p14]

Antibiotics in pregnancy

- Oral penicillin doses in pregnancy are ampicillin 500 mg four times daily (extensively used), amoxicillin 250 mg three times daily and penicillin V 500 mg four times daily. [Baecteruria in diff Situations p16]
- Oral cephalosporin doses in pregnancy are cephalexin 500 mg four times daily (extensively used) and cefaclor 500 mg four times daily. [Baecteruria in diff Situations p16]
- The oral dose of nitrofurantoin in pregnancy is 100 mg four times daily. [Baecteruria in diff Situations p16]
- Chloramphenicol is avoided in pregnancy because it causes grey baby syndrome. [Baecteruria in diff Situations p16]

Pregnancy-related changes in the kidney

- In pregnancy, renal length increases by about 1 cm because renal vascular and interstitial volume increase. Renal histology does not change. [Baecteruria in diff Situations p17]
- Physiological changes in pregnancy include increased renal plasma flow, increased urinary protein excretion and upregulation of the RAAS. [Baecteruria in diff Situations p17]
- Hyperuricosuria, as well as hypercalciuria, occurs in pregnancy. [Baecteruria in diff Situations p17]
- Average BUN falls from 13.0 in non-pregnant women to 9.0 in pregnancy, and average creatinine from 0.7 to 0.5, so values normal outside pregnancy may mean renal insufficiency during pregnancy. [Baecteruria in diff Situations p17]

Pregnancy-related changes in the ureter, bladder and urethra

- Ureteric peristalsis decreases in pregnancy. [Baecteruria in diff Situations p18]
- Right-sided hydroureteronephrosis is commoner because the right ureter crosses the iliac and ovarian vessels at an angle before it enters the pelvis. [Baecteruria in diff Situations p18]
- In pregnancy, the bladder is displaced superiorly and anteriorly and becomes hyperaemic, possibly congested. [Baecteruria in diff Situations p18]
- Bladder capacity increases in pregnancy because progesterone relaxes smooth muscle. [Baecteruria in diff Situations p18]
- Bladder hypertrophy in pregnancy may be caused by oestrogen stimulation. [Baecteruria in diff Situations p18]
- Oestrogen stimulation in pregnancy may cause squamous changes in the urethra. [Baecteruria in diff Situations p18]

Urinalysis (CUE): collection and physical examination

- In women, a midstream sample is taken after cleaning the vulva and separating the labia. [CUE Campbell p1]
- A bag specimen from an infant should be examined within 1 hour. [CUE Campbell p1]
- If examination is delayed, urine should be stored at 5 °C. [CUE Campbell p1]
- Normal urine is pale yellow from urochrome, and brown-black urine suggests alkaptonuria. [CUE Campbell p1]
- Turbid urine is often due to phosphaturia, which clears with acetic acid. [CUE Campbell p1]

- Pyuria gives turbid urine with a pungent smell and needs microscopy. [CUE Campbell p1]
- Other causes of turbid urine are lipiduria, hyperoxaluria and hyperuricosuria. [CUE Campbell p1]
- Normal urine specific gravity is 1.001–1.035. [CUE Campbell p2]
- A fixed specific gravity of 1.010 indicates renal insufficiency (isosthenuria). [CUE Campbell p2]
- Urine osmolality ranges from 50 to 1200 mOsm/L; it is not measured on a dipstick but is a better indicator of renal function. [CUE Campbell p2]
- Normal urine pH is 4.5–8; 4.5–5.5 is acidic and 6.5–8 alkaline. [CUE Campbell p2]
- Urine pH usually reflects blood pH, the exception being renal tubular acidosis. [CUE Campbell p2]
- The standard dipstick tests blood, protein, glucose, ketones, urobilinogen, bilirubin and white cells on a fresh uncentrifuged specimen. [CUE Campbell p2]
- Pyridium (phenazopyridine) makes dipstick examination unreliable. [CUE Campbell p2]
- The dipstick blood test relies on the peroxidase activity of haemoglobin oxidising orthotoluidine, which changes colour. [CUE Campbell p3]
- Discrete coloured dots on the dipstick indicate intact red cells (below about 250 RBC/mL); the dots coalesce above 250/mL, and a uniform field change indicates haemoglobinuria or myoglobinuria. [CUE Campbell p3]
- To separate them, centrifuge a blood sample: pink plasma means haemoglobinuria (free haemoglobin binds haptoglobin), clear plasma means myoglobinuria. [CUE Campbell p3]
- Normal red cell excretion is about 1000 RBC/mL of urine. [CUE Campbell p3]
- Dipstick blood may be falsely positive with menstrual blood, a first-morning sample, dehydration and vigorous exercise. [CUE Campbell p3]

Urinalysis: glucose, ketones, bilirubin, leucocyte esterase

- Glucose appears on the urine dipstick once blood glucose exceeds 180 mg/dL. [CUE Campbell p5]
- The dipstick glucose test is a double oxidation: glucose oxidase forms gluconic acid and hydrogen peroxide, and peroxidase then produces a colour change; it does not cross-react with other sugars. [CUE Campbell p5]
- The ketone bodies are acetoacetic acid, acetone and β-hydroxybutyric acid; the bracket in the notes marks them as not detected on the dipstick (standard sources say only β-hydroxybutyric acid is missed). [CUE Campbell p5]
- Dipstick ketone testing uses sodium nitroprusside, which turns purple with acetoacetic acid. [CUE Campbell p5]
- Bilirubin is absent from normal urine, whereas a small amount of urobilinogen is normal; both are detected by a diazonium salt colorimetric reaction. [CUE Campbell p5]
- The ChemStrip-LN dipstick uses a diazonium salt: neutrophil leucocyte esterase converts an indoxyl carbonic acid ester to indoxyl within about 5 minutes, and indoxyl oxidises the diazonium salt to change colour. [CUE Campbell p6]
- A false-positive leucocyte esterase result is due to contamination. [CUE Campbell p6]

Urinalysis: microscopy

- For microscopy, 10–15 mL of urine is centrifuged for 5 minutes at 3000 rpm, and the sediment (0.01–0.02 mL) is placed on a slide under a coverslip. [CUE Campbell p6]
- The sediment is examined at low power (100×) and high power (400×); the whole area under the coverslip is scanned at low power, with attention to the edges, where casts collect. [CUE Campbell p6]
- Low power shows red cells, white cells, casts, parasites, cystine crystals and macrophages; high power separates circular from dysmorphic red cells and shows other crystals, bacteria and yeast. [CUE Campbell p7]
- Red cell morphology is assessed on high power or by phase contrast; red cells can be confused with yeast (which buds) or fat droplets (which are refractile). [CUE Campbell p7]
- Normal urine white cell counts are 1–2/HPF in males and up to 5/HPF in females. [CUE Campbell p7]
- Glitter cells are white cells whose cytoplasmic granules show glittering movement when specific gravity is below 1.019. [CUE Campbell p7]

- Urine cytology is done by Pap smear or automated flow cytometry. [CUE Campbell p7]
- A cast is a coagulum of Tamm–Horsfall protein in the renal tubule that traps any luminal content within its matrix. [CUE Campbell p7]
- Hyaline casts (mucoprotein) have no pathological significance. [CUE Campbell p7]
- Broad granular or waxy casts indicate chronic kidney disease, and fatty casts indicate nephrotic syndrome. [CUE Campbell p7]
- One high-power field corresponds to about 1/20,000 to 1/50,000 mL of urine. [CUE Campbell p8]
- One bacterium per HPF means more than 30,000 bacteria/mL, and 5 per HPF means more than 100,000 (10^5)/mL, the criterion used in women. [CUE Campbell p8]
- Any bacteria in a male urine sample should be evaluated by urine culture. [CUE Campbell p8]
- Candida albicans in urine can be confused with red cells and calcium oxalate crystals but shows characteristic budding; it indicates contamination, vaginal candidiasis or diabetes. [CUE Campbell p8]
- Parasites seen on urine microscopy include Trichomonas vaginalis and Schistosoma haematobium. [CUE Campbell p8]

Urosepsis

- SIRS and sepsis-like states also have non-infectious causes such as trauma and burns. [Compilation 04-3 p24]
- Gram-negative lipopolysaccharide binds CD14 with TLR4 on the macrophage membrane, whereas Gram-positive teichoic acid and lipoteichoic acid bind CD14 with TLR2. [Compilation 04-3 p24]
- The pro-inflammatory cytokines in sepsis include IL-2 and interferon-gamma, and platelet-activating factor is also released. [Compilation 04-3 p24]
- Laboratory abnormalities in sepsis include leucocytosis or leucopenia, hyperbilirubinaemia, hyperglycaemia and raised CRP. [Compilation 04-3 p24]
- Respiratory alkalosis is an early change in sepsis, and metabolic acidosis appears in advanced shock. [Compilation 04-3 p24]
- Gram-negative urosepsis organisms also include Providencia, Serratia and Enterobacter, and Gram-positive urosepsis is caused by staphylococci and streptococci. [Compilation 04-3 p24]
- Dopamine is listed with noradrenaline as a vasoactive agent in septic resuscitation. [Compilation 04-3 p24]
- Antibiotics in sepsis are continued until the patient has been afebrile for 3–4 days and is clinically stable. [Compilation 04-3 p24]
- Source control in urosepsis may even require nephrectomy, for pyonephrosis or emphysematous pyelonephritis. [Compilation 04-3 p24]

Bladder Infections: Uncomplicated Cystitis

- Dehydration is a UTI risk factor because it reduces urine flow. [Compilation 04-3 p29]
- Factors that facilitate bacterial ascent include faecal incontinence and residual urine with ischaemia of the bladder wall, as well as catheterisation and urinary incontinence. [Compilation 04-3 p29]
- Acute cystitis is by definition a superficial infection of the bladder mucosa, so fever, chills and other signs of dissemination are absent. [Compilation 04-3 p29]
- Foul-smelling urine is a presenting feature of cystitis. [Compilation 04-3 p29]
- Urine culture is not indicated in every patient with uncomplicated cystitis; the indications are a doubtful diagnosis and recent antibiotic therapy. [Compilation 04-3 p30]
- Miscellaneous non-inflammatory causes of urethral discomfort are a differential diagnosis of cystitis. [Compilation 04-3 p30]
- In the cystitis algorithm, patients without pyuria or bacteriuria are assessed for vaginitis or urethritis, and those with complicating factors have a culture before treatment. [Compilation 04-3 p31]
- If symptoms persist on therapy, a culture on therapy is taken: a negative culture needs no further assessment, while a positive culture means unresolved infection, managed by susceptibility testing (new antimicrobials if resistant, urological evaluation if susceptible and compliant). [Compilation 04-3 p31]
- Symptoms with a positive urinalysis after therapy has stopped are classed as recurrence. [Compilation 04-3 p31]

- Fluoroquinolone regimens for uncomplicated cystitis in women are ciprofloxacin 250 mg twice daily or levofloxacin 250 mg once daily for 3 days. [Compilation 04-3 p32]
- For healthy men under 50 years with cystitis, the regimens are TMP-SMX double-strength, ciprofloxacin 500 mg twice daily or levofloxacin 500 mg once daily, each for 7 days. [Compilation 04-3 p32]
- Nitrofurantoin is the best choice for a patient who has recently taken antibiotics, because resistance to it is minimal. [Compilation 04-3 p32]
- Nitrofurantoin does not eliminate E. coli from the vagina. [Compilation 04-3 p32]
- Cephalosporin resistance among cystitis pathogens is below 5%. [Compilation 04-3 p32]
- Cephalosporins eradicate vaginal E. coli but alter the normal vaginal flora only minimally. [Compilation 04-3 p32]

Complicated Cystitis

- Complicated cystitis can range from mild cystitis to life-threatening kidney infection and urosepsis. [Compilation 04-3 p33]
- For mild to moderate complicated cystitis due to E. coli, Proteus, Klebsiella or Pseudomonas, treated as an outpatient without nausea or vomiting, empirical therapy is ciprofloxacin 500 mg twice daily or TMP-SMX one double-strength tablet twice daily for 10–14 days. [Compilation 04-3 p33]
- Severe illness, possible urosepsis or hospitalisation, with organisms such as Serratia, enterococci and staphylococci, is treated with parenteral ampicillin 1 g 6-hourly plus gentamicin 1 mg/kg 8-hourly. [Compilation 04-3 p33]
- Other parenteral options for severe complicated cystitis are ciprofloxacin 400 mg twice daily, levofloxacin 250 mg once daily, aztreonam 1 g two or three times daily, imipenem 250–500 mg three or four times daily, and ticarcillin–clavulanate 3.1 g 6-hourly. [Compilation 04-3 p33]
- After parenteral therapy, complicated cystitis is completed with an oral cephalosporin or TMP-SMX for 10–14 days. [Compilation 04-3 p33]

Unresolved UTI

- Unresolved UTI means initial therapy has failed to eliminate symptoms and/or bacterial growth, with the same pathogen and the same resistance profile. [Compilation 04-3 p35]
- In azotaemia and papillary necrosis, the urinary antibiotic concentration may fall below the MIC. [Compilation 04-3 p35]
- In Munchausen syndrome, patients secretly inoculate the bladder with uropathogens and frequently miss doses. [Compilation 04-3 p35]
- The best initial empirical therapy for unresolved UTI is a fluoroquinolone for 7 days. [Compilation 04-3 p35]
- Follow-up cultures in unresolved UTI are taken during therapy and 7 days after therapy. [Compilation 04-3 p35]

Recurrent UTI: Bacterial Persistence

- Further correctable causes of bacterial persistence are ureteric duplication, infected periurethral glands, unilateral medullary sponge kidney, an infected urachal cyst, an infected communicating cyst of the renal calyces, papillary necrosis and a perivesical abscess with a fistula to the bladder. [Compilation 04-3 p36]
- The infected ureteric stump causing persistence after nephrectomy is typically normal-appearing and non-refluxing. [Compilation 04-3 p36]

Recurrent UTI: Reinfection

- Reinfections usually occur at varying intervals, most often in women and girls, from colonisation by bowel flora. [Compilation 04-3 p37]
- Diaphragms and spermicides containing nonoxynol-9 increase adherence of E. coli to vaginal cells and reduce vaginal lactobacilli. [Compilation 04-3 p37]
- Abnormalities that reduce the formation, transmission and elimination of urine make the patient susceptible to reinfection. [Compilation 04-3 p37]

Management of Recurrent UTI

- In recurrent UTI with a negative culture, causes such as vaginitis, urethritis and interstitial cystitis are evaluated. [Compilation 04-3 p38]
- In reinfection without risk factors and not related to coitus, the choice is low-dose prophylaxis or self-start therapy; reinfection with risk factors goes to urological evaluation. [Compilation 04-3 p38]
- The rationale of low-dose continuous prophylaxis is bactericidal urine drug levels and elimination of pathogenic bacteria from introital and bowel reservoirs, using agents that should not cause bacterial resistance. [Compilation 04-3 p39]
- TMP-SMX for prophylaxis eradicates Gram-negative aerobic flora from the bowel and vagina, at a dose of 40–200 mg daily at night. [Compilation 04-3 p39]
- Nitrofurantoin prophylaxis repeatedly eliminates bacteria from the urine, but colonisation of the vaginal introitus by Enterobacteriaceae continues throughout therapy. [Compilation 04-3 p39]
- Cephalexin prophylaxis is 250 mg once daily at night. [Compilation 04-3 p40]
- Other cephalosporins are expensive and are used for prophylaxis only when the other three drugs cannot be used because of resistance or intolerance. [Compilation 04-3 p40]
- Continuous prophylaxis is usually stopped after 6 months and may be given nightly or on alternate nights. [Compilation 04-3 p40]
- Continuous prophylaxis decreases recurrences by 95%. [Compilation 04-3 p40]
- After stopping prophylaxis, 30% have spontaneous remissions lasting up to 6 months, but many remissions are followed by reinfection, which is treated with low-dose prophylaxis again or another strategy. [Compilation 04-3 p40]
- Patients on prophylaxis are monitored every 1–3 months; a breakthrough infection is treated with a full dose of the same antibiotic or a culture-directed drug, and then prophylaxis is restarted. [Compilation 04-3 p40]
- For self-start intermittent therapy, the patient is given a dip-slide device and performs a urine culture when symptoms occur, then starts a 3-day full-dose empirical course. [Compilation 04-3 p41]
- Fluoroquinolones are ideal for self-start therapy, with nitrofurantoin or TMP-SMX as alternatives, and the next culture is taken 7–10 days after therapy. [Compilation 04-3 p41]
- Post-coital prophylaxis may use a single dose of nitrofurantoin, cephalexin, TMP-SMX or a fluoroquinolone. [Compilation 04-3 p41]
- Hygiene, timing and frequency of voiding, and avoiding hot tubs are other strategies, with no conclusive evidence. [Compilation 04-3 p41]

Urinary Tract Infection: General Considerations

- Renal concentrating problems are a functional abnormality that makes a UTI complicated. [Compilation 04 p1]
- A history of childhood UTI is listed as a factor that makes a UTI complicated. [Compilation 04 p1]
- Bacteroides is listed among the organisms of community-acquired UTI. [Compilation 04 p3]
- Providencia is listed among the organisms of hospital-acquired UTI. [Compilation 04 p3]

UPEC Pathogenesis, Virulence Factors and Bacterial Adherence

- Bacterial virulence factors may be intrinsic to the bacterium (chromosomally encoded) or obtained from mobile genetic elements such as plasmids. [Compilation 04 p5]
- The UPEC-specific autotransporter Sat is toxic to urinary tract cells. [Compilation 04 p6]
- UPEC iron acquisition systems include the IroN system as well as aerobactin. [Compilation 04 p6]
- Bacterial adherence is influenced by the adhesive capacity of the bacterium, the receptive capacity of the host epithelium and the fluid bathing both surfaces. [Compilation 04 p6]
- Adhesins are fimbrial (pili, made of pilin: type 1, P, S and F1C pili) or afimbrial (the Dr adhesin family). [Compilation 04 p7]
- Type 1 pili are composed of repeating FimA subunits and mediate haemagglutination of guinea pig red cells, which is inhibited by mannose. [Compilation 04 p7]
- P pili cause haemagglutination of human red cells, which is not inhibited by mannose. [Compilation 04 p7]

- S pili bind sialic acid residues and are associated with both bladder and kidney infection. [Compilation 04 p8]
- F1C pili bind glycosphingolipids in renal epithelial cells. [Compilation 04 p8]
- Dr adhesins bind to many locations throughout the urinary tract. [Compilation 04 p8]
- Phase variation has a cost: type 1 pili help bacteria adhere to and colonise the bladder mucosa, but they also enhance phagocytosis and killing. [Compilation 04 p8]

Antimicrobial Resistance

- An example of inherited chromosomal resistance is that Proteus and Pseudomonas are always resistant to nitrofurantoin. [Compilation 04 p9]
- Transferable plasmid-mediated resistance is also called R-factor resistance, and all antimicrobial agents can cause it except nitrofurantoin, for which plasmid-mediated resistance has not been reported. Fluoroquinolone resistance is rarely transmitted by plasmids. [Compilation 04 p10]
- Resistance to nitrofurantoin is uncommon, and its mechanism is not fully known. [Compilation 04 p11]

Natural Defence Against UTI

- The normal flora of the introitus, periurethral area and urethra include corynebacteria, coagulase-negative staphylococci and streptococci as well as lactobacilli. [Compilation 04 p13]
- Cervical IgA and vaginal pH, as well as oestrogen, antimicrobials and spermicides, can change the normal flora. [Compilation 04 p13]
- Uromodulin (Tamm–Horsfall protein) saturates the mannose-binding sites of type 1 pili and so blocks bacterial binding to the urothelium. [Compilation 04 p13]
- Bacterial lipopolysaccharide and peptidoglycan are recognised by Toll-like receptors (TLR-4 and TLR-11) on superficial bladder cells, which triggers innate immune and inflammatory responses that kill the bacteria. [Compilation 04 p14]
- Cells of the innate immune response in the bladder are PMN leucocytes (neutrophils), macrophages, eosinophils, NK cells, mast cells and dendritic cells. [Compilation 04 p14]
- PMN leucocytes increase transcription of inducible nitric oxide synthase (iNOS), and the resulting high levels of NO and its breakdown products are toxic to bacteria. [Compilation 04 p14]
- Type 1 piliated bacteria induce exfoliation by programmed cell death of superficial cells. This can allow later invasion of deeper tissue, so it may be a bacterial offence manoeuvre rather than a host defence. [Compilation 04 p15]
- Strong exfoliation leaves no IBCs, whereas mild exfoliation allows IBCs and biofilms to form, which leads to recurrent UTI. [Compilation 04 p15]

Conditions Altering Host Defence

- Amyloidosis and contrast media injection are listed among the causes of renal papillary necrosis. [Compilation 04 p16]
- Diabetes increases asymptomatic and symptomatic UTI in women (the data in men are doubtful), and the infections are more severe and caused by atypical organisms. [Compilation 04 p17]
- In HIV, UTIs are 5-fold more common than in the normal population. They are complicated, tend to recur and need longer treatment. [Compilation 04 p17]

Clinical Features of UTI

- Lower tract symptoms appear several days before upper tract symptoms. [Compilation 04 p19]

Diagnosis of UTI: urine collection

- Circumcised men need no preparation for a voided sample. In uncircumcised men, retract the foreskin and clean the glans with soap and water. [Compilation 04 p20]
- In women, spread the labia and wash the periurethral area with moist gauze, without antiseptics (they give false-negative results), then take a clean-catch midstream sample. [Compilation 04 p20]
- In women, a mid-catheter specimen can be used if voided specimens are contaminated. Because of the risk of iatrogenic infection, give a single dose of antibiotic (such as oral TMP-SMX) in high-risk cases. [Compilation 04 p20]

- In men, catheterisation for a urine sample is not indicated unless the patient cannot void. [Compilation 04 p20]
- Suprapubic aspiration is highly accurate but slightly invasive. With a full bladder, shave the skin, clean it with an alcohol sponge and withdraw 20 mL with an LP needle (5 mL for culture and 15 mL for urinalysis). It is used in neonates and SCI patients. [Compilation 04 p20]
- False-negative results occur early in infection, when bacteria and white cells are few, or when urine is diluted by high fluid intake and diuresis. [Compilation 04 p21]

Urine analysis

- Microscopic bacteria are found in more than 90% of infections with counts of 10^5 CFU/mL. [Compilation 04 p22]
- Microscopy gives false negatives at counts of 10^2–10^4 CFU/mL, because only a limited volume of urine is seen. A negative urinalysis for bacteria never excludes bacteria at 3000/mL or fewer. [Compilation 04 p22]
- Bacteriuria on urinalysis has a sensitivity of 40–70% and a specificity of 85–95% for UTI. [Compilation 04 p22]
- Microscopic haematuria is found in 40–60% of cases of cystitis. [Compilation 04 p23]
- Glomerulonephritis and interstitial cystitis are also causes of sterile pyuria. [Compilation 04 p23]

Urine culture and localisation of UTI

- Direct surface plating uses a split agar plate: blood agar grows both Gram-positive and Gram-negative bacteria, and EMB or deoxycholate agar grows Gram-negative bacteria only. [Compilation 04 p25]
- Dip-slides carry MacConkey or EMB agar on one side (Gram-negative organisms) and soy agar on the other (all bacteria), and the slide is dipped in the urine sample. [Compilation 04 p25]
- Culture usually needs overnight incubation, and the urine should be refrigerated and cultured within 24 hours of collection. [Compilation 04 p25]
- 20–40% of women with symptomatic UTI have counts of 10^2–10^4 CFU/mL. [Compilation 04 p25]
- Clinical symptoms correlate poorly with the site of infection. [Compilation 04 p26]
- For localisation by ureteral catheterisation, irrigate the bladder thoroughly, pass two ureteric catheters into a small residual volume of irrigant and sample through each, then pass the catheters into the renal pelvis and take four serial cultures from each kidney. [Compilation 04 p26]

Imaging in UTI

- Indications for imaging in acute pyelonephritis include a history of calculi, especially struvite stones, and infection with urea-splitting organisms such as Proteus. [Compilation 04 p27]

Emphysematous pyelonephritis: risk factors and etiopathogenesis

- Obstruction from papillary necrosis (as well as from stones) is a risk factor for emphysematous pyelonephritis. [Emphysematous pyelonephritis p1]
- A debilitated or immunocompromised state is a risk factor for emphysematous pyelonephritis. [Emphysematous pyelonephritis p1]
- Clostridium is another gas-forming organism that can cause emphysematous pyelonephritis. [Emphysematous pyelonephritis p1]

Clinical features

- The patient is sick, with features of severe acute pyelonephritis: a triad of fever, flank pain and vomiting. [Emphysematous pyelonephritis p2]
- Pneumaturia occurs if the infection involves the pelvicalyceal system. [Emphysematous pyelonephritis p2]

Investigations

- On plain X-ray KUB, emphysematous pyelonephritis shows mottled gas over the perirenal area and a crescent of gas over the upper pole. [Emphysematous pyelonephritis p2]
- On ultrasound, focal echoes suggest gas. [Emphysematous pyelonephritis p2]

Treatment by CT class

- In class 4 disease (bilateral or solitary kidney), drainage is by bilateral PCN or DJ stents, with follow-up if there is a response and nephrectomy if there is none. [Emphysematous pyelonephritis p3]

Prognosis

- Diabetic ketoacidosis is an additional poor prognostic factor in emphysematous pyelonephritis. [Emphysematous pyelonephritis p4]
- A raised total leucocyte count is an additional poor prognostic factor in emphysematous pyelonephritis. [Emphysematous pyelonephritis p4]

Fungal infections in urology: types and risk factors

- By organ, candidiasis mainly involves the bladder and kidney (prostate less often); mucormycosis and aspergillosis mainly involve the kidney and only occasionally the prostate or bladder. [Fungal Infections p4]
- Blastomycosis is a genitourinary fungal infection that mainly involves the prostate, then the kidney, and only occasionally the bladder. [Fungal Infections p4]
- Cryptococcosis mainly involves the kidney, then the prostate; histoplasmosis mainly involves the kidney, and less often the prostate. [Fungal Infections p4]
- Further risk factors for fungal UTI are old age, recent surgery, dialysis and central venous catheters. [Fungal Infections p4]

Candidiasis: presentation, diagnosis and treatment

- Candida albicans accounts for 50–70% of urinary candidiasis. [Fungal Infections p5]
- Biopsy can provide the definitive diagnosis of fungal infection. [Fungal Infections p5]
- Fungal prostatitis or epididymo-orchitis is treated by surgical drainage and fluconazole 400 mg daily until imaging shows resolution. [Fungal Infections p5]
- A fungal ball causing only partial obstruction is treated with systemic antifungal for 6 weeks. [Fungal Infections p7]
- Streptokinase or urokinase can be added to the local irrigation to stop the fungal ball mechanically blocking the drainage tube. [Fungal Infections p7]

Renal Abscess

- Neurogenic bladder and bladder outlet obstruction are risk factors for ascending Gram-negative renal abscess. [Renal Infections p2]
- Intravenous drug abuse and HIV are risk factors for haematogenous Gram-positive (staphylococcal) renal abscess, along with multiple skin carbuncles. [Renal Infections p2]
- Renal abscess presents with chills, weight loss and malaise, in addition to fever and flank pain. [Renal Infections p2]
- On ultrasound, a renal abscess appears as a low-echodensity space-occupying lesion with increased transmission. [Renal Infections p2]
- On CT, an acute renal abscess shows renal enlargement with focal rounded areas of decreased attenuation, whereas a chronic abscess shows an echo-free or slightly echogenic mass with a higher-attenuation wall that rim-enhances on contrast CT. [Renal Infections p2]
- Radionuclide imaging with gallium or indium can be used in the investigation of renal abscess. [Renal Infections p2]
- A small renal abscess in a clinically unstable patient should be drained (percutaneously or surgically) rather than treated with antibiotics alone; a clinically stable patient gets antibiotics with careful observation, and drainage if CT shows no resolution. [Renal Infections p2]

Perinephric Abscess

- Immunocompromised patients are at risk of perinephric abscess. [Renal Infections p5]
- Leucocytosis, elevated serum creatinine and pyuria are seen in 75% of cases of perinephric abscess. [Renal Infections p5]

- On ultrasound, a perinephric abscess appears as an anechoic mass displacing the kidney or as an echogenic collection. [Renal Infections p5]
- Once the perinephric abscess has been drained, the underlying problem must be treated, for example an enteric communication or pyonephrosis. [Renal Infections p5]

Acute Pyelonephritis

- The combination of fever, chills and flank pain with bacteriuria and pyuria is reasonably specific for acute bacterial infection of the kidney, and the term acute pyelonephritis should not be used if flank pain is absent. [Renal Infections p8]
- The clinical spectrum of acute pyelonephritis ranges from Gram-negative sepsis to a cystitis-like illness with mild flank pain. [Renal Infections p8]
- The classical presentation is chills of abrupt onset, fever of 100.3 F or more, and unilateral or bilateral flank pain and/or costovertebral angle tenderness. [Renal Infections p8]
- Rarely, progression to chronic pyelonephritis may occur in the absence of overt symptoms. [Renal Infections p8]
- About 20% of patients with acute pyelonephritis have urine colony counts below 10^5 CFU/mL. [Renal Infections p8]
- Besides E. coli, causative Gram-negative organisms include Proteus, Klebsiella, Pseudomonas, Enterobacter, Citrobacter and Serratia, especially in patients with recurrent UTI, hospitalisation, catheters or instrumentation. [Renal Infections p8]
- Gram-positive causes of acute pyelonephritis include Enterococcus faecalis, Staphylococcus aureus and Staphylococcus epidermidis. [Renal Infections p8]
- Blood cultures are positive in 25% of cases of acute pyelonephritis. [Renal Infections p8]
- Laboratory findings include raised ESR and CRP, and urinalysis may show granular casts as well as leucocyte casts. [Renal Infections p8]
- CT or ultrasound in acute pyelonephritis shows renal enlargement, hypoechoic or attenuated parenchyma, a compressed collecting system and perirenal fat stranding. [Renal Infections p8]
- The differential diagnosis of acute pyelonephritis includes acute appendicitis, diverticulitis, pancreatitis and herpes zoster. [Renal Infections p8]
- An uncomplicated but ill patient (sepsis, raised white count, high fever, vomiting) should be hospitalised with correction of dehydration and electrolytes. [Renal Infections p8]
- If there is no improvement on parenteral therapy, review cultures and sensitivities, give IV treatment for 14–21 days, and drain any obstruction or abscess. [Renal Infections p8]
- Follow-up urine cultures should be repeated on the 5th–7th day of therapy and 10–14 days after stopping treatment, to ensure the urine is free of bacteria. [Renal Infections p8]
- The relapse rate after a 14-day course of therapy is 10–30%; relapse is treated with a second 14-day course, and rarely a 6-week course is needed. [Renal Infections p8]
- Chronic pyelonephritis develops in a complicated urinary system. [Renal Infections p8]

Chronic Pyelonephritis

- Chronic pyelonephritis occurs in patients with an underlying structural or functional urinary tract abnormality; uncomplicated chronic bacterial pyelonephritis is very rare. [Renal Infections p13]
- Chronic pyelonephritis causes no symptoms unless it produces renal insufficiency, when the symptoms are those of chronic renal failure. [Renal Infections p13]
- Patients may give a history of recurrent episodes of acute pyelonephritis. [Renal Infections p13]
- Grossly the kidney is diffusely contracted, scarred and pitted, and the scars are Y-shaped. [Renal Infections p13]
- The parenchyma is thin and corticomedullary differentiation is lost. [Renal Infections p13]
- Histological changes are patchy, with an interstitial infiltrate of lymphocytes, plasma cells and polymorphonuclear cells, and periglomerular fibrosis. [Renal Infections p13]
- Leucocyte and hyaline casts in the tubules resemble thyroid colloid, which is called renal thyroidization. [Renal Infections p13]

- On pyelography the kidney outlines are asymmetric and irregular, with blunting and dilatation of one or more calyces. [Renal Infections p14]
- CT shows parenchymal thinning or atrophy and calyceal clubbing or asymmetry. [Renal Infections p14]
- Functional assessment is by nuclear renogram. [Renal Infections p14]
- Management is to treat any infection with prolonged therapy, prevent future infections and preserve renal function. [Renal Infections p14]
- Obstruction should be relieved and stones removed, because they may enhance renal damage. [Renal Infections p14]
- Patients need appropriate urological and nephrological evaluation. [Renal Infections p14]

Pyonephrosis

- In pyonephrosis there is total or near-total loss of function of the kidney. [Renal Infections p15]
- Presentation is usually with fever, chills, flank pain and tenderness, but occasionally the patient has only mild symptoms, and the patient may go into sepsis. [Renal Infections p15]
- The routes of infection are ascending and haematogenous. [Renal Infections p15]
- Risk factors are immunosuppression or AIDS, diabetes, obstruction and stones. [Renal Infections p15]
- Investigations show a raised total leucocyte count, and urine and blood should be cultured. [Renal Infections p16]
- Radiological diagnosis depends on demonstrating internal echoes in the dependent portion of the dilated pelvicalyceal system. [Renal Infections p16]
- CT or ultrasound may show non-specific findings of a thickened renal pelvis, perirenal fat stranding and hydronephrosis. [Renal Infections p16]
- Once the patient is stable, the underlying source should be identified and treated. [Renal Infections p16]

Xanthogranulomatous Pyelonephritis

- Anaerobes are among the organisms in XGP, besides Proteus (commonest) and E. coli. [Renal Infections p17]
- Segmental XGP involves only one or more calyceal areas. [Renal Infections p17]
- In diffuse XGP the kidney is enlarged but its shape is preserved. [Renal Infections p17]
- Diffuse XGP can show papillary necrosis and cortical thinning. [Renal Infections p17]
- The foamy histiocytes have small dark nuclei and clear cytoplasm. [Renal Infections p18]
- Foamy lipid-laden histiocytes are not specific for XGP. [Renal Infections p18]
- Patients usually have a history of UTI, previous calculi or urological intervention. [Renal Infections p18]
- Urinalysis shows pus and protein. [Renal Infections p18]
- Azotaemia or renal failure is rare because the disease is usually unilateral. [Renal Infections p18]
- On CT, 50–80% have the triad of a unilateral enlarged kidney, a large renal pelvic calculus and little or no function. [Renal Infections p19]
- Pyonephrosis is distinguished from XGP because it usually has a dilated pelvis. [Renal Infections p19]
- Renal lymphoma is distinguished from XGP because it is usually bilateral and has no stones. [Renal Infections p19]
- The primary obstacle to treatment of XGP is incorrect diagnosis. [Renal Infections p20]
- Open nephrectomy is the best approach for XGP. [Renal Infections p20]
- When RCC versus XGP is a diagnostic dilemma, after nephrectomy the histopathology report must be reviewed. [Renal Infections p20]

Malacoplakia

- Malacoplakia can also affect bones and mesenteric lymph nodes. [Renal Infections p21]
- Michaelis–Gutmann bodies are basophilic inclusion bodies that can be intracytoplasmic or extracytoplasmic. [Renal Infections p21]
- On IVU, bladder and upper tract malacoplakia shows filling defects in the renal pelvis, ureter and bladder. [Renal Infections p22]

- On CT, malacoplakic masses are less dense than the surrounding parenchyma, and CT demonstrates extension beyond the kidney. [Renal Infections p22]

CH 27 · INFLAMMATORY AND PAIN CONDITIONS OF THE LOWER GENITOURINARY TRACT: PROSTATITIS, ORCHITIS, EPIDIDYMITIS AND SCROTAL PAIN

Prostatitis & CPPS: epidemiology

- Prostatitis is the most common urologic diagnosis in men younger than 50 years. [Prostatitis CPPS p1]
- In men older than 50 years, prostatitis is the third most common urologic diagnosis, after BPH and prostate cancer. [Prostatitis CPPS p1]
- Prostatitis-like symptoms have a prevalence of 2.2–16%. [Prostatitis CPPS p1]
- Chronic prostatitis/CPPS has a prevalence of 7%. [Prostatitis CPPS p1]

Etiology

- Klebsiella is one of the commonest Gram-negative organisms in bacterial prostatitis, along with E. coli and Pseudomonas. [Prostatitis CPPS p1]
- Gram-positive organisms cause 5–10% of bacterial prostatitis; they include S. aureus, haemolytic streptococci and coagulase-negative staphylococci. [Prostatitis CPPS p1]
- Candida and viruses are other causative organisms of prostatitis. [Prostatitis CPPS p1]
- The role of anaerobes in prostatitis is unknown. [Prostatitis CPPS p1]
- Altered prostatic host defence allows potentially pathogenic bacteria to colonise or infect the prostate. [Prostatitis CPPS p1]
- Phimosis is a risk factor for bacterial prostatitis. [Prostatitis CPPS p1]
- Acute epididymitis is a risk factor for bacterial prostatitis. [Prostatitis CPPS p1]
- Condom catheters, as well as indwelling catheters, are a risk factor for bacterial prostatitis. [Prostatitis CPPS p1]
- Dysfunction of prostatic secretion is a risk factor for bacterial prostatitis. [Prostatitis CPPS p1]
- Intraprostatic ductal reflux is a risk factor for prostatitis, and it causes prostatic inflammation and prostatic calculi. [Prostatitis CPPS p2]
- High-pressure dysfunctional voiding causes CP/CPPS in two ways: autonomic overstimulation of the perineal and pelvic neural system, which leads to chronic neuropathic pain, and intraprostatic ductal reflux. [Prostatitis CPPS p2]
- Chemical irritation is listed as an aetiological factor in CP/CPPS. [Prostatitis CPPS p2]

Pathogenesis

- The pathogenesis of CP/CPPS runs in five stages: initiation (infection, immunogen, toxin, trauma or stress), response (inflammation and neurological injury), facilitation (neuroendocrine mechanisms), propagation (immunologic and neurologic) and outcome (neuropathic pain). [Prostatitis CPPS p3]

Classification

- The traditional (Drach) classification has four categories: acute bacterial prostatitis, chronic bacterial prostatitis, non-bacterial prostatitis and prostatodynia. [Prostatitis CPPS p3]
- Acute bacterial prostatitis can also cause suprapubic pain, external genital pain and hypotension. [Prostatitis CPPS p3]
- About 5% of men with acute bacterial prostatitis progress to chronic bacterial prostatitis. [Prostatitis CPPS p3]
- NIH category IIIA corresponds to the older term non-bacterial prostatitis, and category IIIB corresponds to prostatodynia. [Prostatitis CPPS p4]
- CP/CPPS pain can also occur in the groin and the back. [Prostatitis CPPS p4]
- Category IV (asymptomatic inflammatory prostatitis) is found in asymptomatic men secondary to BPH, prostate cancer and similar conditions. [Prostatitis CPPS p4]

Evaluation algorithm

- Evaluation of CP/CPPS is tiered: mandatory (history, examination including DRE and pelvic floor, urinalysis and midstream culture), recommended (2-glass test, cystoscopy, flow rate and residual urine, urine cytology, NIH-CPSI and sexual history) and optional tests. [Prostatitis CPPS p5]
- Urethral symptoms (discharge, dysuria or penile pain) call for an STI assessment. [Prostatitis CPPS p5]
- TRUS, CT or MRI are optional when other tests suggest an abnormality. [Prostatitis CPPS p5]
- Major psychopathology calls for a psychological evaluation. [Prostatitis CPPS p5]
- When the 2-glass test localises bacteria, the 4-glass test may follow. [Prostatitis CPPS p5]

NIH-CPSI and physical examination

- The NIH-CPSI has 9 questions and is used as a research tool, a clinical tool and for follow-up. [Prostatitis CPPS p6]
- In the NIH-CPSI, the pain domain has 4 questions (location, severity, frequency and relation to ejaculation), the urinary domain has 2 questions (voiding and storage symptoms), impact of symptoms has 2 questions (effect on work and thinking about symptoms) and quality of life has 1 question. [Prostatitis CPPS p6]
- NIH-CPSI pain domain scores (0–21) are graded mild 0–7, moderate 8–13 and severe 14–21. [Prostatitis CPPS p6]
- In category I, general examination shows tachycardia and hypotension as well as fever, and the prostate is hot; DRE may be painful. [Prostatitis CPPS p6]
- In categories II and III, examination is usually unremarkable; the prostate may be normal or enlarged and boggy, with or without nodules. [Prostatitis CPPS p6]

Localisation tests

- The Meares–Stamey 4-glass test is the gold standard, but it is time-consuming and expensive. [Prostatitis CPPS p7]
- VB1 is the first 10 mL of voided urine and is the urethral specimen. [Prostatitis CPPS p7]
- In the 4-glass test, EPS is collected directly into a sterile container after massage, and VB3 (the first 10 mL after massage) includes any EPS trapped in the prostatic urethra. [Prostatitis CPPS p7]
- In category IIIA, the EPS shows more than 5–10 WBC per high-power field. [Prostatitis CPPS p7]
- The advantages of the 2-glass test are that it is simple and cost-effective. [Prostatitis CPPS p8]

Imaging and markers

- In prostatitis, TRUS is used to detect a medial (midline) prostatic cyst. [Prostatitis CPPS p9]
- TRUS is used to diagnose and drain obstructed seminal vesicles. [Prostatitis CPPS p9]
- Monocyte chemoattractant protein-1 (MCP-1) and macrophage inflammatory protein-1 alpha (MIP-1α) are potential markers that are elevated in categories IIIA and IIIB. [Prostatitis CPPS p9]

Phenotype-directed management

- Heat therapy is used for pelvic floor muscle tenderness and resolves inflammation quickly. [Prostatitis CPPS p10]
- Anxiolytics and antidepressants are used, along with behavioural therapy, for the depression and anxiety phenotype. [Prostatitis CPPS p10]
- Referral to a pain clinic is part of management of the neuropathic or extrapelvic phenotype. [Prostatitis CPPS p10]
- Cernilton is the named pollen-extract phytotherapy used for prostate tenderness or inflammation; phytotherapeutic products have 5-ARI, alpha-blocker and anti-inflammatory effects. [Prostatitis CPPS p10]
- Prostatic massage is a treatment for the prostate tenderness or inflammation phenotype. [Prostatitis CPPS p10]
- For the bladder phenotype, pentosan polysulfate, quercetin and intravesical therapy are used. [Prostatitis CPPS p10]

Minimally invasive therapies and surgery

- Electro-acupuncture, TUMT (microwave thermotherapy) and botulinum toxin A injection into the prostate are minimally invasive therapies for CP/CPPS; almost all of them need further investigation. [Prostatitis CPPS p13]
- For retention in acute bacterial prostatitis, use a suprapubic catheter or a short-term (12 hours) small-calibre urethral catheter, because a large or long-term catheter obstructs the prostatic ducts and can lead to an abscess. [Prostatitis CPPS p13]
- A prostatic abscess that extends beyond the capsule is incised and drained transperineally, through the levator ani muscle. [Prostatitis CPPS p13]
- Laparoscopic seminal vesiculectomy is used for a seminal vesicle abscess that does not respond to conservative therapy. [Prostatitis CPPS p13]

CH 28 · INTERSTITIAL CYSTITIS/BLADDER PAIN SYNDROME AND RELATED DISORDERS

Bladder Pain Syndrome: definition, epidemiology, natural history

- The ICS definition of painful bladder syndrome is the complaint of suprapubic pain related to bladder filling, accompanied by other symptoms such as increased daytime or night-time frequency, in the absence of proven urinary infection or other obvious pathology. [Bladder Pain Syndrome p1]
- About 3% of the female population is affected by BPS. [Bladder Pain Syndrome p1]
- BPS has no deleterious effects on pregnancy. [Bladder Pain Syndrome p1]
- Men with BPS are diagnosed at older ages and have more Hunner lesions. [Bladder Pain Syndrome p1]

Associated disorders and aetiology

- Vulvodynia associated with BPS includes focal vulvitis. [Bladder Pain Syndrome p2]
- Allergies and depression are disorders associated with BPS. [Bladder Pain Syndrome p2]
- Proposed initiating bladder insults in BPS include bladder trauma, bladder overdistension and pelvic floor dysfunction, which damage the bladder epithelium. [Bladder Pain Syndrome p2]
- An antiproliferative factor secreted by bladder epithelial cells stops the bladder from repairing epithelial damage, which allows urine constituents such as potassium to leak into the interstitium. [Bladder Pain Syndrome p3]
- Leaked urine constituents activate C fibres with substance P release, which interacts with mast-cell histamine release and immunogenic and allergic responses, producing progressive bladder injury that feeds back into further leakage. [Bladder Pain Syndrome p3]

Pathology

- Eosinophilic cystitis and radiation cystitis are among the diagnoses that biopsy is used to rule out in suspected BPS. [Bladder Pain Syndrome p4]
- BPS biopsies fall into three patterns: a completely normal specimen, non-ulcerative inflammation, or ulcerative inflammation in which epithelial sloughing is common. [Bladder Pain Syndrome p4]
- Lamina propria findings in BPS include oedema, haemorrhage and congested venules, in addition to increased mast cells and inflammatory infiltrates. [Bladder Pain Syndrome p4]
- Detrusor muscle fibrosis is seen in 10% of patients with BPS. [Bladder Pain Syndrome p4]

Diagnosis

- Mandatory evaluation is clinical history, physical examination and urinalysis; recommended tests are urine culture and cytology, symptom and quality-of-life scores, frequency–volume chart, PSA, PVR, and cystoscopy with hydrodistension; optional tests are ultrasound, urodynamics, X-ray, potassium test and biopsy. [Bladder Pain Syndrome p5]
- Examination standing looks for kyphosis, scars and hernia; supine examination tests hip abduction and adduction and looks for hyperaesthesia. [Bladder Pain Syndrome p5]
- In men, examination includes the scrotum and pelvic floor muscles as well as DRE, and pain mapping of the lower extremities is done in both sexes. [Bladder Pain Syndrome p5]

Treatment: conservative

- Doxycycline is the antibiotic used for the single empirical course if antibiotics have not been given before. [Bladder Pain Syndrome p7]
- Exercise and warm tub baths are conservative measures, along with stress reduction. [Bladder Pain Syndrome p7]
- Biofeedback and soft tissue massage are pelvic therapies used for muscle relaxation. [Bladder Pain Syndrome p7]
- Dietary advice, though unsupported by the literature, includes avoiding artificial sweeteners, hot pepper and beverages that might acidify the urine, as well as caffeine and alcohol. [Bladder Pain Syndrome p7]
- Alkalinisation of the urine may be worth trying. [Bladder Pain Syndrome p7]

Treatment: oral therapies

- Oral therapies do not change the natural course of BPS. [Bladder Pain Syndrome p8]
- Amitriptyline acts through central and peripheral anticholinergic action, serotonin–noradrenaline reuptake inhibition and antihistaminic properties. [Bladder Pain Syndrome p8]
- Side effects of amitriptyline are sedation, constipation, dry mouth, blurred vision, dizziness, weight gain and cardiovascular effects. [Bladder Pain Syndrome p8]
- Unstable angina and premature ventricular contractions are also contraindications to amitriptyline. [Bladder Pain Syndrome p8]
- Pentosan polysulfate is marketed as Cystopen-100 and acts by correcting the defect in epithelial permeability. [Bladder Pain Syndrome p8]
- Side effects of pentosan polysulfate are nausea, diarrhoea, rash and reversible alopecia. [Bladder Pain Syndrome p8]
- Immunomodulatory drugs used in BPS include suplatast tosilate, azathioprine, chloroquine derivatives, mycophenolate mofetil and adalimumab, as well as ciclosporin. [Bladder Pain Syndrome p9]
- Miscellaneous agents with no strong evidence include L-arginine (increases urinary NOS activity), quercetin (anti-inflammatory), methotrexate, montelukast (leukotriene inhibitor), nifedipine, misoprostol (cytoprotective), PDE inhibitors and dextroamphetamine. [Bladder Pain Syndrome p9]
- NSAIDs and acetaminophen are analgesic options, and analgesia is an integral part of managing chronic pain syndromes. [Bladder Pain Syndrome p9]

Treatment: intravesical and intradetrusor therapies

- DMSO (RIMSO-50) is an organosulfur compound derived from lignin in the wood pulp industry, and it is freely miscible with water, lipids and organic solvents. [Bladder Pain Syndrome p10]
- DMSO has analgesic, anti-inflammatory and muscle-relaxant properties and actually desensitises nociceptive (C-fibre) pathways. [Bladder Pain Syndrome p10]
- The intravesical cocktail is 50 mL RIMSO-50 plus 10 mg triamcinolone (Kenalog), 44 mEq sodium bicarbonate and 20,000–40,000 U heparin, given weekly for 6 weeks and then, if the response is good, monthly for 6 months. [Bladder Pain Syndrome p10]
- Doxorubicin and capsaicin are other intravesical agents that have been used in BPS. [Bladder Pain Syndrome p11]
- Botulinum toxin A has analgesic properties as well as causing muscle paralysis: it reduces peripheral sensitisation and affects the sensory feedback loop. [Bladder Pain Syndrome p11]

Neuromodulation and surgical therapy

- TENS is a neuromodulation option for BPS. [Bladder Pain Syndrome p12]
- Uncontrolled trials of neuromodulation report 60–80% success in controlling chronic pelvic pain. [Bladder Pain Syndrome p12]
- Therapeutic hydrodistension acts by damaging mucosal nerve endings. [Bladder Pain Syndrome p12]
- Complications of hydrodistension include haematuria, hypogastric pain and low backache, as well as possible bladder rupture. [Bladder Pain Syndrome p13]
- Transurethral resection of Hunner ulcers is a surgical option. [Bladder Pain Syndrome p13]

- Supratrigonal cystectomy with bowel augmentation for intractable IC may need CIC afterwards. [Bladder Pain Syndrome p13]

Treatment algorithm

- Complicated PBS is BPS with incontinence, haematuria, sterile pyuria, gynaecological symptoms or GI signs and symptoms, and it needs urine cytology, urodynamics, imaging, cystoscopy and appropriate referrals; if these are normal, the patient enters the first-line pathway, and if abnormal, the other abnormalities are treated. [Bladder Pain Syndrome p14]
- In the older six-line algorithm, first line is stress reduction, pain management, patient education and behavioural modification; second line is oral and intravesical drugs; third line is hydrodistension under anaesthesia and treatment of Hunner lesions; fourth line is neuromodulation; fifth line is ciclosporin or botulinum toxin; sixth line is surgery for intractable IC, with pain management continued throughout. [Bladder Pain Syndrome p14]

CH 29 · SEXUALLY TRANSMITTED DISEASES

Classification of epididymitis and orchitis

- Epididymitis and orchitis are classified as acute bacterial (secondary to UTI or to STDs), non-bacterial infectious (viral, fungal or parasitic), non-infectious orchitis (idiopathic, traumatic or autoimmune), chronic orchialgia or epididymalgia, and chronic orchitis or epididymitis. [Compilation 04-2 p4]
- Trauma is a cause of non-infectious orchitis. [Compilation 04-2 p4]

Renal mucormycosis

- The Mucorales fungi are found in decaying vegetables and organic matter. [Fungal Infections p1]
- Besides inhalation and ingestion, mucormycosis can be acquired by traumatic inoculation. [Fungal Infections p1]
- Isolated renal mucormycosis is rare, and disseminated disease is the more common form. [Fungal Infections p1]
- Disseminated mucormycosis has a mortality of 50–80%, and the kidney is involved in 22% of disseminated cases. [Fungal Infections p1]
- Further predisposing conditions are chronic renal failure and viral hepatitis. [Fungal Infections p1]
- Renal mucormycosis can present with hypotension. [Fungal Infections p2]
- Laboratory work-up shows a raised total leucocyte count and should include blood glucose and an ELISA for HIV. [Fungal Infections p2]
- Ultrasound shows an enlarged kidney with a heterogeneous renal mass. [Fungal Infections p2]
- On contrast CT there is a large hypodense mass with loss of fat planes with adjacent structures and lack of contrast excretion. [Fungal Infections p2]
- On renal biopsy the broad aseptate hyphae are shown with PAS stain. [Fungal Infections p2]
- The main differential diagnosis of renal mucormycosis is renal cell carcinoma. [Fungal Infections p2]
- Reversal of immunosuppression can include immune stimulation with GM-CSF. [Fungal Infections p2]
- The cumulative dose of amphotericin B for mucormycosis is about 3 g. [Fungal Infections p3]
- Liposomal amphotericin B has less toxicity than conventional amphotericin B. [Fungal Infections p3]
- Itraconazole is another antifungal option (besides posaconazole). [Fungal Infections p3]

HIV and Urologist: Basics and Diagnosis

- HIV infects dendritic cells as well as T cells. [HIV and urologist p1]
- The commonest age group affected is 25–35 years, followed by 15–25 years and then 35–45 years. [HIV and urologist p1]
- Initial screening is by conventional or rapid enzyme immunoassay (EIA), with results in 30 minutes. [HIV and urologist p1]
- A positive screening result must be confirmed by a supplemental antibody test such as Western blot or indirect immunofluorescence, or by an HIV-1 RNA assay. [HIV and urologist p1]

- HIV-1 is much more common than HIV-2. [HIV and urologist p1]
- The window period is the first 3 months, when the antibody screening test may still be negative. [HIV and urologist p1]

Interaction with Other STDs

- STDs raise HIV transmission and acquisition because bleeding from genital ulcers during intercourse raises infectiousness, HIV is present in genital ulcer exudates, and loss of mucosal integrity raises susceptibility. [HIV and urologist p2]
- HSV infection makes keratinocytes vulnerable to HIV and also increases HIV replication. [HIV and urologist p2]
- Non-ulcerative STDs increase HIV shedding from infected people. [HIV and urologist p2]

Renal Infections

- In HIV, tuberculosis arises from reactivation of a latent focus, causing both renal and other extrapulmonary disease. [HIV and urologist p2]
- Rifampicin is an enzyme inducer and lowers the levels of NNRTIs as well as protease inhibitors. [HIV and urologist p2]
- Other renal infections in HIV are CMV, Aspergillus and Toxoplasma. [HIV and urologist p2]

Prostatitis

- E. coli is still the commonest cause of prostatitis, but in HIV many other organisms cause it: Staphylococcus aureus, Klebsiella, Pseudomonas, Serratia, Salmonella, mycobacteria, CMV and fungi (Candida, Aspergillus, Cryptococcus, Histoplasma), especially when the CD4 count is below 200 cells/µL. [HIV and urologist p3]
- Cultures in HIV prostatitis should cover the usual organisms, atypical bacteria, fungi and Mycobacterium tuberculosis. [HIV and urologist p3]
- After 4–6 weeks of antibiotic therapy, HIV patients may need low-dose antimicrobial suppression to prevent recurrence, and those who stay immunocompromised despite ART may need lifelong suppressive antimicrobials. [HIV and urologist p3]

UTI

- About 40% of HIV patients with bacteriuria are asymptomatic. [HIV and urologist p4]
- Unusual urinary pathogens in HIV include Enterococcus, Proteus, parvovirus B19, Toxoplasma gondii and Aspergillus, as well as CMV, adenovirus, Cryptococcus, Candida and mycobacteria. [HIV and urologist p4]
- UTI treatment in HIV may be complicated by the antibiotics already being given as prophylaxis for other infections. [HIV and urologist p4]

Testis, Epididymis and Seminal Vesicle

- The testis and seminal vesicles appear to be HIV reservoirs; HIV in semen is the main vector of transmission and can persist despite ART. [HIV and urologist p4]
- HIV can cause reduced spermatogenesis and maturation arrest in the testis. [HIV and urologist p4]
- Testicular opportunistic infections in HIV include CMV, Toxoplasma gondii and Mycobacterium avium-intracellulare complex. [HIV and urologist p4]

Renal Function (HIVAN)

- HIVAN causes a rapid decline in renal function to irreversible renal failure. [HIV and urologist p5]
- HIVAN pathology includes global or focal segmental glomerulosclerosis and "fetal variant" glomerulopathy as well as collapsing glomerulopathy. [HIV and urologist p5]
- HIVAN arises from HIV infection of renal epithelial cells, including podocytes, glomerular parietal epithelial cells and tubular cells; the responsible viral genes are vpr and nef. [HIV and urologist p5]
- Lowering viral load reduces HIVAN incidence, and ART reduces HIVAN mortality by 57%. [HIV and urologist p5]

Voiding Dysfunction and Haematuria

- Direct toxic effects of HIV or of ART can cause LUTS, with overactive bladder and detrusor underactivity as the urodynamic findings. [HIV and urologist p5]
- Haematuria has a reported higher incidence in HIV and is investigated as in other patients. [HIV and urologist p5]

Erectile Dysfunction

- ED in HIV may be a manifestation of HIV-induced early ageing, and psychological stress and hypogonadism also contribute. [HIV and urologist p6]
- PDE5 inhibitors are cleared by CYP3A, and NNRTIs inhibit CYP3A, so PDE5 inhibitors are started at the lowest possible dose. [HIV and urologist p6]

Stones

- Lopinavir and nelfinavir have also been associated with stone formation, to a lesser extent than indinavir. [HIV and urologist p6]

Neoplasms

- Kaposi sarcoma is a neoplasm of vascular origin. [HIV and urologist p7]
- Kaposi sarcoma has four types: classical, African endemic, iatrogenic and epidemic (AIDS). [HIV and urologist p7]
- Kaposi sarcoma is the second commonest tumour in HIV-infected people and typically occurs at CD4 counts below 150 cells/µL. [HIV and urologist p7]
- HHV-8 drives proliferation of endothelial and spindle cells and angiogenesis. [HIV and urologist p7]
- Kaposi lesions are pink to purple or brown pigmented lesions, often with oedema and lymph node and visceral involvement. [HIV and urologist p7]
- Kaposi sarcoma treatment depends on the type: ART, then first-line chemotherapy with pegylated liposomal doxorubicin or daunorubicin, and second-line paclitaxel or docetaxel. [HIV and urologist p7]
- HIV increases the risk of RCC. [HIV and urologist p7]
- HIV increases testicular cancer risk, particularly seminoma and non-Hodgkin lymphoma. [HIV and urologist p7]
- Prostate and bladder cancer risk in HIV is the same as in the general population or even lower. [HIV and urologist p7]
- BCG therapy for bladder cancer in HIV carries a theoretical risk of disseminated infection. [HIV and urologist p7]

Post-Exposure Prophylaxis

- Factors affecting transmission risk when assessing PEP are the presence of other STDs, the plasma and anogenital viral load of the source patient, and circumcision status. [HIV and urologist p8]
- If the source's HIV status is difficult to establish, PEP should be started without delay. [HIV and urologist p8]
- Body fluids that may warrant PEP are blood, blood-stained saliva, breast milk, genital secretions, CSF, amniotic, peritoneal, synovial, pericardial and pleural fluid. [HIV and urologist p8]
- Mucous-membrane exposures include sexual exposure, a splash to the eye and the oral cavity. [HIV and urologist p8]
- PEP is not needed when the exposed person is already HIV positive, when the source is established HIV negative, or after exposure to fluids of no significant risk such as tears, urine, sweat and non-blood-stained saliva. [HIV and urologist p8]
- A two-drug PEP regimen is effective, but a three-drug regimen is preferred. [HIV and urologist p8]
- The PEP regimen for children is zidovudine + lamivudine + lopinavir for 28 days. [HIV and urologist p8]

Classification of Anti-HIV Drugs

- NRTIs are zidovudine, lamivudine, stavudine, didanosine, zalcitabine and abacavir. [HIV and urologist p10]
- NNRTIs are nevirapine, delavirdine and efavirenz. [HIV and urologist p10]
- Tenofovir is a nucleotide reverse transcriptase inhibitor. [HIV and urologist p10]

- Protease inhibitors are nelfinavir, amprenavir, saquinavir, indinavir, ritonavir, lopinavir, atazanavir and fosamprenavir. [HIV and urologist p10]
- Enfuvirtide is an entry (fusion) inhibitor. [HIV and urologist p10]

CH 31 · TUBERCULOSIS AND PARASITIC INFECTIONS OF THE GENITOURINARY TRACT

Genitourinary Tuberculosis: Epidemiology

- About one-third of the world's population is infected with latent TB. [GUTB p1]
- In developed countries, GUTB occurs in 2-10% of patients with pulmonary TB. [GUTB p1]
- Two-thirds of patients with GUTB are men, usually adults. [GUTB p1]
- After treatment, pulmonary TB needs only about 2 years of follow-up, compared with 10 years for GUTB. [GUTB p1]

Pathogenesis

- Infection begins with inhalation of MTBC-infected aerosols into the alveoli (rarely by direct inoculation), followed by phagocytosis by alveolar macrophages. [GUTB p2]
- When the macrophages kill all the inhaled bacilli, there is no infection and no immune response. [GUTB p2]
- Granuloma formation is driven by macrophages secreting IL-6, IL-12, IL-1 beta and TNF-alpha as chemoattractants. [GUTB p3]
- The granuloma contains foamy macrophages in addition to macrophages, multinucleated giant cells, epithelioid cells and lymphocytes. [GUTB p3]
- Activated T cells secrete IL-2, TNF-alpha and IFN-gamma, which maintain the granuloma and induce killing of bacilli inside macrophages. [GUTB p3]
- About 5% of infected patients progress to active tuberculosis within 1 year of infection. [GUTB p3]
- Malignancy is a further trigger for reactivation of latent TB. [GUTB p3]

Development of GUTB

- Direct inoculation is the fourth route of GUTB, in addition to haematogenous, ascending (retrograde) and contiguous spread. [GUTB p4]
- GUTB typically becomes evident after a prolonged latency that can be as long as 46 years. [GUTB p4]

Pathological Forms of Renal TB

- Chronic pyelonephritis is one of the pathological forms of renal TB. [GUTB p6]
- In caseo-cavernous autonephrectomy, viable renal tissue is replaced by granulomas and cavities filled with inflammatory exudate, with or without calcification; in the fibrotic type, severe scarring and calcification produce a shrunken kidney. [GUTB p6]

Ureter, Bladder, Urethra and Penis

- Bladder TB usually begins at the ureteric orifices and spreads along lymphatics to other areas, passing through mucosal oedema, ulceration and healing by fibrosis. [GUTB p7]
- Urethral TB is very rare because the urethra is relatively resistant to TB. [GUTB p8]
- Penile TB can present as inflamed papules or a keratotic plaque (lupus vulgaris). [GUTB p8]
- Orificial tuberculosis of the penis carries a poor prognosis. [GUTB p8]
- Prostatic TB can arise by haematogenous spread or by urinary contamination. [GUTB p8]
- Tuberculous prostatitis may present as hard nodules that must be differentiated from malignancy. [GUTB p8]
- Seminal vesicle TB can cause oligospermia as well as low-volume ejaculate and azoospermia. [GUTB p8]

Epididymis

- Tuberculous epididymitis may present as a hard nodular epididymis containing large granulomas. [GUTB p9]

Diagnosis: Criteria and Culture

- Major diagnostic criteria for GUTB are a granulomatous lesion, a positive AFB test and a positive PCR; minor criteria are radiological signs of TB, raised ESR, weight loss and pulmonary changes. [GUTB p11]
- GUTB is diagnosed by 1 major criterion or 2 minor criteria. [GUTB p11]
- Lowenstein-Jensen medium is a solid, egg-based medium. [GUTB p11]
- Middlebrook 7H10 is a solid, agar-based medium that gives results about 1 week earlier than LJ medium. [GUTB p11]
- BACTEC is a liquid culture system, alongside MGIT. [GUTB p11]
- NAAT can detect TB in cases with a low bacterial load. [GUTB p12]
- NAAT sensitivity is 85-95%. [GUTB p12]
- NAAT on urine can give false-negative results because urine contains natural inhibitors that interfere with DNA or RNA amplification. [GUTB p12]
- On histopathology, the bacteria themselves are often not seen. [GUTB p13]
- A TST induration of 10 mm or more is positive in recent arrivals (within 5 years) from high-prevalence countries, laboratory personnel, and patients with diabetes mellitus or chronic renal failure. [GUTB p13]
- A TST induration of 5 mm or more is positive in organ transplant recipients. [GUTB p13]
- The QuantiFERON test is specifically the QuantiFERON-TB Gold In-Tube test (QFT-GIT). [GUTB p13]

Imaging

- Papillary necrosis in TB gives a triangular, ring-like calcification on plain film. [GUTB p14]
- Plain film may show TB-infected renal or ureteric calculi. [GUTB p14]
- Plain film may show calcification of the bladder wall and seminal vesicles (as well as the prostate). [GUTB p14]
- Plain film may show erosion of vertebral bodies. [GUTB p14]
- IVU is the gold standard for imaging early renal tuberculosis. [GUTB p14]
- The earliest IVU finding is loss of sharpness or irregularity of the calyceal edges. [GUTB p14]
- IVU filling defects are caused by tuberculomas rupturing into a calyx or by papillary necrosis. [GUTB p15]
- The commonest IVU findings include calyceal obliteration, infundibular narrowing, hydrocalycosis, segmental or total hydronephrosis and hydroureter. [GUTB p15]
- Kerr's kink is a hiked-up renal pelvis with sharp angulation at the pelviureteric junction. [GUTB p15]
- The tuberculous ureter is rigid, calcified, straightened and tubular (pipe-stem) and lacks peristaltic activity. [GUTB p15]
- A non-functioning tuberculous kidney does not take up contrast on IVU. [GUTB p15]
- CT urography has largely replaced IVU, but early changes may be better appreciated on IVU. [GUTB p16]
- CT urography can detect lesions of 3-4 mm. [GUTB p16]
- Retrograde or antegrade pyelography is used when CT or IVU cannot be done, for example in renal insufficiency or contrast allergy. [GUTB p16]
- MRI is not commonly used in GUTB. [GUTB p16]

Cystoscopy and Ureteroscopy

- Cystoscopy and ureteroscopy have a limited role in diagnosing GUTB; the findings of hyperaemia, mucosal erosion and oedema are non-specific. [GUTB p17]

Medical Treatment: Principles and Regimen

- Second-line drugs are reserved for patients in whom first-line agents fail, cause side effects or meet drug resistance. [GUTB p18]
- Liver function tests should be repeated monthly during therapy. [GUTB p18]

- The intensive phase targets rapidly multiplying bacilli, while the continuation phase aims to eradicate slow, sporadic multipliers and persistent bacilli. [GUTB p18]
- Treatment beyond 6 months (usually 9 months) is needed for extensive pockets of infection, concurrent smear-positive cavitary pulmonary TB, CNS involvement, delayed conversion of positive cultures to negative, or inability to take pyrazinamide. [GUTB p18]

Drugs in GUTB and Dosage

- Rifabutin and rifapentine are listed among the first-line drugs. [GUTB p19]
- Second-line drugs are streptomycin, capreomycin, kanamycin, amikacin, levofloxacin, moxifloxacin, cycloserine, ethionamide, linezolid, bedaquiline and aminosalicylic acid. [GUTB p19]
- Intermittent (three times weekly) adult doses are isoniazid 15 mg/kg (maximum 900 mg), rifampicin 10 mg/kg (maximum 600 mg), pyrazinamide 2500 mg and ethambutol 2000 mg. [GUTB p19]
- Streptomycin is given as 15 mg/kg IM or IV (maximum 1 g). [GUTB p19]

Isoniazid

- Isoniazid is a prodrug activated by mycobacterial catalase-peroxidase; activated isoniazid generates oxygen-derived free radicals and inhibits mycolic acid synthesis in the bacterial cell wall, causing DNA damage and bacterial death (bactericidal). [GUTB p20]
- Isoniazid is metabolised in the liver by acetylation, acts on both extracellular and intracellular organisms, and is an essential component of all regimens. [GUTB p20]
- Isoniazid neuritis can present with paraesthesia, numbness and convulsions. [GUTB p20]
- Isoniazid hepatitis is dose-related and reversible on stopping the drug. [GUTB p20]
- Other isoniazid side effects are arthralgia, acne, pruritus, nausea, vomiting, epigastric pain, headache, insomnia and agitation. [GUTB p20]

Rifampicin

- Rifampicin is bactericidal against both intracellular and extracellular organisms; it blocks DNA-dependent RNA polymerase, inhibiting gene transcription and the synthesis of mRNA and protein. [GUTB p21]
- Rifampicin hepatitis is dose-related, occurs more often with pre-existing hepatitis, and is reversible on stopping the drug. [GUTB p21]
- Rifampicin causes a cutaneous syndrome (rash, flushing, pruritus), a flu-like syndrome (chills, fever, bone pain, myalgia) and an abdominal syndrome (nausea, vomiting, pain, diarrhoea). [GUTB p21]
- Rifampicin also causes orange-coloured body fluids, thrombocytopenia and leukopenia. [GUTB p21]

Pyrazinamide, Ethambutol and Streptomycin

- The active form of pyrazinamide is pyrazinoic acid, which impairs mycolic acid biosynthesis by inhibiting enzymes such as fatty acid synthase I; pyrazinamide is metabolised by the liver. [GUTB p22]
- Other pyrazinamide side effects are nausea, vomiting, anorexia, pruritus, rash, dermatitis and rhabdomyolysis. [GUTB p22]
- Ethambutol inhibits arabinosyl transferase, blocking the conversion of arabinose to arabinogalactan, the principal polysaccharide of the mycobacterial cell wall. [GUTB p22]
- Ethambutol is tuberculostatic. [GUTB p22]
- Ethambutol is renally excreted, so caution is needed when GFR is reduced. [GUTB p22]
- Ethambutol side effects are dose- and time-dependent; its retrobulbar neuritis is reversible and causes blurred vision, reduced visual acuity and scotomas as well as red-green colour loss. [GUTB p23]
- Other ethambutol side effects are peripheral neuritis, GI symptoms (nausea, vomiting, pain), neutropenia, thrombocytopenia, headache, dizziness, confusion, hyperuricaemia or gout, and hypersensitivity. [GUTB p23]
- Streptomycin binds the 16S rRNA of the 30S ribosomal subunit, causing codon misreading and inhibition of protein synthesis (bactericidal). [GUTB p23]
- Streptomycin side effects are vestibular and auditory ototoxicity, nephrotoxicity, injection-site abscess, thrombocytopenia, aplastic anaemia, haemolytic anaemia and agranulocytosis. [GUTB p23]

Surgical Treatment

- Waiting 4-6 weeks after starting medical therapy also allows the lesions to stabilise. [GUTB p24]
- Upper ureteric strictures can be managed endoscopically (with or without balloon dilatation, plus a double-J stent) or by ureteroureterostomy. [GUTB p25]
- Bladder neck contracture can be managed by dilatation as well as transurethral incision. [GUTB p26]
- Tuberculous urethral stricture is managed endoscopically, usually needing repeated procedures. [GUTB p26]
- A tuberculous urethral fistula is managed with medical therapy and a suprapubic catheter, followed by delayed reconstruction. [GUTB p26]

GeneXpert MTB/RIF

- GeneXpert can detect TB even in sputum-negative cases and needs only one sample. [GUTB p27]
- GeneXpert has a specificity of 99% and a sensitivity of 92%. [GUTB p27]
- GeneXpert limitations are the need for an uninterrupted power supply, cost, the continued need for culture and sensitivity, an ambient temperature below 30 C, and annual calibration of the Xpert modules. [GUTB p27]

GUTB in the Renal Transplant Patient

- GUTB in renal transplant recipients is rare (0.35-15%). [GUTB p28]
- Transplant recipients develop GUTB with a shorter interval from infection to presentation, usually within 6 months, though it can occur after more than 7 years. [GUTB p28]
- Transplant recipients have a 50-100 times greater risk of TB. [GUTB p28]
- Transplant recipients may lack typical symptoms because of immunosuppression; they usually present with fever, and urinary symptoms occur in under 20%. [GUTB p28]
- In transplant recipients, the chest radiograph commonly shows bilateral involvement, while fibrosis and cavitation are uncommon because of immune deficiency. [GUTB p28]
- In transplant recipients, culture and sensitivity are required for diagnosis, and imaging shows little or no change because immunosuppression prevents its development. [GUTB p28]
- Rifampicin increases the metabolism of sirolimus, ciclosporin and prednisolone, and graft loss occurs in 20-30% of patients. [GUTB p29]
- Rifampicin is usually not given with ciclosporin; if rifampicin is used, the prednisolone dose is doubled. [GUTB p29]
- A suggested transplant regimen is 3 months of pyrazinamide, ofloxacin, isoniazid and ethambutol, then 6 months of ofloxacin, isoniazid and ethambutol, then 9 months of isoniazid and ethambutol (3 ZOHE + 6 OHE + 9 HE). [GUTB p29]
- Isoniazid prophylaxis (10 mg/kg, duration without consensus) should be considered after 18 months of immunosuppression in recipients with a prior history of TB or whose donor had prior TB. [GUTB p30]
- In transplant recipients, drug side effects are more common than in the general population, with more neuropathy (calcineurin inhibitors plus isoniazid) and more isoniazid hepatotoxicity. [GUTB p30]
- Ciclosporin aggravates the hyperuricaemia caused by pyrazinamide and ethambutol. [GUTB p30]

MDR-TB, XDR-TB and Pregnancy

- The cure rate of MDR-TB is 50-60%. [GUTB p31]
- XDR-TB is extremely difficult to cure, with a cure rate of 30-50%. [GUTB p31]
- Women of child-bearing age with GUTB should avoid pregnancy. [GUTB p31]
- GUTB discovered during pregnancy should be treated promptly, because the risk to the fetus from TB outweighs the risk of adverse drug effects. [GUTB p31]

GUTB and HIV

- About 25% of HIV deaths are due to TB. [GUTB p32]
- HIV-positive patients have more lymph node enlargement and more bilateral renal disease. [GUTB p32]
- Treatment should be started promptly in HIV-positive patients, and mortality is high. [GUTB p32]

Hydatid Disease

- Humans are an accidental, dead-end host of Echinococcus granulosus; sheep and goats are the true intermediate hosts. [Parasites 3 p1]
- Life cycle: the adult worm lives in the small intestine of dogs and passes embryonated eggs in the faeces; sheep and goats ingest the eggs while grazing; the larva (oncosphere) hatches in the small intestine, penetrates the intestinal wall and enters the bloodstream; dogs are infected by eating sheep or goat organ meat containing protoscolices from the cyst. [Parasites 3 p1]
- The liver is the first filter and the lung the second filter for the circulating larvae, but larvae can reach any organ and develop into a hydatid cyst. [Parasites 3 p1]
- Humans are infected through contact with infected dogs or by eating raw, uncooked vegetables; the oncosphere penetrates the intestine (duodenum) and enters the circulation. [Parasites 3 p1]
- Hydatid cysts grow slowly and may take 5-20 years to become symptomatic. [Parasites 3 p2]
- Symptoms of renal hydatid may include haematuria, malaise and fever in addition to a palpable mass and flank pain. [Parasites 3 p2]
- Symptoms depend on location, for example adrenal, adnexal or pelvic cysts. [Parasites 3 p2]
- CT is more sensitive than ultrasound for hydatid cysts and helps in surgical planning; other investigations include IVU, plain X-ray KUB and MRI. [Parasites 3 p2]
- Eosinophilia occurs in 20-50% of cases of hydatid disease. [Parasites 3 p3]
- Other serological tests for hydatid disease include the complement fixation test, immunoelectrophoresis and indirect haemagglutination; ELISA is the one commonly used. [Parasites 3 p3]
- Serology has a role in follow-up: titres should fall after definitive medical or surgical therapy. [Parasites 3 p3]
- Medical therapy is used as an adjunct to surgical excision, or as primary therapy in poor surgical candidates or disseminated echinococcosis. [Parasites 3 p3]
- Preoperative albendazole with or without praziquantel (50 mg/kg) is given for 4 weeks before surgery; it kills the scolices, decreases cyst antigenicity, renders the cyst inactive and reduces the risk of intraoperative spillage. [Parasites 3 p3]
- Surgery is the mainstay of treatment specifically for active, expanding, symptomatic and infected cysts. [Parasites 3 p3]
- Side effects of albendazole are hepatotoxicity, leucopenia, allergy and alopecia. [Parasites 3 p4]
- Praziquantel 40 mg/kg/day is given for 1-2 weeks postoperatively to eradicate infective particles spilled at the time of surgery. [Parasites 3 p4]
- The surgical approach to a renal hydatid cyst may be transperitoneal or retroperitoneal, open or laparoscopic. [Parasites 3 p4]
- Scolicidal agents include 10% povidone-iodine, 95% ethanol, 10% cetrimide and 0.5% silver nitrate, in addition to hypertonic saline. [Parasites 3 p4]
- When a scolicidal agent is injected into the cyst before aspiration, it is left for 10 minutes. [Parasites 3 p4]
- The Palanivelu hydatid system is a specially designed laparoscopic controlled suction-irrigation system, described for hepatobiliary hydatid, that allows controlled removal of cyst contents and endocyst without excising the entire cyst, with minimal risk of intraperitoneal spillage. [Parasites 3 p5]
- Deroofing of a renal hydatid cyst involves enucleation of the contents, with removal of the endocyst and ectocyst. [Parasites 3 p5]

Classification of Hydatid Cyst

- The Gharbi ultrasound classification has five types: type 1, pure fluid collection (unilocular cyst); type 2, fluid collection with a split wall and floating endocyst (water-lily sign); type 3, multivesicular, multiseptate honeycomb pattern; type 4, heterogeneous hypo- or hyperechoic cyst without daughter vesicles; type 5, cyst with partial or complete wall calcification. Types 4 and 5 are inactive. [Parasites 3 p6]

- The WHO ultrasound classification includes a CL stage: a unilocular cyst without internal echoes or septa. [Parasites 3 p6]
- WHO CE2 corresponds to Gharbi type 3, WHO CE3 to Gharbi type 2, and WHO CE4 and CE5 to Gharbi types 4 and 5. [Parasites 3 p6]

Schistosomiasis

- In schistosomiasis the snail is the intermediate host and man is the definitive host. [Parasites 3 p7]
- After penetrating the skin the cercariae lose their tail and become schistosomulae. [Parasites 3 p7]
- Schistosomulae migrate to the portal blood in the liver and become adults over 1-4 weeks. [Parasites 3 p7]
- Some S. haematobium eggs are also shed in the faeces as well as the urine. [Parasites 3 p7]
- Eggs hatch 1-3 hours after reaching fresh water and release miracidia. [Parasites 3 p7]
- Within the snail, miracidia develop into primary sporocysts, then daughter sporocysts, then cercariae. [Parasites 3 p7]
- Cercariae released into water can live for 1-4 days. [Parasites 3 p7]
- Schistosomiasis is associated with poor sanitation and poverty. [Parasites 3 p8]
- Acute schistosomiasis may present with a maculopapular rash, and the acute phase may be silent, especially in people living in endemic areas, or go unrecognised because of non-specific features. [Parasites 3 p8]
- Katayama fever comprises fever, dry cough, fatigue, headache, diarrhoea, neck pain and urticaria. [Parasites 3 p8]
- Katayama fever is commonest with S. japonicum, then S. mansoni, and is much less common with S. haematobium. [Parasites 3 p8]
- Katayama fever develops 2-8 weeks after primary infection. [Parasites 3 p8]
- Chronic urinary schistosomiasis can cause dysuria, frequency and proteinuria in addition to haematuria. [Parasites 3 p9]
- Female genital schistosomiasis results from egg deposition in the ovaries, tubes, uterus, cervix, vagina and vulva, causing dyspareunia, pelvic or abdominal pain, vaginal bleeding and infertility. [Parasites 3 p9]
- In men, high numbers of eggs are deposited in the seminal vesicles and ejaculatory ducts, causing testicular mass, scrotal pain and infertility. [Parasites 3 p9]
- The chronic stage has a late chronic active stage, in which tissue egg burdens peak, which over years becomes a chronic inactive phase. [Parasites 3 p10]
- In the late chronic active stage the bladder may contract to less than 50 mL from fibrosis, with chronic suprapubic and pelvic pain and irritative LUTS. [Parasites 3 p10]
- Schistosome eggs are usually seen in urine or stool 8-12 weeks after infection. [Parasites 3 p11]
- Tissue biopsy is taken when eggs are not found but clinical suspicion remains high and serology is positive. [Parasites 3 p11]
- PCR for schistosomiasis can be performed on serum, urine or stool. [Parasites 3 p11]
- Circulating antigen tests detect circulating anodic antigen (CAA) as well as circulating cathodic antigen (CCA); both are specific for active infection because only adult worms release them. [Parasites 3 p11]
- Antibody detection uses the Falcon Assay Screening Test-ELISA (FAST-ELISA) combined with Western blot, with 90% sensitivity. [Parasites 3 p12]
- Antibodies are detectable 4-6 weeks after infection, so serology is not useful in endemic areas. [Parasites 3 p12]
- VCUG shows vesicoureteric reflux in 25% of cases of urinary schistosomiasis. [Parasites 3 p12]
- Schistosomal ureteric calcification is intramural with a dilated ureter, whereas tuberculous ureteric calcification occurs in a non-dilated ureter. [Parasites 3 p12]
- Corticosteroids are given for Katayama fever. [Parasites 3 p13]
- Side effects of praziquantel are abdominal pain, nausea, headache and dizziness, which are mild and self-limited; the pills are large and bitter. [Parasites 3 p13]

- Artemisinin and its analogues (artemether, artesunate) specifically target schistosomules but are poor treatment for established infection. [Parasites 3 p13]
- Praziquantel and artemisinin are not usually combined, because artemisinin use induces malaria resistance. [Parasites 3 p13]
- Immediate surgical intervention in schistosomiasis is needed for severe bladder haemorrhage; delayed surgery is for complications that have not responded to medical therapy. [Parasites 3 p14]
- Options for the end-stage schistosomal bladder also include hydrodistension and vesical denervation, and cancer surgery is needed for malignancy. [Parasites 3 p14]
- Prevention includes improved sanitation and hygiene, and heating water (above 125 degrees for 5 minutes) and chlorination, which kill cercariae; mass drug treatment is the WHO strategy. [Parasites 3 p14]
- Most patients have mild infection with good prognosis, but severe disease carries 50% mortality in 2-5 years. [Parasites 3 p14]

Filariasis

- Filariasis is a vector-borne disease in which humans are the definitive host. [Parasites 3 p15]
- Eight filarial nematodes use humans as definitive hosts: lymphatic filariasis (Wuchereria bancrofti, Brugia malayi, Brugia timori); subcutaneous filariasis (Loa loa, Onchocerca volvulus, Mansonella streptocerca); and serous cavity filariasis (Mansonella perstans, Mansonella ozzardi). [Parasites 3 p15]
- Filariasis is commoner in Africa and Asia, and one third of infected patients develop clinical disease. [Parasites 3 p15]
- Life cycle of W. bancrofti: adult worms in human lymphatics produce sheathed microfilariae; a mosquito ingests them with a blood meal; the microfilariae shed their sheaths, penetrate the mosquito midgut and migrate to the thoracic muscles, develop from L1 to L3 larvae, then migrate to the head and proboscis and enter human skin at the next blood meal. [Parasites 3 p16]
- Adult filarial worms live in afferent lymphatics, especially of the lower limbs. [Parasites 3 p16]
- Subclinical filarial infection shows dilated lymphatics, scrotal lymphangiectasia, and microscopic haematuria or proteinuria. [Parasites 3 p17]
- Acute adenolymphangitis (ADL) is often the first clinical manifestation, with fever, lymphadenitis, lymphangitis and oedema lasting days to a week, and may cause funiculitis, epididymitis and scrotal pain and tenderness. [Parasites 3 p17]
- Dermatolymphangioadenitis (DLA) presents with fever, chills, myalgia and headache, with plaques, vesicles or ulcers on the skin. [Parasites 3 p17]
- Filarial lymphoedema is commoner in the lower limbs, is usually asymmetrical, and results from progressive scarring and obliteration of lymphatics. [Parasites 3 p17]
- Genitourinary filarial manifestations are commoner in males and with W. bancrofti. [Parasites 3 p18]
- Funiculoepididymitis is usually self-limiting but can recur or lead to chronic lymphoedema, and causes no sterility or orchitis. [Parasites 3 p18]
- Filarial hydroceles are usually large. [Parasites 3 p18]
- Scrotal and penile elephantiasis present late, in those with poor medical care. [Parasites 3 p18]
- Tropical pulmonary eosinophilia presents with cough, fever, wheeze and adenopathy, with eosinophilia and raised IgE; it is an allergic response and microfilaraemia is usually absent. [Parasites 3 p18]
- Microfilariae are best detected on a Giemsa-stained blood smear. [Parasites 3 p19]
- Microfilaraemia is found in only 30-40% of all filarial infections. [Parasites 3 p19]
- Filarial antigen detection methods include ELISA, PCR and the immunochromatographic card test (ICT). [Parasites 3 p19]
- Filarial antibody detection is by IgG4 antibodies. [Parasites 3 p19]
- Lymphoscintigraphy shows lymphatic anatomy but is not specific for filariasis. [Parasites 3 p19]
- DEC is marketed as Hetrazan-100. [Parasites 3 p20]
- For tropical pulmonary eosinophilia, DEC is given for 2-3 weeks. [Parasites 3 p20]

- Side effects of DEC are fever, chills, arthralgia, headache, nausea, vomiting and skin nodules. [Parasites 3 p20]
- Doxycycline augments the effect of DEC and other anthelminthics, and benefits lymphoedema and hydrocele even without active infection. [Parasites 3 p20]
- Chronic filarial lymphoedema is managed with good hygiene, prevention of secondary infection, elastic stockings, elevation and physiotherapy; antiparasitic therapy is reserved for those with active infection. [Parasites 3 p21]
- Surgery for chronic lymphoedema, lymphovenous anastomosis or nodovenous anastomosis, is challenging and usually unnecessary. [Parasites 3 p21]
- Genital elephantiasis is rarely amenable to surgery. [Parasites 3 p21]
- More radical surgery is needed for recurrent, painful funiculoepididymitis. [Parasites 3 p21]

OTHER TOPICS IN THIS BOOK

Dimethyl Sulfoxide (DMSO)

- DMSO is a membrane penetrant, and it is also bacteriostatic, diuretic, a cholinesterase inhibitor, a collagen solvent and a vasodilator. [Compilation 03 p49]
- DMSO gives good results only in interstitial cystitis and is not useful in neurogenic or idiopathic detrusor overactivity. [Compilation 03 p49]

Fournier Gangrene

- Fournier gangrene is also known as idiopathic gangrene of the scrotum, streptococcal scrotal gangrene, perineal phlegmon and spontaneous fulminant gangrene of the scrotum. [Compilation 04-3 p45]
- Mixed organisms are cultured: facultative organisms (E. coli, Klebsiella, Enterococci) along with anaerobes (Bacteroides, Clostridium, microaerophilic streptococci, Fusobacterium). [Compilation 04-3 p45]
- Further risk factors are local trauma, paraphimosis, perirectal or perianal infections, and circumcision or herniorrhaphy. [Compilation 04-3 p45]
- It starts as cellulitis adjacent to the portal of entry, with swelling, erythema and tenderness, then progresses to dark purple areas and frank gangrene. [Compilation 04-3 p46]
- Systemic features include tachycardia, tachypnoea and mental status alterations. [Compilation 04-3 p46]
- Investigations commonly show anaemia, hyponatraemia, hypocalcaemia and usually an elevated serum creatinine. [Compilation 04-3 p46]
- Scrotal ultrasound can be useful in diagnosis. [Compilation 04-3 p46]
- Differentiating cellulitis from necrotising fasciitis may be difficult initially, but marked systemic toxicity out of proportion to the local findings should raise the suspicion. [Compilation 04-3 p47]
- Ampicillin-sulbactam is an alternative broad-spectrum antibiotic option to piperacillin-tazobactam. [Compilation 04-3 p47]
- Used along with antibiotics and debridement, hyperbaric oxygen shortens hospital stay, improves wound healing and reduces gangrenous spread. [Compilation 04-3 p48]
- Reconstructive flap cover is done once the patient is stable and wound healing is complete. [Compilation 04-3 p48]

Haemorrhagic Cystitis

- Haemorrhagic cystitis often occurs in elderly patients with medical comorbidities. [Compilation 04-3 p49]
- Infective causes include fungal and parasitic infections as well as bacterial and viral. [Compilation 04-3 p49]
- Trauma (external, or after TURBT) is a cause of haemorrhagic cystitis. [Compilation 04-3 p49]
- Malignancy causes include bladder cancer and invasion or metastasis from other tumours. [Compilation 04-3 p49]
- Vascular malformations (arteriovenous malformation and telangiectasia) are a cause. [Compilation 04-3 p49]

- Aniline dyes are a chemical cause, along with cyclophosphamide, ifosfamide and busulfan. [Compilation 04-3 p49]
- Radiation-induced haemorrhagic cystitis typically follows radiotherapy for carcinoma of the prostate or cervix. [Compilation 04-3 p49]
- Drug causes include bleomycin, danazol and allopurinol, and penicillin causes it through an immune reaction. [Compilation 04-3 p49]
- Systemic illnesses that can manifest as haemorrhagic cystitis include rheumatoid arthritis and Crohn's disease, besides amyloidosis. [Compilation 04-3 p49]
- Coagulation disorders are a cause of haemorrhagic cystitis. [Compilation 04-3 p49]
- In the management algorithm, the presence of renal failure decides the intravesical agent: alum if no renal failure; aminocaproic acid (Amicar), silver nitrate or prostaglandin if renal failure is present. [Compilation 04-3 p50]
- In a clinically stable patient with continued bleeding, a history of radiotherapy or cyclophosphamide leads to hyperbaric oxygen with or without formalin, whereas without such history formalin is used. [Compilation 04-3 p50]
- If bleeding continues after angioembolisation, temporary bilateral nephrostomies, then conduit diversion with or without cystectomy, follow. [Compilation 04-3 p50]
- Other listed options are hydrostatic pressure (bladder distension) and Cystopen(?) and pentosan polysulfate. [Compilation 04-3 p50]

Antibiotics: Quinolones and Cotrimoxazole

- Quinolone side effects also include bad taste, dizziness, insomnia, anxiety, and skin rash or hypersensitivity. [Compilation 04-3 p77]
- Sulfamethoxazole inhibits folate synthase (para-aminobenzoic acid to dihydrofolic acid) and trimethoprim inhibits dihydrofolate reductase (dihydrofolic acid to tetrahydrofolic acid), giving sequential blockade of folate metabolism. [Compilation 04-3 p77]
- Sulfonamides are bacteriostatic, but bactericidal concentrations may be obtained in urine. [Compilation 04-3 p77]
- Cotrimoxazole covers Gram-positive and Gram-negative organisms but not anaerobes. [Compilation 04-3 p77]
- Cotrimoxazole side effects include nausea, vomiting and stomatitis, and rarely megaloblastic anaemia and blood dyscrasias. [Compilation 04-3 p77]
- In AIDS patients treated for Pneumocystis infection, cotrimoxazole causes fever, rash and bone-marrow hypoplasia. [Compilation 04-3 p77]
- Cotrimoxazole tablets contain TMP:SMX 80:400 mg (single strength) or 160:800 mg (DS tablet). [Compilation 04-3 p77]

Beta-lactam Antibiotics and Cephalosporins

- Beta-lactam antibiotics (having a beta-lactam ring) comprise penicillins, cephalosporins, monobactams and carbapenems. [Compilation 04-3 p78]
- Cephalosporins are semisynthetic antibiotics obtained from the fungus Cephalosporium. [Compilation 04-3 p78]
- Cephalosporins act like penicillins by inhibiting cell wall synthesis, but bind to different proteins from those that bind penicillins. [Compilation 04-3 p78]
- Second-generation cephalosporins (including oral cefuroxime axetil) are weaker than first-generation agents against Gram-positive organisms. [Compilation 04-3 p78]
- Third-generation cephalosporins are resistant to beta-lactamases and less active against Gram-positive organisms; the group also includes ceftizoxime, cefoperazone (anti-pseudomonal) and cefdinir. [Compilation 04-3 p78]
- Cefpirome is a fourth-generation cephalosporin alongside cefepime, with a spectrum similar to the third generation. [Compilation 04-3 p78]
- Side effects of cephalosporins include diarrhoea, disulfiram-like reaction, thrombophlebitis, hypoprothrombinaemia, pain on injection, nephrotoxicity and fungal superinfection. [Compilation 04-3 p79]

Penicillins

- Penicillin G is the natural prototype penicillin. [Compilation 04-3 p79]
- Penicillins bind penicillin-binding proteins and inhibit the transpeptidase enzyme, blocking cross-linking during cell wall synthesis. [Compilation 04-3 p79]
- Penicillin adverse effects are local irritation, hypersensitivity in 1-10% including anaphylaxis, and rarely superinfection. [Compilation 04-3 p79]
- Phenoxymethylpenicillin (penicillin V) is the acid-resistant alternative to penicillin G. [Compilation 04-3 p80]
- Penicillinase-resistant penicillins are methicillin, cloxacillin and dicloxacillin. [Compilation 04-3 p80]
- Extended-spectrum penicillins are aminopenicillins (ampicillin, amoxycillin, bacampicillin), carboxypenicillins (carbenicillin, ticarcillin) and ureidopenicillins (piperacillin, mezlocillin). [Compilation 04-3 p80]
- The beta-lactamase inhibitors are clavulanic acid, sulbactam and tazobactam. [Compilation 04-3 p80]

Piperacillin + Tazobactam

- Piperacillin has good activity against E. coli, Klebsiella and some Bacteroides, as well as Gram-positive organisms, and is frequently used for serious Gram-negative infections. [Compilation 04-4 p1]
- Concurrent gentamicin or tobramycin is advised with piperacillin. [Compilation 04-4 p1]
- Tazobactam is paired with piperacillin because its pharmacokinetics match those of piperacillin. [Compilation 04-4 p1]
- Side effects of piperacillin–tazobactam are nausea, constipation or diarrhoea, allergic reactions including anaphylaxis, and thrombocytopenia. [Compilation 04-4 p1]
- The dose of piperacillin–tazobactam is 4.5 g (4 g/500 mg) IV every 6–8 hours. [Compilation 04-4 p1]
- Piperacillin–tazobactam needs dose adjustment when creatinine clearance is below 40 mL/min. [Compilation 04-4 p1]

Imipenem

- Imipenem undergoes rapid hydrolysis by dehydropeptidase-I on the brush border of the renal tubular cells. [Compilation 04-4 p3]
- Imipenem is usually given with cilastatin, a reversible inhibitor of dehydropeptidase-1, which prevents its renal metabolism. [Compilation 04-4 p3]
- The spectrum of imipenem covers Gram-positive cocci, Enterobacteriaceae, Pseudomonas and Listeria, as well as anaerobes such as B. fragilis and C. difficile. [Compilation 04-4 p3]
- The dose of imipenem–cilastatin is 0.5 g IV every 6 hours, combined with gentamicin for patients with Pseudomonas infection. [Compilation 04-4 p4]
- Side effects of imipenem are nausea, vomiting, diarrhoea, skin rashes and hypersensitivity reactions, and seizures at higher doses in predisposed individuals. [Compilation 04-4 p4]

Meropenem and other carbapenems

- Meropenem is not hydrolysed by renal peptidase, so it needs no cilastatin. [Compilation 04-4 p4]
- Meropenem is a reserve drug for serious nosocomial infections and is less likely than imipenem to cause seizures. [Compilation 04-4 p4]
- The dose of meropenem is 0.5–2 g IV every 8 hours. [Compilation 04-4 p4]
- Other carbapenems include faropenem (oral) and doripenem. [Compilation 04-4 p4]

Aminoglycosides

- Aminoglycosides act in two steps: transport of the drug through the bacterial cell wall and cytoplasmic membrane, then binding to the ribosome, which inhibits protein synthesis. [Compilation 04-4 p5]
- Aminoglycosides bind the 16S rRNA of the 30S ribosomal subunit and cause codon misreading. [Compilation 04-4 p5]
- Aminoglycoside ototoxicity can be vestibular or cochlear, and nephrotoxicity is due to tubular damage. [Compilation 04-4 p5]

- Neuromuscular blockade is a side effect of aminoglycosides. [Compilation 04-4 p5]
- In renal impairment, aminoglycosides are given at 70% of the adult dose at a creatinine clearance of 70, 50% at 50 and 30% at 30; below 30, reduced doses are given on alternate days. [Compilation 04-4 p5]

Nitrofurantoin

- Nitrofurantoin is rapidly metabolised in the liver and other tissues, but is excreted unchanged in the urine. [Compilation 04-4 p6]
- The acute reaction to nitrofurantoin includes chills, fever and leucopenia. [Compilation 04-4 p6]

4 Paediatric Urology 1

CH 41 · MANAGEMENT STRATEGIES FOR VESICOURETERAL REFLUX

VUR: Demographics

- In patients with UTI, VUR prevalence by age is 75% under 1 year, 25% at 4 years, 15% at 12 years and 5.2% in adults. [vur p1]
- In infants, VUR is more common in males (about 75%). [vur p1]
- In older children with persisting reflux, 85% are female. [vur p1]
- Fetal hydronephrosis is common but often resolves and has low specificity for postnatal VUR; about 35% have reflux in postnatal life. [vur p1]
- VUR may show an autosomal dominant inheritance pattern. [vur p1]

Radionuclide cystography grading

- RNC grade 1 corresponds to VCUG grade 1, RNC grade 2 (low-grade reflux) to VCUG grades 2 and 3, and RNC grade 3 (high-grade reflux) to VCUG grades 4 and 5. [vur p3]

Functional anatomy of antireflux mechanism

- At the extravesical hiatus the ureter has three muscle layers. [vur p5]
- Some fibres of the inner ureteral muscle (which forms the superficial trigone) pass medially to form Mercier's bar. [vur p5]
- The VUJ opens by active contraction of the longitudinal muscle within the tunnel, which draws the extravesical and intravesical points of the intramural ureter together, shortening and widening the tunnel. [vur p5]
- VUJ closure results both from compression of the intramural ureter and from return to its full tunnel length as the ureteral muscle relaxes. [vur p5]
- No true space separates the ureter from the bladder. [vur p5]

Pathophysiology of VUR

- Abnormalities of ureteral smooth muscle contribute to reflux. [vur p6]
- Abnormal extracellular matrix composition contributes to reflux. [vur p6]
- Abnormal neural function contributes to reflux. [vur p6]
- A prolapsing ureterocele in females is an anatomical cause of secondary VUR. [vur p6]

Pathophysiology of scarring

- A larger renal scar appears as a hyperechoic, shrunken area on ultrasound. [vur p7]
- Children under 4 years are more prone to scarring after UTI. [vur p7]
- Bacterial virulence is a factor in renal scarring. [vur p7]
- Host susceptibility and response are factors in renal scarring. [vur p7]
- Hypertension in reflux arises from post-infection scarring nephropathy, from congenital dysmorphism associated with reflux, or from both combined. [vur p7]
- Reflux correction is a poor predictor of catch-up renal growth. [vur p7]
- The somatic growth curve is one of the best predictors of renal function in children. [vur p7]

Associated anomalies: PUJO

- VUR is found in 9–18% of patients with PUJO. [vur p10]
- VUR found in PUJO patients is mostly incidental, low grade and resolves spontaneously. [vur p10]
- In VUR associated with PUJO, the ureter is minimally dilated but the pelvis is significantly dilated. [vur p10]
- PUJO is found in 0.75–3.6% of patients with VUR. [vur p10]
- PUJO diagnosed in patients with VUR is usually associated with high-grade reflux. [vur p10]
- On VCUG, a sign of coexisting PUJO is that the pelvis shows little or no filling. [vur p10]
- On VCUG, a sign of coexisting PUJO is that contrast is diluted in the pelvis and shows reduced radiodensity. [vur p10]
- Hollowell and colleagues described three categories of UPJO with VUR: primary PUJO with incidental low-grade reflux, secondary PUJO with high-grade reflux (pyeloplasty recommended in both), and significant upper tract dilatation only. [vur p10]
- Pyeloplasty comes before reflux correction because reflux correction alone risks amplifying upper tract dilatation during the phase of postoperative oedema. [vur p10]
- Pyeloplasty comes first because reflux correction risks introducing infection into an obstructed renal pelvis. [vur p10]
- Pyeloplasty comes first because sterile reflux can be observed, whereas obstruction can jeopardise renal function. [vur p10]
- Pyeloplasty comes first because the secondary insult to the UPJ from reflux is an evolving process that cannot be corrected by repair of the reflux itself. [vur p10]
- Pyeloplasty combined with endoscopic injection can be done at the same time. [vur p10]

Associated anomalies: ureteral duplication and bladder diverticula

- VUR is the most common anomaly associated with complete ureteral duplication. [vur p12]
- VUR in duplication is mostly low grade, with no heightened risk of infection or scarring. [vur p12]
- High-grade duplex reflux in females is more prone to infection and scarring and may need aggressive management. [vur p12]
- A diverticulum projecting into the bladder neck can incite secondary reflux. [vur p12]
- VUR with a bladder diverticulum is managed according to the prevailing indications for the reflux itself, irrespective of the diverticulum. [vur p12]

Associated anomalies: renal anomalies and megacystis-megaureter

- With renal agenesis, contralateral renal pathology is present in about 46%. [vur p13]
- Contralateral VUR with renal agenesis has less tendency to resolve spontaneously than with MCDK. [vur p13]
- Megacystis-megaureter syndrome is non-obstructive, unlike PUV. [vur p13]
- Megacystis-megaureter syndrome carries a high chance of infection. [vur p13]
- In megacystis-megaureter syndrome, vesicostomy can be considered before ureteric reimplantation. [vur p13]

Other anomalies associated with VUR

- VACTERL association (vertebral, anal, cardiac, tracheo-oesophageal, renal and limb anomalies) is listed among the anomalies associated with VUR. The corpus mentions VACTERL only with renal agenesis and spinal dysraphism. [vur p14]
- CHARGE syndrome (coloboma, heart disease, atresia of the choanae, retarded development, genital hypoplasia and ear anomalies) is listed among the anomalies associated with VUR. [vur p14]

Pregnancy and reflux

- In pregnancy, bladder tone decreases, which predisposes to bacteriuria. [vur p14]
- Women with prior urinary infections have a higher incidence of bacteriuria in pregnancy. [vur p14]
- Pregnant women with renal scarring have a 3.3-fold higher incidence of hypertension and a 7.6-fold higher risk of pre-eclampsia. The corpus gives only rounded figures of about 3 times and 8 times

(Book 04 p4; Paper 3). [vur p14]
- Pregnant women with renal scarring have a higher rate of obstetric intervention. [vur p14]
- Women with reflux but normal kidneys have an increased risk of hypertension in the third trimester. [vur p14]

Diagnosis and evaluation of VUR: confirming UTI

- The urine sample should be stored and delivered at 4°C. [vur p15]
- Colonisation (bacteriuria without pyuria) alone, in the presence of reflux, still poses a threat to the upper tracts. [vur p15]
- Radiographic investigation for VUR is traditionally directed to (i) all children with a febrile UTI, (ii) any male child with a UTI, regardless of age or fever, and (iii) all children under 5 years with a UTI. [vur p15]

Ultrasound-first pathway

- In older children without fever, detecting reflux is of questionable significance. [vur p16]

Assessment of the lower urinary tract: cystography

- Direct cystographic ultrasound can be done with colour Doppler as well as with contrast-enhanced ultrasound. [vur p16]
- Indirect cystography has a high false-positive rate. The corpus says only that it is insensitive and not routine (Book 05 p14). [vur p16]
- Three factors affect direct cystographic imaging: bladder contraction during voiding, the volume of fluid instilled, and the presence of infection and inflammation at the VUJ. [vur p16]
- Cyclic cystography (2nd and 3rd fill–void cycles) is commonly used with radionuclide cystography. [vur p16]
- During VCUG, bladder capacity is recorded at the point when inflow of contrast stops. [vur p16]

Radionuclide cystography

- Radionuclide cystography gives about 1% of the radiation dose of VCUG. The corpus says only "lower dose". [vur p18]
- Radionuclide cystography is ideal for screening as well as for monitoring natural history and surgical follow-up. [vur p18]
- Direct radionuclide cystography uses Tc-99m pertechnetate, with gamma camera detection. [vur p18]

Cystoscopy and PIC technique

- Cystoscopy is not part of routine VUR evaluation. Done immediately before surgery, it can show orifice position, duplication, proximity to a diverticulum and urethral patency. [vur p18]
- PIC technique: after the bladder is emptied, the beak of a 9.5–14 F rigid cystoscope is placed at the ureteric orifice. Contrast is instilled from a height of 1 m, with simultaneous fluoroscopic spot imaging to detect reflux. [vur p18]
- The PIC technique is not yet standardised, and high pressures may create iatrogenic reflux. [vur p18]

Assessment of the upper tract

- Renal ultrasound in VUR shows renal dimensions, degree of corticomedullary differentiation and cortical abnormalities. It is better for large defects and may miss small ones. [vur p19]
- Doppler ultrasound may show perfusion abnormalities, with a raised resistive index in higher grades of reflux and in pyelonephritis. [vur p19]
- The DMSA scan gives a proportional representation of GFR (as written in the notes). [vur p19]
- In the top-down approach, a child with a normal DMSA needs no further evaluation unless UTIs recur, in which case a VCUG is done. [vur p19]
- The top-down approach detects high-grade VUR poorly, with a sensitivity of 79% and a specificity of 53%. [vur p19]

Management of VUR: principles and natural history

- Walker's essential tenets of reflux management are: (i) spontaneous resolution is very common, (ii) high-grade reflux is less likely to resolve spontaneously, (iii) sterile reflux is benign, (iv) extended use of prophylactic antibiotics is benign, and (v) open surgical correction is very successful. The named framework and tenet (iv) are not in the corpus. [vur p21]
- Spontaneous resolution by grade is 100% for grade I and 9–25% for grades IV/V. The corpus says "most" for grades I–II and "few" for grades IV–V. [vur p21]
- When reflux resolves, it does so within 5 years. The corpus says about 90% of grade III resolutions occur within 4 years. [vur p21]

Medical treatment (watchful waiting)

- Prophylaxis is given at night so that the antibiotic stays concentrated in bladder urine for the longest period of physiological retention, when infection is most likely to develop. [vur p22]
- Many practitioners withdraw prophylaxis at age 5. [vur p22]
- When radiographic resolution of reflux is documented, prophylaxis is usually stopped a few days after the VCUG. [vur p22]
- When a breakthrough infection is caused by an organism sensitive to the prophylactic drug, the cause is either non-compliance or too low a dose. [vur p22]
- When a breakthrough infection is caused by a resistant organism, the notes give the cause as the dose being too high (as written) or the residual bladder volume being too high. [vur p22]

Landmark studies

- The International Reflux Study in Children was a cooperative study with a North American arm and a European arm. [vur p23]

Surgical management

- Principles of reflux correction include attention to the ureter's point of entry into the bladder, the direction of the tunnel and the ureteromucosal anastomosis, and to the muscular backing of the ureter. [vur p25]
- The bladder is handled gently to reduce postoperative haematuria and bladder spasm. [vur p25]
- Extravesical procedures include the Lich–Gregoir and the modified Lich–Gregoir. [vur p25]
- Gil-Vernet is listed among the laparoscopic antireflux procedures. The corpus describes only an open intravesical Gil-Vernet trigonoplasty. [vur p25]
- For an intravesical reimplant, the bladder is opened up to 2 cm proximal to the bladder neck. [vur p25]
- A 3–5 Fr feeding tube is placed in the ureter and sutured to the bladder mucosa at the inferior edge of the ureteric orifice. [vur p25]
- A needle-point cautery makes a circumscribing incision in the bladder mucosa about 1–2 mm from the ureteric orifice. The ureter is then mobilised into the bladder with tenotomy scissors and gentle traction on the feeding tube. [vur p25]
- In the Politano–Leadbetter technique, the ureter is brought into the bladder through a new hiatus about 2.5 cm superior and somewhat medial to the original one. [vur p25]

VUR management by sex and age

- In girls, management is to wait until 5 years of age on low-dose antibiotic prophylaxis unless an indication for correction arises. [vur p38]
- Global dysmorphism of one or both kidneys is an indication for earlier correction in girls. [vur p38]
- After 5 years of age, girls should have surgical correction even when asymptomatic, to reduce fetal and maternal morbidity in pregnancy. [vur p38]
- In boys over 5 years, bilateral high-grade reflux (as well as recurrent pyelonephritis) is an indication for surgery (Paper 3 p.45 calls treating the asymptomatic older boy with bilateral high-grade VUR controversial). [vur p38]

Sibling reflux

- Older siblings of children with VUR have a reflux prevalence of about 7%. [vur p39]
- In siblings over 5 years, ultrasound is done; a normal kidney needs no further screening, while renal abnormalities prompt a check of voiding habits and proximate urological history, with screening only if these are positive. [vur p39]

CH 49 · ANOMALIES OF THE KIDNEY AND COLLECTING SYSTEM

Calyceal diverticulum: definition and classification

- A calyceal diverticulum is a non-secretory, smooth-walled, spherical cavity in the corticomedullary region, usually in the upper or lower pole. [Anomalies p1]
- A calyceal diverticulum is also called a pyelogenic cyst, especially when it arises from the renal pelvis. [Anomalies p1]
- Dretler classification: type I has an open mouth and short neck (suits SWL); type II has a closed mouth and short neck (suits URS). [Anomalies p1]
- Dretler type III has a closed mouth and long neck, and type IV has an obliterated neck; both suit PCNL. [Anomalies p2]

Calyceal diverticulum: embryology and causes

- Congenital calyceal diverticula arise at the 5 mm stage of the embryo, when the third- and fourth-generation ureteric branches that normally degenerate persist as isolated branches. [Anomalies p2]
- Other acquired causes include obstruction from stones or infection within a calyx, and spasm or dysfunction of the supposed sphincters around a minor calyx. [Anomalies p2]

Calyceal diverticulum: diagnosis and indications

- On IVP a diverticulum is a spherical collection of contrast adjacent to a papilla, the connecting channel is difficult to show, and upright films show a contrast–fluid level. [Anomalies p3]
- Progressive enlargement of the diverticulum with pain or infection, and urosepsis, are indications for surgery. [Anomalies p3]

Bilateral renal agenesis

- Bilateral renal agenesis was described extensively by Potter, and only about 500 cases have been reported. [Anomalies p6]
- Increasing maternal age is a risk factor for bilateral renal agenesis. [Anomalies p6]
- An autosomal recessive pattern may exist, and an autosomal dominant trait with variable penetrance has also been suggested. [Anomalies p6]
- In families of an index child with BRA, 4.5% of siblings and parents had unilateral renal agenesis and 3.5% had BRA. [Anomalies p6]
- Syndromes associated with BRA include oesophageal atresia (VATER), Fraser syndrome (cryptophthalmos), Klinefelter syndrome and Kallmann syndrome. [Anomalies p6]
- BRA results from absence of the nephrogenic ridge or failure of the ureteric bud to develop from the nephric duct, from altered molecular development or a mutation of the control genes on both sides. [Anomalies p6]
- In BRA the renal vessels are completely absent in about 25%. [Anomalies p7]
- The bladder is absent or hypoplastic in BRA because of lack of stimulation by fetal urine and because the ureteric bud and Wolffian duct migrating to the cloaca are needed for bladder development. [Anomalies p7]
- Infants with BRA have low birth weight or intrauterine growth restriction. [Anomalies p7]
- Potter facies: a prematurely senile look; a prominent skin fold starting over each eye and curving in a semicircle over the inner canthus onto the cheek; a blunted nose; a prominent depression between lower lip and chin; and low-set, forward-drawn ears pressed against the head with unusually broad, large lobes. Periauricular pits and tags may be seen. [Anomalies p7]
- Other features of BRA include excessively dry, loose skin and large claw-like hands. [Anomalies p7]

- Other features of BRA include bowed, clubbed legs with excessive flexion at the hip and knee, occasional fused lower limbs (sirenomelia), and meningocele. [Anomalies p8]
- In BRA the penis and vas are usually normal, but absence of the scrotum and clitoral hypertrophy are described. [Anomalies p8]
- BRA has a high incidence of Müllerian duct anomalies: absent or hypoplastic ovaries, rudimentary or bicornuate uterus, and absent vagina or a short blind-ending pouch. [Anomalies p8]
- In the first step of lung development the kidneys produce proline, which is needed for collagen formation in the bronchiolar tree. [Anomalies p8]
- Amnion nodosum (small white keratinised nodules on the amniotic surface) is the placental hallmark of severe oligohydramnios, but it is not pathognomonic of BRA. [Anomalies p9]
- DMSA is used if ultrasound is inconclusive, and absent radionuclide uptake above background in both renal fossae and any ectopic site confirms BRA. [Anomalies p9]
- Prenatal and postnatal ultrasound is recommended when either parent or a first-degree relative has a solitary congenital kidney. [Anomalies p9]

Unilateral renal agenesis

- Unilateral renal agenesis is more frequent on the left side and shows an autosomal dominant familial tendency. [Anomalies p10]
- Poland syndrome and Fraser syndrome are associated with unilateral renal agenesis. [Anomalies p10]
- About 30% of patients with VACTERL have unilateral renal agenesis. [Anomalies p10]
- Ectopia or malrotation of the contralateral kidney is associated with unilateral renal agenesis. [Anomalies p11]
- In unilateral renal agenesis the ipsilateral adrenal may be flattened, and adrenal agenesis occurs in under 10%. [Anomalies p11]
- In men with unilateral renal agenesis, the ampulla and ejaculatory duct may also be absent, and Wolffian structures are sometimes ectopic or rudimentary. [Anomalies p12]
- 43% of women with genital abnormalities have unilateral renal agenesis. [Anomalies p12]
- The atypical (type II) form of MRKH is more common than the typical form and is associated with unilateral renal agenesis, ectopia of one or both kidneys and horseshoe kidney. [Anomalies p13]
- Other anomalies associated with unilateral renal agenesis are cardiovascular (15%), gastrointestinal (9%), haematological (6%) and neurological (3%). [Anomalies p13]
- About 5% of apparent unilateral agenesis cases are dysplastic kidneys or MCDKs that involuted before birth. [Anomalies p13]
- On a plain abdominal film, the splenic or hepatic flexure lying in its normal place, rather than in the empty renal fossa as in true agenesis, suggests an involuted kidney. An MCDK shows curvilinear calcification on X-ray or CT. [Anomalies p13]
- Complex anomalies with renal agenesis need comprehensive imaging of all organs, including the spine. [Anomalies p13]
- Postnatal colour Doppler ultrasound shows absent ipsilateral renal vessels. [Anomalies p13]

Prognosis of the solitary kidney

- The notes record a 50% probability of needing dialysis by age 30. [Anomalies p14]

Supernumerary kidney

- The supernumerary (third) kidney is usually smaller. [Anomalies p14]
- Only about 100 cases of supernumerary kidney have been reported. The sexes are equally affected, and the left side is involved more often. [Anomalies p15]
- Embryologically, a second ureteric bud or an extra branch of the first bud forms, and then the nephrogenic anlage divides into two metanephric tails. [Anomalies p15]
- The supernumerary kidney is a distinct mass, either completely separate or attached to the main kidney only by loose areolar tissue. [Anomalies p15]
- In 60% the supernumerary kidney lies caudal to the main kidney, but when it has a separate, distinct ureter it is likely to lie cranial to the dominant kidney. [Anomalies p15]

- The ureter of a supernumerary kidney joins the main ureter to share a single orifice in 50% (a "bud off a bud") and is fully independent in 50%. [Anomalies p15]
- The supernumerary kidney's blood supply is anomalous and depends on its position. [Anomalies p15]
- The supernumerary kidney is rarely symptomatic. Complaints are pain, fever, hypertension or a mass, and it is usually diagnosed on imaging done for other reasons. [Anomalies p15]
- Pre-operative evaluation of a supernumerary kidney uses MRU or retrograde pyelography for anatomy, radionuclide studies for function, and cystoscopy for the orifices. [Anomalies p16]

Crossed renal ectopia: definition and presentation by age

- Crossed renal ectopia is diagnosed in children as part of other malformations, in young adults during evaluation at menarche, and in the elderly as an incidental finding. [Fused ectopia p1]

Embryology

- The cause of renal ectopia is unknown. [Fused ectopia p2]
- Cook and Stephens proposed that crossover results from malalignment and abnormal rotation of the caudal end of the fetus, with the distal curled end of the vertebral column displaced to one side, so that the Wolffian duct structures or cloaca lie to one side of the vertebral column. One ureter then crosses the midline and enters the opposite nephrogenic blastema, or the kidney and ureter are carried to the opposite side during normal ascent. [Fused ectopia p2]
- Kelalis and colleagues held teratogenic factors responsible, and genetic factors may also play a role. [Fused ectopia p2]
- Ascent of both kidneys begins at the same time, but the ectopic unit lags behind because of the time taken to cross over. [Fused ectopia p2]
- Ascent continues until the uncrossed kidney reaches its normal position or a retroperitoneal structure prevents further ascent. [Fused ectopia p2]
- The final shape of a crossed fused kidney depends on the time of fusion, the extent of fusion and the degree of renal rotation. [Fused ectopia p3]

Crossed ectopia without fusion, solitary and bilateral crossed ectopia

- Crossed ectopia without fusion has a male-to-female ratio of 2:1. [Fused ectopia p3]
- In crossed ectopia without fusion, the uncrossed kidney usually lies in its normal location, and the ectopic kidney lies inferiorly in a diagonal or horizontal position with an anterior pelvis. [Fused ectopia p3]
- In crossed ectopia without fusion, each kidney has its own capsule and Gerota's fascia. [Fused ectopia p3]
- Solitary crossed ectopia is very rare (about 34 patients reported), with a male-to-female ratio of 2:1. [Fused ectopia p3]
- In solitary crossed ectopia, migration of the left kidney to the right with absence of the right kidney is twice as common as the opposite. [Fused ectopia p3]
- In solitary crossed ectopia the kidney usually shows incomplete rotation and ascent. [Fused ectopia p3]
- Bilateral crossed ectopia has been described in only about 5 patients. [Fused ectopia p3]
- In bilateral crossed ectopia the kidneys and renal pelves may look normal, but the ureters cross the midline at the level of the lower lumbar vertebrae. [Fused ectopia p4]

Crossed fused forms

- In the S-shaped (sigmoid) kidney, the crossed kidney lies inferiorly and is fused to the lower pole of the uncrossed kidney. Fusion is late, after rotation is complete, and the convex lateral borders of the two kidneys face opposite ways, giving the S shape. [Fused ectopia p4]
- The lump (cake) kidney usually ascends only as far as the sacral promontory, and both pelves are anterior and drain separate areas of parenchyma. [Fused ectopia p4]
- In the L-shaped kidney, the crossed kidney lies in the midline or the contralateral paramedian space. [Fused ectopia p5]
- In the disc kidney, more extensive medial fusion produces a disc or shield shape, and both pelves are anterior. [Fused ectopia p5]
- In superior ectopia both pelves are anterior, which indicates very early fusion. [Fused ectopia p5]

Vascular supply

- The ectopic kidney is supplied by one or more branches of the aorta or the common iliac artery. [Fused ectopia p5]
- The normal (uncrossed) kidney often has an anomalous blood supply, with multiple renal arteries from various levels of the aorta. [Fused ectopia p5]

Associated anomalies

- The trigone is normal in crossed ectopia except in solitary crossed ectopia. [Fused ectopia p5]
- Associated anomalies include ureterocele, ectopic ureteric orifice, imperforate anus, vertebral or orthopaedic anomalies and cardiovascular anomalies. [Fused ectopia p5]
- Solitary crossed ectopia has the highest rate of associated anomalies: 40–50% have skeletal and genital abnormalities, most often cryptorchidism, absent vas, vaginal atresia, unilateral uterine abnormality and imperforate anus. [Fused ectopia p5]

Symptoms, diagnosis and prognosis

- Most crossed ectopic kidneys are asymptomatic and found incidentally. [Fused ectopia p6]
- In about one-third of cases there is an asymptomatic abdominal mass. [Fused ectopia p6]
- Investigations are ultrasound, DMSA scan and CT urography, with MRU or MRA for pre-operative planning, and retrograde pyelography. [Fused ectopia p6]
- Most patients with crossed renal ectopia have normal longevity, and prognosis also depends on the associated anomalies. [Fused ectopia p7]

CH 50 · RENAL DYSGENESIS AND CYSTIC DISEASE OF THE KIDNEY

Autosomal recessive polycystic kidney disease

- The PKHD1 gene product, fibrocystin, is expressed in the kidney, liver and pancreas. [Cystic Disease p2]
- In ARPKD the enlarged kidneys keep a reniform shape. [Cystic Disease p2]
- In ARPKD the increased renal echogenicity on ultrasound is due to diffuse microcysts. [Cystic Disease p3]
- CT is more sensitive than ultrasound for ARPKD and can be done if the diagnosis is in doubt. [Cystic Disease p3]
- Once ARPKD is strongly suspected, genetic evaluation and counselling should be done, with a detailed family history covering at least 3 generations. [Cystic Disease p3]
- Ultrasound differentiates ARPKD from bilateral hydronephrosis (enlarged kidneys with hypoechoic calyces) and from multicystic kidney (hypoechoic cysts and a non-reniform mass). [Cystic Disease p3]
- Wilms tumour and bilateral mesoblastic nephroma are in the differential diagnosis of ARPKD. [Cystic Disease p3]
- In ARPKD, respiratory care can extend the child's life. [Cystic Disease p4]
- Surviving ARPKD patients may need treatment for congestive cardiac failure as well as hypertension, renal and hepatic failure and portal hypertension. [Cystic Disease p4]
- The diagnosis of ARPKD rests on the overall clinical data: a positive family history with a recessive mode of inheritance, a positive liver biopsy, and the absence of the extrarenal malformations seen with other cystic diseases. [Cystic Disease p4]

Medullary sponge kidney

- Although medullary sponge kidney is usually non-hereditary, some forms are inherited in an autosomal dominant fashion. [Cystic Disease p12]
- Embryologically, MSK results from disruption at the ureteric bud–metanephric blastema (RET/GDNF) interface. [Cystic Disease p12]
- MSK may occur with Beckwith–Wiedemann syndrome, Caroli disease and Ehlers–Danlos syndrome. [Cystic Disease p12]
- MSK has three clinical pictures: a typical form (recurrent calcium stones, renal tubular acidosis and metabolic bone disease), an indolent form (no or few stones and no tubular defects), and a rare but very severe form dominated by excruciating renal pain. [Cystic Disease p13]

- Histologically, MSK has dilated intrapapillary collecting ducts and small medullary cysts 1–8 mm in diameter, giving a sponge-like cut surface. [Cystic Disease p13]
- MSK is bilateral in 70%. [Cystic Disease p13]
- The concretions in the cysts and dilated ducts are mostly calcium phosphate. [Cystic Disease p13]
- Single-slice helical CT, MRI and ultrasound cannot detect the findings specific to MSK, whereas classical IVU and 3D volume-rendered images from the urographic phase of multidetector CT urography are sensitive. [Cystic Disease p13]
- Some MSK patients with infections need long-term antibiotic prophylaxis. [Cystic Disease p14]

Multicystic dysplastic kidney

- An MCDK has no reniform shape and little stroma. [Cystic Disease p15]
- In MCDK there is no connection between the glomeruli and the calyces. [Cystic Disease p15]
- MCDK size varies from a small nubbin to a large palpable mass. [Cystic Disease p15]
- There are two types of MCDK: the infundibulopelvic (pyeloinfundibular atresia) type, with atresia of both renal pelvis and ureter, and the less common hydronephrotic type, in which only the upper ureter is atretic. [Cystic Disease p15]
- MCDK is more common in males. [Cystic Disease p15]
- Proposed theories of MCDK are an extreme form of obstructive hydronephrosis from ureteric or pelvic atresia; abnormal interaction of the ureteric bud with the metanephric mesenchyme (mutations in genes such as GDNF, WNT and PAX2); and failure of union of the ureteric bud with the metanephric mesenchyme. [Cystic Disease p16]
- Angiography in MCDK shows an absent or small renal artery, but is rarely indicated. [Cystic Disease p17]
- Cystoscopy in MCDK shows a hemitrigone with an absent ureteric orifice, or, if the orifice is present, retrograde pyelography shows an atretic ureter; it is rarely indicated. [Cystic Disease p17]
- An MCDK can be ignored until its bulk becomes inconvenient. [Cystic Disease p18]

Genes in kidney and ureter development

- More than 30 genes have been identified in mammalian kidney development, but only a few (Agtr2, Bmp4, FoxC1, Pax2, Eya1) are currently known to cause kidney and ureteric anomalies. [Embryology p12]
- Renal–coloboma syndrome (PAX2 mutation) is autosomal dominant and comprises optic nerve coloboma, VUR and renal anomalies. [Embryology p12]
- Branchio-oto-renal syndrome (EYA1 mutation) is autosomal dominant, and its renal features are duplex system, hypoplasia, dysplasia and renal agenesis. [Embryology p12]
- The renin–angiotensin system is active in fetal life, acting through the AGTR2 (angiotensin II receptor type 2) gene, and maintains fetal glomerular filtration, ensures adequate urine production and supports normal growth and development of the kidney and ureter. [Embryology p12]
- ACE inhibitors in pregnancy cause oligohydramnios, fetal hypotension and anuria. [Embryology p12]
- Both AT1 and AT2 receptors are found in the developing mesonephros and metanephros; AT2 acts in embryonic renal development, while AT1 modulates later renal vascular development and the classical angiotensin II effects of vasoconstriction and sodium reabsorption. [Embryology p13]
- FOXC1 mutation causes a duplex kidney with dilated upper pole ureters that may open into the seminal vesicle or vas. [Embryology p13]
- BMP4 and BMP5 mutations cause hydronephrosis and hydroureter, because these genes control smooth muscle formation in the proximal ureter and pelvis. [Embryology p13]

CH 52 · ECTOPIC URETER, URETEROCELE AND URETERAL ANOMALIES

Ectopic ureter and ectopic vas

- A vas deferens connected to the ureter (from anomalous development of the common excretory duct) should be considered in males with recurrent epididymitis and ipsilateral hydronephrosis. [Embryology p11]

Ureterocele: classification

- In an extravesical (ectopic) ureterocele, the orifice may lie in the bladder, at the bladder neck or in the urethra. [Ureterocele etc p1]
- In Stephens' classification, the cecoureterocele is the most important subgroup. [Ureterocele etc p2]
- In a cecoureterocele the orifice lies within the bladder, while the cavity of the ureterocele extends beyond the bladder neck into the urethra. [Ureterocele etc p2]
- Ureterocele disproportion is defined as a non-obstructive ureterocele with duplication; the affected upper pole is non-dilated and so dysplastic that most imaging does not readily detect it. [Ureterocele etc p2]
- In Churchill's classification, the contralateral system is at risk because of reflux or bladder outlet obstruction (BOO). [Ureterocele etc p2]

Embryology and aetiology of ectopic ureter and ureterocele

- The specific mechanisms responsible for ectopic ureter and ureterocele remain unidentified. [Ureterocele etc p4]
- In ureterotrigonal development, the classical view holds that the common nephric duct (mesoderm) forms the trigone, whereas the revised model holds that the common nephric duct regresses by apoptosis and the trigone forms from endoderm, like the bladder. [Ureterocele etc p4]
- A mature ureterovesical connection depends on ureteral maturation and apoptosis of the common nephric duct: a loop made of the remaining common nephric duct and the distal ureter contacts the urogenital sinus, and this contact sets the position of the future ureteric orifice. [Ureterocele etc p5]
- Merger of this loop with the urogenital sinus, and removal of the remaining common nephric duct by apoptosis, creates the patent ureter–bladder connection, which at this stage lies at the level of the verumontanum; later growth of the bladder and urethra separates the mesonephric duct from the distal ureter. [Ureterocele etc p5]
- During development the ureter is a solid cord that canalises from its centre towards both ends, upwards and downwards. [Ureterocele etc p6]
- The Chwalla membrane is formed by cells at the distal end of the ureter and undergoes apoptosis shortly before birth to create a patent ureterovesical communication; persistence or abnormality of the membrane leads to ureterocele. [Ureterocele etc p6]
- Causes of a single-system ectopic ureter include misalignment of the loop with the distal urogenital sinus, which alters where the ureteral orifice inserts, and delayed or defective insertion of the nephric duct into the cloaca. [Ureterocele etc p6]
- Defective ureteral maturation (under-rotation) is a further cause of single-system ectopic ureter. [Ureterocele etc p7]
- Abnormal apoptosis may preserve the connection between the ureter and the Wolffian duct, giving drainage into the vas or ejaculatory duct in males and a Gartner duct cyst in females. [Ureterocele etc p7]
- Partial regression of the common nephric duct makes the ureter join the bladder at an abnormal posterior site, close to or within the urethra. [Ureterocele etc p7]
- The final position of the ureter depends on the extent of common nephric duct remodelling, which the Ret gene controls: with no remodelling the ureter stays joined to the sex ducts, and with partial remodelling it joins the urogenital sinus close to the nephric duct, at the level of the verumontanum. [Ureterocele etc p8]

Evaluation

- Evaluation has three parts: anatomical assessment (physical examination, ultrasound, MRI), functional assessment and endoscopic evaluation. [Ureterocele etc p10]
- A palpable dilated upper pole is a possible finding on physical examination. [Ureterocele etc p10]

Ureterocele and Ectopic Ureter: Functional Assessment

- Ultrasound gives an idea of parenchymal thickness and corticomedullary differentiation (CMD) of the affected moiety. [Ureterocele etc p11]
- IVP gives only qualitative function and is a less useful baseline study. [Ureterocele etc p11]

Endoscopic Assessment

- Endoscopic assessment is not always essential. [Ureterocele etc p12]

Algorithm for Treatment of Ureterocele (Duplex System)

- In a stable child with no hydronephrosis and normal function but low-grade reflux, the plan is antibiotic prophylaxis and follow-up. [Ureterocele etc p14]

Upper Pole Partial Nephrectomy

- A second (Gibson) incision may be needed to resect the upper pole ureter in its entirety when there is reflux into it. [Ureterocele etc p16]
- Doppler sonography is the best postoperative evaluation after upper pole partial nephrectomy. [Ureterocele etc p16]
- Postoperative Doppler sonography shows normal postoperative anatomy, the absence of a urinoma and normal blood flow to the lower pole. [Ureterocele etc p16]

Upper Pole Partial Nephrectomy: Outcome and Complications

- After upper pole partial nephrectomy for ureterocele, pre-existing lower pole reflux resolves in 20%. [Ureterocele etc p17]
- Inferior vena cava (IVC) laceration is a complication of upper pole partial nephrectomy. [Ureterocele etc p17]
- Duodenal perforation is a complication of upper pole partial nephrectomy. [Ureterocele etc p17]
- Conversion to (or loss of the kidney needing) total nephrectomy is a complication of upper pole partial nephrectomy. [Ureterocele etc p17]
- Peritoneal tears are a complication of upper pole partial nephrectomy. [Ureterocele etc p17]

Ureterocele Excision and Common Sheath Reimplant

- Through a transvesical approach, excision of the ureterocele begins with a transverse incision over it. [Ureterocele etc p17]
- Total reconstruction consists of upper pole partial nephrectomy plus ureterocele excision and common sheath reimplantation of the lower pole ureter; it is definitive surgery but an extensive operation. [Ureterocele etc p17]

Pyeloureterostomy or Ureteroureterostomy

- When the upper pole drains into a ureterocele and is preserved by pyeloureterostomy or ureteroureterostomy, resection of the ureterocele may be avoided as long as it is well decompressed. [Ureterocele etc p18]

Alternative Approach for Ureterocele Resection

- Marsupialisation of a ureterocele removes its thin intravesical aspect and sutures the edges. [Ureterocele etc p19]
- Marsupialisation gives satisfactory results, but once open surgery is embarked upon it is better to do a definitive surgical correction. [Ureterocele etc p19]

CH 53 · SURGERY OF THE URETER IN CHILDREN

Intravesical reimplantation: Politano–Leadbetter and Glenn–Anderson

- In the Politano–Leadbetter repair, a second submucosal tunnel can be made so that the ureteral opening sits closer to the bladder neck, especially when the original orifice lies too laterally. [vur p27]
- The sure sign of a non-obstructed reimplanted ureter is a jet of urine from the orifice, and a 5 Fr feeding tube can also be passed up the ureter to confirm patency. [vur p27]
- The Glenn–Anderson (infrahiatal) technique has a success rate of about 98%, similar to the other reimplant procedures. [vur p27]

Cohen cross-trigonal technique

- The Cohen technique overcomes the main problem of the Glenn–Anderson technique, which is inadequate tunnel length. [vur p28]

- In a bilateral Cohen repair, the tunnel for the more laterally displaced ureter is directed superior to the contralateral orifice, and the second tunnel is directed towards the inferior edge of the laterally displaced ureter. [vur p28]
- To catheterise a superolaterally placed orifice after a Cohen repair, a suprapubic cystostomy (SPC) can be used to place the wire or ureteric catheter, which is then directed cystoscopically into the orifice. [vur p28]
- A curved-tip vascular access catheter can be used to cannulate the orifice after a Cohen repair. [vur p28]
- Flexible ureteroscopy is another solution for upper-tract access (e.g. for nephrolithiasis) after a Cohen repair. [vur p28]

Extravesical procedures

- Extravesical reimplantation is simple to learn. [vur p29]
- Transient voiding inefficiency after bilateral extravesical reimplantation occurs especially in boys younger than 3 years. [vur p29]
- Children with transient voiding inefficiency after extravesical reimplantation usually void normally within 1–2 weeks. [vur p29]
- The distal ureter is identified where it crosses medial to the point at which the obliterated umbilical artery arises from the internal iliac artery. [vur p29]

Laparoscopic and endoscopic procedures for VUR

- The Gil-Vernet procedure is the least successful laparoscopic procedure for reflux. [vur p30]
- Laparoscopic extravesical reimplantation is the most common laparoscopic procedure for reflux. [vur p30]
- Laparoscopic extravesical reimplantation has a steep learning curve: it is initially difficult to develop the extravesical tunnel without injuring the urothelium, operative times are long, and elevating the detrusor flaps is difficult. [vur p30]
- Endoscopic (transvesical) cross-trigonal reimplantation is similar to the Cohen operation and is done with CO_2 insufflation of the bladder. [vur p30]

Post-operative evaluation

- After reimplantation, ultrasound and VCUG are done at 3 months; if there is no reflux, antibiotic prophylaxis is stopped and no further VCUG is needed. [vur p31]
- If reflux persists at 3 months, the VCUG is repeated at 1 year, and persistent reflux and/or recurrent UTI at that point means the failed surgery is corrected. [vur p31]

Early complications of reimplantation

- Persistent reflux after reimplantation is more common in high-grade cases, and high-grade reflux is one cause of long-term persistent reflux. [vur p32]
- The cause of new contralateral reflux after unilateral reimplantation is not clearly known. [vur p32]

Long-term complications of reimplantation

- Late obstruction is classified by location: suprahiatal (twist and ischaemia), hiatal (angulation of the ureter), tunnel (improper tunnel creation) and orifice. [vur p33]
- At the orifice, the most vulnerable point for obstruction is the apex of the ureteral spatulation. [vur p33]
- Obstruction at the tunnel level can be treated by endoscopic unroofing of the distal few millimetres of the tunnel. [vur p33]

Endoscopic treatment: injection technique

- Additional volume is injected until the ureteric orifice becomes crescent-shaped or slit-shaped. [vur p34]
- The needle is kept in place for 1 minute after injection so the implant is not dislodged, although this is not necessary with Deflux. [vur p34]

Double HIT, follow-up and ideal material

- If reflux persists after endoscopic injection, the injection is repeated at 6 months, and surgery follows if there is still no resolution. [vur p35]
- The ideal injectable material is non-toxic, easy to inject, causes minimal inflammation, becomes encapsulated by normal fibrous tissue, is viscous enough to prevent leakage, and maintains its injected volume. [vur p35]

Classification of injectable materials

- Injectable materials are classified as autologous or non-autologous, and as particulate or degradable; the concern with particulate agents is migration. [vur p36]
- Calcium hydroxyapatite (Coaptite) and cross-linked bovine collagen are non-autologous agents used for VUR injection (the corpus lists them only as urethral bulking agents for SUI). [vur p36]
- Autologous injectable materials include fat, collagen, muscle and chondrocytes. [vur p36]
- The main disadvantage of autologous injectables is volume loss. [vur p36]
- Autologous injectables need harvesting and expansion. [vur p36]

Deflux

- After Deflux injection, the carrier gel is reabsorbed and the dextranomer microspheres are encapsulated by fibroblasts and collagen. [vur p37]
- The overall success rate of Deflux injection is 65–90%. [vur p37]
- Deflux is a natural product. [vur p37]
- Deflux does not need a ratcheted syringe because a small-gauge needle is used. [vur p37]

CH 56 · HYPOSPADIAS

Hypospadias: definition, epidemiology and syndromic associations

- First-degree relatives of a boy with hypospadias have a 13 times greater risk of hypospadias. [Hypospadias p1]
- About 90% of hypospadias is isolated, a clinical diagnosis needing no imaging, while about 10% is associated with syndromes or other abnormalities. [Hypospadias p1]
- Syndromes associated with hypospadias include WAGR syndrome, G syndrome, Wolf–Hirschhorn syndrome, 13q deletion syndrome and hand–foot–uterus syndrome (as well as Smith–Lemli–Opitz syndrome). [Hypospadias p1]
- When hypospadias occurs with an undescended testis as part of a DSD, the commonest underlying diagnoses are mixed gonadal dysgenesis or ovotesticular DSD. [Hypospadias p1]

Timing of repair and pre-operative androgen stimulation

- Pre-operative androgens increase penile length as well as glans circumference. [Hypospadias p2]
- Study results on pre-operative androgen are controversial, with some studies reporting increased rather than decreased complications. [Hypospadias p2]

Ventral curvature (chordee)

- In distal hypospadias only about 10% have chordee (curvature, usually under 30°) after degloving. [Hypospadias p3]
- In proximal hypospadias about 50% have no curvature or curvature under 30°, and the other 50% have curvature over 30°. [Hypospadias p3]
- For curvature over 30°, three transverse ventral corporotomies (from the 4 to 8 o'clock position) without grafting can be made: one at the point of greatest curvature, one 4 mm above and one 4 mm below it. [Hypospadias p3]
- Multiple plications for curvature over 30° may give results that are not durable into adulthood. [Hypospadias p3]
- A single 3-to-9 o'clock corporotomy needs a graft, which is undesirable because it leaves only a flap urethroplasty as the option for the urethra (a graft will not take over a graft). [Hypospadias p3]

Techniques for distal hypospadias: MAGPI

- MAGPI is indicated for glanular or coronal hypospadias with a flat or convex glans, with healthy, thick skin proximal to the meatus and no chordee. [Hypospadias p4]
- MAGPI steps: subcoronal incision with circumferential degloving; a vertical incision in the glanular groove is closed transversely so that the dorsal meatus is advanced (meatal advancement); the ventral lip of the meatus is then pulled distally and the glans is closed beneath it (glanuloplasty); the skin is closed as for a circumcision. [Hypospadias p4]

Mathieu (flip-flap) technique

- Mathieu is used for distal hypospadias that is too proximal for MAGPI. [Hypospadias p5]
- Mathieu steps: measure the urethral plate from meatus to glans tip and mark an equal length on the proximal shaft skin; incise the glans and flap and deglove the shaft; dissect the flap's subcutaneous tissue carefully so it can be flipped over the plate, with a feeding tube between the two into the bladder; suture the flap edges to the plate edges; take a dartos flap from the dorsal degloved tissue, divide it in the midline and bring it round on either side to cover the suture lines; then reapproximate the glans wings and close the skin. [Hypospadias p5]

TIP (Snodgrass) repair

- In TIP, degloving uses a ventral incision placed 2 mm below the meatus. [Hypospadias p6]
- The ventral dartos flap is split, when possible, into two longitudinal segments that are crossed over the neourethra to give a two-layer cover. [Hypospadias p6]
- Technical variations of TIP include a single- or two-layer neourethra and epithelial or subepithelial suturing. [Hypospadias p6]

Proximal hypospadias: onlay preputial flap

- Onlay flap steps: hold the corners of the dorsal prepuce with stay sutures; harvest a 10 mm wide strip of inner prepuce, preserving its dartos blood supply; extend the pedicle dissection to the penopubic junction to avoid tension when the flap is moved ventrally round the penis; pass a 6 Fr stent into the bladder and sew the flap to the urethral plate on each side with subepithelial sutures; then glansplasty and skin closure. [Hypospadias p9]
- The onlay preputial flap has a complication rate of 27–45%. [Hypospadias p9]

Single-stage tubularised preputial flap

- After degloving, a 12–15 mm wide inner preputial flap is raised, preserving the underlying dartos blood supply. [Hypospadias p10]
- Option 1: tubularise the flap over a 6 Fr stent in two layers of 7-0 Vicryl subepithelial sutures, move it ventrally and sew it to the native meatus proximally and the glans wings distally. [Hypospadias p10]
- Option 2: bring the flap ventrally before tubularising, sew one edge to the underlying corpora to create a pseudoplate, and then tubularise it. [Hypospadias p10]
- The single-stage tubularised preputial flap has a complication rate of 14–33%. [Hypospadias p10]

Two-stage graft repair

- Choice of graft: use prepuce if circumcision is wanted, and oral mucosa when preputioplasty is desired. [Hypospadias p11]
- In stage 1 the urethral plate is transected distally at the coronal level and excised off the corpora, moving proximally to the meatus and beyond towards the membranous portion. [Hypospadias p11]
- The native urethra is then gently stretched distally and anchored to the corpora, which reduces the length of graft needed. [Hypospadias p11]
- A complete proximal urethrostomy is made by suturing the native meatus to the corpora at 10, 12 and 2 o'clock and to the penile skin or scrotum at 4, 6 and 8 o'clock. [Hypospadias p11]
- The dorsal shaft skin is sutured to the preputial collar, and the inner preputial graft is placed ventrally and stitched to the glans at the level of the corona and to the skin on either side. [Hypospadias p12]

- The proximal end of the graft is split in the midline to extend on either side of the urethrostomy, sutured at 10, 12 and 2 o'clock medially and to the shaft skin or scrotum laterally. [Hypospadias p12]
- After quilting, the boy is catheterised and a tie-over bandage is applied for 7 days to immobilise the graft. [Hypospadias p12]
- At stage 2, the penile and scrotal skin is closed in the midline with subepithelial sutures. [Hypospadias p12]

Byars flaps

- Few outcomes of Byars two-stage flap repair are reported; Snodgrass had complications in all 9 of his cases and does not recommend the operation. [Hypospadias p13]

Reoperations

- A dorsal inlay graft is indicated at reoperation when the urethral plate has been removed but a strip of grossly healthy skin remains in its place. [Hypospadias p14]
- The dorsal inlay graft is also used for neourethral strictures. [Hypospadias p14]
- Further indications for a two-stage oral mucosa graft at reoperation are strictures that nearly obliterate the lumen and hair in the neourethra. [Hypospadias p14]

Complications: risk factors and fistula

- A glans width under 14 mm calls for extended dissection of the glans wings. [Hypospadias p16]

Glans dehiscence and meatal stenosis

- Glans dehiscence is complete separation of the glans wings, with or without a band of skin bridging the gap between them, and causes functional impairment with a splaying stream. [Hypospadias p17]
- To prevent glans dehiscence, use polyglactin sutures (chromic has been abandoned). [Hypospadias p17]
- To prevent meatal stenosis, limit the plate incision to the urethral plate without extending it distally into the glans. [Hypospadias p17]
- To prevent meatal stenosis, do not suture the glans wings to the neourethra. [Hypospadias p17]

Meatal stenosis treatment, neourethral stricture, diverticulum and BXO

- Meatal stenosis is treated by incising the neomeatus dorsally, which enlarges it without recreating the hypospadias defect. [Hypospadias p18]
- A neourethral stricture under 1 cm can be treated by direct vision internal urethrotomy, but not when the neourethra is a tubularised flap or graft. [Hypospadias p18]
- A diverticulum is exposed through a ventral median raphe incision and opened, excess tissue is de-epithelialised, and the neourethra is sutured in two layers over a catheter in a vest-over-pants fashion. [Hypospadias p18]
- Medical treatment of BXO after hypospadias repair is topical steroid or tacrolimus. [Hypospadias p18]

CH 57 · ETIOLOGY, DIAGNOSIS AND MANAGEMENT OF THE UNDESCENDED TESTIS

Testicular Descent (five phases)

- In phase I of descent (about 5 weeks), the future gubernaculum comes into contact with the mesonephros at the internal inguinal ring. [UDT p1]
- In phase II (8–10 weeks), the genitofemoral nerve accompanies the newly formed gubernaculum and processus vaginalis, with growth of the gubernaculum, deepening of the processus vaginalis and extension of the cremaster into the gubernaculum. [UDT p1]
- Phase III (10–14 weeks) is marked by growth of the testis with regression of the Müllerian ducts and the mesonephros, and the gubernaculum begins its swelling phase. [UDT p1]
- Phase IV (14–20 weeks) comprises continued swelling of the gubernaculum, further development of the cremasteric muscle, and migration of the processus vaginalis that widens the inguinal canal. [UDT p1]

Regulation of Testicular Descent

- Neuromuscular mechanisms are a third regulator of the inguinal phase, alongside androgens and the genitofemoral nerve/CGRP. [UDT p2]
- The formation of the inguinal bursa is listed as one of the factors responsible for descent, together with differential growth of the body wall and the processus vaginalis. [UDT p2]

Incidence and Non-syndromic Aetiology

- Undescended testis is more common on the right side. [UDT p4]
- Breech presentation is a risk factor for non-syndromic congenital cryptorchidism. [UDT p4]
- Being small for gestational age is a risk factor for cryptorchidism. [UDT p4]
- Maternal obesity is a risk factor for cryptorchidism. [UDT p4]
- Maternal cola consumption is listed as a risk factor for cryptorchidism. [UDT p4]

Genetic, Environmental and Syndromic Cryptorchidism

- Familial cryptorchidism may follow an autosomal dominant pattern with reduced penetrance. [UDT p5]
- The recurrence risk ratio is highest in twins (10), then brothers (3.5), then the father (2.3). [UDT p5]
- Maternal use of acetaminophen (paracetamol) or other analgesics in pregnancy is an environmental risk factor for cryptorchidism. [UDT p5]
- Syndromic cryptorchidism (diseases with reduced androgen production and/or action, such as DSD) spans about 500 syndromes and 200 genes. [UDT p5]
- Down syndrome is among the syndromes associated with cryptorchidism. [UDT p5]

Diagnosis

- The superficial inguinal pouch lies anterior to the rectus abdominis and medial to the pubic tubercle (as written in the notes). [UDT p6]
- Perirenal is listed among the ectopic sites off the normal line of descent (as written in the notes). [UDT p6]
- Scrotal asymmetry should be noted on examination. [UDT p6]
- An extra-abdominal testis may be impalpable because of small size, obesity or a non-cooperative child. [UDT p6]

Role of Imaging

- MRI is indicated when an ectopic testis has not been localised by laparoscopy. [UDT p7]

Fertility Risk and Catch-up Growth

- After orchidopexy before 9 months of age catch-up testicular growth may occur, between 9 months and 3 years it is variable, and after 3 years it does not occur. [UDT p8]

Torsion Risk and Aims of Surgery

- The undescended testis is prone to torsion because of anomalous attachment of the tunica vaginalis to the testis and partial or complete non-fusion between the testis and epididymis. [UDT p9]
- The aims of orchidopexy also include cosmesis and earlier detection of malignancy (by placing the testis where it can be examined), in addition to preserving fertility, reducing malignancy risk and lowering the risk of hernia or torsion. [UDT p9]

Age-based management of UDT

- In men older than 50 years with an undescended testis, the risk of malignancy is not defined. [UDT p11]
- In pubertal or post-pubertal boys, orchidectomy aims to prevent both malignancy and torsion, because by this age fertility potential of the undescended testis is already lost. [UDT p11]

Approach to palpable testis: inguinal orchiopexy

- When length is short, the transversalis fascia is incised along the medial border of the internal ring as part of gaining length (with medialisation of the cord beneath the inferior epigastric vessels, the

Prentiss manoeuvre). [UDT p12]

Testis not reaching the scrotum; transscrotal orchiopexy

- If the testis cannot be brought to the scrotum, gradual tension can be applied by fixing it to the scrotum with a non-absorbable suture and then applying gentle pull at regular intervals. [UDT p13]
- In a two-stage procedure, a PTFE membrane can be wrapped around the testis to help identify it at the second stage. [UDT p13]
- Incisions for transscrotal orchiopexy include, besides the Bianchi (superior scrotal) incision, a transverse low scrotal incision and a midline scrotal incision. [UDT p13]

Laparoscopic orchidopexy for the non-palpable testis

- For a unilateral non-palpable testis, ports are a 5 mm infraumbilical port plus additional 2 or 3 mm trocars in the right and left lower quadrants. [UDT p14]
- For bilateral non-palpable testes, ports are an infraumbilical port plus one port on each side at the level of the umbilicus in the midclavicular line. [UDT p14]
- Laparoscopy in children is harder because of the smaller abdominal space and difficult port placement, and it needs miniature instruments. [UDT p14]
- Laparoscopy makes the diagnosis in about 95% of non-palpable testes. [UDT p15]
- MRI sensitivity for locating the non-palpable testis is about 65% (ultrasound about 45%). [UDT p15]
- The patient is placed supine with 10 degrees of head-down tilt for laparoscopic orchidopexy. [UDT p15]
- The anaesthetist should not use nitrous oxide during paediatric laparoscopy because it causes bowel distension. [UDT p15]
- With a vanishing testis, the vas and spermatic vessels are seen ending blindly near the psoas muscle. [UDT p15]
- When the internal inguinal ring is closed, 97% will have a vanished testis. [UDT p15]

Intra-abdominal testis: technical points

- Mobilisation is adequate when the testis can be brought across towards the contralateral internal ring. [UDT p16]
- The new ring (hiatus) is created medial to the inferior epigastric vessels and lateral to the medial umbilical ligament. [UDT p16]

Management of UDT in a post-pubertal male (unilateral)

- In a post-pubertal male with a palpable unilateral UDT, counsel first and then do inguinal exploration with orchidectomy, especially if the testis is hypoplastic; if the patient declines orchidectomy, do orchiopexy after a clear explanation of the malignancy risk and of the lack of contribution to fertility. [UDT p18]
- In a post-pubertal male with a non-palpable testis whose vas and vessels enter the inguinal canal at laparoscopy, do orchidectomy, or orchiopexy if the patient declines, after explaining the malignancy and fertility issues. [UDT p18]

Management of bilateral UDT in a post-pubertal male

- In bilateral post-pubertal UDT the considerations are no fertility potential, risk of torsion and malignancy, and endocrine dysfunction if both testes are removed. [UDT p19]
- Bilateral palpable post-pubertal UDT is managed by bilateral inguinal exploration: bilateral orchiopexy if both testes are healthy, and orchidectomy on the hypoplastic side with orchiopexy on the healthy side if only one is healthy. [UDT p19]
- If both testes are hypoplastic, do karyotyping and testicular biopsy, then either bilateral orchidectomy with hormone replacement therapy or preserve the better-functioning testis. [UDT p19]
- Bilateral non-palpable post-pubertal UDT, once DSD is ruled out, is managed by bilateral orchidectomy with hormone replacement therapy. [UDT p19]
- With one palpable and one non-palpable testis in a post-pubertal male, remove the abdominal testis (greater chance of loss of function and malignancy) and do orchiopexy on the inguinal testis to preserve some endocrine function, explaining the malignancy and fertility issues. [UDT p19]

General points

- Microsurgical (autotransplant) orchiopexy should be done by experts only. [UDT p20]
- Microsurgical orchiopexy can be used for a single (solitary) intra-abdominal testis. [UDT p20]

CH 63 · PAEDIATRIC UROLOGIC ONCOLOGY: RENAL AND ADRENAL

Neuroblastoma

- Neuroblastoma arises from neural crest cells, at sites in the neck, thorax, retroperitoneum, pelvis or adrenal. [Neuroblastoma p1]
- Neuroblastoma accounts for 8–10% of all childhood cancers. [Neuroblastoma p1]
- Annual incidence is 10 per 1 million live births, which makes it more common than Wilms tumour. [Neuroblastoma p1]

Genetics

- Neuroblastoma is mostly sporadic; less than 1% is familial, and familial cases may be bilateral and present early. [Neuroblastoma p1]
- Siblings and offspring of an affected child have a risk of less than 6%. [Neuroblastoma p1]
- The hereditary neuroblastoma predisposition genes are PHOX2B and ALK (the notes annotate these as "chromosome 11"; for checking, PHOX2B lies on 4p and ALK on 2p). [Neuroblastoma p1]

Pathology

- The tumours form a spectrum of histological maturation and differentiation: neuroblastoma, ganglioneuroblastoma and ganglioneuroma. [Neuroblastoma p2]
- Histology is classified by the Shimada classification and the International Neuroblastoma Pathology Classification (INPC, the modified Shimada classification). [Neuroblastoma p2]
- The Shimada classification is an age-linked histopathological classification; stroma-poor tumours (less than 50% stroma) have a poor prognosis. [Neuroblastoma p2]
- Ganglioneuroma is diagnosed in older children and is uncommon in the adrenal. [Neuroblastoma p2]
- Ganglioneuroma should be resected only when symptomatic, because survival is not influenced by the extent of resection, so aggressive resection is avoided. [Neuroblastoma p2]
- Favourable histology accounts for 90% and unfavourable (anaplastic) histology for 10% of Wilms tumours. [Wilms tumor p3]
- The epithelial component of the triphasic tumour forms tubules, rosettes and glomeruloid structures. [Wilms tumor p3]
- Favourable histology has no anaplasia and no sarcomatous features. [Wilms tumor p3]
- After pre-operative chemotherapy the relative proportions of the histological components change, and stromal and epithelial tumours are seen more often. [Wilms tumor p3]
- Blastemal components are decreased after chemotherapy. [Wilms tumor p3]

Clinical presentation and pattern of spread

- Metastatic disease presents with bone or joint pain, periorbital ecchymosis and proptosis (raccoon eyes), and cutaneous metastases (blueberry muffin lesions). [Neuroblastoma p3]
- Direct extension causes neurological signs (spinal cord), cough and dyspnoea (thoracic tumours), and urinary retention or constipation (pelvic tumours at the organ of Zuckerkandl). [Neuroblastoma p3]
- Opsoclonus–myoclonus is described as "dancing eyes, dancing feet" (acute myoclonic encephalopathy). [Neuroblastoma p3]
- Catecholamine secretion can cause paroxysmal hypertension and palpitations. [Neuroblastoma p3]
- VIP secretion causes hypokalaemia in addition to watery diarrhoea. [Neuroblastoma p3]

Investigations

- Anaemia results from bone marrow involvement, and two bone marrow aspirates with two biopsies are recommended. [Neuroblastoma p4]
- Plain X-ray shows stippled calcification in an abdominal or posterior mediastinal mass. [Neuroblastoma p4]

- Extension into the spinal canal also helps to distinguish neuroblastoma from Wilms tumour. [Neuroblastoma p4]
- MIBG can be done with 131I or 123I; 123I-MIBG is the more sensitive. [Neuroblastoma p4]
- Routine bloods include serum calcium and BUN in addition to CBC, LFT and RFT. [Wilms tumor p2]

Staging (INSS)

- INSS stage 1 is a localised tumour with complete gross resection. [Neuroblastoma p5]
- INSS stage 2A is a localised tumour with incomplete gross resection; stage 2B has positive ipsilateral lymph nodes, with or without complete gross resection. [Neuroblastoma p5]
- INSS stage 3 is an unresectable unilateral tumour infiltrating across the midline with or without regional nodes, or a localised unilateral tumour with contralateral nodal involvement, or a midline tumour with bilateral extension by infiltration or by nodal involvement. [Neuroblastoma p5]
- INSS stage 4 is dissemination to distant lymph nodes, bone, bone marrow, liver, skin or other organs. [Neuroblastoma p5]
- In stage 4S the primary is localised (stage 1, 2A or 2B). [Neuroblastoma p5]

Treatment

- Low-risk tumours are stage 1, stage 4S, and stage 2 without MYCN amplification or unfavourable histology. [Neuroblastoma p6]
- In stage 4S, chemotherapy is indicated for extensive liver disease (the notes write "more than 2 months" of age) or for MYCN amplification; otherwise observe. [Neuroblastoma p6]
- In low-risk stage 2, radiotherapy is used for residual disease in the tumour bed or for relapse. [Neuroblastoma p6]
- Intermediate- and high-risk disease includes stage 3, stage 4 and stage 2 with MYCN amplification and unfavourable histology. [Neuroblastoma p7]
- Induction chemotherapy is etoposide, cisplatin, cyclophosphamide and doxorubicin. [Neuroblastoma p7]
- A tumour still inoperable after induction is treated with radiotherapy. [Neuroblastoma p7]
- Consolidation chemotherapy is busulfan plus melphalan. [Neuroblastoma p7]
- Fenretinide, which induces apoptosis, is a biological therapy for minimal residual disease. [Neuroblastoma p8]
- Radiotherapy should be avoided for spinal cord compression because it impairs spinal growth. [Neuroblastoma p8]

Prognostic factors

- Serum ferritin above 143 ng/mL is unfavourable (below 143 ng/mL favourable). [Neuroblastoma p9]
- A urine VMA/HVA ratio above 1 is unfavourable (below 1 favourable). [Neuroblastoma p9]
- Neuron-specific enolase (NSE) above 100 is unfavourable (below 100 favourable). [Neuroblastoma p9]
- The increased relapse risk with 16q/1p LOH is independent of histology and stage. [Wilms tumor p10]
- High telomerase activity is an unfavourable biological prognostic factor. [Wilms tumor p10]
- Increased VEGF expression is an unfavourable biological prognostic factor. [Wilms tumor p10]
- Aneuploidy is an unfavourable biological prognostic factor. [Wilms tumor p10]

Wilms Tumor: epidemiology and clinical presentation

- The incidence of Wilms tumour is 8 per million children. [Wilms tumor p1]
- About 90% present as an incidentally detected, asymptomatic abdominal mass in the 2–5 year age group. [Wilms tumor p1]
- Constitutional symptoms include fever, anorexia and weight loss. [Wilms tumor p1]
- Acute abdominal pain can occur due to tumour rupture. [Wilms tumor p1]
- Clinically the mass is firm, non-tender and does not cross the midline. [Wilms tumor p1]
- Congestive cardiac failure (with hypertension, varicocele and hepatomegaly) is a sign of venous involvement. [Wilms tumor p1]
- The classic triad is fever, lump and haematuria (margin note). [Wilms tumor p1]

Nephrogenic Rests

- In nephrogenesis the ureteric bud meets the metanephric blastema at 5 weeks, fetal lobulations appear by 28 weeks, and nephrogenesis is complete at 36 weeks. [Wilms tumor p4]
- A nephrogenic rest is defined as persistence of metanephric blastema in the kidney beyond 36 weeks of gestation, when nephrogenesis is almost complete. [Wilms tumor p4]
- Nephrogenic rests are found in 1% of infant autopsies. [Wilms tumor p4]
- Nephrogenic rests are typically clinically occult and are detected only incidentally in conjunction with a Wilms tumour. [Wilms tumor p4]
- Diffuse nephroblastomatosis is the presence of multiple nephrogenic rests that enlarge the kidney but preserve its shape. [Wilms tumor p4]
- Diffuse nephroblastomatosis is prone to develop Wilms tumour, and bilateral lesions are common. [Wilms tumor p4]
- Perilobar rests make up 90% and intralobar rests 10% of nephrogenic rests. [Wilms tumor p5]
- Perilobar rests are multiple, well circumscribed and usually smooth. [Wilms tumor p5]
- Intralobar rests are single or multiple, with irregular margins that may interdigitate with the normal adjacent renal interstitium. [Wilms tumor p5]
- The predominant tissue in perilobar rests is blastema, whereas intralobar rests are predominantly stroma and epithelium. [Wilms tumor p5]
- Perilobar rests have less chance, and intralobar rests more chance, of malignant transformation. [Wilms tumor p5]
- Besides Beckwith–Wiedemann syndrome, perilobar rests are associated with Perlman syndrome and trisomy 18. [Wilms tumor p5]
- Besides WAGR and Denys–Drash syndromes, intralobar rests are associated with sporadic aniridia. [Wilms tumor p5]
- Grossly, nephrogenic rests appear as tan nodules or white plaques. [Wilms tumor p5]
- Histologically, nephrogenic rests may be sclerosing, hyperplastic, dormant or neoplastic (transforming into Wilms tumour). [Wilms tumor p5]
- On ultrasound, nephrogenic rests give a nodular appearance with loss of corticomedullary differentiation and a diffusely echogenic kidney. [Wilms tumor p6]
- On CT, rests enhance less than the renal parenchyma, and the renal surface may appear lobulated. [Wilms tumor p6]
- MRI is the best modality: rests are hypointense on T1-weighted and hyperintense on T2-weighted images. [Wilms tumor p6]
- On angiography rests do not opacify, but contrast may puddle in the intervening normal parenchyma. [Wilms tumor p6]
- Treatment of nephrogenic rests is controversial: some recommend chemotherapy, whereas others hold that close radiological surveillance of an enlarging mass is sufficient. [Wilms tumor p6]
- The greater the number of nephrogenic rests, the more likely a Wilms tumour will develop. [Wilms tumor p6]
- The highest risk is with diffuse perilobar nephrogenic rests, about 35% (the notes put a "?" beneath this figure). [Wilms tumor p6]

Genetics and Syndromes

- Familial Wilms tumour has an earlier onset and more often bilateral disease. [Wilms tumor p7]
- About 10% of Wilms tumours are associated with syndromes or congenital anomalies. [Wilms tumor p7]
- WT1 (11p13), WT2 (11p15) and TP53 (chromosome 17) show both somatic and germline mutations. [Wilms tumor p7]
- WTX and CTNNB1 mutations are somatic only. [Wilms tumor p7]
- FWT1 and FWT2 are the familial Wilms tumour genes. [Wilms tumor p7]
- Beckwith–Wiedemann syndrome is an overgrowth disorder with macroglossia and abdominal wall defects (the notes also list "microcephaly"; standard texts give macrosomia). [Wilms tumor p7]

- After nephrectomy for unilateral Wilms tumour, the contralateral kidney should be screened with 3–4 monthly ultrasound until 5 years of age, especially in children under 12 months with nephrogenic rests in the resected specimen (greatest risk). [Wilms tumor p8]

Clinical Staging (pre-chemotherapy, NWTS/COG)

- The stage distribution is stage I 43%, stage II 20%, stage III 21%, stage IV 11% and stage V 5%. [Wilms tumor p9]

Treatment: Surgery and NWTSG/COG approach

- Exploration of the contralateral kidney is not needed if pre-operative CT or MRI shows a normal contralateral kidney. [Wilms tumor p11]
- Laparoscopic nephrectomy for Wilms tumour is usually done in conjunction with pre-operative chemotherapy. [Wilms tumor p11]
- Under NWTSG/COG, pre-operative chemotherapy is also given when the tumour is found inoperable at exploration (in addition to planned renal-sparing surgery, bilateral tumours and IVC extension above the hepatic veins). [Wilms tumor p11]
- After pre-operative chemotherapy, imaging is repeated at 6 weeks; adequate shrinkage leads to surgery followed by chemotherapy with or without radiotherapy, whereas progressive disease needs more intense chemotherapy and carries a poor prognosis. [Wilms tumor p11]

SIOP and UKCCSG approaches; Bilateral Wilms Tumour

- SIOP pre-operative chemotherapy decreases morbidity, particularly the late effects of radiotherapy. [Wilms tumor p12]
- The UKCCSG (United Kingdom Children's Cancer Study Group) gives pre-operative chemotherapy like SIOP but takes a biopsy before starting chemotherapy, which avoids giving chemotherapy to benign tumours and other histologies. [Wilms tumor p12]
- In bilateral Wilms tumour, the kidney with the lower tumour burden is operated on first. [Wilms tumor p13]
- Renal-sparing surgery may be a partial nephrectomy or a wedge excision of the tumour, aiming for negative surgical margins. [Wilms tumor p13]

Post-operative Chemotherapy (NWTSG/COG)

- Favourable histology stage I/II receives vincristine and actinomycin D (VA) for 18 weeks. [Wilms tumor p14]
- Favourable histology stage III/IV receives vincristine, doxorubicin and actinomycin D (VDA) for 24 weeks plus 10.8 Gy radiotherapy to the tumour site. [Wilms tumor p14]
- Focal anaplasia stage I receives VA for 18 weeks with or without doxorubicin. [Wilms tumor p14]
- Focal anaplasia stage II, III and IV receives VDA for 24 weeks plus 10.8 Gy radiotherapy to the tumour bed. [Wilms tumor p14]
- Diffuse anaplasia stage I receives VA for 18 weeks with or without doxorubicin. [Wilms tumor p14]
- Diffuse anaplasia stage II, III and IV receives VDA plus etoposide and cyclophosphamide for 24 weeks plus radiotherapy. [Wilms tumor p14]
- In stage IV with pulmonary metastases, an additional 12 Gy is given to the chest ("one dozen to chest"). [Wilms tumor p15]
- Whole-abdomen radiotherapy is indicated for tumour rupture or spill and for peritoneal metastasis. [Wilms tumor p15]
- Radiotherapy should start within 14 days of nephrectomy. [Wilms tumor p15]

SIOP 93-01 (Favourable histology)

- In SIOP 93-01, stage I receives pre-operative VA for 4 weeks and post-operative VA for 4 weeks, with no radiotherapy. [Wilms tumor p16]
- In SIOP 93-01, stage II receives pre-operative VA for 4 weeks, post-operative VAD for 27 weeks, and 15 Gy radiotherapy if nodes are positive. [Wilms tumor p16]

- In SIOP 93-01, stage III receives pre-operative VA for 4 weeks, post-operative VDA for 27 weeks, and 15 Gy radiotherapy. [Wilms tumor p16]
- In SIOP 93-01, stage IV receives pre-operative VDA for 6 weeks, then post-operative chemotherapy for 27 weeks (written "VCA"); if there is no complete response after 9 weeks, ICED (ifosfamide, carboplatin, etoposide, doxorubicin) is given for 34 weeks. [Wilms tumor p16]
- In SIOP 93-01 stage IV, radiotherapy is given only if the metastatic lesion has not disappeared by 9 weeks. [Wilms tumor p16]

Pros and Cons of NWTS / SIOP Protocols

- Upfront surgery (NWTS) avoids giving chemotherapy to benign disease and to chemotherapy-resistant histologies, whereas SIOP pre-operative chemotherapy may be given to benign tumours and other histological types. [Wilms tumor p17]
- A disadvantage of upfront surgery (NWTS) is that surgical complications may delay chemotherapy. [Wilms tumor p17]
- Disadvantages of SIOP pre-operative chemotherapy are modification of tumour histology and loss of staging information. [Wilms tumor p17]
- An advantage of SIOP pre-operative chemotherapy is that micrometastases are treated early. [Wilms tumor p17]

Recent Advances in Wilms Tumour

- Recent advances are listed as screening protocols, pre-treatment biopsy, risk stratification, management (laparoscopic partial nephrectomy, new chemotherapy protocols) and tumour markers. [Wilms tumor p18]
- The UKCCSG pre-treatment biopsy avoids giving chemotherapy to all cases, as about 12% of presumed Wilms tumours are benign. [Wilms tumor p18]
- A disadvantage of pre-treatment biopsy is tumour track seeding. [Wilms tumor p18]
- In NWTSG/COG risk stratification by histology, low risk is mesoblastic nephroma, intermediate is favourable histology, and high risk is anaplastic, clear cell sarcoma and rhabdoid tumour. [Wilms tumor p19]
- In SIOP risk stratification, clear cell sarcoma and rhabdoid tumour are high risk alongside diffuse anaplasia. [Wilms tumor p19]
- Neoadjuvant chemotherapy followed by partial nephrectomy gives oncological outcomes equal to radical nephrectomy, so partial nephrectomy is now extended to unilateral tumours. [Wilms tumor p19]
- Previously, partial nephrectomy was reserved for bilateral Wilms tumour, a solitary kidney and impaired renal function. [Wilms tumor p19]

Tumour Markers

- Serum amyloid A is a tumour marker for Wilms tumour. [Wilms tumor p20]
- Apolipoprotein C-III is a tumour marker for Wilms tumour. [Wilms tumor p20]
- MicroRNA markers allow a targeted biopsy before chemotherapy. [Wilms tumor p20]
- Loss of heterozygosity at 11p15 and WT1 gene mutation (11p13) are newer genetic markers that predict relapse in very-low-risk Wilms tumour. [Wilms tumor p20]

OTHER TOPICS IN THIS BOOK

Paediatric versus adult ACC

- In paediatric ACC the female-to-male ratio is 1:1 under 12 years and 6:1 between 12 and 20 years. [Malignant Adrenal Lesions p11]
- Li–Fraumeni syndrome is the commonest association in paediatric ACC. [Malignant Adrenal Lesions p11]
- Paediatric ACC presents at a low stage and has an improved survival of 54.2%, whereas adult ACC presents at an advanced stage. [Malignant Adrenal Lesions p11]
- Paediatric staging: stage I is <200 g and <5 cm, completely resected; stage II is >200 g or >5 cm, completely resected; stage III is nodal, surrounding-organ or vessel involvement or incomplete resection; stage IV is haematogenous metastasis. [Malignant Adrenal Lesions p11]

MCUG: procedure

- If the child is not already on antibiotics, give a 3-day course of prophylaxis on day −1, day 0 and day +1 (trimethoprim or cephalexin). [MCUG p1]
- Cephalexin syrup is available as 125 mg/5 mL and 250 mg/5 mL (Sporidex); the dose is 25–50 mg/kg/day divided into 3–4 doses. [MCUG p1]
- Ask the child to empty the bladder if possible before the study. [MCUG p1]
- The feeding tube (5–8 Fr) is inserted under sterile conditions with 2% xylocaine (lignocaine) jelly, and any residual urine is measured and collected. [MCUG p1]
- Contrast is used as 10 mL diluted 1:5. [MCUG p1]
- The bladder is filled until dripping of contrast stops or the child desires to void; this volume is the bladder capacity, which is compared with the formula-based expected capacity to judge whether capacity is normal for age. [MCUG p1]

MCUG: images, sedation and cyclic study

- Images during early filling reveal a ureterocele, and full-bladder images show bladder wall appearance and capacity. [MCUG p2]
- Voiding images are taken supine in females (2–4 images), whereas males are turned into the right or left anterior oblique position (2–4 images). [MCUG p2]
- Infants may need tapping (with a wooden object or scale) over the bladder to initiate voiding. [MCUG p2]
- Sedation or anaesthesia should be avoided if possible; if necessary in certain situations, it must not interfere with voiding. [MCUG p2]
- Cyclic MCUG (at least 2 voiding cycles) also helps identify a potential ectopic ureter or intermittent VUR, and the child can void around the catheter for the first void. [MCUG p2]

How to read an MCUG

- On the scout image, look at the bony structures (especially symphysis, spine and sacrum), any other radio-opaque density and the bowel gas pattern. [MCUG p3]
- On the cystogram phase, assess bladder shape and contour, filling defects, trabeculation, diverticula, capacity and position of the bladder neck. [MCUG p3]
- On the voiding phase, record whether reflux is unilateral or bilateral, and look at the bladder neck, posterior urethra and any reflux into the ejaculatory ducts. [MCUG p3]

Complications and indications

- Complications also include haematuria, rare bladder perforation and rare adverse reaction to contrast due to its absorption. [MCUG p4]
- Further indications for MCUG are enuresis, prune belly syndrome and urethral diverticulum. [MCUG p4]

5 Paediatric Urology 2

CH 32 · EMBRYOLOGY OF THE HUMAN GENITOURINARY TRACT

Embryology of kidney: origin from intermediate mesoderm

- As the notochord and neural tube develop, the mesoderm on either side of the midline divides into paraxial (somite), intermediate and lateral plate mesoderm. [Embryology p1]
- With transverse folding of the embryo, the intermediate mesoderm migrates towards the intra-abdominal coelom (future peritoneum). [Embryology p1]
- The intermediate mesoderm forms bilateral longitudinal mesodermal masses, the nephrogenic cords, which develop progressively in a craniocaudal direction. [Embryology p1]
- Each nephrogenic cord bulges from the posterior wall of the coelomic cavity, producing the urogenital ridge. [Embryology p1]

Pronephros and mesonephros

- The pronephros develops in the region of the future neck and thorax as five to seven paired segments. [Embryology p2]
- The mesonephros serves as the excretory organ of the embryo while the metanephros begins its development. [Embryology p2]
- The mesonephros develops at about the 9th to 10th somite levels. [Embryology p2]
- On day 24, the nephric (mesonephric, Wolffian) ducts appear as a pair of solid longitudinal tissue condensations parallel to the nephrogenic cords, in the dorsolateral aspect of the embryo, before the mesonephric tubules develop. [Embryology p2]
- On day 28, the blind distal end of the nephric duct fuses with the cloaca, and canalisation of the duct then progresses cranially (in reverse direction) to give definitive nephric ducts with excretory capability. [Embryology p2]
- Mesonephric vesicles begin to form in the 4th week, and 40–42 pairs of mesonephric tubules form in all. [Embryology p2]
- Only about 30 pairs of mesonephric tubules are seen at any one time, because the cranial tubules start to degenerate at about the 5th week. [Embryology p3]
- Each mesonephric tubule differentiates into an excretory unit whose medial end forms a renal corpuscle and whose lateral end joins the nephric duct. [Embryology p3]
- By the end of the 5th week the mesonephros has almost completely disappeared, apart from its derivatives. [Embryology p3]
- In females, mesonephric tubule remnants persist as small non-functional mesosalpingeal structures, the epoophoron and the paroophoron. [Embryology p3]

Metanephros and reciprocal induction

- The tip of each dividing ureteric bud branch is called the ampulla; it interacts with the metanephric mesenchyme to induce future nephrons. [Embryology p4]
- Each new ampulla acquires a cap-like condensation of metanephric mesenchyme, which gives the developing kidney a lobulated appearance. [Embryology p4]

Stages of nephron development (Larsson et al)

- In Larsson stage I, mesenchymal condensation forms a pretubular aggregate (a detached oval mass), which becomes the renal vesicle, still unconnected to the ureteric bud. [Embryology p4]
- Larsson stage II is marked by the development of two clefts. [Embryology p5]
- In Larsson stage III, the glomerular (Bowman) capsule is oval, proximal and distal tubules are identifiable, and endothelial and mesangial cells are recruited to form the glomerulus. [Embryology p5]
- In Larsson stage IV, the glomerulus is round and resembles a mature renal corpuscle, the proximal tubule appears mature, and the distal segments are still primitive. [Embryology p5]
- In humans, all nephrons at birth are at varying steps of Larsson stage IV. [Embryology p5]
- The more differentiated nephrons lie near the juxtamedullary region and the less differentiated ones at the periphery. [Embryology p5]
- Renal maturation continues after birth even though nephrogenesis is complete before birth. [Embryology p5]

Collecting system

- Dichotomous branching of the ureteric bud determines the eventual pelvicalyceal pattern. [Embryology p6]
- Dichotomous branching is complete by 20–22 weeks, and the cortical and medullary domains are established by 22–24 weeks. [Embryology p6]
- At birth, the kidney is 70% cortex and 30% medulla. [Embryology p6]
- The apex of each medullary cone, formed by convergence of the collecting ducts, is the papilla. [Embryology p6]
- The calyces and pelvis form from the most central segments of the collecting system, derived from the first five generations of ureteric bud branching. [Embryology p6]

Renal ascent

- Renal ascent is attributed to differential growth of the lumbar and sacral regions of the embryo, though the exact mechanism is not known. [Embryology p6]
- During ascent, transient aortic sprouts arise at progressively higher levels and degenerate; the final pair forms in the upper lumbar region and becomes the definitive renal arteries. [Embryology p6]

Cadherins in nephrogenesis

- Proper mesenchymal condensation and tubulogenesis need expression of R-cadherin, E-cadherin and cadherin 6, and suppression of cadherin 11. [Embryology p7]

Formation of the urogenital sinus

- In the 4th week, embryonic folding carries the cloacal membrane to the ventral aspect of the embryo, and the terminal portion of the endoderm-lined yolk sac dilates to become the cloaca. [Embryology p8]
- In the classic Rathke and Tourneux theory, two lateral ridges fuse in the midline while the descending urorectal septum fuses with the cloacal membrane. [Embryology p8]
- The challenge to the classic view holds that there are no lateral ridges and no descending septum. [Embryology p8]

Formation of the trigone

- The common excretory duct stage of trigone formation is at about day 33. [Embryology p9]
- In the classical view, the right and left common excretory ducts fuse in the midline as a triangular area to form a primitive trigone that is structurally different from the bladder and urethra. [Embryology p9]
- On day 37, the ureteric orifices open into the bladder and migrate cranially and laterally. [Embryology p10]
- As the ureteric orifices migrate, the nephric duct orifices move caudally and flank the paramesonephric ducts at the level of the urogenital sinus, at the site of the future verumontanum (or vaginal canal). [Embryology p10]

Development of the ureter

- The ureter begins as a simple cuboidal epithelial tube surrounded by loose mesenchymal cells. [Embryology p11]
- The ureteric lumen is complete by day 28. [Embryology p11]
- The ureteric lumen is transiently obstructed between days 37 and 40, followed by bidirectional recanalisation that begins in the mid ureter. [Embryology p11]
- The Chwalla membrane, present at about days 37–39, is a two-cell-thick layer over the ureteric orifice that may cause physiological ureteric obstruction. [Embryology p11]
- The ureter acquires transitional epithelium by 14 weeks. [Embryology p11]
- Ureteric muscularisation and elastic fibre development begin at 12 weeks of gestation. [Embryology p12]
- Bladder smooth muscle develops before ureteric muscle; it appears first in the subserosal region of the bladder dome and extends towards the base and urethra. [Embryology p12]
- Ureteric smooth muscle begins in the subepithelial region at the vesicoureteric junction and ascends towards the intrarenal collecting system. [Embryology p12]

Development of bladder and continence mechanism

- At 10 weeks the bladder is a cylindrical tube lined by a single layer of cuboidal cells and surrounded by loose connective tissue, with its apex tapering into the urachus, which is continuous with the allantois. [Embryology p13]
- Between 7 and 12 weeks the surrounding connective tissue condenses and smooth muscle fibres begin to appear. [Embryology p13]
- By 21 weeks the bladder epithelium has four to five cell layers, similar to fully differentiated urothelium. [Embryology p13]

- Bladder compliance changes during development with fetal urine production (mechanical distension), increasing wall muscle thickness and a falling ratio of thick to thin collagen fibres. [Embryology p14]
- Fetal bladder compliance is very low in early gestation and gradually increases; the exact cause is not known. [Embryology p14]
- There are no detailed studies on the development of continence. [Embryology p14]
- After division of the cloaca and rupture of the urogenital membrane, a mesenchymal condensation forms around the caudal end of the urogenital sinus. [Embryology p14]
- Striated muscle fibres are seen by 15 weeks, when the smooth muscle becomes thicker at the bladder neck. [Embryology p14]
- The urethral sphincter (central smooth muscle and peripheral striated fibres) develops first in the anterior urethral wall and later extends to the posterior wall; its fibres project to the lateral wall of the prostate in males and the lateral wall of the vagina in females. [Embryology p14]

CH 43 · EXSTROPHY–EPISPADIAS COMPLEX

Exstrophy–Epispadias Complex: Incidence and Inheritance

- The risk of recurrence of bladder exstrophy within a given family is approximately 1 in 100. [Exstrophy p1]
- The 1 in 70 risk for offspring of a parent with bladder exstrophy–epispadias is about 500 times the risk in the general population. [Exstrophy p1]
- A breakpoint mutation at the 5' region of the CASPR3 gene on chromosome 9 was the first finding to suggest a genetic basis for exstrophy. [Exstrophy p1]
- Advanced maternal age is a suspected risk factor for exstrophy. [Exstrophy p1]

Classic Bladder Exstrophy: Skeletal Defects

- On 3D CT, the coronal rotation of the sacroiliac joint gives an angle about 10 degrees larger than normal. [Exstrophy p2]
- Convergence of the iliac wings is one of the rotational pelvic abnormalities seen on 3D CT. [Exstrophy p2]

Classic Bladder Exstrophy: Pelvic Floor Defects

- The puborectalis sling supports about two times more body cavity area than normal and is more flattened, but its length and thickness are normal. [Exstrophy p3]
- The levator ani group is less dome-shaped and more irregular than normal, while its length and thickness are usually the same as in children without exstrophy. [Exstrophy p3]
- Closure reshapes the pelvic floor from a box-like configuration into a more inwardly rotated hammock. [Exstrophy p3]
- Closure redistributes a significant proportion of the levator ani group into the anterior compartment and gives a smooth, uniform contour to the pelvic floor. [Exstrophy p3]

Classic Bladder Exstrophy: Abdominal Wall Defects

- Premature rupture of the abnormal cloacal membrane leaves a triangular fascial defect occupied by the exstrophied bladder and posterior urethra. [Exstrophy p3-4]
- The inferior limit of the fascial defect is the intersymphyseal band, which represents the divergent urogenital membrane. [Exstrophy p3-4]
- The anterior rectus sheath has a fan-like extension behind the urethra and bladder neck that inserts into the intersymphyseal band. [Exstrophy p3-4]
- The bladder musculature extends laterally and interdigitates with fibres of the rectus fascia; these fibres must be incised lateral to the urethral plate down to the inferior pubic ramus and levator hiatus. [Exstrophy p3-4]
- A small umbilical hernia may be present and is repaired at the initial exstrophy closure. [Exstrophy p3-4]
- A small omphalocele occurs rarely in classic exstrophy and is closed at the time of bladder closure. [Exstrophy p3-4]

- Indirect inguinal hernias occur in about 80% of boys and 10% of girls with exstrophy, and are bilateral in 81.8%. [Exstrophy p3-4]
- Inguinal hernias arise from a persistent processus vaginalis, large internal and external inguinal rings, and lack of obliquity of the inguinal canal. [Exstrophy p3-4]

Classic Bladder Exstrophy: Anorectal Defects

- Anorectal anomalies are 72-fold more likely in exstrophy, with an incidence of 1.8%. [Exstrophy p5]
- The anus lies directly behind the urogenital diaphragm, and the perineum is short and broad. [Exstrophy p5]
- The commonest anorectal anomalies are imperforate anus (most common), rectal stenosis and congenital rectal prolapse. [Exstrophy p5]
- Congenital rectal prolapse is due to the divergent levator ani and puborectalis, and its appearance is an indication for definitive repair of the exstrophied bladder. [Exstrophy p5]

Classic Bladder Exstrophy: Male Genital Defects

- The male genital defect is the most severe and troublesome aspect of surgical reconstruction. [Exstrophy p5-6]
- The posterior corporal length is the same as normal, and the angle between the corpora cavernosa is unchanged. [Exstrophy p5-6]
- The prostate is normal and PSA is below the upper limit of normal. [Exstrophy p5-6]
- The urethra usually lies anterior to the prostate. [Exstrophy p5-6]
- The vas deferens, ejaculatory ducts and seminal vesicles are normal. [Exstrophy p5-6]
- The testes are mostly retractile. [Exstrophy p5-6]

Classic Bladder Exstrophy: Female Genital Defects

- Female genital reconstruction is less complex than male reconstruction. [Exstrophy p6-7]
- The vagina is hardly more than 6 cm deep, but its calibre is normal. [Exstrophy p6-7]
- The labia, mons pubis and clitoris are divergent. [Exstrophy p6-7]
- The ovaries and fallopian tubes are normal. [Exstrophy p6-7]
- The levator hiatus has a normal anteroposterior length but is two times wider than normal. [Exstrophy p6-7]

Classic Bladder Exstrophy: Urinary Defects

- The bladder mucosa is usually normal at birth but may carry hamartomatous polyps. [Exstrophy p7-8]
- The exstrophic bladder has a normal density of muscarinic receptors with normal binding affinity. [Exstrophy p7-8]
- The collagen-to-smooth-muscle ratio is increased, with a threefold increase in type III collagen. [Exstrophy p7-8]
- The average number of myelinated nerve fibres per field is significantly reduced, which may mean the newborn exstrophy bladder is at an earlier stage of development, with potential for normal development after successful initial closure. [Exstrophy p7-8]
- Bladder polyps are of two types, fibrotic and oedematous, and both show squamous metaplasia in about 50% of cases. [Exstrophy p7-8]
- Cystitis cystica and von Brunn nests can occur in the exstrophy bladder, in addition to cystitis glandularis, and surveillance uses cytology and cystoscopy. [Exstrophy p7-8]
- Caveolae (intracellular structures for cell-to-cell signalling) are usually normal in patients with a successful closure but are significantly lacking in patients who eventually need augmentation cystoplasty. [Exstrophy p7-8]
- Exstrophy detrusor shows lower intracellular calcium. [Exstrophy p7-8]

Classic Bladder Exstrophy: Upper Tract

- Upper-tract anomalies in classic exstrophy include duplicated collecting system (most common), hypoplastic or absent kidney, pelvic kidney, PUJ obstruction and multicystic dysplastic kidney. [Exstrophy p8-9]

- The pouch of Douglas between bladder and rectum is unusually deep and enlarged, so the distal ureter approaches the bladder from a point inferior and lateral to the orifice with little or no obliquity. [Exstrophy p8-9]

Exstrophy Variants

- In superior vesical fissure, the resulting superior vesical fistula resembles a vesicostomy. [Exstrophy p9-10]
- In anteroposterior duplication (duplicate exstrophy), the ureters attach to the closed pelvic bladder, so the superficial everted mucosa is dry. [Exstrophy p9-10]
- In side-by-side bladder duplication with exstrophy, the two bladder halves lie left and right with a muscle-containing midline septum, and each has its own ureter and intact sphincter. [Exstrophy p9-10]

Management at Birth and Selection for Closure

- Ectopic bowel within the exstrophied bladder is a relative contraindication to newborn closure. [Exstrophy p12]

Osteotomy

- Other osteotomy types are bilateral osteotomy of the superior pubic ramus and diagonal osteotomy of the iliac wing. [Exstrophy p13-15]
- The vertical iliac osteotomy is made just lateral to the sacroiliac joint, leaving the posterior iliac cortex intact. [Exstrophy p13-15]
- The combined anterior innominate and vertical iliac osteotomy allows placement of the external fixator and intrafragmentary pins under direct vision and gives a superior cosmetic appearance. [Exstrophy p13-15]
- Osteotomy can be done at any stage of exstrophy repair if the diastasis prevents the urological goals being reached. [Exstrophy p13-15]
- Osteotomy is usually not needed in newborns because laxity of the sacroiliac ligaments allows closure without undue tension in the first 48–72 hours. [Exstrophy p13-15]
- Osteotomy at initial closure removes the need for fascial flaps. [Exstrophy p13-15]
- Osteotomy brings the large pelvic floor muscles near the midline, where they can support the bladder neck and improve continence. [Exstrophy p13-15]
- The No. 2 nylon horizontal mattress suture is placed between the fibrous cartilage of the pubic rami, and No. 2 nylon has the strongest load-to-failure ratio. [Exstrophy p13-15]
- Tunnelled epidural catheters for 2–3 weeks control pain and movement after closure and so prevent the effects of inadequate immobilisation. [Exstrophy p13-15]

Surgical Options in the Newborn with Classic Exstrophy

- In the Warsaw approach, bladder, posterior urethra, pubis and abdominal wall are closed at primary closure with or without osteotomy, followed by a spica cast for 3 weeks and then an elastic bandage for 3 weeks. [Exstrophy p16-18]
- In the Warsaw approach, bladder neck and epispadias repair are done together once the bladder has a suitable capacity and the child is interested in continence, and complications (fistula or urethral stricture) occur in 10%. [Exstrophy p16-18]
- In the Kelly repair, the newborn stage is bladder dome closure with hernia repair. [Exstrophy p16-18]
- In Kelly radical soft-tissue mobilisation, the incision is taken onto the pubis to raise a flake of bone and carried down through the levator muscles, which are then wrapped around the reconstructed proximal urethra to provide continence. [Exstrophy p16-18]
- In the Kelly repair, the penile dorsum is dissected from the corporal bodies as in penile disassembly, the meatus is brought to the penoscrotal junction, and the resulting penoscrotal hypospadias is repaired at 3 years. [Exstrophy p16-18]
- After the Mitchell complete repair, 30–70% need a later hypospadias repair. [Exstrophy p16-18]
- The Mainz ureterosigmoidostomy is used for failed cases and for patients with small bladder templates. [Exstrophy p16-18]

- In the Mainz repair, the penis is reconstructed during the same procedure, and one further operation may be needed for cosmetic correction. [Exstrophy p16-18]
- The classical Erlangen repair includes bladder closure with bilateral reimplantation, bladder neck repair, bilateral groin exploration, epispadias repair and pubic closure without osteotomy, with an epidural catheter for 5 days. [Exstrophy p16-18]
- In the Erlangen approach, osteotomy is used only for cloacal exstrophy or a failed repair. [Exstrophy p16-18]

Management After Primary Closure

- If the first renal ultrasound after closure shows good drainage, it is repeated 3 months after discharge and then every 6 months to 1 year for the next 2–3 years, to detect upper-tract changes from reflux, infection or obstruction. [Exstrophy p19-20]
- A urine culture is sent before the patient leaves hospital and at later intervals to detect infection. [Exstrophy p19-20]
- Abdominal distension is one of the causes of dehiscence after exstrophy closure. [Exstrophy p19-20]

Failed Closure: Bladder Prolapse, Neourethral Stricture and Vesicocutaneous Fistula

- Before repair of a significant bladder prolapse or dehiscence, intramuscular testosterone is given 5 weeks and 2 weeks before the operation. [Exstrophy p21-22]
- If there is major soft-tissue loss at reclosure, buccal mucosal or full-thickness skin grafts may be needed. [Exstrophy p21-22]
- Neourethral stricture is usually associated with the use of paraexstrophy skin flaps. [Exstrophy p21-22]
- Prolonged dry intervals and a detectable post-void residual are warning signs of neourethral stricture. [Exstrophy p21-22]
- Significant outlet obstruction with upper-tract deterioration may need upper-tract diversion if dilatation, incision or open urethroplasty is not enough. [Exstrophy p21-22]
- Bladder neck repair should not be done until a posterior urethral stricture has been clearly repaired. [Exstrophy p21-22]
- A vesicocutaneous fistula shows as a fistulous tract in the midline lower abdomen, pubic diastasis wider than the preclosure measurement, and on cystoscopy a bladder lying anteriorly just under the abdominal skin. [Exstrophy p21-22]

Failed Bladder Neck Repair

- Incontinence after bladder neck repair is due to inadequate outlet resistance, small bladder capacity, decreased compliance, or a combination of these. [Exstrophy p22-23]
- When the dry interval is nearly acceptable (more than 2 hours), urethral bulking agents can be used, such as repeated collagen injections or a dextranomer-based agent. [Exstrophy p22-23]
- An artificial urinary sphincter has been used in some patients with good bladder capacity. [Exstrophy p22-23]

Failed Genitourinary Reconstruction

- After loss of glans, corpora, penile skin or urethral plate from penile disassembly, the options include tissue expansion, skin grafting and buccal mucosal grafting before neophalloplasty. [Exstrophy p23-24]
- Penile scars are managed by scar excision followed by flaps, grafts or tissue expanders. [Exstrophy p23-24]

Alternative Techniques of Reconstruction (Diversion)

- Standard ureterosigmoidostomy was the first form of urinary diversion used in exstrophy patients. [Exstrophy p24-25]
- Rectal incontinence and ureteral obstruction are complications of ureterosigmoidostomy. [Exstrophy p24-25]
- In the Mainz sigma pouch, the ureters are reimplanted non-refluxing, the abdominal wall is reconstructed, the bladder is closed as a seminal receptacle above the prostate, and the penis is

reconstructed later. (The notes on p18 say the penis is reconstructed in the same procedure.) [Exstrophy p24-25]

- After ureterosigmoidostomy, lifelong surveillance includes a yearly renal ultrasound as well as colonoscopy in adult life. [Exstrophy p24-25]
- A non-refluxing colonic conduit protects the kidneys from reflux and allows undiversion at an older age when clinically indicated. [Exstrophy p24-25]

Sexuality in Exstrophy Patients

- Male sexual concerns relate to the length, appearance and chordee of the penis. [Exstrophy p26]
- The penis is lengthened by degloving and resecting and releasing all residual scar tissue. [Exstrophy p26]
- Tissue expanders for 4–6 weeks give local skin cover, or full-thickness grafts may be needed. [Exstrophy p26]
- In cloacal exstrophy the corporal structures are very small, so flap phalloplasty may be the only option. [Exstrophy p26]
- Men with exstrophy can lead a sexual life as satisfactory as that of their peers. [Exstrophy p26]
- Female sexual concerns relate to the appearance of the external genitalia, the adequacy of the vaginal opening and uterine prolapse. [Exstrophy p26]
- Initial female external genital correction is done at birth, and "touch-up" surgery may be needed at puberty. [Exstrophy p26]
- Female options include tissue expanders, an inverted U flap of perineal skin to widen the vaginal opening, and slings for prolapse. [Exstrophy p26]

Fertility Issues

- Male infertility can result from iatrogenic injury to the verumontanum during functional closure or bladder neck reconstruction. [Exstrophy p27]
- Gamete intrafallopian transfer (GIFT) is an assisted reproduction option for exstrophy men. [Exstrophy p27]
- Most women with exstrophy can complete a pregnancy successfully. [Exstrophy p27]
- Both vaginal delivery and caesarean section have been performed, but deliveries are high risk for mother and baby and should take place in a tertiary care centre. [Exstrophy p27]

Male Epispadias

- The male-to-female ratio of epispadias is 3:1 to 5:1. [Exstrophy p28-30]
- In frequency, penopubic epispadias is commoner than penile, which is commoner than glanular. [Exstrophy p28-30]
- Associated anomalies in male epispadias are usually confined to the external genitalia, the pubic diastasis and the continence mechanism. [Exstrophy p28-30]
- The only renal anomaly seen in some cases of male epispadias is agenesis of the left kidney. [Exstrophy p28-30]
- Reflux in complete epispadias is due to an inherently deficient ureterovesical junction. [Exstrophy p28-30]
- Inguinal hernias are significantly less common in epispadias than in exstrophy. [Exstrophy p28-30]
- Urethral reconstruction options in epispadias include a transverse island flap. [Exstrophy p28-30]
- Complete penile disassembly can be combined with reapproximation of the perineal muscle complex, in an attempt to avoid bladder neck reconstruction. [Exstrophy p28-30]
- Endoscopic submucosal injection of plastic microspheres is an option at the bladder neck. [Exstrophy p28-30]

Female Epispadias

- In the intermediate (subsymphyseal) Davis type, the urethra is split dorsally along most of its length; in the most severe (retrosymphyseal) type, the cleft involves the entire urethra and the sphincter mechanism. [Exstrophy p31-33]

- In female epispadias the labia minora are usually poorly developed and end anteriorly at the corresponding half of the clitoris. [Exstrophy p31-33]
- The symphysis in female epispadias may be represented by a narrow fibrous band. [Exstrophy p31-33]
- The vagina and internal genitalia are usually normal in female epispadias. [Exstrophy p31-33]
- In female epispadias repair, the inverting urethral closure starts near the bladder neck and progresses distally. [Exstrophy p31-33]
- The medial halves of the labia minora and clitoris are denuded before being brought together, which adds urethral resistance. [Exstrophy p31-33]
- If bladder neck reconstruction is done at the same time, the Foley catheter is not left in the urethra. [Exstrophy p31-33]
- Older continence operations for female epispadias (transvaginal plication of the urethra and bladder neck, muscle transplantation, urethral twisting, urethral cauterisation, bladder flap and Marshall–Marchetti vesicourethral suspension) raise urethral resistance but do not correct the incontinence or the malformed anatomy, and their results are disappointing. [Exstrophy p31-33]
- Options for a small bladder include bladder augmentation with PTFE injection at the bladder neck, or simultaneous bladder neck reconstruction and augmentation. [Exstrophy p31-33]

Cloacal Exstrophy

- Vaginal reconstruction is a planned step of modern functional reconstruction of cloacal exstrophy. [Exstrophy p34]

CH 44 · PRUNE-BELLY SYNDROME

Prune Belly Syndrome: Synonyms and Genetics

- Prune belly syndrome is also called the abdominal musculature syndrome (in addition to triad and Eagle–Barrett syndrome). [Prune belley p1]
- Chromosomal abnormality is the exception rather than the rule, but reported syndromic associations include Turner syndrome, trisomy 13 and 18, monosomy 16 and Beckwith–Wiedemann syndrome. [Prune belley p1]

Epidemiology

- The incidence of prune belly syndrome is similar to that of bladder exstrophy. [Prune belley p2]
- Incidence is higher in children born to younger mothers. [Prune belley p2]

Genitourinary Anomalies: Kidneys, Ureters, Bladder

- Pelviureteric junction obstruction can occur in prune belly kidneys, either primary or secondary. [Prune belley p2-3]
- Poor ureteric peristalsis results from ineffective coaptation caused by a reduced number of thick and thin myofibrils. [Prune belley p2-3]
- On the splayed trigone the ureteric orifices lie laterally and superiorly, which is possibly the reason for reflux. [Prune belley p2-3]
- During voiding the bladder neck opens widely into the dilated prostatic urethra. [Prune belley p2-3]

Prostate, Accessory Sex Organs and Anterior Urethra

- Prostatic histology shows increased connective tissue with decreased epithelial and smooth muscle cells. [Prune belley p4]
- Obstructive lesions at the distal posterior urethra include atresia, valve, stenosis, membrane and diverticulum. [Prune belley p4]
- The vas deferens and seminal vesicles are often atretic. [Prune belley p4]
- The epididymis is poorly attached to the testis. [Prune belley p4]
- Retrograde ejaculation in prune belly syndrome is due to an incompetent bladder neck. [Prune belley p4]
- The anterior urethra is usually normal. [Prune belley p4]
- In fusiform megalourethra the whole penis dilates with voiding because of mesenchymal deficiency of the urethral folds; in scaphoid megalourethra there is ventral dilatation with voiding because of

mesenchymal deficiency of the urethral supportive tissue. [Prune belley p4]

Testis and Infertility

- The intra-abdominal testes lie adjacent to the dilated ureters. [Prune belley p4-5]
- The undescended testes are probably caused by the distended bladder and raised intra-abdominal pressure. [Prune belley p4-5]
- Testicular histology is usually similar to that of other patients with undescended testes. [Prune belley p4-5]
- Infertility results from a combination of histological testicular abnormalities, structural defects of the ducts and prostatic abnormalities. [Prune belley p4-5]

Abdominal Wall Defect

- The medial and inferior abdominal musculature is most commonly deficient; complete absence of the abdominal wall musculature is rare. [Prune belley p5]
- In the neonate the skin is wrinkled and redundant with flank bulges; in the growing child the wrinkles lessen and the abdomen becomes pot-bellied. [Prune belley p5]
- Flaring of the costal margin due to poor support of the lower chest wall contributes to respiratory illness. [Prune belley p5]
- Wound healing is good, with no tendency towards wound infection or incisional hernia. [Prune belley p5]

Cardiac and Pulmonary Anomalies

- The cardiac anomalies seen are patent ductus arteriosus, atrial septal defect, ventricular septal defect and tetralogy of Fallot. [Prune belley p5]
- Pneumomediastinum (as well as pneumothorax) can occur, with or without pulmonary hypoplasia. [Prune belley p5]
- Restrictive lung disease can develop secondary to the musculoskeletal disorders. [Prune belley p5]

Gastrointestinal Anomalies

- Gastrointestinal anomalies result from incomplete rotation of the midgut and include volvulus, atresia and stenosis as well as malrotation. [Prune belley p6]
- Omphalocele, gastroschisis and anorectal abnormalities are associated. [Prune belley p6]
- Splenic torsion can occur because of abnormal mesenteric fixation. [Prune belley p6]

Prenatal Diagnosis and Management

- Fetal hydronephrosis can be correctly diagnosed in the second trimester. [Prune belley p6]
- Routine termination of pregnancy is not recommended partly because the aetiology of prenatal hydronephrosis cannot be diagnosed precisely. [Prune belley p6]

Neonatal and Later Presentation

- Woodard described the spectrum (categories I–III) in 1985. [Prune belley p7]
- Category I infants have Potter features in addition to dysplasia, oligohydramnios, pulmonary hypoplasia and urethral atresia. [Prune belley p7]
- Incomplete forms, and patients without the abdominal wall anomaly, may present late in adulthood with urinary infection, hypertension or renal failure. [Prune belley p7]

Evaluation

- Serum BUN and electrolytes are checked to detect systemic acidosis or electrolyte imbalance. [Prune belley p8]

Surgical Management: Urinary Tract

- Supravesical urinary diversion is used only in rare instances of pelviureteric or vesicoureteric junction obstruction. [Prune belley p10-11]
- Internal urethrotomy can be considered in prune belly children with high post-void residual, increasing hydroureteronephrosis, or VUR with recurrent upper tract infections. [Prune belley p10-11]

- Internal urethrotomy is done at the distal end of the prostatic urethra and does not produce incontinence in this population. [Prune belley p10-11]
- Reduction cystoplasty remodels the bladder into a more spherical shape to improve contractility and emptying. [Prune belley p10-11]
- Reduction cystoplasty ranges from simple excision of the urachal diverticulum to excision of redundant mucosa with creation of an overlap between flaps. [Prune belley p10-11]
- Urethral atresia or hypoplasia is treated by simple dilatation, urethroplasty or hydrodistension. [Prune belley p10-11]
- Megalourethra repair uses a subcoronal circumferential incision with penile degloving, excision or infolding of the redundant urethra, and reconstruction over a catheter. [Prune belley p10-11]

Orchidopexy

- One reason for early orchidopexy is the potential for testicular carcinoma. [Prune belley p12]
- When a transabdominal approach is not possible, microvascular autotransplantation is an option alongside single-stage and staged Fowler–Stephens orchidopexy. [Prune belley p12]

Abdominal Wall Reconstruction

- Abdominoplasty is done for moderate to severe abdominal wall laxity. [Prune belley p12-13]
- The Randolph technique is based on electromyographic mapping, which showed the infraumbilical regions to be most severely affected and the supraumbilical and lateral regions least affected. [Prune belley p12-13]
- In the Randolph procedure the incision runs from the 12th rib to the pubic symphysis and on to the opposite 12th rib, with full-thickness removal of skin, lower abdominal musculature and peritoneum; the healthy fascia is then approximated to the anterior iliac spines, pubic tubercle and inferior fascia. [Prune belley p12-13]
- In the Ehrlich procedure the umbilicus is preserved on the inferior epigastric artery, skin and subcutaneous tissue are elevated off the muscle and fascia, and each side is advanced vest-over-pants to the contralateral flank. [Prune belley p12-13]

CH 46 · NEUROMUSCULAR DYSFUNCTION OF THE LOWER URINARY TRACT IN CHILDREN

Meningomyelocele: Embryology and Risk Factors

- Formation of the spinal cord and vertebral column begins at day 18 of gestation, and closure of the canal proceeds in a caudal direction. [MMC p1]
- The mechanism of neural tube closure, and what produces dysraphism, are not known. [MMC p1]
- Maternal risk factors for meningomyelocele include young and advanced maternal age, maternal obesity and diabetes, caffeine and smoking, low education and socioeconomic status, maternal occupational exposures, a maternal preconceptional stress event and low maternal weight gain. [MMC p1]
- Folate deficiency, including use of folic acid antagonists such as valproate, is a risk factor for meningomyelocele. [MMC p1]
- Higher birth order and a preceding history of maternal miscarriage are risk factors for meningomyelocele. [MMC p1]

Spinal Level of Meningomyelocele and Bladder Effects

- Cervical to high thoracic lesions make up 2% and low thoracic lesions 5% of meningomyelocele. [MMC p2]
- About 10% of children with meningomyelocele have no urological deficit. [MMC p2]
- An upper motor neuron lesion in meningomyelocele gives an overactive bladder, exaggerated sacral reflexes, absent voluntary control over the sphincter, detrusor–sphincter dyssynergia, a thick-walled bladder and a closed bladder neck. [MMC p2]
- A lower motor neuron lesion in meningomyelocele gives an acontractile bladder, absent sacral reflexes, a small smooth-walled bladder, an open bladder neck and partial or complete denervation

of the external sphincter. [MMC p2]
- The bulbocavernosus reflex (S2–S4) helps to distinguish an upper from a lower motor neuron lesion. [MMC p2]

Neonatal Assessment

- Neonatal lower urinary tract dynamics fall into three categories: synergic (26%), dyssynergic with or without poor compliance (37%) and complete denervation (36%). [MMC p3]
- The dyssynergic and complete denervation categories are the at-risk categories. [MMC p3]

Management Principles

- After puberty, ultrasound surveillance is done every 2 years. [MMC p4]
- VCUG is repeated if there are recurrent infections. [MMC p4]
- A child with reflux who cannot empty the bladder needs CIC, whatever the reflux grade. [MMC p4]
- A child with grade 1–3 reflux who is able to void is managed with antibiotic prophylaxis. [MMC p4]
- Grade 4–5 reflux in meningomyelocele is managed with CIC. [MMC p4]
- Reflux with associated poor detrusor compliance is treated with anticholinergics. [MMC p4]
- Vesicostomy is indicated when the child is a poor candidate for augmentation. [MMC p5]
- Antireflux surgery is indicated for reflux that persists into puberty. [MMC p5]
- Antireflux surgery is indicated for reflux in a child undergoing surgery to increase outlet resistance. [MMC p5]

Other Treatment Options

- The artificial somatic–autonomic reflex pathway is a novel approach, with limited data, that aims to restore bladder function in children. [MMC p6]
- The artificial somatic–autonomic reflex pathway is made by a limited laminectomy and microanastomosis of a lumbar ventral root to the S3 ventral root. [MMC p6]

CH 47 · FUNCTIONAL DISORDERS OF THE BLADDER AND BOWEL IN CHILDREN

Nocturnal Enuresis

- The American Psychiatric Association defines enuresis as repeated urination into bed or clothes at least twice a week for at least 3 consecutive months in a child of at least 5 years, not due to a drug side effect or a medical condition. [Compilation 03-4 p37]
- Nocturnal enuresis affects boys and girls in a ratio of 2:1. [Compilation 03-4 p37]
- Nocturnal enuresis affects 15% of children at 5 years and 1% at 15 years. [Compilation 03-4 p37]
- Autosomal dominant inheritance of enuresis has been reported, and chromosomes 12, 13 and 22 have been implicated. [Compilation 03-4 p37]
- Concordance for enuresis is higher in monozygotic than in dizygotic twins. [Compilation 03-4 p37]
- The chance of enuresis in the offspring is 43% if one parent had prolonged enuresis, 77% if both parents did, and 15% if neither did. [Compilation 03-4 p37]
- Enuresis is thought to stem from a maturational delay in the development of bladder control. [Compilation 03-4 p38]
- Urodynamics during sleep show an increased rate of bladder contractions; if they are accompanied by pelvic floor activity there is no enuresis, whereas without pelvic floor activity the child wets. [Compilation 03-4 p38]
- Nocturnal polyuria in enuresis may also be due to increased evening fluid intake, increased evening dietary solute load, abnormal renal sodium handling, an abnormal circadian rhythm of GFR, and abnormal sodium and calcium excretion. [Compilation 03-4 p38]
- Children with enuresis are often described by their parents as deep sleepers. [Compilation 03-4 p39]
- Enuretic episodes may occur primarily during non-REM sleep. [Compilation 03-4 p39]
- Stressful events that precipitate secondary enuresis include parental divorce and sexual abuse. [Compilation 03-4 p39]
- Conduct disorder is a psychological cause of secondary enuresis, alongside ADHD. [Compilation 03-4 p39]

- The examination in enuresis includes the lower limbs and the sacral reflex arc. [Compilation 03-4 p40]
- MRI and cystoscopy are among the investigations rarely needed in enuresis. [Compilation 03-4 p40]
- The decision to start treatment should be guided by the degree of concern and motivation of the child rather than of the parents. [Compilation 03-4 p40]
- After failure of combined alarm, desmopressin or anticholinergic therapy, a tricyclic antidepressant may be given with desmopressin or alone. [Compilation 03-4 p41]
- As behavioural therapy, the child should void regularly during the day and just before going to bed, 6–7 times in total daily. [Compilation 03-4 p42]
- Children with enuresis should avoid high-sugar and caffeine-based drinks. [Compilation 03-4 p42]
- The proposed mechanisms of the enuresis alarm are suppression of bladder emptying during sleep, an increase in nocturnal bladder volume, and waking the child to void by signalling when they urinate. [Compilation 03-4 p42]
- The alarm's moisture detector may be wired and placed on a bed pad, or wireless and placed in the undergarments; the arousal device is an auditory alarm with or without a vibratory belt. [Compilation 03-4 p42]
- About two-thirds of children become dry during active alarm treatment, and about 50% remain dry after treatment is completed. [Compilation 03-4 p42]
- The alarm must be used every night without holidays for 6 weeks, and if there is no improvement in 6 weeks, treatment should shift to desmopressin or combination therapy. [Compilation 03-4 p43]
- Relapse after alarm treatment is defined as more than 2 wet nights in 2 weeks, and is treated by reusing the alarm with overlearning. [Compilation 03-4 p43]
- Oral desmopressin has a half-life of 2–3 hours. [Compilation 03-4 p44]
- Desmopressin produces significant decrease in night-time wetting in 40%, and 60–70% relapse after discontinuation. [Compilation 03-4 p44]
- When desmopressin is given daily, a drug holiday of 1 week every 3–6 months is taken, and the child is then reassessed. [Compilation 03-4 p44]
- If desmopressin gives a negative response, options are increasing the dose, combination therapy, or second- or third-line treatment. [Compilation 03-4 p44]
- Anticholinergics (oxybutynin, tolterodine) are not effective as monotherapy for enuresis, and act by reducing nocturnal detrusor overactivity. [Compilation 03-4 p45]
- Response to imipramine is assessed after 1 month; if there is no improvement after 3 months, it is tapered and discontinued. [Compilation 03-4 p45]
- Imipramine is given with a drug holiday every 3–6 months, and the dose is gradually tapered over a 2-week period. [Compilation 03-4 p46]
- Investigational alternative therapies for enuresis include indomethacin, diclofenac, diazepam, hypnosis, psychotherapy and acupuncture. [Compilation 03-4 p46]

6 Female Urology

CH 101 · SURGICAL, RADIOGRAPHIC AND ENDOSCOPIC ANATOMY OF THE FEMALE PELVIS

Female Urethra

- The lumen of the female urethra is lined by urothelium proximally and by non-keratinised stratified squamous epithelium distally. [Compilation 03-2 p71]
- The muscularis propria of the female urethra has an inner longitudinal smooth muscle layer, an outer circular smooth muscle layer and then skeletal muscle (the sphincter). [Compilation 03-2 p71]
- The outermost layer of the female urethra is the serosa, or periurethral fascia. [Compilation 03-2 p72]
- The arterial supply of the female urethra is from the internal pudendal and vaginal arteries. [Compilation 03-2 p72]

Continence in Females

- The compressor urethrae and urethrovaginal sphincter are longitudinally arranged striated muscle fibres that arise from the vagina and the perineal membrane. [Compilation 03-2 p74]
- Anteriorly, the pubourethral ligament fixes the mid-urethra and prevents transmission of raised intra-abdominal force to the rest of the urethra. [Compilation 03-2 p75]

CH 108 · URINARY INCONTINENCE AND PELVIC PROLAPSE: EPIDEMIOLOGY AND PATHOPHYSIOLOGY

Risk Factors

- Obesity is a risk factor for early recurrence after POP surgery as well as for primary POP. [POP p4]

Pathophysiology of POP (DeLancey levels)

- The vertical fibres of level I support are the critical factor that distinguishes vaginal eversion from a post-hysterectomy cystocele, rectocele or enterocele, in which the vaginal apex stays well suspended. [POP p5]

CH 105 · EVALUATION AND MANAGEMENT OF WOMEN WITH URINARY INCONTINENCE AND PELVIC ORGAN PROLAPSE

Symptom Quantification Instruments and Supplemental Evaluation of Female UI

- Incontinence questionnaires complement the history and avoid physician bias in assessing symptoms. [Compilation 03-3 p1-2]
- Dye testing is done to confirm that the leaked fluid is urine and not vaginal discharge or peritoneal fluid. [Compilation 03-3 p1-2]
- Indications for VCUG in the evaluation of female incontinence include recurrent UTI and fluoroscopic demonstration of leakage, in addition to urethral diverticulum, VUR and bladder contour. [Compilation 03-3 p1-2]

Risk Factors: Female Incontinence

- The prevalence of adult urinary incontinence is 10–25% for SUI, 5–10% for MUI and 3–10% for UUI. [Compilation 03-2 p76]
- Carbonated beverages, as well as caffeine, are a dietary risk factor for UUI but not for SUI. [Compilation 03-2 p76]
- Depression is listed with diabetes as a medical condition that is a risk factor for female urinary incontinence. [Compilation 03-2 p76]

SUI Pathophysiology in Women

- In the hammock hypothesis, loss of the posterior supports while the anterior supports are preserved causes incontinence by creating a shearing effect, so the urethra can no longer be compressed. [Compilation 03-2 p77]

Evaluation of Prolapse

- A careful POP-Q examination, with reassessment at the time of surgery if necessary, remains the cornerstone of POP assessment. [POP p3]
- Investigations are rarely needed in POP; the options listed are VCUG, MRI and defecography (VCUG is not mentioned for POP in the corpus). [POP p4]

CH 114 · CONSERVATIVE MANAGEMENT OF URINARY INCONTINENCE: BEHAVIOURAL AND PELVIC FLOOR THERAPY, URETHRAL AND PELVIC DEVICES

Conservative Management of Incontinence: Behavioural Treatments

- Of the behavioural treatments, behavioural training with urge and stress strategies, bladder training with urge suppression, and pelvic floor muscle training are the three most established programmes. [Behavioural Rx p1]

- A non-relaxing striated sphincter is listed among the conditions that benefit from behavioural treatment. [Behavioural Rx p1]

Assessment and Patient Education

- Urine examination (CUE) is part of the assessment before behavioural treatment, to rule out infection. [Behavioural Rx p2]
- Behavioural treatments are based on the principles of learning, which is why patient education comes first. [Behavioural Rx p2]

Lifestyle Modifications

- Patients should restrict fluids at times when toilet access will be limited. [Behavioural Rx p3-4]
- Caffeine reduction improves both stress and urge incontinence. [Behavioural Rx p3-4]
- Patients with incontinence are advised to reduce alcohol, spicy foods and citrus products, though the evidence for this is weak. [Behavioural Rx p3-4]
- Chronic constipation, defined as fewer than 3 stools per week, is a risk factor for incontinence through chronic straining. [Behavioural Rx p3-4]

Pelvic Floor Muscle Training: Assessment and Regimen

- Pelvic floor muscle function can be assessed by digital (vaginal or rectal) examination, visual observation, EMG, manometry and ultrasound; digital assessment is the most common. [Behavioural Rx p5-6]
- The Laycock PERFECT assessment is a digital scoring scheme for pelvic floor muscle function, used alongside the Brink score. [Behavioural Rx p5-6]
- Daily PFM practice both increases muscle strength and improves motor skills. [Behavioural Rx p5-6]
- The patient should empty the bladder before each exercise session. [Behavioural Rx p5-6]

Vaginal Cones and Evidence for PFMT

- With vaginal cones, the woman first inserts the heaviest cone she can retain while standing, then while walking about; when she succeeds, she moves to the next heaviest cone. [Behavioural Rx p7]
- Vaginal cones may be better than no active treatment, but further studies are needed. [Behavioural Rx p7]
- PFMT gives a cure rate of 16–27% and an improvement rate of 48–80%. [Behavioural Rx p7]
- For long-term incontinence after prostatectomy, there is little research on PFMT. [Behavioural Rx p7]

Behavioural Training with Urge Suppression

- In urge suppression, after squeezing the pelvic floor repeatedly the patient waits about a minute, and uses distraction if the urgency continues. [Behavioural Rx p8]

Biofeedback

- Biofeedback teaches control over physical responses by giving precise, instantaneous feedback; outside urology it is used for headache, muscle spasm and hypertension. [Behavioural Rx p9]
- The perineometer, the first biofeedback device for the pelvic floor, was developed by Kegel. [Behavioural Rx p9]
- Computerised biofeedback can give visual feedback on a monitor (from vaginal or anal probes or skin electrodes) or auditory feedback through speakers. [Behavioural Rx p9]

Bladder Training

- Urgency-suppression techniques in bladder training include deep breathing with relaxation, in addition to quick pelvic floor contractions and distraction. [Behavioural Rx p11]
- Reinforcement in bladder training is given through consistent encouragement and positive feedback. [Behavioural Rx p11]

Behavioural Treatment for Voiding and Pelvic Floor Dysfunction

- Up-training (teaching the muscle to contract) is easier than down-training (teaching it to relax). [Behavioural Rx p14]
- In down-training, patients are taught, using biofeedback, to tell apart the sensation of muscle tension from the sensation of an active muscle contraction. [Behavioural Rx p14]

- Patients with high-tone pelvic floor dysfunction are encouraged to create a relaxing environment and to plan adequate time for voiding. [Behavioural Rx p14]

Mechanical Vaginal and Urethral Devices

- Intravaginal devices are also used for urgency and mixed incontinence, not only stress incontinence. [Behavioural Rx p15]
- Intravaginal devices restore the urethra to above the level of the pelvic floor, where intra-abdominal pressure can help its closure. [Behavioural Rx p15]
- Continence pessaries are made of silicone or plastic. [Behavioural Rx p15]
- Ring and dish pessaries each come with or without support. [Behavioural Rx p15]
- Intraurethral devices are not useful in a fibrotic urethra. [Behavioural Rx p15]
- Contraindications to intraurethral devices are urge incontinence (the device is not retained), active UTI and urethral stricture. [Behavioural Rx p15]
- The FemSoft insert is a single-use intraurethral device made of a silicone tube in a mineral-oil-filled silicone sleeve. [Behavioural Rx p15]
- The FemSoft insert must be removed for voiding and should be changed every 6 hours. [Behavioural Rx p15]
- The FemSoft insert comes in 16, 18 and 20 Fr sizes and in lengths of 3.5 and 4.5 cm. [Behavioural Rx p15]

Conservative Management of Incontinence: Lifestyle Modifications

- The ICI recommends decreasing caffeine intake to improve incontinence. [Compilation 03-3 p21]

CH 116 · RETROPUBIC SUSPENSION SURGERY FOR INCONTINENCE IN FEMALES

Retropubic Suspension for Incontinence in Women

- In SUI with pan-pelvic floor weakness, retropubic colposuspension does not always correct the associated vaginal prolapse, and a pelvic floor repair may also be needed. [Compilation 03-4 p69]
- Inadequate vaginal length or mobility from prior radiotherapy, as well as prior surgery, is a potential contraindication to retropubic suspension. [Compilation 03-4 p69]
- MMK carries a risk of urethral sphincter dysfunction, and is nowadays not recommended for the treatment of SUI. [Compilation 03-4 p70]
- The vagino-obturator shelf is a variant of the paravaginal procedure that combines the principles of paravaginal repair and Burch colposuspension, with sutures placed laterally into the internal obturator fascia rather than hitching the vagina to the iliopectineal line. [Compilation 03-4 p71]
- The iliopectineal line is a compound structure formed by the arcuate line of the ilium and the pectineal line of the pubis. [Compilation 03-4 p71]
- In the hybrid modification of Burch colposuspension, the vaginal wall is approximated to Cooper's ligament and the vaginal tissue is also approximated to the internal obturator fascia. [Compilation 03-4 p72]
- After Burch colposuspension a drain may be placed, and a voiding trial is done on the 5th postoperative day. [Compilation 03-4 p73]
- Reasons for voiding difficulty after colposuspension include detrusor denervation, associated detrusor dysfunction and over-correction of the urethral axis, as well as over-tight sutures. [Compilation 03-4 p74]
- Bladder overactivity is associated with SUI preoperatively in 30% of cases. [Compilation 03-4 p74]
- Prophylactic obliteration of the pouch of Douglas can be considered during Burch colposuspension to prevent posterior vaginal wall weakness and enterocele. [Compilation 03-4 p74]
- Systemic complications of colposuspension include DVT, PE, atelectasis and wound infection. [Compilation 03-4 p75]
- Laparoscopic colposuspension is the most popular laparoscopic incontinence procedure; its problems are technical difficulty, longer operating times and higher cost. [Compilation 03-4 p75]

CH 118 · SLINGS: AUTOLOGOUS, BIOLOGIC, SYNTHETIC AND MID- URETHRAL

Injection therapy for UI (urethral bulking agents)

- Porcine collagen, as well as bovine collagen, was used as a heterologous biological bulking agent; collagen was discontinued in about 2010 (date marked with "?" in the notes). [Compilation 03-4 p7]
- Investigational cell-based injection therapies for SUI include adipose tissue-derived stem cells and minced autologous striated muscle (in addition to autologous muscle-derived stem cells). [Compilation 03-4 p7]
- Uroving (Urovive) microballoons are a balloon-type urethral injectable that is no longer used. [Compilation 03-4 p7]
- Bulking agents are indicated in mild SUI on exercise. [Compilation 03-4 p8]
- Bulking agents are an option in women with SUI who cannot stop anticoagulants. [Compilation 03-4 p8]
- Severe scarring of the urethral mucosa is a contraindication to bulking, because the tissue cannot expand. [Compilation 03-4 p8]
- Hypersensitivity to the agent is a contraindication to bulking. [Compilation 03-4 p8]
- Bulking agents give a rapid response. [Compilation 03-4 p9]
- Bulking augments or restores mucosal coaptation, producing a "hermetic seal" (airtight seal) effect. [Compilation 03-4 p9]
- Bulking improves ISD by preventing bladder neck opening during stress. [Compilation 03-4 p9]
- The transurethral route is more common than the periurethral route for bulking injection. [Compilation 03-4 p9]
- The bulking agent is implanted circumferentially into the submucosa/lamina propria at the 3 and 9 o'clock positions (or 4 and 8 o'clock). [Compilation 03-4 p9]
- The Stamey grading system for incontinence outcome is: grade 0, continent; grade 1, urine loss with sudden increase in abdominal pressure but not when supine; grade 2, urine loss with physical stress such as walking or change of position; grade 3, total incontinence unrelated to physical activity. [Compilation 03-4 p10]
- Migration of bulking agent is commonest with Teflon. [Compilation 03-4 p10]
- Urethral mucosal prolapse is a complication of bulking agent injection. [Compilation 03-4 p10]

Adjustable Continence Therapy (ACT / ProACT)

- Adjustable continence therapy was developed as an alternative to bulking agents that could not migrate. [Compilation 03-4 p11]
- The device consists of two inflatable silicone balloons attached to silicone tubing with a titanium and silicone port. [Compilation 03-4 p11]
- In women, the balloons are placed in the periurethral space at the bladder neck with an introducer through two 1 cm incisions in the labial sulci at the level of the vaginal introitus. [Compilation 03-4 p11]
- The balloons are inflated after placement and the ports are buried in subcutaneous tissue. [Compilation 03-4 p11]
- Balloon volume re-adjustment (reinjection) may be required at 6-weekly intervals. [Compilation 03-4 p11]
- Adjustable continence therapy is better suited after surgical failure of SUI treatment, and long-term data are needed. [Compilation 03-4 p11]
- Complications of adjustable continence therapy include infection and voiding problems (retention, urgency), in addition to perforation and migration. [Compilation 03-4 p11]

Graft Materials

- An ideal graft material should be biocompatible, chemically inert, non-carcinogenic, mechanically strong, sterile, easily available, affordable, cause minimal allergic reaction and resist shrinkage. [Compilation 03-4 p33]
- Dura mater is an allograft material used as a graft (alongside cadaveric fascia lata and dermis). [Compilation 03-4 p34]
- Bovine dermis is a xenograft material, alongside porcine SIS, porcine dermis and bovine pericardium. [Compilation 03-4 p34]

- Allografts and xenografts are processed for sterilisation by freeze-drying, solvent dehydration or gamma irradiation. [Compilation 03-4 p34]
- Absorbable synthetic grafts promote fibroblast proliferation, cause fewer infections, are not rejected and are less harmful to the viscera. [Compilation 03-4 p34]
- Pore size of a synthetic graft also determines its flexibility, in addition to infection risk and fibroblast tissue ingrowth. [Compilation 03-4 p34]
- Knitted and woven meshes are multifilament and have small pores and interstices. [Compilation 03-4 p34]

Urethral Slings in SUI: Pre-operative Assessment

- Urgency elicited on the preoperative history predicts a poor postoperative outcome after sling surgery. [Slings in SUI p1]
- Additional indications for urodynamics before sling surgery are a negative stress test result, an unexplained abnormal urinalysis and evidence of dysfunctional voiding. [Slings in SUI p1]

Pubo-vaginal Slings: Indications

- A pubovaginal sling can aid reconstruction of the urethra after damage from iatrogenic hypospadias, as well as after trauma or mesh perforation. [Slings in SUI p2]

Pubo-vaginal Slings: Mechanism and Sling Materials

- The present pubovaginal sling concept is a modified McGuire technique that uses a shorter free graft whose tension can be adjusted. [Slings in SUI p3]
- Disadvantages of autologous fascial grafts include increased operative time and longer hospital stay. [Slings in SUI p3]
- Seroma is a recognised harvest-site disadvantage of autologous fascial grafts, alongside postoperative pain and hernia. [Slings in SUI p3]
- Acellular human dermis is an allograft sling material, alongside cadaveric fascia. [Slings in SUI p3]
- Allograft sling materials are pliable, easy to use and available in many sizes. [Slings in SUI p3]
- Allograft sling materials carry a potential risk of transmitting HIV, hepatitis and Creutzfeldt–Jakob disease. [Slings in SUI p3]

Synthetic Materials and the Amid Classification

- Advantages of synthetic (prosthetic) sling materials are that they are uniform, more consistent, more durable, sterile and biocompatible, and show the least degradation or disruption of the graft. [Slings in SUI p4]
- Amid type I (macroporous, >75 µm) brands include Prolene, Natural Mesh and Marlex. [Slings in SUI p4]
- Amid type II (microporous, <10 µm) brands include Gore-Tex, Surgical Membrane and Dual Mesh, all made of expanded PTFE (ePTFE). [Slings in SUI p4]
- Amid type III (macroporous with microporous or multifilamentous components) brands include Teflon (PTFE) and Surgipro (polypropylene), in addition to Mycromesh (ePTFE) and Mersilene (Dacron). [Slings in SUI p4]
- Amid type IV (submicronic pore size) brands include Silastic (silicone) and Celgard (polypropylene–Silastic sheet). [Slings in SUI p4]

Outcome of PVS

- Autologous pubovaginal slings give 15-year continence rates of 60–97%. [Slings in SUI p5]
- Data on allograft pubovaginal slings are limited, and failure rates are 6–36% for frozen or freeze-dried grafts. [Slings in SUI p5]
- Synthetic pubovaginal slings give results comparable to modified Burch colposuspension, but with mesh or sling related problems. [Slings in SUI p5]

Complications of PVS

- PVS complications can be grouped as vaginal-incision related (urethral injury, bladder injury, infection), abdominal-incision related (seroma, infection, hernia), synthetic-sling related (perforation, exposure, infection) and general (DVT, PE). [Slings in SUI p6]

PVS Procedure

- Pubovaginal sling surgery is performed under spinal or general anaesthesia. [Slings in SUI p7]
- An 18 Fr Foley catheter is placed to empty the bladder before the pubovaginal sling procedure. [Slings in SUI p7]
- The inverted-U vaginal incision for a pubovaginal sling starts 2 cm from the meatus and extends to the bladder neck. [Slings in SUI p7]
- The endopelvic fascia is perforated with Metzenbaum scissors directed towards the ipsilateral shoulder, just medial to and immediately under the ischiopubic ramus. [Slings in SUI p7]
- The fascial graft is sutured to the periurethral tissue before the vaginal incision is closed. [Slings in SUI p8]
- A vaginal gauze pack soaked in oestrogen is placed at the end of the pubovaginal sling procedure and removed on postoperative day 1 with the Foley catheter. [Slings in SUI p8]

Mid-Urethral Slings: Mechanics

- The MUS impedes movement of the posterior urethral wall above the sling and directs its motion in an antero-inferior or anterior direction. [Slings in SUI p9]
- Inward movement of the posterior urethral wall against the sling narrows the urethral lumen. [Slings in SUI p9]

MUS Procedure: Retropubic, TOT and Single-Incision Slings

- The retropubic MUS kit consists of two specially curved 5 mm trocars attached to a 40 cm segment of polypropylene mesh that is 1.1 cm wide. [Slings in SUI p10]
- An 18 Fr Foley catheter is passed and the bladder emptied at the start of the bottom-up retropubic MUS. [Slings in SUI p10]
- In the top-down retropubic technique, the tip of the surgeon's opposite fingers in the vagina is used as a guide for the trocar. [Slings in SUI p11]
- In the outside-in TOT, the skin entry point for the trocar is at the level of the clitoris, lateral at the adductor longus insertion. [Slings in SUI p11]
- The single-incision sling is polypropylene mesh with two harder pointed ends that anchor it on either side in obturator internus. [Slings in SUI p12]
- A single-incision sling can be placed in a hammock position or a U-position. [Slings in SUI p12]
- Single-incision slings show decreased efficacy on longer follow-up. [Slings in SUI p12]
- The MUS has reasonable efficacy in mixed urinary incontinence, but cure rates are lower than in pure SUI. [Slings in SUI p12]

MUS Complications

- Management of MUS vaginal mesh exposure is not standardised. [Slings in SUI p13]
- One option for MUS vaginal exposure is reapproximation of the vaginal mucosa over the mesh after irrigation with antibiotic saline. [Slings in SUI p13]
- Conservative management is not an option for mesh perforation of the urethra or bladder. [Slings in SUI p13]
- Voiding dysfunction after MUS can occur without obstruction. [Slings in SUI p15]
- There are no clear guidelines for the diagnosis of voiding dysfunction after MUS. [Slings in SUI p15]
- Cystoscopy for voiding dysfunction after MUS looks for a hypersuspended bladder neck, as well as bladder pathology and urethral mesh perforation. [Slings in SUI p15]

CH 117 · VAGINAL AND ABDOMINAL RECONSTRUCTIVE SURGERY FOR PELVIC ORGAN PROLAPSE

Types of POP Surgeries

- Restorative POP surgery suits women with discrete defects in the fibromuscular layer of the endopelvic fascia, and the defined defects are corrected individually. [POP p1]

Surgery for Prolapse: Anterior Compartment

- Anterior colporrhaphy is also known as Kelly's (Kelley's) procedure. [POP p6]
- Materials used for mesh-augmented anterior repairs are polypropylene mesh (better results), cadaveric fascia and porcine dermis. [POP p7]
- Abdominal paravaginal repair is performed through a low midline or a transverse incision. [POP p7]

Apical Compartment: High Uterosacral Vault Suspension

- The uterosacral ligament runs from S1–S3 to the cervical os and the proximal posterior vagina, spreading fan-like, and has three parts: sacral, intermediate and cervical. [POP p9]

Apical Compartment: Iliococcygeus Suspension

- Iliococcygeus suspension keeps the vagina in a more normal axis than sacrospinous fixation. [POP p10]

Abdominal Sacrocolpopexy

- An active lifestyle is an indication to consider abdominal sacrocolpopexy. [POP p10]
- Open sacrocolpopexy is done through a lower midline or Pfannenstiel incision, with the peritoneum entered and the bowel packed away from the field. [POP p11]
- Halban culdoplasty at sacrocolpopexy obliterates the cul-de-sac with sutures running from the posterior peritoneum onto the outer surface of the rectum up to the vaginal cuff. [POP p11]

Colpocleisis

- In total colpocleisis, purse-string sutures are placed circumferentially around the prolapse, incorporating the pubocervical and rectovaginal fascia. [POP p13]

Uterine Prolapse: Hysteropexy

- Vaginal uterus-sparing options include mesh hysteropexy as well as sacrospinous hysteropexy. [POP p14]

Vaginal Kits for POP

- Synthetic transvaginal mesh (TVM) kits for POP are classified as trocar-based or non-trocar-based. [POP p14]

CH 119 · GENITOURINARY COMPLICATIONS OF MESH AND THEIR REPAIR

Complications Related to Mesh in Pelvic Reconstructive Surgery

- The causes of mesh complications can be classified as modifiable factors and non-modifiable factors. [Slings in SUI p17]
- Sutures that are not selected properly lead to wound dehiscence and mesh exposure. [Slings in SUI p17]
- Improper mesh placement can injure the ilioinguinal nerve and the genital branch of the genitofemoral nerve. [Slings in SUI p17]
- Bacterial contamination of mesh is a non-modifiable factor that may not be adequately prevented by sterile techniques. [Slings in SUI p17]
- Osteomyelitis is a recognised complication of pelvic mesh. [Slings in SUI p17]
- Mesh complications can be grouped anatomically as neurologic, musculoskeletal and organ injury. [Slings in SUI p17]
- Vaginoscopy and colonoscopy are used as required in the evaluation of mesh complications. [Slings in SUI p18]
- A voiding cystourethrogram (VCUG) shows the location of an anatomic obstruction after mesh surgery. [Slings in SUI p18]
- In the treatment algorithm, after history and examination the patient is assigned to one of four groups: normal postoperative findings get reassurance and routine follow-up; bleeding or de novo voiding or defecatory dysfunction or obstruction get urine culture, PVR and evaluation for perforation; mesh exposure, extrusion or vaginal atrophy get vaginal oestrogen, revision or excision; pain alone gets symptom control and follow-up. [Slings in SUI p18]

- Patients with persistent or additional symptoms, or unsuccessful treatment, return to continued evaluation and re-enter the algorithm. [Slings in SUI p18]

CH 121 · URINARY TRACT FISTULAE

Vesicovaginal Fistula — Causes

- Congenital VVF is listed as a separate cause of vesicovaginal fistula. [Fistulae p1]
- Post-surgical causes of VVF include colporrhaphy, vaginal biopsy, bladder biopsy, endoscopic or laser therapy of the bladder, and vascular or pelvic rectal surgery. [Fistulae p1]
- External trauma causing VVF includes penetrating injuries, pelvic fracture and sexual trauma. [Fistulae p1]
- Obstetric causes of VVF, besides obstructed labour, include forceps laceration, uterine rupture and bladder injury during LSCS. [Fistulae p1]
- VVF is 3 times more common after total abdominal hysterectomy than after vaginal hysterectomy. [Fistulae p1]

VVF — Evaluation algorithm

- If the VVF cannot be located on examination and dye testing, do a VCUG or cystography; if still not found, evaluate for a ureterovaginal fistula, and if that is also absent, consider other causes of wetness such as incontinence or vaginitis. [Fistulae p2]

VVF — Clinical features and dye tests

- Pain in VVF arises from skin irritation by leaking urine, or occurs in radiation fistulae. [Fistulae p3]
- In the double dye test, oral pyridium (phenazopyridine) is given as 200 mg three times daily. [Fistulae p3]
- In the double dye test, an IV dye can be given in place of the oral dye. [Fistulae p3]

VVF — Three-swab test and investigations

- In the three-swab test the swabs are removed after 10 minutes. [Fistulae p4]
- In the three-swab test, staining of only the lowest swab indicates a urethrovaginal fistula or backflow of dye into the introitus. [Fistulae p4]
- Cystoscopy in VVF distinguishes an immature from a mature fistula. [Fistulae p4]
- On VCUG for VVF, voiding images are necessary. [Fistulae p4]

VVF — Other investigations

- Urine routine examination, culture and, when indicated, urine cytology are part of the VVF work-up. [Fistulae p5]
- When urine leak is doubtful, both creatinine and potassium levels in the vaginal fluid are higher than in serum. [Fistulae p5]

VVF — Treatment algorithm

- Small (<3–5 mm) fistulae suitable for a conservative trial are typically oblique fistulae. [Fistulae p6]
- Anticholinergics are given during the 2–3 week conservative catheter trial. [Fistulae p6]
- Nd:YAG laser is listed as a minimally invasive option for small VVF, alongside fibrin sealant and bovine collagen. [Fistulae p6]

VVF — Immediate vs delayed repair; abdominal vs vaginal

- The best opportunity for repair is an early initial operation if the injury is recognised within 24–48 hours. [Fistulae p7]
- A reason for delaying repair is to optimise the pliability of the tissues. [Fistulae p7]
- Transvaginal repair can be done as soon as 2–3 weeks after injury, whereas abdominal repair is often done after a 3–6 month delay. [Fistulae p7]
- Transvaginal repair has a shorter operating time than abdominal repair. [Fistulae p7]

VVF — Vaginal repairs

- In flap-splitting repair, the perivesical fascia is closed with Lembert-type sutures perpendicular to the first suture line. [Fistulae p8]
- In vaginal cuff excision, the funnel-shaped defect left after excising the tract is closed in 3 or 4 layers. [Fistulae p8]

VVF — Abdominal techniques

- In the O'Conor (suprapubic) repair, the vagina is closed with a running absorbable suture. [Fistulae p9]
- Transvesical repair uses a vertical anterior wall cystotomy without bivalving the bladder; the tract is excised transvesically, the vaginal edges are mobilised from the bladder, the vagina and then the bladder are closed separately, and an SPC is placed before closing the anterior bladder wall. [Fistulae p9]

VVF — Martius flap

- The Martius flap is named after Heinrich Martius. [Fistulae p10]
- The lateral (obturator) blood supply of the Martius flap is sacrificed during mobilisation. [Fistulae p10]

VVF — Omentum and other flaps

- Advantages of omentum are its inherent lymphatic properties, its ability to heal even in infection, and easy epithelialisation over its surface. [Fistulae p11]
- Other interposition options for VVF include labial myocutaneous flaps, seromuscular intestinal flaps and a bladder mucosa free graft. [Fistulae p11]

Ureterovaginal Fistula — Causes

- Gynaecological causes of ureterovaginal fistula include radical hysterectomy, LSCS and cystocele repair; other pelvic causes are vascular surgery, colon surgery and bladder neck suspension. [Fistulae p13]
- Further causes of ureterovaginal fistula are pelvic trauma, pelvic inflammatory disease and actinomycosis. [Fistulae p13]

Vesicouterine Fistula

- Vesicouterine fistula usually results from simultaneous injury to the bladder and uterus. [Fistulae p15]
- Other causes of vesicouterine fistula are vaginal operative delivery, vaginal birth after caesarean (VBAC), placenta percreta, an IUD (foreign body), brachytherapy, uterine artery embolisation and obstructed labour (uterine rupture with bladder injury or ischaemia). [Fistulae p15]
- Small immature vesicouterine fistulae may resolve with prolonged catheter drainage, with or without fulguration. [Fistulae p16]
- Hormonal induction of menopause can be used conservatively, as it induces involution of the puerperal uterus and may close a vesicouterine fistula. [Fistulae p16]
- Intraoperative ureteric stents can be placed for identification of the ureters during vesicouterine fistula repair. [Fistulae p16]

Urethrovaginal Fistula

- Behçet syndrome (through vasculitis and necrosis) is a cause of urethrovaginal fistula. [Fistulae p17]
- A small urethrovaginal fistula causes minimal leakage, whereas a large one causes continuous leakage. [Fistulae p17]
- Urethrovaginal fistula repair is often more difficult than VVF repair, because of extensive soft tissue defects, lack of local viable tissue for multilayer closure, and the need for rotational vaginal flaps or bladder flap tubes. [Fistulae p18]
- When repair is delayed, a 2–6 month interval is used (timing is controversial; the alternative is immediate repair once vaginal tissues are healthy). [Fistulae p18]
- A distal urethrovaginal fistula without incontinence can be managed by observation or extended meatotomy. [Fistulae p18]
- In the vaginal flap repair, the inverted-U incision is placed with the base of the U at the proximal margin of the fistula. [Fistulae p18]

- Fibrin glue and free labial skin grafts are listed as adjuncts for urethrovaginal fistula repair. [Fistulae p19]

Vesicoenteric Fistula

- Other causes of vesicoenteric fistula include trauma and foreign body. [Fistulae p20]
- Vesicoenteric fistula may present with fever, chills, haematuria, non-specific GI symptoms and orchitis. [Fistulae p20]
- The CT triad of vesicoenteric fistula is bladder wall thickening adjacent to a thickened colon loop, air in the bladder, and colonic diverticula. [Fistulae p20]
- CT for vesicoenteric fistula should be done after oral contrast and before IV contrast, so that enteric contrast in the bladder can be detected. [Fistulae p20]
- Cystography and barium enema have a low yield for vesicoenteric fistula and are adjuncts to evaluate malignancy or other colonic disease. [Fistulae p21]
- The Bourne test is performed after a non-diagnostic barium enema: the first voided urine is immediately centrifuged and X-rayed, and radiodense particles indicate a positive test. [Fistulae p21]
- Oral activated charcoal appearing as black particles in the urine indicates a vesicoenteric fistula. [Fistulae p21]
- Non-operative management of vesicoenteric fistula consists of IV TPN, bowel rest, antibiotics and treatment of Crohn disease. [Fistulae p21]
- For complex enterovesical fistulae a three-stage procedure is described: stage I proximal defunctioning and distal drainage of both GI and urinary fistula with eradication of sepsis; stage II recovery (TPN, nursing, radiological planning); stage III joint urological and GI reconstruction. [Fistulae p22]
- Preoperative preparation for vesicoenteric fistula surgery includes bowel preparation, nutritional supplementation and appropriate antibiotics. [Fistulae p22]

Ureteroenteric Fistula

- The commonest cause of ureteroenteric fistula is Crohn disease involving the right ureter and terminal ileum, so most are right-sided. [Fistulae p23]
- Ureteroenteric fistula is diagnosed by RGP, CT or MR. [Fistulae p23]
- Ureteroenteric fistula is treated by ureterolysis with stenting and possible bowel resection; ureteric resection may not be necessary if the ureter can be separated from the involved bowel. [Fistulae p23]

Pyeloenteric Fistula

- Pyeloenteric fistula most commonly involves the duodenum on the right and the descending colon on the left. [Fistulae p23]
- Pyeloenteric fistula presents with flank pain, fever, pyuria (60–70%), frequency and flank mass; iatrogenic ones cause minimal symptoms. [Fistulae p23]
- Pyeloenteric fistula is diagnosed by nephrostogram, urography, RGP and barium swallow or enema. [Fistulae p23]
- Non-functioning kidney with pyeloenteric fistula is treated, like XGP, by nephrectomy with GI reconstruction with the help of a GI surgeon. [Fistulae p24]
- Other options for pyeloenteric fistula are internal stenting of the urinary tract, or a large nephrostomy tube positioned in the colon with bowel rest or enteric suction. [Fistulae p24]

Urethrorectal / Prostatorectal Fistula

- Further causes of rectourethral fistula are anorectal surgery, urethral instrumentation and a ruptured prostatic abscess. [Fistulae p24]
- Rectourethral fistula may present with haematuria, nausea, vomiting and fever, and rarely with peritonitis. [Fistulae p25]
- Conservative management of RUF after open or laparoscopic prostatectomy includes bowel rest and TPN in addition to catheter drainage. [Fistulae p25]
- Staged repair of RUF is chosen when bowel preparation is poor, in addition to large radiation fistulae, infection or abscess. [Fistulae p25]

- A transanal approach (for example a Latzko-type procedure) can be used for RUF, but its problem is poor exposure. [Fistulae p26]

Urovascular Fistula

- Ureterovascular fistulae are mostly ureteroiliac artery fistulae, followed by ureteroiliac vein fistulae; ureteroaortic fistula is very rare. [Fistulae p27]
- Other causes of ureterovascular fistula are penetrating trauma, pregnancy and prior ureterolithotomy. [Fistulae p27]
- Reno- or pyelovascular fistula after PCNL presents with haematuria before or after nephrostomy removal; temporary control is by replacing the nephrostomy tube. [Fistulae p27]
- In a stable patient with ureterovascular fistula, full radiological evaluation is done for both diagnosis and planning of reconstruction. [Fistulae p28]
- Eventual management of ureterovascular fistula addresses both sides: on the vascular side embolisation, endovascular stent, primary repair, or ligation with or without bypass; on the urinary side ureteroureterostomy, transureteroureterostomy, cutaneous ureterostomy, or PCN with ureteric ligation. [Fistulae p28]

CH 122 · BLADDER AND FEMALE URETHRAL DIVERTICULA

Bladder Diverticulum: definition and classification

- Histologically, the diverticulum wall consists of mucosa, lamina propria, some scattered thin muscle fibres and an adventitial layer, and the smooth muscle present is disorganised and dysfunctional. [Bladder Diverticulum p1]
- Fetal alcohol syndrome is a syndromic association of congenital bladder diverticula. [Bladder Diverticulum p1]

Presentation and imaging

- Congenital diverticula can present with enuresis, as well as with acute UTI, haematuria and pyelonephritis. [Bladder Diverticulum p2]
- The VCUG needs AP, lateral and oblique films. [Bladder Diverticulum p2]
- Cross-sectional imaging (CT or MRI) is particularly needed when the diverticular neck is not patent or is obstructed by a tumour, so that contrast studies do not fill the sac. [Bladder Diverticulum p2]
- The pelvic ureter is usually deviated medially by a diverticulum, but lateral deviation can also occur. [Bladder Diverticulum p2]

Endoscopic evaluation

- Cellules and saccules are small outpouchings between hypertrophied bands of bladder muscle, and there are no size criteria to distinguish cellule, saccule and diverticulum, so the distinction is subjective and arbitrary. [Bladder Diverticulum p3]

Malignancy in diverticulum

- Stage at presentation is the most important prognostic factor for diverticular tumours: 5-year survival is 85% for superficial disease and 45% for extradiverticular disease. [Bladder Diverticulum p7]

Female Urethral Diverticula: Pathophysiology and Location

- Besides the commoner ventral (anterior vaginal wall) position, urethral diverticula can lie dorsal or lateral to the urethra. [Female Urethral Diverticulum p1]
- The urethropelvic ligament is formed by the endopelvic fascia and the periurethral fascia and is attached laterally to the tendinous arc of the obturator; the diverticulum lies within it, above the vaginal wall. [Female Urethral Diverticulum p1]

Anatomy of UD

- The lining of a urethral diverticulum may be urothelial, squamous, columnar or mixed epithelium, or the epithelium may be absent, leaving only fibrous tissue. [Female Urethral Diverticulum p2]
- About two-thirds of urethral diverticula show inflammatory change. [Female Urethral Diverticulum p2]

- Urethral diverticula range in size from a few millimetres to several centimetres. [Female Urethral Diverticulum p2]

Presentation

- Hesitancy and a sensation of incomplete voiding are symptoms of urethral diverticulum. [Female Urethral Diverticulum p3]
- Suprapubic fullness or urinary retention can be a sign of urethral diverticulum. [Female Urethral Diverticulum p3]

Evaluation: Urine Studies

- On urine culture the commonest organism is E. coli; others are Neisseria gonorrhoeae, Chlamydia, Streptococcus and Staphylococcus. [Female Urethral Diverticulum p3]
- Urine cytology is sent if irritative symptoms are present. [Female Urethral Diverticulum p3]

Evaluation: Urodynamics and Imaging

- Synthetic materials are avoided in concomitant anti-incontinence surgery because they raise the risk of erosion and infection. [Female Urethral Diverticulum p4]
- Double-balloon positive-pressure urethrography uses the Trattner catheter and is not widely performed clinically. [Female Urethral Diverticulum p4]
- On MCUG, both voiding and post-void films are needed to show the diverticulum. [Female Urethral Diverticulum p4]
- An IVP (intravenous pyelogram) may also show a urethral diverticulum. [Female Urethral Diverticulum p4]
- Transvaginal ultrasound gives the size and location of the diverticulum, which may appear anechoic or hypoechoic. [Female Urethral Diverticulum p5]
- On MRI a urethral diverticulum has low signal intensity on T1 (and high signal on T2). [Female Urethral Diverticulum p5]
- Advantages of MRI are no radiation and no dependence on voiding; both surface-coil and endoluminal-coil MRI are superior to VCUG and double-balloon urethrography. [Female Urethral Diverticulum p5]

Treatment Algorithm and Indications

- In a woman with UD and LUTS but no SUI, other urodynamic findings or LUTS are treated and then the diverticulum is repaired transvaginally. [Female Urethral Diverticulum p6]
- The natural history of urethral diverticulum is not known, so there are no clear guidelines on indications for surgery. [Female Urethral Diverticulum p7]
- Surgery is also indicated for UD with SUI and when the patient asks for surgery. [Female Urethral Diverticulum p7]

Surgical Procedures

- Other (less used) procedures for urethral diverticulum are endoscopic unroofing, fulguration, incision with obliteration using oxidised cellulose, and coagulation. [Female Urethral Diverticulum p7]
- The urethra is closed with vertical sutures and the periurethral fascia with horizontal sutures. [Female Urethral Diverticulum p8]
- Most extravasation seen on the 14–21-day pericatheter VCUG resolves with weekly repeat studies. [Female Urethral Diverticulum p8]
- Female hypospadias is a complication of urethral diverticulectomy. [Female Urethral Diverticulum p8]
- Vaginal scarring or narrowing after diverticulectomy is the cause of postoperative dyspareunia. [Female Urethral Diverticulum p8]

OTHER TOPICS IN THIS BOOK

Female urethral occlusive devices

- The first-generation urethral insert was the Reliance (no longer used) and the second generation is the FemSoft insert. [Compilation 03-5 p9]

- Urethral inserts suit mild SUI during sports activities and women unwilling to have SUI surgery. [Compilation 03-5 p9]

Intravaginal continence devices

- Intravaginal continence devices support the anterior vaginal wall and urethrovesical junction to improve urethral closure; options include the contraceptive diaphragm and tampons as well as pessaries. [Compilation 03-5 p9]
- Intravaginal devices have a mild side-effect profile but low efficacy, need regular cleaning, cannot address a fibrotic urethra or ISD, and may dislodge or cause irritation. [Compilation 03-5 p9]

7 Voiding Dysfunction

CH 103 · PHYSIOLOGY AND PHARMACOLOGY OF THE BLADDER AND URETHRA

Functional classification of LUT dysfunction

- Urethral instability is listed, with intrinsic sphincter dysfunction and urethral hypermobility, as an outlet cause of failure to store. [Compilation 03-2 p9]
- Severe infection is listed, with neurogenic disorders, overdistension and drugs, as a bladder cause of failure to void. [Compilation 03-2 p9]

Bladder mechanics: accommodation

- During bladder accommodation the lamina propria thins at a faster rate than the muscle wall. [Compilation 03-2 p25]
- Bladder wall thinning on filling is due to rearrangement of muscle bundles, which shift from a "top to bottom" to a "side to side" configuration. [Compilation 03-2 p25]
- Bladder wall thinning on filling is also due to collagen coils becoming extended and taut, with the fibres oriented parallel to the lumen. [Compilation 03-2 p25]

Bladder compartments: urothelium structure

- The urothelium is about 7 layers thick, with basal cells on the basal lamina, intermediate cells and apical umbrella cells, and a glycosaminoglycan layer on the luminal surface. [Compilation 03-2 p29]
- Umbrella cells are the largest urothelial cells, hexagonal in shape with microvilli. [Compilation 03-2 p29]
- Uroplakins are assembled in the apical membrane in specialised areas called plaques, whose outer and inner membrane leaflets differ in thickness; this is the asymmetric unit membrane (AUM). [Compilation 03-2 p29]
- The membrane area between the uroplakin plaques is symmetrical and is called the hinge region. [Compilation 03-2 p29]

Lamina propria, vasculature and stroma

- The lamina propria coordinates the activities of the urothelium and the detrusor muscle. [Compilation 03-2 p30]
- The lamina propria nerve plexus consists of unmyelinated fibres, and its microvasculature forms mucosal and subepithelial plexuses. [Compilation 03-2 p30]
- Intramural tension determines blood flow in the bladder wall; in normal filling, blood flow adapts to the large increase in surface area until bladder pressure rises. [Compilation 03-2 p30]
- Bladder stroma is collagen and elastin in a proteoglycan matrix that fills the spaces among muscle bundles, vessels and nerves; it is a passive, low-metabolic tissue responsible, with the detrusor, for bladder viscoelasticity. [Compilation 03-2 p30]

Bladder wall collagen, elastin and matrix

- Ageing causes a relative decrease of bladder smooth muscle. [Compilation 03-2 p31]
- Elastin is amorphous and is surrounded by a microfibrillar component containing glycoproteins. [Compilation 03-2 p31]

- Mature elastic fibres are 90% elastin. [Compilation 03-2 p31]
- The non-fibrillar bladder matrix is composed of proteoglycans and water. [Compilation 03-2 p31]
- Elastin is found in all layers of the bladder. [Compilation 03-2 p31]

Detrusor smooth muscle structure

- Detrusor myofibrils are arranged into fascicles running in random directions, unlike the circular and longitudinal layers of gut or ureteric smooth muscle. [Compilation 03-2 p32]
- Detrusor cells are connected to form a functional syncytium. [Compilation 03-2 p32]
- Detrusor smooth muscle cells are small and spindle-shaped with a single central nucleus, no cross-striations, and caveolae (flask-like invaginations of the membrane). [Compilation 03-2 p32]

Urothelium physiology

- The urothelium modifies the final urinary concentration of solutes depending on hydration status and protein. [Compilation 03-2 p33]
- The urothelium is permeable to water and expresses aquaporins. [Compilation 03-2 p33]
- Breakdown of apical cells increases water and urea permeability, producing the features of cystitis. [Compilation 03-2 p33]
- Urothelial tight junction proteins include zona occludens-1, occludin, and claudins 4, 8 and 12; the tight junctions adapt to filling and voiding without affecting permeability. [Compilation 03-2 p33]
- The barrier role of the GAG layer is controversial; it may matter in bacterial antiadherence and in preventing urothelial damage by large macromolecules. [Compilation 03-2 p33]
- The main urothelial GAG is chondroitin sulphate. [Compilation 03-2 p33]
- The apical urothelial membrane has a high electrical resistance, and the basolateral membrane a 10-fold lower resistance. [Compilation 03-2 p34]
- Na+ channels are present on the apical surface of umbrella cells and in cytoplasmic vesicles. [Compilation 03-2 p34]
- Na+, K+, Cl- and urea all cross the urothelium, allowing modification of their urinary concentration. [Compilation 03-2 p34]
- Substance P is among the mediators released by urothelial nociceptor and mechanoreceptor stimulation (with NO, ATP and ACh). [Compilation 03-2 p34]
- Urothelial ATP may have a role in regulating membrane trafficking. [Compilation 03-2 p34]
- Suburothelial interstitial cells (myofibroblasts) lie in the lamina propria just below the basal layer of the urothelium and mediate interaction between urothelial cells and afferent nerves. [Compilation 03-2 p35]

Components of detrusor muscle for contraction

- Detrusor thick filaments are 15 nm in diameter and made of myosin II, which has two intertwined heavy chains, each with a rod, a head and an intervening hinge. [Compilation 03-2 p37]
- Each myosin heavy chain head carries the light chains MLC20 and MLC17. [Compilation 03-2 p37]
- Thin filaments are 6–8 nm and made of actin, to which tropomyosin and caldesmon are bound; these control the access points on actin that let myosin II heads form cross-bridges. [Compilation 03-2 p37]
- Intermediate filaments are 10 nm and made of desmin and vimentin. [Compilation 03-2 p37]
- Interstitial cells or myofibroblasts within the detrusor have been proposed to have a pace-making role in spontaneous bladder activity. [Compilation 03-2 p37]

Anticholinergic (Antimuscarinic) Drugs: Mechanism

- Antimuscarinic drugs are competitive antagonists at muscarinic receptors. [LUT Drugs p3]

Muscarinic Receptors

- In the bladder, M2 and M3 receptors are present in a ratio of about 3:1. [LUT Drugs p4]
- In the GI tract, M2 exceeds M3 (about 4:1) and muscarinic stimulation increases motility. [LUT Drugs p4]
- In the eye, M1-M3 receptors mediate iris sphincter contraction; in the brain, M1-M5 are present (M3 sparse) and are involved in cognition; salivary and parotid glands carry M1 and M3 receptors. [LUT Drugs p4]

Neural control of bladder and urethra/EUS: efferent pathway

- Sympathetic preganglionic fibres from the lumbar cord pass through the sympathetic chain ganglia; postganglionic fibres relay via the inferior mesenteric (hypogastric) ganglia into the hypogastric nerves and then the pelvic ganglia. [Neural Control of LUT p1]

Afferent signals

- C fibres act both as volume sensors (responding to stretch) and as pain sensors. [Neural Control of LUT p2]
- C fibres are more numerous at the bladder neck and urethra than in the rest of the bladder. [Neural Control of LUT p2]
- Aδ fibres are distributed evenly throughout the detrusor smooth muscle, which allows a coordinated bladder contraction. [Neural Control of LUT p2]

Modulators of afferent activity

- Urothelial ATP released on stretch acts on P2X2 as well as P2X3 receptors to increase afferent signalling. [Neural Control of LUT p4]
- If NO rises while ATP stays normal, the ATP:NO balance tips towards detrusor underactivity (the reverse, high ATP with normal NO, gives detrusor overactivity). [Neural Control of LUT p4]
- Besides TRPV1 and TRPM8, TRPV2, TRPV4 and TRPA1 channels are also expressed in the bladder. [Neural Control of LUT p4]
- Bladder wall stretch increases afferent impulses either directly or through activation of the neurokinin A–NK2 receptor complex. [Neural Control of LUT p5]
- Bladder smooth muscle modulates afferent activity either directly or by secreting nerve growth factor. [Neural Control of LUT p5]
- Cannabinoid receptors are present in the urothelium and detrusor (CB1 more than CB2) and modulate sensory afferent signalling. [Neural Control of LUT p5]
- Cannabis-based extracts improve symptoms of urgency urinary incontinence and detrusor overactivity. [Neural Control of LUT p5]

Efferent signals: neurotransmitters

- Duloxetine (serotonin and noradrenaline reuptake inhibitor), by increasing EUS activity and reducing bladder activity, has been proposed as treatment for urge as well as stress incontinence. [Neural Control of LUT p6]
- Blockade of α-adrenoceptors increases pelvic blood flow, in addition to improving voiding. [Neural Control of LUT p6]

Central and peripheral circuit (diagram)

- The limbic system and hypothalamus (emotion and safety) feed into the PAG and PMC alongside the prefrontal cortex (social aspect). [Neural Control of LUT p7]
- Ascending bladder afferents reach the prefrontal cortex via the thalamus and insula. [Neural Control of LUT p7]
- Bladder afferents run in the hypogastric nerve to the lumbar dorsal root ganglia, and in the pelvic and pudendal nerves to the sacral dorsal root ganglia. [Neural Control of LUT p7]

CH 106 · EVALUATION AND MANAGEMENT OF MALES WITH URINARY INCONTINENCE

Continence in Males

- The internal (lisso)sphincter extends from the bladder neck to the distal verumontanum. [Post Prostatectomy Incontinence p2]
- The prostate itself adds to bladder outlet resistance. [Post Prostatectomy Incontinence p2]

Components of the External Sphincter

- In men the rhabdosphincter is supported anteriorly by the pubourethral ligaments. [Post Prostatectomy Incontinence p3]

- In men the sphincter is supported posteriorly by musculofascial elements that condense and fuse at the perineal body. [Post Prostatectomy Incontinence p3]

CH 107 · URODYNAMIC AND VIDEOURODYNAMIC EVALUATION OF THE LOWER URINARY TRACT

Urodynamic Study: Indications and Equipment

- Indications for urodynamics (ICS) include: obtaining information on aspects of LUT function not expressed as a symptom or sign, predicting the outcome of a contemplated treatment, confirming the effects of an intervention, and understanding why a previous treatment failed. [Compilation 03-3 p65-67]
- Intravesical catheters are 4–10 Fr. They are dual-lumen or triple-lumen, and the triple-lumen type has channels for filling, Pves and UPP. [Compilation 03-3 p65-67]
- Abdominal pressure catheters are 8–12 Fr. They are usually a rectal balloon catheter, or a balloon in the vagina or a stoma. [Compilation 03-3 p65-67]
- Sterile water can be used as the filling medium, as well as saline or radiographic contrast (if fluoroscopy is used). [Compilation 03-3 p65-67]

Filling/Storage Phase

- Normal cystometric capacity is 370–450 mL (±100 mL). [Compilation 03-3 p68]

Voiding Phase

- Normal voiding detrusor pressure is 40–60 cmH2O in men and 30–50 cmH2O in women. [Compilation 03-3 p73]
- Normal detrusor function is a voluntarily initiated continuous contraction that empties the bladder completely within a normal time span. [Compilation 03-3 p73]

CH 109 · NEUROMUSCULAR DYSFUNCTION OF THE LOWER URINARY TRACT

Bashful syndrome

- Bashful syndrome is low-flow, low-pressure voiding in young men who report marked hesitancy when trying to start micturition in the presence of others. [Compilation 03-5 p2]
- Bashful syndrome is psychological, reflecting a tendency to overcontrol the process of micturition. [Compilation 03-5 p2]
- Bashful syndrome is treated with behavioural therapy or drugs. [Compilation 03-5 p2]

Hinman syndrome

- Synonyms for Hinman syndrome include occult voiding dysfunction and occult neuropathic bladder. [Compilation 03-5 p2]

Urinary retention in young women (Fowler syndrome)

- Fowler syndrome typically affects women younger than 30 years. [Compilation 03-5 p3]
- Patients cannot void for one or more days but feel no urgency (painless retention); mild lower abdominal discomfort may be present. [Compilation 03-5 p3]
- Diagnostic criteria are bladder capacity over 1 L without urgency, no neurological abnormality, and a normal MRI of the brain and spinal cord. [Compilation 03-5 p3]
- Fowler syndrome is associated with polycystic ovaries and may be a focal, hormonally dependent channelopathy. [Compilation 03-5 p3]
- Detrusor acontractility and impaired sphincter relaxation are urodynamic features of Fowler syndrome (with high urethral closure pressure). [Compilation 03-5 p3]
- In the pathogenesis, abnormal urethral sphincter activity drives overactive urethral afferents, which inhibit bladder afferents and efferents and produce an acontractile bladder. [Compilation 03-5 p4]
- Opioids make Fowler syndrome worse. [Compilation 03-5 p4]

- Apart from sacral neuromodulation, other treatments (hormonal therapy, drugs, botulinum toxin injection) give poor outcomes in Fowler syndrome. [Compilation 03-5 p4]

Neuromuscular dysfunction of LUT: CVA

- Uncommon findings after recovery from cerebral shock in stroke include loss of bladder sensation and persistence of detrusor areflexia. [Compilation 03 p61]

Parkinson disease

- LUTS in Parkinson disease typically begin about 5 years after the onset of the disease. [Compilation 03 p63-64]
- Dopaminergic drugs used for Parkinson disease have no impact on LUTS. [Compilation 03 p63-64]

Other diseases at or above the brainstem

- Posterior fossa brain tumours cause urinary retention. [Compilation 03 p65]
- In cerebellar ataxia, detrusor overactivity and incontinence are common, and spinal cord involvement adds DSD or areflexia. [Compilation 03 p65]
- Cerebral palsy is a non-progressive brain injury occurring in the first year of life, and most patients have normal storage and emptying. [Compilation 03 p65]
- Multiple system atrophy is characterised by cell loss and gliosis in widespread areas of the brain. [Compilation 03 p65]
- Acute traumatic brain injury causes urinary retention from detrusor areflexia. [Compilation 03 p65]

Spinal cord anatomy

- There are 31 spinal cord segments, each with a pair of ventral and dorsal roots that combine on each side to form a spinal nerve. [Compilation 03 p66]
- Vertebral levels correspond to cord levels as follows: C1–C7 vertebrae to C1–C8 cord, T1–T8 to T1–T12, T9–T11 to L1–L5, and T12–L1 to S1–S5. [Compilation 03 p66]
- The conus medullaris (tapered distal end of the cord) lies at L1–L2, the filum terminale is pia mater extending to the coccyx that stabilises the cord, and the cauda equina is formed by the L2–S5 and coccygeal nerve roots. [Compilation 03 p66]

Multiple sclerosis

- MS most commonly involves the lateral corticospinal (pyramidal) and reticulospinal columns of the cervical spinal cord, more than the lumbar and sacral cord. [Compilation 03 p67]

Spinal cord injury and spinal shock

- Causes of spinal cord injury are road traffic accidents, falls, violence and sports injuries, and 70–80% of patients are male. [Compilation 03 p68]
- Leading causes of death after spinal cord injury are pneumonia, septicaemia, pulmonary embolism, heart disease, accidents and suicide. [Compilation 03 p68]
- Spinal shock is defined as decreased excitability of spinal cord segments at and below the level of the lesion, with absent somatic reflex activity and flaccid muscle paralysis below the lesion. [Compilation 03 p68]
- Spinal shock lasts only a short period in incomplete suprasacral spinal cord injury. [Compilation 03 p68]
- During spinal shock the striated sphincter has no voluntary control and the guarding reflex is absent. [Compilation 03 p68]
- Musculoskeletal signs of suprasacral SCI include impaired superficial and deep sensation and an abnormal plantar response. [Compilation 03 p69]
- In sacral SCI, emptying by Credé manoeuvre or Valsalva may cause obstruction at the closed bladder neck or at the distal sphincter when its tone is fixed. [Compilation 03 p71]

Lesions of peripheral reflex arc

- Spinal stenosis is a cause of peripheral reflex arc lesions causing storage and emptying dysfunction. [Compilation 03 p72]

Autonomic dysreflexia

- Autonomic dysreflexia occurs in 20% of thoracic lesions (compared with 60% of cervical lesions). [Compilation 03 p73]
- A prerequisite for autonomic dysreflexia is a viable distal spinal cord. [Compilation 03 p73]
- Triggers of autonomic dysreflexia include sexual activity, clot retention and electrocoagulation. [Compilation 03 p73]
- Bradycardia is common in autonomic dysreflexia, while tachycardia is rare. [Compilation 03 p73]
- Symptoms of autonomic dysreflexia resolve quickly after the stimulus is removed. [Compilation 03 p73]
- For prophylaxis, sublingual nifedipine 10 mg may be given 30 minutes before cystoscopy. [Compilation 03 p74]
- Other, less commonly used drugs for autonomic dysreflexia are captopril, hydralazine and diazoxide. [Compilation 03 p74]
- Long-term prophylaxis is terazosin 5 mg once daily for 3 months, with careful monitoring still needed during any procedure. [Compilation 03 p74]
- Neuroablative procedures for drug-refractory autonomic dysreflexia also include sacral neurectomy and dorsal root ganglionectomy. [Compilation 03 p74]

Complications of SCI with respect to the urinary tract

- The incidence of UTI in spinal cord injury patients is 57%. [Compilation 03 p75]
- In SCI, choose an antimicrobial with little or no effect on normal flora where possible. [Compilation 03 p75]
- In an institutional setting, hydrophilic-coated catheters reduce the infection rate by 21%. [Compilation 03 p75]
- VUR in SCI is more common with suprasacral lesions. [Compilation 03 p75]
- Skin complications and depression or suicide are among the complications of spinal cord injury. [Compilation 03 p76]
- Bladder cancer in SCI has an earlier age of onset than in the general population. [Compilation 03 p76]
- Neither cystoscopy nor cytology is an entirely reliable diagnostic tool for bladder cancer in SCI patients. [Compilation 03 p77]

Bladder Neck Dysfunction

- Bladder neck dysfunction is also called smooth sphincter dyssynergia, proximal urethral obstruction or dysfunctional bladder neck, besides primary bladder neck obstruction. [Compilation 03-4 p79]
- Bladder neck dysfunction is a poorly understood non-neurogenic condition with incomplete opening of the bladder neck during voluntary and involuntary voiding; its exact cause is unknown. [Compilation 03-4 p79]
- The proposed myogenic theory of bladder neck dysfunction is an abnormal arrangement of the musculature in the bladder neck region, causing narrowing of the bladder neck during detrusor contraction. [Compilation 03-4 p79]
- The proposed neurogenic theory of bladder neck dysfunction is sympathetic hyperactivity in young, anxious individuals. [Compilation 03-4 p79]
- Men with bladder neck dysfunction characteristically report long-standing voiding and storage symptoms. [Compilation 03-4 p79]

Detrusor sphincter dyssynergia

- In DSD, interruption of the spinobulbospinal pathways causes failure of inhibition of the guarding reflex and excitation of Onuf's nucleus, so the EUS contracts during detrusor contraction. [Compilation 03-2 p1]
- The continuous versus intermittent DESD division is the Weld classification. [Compilation 03-2 p2]
- DSD presents with mixed storage and voiding symptoms together with neurological symptoms. [Compilation 03-2 p3]

- Ureterovesical obstruction is a further complication of DSD (along with upper tract deterioration, VUR, urolithiasis, hydronephrosis and sepsis). [Compilation 03-2 p3]
- Examination in suspected DSD includes perianal sensation, the bulbocavernosus reflex and the cremasteric reflex, as well as anal tone on DRE. [Compilation 03-2 p3]
- On EMG, DSD is elevated sphincter activity during bladder contraction in the absence of raised intra-abdominal pressure (Valsalva), which separates it from straining. [Compilation 03-2 p3]

Lesion level versus bladder and sphincter pattern (table)

- Bladder sensation is normal with lesions above the brainstem, but absent with complete suprasacral cord lesions, complete sacral cord lesions and interruption of the peripheral reflex arc. [Compilation 03-2 p5]
- The bladder neck is closed (competent) during spinal shock. [Compilation 03-2 p5]

Lapides classification of neurogenic bladder

- Lapides sensory neurogenic bladder (diabetes, tabes dorsalis, pernicious anaemia) results from interruption of sensory fibres between bladder and spinal cord or of the afferent tracts to the brain; sensation is impaired early, and the bladder decompensates late into a large-capacity atonic bladder. [Compilation 03-2 p6]
- Lapides motor paralytic neurogenic bladder follows extensive pelvic surgery or pelvic trauma that destroys the parasympathetic supply; early it gives painful retention or inability to initiate or maintain micturition, and at end stage bladder decompensation. [Compilation 03-2 p6]
- Lapides uninhibited neurogenic bladder results from lesions in or above the brainstem, such as stroke, Parkinson disease and brain tumour. [Compilation 03-2 p6]
- Lapides reflex neurogenic bladder results from suprasacral spinal cord injury or transverse myelitis. [Compilation 03-2 p7]
- Lapides autonomous neurogenic bladder is a lower motor neuron lesion from any disease destroying the sacral cord, sacral roots or pelvic nerves, giving complete motor and sensory separation of the bladder from the sacral cord. [Compilation 03-2 p7]

CH 110 · OVERACTIVE BLADDER

Detrusor Instability

- In older terminology, idiopathic detrusor overactivity on urodynamics was called detrusor instability, and neurogenic detrusor overactivity was called detrusor hyperreflexia. [Compilation 03-4 p55]

Overactive bladder: aetiology

- OAB is more common in African-Americans than in white people. [OAB p1]
- The female excess of OAB is noted as greater in the young than in the old ("young > old"). [OAB p1]
- Neurological diseases that cause OAB include Parkinsonism, multiple sclerosis and stroke, probably through loss of cortical inhibitory control. [OAB p1]
- OAB may be related to fibromyalgia. [OAB p1]

Prevalence and natural history

- "OAB dry" is more common in men, and "OAB wet" is more common in women because of relative weakness of the female urethral sphincter and bladder neck. [OAB p2]
- OAB is generally progressive, but it can stay stable for prolonged periods and remission can occur. [OAB p2]

Pathophysiology: hypotheses

- In the neurogenic hypothesis, DO arises from generalised nerve-mediated excitation of the detrusor during filling, due to loss of cerebral inhibition, re-emergence of primitive reflexes, acquisition of new reflexes or sensitisation of afferents. [OAB p2]
- Inappropriate sensory activity may come from abnormally numerous, as well as pathologically sensitised, afferent nerve endings. [OAB p2]

Evaluation: OABSS questionnaire

- The OAB Symptom Score (OABSS, Homma et al) asks about urinary condition during the last week in four questions. [OAB p5]
- OABSS Q1, daytime voids (waking to sleeping): 7 or fewer scores 0, 8–14 scores 1, 15 or more scores 2. [OAB p5]
- OABSS Q2, night-time voids (sleeping to waking): 0 scores 0, 1 scores 1, 2 or more scores 2. [OAB p5]
- OABSS Q3, urgency (sudden desire difficult to defer): not at all 0, less than once a week 1, once a week or more 2, about once a day 3, 2–4 times a day 4, 5 or more times a day 5. [OAB p5]
- OABSS Q4, urgency incontinence (leak because the sudden desire cannot be deferred): same options, scored 0–5. [OAB p5]

Behavioural therapy and drugs

- Bladder irritants to avoid include alcohol, tomato, chocolate and nuts, as well as caffeine. [OAB p6]
- Bladder training intervals are stepped up from half an hour to 1 hour, 2 hours and then 3–4 hours. [OAB p6]
- Pelvic floor exercises are done for 30–50 minutes daily for 8 weeks. [OAB p6]
- Behavioural therapy includes treating coexisting conditions such as diabetes and stones. [OAB p6]
- The detrusor M2:M3 receptor ratio is 3:1. [OAB p6]

OAB refractory to conservative or drug therapy

- Phasic DO, the characteristic pattern in most idiopathic DO, shows contractions of increasing amplitude as the bladder fills and may or may not cause incontinence. [OAB p7]
- Terminal DO is a single involuntary detrusor contraction at cystometric capacity, which often empties the bladder completely. [OAB p7]
- Increased filling sensation on urodynamics is shown by an early desire to void. [OAB p7]
- On ambulatory urodynamics DO may appear in up to 60% of asymptomatic women. [OAB p8]
- Containment options for intractable OAB are appliances, catheters, urethral closure and urinary diversion. [OAB p8]

CH 111 · UNDERACTIVE DETRUSOR

Underactive Detrusor: Epidemiology

- The population prevalence of detrusor underactivity is unknown because there is no non-invasive marker for it. [Underactive Bladder p1]

Pathophysiology and Aetiology

- Detrusor underactivity can also arise from a combined myogenic and neurogenic pathophysiology. [Underactive Bladder p2]
- Idiopathic detrusor underactivity is attributed to ageing and, in men, to unknown factors. [Underactive Bladder p2]
- Neurosyphilis is a neurogenic cause of detrusor underactivity. [Underactive Bladder p2]
- AIDS is listed as a neurogenic cause of detrusor underactivity (Paper 2 p86 mentions areflexia only with CMV in HIV/AIDS). [Underactive Bladder p2]
- Pelvic fracture and pudendal nerve injury are listed as neurogenic causes of detrusor underactivity. [Underactive Bladder p2]
- Radical prostatectomy is listed among the radical pelvic operations that cause iatrogenic detrusor underactivity. [Underactive Bladder p3]
- Detrusor myomectomy is an iatrogenic cause of detrusor underactivity. [Underactive Bladder p3]
- Pelvic radiation therapy is an iatrogenic cause of detrusor underactivity. [Underactive Bladder p3]
- Antispasmodics are among the drugs that cause detrusor underactivity. [Underactive Bladder p3]

Diagnosis and Diagnostic Methods

- Current urodynamic methods neglect the velocity and sustainability of the detrusor contraction (the corpus says only that nearly all measure strength). [Underactive Bladder p3]

- The Watts factor expresses detrusor power per unit area of bladder surface; it needs a lengthy, complex calculation and is not validated. [Underactive Bladder p4]
- Detrusor shortening velocity can be measured because a slow contraction may cause symptoms, but velocity is not part of the ICS definition of detrusor underactivity. [Underactive Bladder p4]
- Occlusion testing includes the voluntary stop test, the mechanical stop test and continuous occlusion, all measuring isovolumetric pressure by obstructing outflow. [Underactive Bladder p4]
- Occlusion tests are impractical and are uncomfortable or painful for some patients. [Underactive Bladder p4]
- Contractility indices such as the BCI are valued because they are simple to use. [Underactive Bladder p4]
- Simple cut-offs of urodynamic measurements, for example Pdet@Qmax under 40 cmH2O and Qmax under 15 mL/s, are easy to use but have no widely accepted ranges for diagnosing detrusor underactivity. [Underactive Bladder p5]

Management

- A goal of management is to reduce complications including vesicoureteric reflux and skin damage from overflow incontinence (besides UTI and bladder stones). [Underactive Bladder p6]
- Carbachol is a muscarinic receptor agonist used as a parasympathomimetic in detrusor underactivity (alongside bethanechol). [Underactive Bladder p6]
- Parasympathomimetics are most likely to work when the problem is a reduced or absent contractile stimulus, that is reduced efferent input, impaired acetylcholine release or increased acetylcholine breakdown. [Underactive Bladder p6]
- Further side effects of parasympathomimetics are nausea, abdominal cramps and visual disturbance. [Underactive Bladder p6]
- Muscle relaxants are listed with alpha-blockers as routine therapy to decrease outlet resistance in detrusor underactivity. [Underactive Bladder p7]

Intravesical Electrotherapy

- In intravesical electrotherapy the bladder is filled with saline and current passes through an electrode at the catheter tip, with a skin pad as the return electrode. [Underactive Bladder p8]
- Intravesical electrotherapy is given in daily sessions, usually of 1 hour or more, and 10–15 sessions are taken as the trial period. [Underactive Bladder p8]
- Intravesical electrotherapy activates A-delta afferent fibres, giving central reflex activation of the detrusor. [Underactive Bladder p8]
- The drawbacks of intravesical electrotherapy are that it is very time-consuming, needs resources, and has no definitive studies on safety and application. [Underactive Bladder p8]
- Intravesical electrotherapy has shown promising early results, mainly in children. (Book 07 p29 notes a negative sham-controlled trial.) [Underactive Bladder p8]

CH 113 · PHARMACOLOGIC MANAGEMENT OF LOWER URINARY TRACT STORAGE AND EMPTYING FAILURE

Medical Management of SUI in Females

- Oestrogen has little effect in managing urodynamic stress incontinence, although two meta-analyses showed significant subjective improvement. [Compilation 03-3 p37-39]
- The oestrogen-sensitive tissues of the lower tract are the bladder, the pelvic floor and the urethra (its epithelium, vasculature, connective tissue and muscle). [Compilation 03-3 p37-39]
- Midodrine and methoxamine have some α1 selectivity, but side effects limit their clinical use in SUI. [Compilation 03-3 p37-39]
- Ephedrine and phenylpropanolamine have been used for SUI but were no better than placebo. [Compilation 03-3 p37-39]
- There are no RCTs of imipramine in SUI. [Compilation 03-3 p37-39]
- Duloxetine is licensed in Europe at 40 mg twice daily for moderate to severe SUI, defined as 15 or more incontinence episodes per week. [Compilation 03-3 p37-39]

- Other uses of duloxetine are fibromyalgia, chronic musculoskeletal pain and generalised anxiety disorder, as well as major depressive disorder. [Compilation 03-3 p37-39]
- Further side effects of duloxetine are diarrhoea, fatigue, headache and somnolence. [Compilation 03-3 p37-39]

Bethanechol

- Bethanechol is not hydrolysed by cholinesterase. [Compilation 03 p49]
- Further side effects of bethanechol are abdominal cramps, sweating, headache and difficulty in visual accommodation. [Compilation 03 p49]
- Bowel obstruction and hyperthyroidism are contraindications to bethanechol. [Compilation 03 p49]
- The bethanechol dose is 10-50 mg three or four times daily. [Compilation 03 p49]

Flavoxate

- Flavoxate has mixed actions and an unknown mechanism; it acts as a calcium antagonist, inhibits phosphodiesterase and has a local anaesthetic property. [Compilation 03 p49]
- Flavoxate is used in patients with detrusor overactivity or OAB, but no RCTs are available. [Compilation 03 p49]

Baclofen

- Baclofen acts at the GABA-B receptor (GABA being an inhibitory neurotransmitter) and has an inhibitory action in the spinal cord that controls micturition disorders and lowers outlet resistance. [Compilation 03 p49]
- Baclofen is not effective orally for voiding dysfunction because it does not cross the blood-brain barrier. [Compilation 03 p49]

Drug Therapy for LUT Storage and Emptying Failure

- Among alpha-blockers, silodosin (as well as naftopidil) also improves OAB symptoms. [LUT Drugs p1]
- COX inhibitors, acting by prostaglandin inhibition, have been tried to facilitate storage but are not a therapeutic option at present. [LUT Drugs p1]

Increasing Outlet Resistance and Facilitating Emptying

- Drugs for SUI in men are not well studied; duloxetine is used off-label. [LUT Drugs p2]
- Prostaglandins to increase bladder contractility have only a paucity of literature. [LUT Drugs p2]
- Intrathecal baclofen is an option to decrease outlet resistance, with few studies available. [LUT Drugs p2]

Pharmacological Properties of Antimuscarinics

- Antimuscarinics are grouped by lipophilicity, molecular charge and molecular size into tertiary amines and quaternary amines. [LUT Drugs p5]
- Tertiary amines (darifenacin, solifenacin, fesoterodine, oxybutynin, tolterodine, atropine) have higher lipophilicity and are more likely to cross the blood-brain barrier. [LUT Drugs p5]
- Quaternary amines (propantheline, trospium) have lower lipophilicity and charge, are not well absorbed and pass poorly into the CNS, so they cause fewer CNS side effects. [LUT Drugs p5]
- Enzyme inducers such as phenytoin and phenobarbitone lower antimuscarinic drug levels, and enzyme inhibitors such as ketoconazole and cimetidine raise them. [LUT Drugs p5]

Clinical Use and Adverse Effects of Antimuscarinics

- Cardiac adverse effects of antimuscarinics include torsades de pointes (as well as QT prolongation and raised heart rate). [LUT Drugs p6]
- Other listed antimuscarinic adverse effects are memory impairment, dizziness, fatigue, headache and photophobia. [LUT Drugs p6]

Darifenacin, Solifenacin, Tolterodine, Trospium

- Darifenacin has a half-life of 13-19 hours. [LUT Drugs p7]
- Tolterodine IR may be given as 1 mg BD (as well as 2 mg BD), and tolterodine ER as 2 mg OD (as well as 4 mg OD). [LUT Drugs p7]

- Trospium has an oral bioavailability below 10% and a half-life of about 20 hours. [LUT Drugs p7]
- About 60% of trospium is eliminated unchanged in urine, and it is not metabolised by CYP450. [LUT Drugs p7]

Oxybutynin Chloride

- Oxybutynin has a half-life of 2 hours. [LUT Drugs p8]
- Oxybutynin has three actions: antimuscarinic (higher affinity for M1 and M3), local anaesthetic, and direct muscle relaxant (relevant to intravesical use). [LUT Drugs p8]
- Oxybutynin ER is given as 5-10 mg OD. [LUT Drugs p8]
- Transdermal oxybutynin reduces desethyloxybutynin production even further, but causes pruritus and erythema. [LUT Drugs p8]
- Oxybutynin is also available as a topical gel and for intrarectal use (in addition to intravesical). [LUT Drugs p8]

Beta-adrenoceptor Agonists (Mirabegron): Mechanism

- All three beta subtypes (beta1, beta2, beta3) are present in detrusor and urothelium, with beta3 predominant. [LUT Drugs p9]
- Potassium channels, particularly BKCa, may also be involved in (or be more important for) beta-receptor-mediated detrusor relaxation. [LUT Drugs p9]
- Mirabegron is rapidly absorbed orally and metabolised in the liver by CYP450. [LUT Drugs p9]

Mirabegron: Pharmacology, Adverse Effects, Dose

- In plasma, mirabegron circulates unchanged and as glucuronide conjugates and other inactive metabolites; its terminal elimination half-life is about 24 hours. [LUT Drugs p10]
- Mirabegron does not prolong the QTc interval at therapeutic or supratherapeutic doses. [LUT Drugs p10]

Botulinum Toxin: Preparations and Mechanism

- Botulinum toxin brands are wrapped by different proteins that modify their relative potency: Botox (onabotulinumtoxinA), Dysport (abobotulinumtoxinA), Xeomin (incobotulinumtoxinA) and Prosigne. [LUT Drugs p11]
- Botox is about 3 times more potent than the other preparations. [LUT Drugs p11]
- The toxin binds SV2 on the nerve terminal and enters by endocytosis before the light chain cleaves SNAP-25; the SNARE proteins are synaptobrevin, SNAP-25 and syntaxin. [LUT Drugs p11]

Botulinum Toxin: Other Mechanisms, Adverse Effects

- The heavy and light chains of BoNT-A are joined by a disulfide bond. [LUT Drugs p12]
- BoNT-A also inhibits spinal cord release of glutamate from sensory nerves (with substance P and CGRP). [LUT Drugs p12]
- Bladder pain is among the commonest adverse effects of intravesical botulinum toxin (with UTI). [LUT Drugs p12]
- Advantages of intravesical administration are a high local concentration and minimal systemic side effects. [LUT Drugs p12]

Botulinum Toxin: Indications, Contraindications, Preparation

- Refractory bladder outlet obstruction is listed among the urological indications for botulinum toxin. [LUT Drugs p13]
- Contraindications include hypersensitivity to BoNT and a history of acute urinary retention in a patient not on CIC. [LUT Drugs p13]
- Relative contraindications include children and patients with low pulmonary reserve (as well as myasthenia gravis and aminoglycosides). [LUT Drugs p13]
- Botox comes in 100 U and 200 U vials; once reconstituted it is stored at 2-8 °C and used within 24 hours. [LUT Drugs p13]

Botulinum Toxin: Injection Technique

- Injections are given through a cystoscope with a 4 mm ultrafine endoscopic needle inserted about 2 mm into the bladder wall, 30 injections 1 cm apart over the dome and the posterior, right and left walls. [LUT Drugs p14]
- After the injections the bladder is emptied and the patient observed for 30 minutes. [LUT Drugs p14]

CH 115 · ELECTRICAL STIMULATION AND NEUROMODULATION IN STORAGE AND EMPTYING FAILURE

Sacral Neuromodulation: Indications

- Idiopathic non-obstructive urinary retention treated by SNM includes cases due to DSD and functional outlet resistance, as well as Fowler syndrome. [Compilation 03 p53]
- Neurogenic bladder in multiple sclerosis is listed as an indication for SNM. [Compilation 03 p53]
- Constipation (as well as faecal incontinence) is an indication for SNM. [Compilation 03 p53]

Sacral Neuromodulation: staged trial and lead positioning

- The test-stimulation trial lasts 3–4 weeks for urinary retention and 1–2 weeks for overactive bladder. [Compilation 03 p54]
- The sensory check of correct lead position is stimulation felt between the rectum and vagina in women and a penoscrotal sensation in men. [Compilation 03 p55]

Sacral Neuromodulation: mechanism of action

- In overactive bladder, sacral nerve stimulation activates pudendal afferents, which inhibit bladder C-fibre afferents and their sensory output to the PMC, so detrusor overactivity is inhibited and bladder sensation improves. [Compilation 03 p55-56]
- Aδ bladder afferents are not inhibited by sacral nerve stimulation, which is why normal voiding is preserved. [Compilation 03 p55-56]
- In functional retention (DSD, Fowler syndrome), abnormal pudendal afferent activity inhibits detrusor contraction; sacral nerve stimulation restores normal pudendal afferent activity, so sensation reaches the PAG/PMC and parasympathetic efferents produce detrusor contraction and voiding. [Compilation 03 p55-56]

Sacral Neuromodulation: complications and troubleshooting

- For pain at the IPG site, first turn the device off; if discomfort persists the cause is a pocket complication (erosion, infection, seroma, or a pocket that is too loose or too tight), which is addressed accordingly. [Compilation 03 p57]
- If IPG-site discomfort resolves when the device is turned off, change from a unipolar to a bipolar setting, or, if current leakage is the cause, revise the device. [Compilation 03 p57]
- When stimulation is felt in the wrong location, perform unipolar or bipolar reprogramming, and revise the leads if this fails. [Compilation 03 p58]
- When no stimulation is felt, first exclude inadvertent switching off and IPG end of life, then check impedances: abnormal on all electrodes means revision, some abnormal means programming with the normal electrodes, and equal normal impedances on all electrodes suggest current leakage, which needs revision. [Compilation 03 p58]
- With intermittent stimulation, check impedances: normal impedances indicate a positional problem and abnormal impedances a mechanical problem. [Compilation 03 p58]
- Relative contraindications to sacral neuromodulation include pregnancy and significant spine abnormalities. [Compilation 03 p58]

OTHER TOPICS IN THIS BOOK

Detrusor Leak Point Pressure

- DLPP is useful in lower motor neurone lesions causing decentralisation of the bladder, in addition to upper motor neurone lesions with high storage pressure and non-neurogenic low-compliance

bladders. [ALPP DLPP p2]

Augmentation cystoplasty: goals and indications

- In poor bladder compliance, augmentation cystoplasty decreases maximum detrusor pressure by 65–82%. [Augmentation Cystoplasty p1]
- Reduced bladder capacity, which causes frequent micturition and incontinence, is an indication, and augmentation increases cystometric capacity to over 500 mL. [Augmentation Cystoplasty p2]
- Detrusor overactivity refractory to drugs, neuromodulation and botulinum toxin is an indication, particularly in patients who fear incontinence. [Augmentation Cystoplasty p2]

Pressure-flow studies (BOOI / BCI)

- The bladder outlet obstruction index (PdetQmax − 2Qmax) is also called the Abrams–Griffiths number. [Compilation 03-3 p74]

Co-ordination (DESD, dysfunctional voiding, non-relaxing sphincter)

- A non-relaxing urethral sphincter is classically seen in meningocele, and it is actually a failure of urethral relaxation rather than active dyssynergia. [Compilation 03-3 p75]

VUDS

- VUDS is indicated in children with abnormal voiding in whom invasive therapy is being contemplated. [Compilation 03-3 p76]
- VUDS is indicated in impaired renal function where urethral overactivity may be the cause (for example with VUR). [Compilation 03-3 p76]

Ambulatory UDS

- Ambulatory UDS records three micturition cycles: a resting cycle (patient sits on a chair), an ambulant cycle (patient moves around) and an exercising cycle (patient performs specific incontinence-provoking acts). [Compilation 03-3 p77]
- Ambulatory UDS is indicated to determine whether detrusor overactivity or sphincter weakness is the main cause of incontinence. [Compilation 03-3 p77]
- Limitations of ambulatory UDS are that it is time-consuming and technically challenging, normal and abnormal standards are not yet established, and more data are needed on its reproducibility. [Compilation 03-3 p77]

Urethral pressure profile

- Urethral pressure is defined as the fluid pressure required to just open a closed urethra. [Compilation 03-3 p78-79]
- UPP method: a small catheter with lateral apertures infuses fluid while bladder and urethral pressures are measured simultaneously as the catheter is slowly withdrawn along the urethra. [Compilation 03-3 p78-79]
- The urethral closure pressure profile is the difference between urethral and intravesical pressure along the urethra (written in the notes as intravesical pressure minus urethral pressure). [Compilation 03-3 p78-79]
- Maximum urethral pressure (MUP) is the highest pressure measured along the urethral pressure profile. [Compilation 03-3 p78-79]
- Functional profile length is the length of urethra along which urethral pressure exceeds intravesical pressure, about 3 cm in women. [Compilation 03-3 p78-79]
- Total profile length is the full length of the urethral pressure profile tracing (shown on the UPP diagram alongside MUCP, MUP and bladder pressure). [Compilation 03-3 p78-79]

Uroflowmetry

- Flow time is the time over which measurable flow occurs. [Compilation 03-3 p80]
- Average flow rate is voided volume divided by flow time. [Compilation 03-3 p80]
- Time to maximum flow is the time elapsed from the onset of flow to maximum flow. [Compilation 03-3 p80]

PBNO and functional obstruction: treatment and terminology

- TUIP or TURP is a further treatment option for primary bladder neck obstruction in men, after alpha-blockers (partial relief) and bladder neck incision (definitive relief). [Compilation 03-5 p1]
- In women with bladder neck obstruction, pelvic floor therapy or behavioural treatment and drugs come first, because surgery carries a high risk of stress urinary incontinence. [Compilation 03-5 p1]

Ch 87 Additional therapies: storage failure at the outlet

- Additional therapies for storage failure at the outlet are myoplasty, male urethral compression devices, female urethral occlusive devices, intravaginal continence devices and bladder outlet closure (functional or complete). [Compilation 03-5 p5]

Bladder outlet closure

- Functional bladder outlet closure uses an autologous pubovaginal sling to create obstruction at the bladder neck, which still allows a pop-off for leakage at higher pressures. [Compilation 03-5 p7]
- Complete closure is used when a functional outlet cannot be preserved; it is a multilayered bladder neck closure (transvaginal, transabdominal or combined) with an abdominal stoma for CIC. [Compilation 03-5 p7]
- Bladder outlet closure is indicated after failure of other surgical options or with urethral or bladder neck erosion or destruction. [Compilation 03-5 p7]

Myoplasty

- Myoplasty for storage failure is an autologous gracilis muscle transfer to form a neosphincter around the urethra and bladder neck; large studies are lacking. [Compilation 03-5 p8]

Continent catheterisable stomas

- Further indications for a continent catheterisable stoma are urethral or bladder neck destruction (erosion from a chronic catheter, radiation, trauma) and refractory urethral fistula or stricture. [Compilation 03-5 p10]
- Stomal prolapse and stricture, which may need revision, are complications of continent catheterisable stomas. [Compilation 03-5 p10]

Catheterisation: long-term indwelling catheter and suprapubic catheter

- Indwelling catheters are generally for short-term drainage; for long-term drainage (such as spinal cord injury) CIC is preferred. [Compilation 03-5 p10]
- Long-term indwelling catheters cause poor bladder compliance (exact cause unknown), with VUR, pyelonephritis, renal scars and calculi. [Compilation 03-5 p11]
- Other complications of long-term indwelling catheters are periurethral abscess, catheter blockage and haematuria. [Compilation 03-5 p11]
- The long-term risk of carcinoma with an indwelling catheter is 8–10%, from chronic inflammation and nitrosamines. [Compilation 03-5 p11]
- A suprapubic catheter preserves sexual function and has a low risk of urethral stricture and epididymitis. [Compilation 03-5 p11]
- Disadvantages of a suprapubic catheter are that it is invasive and incontinence may persist with a very weak sphincter or reflex detrusor overactivity. [Compilation 03-5 p11]

Clean intermittent catheterisation

- Adequate urethral exposure is a prerequisite for CIC, along with dexterity, cognition and compliance. [Compilation 03-5 p12]
- CIC should be done more often in patients with low bladder capacity. [Compilation 03-5 p12]
- A short (6-inch) catheter is used for women. [Compilation 03-5 p12]
- The catheter is washed with soap and water and air-dried before storage. [Compilation 03-5 p12]
- Advantages of CIC are no stoma bag, preservation of sexual function and no cancer risk, with fewer complications than an indwelling catheter. [Compilation 03-5 p12]

Increasing intravesical pressure: Credé and Valsalva

- Credé and Valsalva may succeed in an areflexic bladder with low outlet resistance (after surgery or botulinum toxin); these patients have stress incontinence and the manoeuvre lets them empty at a time when they choose. [Compilation 03-5 p14]
- Credé manoeuvre can generate an intravesical pressure of more than 50 cm H2O. [Compilation 03-5 p14]
- Credé is done by an open-handed method or a closed-fist method, or with a rolled-up towel. [Compilation 03-5 p14]
- For Valsalva voiding the patient sits and lets the abdomen protrude onto the thighs; hugging the knees and legs may help prevent bulging of the abdomen. [Compilation 03-5 p14]

Promotion or initiation of reflex contractions

- In spinal cord injury patients with intact spinal cord reflexes, rhythmic suprapubic manual pressure (seven or eight pushes every 3 seconds) has a summation effect on bladder-wall tension receptors and activates the bladder reflex arc. [Compilation 03-5 p15]
- Other triggering methods are pulling the skin or hair of the pubis, scrotum or thigh, digital rectal stimulation and squeezing the clitoris. [Compilation 03-5 p15]

Circumventing the problem: collection devices and absorbent products

- There are no effective external collection devices for women; men can use a condom catheter, penile sheath or Texas catheter. [Compilation 03-5 p16]
- Absorbent products include pads, shields, drip collectors, guards, briefs, undergarments and underpads. [Compilation 03-5 p16]
- Incontinence-associated dermatitis is inflammation of the skin surface with redness, oedema and sometimes bullae containing clear exudate; it occurs mainly in skin folds and may develop candidiasis or bacterial skin infection. [Compilation 03-5 p16]

Diabetes and Urology

- The urological complications of diabetes can be remembered as FRIENDS: Fournier gangrene and soft-tissue infection, Renal failure, Infections and UTI, Enhanced wound complications and EPN, Necrosis (papillary), Diabetic cystopathy, and Stones (low urine pH) and Sexual dysfunction. [Compilation 03 p80]

Uroflowmetry: normal flow pattern

- The ideal voided volume for a uroflow study is 200–400 mL. [Compilation 03-4 p1]
- Curve tracing depends on voided volume and on paper speed, which is 0.25 cm/s. [Compilation 03-4 p1]
- Age, sex and voided volume must be taken into account when interpreting a flow pattern. [Compilation 03-4 p1]
- Maximum flow should be reached within 5 seconds from the start of flow. [Compilation 03-4 p1]
- Age-specific normal flow values (minimum voided volume; Qmax male / female): age 4–7, 100 mL, 10 / 10 mL/s; age 8–13, 100 mL, 12 / 15 mL/s; age 14–45, 200 mL, 21 / 18 mL/s; age 46–65, 200 mL, 12 / 15 mL/s; age 66–80, 200 mL, 9 / 10 mL/s. [Compilation 03-4 p1]

Abnormal flow patterns

- In detrusor overactivity, the flow trace rises very rapidly to maximum in an abnormally short time (1–3 seconds). [Compilation 03-4 p2]
- In detrusor underactivity, the time to reach maximum flow is very variable and maximum flow may occur in the second half of the trace. [Compilation 03-4 p2]
- Bladder outlet obstruction curves are divided into compressive (BPH) and constrictive (stricture) types. [Compilation 03-4 p2]
- In compressive (BPH) obstruction, the first third of the flow curve may appear relatively normal with a reduced Qmax, but the later part is elongated into a tail of reducing flow rate. [Compilation 03-4 p2]
- In constrictive (stricture) obstruction, there is little difference between Qmax and Qaverage, giving a constant flow rate due to fixed resistance. [Compilation 03-4 p2]

- Interrupted flow from abdominal straining gives variable traces and may occur with or without obstruction or detrusor contraction. [Compilation 03-4 p3]
- Interrupted flow can be secondary to urethral overactivity, in which the flow rate changes faster than it does with straining. [Compilation 03-4 p3]
- A poorly sustained or fluctuating detrusor contraction (hypotonic bladder) gives a low, undulating interrupted flow curve. [Compilation 03-4 p3]

Diabetes and LUTD: mechanism

- The peripheral and autonomic neuropathy of diabetic cystopathy is characterised by axonal degeneration with segmental demyelination. [Diabetic Cystopathy p1]
- The sequence is impaired sensory afferent pathway, impaired bladder sensation, a gradual increase in the interval between voids, overdistension of the bladder, impaired detrusor function and finally decompensation (end-stage bladder). [Diabetic Cystopathy p1]
- The early-diuresis explanation of diabetic bladder dysfunction is called the "temporal theory": hyperglycaemia causes osmotic diuresis, overstretching of the bladder wall and raised intravesical pressure, then compensatory hypertrophy with storage symptoms, and later progression from accumulation of toxic products of oxidative stress. [Diabetic Cystopathy p2]
- Diabetic bladder dysfunction may also result from detrusor myocyte dysfunction, by a poorly understood mechanism. [Diabetic Cystopathy p2]
- 5–59% of patients with diabetes report lower urinary tract dysfunction. [Diabetic Cystopathy p2]

Evaluation

- Evaluation of diabetic cystopathy includes random or fasting blood sugar, HbA1c, urine routine examination, renal function tests and a bladder diary. [Diabetic Cystopathy p2]
- Ultrasound shows a high PVR and bladder wall thickening and assesses the upper tracts, and uroflowmetry shows an impaired flow rate and pattern. [Diabetic Cystopathy p2]

Urodynamics and treatment

- On urodynamics in diabetic cystopathy both sphincters are usually synergistic, though pseudodyssynergia may be seen on EMG. [Diabetic Cystopathy p3]
- Treatment begins with glycaemic control and control of infection. [Diabetic Cystopathy p3]
- In early disease, scheduled voiding and double voiding are used. [Diabetic Cystopathy p3]
- Bladder outlet resistance may be reduced with drugs or surgery. [Diabetic Cystopathy p3]
- For the end-stage bladder, the options are CIC or urinary diversion. [Diabetic Cystopathy p3]

8 Nocturia & Male Incontinence

CH 112 · NOCTURIA

Nocturia: Definition and Rationale for Evaluation

- The prevalence of nocturia is 28.4% in the 30–80 year age group. It is commoner in women than in men and rises with age. [Compilation 03-3 p51]
- Sleep latency is the time taken to go from fully awake to fully asleep, and nocturia affects both sleep latency and sleep efficiency. [Compilation 03-3 p51]
- Sleep efficiency is the actual sleep time divided by the intended sleep time (in minutes), and a value above 85% is normal. [Compilation 03-3 p51]
- Nocturia costs society through lost productivity. [Compilation 03-3 p51]

Nocturia: Causes

- Causes of low nocturnal or global bladder capacity include ureteric or bladder calculi, carcinoma of the bladder, prostate or urethra, nocturnal detrusor overactivity, anxiety disorders, learned voiding dysfunction, radiation and tuberculosis. [Compilation 03-3 p52]

Evaluation of Nocturia and Algorithm for Treatment; Global Polyuria

- In global polyuria, an overnight water deprivation test is done. A first-morning urine osmolality above 800 mOsm/kg H2O is normal and points to primary polydipsia (an appropriate renal response and ADH secretion). [Compilation 03-3 p53-55]
- If first-morning urine osmolality is below 800 mOsm/kg, give 40 µg intranasal desmopressin or 0.4 mg orally and recheck urine osmolality 3–5 hours later. A value of 800 or more means central DI, and a value below 800 means nephrogenic DI. [Compilation 03-3 p53-55]
- Further causes of central DI (pituitary or hypothalamic) are metastasis, sarcoidosis, infarction and infection. [Compilation 03-3 p53-55]
- Extrarenal and other causes of DI listed are PGE2, ANP, lithium, tetracycline, hypercalcaemia and hypokalaemia. [Compilation 03-3 p53-55]
- Primary polydipsia may be dipsogenic or psychogenic. [Compilation 03-3 p53-55]
- Patients with dipsogenic polydipsia have a history of a central neurological abnormality or radiation, and are treated if a cause is found. [Compilation 03-3 p53-55]
- Psychogenic polydipsia is treated with behavioural therapy or psychotherapy. [Compilation 03-3 p53-55]
- In diabetes mellitus causing global polyuria, treatment targets the glycosuria. Nephrogenic DI has no specific treatment. [Compilation 03-3 p53-55]

Nocturnal Polyuria: Treatment (Imipramine)

- Imipramine has variable effects in nocturnal polyuria. [Compilation 03-3 p56]
- Imipramine (a TCA) carries a black box warning for increased suicidal tendency in depressed patients. [Compilation 03-3 p56]
- The cardiac effects of imipramine include prolonged PR, QRS and QTc intervals and torsades de pointes, as well as sudden death. Other side effects are dizziness, low blood pressure and blurred vision. [Compilation 03-3 p56]
- Imipramine has complex pharmacology. It is an SNRI and a non-selective antimuscarinic, and it modulates AVP release and proximal tubular sodium and water reabsorption. [Compilation 03-3 p56]

Diminished Global and Nocturnal Bladder Capacity

- Treating BOO improves nocturia by lowering the PVR, increasing functional bladder capacity and lowering frequency. [Compilation 03-3 p57]
- Other agents tried for nocturia are COX-2 inhibitors (which block PGE2-mediated diuresis), Pygeum africanum and Cernilton (rye-derived), and none has conclusive evidence. [Compilation 03-3 p57]

Nocturia Index

- When Ni is above 1 (night urine volume exceeds maximum bladder capacity), the result is nocturia, or enuresis if the patient does not wake. [Compilation 03-3 p59]

AVP (ADH)

- In collecting-duct cells, water leaves across the basolateral membrane through aquaporin-3 and aquaporin-4, while aquaporin-2 sits in recycling vesicles that are inserted into the luminal membrane and retrieved by endocytosis. [Compilation 03-3 p60]
- Stimuli for ADH release are hyperosmolality, angiotensin II, reduced atrial receptor firing and sympathetic stimulation. [Compilation 03-3 p60]

Desmopressin

- In children with nocturnal enuresis, desmopressin gives 2.2 fewer wet nights per week, and children taking it are 4.5 times more likely than with placebo to sleep without disruption. [Compilation 03-3 p61-62]
- US regulators banned desmopressin nasal sprays for nocturnal enuresis after 2 patients died and 59 had seizures from hyponatraemia. Tablets are still considered safe in otherwise healthy children. [Compilation 03-3 p61-62]
- Desmopressin raises von Willebrand factor and factor VIII levels 3- to 5-fold. [Compilation 03-3 p61-62]

- Desmopressin is used in mild haemophilia A and thrombocytopenia as well as von Willebrand disease. It is not effective in haemophilia B or severe haemophilia A. [Compilation 03-3 p61-62]
- Desmopressin is contraindicated in type IIb von Willebrand disease. [Compilation 03-3 p61-62]
- Facial flushing is a side effect of desmopressin. [Compilation 03-3 p61-62]
- Vasopressin stimulates ACTH release, but desmopressin does not (except in Cushing's disease), so it does not directly raise blood pressure. [Compilation 03-3 p61-62]
- Desmopressin should be stopped during severe vomiting and diarrhoea, fever or flu, and used with caution in hot weather or after strenuous exercise. [Compilation 03-3 p61-62]
- The initial oral dose is 0.2 mg (1 tablet) in the late evening, which can be raised to 0.4 mg after 10–14 days if needed. The melt is started at 120 µg and raised to 240 µg after 10–14 days if needed. [Compilation 03-3 p61-62]
- To prevent dilutional hyponatraemia, water intake is limited to 240 mL from 1 hour before until 8 hours after dosing. [Compilation 03-3 p61-62]
- Response is assessed at 1–2 weeks. If the response is positive, treatment can be continued for 3 months. [Compilation 03-3 p61-62]
- After the early checks, sodium is monitored every 6 months. [Compilation 03-3 p61-62]

CH 123 · SURGICAL PROCEDURES FOR SPHINCTERIC INCONTINENCE IN THE MALE: ARTIFICIAL URINARY SPHINCTER AND PERINEAL SLINGS

Incontinence in the Male: Sphincteric Incontinence, Causes and Device Mechanisms

- Causes of internal (bladder neck) sphincter deficiency in men are pelvic surgery, bladder neck injury, sympathetic neuropathy and embryological disruption. [Compilation 03-3 p5]
- Causes of external sphincter deficiency in men are radical prostatectomy (commonest), pelvic fracture, myelopathy and congenital conditions (sacral agenesis, spinal dysraphism and the exstrophy-epispadias complex). [Compilation 03-3 p5]
- The bone-anchored male sling works by compression of the urethra. [Compilation 03-3 p5]

Male Incontinence: Management Algorithm

- If urodynamics or an accurate voiding diary shows significant OAB or reduced compliance, antimuscarinics are given first; if OAB resolves the patient returns to the sphincter pathway, and if not, the bladder dysfunction is evaluated and treated before sphincter surgery. [Compilation 03-3 p6-7]
- Gravitational (continuous) incontinence is an indication for an AUS rather than a sling, and in gravitational incontinence surgery need only be delayed about 6 months after prostatectomy. [Compilation 03-3 p6-7]
- When AUS revision options (smaller cuff, transcorporal cuff, double cuff, bladder neck cuff) still leave persistent incontinence, the final options are bladder neck closure or urinary diversion. [Compilation 03-3 p6-7]

AUS versus Sling: Advantages and Disadvantages

- The AUS gives a reproducible outcome. [Compilation 03-3 p9]
- AUS revision rate is about 16% at 2 years and 28% at 5 years. [Compilation 03-3 p9]
- A male sling is an option for men who refuse an AUS for fear of its complications. [Compilation 03-3 p9]

AUS (AMS 800) Placement Technique

- The patient is placed in the low lithotomy position for AUS placement. [Compilation 03-3 p10]
- The pressure-regulating balloon is placed slightly laterally in the prevesical space through a horizontal lower-quadrant abdominal incision; scrotal and perineal routes are alternatives. [Compilation 03-3 p10]
- The pump is placed in the anterior scrotum through the same abdominal incision, and the tubing connections are made in the abdominal incision; scrotal and perineal approaches are alternatives. [Compilation 03-3 p10]

AUS Variations

- In the transcorporal cuff the plane of dissection passes through the septum of the corpora cavernosa from one side to the other, and it reduces urethral injury during dissection. [Compilation 03-3 p11]
- In the transscrotal AUS all components are placed through a single scrotal incision, but continence results are inferior. [Compilation 03-3 p11]

Male Sling Placement

- The transobturator bulbourethral sling is fixed to the corpus spongiosum with 4–6 permanent or absorbable sutures. [Compilation 03-3 p12]
- The bone-anchored bulbourethral sling (InVance) is placed through a perineal incision that exposes the bulbospongiosus muscle and the medial aspect of the inferior pubic ramus. [Compilation 03-3 p12]
- In the InVance sling, the sling lies over the bulbospongiosus muscle and is fixed with 3 screws on each side to the inferior pubic ramus. [Compilation 03-3 p12]

Complications of AUS

- Urinary retention after AUS is rare, and the suprapubic catheter for retention beyond 48 hours is placed under ultrasound guidance. [Compilation 03-3 p13-14]
- AUS infection presents with scrotal pain, erythema, oedema and frank purulence. [Compilation 03-3 p13-14]
- Immediate reimplantation of an infected non-eroded AUS is done with antiseptic irrigation and device replacement; otherwise reimplantation is delayed. [Compilation 03-3 p13-14]
- After AUS erosion, urethral healing is confirmed by RGU or cystoscopy at 3–6 months before reimplantation. [Compilation 03-3 p13-14]
- Urethral atrophy from chronic cuff compression is the commonest reason for AUS revision. [Compilation 03-3 p13-14]

Complications and Outcomes of Slings; History of AUS

- Complications of male slings include perineal pain, urinary retention, infection, erosion and osteomyelitis (bone-anchored slings). [Compilation 03-3 p15]
- Bone-anchored male slings give cure or improvement in 67–92%, with a failure rate of 8–33%. [Compilation 03-3 p15]
- Continence devices evolved from the Kaufman I, II and III devices, then slings, then the AMS 721, 742, 791 and 792 sphincters, and finally the AMS 800. [Compilation 03-3 p15]

Urinary Incontinence – Men

- Residence in old-age (nursing) homes is a risk factor for male urinary incontinence. [Post Prostatectomy Incontinence p1]
- Preoperative bladder function is a risk factor for incontinence after prostatectomy. [Post Prostatectomy Incontinence p1]

Post-Prostatectomy Urinary Incontinence: Aetiology

- Sphincter dysfunction or injury, the commonest cause of post-prostatectomy incontinence, accounts for more than two-thirds of patients. [Post Prostatectomy Incontinence p4]

Evaluation

- The 24-hour pad test is the most accurate pad test, whereas the 1-hour pad test is the most commonly used. [Post Prostatectomy Incontinence p5]

Urethral Bulking Agents

- Deflux is listed among the urethral bulking agents used for post-prostatectomy incontinence (with Durasphere and Macroplastique). [Post Prostatectomy Incontinence p7]
- The usual injected volume of bulking agent is 2.5–5 mL. [Post Prostatectomy Incontinence p7]
- Urethral inflammation leading to a "frozen" urethra is a complication of bulking agents. [Post Prostatectomy Incontinence p7]

- In men, bulking agents can be injected by an antegrade or a retrograde route (in addition to transurethral or periurethral). [Post Prostatectomy Incontinence p7]

Male Slings

- Detrusor overactivity is a contraindication to a male sling (along with prior radiotherapy). [Post Prostatectomy Incontinence p7]
- Bone-anchored slings are one of the three types of male sling (with transobturator and quadratic-fixation slings). [Post Prostatectomy Incontinence p7]

OTHER TOPICS IN THIS BOOK

Male urethral compression devices

- External male compression devices include the Hiesler metal clamp (no longer used), the Cunningham clamp, the C-3 and the U-Tex. [Compilation 03-5 p8]
- Berry's implantable prosthetic device was an internal compression device that is no longer used. [Compilation 03-5 p8]
- Kaufman I and II compressed the bulbar urethra using a cavernous–crural crossover (I) or approximation (II); Kaufman III was a silicone gel prosthesis implanted perineally to compress the bulbar urethra. [Compilation 03-5 p8]
- External compression devices cause pressure necrosis, so they should be released every 3–4 hours. [Compilation 03-5 p8]
- External compression devices cannot be used in patients with impaired penile sensation. [Compilation 03-5 p8]

9 Andrology & Sexual Medicine

CH 69 · MALE INFERTILITY

Genetic disorders of male infertility: classification

- Genetic causes of male infertility can be classified into three groups: single-gene disorders, chromosomal disorders and non-Mendelian disorders, the last accounting for most human disease. [Infertility p1]
- A single-gene disorder is caused by a mutant allele or a pair of alleles at a single genetic locus. [Infertility p1]
- A truncated Y chromosome is a chromosomal cause of male infertility. [Infertility p1]

Single-gene disorder: CBAVD

- About 1% of infertile men have congenital bilateral absence of the vas deferens (CBAVD). [Infertility p2]
- The CFTR protein maintains secretion viscosity through sodium and chloride channel balance. [Infertility p2]
- In the non-CF form of CBAVD (noted as "Potter syndrome"), abnormal differentiation of the mesonephric duct before 7 weeks of gestation causes CBAVD with renal agenesis or renal ectopia, and genetic screening is essential for diagnosis. [Infertility p2]

Chromosomal disorders: Klinefelter syndrome

- Klinefelter syndrome carries a higher risk of non-Hodgkin lymphoma. [Infertility p3]
- Men with Klinefelter syndrome typically present with infertility despite normal libido and erectile function. [Infertility p3]

Chromosomal disorders: 46,XX male

- In 46,XX males, SRY may be translocated to an autosome as well as to the X chromosome. [Infertility p4]

Y chromosome microdeletion

- The Y chromosome is present as a single copy, so loss of any part of it has reproductive consequences. [Infertility p4]
- The AZFa region has a separate, distinct location on Yq, while the AZFb and AZFc regions overlap. [Infertility p4]

CH 70 · SURGICAL MANAGEMENT OF MALE INFERTILITY

Sperm Retrieval Techniques

- Sperm can be retrieved fresh for immediate use when the female partner's IVF cycle has been prepared in step with the retrieval, as an alternative to retrieving for cryopreservation. [Sperm ART etc p1]
- After conventional TESE, about 45% of couples achieve pregnancy, with an abortion (miscarriage) rate of 19%. [Sperm ART etc p1]
- The TESA puncture site is the superomedial or superolateral aspect of the testis. [Sperm ART etc p1]
- In post-mortem sperm retrieval, up to 24 million sperm with 70% grade I motility can be retrieved within 13 hours of death. [Sperm ART etc p1]

Pre-requisites for IVF

- Before IVF, the female partner is screened for HBsAg, anti-HCV, HIV I and II, HTLV I and II, and syphilis by rapid plasma reagin (RPR) test. [Sperm ART etc p5]
- Female pre-IVF tests also include a Pap smear, blood group, Rh type and an antibody screen. [Sperm ART etc p5]
- The reproductive hormone assay before IVF measures FSH and oestrogen. FSH rises when the ovary cannot produce normal eggs, and oestrogen influences FSH secretion and relates to egg quality. [Sperm ART etc p5]
- A sonohysterogram assesses the endometrial cavity before IVF, looking for polyps, scarring and fibroids. [Sperm ART etc p5]
- The male partner is screened for HIV, HBsAg, anti-HCV and RPR along with his semen evaluation. [Sperm ART etc p5]

Semen Processing Techniques

- The simplest semen processing method is a single or double wash with culture medium followed by centrifugation. It is simple and effective, and it fails mainly because it generates reactive oxygen species (ROS). [Sperm ART etc p6]
- In the swim-up technique, sperm wash medium is layered over liquefied semen and the tube is incubated at a 45 degree angle for 60 minutes. Viable spermatozoa swim up into the medium and are aspirated, while non-viable sperm stay behind. [Sperm ART etc p6]
- In the swim-down technique, the semen sample is layered over bovine serum albumin medium and incubated at 37 degrees C for 1 hour. The most motile sperm move downward. [Sperm ART etc p6]
- Density gradient centrifugation separates sperm by density, so that each spermatozoon settles at the gradient level matching its own density. Morphologically normal mature spermatozoa have a density of 1.10 g/mL. [Sperm ART etc p6]
- For density gradient centrifugation, semen is layered over a lower-density and then a higher-density solution and spun for 20 minutes at 1600 rpm. From top to bottom the tube then holds seminal plasma, white cells and debris, abnormal and non-motile sperm, the interface, and viable motile spermatozoa in the pellet. [Sperm ART etc p6]
- The drawback of density gradient centrifugation is that in severe oligozoospermia sperm are lost in the low-density fraction or at the interface. [Sperm ART etc p6]
- Filtration techniques use glass wool, glass beads or cross-linked dextran gel, relying on gravity to help pull motile spermatozoa through the column. They are not used routinely. [Sperm ART etc p6]

Sperm Cryopreservation

- Sperm cryopreservation is also indicated in men with autoimmune disorders or diabetes. [Sperm ART etc p8]
- Sperm cryopreservation is used in donor insemination programmes. [Sperm ART etc p8]
- Sperm cryopreservation can overcome the stress of producing a semen sample on the day of treatment. [Sperm ART etc p8]
- Sperm can be banked before vasectomy, or at the time of the procedure. [Sperm ART etc p8]
- Men exposed to potentially toxic agents that interfere with gametogenesis should be offered sperm cryopreservation. [Sperm ART etc p8]
- Cryoprotectants protect spermatozoa from freeze damage by ice crystallisation. They are glycerol (the most commonly used), ethylene glycol, dimethyl sulphoxide and 1,2-propanediol. [Sperm ART etc p8]
- Slow freezing, manual or automated, takes 2 to 4 hours in total. The semen is cooled while cryoprotectant is added: from room temperature to 5 degrees C at 0.5 to 1 degree C per minute, then from 5 to minus 80 degrees C at 1 to 10 degrees C per minute, then stored in liquid nitrogen at minus 196 degrees C. [Sperm ART etc p8]
- In fast freezing, the sample is mixed with an equal volume of cold cryoprotectant and loaded into straws. The straws are held at 4 degrees C for 10 minutes, then 15 cm above liquid nitrogen (at minus 80 degrees C) for 15 minutes, and then immersed in liquid nitrogen at minus 196 degrees C. [Sperm ART etc p8]
- Freezing damages sperm by thermal shock, by ice-crystal formation (which impairs organelle function and survival), by osmotic shock, and by cellular dehydration (toxic damage from high solute content). [Sperm ART etc p8]
- Freezing injures the plasma and mitochondrial membranes and activates caspases 3, 8 and 9. This lowers ATP generation and so reduces motility. [Sperm ART etc p8]
- Cryopreservation causes sperm DNA damage through oxidative stress, thermal denaturation and defects in DNA repair enzymes. [Sperm ART etc p8]
- Thawing is done gradually so that cells recover without abrupt temperature change, routinely at 37 degrees C. The sample stands at room temperature for 10 minutes, then in a 37 degree C thermostat for 10 minutes, then at room temperature for 15 minutes, and is then washed and centrifuged before use. [Sperm ART etc p8]
- Frozen sperm can be stored for up to 28 years. [Sperm ART etc p8]
- Long-term follow-up shows no increase in birth defects or chromosomal anomalies in children from cryopreserved sperm compared with the general population. [Sperm ART etc p8]

Assisted Reproductive Technology and IUI

- ART is defined as methods that manipulate gametes and embryos outside the woman's body and return the result to the female genital tract to achieve pregnancy. It comprises IUI, IVF (including ICSI) and third-party assisted ART: sperm donation, egg donation, and surrogates or gestational carriers. [Sperm ART etc p12]
- In IUI, the sperm are placed in the uterus using a Wallace catheter. [Sperm ART etc p12]
- IUI is effective for women with a scarred or defective cervix and for men with refractory erectile dysfunction. [Sperm ART etc p12]
- IUI can be natural or stimulated. Stimulation uses human menopausal gonadotrophin (hMG), FSH, LH, hCG, clomiphene citrate or letrozole. [Sperm ART etc p12]
- The success of monthly IUI is about 20% per cycle and depends on the cause of the couple's infertility. [Sperm ART etc p12]

In-Vitro Fertilisation

- IVF has four steps: stimulation of egg maturation (ovarian stimulation or induction), egg retrieval, fertilisation and embryo transfer. [Sperm ART etc p13]
- During ovarian stimulation, egg development is monitored by transvaginal ultrasound. Maturity is judged by follicle size and oestrogen level, an hCG injection then triggers ovulation, and the eggs are retrieved 36 hours after that injection. [Sperm ART etc p13]

- Egg retrieval needs mild sedation and takes about 30 minutes. Under transvaginal ultrasound, a needle is passed through the vaginal wall into the ovarian follicles and the eggs are aspirated by suction. [Sperm ART etc p13]
- In conventional IVF, sperm and egg are placed together in a dish and left overnight in an incubator, and fertilisation usually occurs on its own. Genetic testing should be considered if ICSI is needed. [Sperm ART etc p13]
- The embryo is placed in the uterus 1 to 6 days after retrieval, through the vagina with a long thin tube, and should implant into the uterine lining 6 to 10 days after retrieval. [Sperm ART etc p13]
- Assisted hatching makes a hole in the zona pellucida just before embryo transfer to help the embryo hatch. It is used in couples with previous unsuccessful IVF attempts. [Sperm ART etc p13]
- GIFT (gamete intrafallopian transfer) places the gametes laparoscopically into normal fallopian tubes. [Sperm ART etc p13]
- ZIFT (zygote intrafallopian transfer) places the fertilised egg (zygote) into the fallopian tube. [Sperm ART etc p13]

Third-Party Assisted ART

- Donor sperm is indicated when the husband has a genetic disease. [Sperm ART etc p15]
- Egg donation is used when the woman cannot produce eggs. Ova are taken from another woman after ovarian stimulation, fertilised by IVF, and the embryo is placed in the wife's uterus. [Sperm ART etc p15]
- Egg donation is indicated in ovarian insufficiency, after chemotherapy or radiotherapy, after oophorectomy, in women born without ovaries, in carriers of known genetic diseases, and in infertility due to poor egg quality. [Sperm ART etc p15]
- Surrogates and gestational carriers are used when the wife cannot carry a pregnancy to term. [Sperm ART etc p15]
- A surrogate is inseminated with the male partner's sperm, so the child is biologically related to the surrogate and the male partner. [Sperm ART etc p15]
- A gestational carrier receives an embryo made by IVF from the husband's sperm and the wife's ova, so the child is not biologically related to the carrier. [Sperm ART etc p15]

Risks of IVF/ART

- Ovarian hyperstimulation, in which the ovaries become swollen and painful, is a risk of IVF. [Sperm ART etc p17]
- Preterm delivery is a risk of ART, along with multiple pregnancy. [Sperm ART etc p17]
- Ectopic pregnancy and bleeding are risks of IVF/ART. [Sperm ART etc p17]

Intracytoplasmic Sperm Injection

- ICSI may not be safe in HBsAg-positive mothers because of vertical transmission. [Sperm ART etc p18]
- After ICSI, follow-up is regular prenatal care with prenatal aneuploidy screening based on maternal age, nuchal translucency scan and biomarkers. [Sperm ART etc p18]
- For ICSI, mature sperm can be selected by their binding to hyaluronan, the main constituent of the gel layer around the oocyte, and a commercial device is available for this. [Sperm ART etc p18]
- ICSI procedure: a holding micropipette steadies the mature oocyte with gentle suction from a micro-injector. A single sperm is immobilised by cutting its tail and drawn into an injection micropipette. The pipette pierces the oolemma, the sperm is released into the cytoplasm, and the oocyte is placed in culture and watched for signs of fertilisation before embryo transfer. [Sperm ART etc p18]

CH 71 · PHYSIOLOGY OF PENILE ERECTION AND PATHOPHYSIOLOGY OF ERECTILE DYSFUNCTION

Mechanism of Erection: Anatomic

- The corpora cavernosa provide support to the corpus spongiosum and glans. [Erection etc p1]

- The cavernous smooth muscle is described as two components, periarteriolar and perisinusoidal, which together regulate blood flow into the sinusoids. [Erection etc p1]
- The ischiocavernosus muscle pumps blood distally to hasten erection, in addition to giving extra rigidity in the rigid erection phase. [Erection etc p1]
- The corpus spongiosum constricts the urethral lumen to allow forceful expulsion of semen. [Erection etc p2]
- Functions of the glans: it acts as a cushion that lessens the impact of the penis on the female organs, it provides sensory input for erection and pleasure, and its cone shape facilitates intromission. [Erection etc p2]

Mechanism of Erection: Hemodynamics

- In the onset of erection, arterioles and arteries dilate with increased flow in both the systolic and the diastolic phases. [Erection etc p2]

Mechanism of Erection: Neurogenic

- Among brain centres, the forebrain controls sexual behaviour and the hypothalamus is responsible for sexual partner recognition; both are involved in penile erection. [Erection etc p3]
- Adrenaline (as well as noradrenaline) is listed among the inhibitory neurotransmitters of erection. [Erection etc p4]
- The renin-angiotensin system has mixed effects, with both pro-erectile and pro-detumescence actions depending on receptor type and location (the corpus mentions only the contractile angiotensin II/AT1 arm). [Erection etc p4]
- The afferent pathway of erection runs from free nerve endings and corpuscular receptors in the penis, via the dorsal nerve of the penis and pudendal nerve, to the S2–S4 segments, then ascends in the spinothalamic and spinoreticular pathways to the thalamus, hypothalamus and frontal cortex. [Erection etc p4]

Types of Erection and Phases

- Nocturnal (REM sleep) erections are associated with increased activity in the pontine area, the amygdalae and the anterior cingulate gyrus (the corpus mentions only the pontine tegmentum). [Erection etc p6]

CH 72 · EVALUATION AND MANAGEMENT OF ERECTILE DYSFUNCTION

Evaluation of Erectile Dysfunction: Special Vascular Investigations

- Historical and investigational studies of penile blood flow include the penile brachial pressure index, penile plethysmography, radioisotope penography, penile MRI, penile near-infrared spectrophotometry and cavernous smooth muscle content on biopsy. [Erection etc p8]

Evaluation of Erectile Dysfunction: Hormonal

- High serum prolactin also interferes with the peripheral conversion of testosterone to DHT, in addition to suppressing GnRH and so impairing LH secretion. [Erection etc p9]
- Hyperthyroidism causes ED by increasing aromatization of testosterone into oestrogen and by increasing alpha-adrenergic tone in the cavernous smooth muscle. [Erection etc p9]

Evaluation of Erectile Dysfunction: Neurological and Psychophysiologic

- Somatic nervous system testing includes the genitocerebral evoked potential (in addition to biothesiometry, sacral evoked response/BCR latency and dorsal nerve conduction velocity). [Erection etc p10]
- Autonomic nervous system testing includes heart rate variability with sympathetic skin response, penile thermal sensory testing, and corpus cavernosum EMG with single potential analysis of cavernous electrical activity. [Erection etc p10]
- Normal NPTR criteria include a mean erection duration of more than 30 minutes and an increase in circumference of more than 3 cm at the base and more than 2 cm at the tip. [Erection etc p10]

- Other psychophysiologic tests are audiovisual and vibratory stimulation, and neuroimaging (functional MRI and PET). [Erection etc p10]

Treatment of ED: Alternative Therapies

- Alternative therapies for ED include herbs, dietary supplements, ginkgo biloba, L-arginine (the substrate for NO synthesis) and Korean red ginseng. [Erection etc p11]

Hormonal Therapy: Testosterone Replacement

- Testosterone replacement should start with short-acting preparations. [Erection etc p12]
- The efficacy of testosterone replacement is best judged by clinical response. [Erection etc p12]
- Testosterone enanthate or cypionate can alternatively be given as 100 mg weekly (instead of 200–250 mg every 2–3 weeks). [Erection etc p12]
- Testosterone propionate is given as 200 mg every 2–3 days because of its short half-life. [Erection etc p12]
- Subcutaneous testosterone pellets (Testopel) contain 75 mg testosterone each; the dose is 2–6 pellets every 3–6 months. [Erection etc p12]
- Transdermal testosterone patches (Testoderm TTS, Androderm) deliver 5 mg daily and can be applied on the arm, back or buttock. [Erection etc p13]
- Problems with testosterone patches are itching and chronic dermatitis, treated with cortisone cream. [Erection etc p13]
- Testosterone gels Androgel (1%) and Testim (1%) are applied to the shoulder, upper arm or abdomen; Axiron (2%) is applied to the axilla, a concealed site with high permeability. [Erection etc p13]
- Buccal testosterone (Striant) is a mucoadhesive 30 mg tablet applied over the gum twice daily. [Erection etc p13]
- Oral testosterone needs large doses (more than 200 mg/day) because first-pass liver metabolism inactivates most of it, and the notes state that it is not approved in the US (an older statement; oral testosterone undecanoate has since been FDA-approved). [Erection etc p13]
- Side effects of testosterone replacement include urinary symptoms, gynaecomastia and acne (in addition to erythrocytosis, sleep apnoea and progression of prostate cancer). [Erection etc p13]

Alternative Hormonal Treatments and Hyperprolactinaemia

- DHT gel is a pure androgen that undergoes no peripheral aromatization, so it has no oestrogenic side effects. [Erection etc p14]
- DHEA and hCG offer only limited benefit on sexual function. [Erection etc p14]
- Oestrogens are among the offending drugs to stop in hyperprolactinaemia (with morphine). [Erection etc p14]
- Bromocriptine, besides lowering prolactin, restores normal testosterone and shrinks the pituitary tumour. [Erection etc p14]

Pharmacological Therapy: Intraurethral and Topical

- Contraindications to intraurethral alprostadil are hypersensitivity to alprostadil, abnormal penile anatomy and an increased risk of priapism. [Erection etc p15]
- Intraurethral therapy avoids needle injections; the drug is absorbed through the urethral mucosa into the corpus spongiosum and passes through small vascular channels into the corpora cavernosa. [Erection etc p15]
- MUSE (Medicated Urethral System for Erection) uses semisolid PGE1 pellets measuring 1 x 3 mm, inserted into the urethra 3 cm from the meatus, with an optional adjustable penile constriction band for better local retention. [Erection etc p15]
- ALIBRA is an intraurethral combination of alprostadil and prazosin. [Erection etc p15]
- Further problems of intraurethral therapy are minor urethral bleeding, hypotension, dizziness and priapism, and it is overall inferior to PDE5 inhibitors and intracavernosal injection. [Erection etc p15]
- Topical alprostadil 0.3% gives significantly more successful intercourse than placebo. [Erection etc p15]
- Topical prostaglandin E1 ethyl ester is a prodrug option. [Erection etc p15]
- Topical 2% nitroglycerine causes headache and only rarely gives a rigid penis. [Erection etc p15]

- Topical papaverine has minimal absorption because of its large molecular size. [Erection etc p15]
- Topical and transdermal agents still need further trials. [Erection etc p15]

Intracavernosal Injection

- Bimix concentrations are alprostadil 20 µg/mL with phentolamine 0.5 mg/mL, or papaverine 30 mg/mL with phentolamine 0.5 mg/mL. [Erection etc p16]
- Invicorp is an intracavernosal combination of VIP and phentolamine, with about 80% efficacy. [Erection etc p16]
- Intracavernosal alprostadil is 96% metabolised locally within 1 hour. [Erection etc p16]
- Haematoma is a side effect of intracavernosal alprostadil. [Erection etc p16]
- Papaverine is derived from the opium poppy. [Erection etc p16]
- Papaverine has a half-life of 1–2 hours and is metabolised in the liver. [Erection etc p16]
- Phentolamine causes erection by postsynaptic alpha-1 receptor blockade, but its presynaptic alpha-2 receptor inhibition antagonises this effect. [Erection etc p17]
- Phentolamine has a half-life of 30 minutes. [Erection etc p17]
- Side effects of phentolamine also include nasal congestion and GI upset. [Erection etc p17]
- VIP is a 28-amino-acid hormone that causes vasodilation by activating adenylate cyclase and raising cAMP, and it is not used as monotherapy. [Erection etc p17]

Oral Therapy: PDE5 Inhibitors

- An arrhythmia within the past 6 months is listed with MI and stroke as a contraindication to PDE5 inhibitors. [Erection etc p19]
- PDE5 inhibitors need caution in left ventricular outflow obstruction and hypovolaemia (in addition to aortic stenosis, hypotension and QTc prolongation). [Erection etc p19]
- Photophobia is one of the visual side effects of PDE5 inhibitors. [Erection etc p20]
- A causal link between PDE5 inhibitors and NAION (nonarteritic anterior ischaemic optic neuropathy) has not been proven by studies. [Erection etc p20]

Oral Therapy: Non-PDE5 Drugs

- Oral phentolamine is a nonspecific alpha-1 and alpha-2 antagonist that relaxes corporal smooth muscle by blocking postsynaptic alpha-1 receptors; it gives about a 40% response in mild ED and needs more study. [Erection etc p20]
- Yohimbine acts centrally as an alpha-2 antagonist; its use is best limited to men with psychogenic ED, as it does not enable successful intercourse in organic ED. [Erection etc p20]
- Apomorphine has a role in psychogenic ED; it is given sublingually at 2, 4 or 6 mg and has no erectile efficacy if swallowed. [Erection etc p21]
- The active metabolite of trazodone activates the pro-erectile 5-HT2C receptor and antagonises the anti-erectile 5-HT1A receptor. [Erection etc p21]
- Early studies of melanocortin receptor agonists show efficacy in inducing erectile responses. [Erection etc p21]

Medical Devices: Vacuum Erection Device

- Problems with the vacuum erection device include local pain or discomfort, and rarely penile pain, numbness, difficult ejaculation, ecchymosis, skin necrosis, urethral varicosities and Fournier gangrene. [Erection etc p23]

CH 73 · PRIAPISM

Ischemic Priapism: Histological Changes

- At 12 hours of ischaemic priapism, histology shows interstitial oedema in the corpora. [Priapism p1]
- At 24 hours, the sinusoidal endothelium is destroyed, exposing the basement membrane, and platelets adhere to it. [Priapism p1]
- At 48 hours, thrombus forms in the sinusoidal spaces, with smooth muscle necrosis and transformation of smooth muscle into fibroblast-like cells. [Priapism p1]

Etiology

- Haematological causes besides sickle cell disease include G6PD deficiency, hereditary spherocytosis, paroxysmal nocturnal haemoglobinuria (PNH) and congenital dyserythropoietic anaemia. [Priapism p2]
- Thrombotic causes include asplenia, erythropoietin use and haemodialysis with heparin use. [Priapism p2]
- After intracavernosal injection, a prolonged erection is more common than true priapism. [Priapism p2]
- With intracavernosal PGE1 (Caverject, alprostadil), prolonged erection occurs in 5% and priapism in 1%. [Priapism p2]
- Priapism with oral PDE5 inhibitors is rare and usually occurs in men who have other risk factors. [Priapism p2]
- Intracavernosal Trimix (papaverine, phentolamine, alprostadil) causes priapism in 5–35%. [Priapism p2]
- Drugs for ADHD, atomoxetine and methylphenidate, can cause priapism. [Priapism p2]
- The spider implicated is Phoneutria nigriventer, whose neurotoxin causes paralysis, asphyxia and priapism. [Priapism p2]
- Priapism can be idiopathic. [Priapism p2]

Non-Ischemic Priapism (Arterial or High-Flow): Etiology

- Iatrogenic causes of high-flow priapism include optical internal urethrotomy (OIU), Nesbit corporoplasty and penile revascularisation procedures. [Priapism p3]

Diagnosis

- Abdominal and genital examination should look for malignancies as a cause of priapism. [Priapism p4]
- In ischaemic priapism the first aspirate is dark, like crankcase oil, and later samples become bright red. [Priapism p5]
- When a full or partial erection persists after treatment for ischaemic priapism, always consider colour Doppler; the differential is resolved ischaemia with penile oedema, persistent ischaemia, or conversion to high flow. [Priapism p5]
- Penile arteriography is too invasive for routine diagnosis, and blood gas data obtained after arteriography are confusing. [Priapism p6]
- MRI has three possible roles in priapism (from some reports), the first being imaging of a well-established arteriovenous fistula. [Priapism p6]
- The second possible role of MRI is imaging of corporal malignancy or metastases. [Priapism p6]
- The third possible role of MRI is showing the presence and extent of tissue thrombosis or corporal smooth muscle infarction. [Priapism p6]

Management of Ischemic Priapism

- Aspiration should be repeated until fresh bright red blood is obtained. [Priapism p7]
- The exact timing of early penile implantation after priapism is not known. [Priapism p7]
- A semirigid (malleable) implant is preferable when there is corporal fibrosis. [Priapism p7]

Sympathomimetic Drugs

- Metaraminol is a pure α-agonist. [Priapism p8]
- Etilefrine, phenylephrine and epinephrine are listed in the notes as mixed α- and β-agonists. [Priapism p8]
- Terbutaline is a pure β-agonist. [Priapism p8]
- α-agonists contract cavernous smooth muscle, so sinusoidal blood leaves through the subtunical veins. [Priapism p8]
- β-agonists relax cavernous smooth muscle and dilate the cavernous artery and arterioles, which improves oxygenation. [Priapism p8]
- Aspirate the penis between successive phenylephrine injections while tightly pinching the shaft at the penoscrotal junction, just below the needle insertion. [Priapism p8]
- Further side effects of phenylephrine are dizziness and tachycardia. [Priapism p8]

- Haemorrhagic stroke can occur with phenylephrine doses of 2 mg or more. [Priapism p8]

Shunts

- Shunting is also indicated when aspiration and sympathomimetic injection have caused significant cardiovascular problems. [Priapism p9]
- Early shunting (rather than prolonged sympathomimetic dosing) is advised in patients with poorly controlled hypertension and in those taking MAO inhibitors. [Priapism p9]
- The objective of a shunt is to re-oxygenate cavernous smooth muscle by relieving venous outflow obstruction and so re-establishing corporal inflow. [Priapism p9]
- The Ebbehoj shunt is a straight incision into the glans and corpus cavernosum using a No. 11 blade. [Priapism p9]
- The T-shunt uses a No. 10 blade inserted as for Ebbehoj, then rotated 90 degrees outward and withdrawn. [Priapism p9]
- After a percutaneous distal shunt, deoxygenated blood is milked out and the incisions are closed once bright red blood comes out. [Priapism p9]
- The Al-Ghorab shunt excises a 5 mm circular core of distal tunica albuginea, and the corporal snake is done after excising a 5 mm tunica core. [Priapism p9]
- Quackels/Sacher proximal shunts are bilateral, placed through a transscrotal or transperineal approach. [Priapism p10]
- Urethral stricture is a complication of shunt surgery. [Priapism p10]
- Shunt patency is assessed by bright red blood in the corporal aspirate, corporal blood gas, colour Doppler, a fall in intracavernous pressure, and a penile compression (squeeze and release) manoeuvre in which the penis should detumesce and refill. [Priapism p10]
- Shunt obstruction is prevented by avoiding compressive penile dressings and by periodic penile squeeze and release (in addition to anticoagulation). [Priapism p10]

Management of High Flow Priapism

- Spontaneous resolution with conservative care (ice to the perineum and site-specific compression) is due to spontaneous closure of the fistula. [Priapism p11]
- Active intervention (selective internal pudendal catheterisation, angiography and embolisation) is for patients who seek immediate relief. [Priapism p11]
- Expectant management is recommended especially in children. [Priapism p11]
- Side effects of angioembolisation include penile gangrene, gluteal ischaemia, purulent cavernositis and perineal abscess. [Priapism p12]
- The commonest notable side effect of bilateral embolisation is erectile dysfunction. [Priapism p12]
- After embolisation, erections are normally preserved in 75–85% of patients. [Priapism p12]
- Surgery is reserved for patients who do not want conservative management and are poor candidates for angioembolisation, who refuse embolisation, in whom embolisation is not available, or in whom it has failed. [Priapism p12]
- Surgical ligation uses a transcorporeal approach with intraoperative Doppler ultrasound. [Priapism p12]
- Surgery should be done weeks to months after trauma, which allows a pseudocapsule to form over the fistula. [Priapism p12]
- If surgery is attempted before the pseudocapsule forms, the cavernous artery may be ligated instead of selectively ligating the fistula. [Priapism p12]

Stuttering Priapism

- A past history of ischaemic priapism is a cause of stuttering priapism. [Priapism p13]
- Precipitants of full-blown sickle cell priapism are stuttering nocturnal or early-morning erections, dehydration, fever and exposure to cold. [Priapism p13]
- Patients typically wake with an erection that lasts up to 4 hours, becomes progressively painful from ischaemia, and finally remits. [Priapism p13]
- Hormonal prophylaxis also includes diethylstilbestrol (DES, by feedback inhibition) and androgen synthesis inhibitors such as ketoconazole. [Priapism p13]

- Baclofen inhibits penile erection and ejaculation through GABA receptor activity. [Priapism p14]
- Intrathecal baclofen pumps carry a risk of withdrawal syndrome if the pump fails. [Priapism p14]
- Long-term PDE5 inhibitor therapy, used apart from erection-stimulating conditions, reduces recurrent episodes without affecting normal erectile capacity. [Priapism p14]
- The effect of PDE5 inhibitor prophylaxis appears after a week or more of administration. [Priapism p14]
- Sildenafil is started at 25 mg once daily and increased to 50 mg once daily; tadalafil 5 or 10 mg is then given three times weekly. [Priapism p14]

Sickle Cell Disease & Priapism

- In sickle cell disease, haemolysis and reduced nitric oxide are responsible for pulmonary hypertension, priapism, leg ulcers and stroke. [Priapism p15]
- Increased blood viscosity in sickle cell disease is responsible for painful crises, osteonecrosis and acute chest syndrome. [Priapism p15]
- ASPEN syndrome is the association of sickle cell disease, priapism, exchange transfusion and neurological events. [Priapism p15]
- The sickle cell mutation substitutes valine for glutamic acid at position 6 of the β-globin chain of haemoglobin. [Priapism p15]
- Hydroxyurea is part of the haematological therapy of sickle cell disease and is used for sickle cell crisis. [Priapism p15]
- In stuttering priapism, parents and emergency staff should be educated. [Priapism p16]

Box 28-1 Causes of Priapism (printed box pasted in notes)

- The antianxiety agent hydroxyzine can cause priapism. [Priapism p16]
- Antidepressants and antipsychotics besides trazodone can cause priapism: bupropion, fluoxetine, sertraline, lithium, clozapine, risperidone, olanzapine, chlorpromazine, thioridazine and phenothiazines. [Priapism p16]
- Antihypertensives that can cause priapism are hydralazine, guanethidine and propranolol. [Priapism p16]
- Recreational causes include alcohol, crack cocaine and marijuana. [Priapism p16]
- Genitourinary causes include arteriovenous or arteriocavernous bypass surgery and urinary retention. [Priapism p16]
- Further haematological causes are thalassaemia, granulocytic, myeloid and lymphocytic leukaemia, multiple myeloma, the haemoglobin Olmsted variant, and fat emboli associated with hyperalimentation. [Priapism p16]
- Hormonal causes are gonadotropin-releasing hormone and testosterone. [Priapism p16]
- Infectious causes include rabies and malaria (with scorpion sting and spider bite). [Priapism p16]
- Metabolic causes are amyloidosis, Fabry disease and gout. [Priapism p16]
- Neoplastic infiltration can come from the urethra, testis, lung and kidney as well as prostate, bladder and rectum. [Priapism p16]
- Neurogenic causes are syphilis, spinal cord injury, cauda equina compression, autonomic neuropathy, lumbar disc herniation, spinal stenosis, cerebrovascular accident, brain tumour, spinal anaesthesia and cauda equina syndrome. [Priapism p16]

CH 74 · DISORDERS OF MALE ORGASM AND EJACULATION

Premature Ejaculation: Classification and Causes

- The prevalence of premature ejaculation is about 20%, but the exact figure is not known. [Ejaculation p2]
- Lifelong PE makes up about 25% and acquired PE about 75% of cases. [Ejaculation p2]
- In lifelong PE, ejaculation occurs within 30–60 seconds in the majority (80%) and between 1 and 2 minutes in the rest (20%). [Ejaculation p2]
- Acquired PE is often situational. [Ejaculation p2]
- Subjective PE is also called premature-like ejaculatory dysfunction. [Ejaculation p2]

Physiology of Ejaculation

- The supraspinal ejaculatory pathway runs from the sensory cortex to the hypothalamus (which also receives thalamic input) and the brainstem, then down to the lumbosacral spinal cord, with pudendal and sympathetic nerves to the penis; tactile penile input ascends as somatosensory input, and post-ejaculation afferent input returns to the thalamus. [Ejaculation p3]

Premature Ejaculation: Evaluation

- The medical history for PE should include the partner. [Ejaculation p4]
- Optional diagnostic questions for PE cover five areas: differentiating lifelong from acquired PE (when it was first experienced, and whether it has occurred since the first sexual experience on every or almost every attempt and with every partner); erectile function (is the erection hard enough to penetrate, difficulty maintaining it until ejaculation, rushing intercourse to avoid losing the erection); relationship impact (how upset the partner is, whether the partner avoids intercourse, whether PE affects the overall relationship); previous treatment; and quality of life (avoiding intercourse from embarrassment, feeling anxious, depressed or embarrassed). [Ejaculation p4]
- On the full IIEF, a normal erectile function domain score is 26 or more. [Ejaculation p5]

Premature Ejaculation: Pharmacotherapy

- 5-HT2C receptor stimulation inhibits ejaculation but facilitates erection. [Ejaculation p7]
- SSRIs block 5-HT reuptake by serotonin transporters, raising synaptic 5-HT; this leads to downregulation of 5-HT1A receptors, which increases the IELT. [Ejaculation p7]
- The main problem with topical lignocaine/prilocaine is penile hypoaesthesia. [Ejaculation p8]
- Intracavernosal alprostadil has been used for PE but has a very low level of evidence. [Ejaculation p8]
- Paroxetine gives the greatest increase in IELT, about 8.8-fold. [Ejaculation p8]
- The brand name of dapoxetine is Priligy. [Ejaculation p9]
- Dapoxetine increases IELT 2.5–3 times and also improves ejaculatory control, reduces distress and increases satisfaction. [Ejaculation p9]
- Side effects of dapoxetine are nausea, diarrhoea, headache, dizziness, dry mouth and insomnia. [Ejaculation p9]
- Dapoxetine has no suicidal tendencies, no withdrawal symptoms and negligible effect on sexual function. [Ejaculation p9]
- Dapoxetine is contraindicated in severe hepatic impairment, in men taking CYP3A4 inhibitors such as ketoconazole, and in heart failure or significant ischaemic heart disease. [Ejaculation p9]
- Adverse effects of other SSRIs and TCAs include nausea, diarrhoea, perspiration, yawning, agitation, weight gain and an increased risk of type 2 diabetes. [Ejaculation p9]

Premature Ejaculation: Behavioural Therapy and Surgery

- Pelvic floor exercises are a physical technique for PE, alongside stop–start and squeeze. [Ejaculation p10]
- Behavioural and psychological therapy for PE has a success rate of 45–60%. [Ejaculation p10]
- Surgery for PE is dorsal penile nerve division, which reduces penile sensation and increases IELT. [Ejaculation p10]

Delayed Ejaculation, Anejaculation and Anorgasmia

- Investigations for DE include urine routine examination, urine culture, urine cytology, PSA, ultrasound scanning and testosterone level. [Ejaculation p12]
- The recommended diagnostic questions for DE are: how often he can ejaculate during intercourse; how long after penetration he ejaculates or stops intercourse; how often he feels close to ejaculation when he cannot ejaculate; why he stops intercourse if he cannot ejaculate; whether he ever feels he has ejaculated but fails to release semen; whether he feels bothered, annoyed or frustrated by DE; and how often he can ejaculate during masturbation alone or with his partner. [Ejaculation p12]
- Optional questions for DE cover the same five areas as for PE: lifelong versus acquired, erectile function, relationship impact, previous treatment and quality of life. [Ejaculation p13]

- Psychoeducation or psychosexual therapy for DE should involve both patient and partner. [Ejaculation p13]
- Lifestyle changes for DE are enjoying more time together, minimising alcohol and making love when not tired. [Ejaculation p13]
- Men with DE should practise techniques that maximise penile stimulation, such as pelvic floor training. [Ejaculation p14]

Retrograde Ejaculation

- The RE rate after TUIP depends on whether one or both sides are incised and whether the incision is on the bladder neck only or extends to the verumontanum. [Ejaculation p16]
- Other causes of RE are idiopathic and congenital bladder neck abnormality. [Ejaculation p16]
- Before urinary sperm retrieval, urine can be diluted by oral fluid intake as an alternative to alkalinising it with oral sodium bicarbonate. [Ejaculation p17]
- Retrograde ejaculation causes less than 2% of cases of infertility. [Ejaculation p17]

CH 75 · SURGERY FOR ERECTILE DYSFUNCTION

Surgical Therapy: Penile Revascularization

- Further arterial revascularization options are inferior epigastric artery to deep dorsal vein combined with venous ligation (arterialization with venous reconstruction), or anastomosis directly to the corpus cavernosum. [Erection etc p22]
- Revascularization gives the best results in young men under 30 years with isolated arterial stenosis after trauma. [Erection etc p22]

Penile Prosthesis: Types

- A disadvantage of semirigid prostheses is that they do not increase penile length. [Erection etc p24]
- Malleable prosthesis examples are the AMS 600 and the Shah penile prosthesis with a hinge. [Erection etc p24]
- The positional (mechanical) prosthesis has a central metal cable with a series of polyethylene discs over it, which holds upward and downward positions better; the example given is Dura-II (written "Duva-II"). [Erection etc p24]
- Two-piece inflatable prostheses need no separate reservoir and inflate fully with few pump squeezes, but the small hand pump is hard to manipulate and a larger tunical incision is needed. [Erection etc p24]
- Coloplast Excel is a two-piece inflatable device (with AMS Ambicor). [Erection etc p24]

Penile Prosthesis: Indications and Contraindications

- Psychological impotence after failure of all other treatments is an indication for a penile prosthesis. [Erection etc p25]
- Situational ED and ED arising from relationship conflict are contraindications to a penile prosthesis. [Erection etc p25]
- Poor hygiene and skin cleanliness, and non-compliance with concurrent medications (for example for hypertension or diabetes), are contraindications to a penile prosthesis. [Erection etc p25]

Penile Prosthesis: Procedure

- A chlorhexidine soap wash is used for 3 days before prosthesis surgery. [Erection etc p26]
- The longitudinal corporotomy on the tunica albuginea is about 2 cm long. [Erection etc p26]
- Dilatation and length measurement can use the dilators supplied with the prosthesis (Dilamezinsert dilators). [Erection etc p26]
- For the reservoir in the space of Retzius, a defect is created between the crus of the penis medially and the spermatic cord laterally, with a 0.5 cm defect made in the floor of the inguinal canal. [Erection etc p26]
- Previous hernia surgery, abdominoperineal resection and a history of pelvic fracture with pelvic surgery are further reasons not to place the reservoir in the space of Retzius. [Erection etc p27]

- The flat submuscular reservoir is placed above the transversalis fascia. [Erection etc p27]
- After prosthesis surgery the patient avoids heavy lifting and sitting on the scrotum in the first week, has sutures removed at the 2-week review, and is followed up at 3 months, 6 months and then annually. [Erection etc p27]
- Sexual activity is avoided for 1 month after surgery and resumed after cycling, once the patient is comfortable with the device. [Erection etc p28]

Penile Prosthesis: Complications

- Damage to the device is an intra-operative complication. [Erection etc p28]
- Risk factors for prosthesis infection include spinal cord injury and immunosuppression (patient factors), prolonged surgical time over 2 hours and repeated device-skin contact (intra-operative factors), and prolonged hospitalisation (post-operative factor). [Erection etc p28]
- Organisms in prosthesis infection are skin organisms (Staphylococcus epidermidis, Staphylococcus aureus and Candida, which form biofilm) and others (Pseudomonas, E. coli and Enterococcus). [Erection etc p29]
- Precautions against infection include opening the device only at the last moment, using minimal instruments, controlling theatre traffic and using laminar flow ventilation. [Erection etc p29]
- The hydrophilic coating on the Coloplast Titan is polyvinylpyrrolidone, which absorbs and elutes any antibiotic solution. [Erection etc p29]
- Conservative management with antibiotics is reasonable when infection appears within 2 weeks, the patient is improving and the device itself is not infected. [Erection etc p30]
- An infected device, clinical deterioration or a rising total leucocyte count calls for aggressive salvage before systemic symptoms develop. [Erection etc p30]
- Immediate salvage replacement is contraindicated in Enterococcus infection, tissue necrosis, sepsis, diabetic ketoacidosis and cylinder erosion into the urethra. [Erection etc p30]
- After explantation, a spacer of antibiotic cast made of synthetic high-purity calcium sulphate can be placed in the corpora; it delivers antibiotic continuously for 30–60 days as it is absorbed, prevents fibrosis, and the device is then reimplanted. [Erection etc p30]
- Mechanical failure modes include cracks in the silicone tubing and leaks where the tubing connects to the pump. [Erection etc p31]
- When malfunction occurs within a few months of implantation, only the defective component is replaced. [Erection etc p31]
- After intra-operative corporal perforation, one option is to delay the prosthesis to 3 months instead of repairing and proceeding. [Erection etc p31]
- Poor glans support (supersonic transporter deformity) is caused by undersized cylinders; options are inserting longer cylinders or a coronal hitch or plication. [Erection etc p32]
- Scrotal haematoma is a complication of penile prosthesis surgery. [Erection etc p32]
- S-shaped penile deformity from overlong cylinders needs device replacement. [Erection etc p32]

Penile Implant After Radical Prostatectomy

- After radical prostatectomy, retropubic fibrosis puts the bladder and vena cava (as well as bowel) at risk during reservoir placement. [Erection etc p32]
- A two-piece device (no reservoir) is an option after radical prostatectomy. [Erection etc p32]
- If a separate incision is used for the reservoir (when a scrotal incision was used for the prosthesis), it is preferably on the right side to avoid the sigmoid colon. [Erection etc p32]

Penile Prosthesis: Outcome

- Satisfaction rates are higher with newer devices. [Erection etc p33]
- Satisfaction rates are lower in Peyronie disease with ED, after radical prostatectomy and with BMI over 30. [Erection etc p33]

Penile prosthesis with straightening manoeuvres

- An inflatable penile prosthesis gives superior correction of curvature with improved girth. [Peyronie Disease p11]

CH 76 · DIAGNOSIS AND MANAGEMENT OF PEYRONIE DISEASE

Treatment options for Peyronie disease: oral agents

- Reassurance alone is enough if the patient is not bothered by PD, because PD carries no risk of cancer. [Peyronie Disease p1]
- Potaba (potassium aminobenzoate) increases MAO activity and lowers serotonin levels, which increases the antifibrotic properties of tissues. [Peyronie Disease p1]
- In studies, Potaba decreased plaque size but gave no other benefit. Its adverse effect is acute hepatitis. [Peyronie Disease p1]
- Vitamin E acts as an antioxidant that limits oxidative stress, and it is the most frequently used oral agent. [Peyronie Disease p1]
- Tamoxifen raises the concentration of TGF-beta, which causes macrophage deactivation and T-lymphocyte suppression and so prevents fibrogenesis. Its adverse effect is thromboembolism. [Peyronie Disease p1]
- Colchicine causes microtubule depolymerisation, inhibits transcellular movement of collagen and stimulates collagenase production. Its adverse effect is GI upset. [Peyronie Disease p1]
- Carnitine increases mitochondrial respiration and decreases free radical formation. Its adverse effects are GI upset and seizures. [Peyronie Disease p1]
- Pentoxifylline blocks TGF-beta1-mediated inflammation, prevents deposition of type I collagen, decreases platelet-activating factor, and acts as a non-specific PDE5 inhibitor. [Peyronie Disease p1]
- In studies, pentoxifylline decreased curvature (by 23 degrees). Its adverse effects are GI upset, headache, flushing and dizziness. [Peyronie Disease p1]
- PDE5 inhibitors have antifibrotic effects and cause septal scar remodelling in PD, as well as treating ED. Their adverse effects are like those of pentoxifylline (hypotension). [Peyronie Disease p1]

Intralesional agents

- Verapamil inhibits the incorporation of proline into extracellular matrix protein. [Peyronie Disease p2]
- Intralesional verapamil reduces plaque-associated penile narrowing and improves the quality of erection. [Peyronie Disease p2]
- Adverse effects of intralesional verapamil are nausea, lightheadedness, penile pain and ecchymosis. [Peyronie Disease p2]
- Nicardipine is a dihydropyridine calcium channel blocker. It reduces glycosaminoglycan (GAG) biosynthesis and extracellular matrix production, and the notes rate it better than verapamil. [Peyronie Disease p2]
- With nicardipine the reduction in pain and plaque size leads to improved erections, and there are no severe side effects. [Peyronie Disease p2]
- Adverse effects of interferon alfa-2b include sinusitis, ecchymosis and penile swelling, as well as flu-like symptoms. [Peyronie Disease p2]

External force application

- Electromotive drug administration (EMDA) bypasses hepatic metabolism and raises the drug concentration in the target tissue. [Peyronie Disease p3]
- With EMDA, verapamil alone gave no benefit, but verapamil plus dexamethasone decreased plaque volume and penile curvature. The adverse effect is temporary erythema at the electrode site. [Peyronie Disease p3]
- ESWT works by direct damage to the plaque, which induces inflammation and increased vascularity and leads to lysis of the plaque. [Peyronie Disease p3]
- ESWT improves quality of life and erections. It can make curvature worse, and it can cause local ecchymosis. [Peyronie Disease p3]
- Penile traction works by mechanotransduction: it decreases alpha-smooth muscle actin and increases matrix metalloproteinases. [Peyronie Disease p3]
- Penile traction increases girth and shrinks the plaque. Overall satisfaction is 85%, and the adverse effect is erythema. [Peyronie Disease p3]

- The mechanism of vacuum therapy is unknown, but it is probably mechanical, like traction. It reduces curvature. [Peyronie Disease p3]
- Adverse effects of vacuum therapy include development of PD, ecchymosis, skin necrosis and urethral bleeding. [Peyronie Disease p3]
- Radiotherapy for PD acts through anti-inflammatory effects and gives no clinical benefit. Its risks are malignancy and ED in the elderly. [Peyronie Disease p3]

Topical drugs and combination therapy

- Topical drugs that have been tried are beta-aminopropionitrile, superoxide dismutase, verapamil and trifluoperazine. None is effective. [Peyronie Disease p4]
- The "three-armed approach" combines daily pentoxifylline plus L-arginine, verapamil injections twice weekly, and daily traction. It offers the best opportunity for deformity correction. [Peyronie Disease p4]

Surgery for PD: algorithm

- In the notes' algorithm, a man whose erection is inadequate may have a pre-prosthesis plication followed by a penile prosthesis. [Peyronie Disease p5]

Pre-operative consent and indications of surgery

- After PD surgery, decreased sexual sensation rarely compromises orgasm or ejaculation, and it resolves in 1-6 months. [Peyronie Disease p6]
- Another indication for surgery is the patient's desire for the most rapid and reliable result. [Peyronie Disease p6]
- Another indication for surgery is pain for the patient or partner during sexual intercourse. [Peyronie Disease p6]
- Another indication for surgery is patient anxiety. [Peyronie Disease p6]

Types of surgery: tunical shortening procedures

- In the Levine modification of Duckett-Baskin tunica albuginea plication (TAP), two partial-thickness transverse incisions are made 0.5-1 cm apart on the most convex part. They involve only the longitudinal fibres of the tunica albuginea. The longitudinal fibres between the two incisions are reduced to remove bulk, and the defect is sutured transversely. [Peyronie Disease p7]
- One problem with tunical shortening procedures is that they leave the plaque in situ. [Peyronie Disease p8]
- Tunical shortening can cause pain from the knots and suture granulomas. [Peyronie Disease p8]
- New ED after tunical shortening procedures occurs in 0-38%. [Peyronie Disease p8]
- Other complications of tunical shortening are haematoma, urethral injury and phimosis. [Peyronie Disease p8]
- Advantages of plication are a short surgical time, a good cosmetic outcome and minimal effect on rigidity. [Peyronie Disease p8]
- Plication gives effective straightening in 80-100% of patients and patient satisfaction of 65-100%. [Peyronie Disease p8]

Tunical lengthening procedures

- Shaft narrowing and extensive plaque calcification are among the features of complex disease that indicate plaque incision or partial excision with grafting. [Peyronie Disease p9]
- Plaque incision or partial excision with grafting gives satisfactory straightening in 75-100%. [Peyronie Disease p9]
- Tutoplast (processed human and bovine pericardium) and porcine SIS are the most commonly used grafts. [Peyronie Disease p9]
- The ideal graft for PD has roughly the strength and elasticity of normal tunica albuginea. It has minimal morbidity and tissue reaction, is readily available and inexpensive, is not too thick, is pliable, is easy to size and suture, resists infection, and preserves erectile capacity. [Peyronie Disease p10]
- After grafting surgery, the dressing is removed after 3 days. [Peyronie Disease p10]

- At 2 weeks after grafting, massage and stretch therapy is started (5 minutes twice daily for 2-4 weeks). [Peyronie Disease p10]
- After grafting, PDE5 inhibitors are given to improve vascularity and reduce ED. [Peyronie Disease p10]
- After grafting, penile traction is used for 3 or more hours daily for 3 months, starting 3-4 weeks after surgery. [Peyronie Disease p10]

Etiopathogenesis: definition and associated conditions

- PD can be defined as a wound-healing disorder of the tunica albuginea that forms an exuberant scar, presumably after an injury to the penis. [Peyronie Disease p12]
- Between 8% and 20% of diabetics have PD, and the risk is related to the duration of diabetes and poor glucose control. [Peyronie Disease p12]
- Between 38% and 58% of men with PD may have ED, partly because of associated comorbidities. [Peyronie Disease p12]
- PD is frequently associated with anxiety, diminished self-esteem, shame, loss of confidence and depression, because of penile shortening and inability to have sex. [Peyronie Disease p12]
- Radical prostatectomy is a possible risk factor for PD. Possible reasons are perioperative penile trauma, neurological consequences and local release of cytokines. [Peyronie Disease p12]
- Hypogonadism increases the severity of PD. [Peyronie Disease p13]
- PD is also associated with tympanosclerosis. [Peyronie Disease p13]
- The trigger can be repetitive microtrauma or a single event in a susceptible man, in the erect or even the flaccid state. [Peyronie Disease p13]
- In PD, microfractures (chronic) or delamination (acute) of the tunical layers (outer longitudinal and inner circular) create clot-filled spaces, which heal abnormally to form the plaque. [Peyronie Disease p13]
- Genetic predisposition, autoimmune factors and infection are probably responsible for PD. [Peyronie Disease p13]

Pathophysiology

- In the role of reactive oxygen species (ROS) and oxidative stress, injury with extravasation of blood leads to thrombus formation. Fibronectin and fibrin are deposited, and inflammatory cells accumulate. ROS production follows, which causes overexpression of fibrogenic cytokines and increased collagen synthesis. [Peyronie Disease p14]
- TGF-beta1 promotes ROS production, directly inhibits collagenase and promotes collagen synthesis. [Peyronie Disease p14]
- Myofibroblasts normally undergo apoptosis when tension or trauma in the tissue decreases. Persistent tension or trauma during rigid erections causes failure of apoptosis. Myofibroblasts, activated by TGF-beta1, then deposit excess extracellular matrix and collagen, which leads to inappropriate wound healing and fibrosis. [Peyronie Disease p14]
- TGF-beta1 is a strong activator of myofibroblasts and stimulates extracellular matrix deposition (it is a fibrogenic growth factor). [Peyronie Disease p15]
- TGF-beta1 decreases the production of matrix-degrading proteases and increases the production of their inhibitors. [Peyronie Disease p15]
- TGF-beta1 modulates integrin expression. [Peyronie Disease p15]
- TGF-beta1 upregulates myostatin, an inducer of fibrosis. [Peyronie Disease p15]
- Genes overexpressed in PD plaques are PTN (OSF-1, which recruits fibroblasts and osteoblasts), MCP-1 (inflammatory response and ossification), MMPs (especially MMP-2 and MMP-9, which break down collagen) and genes related to myofibroblast formation. [Peyronie Disease p15]
- Genes downregulated in PD plaques are Id-2 (an inhibitor of myofibroblast differentiation) and apoptotic genes. [Peyronie Disease p15]
- Nitric oxide has an antifibrotic role. Wound cells (monocytes, macrophages, fibroblasts) synthesise NO, which reacts with ROS and lowers ROS levels, so fibrosis decreases. [Peyronie Disease p16]

Evaluation of PD

- The history should include ejaculatory problems. [Peyronie Disease p17]

- The Peyronie's Disease Questionnaire has 15 questions in 3 domains, and a higher score means a greater negative impact. [Peyronie Disease p17]
- Examination includes the femoral pulses. [Peyronie Disease p17]
- Examination includes the hands and feet (for Dupuytren and Ledderhose disease). [Peyronie Disease p17]
- The penis should be examined in both the flaccid and the erect state if possible, including pain on palpation and nodules. [Peyronie Disease p17]
- Penile duplex ultrasonography also identifies corporal fibrosis. [Peyronie Disease p18]

OTHER TOPICS IN THIS BOOK

Retrograde Ejaculation

- The wide range of retrograde ejaculation after TUIP (5–45%) depends on technique: a unilateral (one-sided) versus bilateral (two-sided) incision, and an incision confined to the bladder neck versus one carried down to the verumontanum. [Retrograde Ejaculation p1]
- Drug therapy (alpha-adrenergic agents) is described as the most promising treatment approach for retrograde ejaculation. [Retrograde Ejaculation p1]

Sperm Cryopreservation (Sperm Banking)

- Cryopreservation is the procedure that makes it possible to stabilise and store sperm at cryogenic temperature. [Sperm Banking p1]
- A sperm bank is a place where semen is cryopreserved for later use in artificial insemination. [Sperm Banking p1]

Indications for sperm banking

- Autoimmune disorders and diabetes are indications for sperm banking. [Sperm Banking p1]
- Donor insemination programmes are an indication for sperm cryopreservation. [Sperm Banking p1]
- Sperm banking can be used to overcome stress at the time of semen collection. [Sperm Banking p1]
- Sperm can be banked before vasectomy, or during the procedure. [Sperm Banking p1]
- Men exposed to potentially toxic agents that interfere with gametogenesis are an indication for sperm banking. [Sperm Banking p1]

Cryoprotectants

- Cryoprotectants protect spermatozoa from freeze damage caused by ice crystallisation. [Sperm Banking p1]
- The cryoprotectants used are glycerol, ethylene glycol, dimethyl sulphoxide and 1,2-propanediol. [Sperm Banking p1]
- Glycerol is the most commonly used cryoprotectant. [Sperm Banking p1]

Freezing techniques

- Slow freezing can be manual or automated and takes a total of 2–4 hours. [Sperm Banking p2]
- In slow freezing, the temperature of the semen is lowered progressively while the cryoprotectant is added. [Sperm Banking p2]
- Slow freezing step 1: cool from room temperature to 5 °C at 0.5–1 °C per minute. [Sperm Banking p2]
- Slow freezing step 2: cool from 5 °C to −80 °C at 1–10 °C per minute. [Sperm Banking p2]
- Slow freezing step 3: store in liquid nitrogen at −196 °C. [Sperm Banking p2]
- Fast freezing step 1: mix the sample with an equal volume of cold cryoprotectant. [Sperm Banking p2]
- Fast freezing step 2: load the sample into straws and incubate at 4 °C for 10 minutes. [Sperm Banking p2]
- Fast freezing step 3: hold the straws 15 cm above liquid nitrogen, at −80 °C, for 15 minutes. [Sperm Banking p2]
- Fast freezing step 4: immerse the straws in liquid nitrogen at −196 °C. [Sperm Banking p2]

Effect of cryopreservation on sperm integrity

- Cryopreservation damages sperm by thermal shock. [Sperm Banking p2]
- Ice-crystal formation impairs the function of cell organelles and reduces cell survival. [Sperm Banking p2]

- Osmotic shock is a mechanism of cryo-injury to sperm. [Sperm Banking p2]
- Cellular dehydration causes toxic damage because of the high solute content. [Sperm Banking p3]
- Plasma and mitochondrial membranes are susceptible; their damage activates caspases 3, 8 and 9, which reduces ATP generation and so reduces sperm motility. [Sperm Banking p3]
- Sperm DNA damage after freezing results from oxidative stress, thermal denaturation and defects in DNA repair enzymes. [Sperm Banking p3]

Thawing procedure

- Thawing is done gradually so that cells recover normal biological activity and abrupt thermal changes are avoided. [Sperm Banking p3]
- The routinely used thawing temperature is 37 °C. [Sperm Banking p3]
- Thawing technique: 10 minutes at room temperature, then 10 minutes in a thermostat at 37 °C, then 15 minutes at room temperature, then washing and centrifugation before the sperm is used. [Sperm Banking p3]

Outcome of cryopreserved sperm

- Sperm can be stored for up to 28 years. [Sperm Banking p4]
- In the long term, offspring from cryopreserved sperm show no increase in birth defects or chromosomal anomalies over the general population. [Sperm Banking p4]
- Frozen sperm give better results with ICSI than with IVF or IUI, because ICSI needs only a small number of motile spermatozoa. [Sperm Banking p4]
- Pregnancy rates with cryopreserved sperm vary with the source (testicular versus epididymal sperm) and the method of ART used. [Sperm Banking p4]

Testicular Biopsy: indications

- A negative antisperm antibody assay is listed as one of the conditions for diagnostic biopsy, along with normal testes, palpable vasa and normal FSH. [Testicular Biopsy p1]
- Biopsy is done on both sides whatever the size difference, because a small firm testis can sometimes have good spermatogenesis and a large healthy testis may show maturation arrest. [Testicular Biopsy p1]
- Fresh unfixed tissue is examined for sperm with tails, and several samples are examined if no sperm are found at first. [Testicular Biopsy p1]

Open (microsurgical) technique

- The wound should be dry before the tunica albuginea is incised, so that blood does not contaminate the biopsy sample. [Testicular Biopsy p2]
- A touch prep is made by blotting the cut surface of the testis several times on a glass slide with a drop of saline, then examining it under high power on a light microscope. [Testicular Biopsy p2]
- A wet squash prep is made by placing a second specimen on a slide with a drop of saline and crushing it under a coverslip. [Testicular Biopsy p2]
- If the squash prep shows no sperm, another area is biopsied through the same skin incision. [Testicular Biopsy p2]

Percutaneous biopsy, aspiration, micro-biopsy and complications

- Micro-biopsy uses a 26-gauge needle to take a single seminiferous tubule (SST). [Testicular Biopsy p3]
- An inadvertent biopsy of the epididymis does no harm if that tubule contains no sperm. [Testicular Biopsy p3]

Interpretation

- In a normal biopsy, several stages of spermatogenesis appear within a single tubular section (the patchwork pattern). [Testicular Biopsy p4]
- Azoospermia with a normal biopsy and dilated tubules indicates obstruction. [Testicular Biopsy p4]
- In maturation arrest the testicular architecture is normal. [Testicular Biopsy p4]

Germ cell aplasia and end-stage testis

- In end-stage testis, both germ cells and Sertoli cells are absent. [Testicular Biopsy p5]
- End-stage testis shows a thickened basement membrane with tubular and peritubular sclerosis. [Testicular Biopsy p5]

Unconsummated Marriages

- To consummate a marriage means to complete the married relationship by having sexual intercourse; the corpus has no section on unconsummated marriage. [Unconsummated M p1]
- Inability to consummate a marriage is common; many sexologists report that about every third patient they see has this problem. [Unconsummated M p1]
- The effects of an unconsummated marriage are severe stress and frustration, hostility, even divorce, and ruined lives. [Unconsummated M p1]
- Aetiology 1, ignorance: many young couples, despite being educated, are ignorant about sexual practices, and false expectations or incorrect technique lead to failure. [Unconsummated M p1]
- Aetiology 2, fear of pain: many women have a preconditioned notion that sex is very painful, which leads to severe vaginismus that makes intercourse impossible. [Unconsummated M p1]
- Aetiology 3, performance anxiety leading to erectile failure, may stem from myths and misconceptions such as masturbation guilt and Dhat syndrome. [Unconsummated M p2]
- Dhat syndrome is a culture-bound condition in which a man complains of premature ejaculation or impotence and believes he passes semen in his urine, with no organic cause. [Unconsummated M p2]
- Aetiology 4, personality disorders: phobias, fear and depression affect the couple's sexual life. [Unconsummated M p2]
- Aetiology 5, physical defects: local inflammatory pathology and, rarely, a tough hymen in women; severe penile curvature or veno-occlusive disease causing inadequate erections in men. [Unconsummated M p2]
- Aetiology 6, social factors: family pressures, lack of privacy and job stress can generate so much stress that sexual intimacy is not possible. [Unconsummated M p2]
- Unconsummated marriage should be treated as a semi-emergency, because delay builds frustration and hostility that make treatment and understanding difficult. [Unconsummated M p2]
- Treatment 1 is basic sex counselling to correct misconceptions, allay fears and teach the right technique. [Unconsummated M p3]
- Treatment 2 is social counselling to correct excessive interference by in-laws, create privacy, minimise stress and improve communication between wife and husband. [Unconsummated M p3]
- Treatment 3 is vaginal self-dilatation when fear of pain is the main problem. [Unconsummated M p3]
- Treatment 4 is surgical hymenectomy when there is a tough hymen. [Unconsummated M p3]
- Treatment 5 is temporary medical support (sildenafil or intrapenile injections) for men with performance anxiety; after a few successful intercourses they gain confidence and usually no longer need medication. [Unconsummated M p3]
- Treatment 6 is correction of any physical defect, such as severe chordee, circumcision for phimosis, penile revascularisation or a penile prosthesis. [Unconsummated M p3]
- Cases with deep-rooted psychopathology are difficult to treat and are the ones that most often fail. [Unconsummated M p3]

10 Testis, Penis & Urethra

CH 79 · NEOPLASMS OF THE TESTIS

Testicular Tumors: Risk Factors

- In unilateral cryptorchidism, the contralateral normally descended testis is also at slightly increased risk of cancer (1.5-2 times). [Ca testis part 1 p1]

- A family history of testicular cancer carries a 10-fold increased risk, and affected relatives present 2-3 years earlier. [Ca testis part 1 p2]
- Incidence is lowest in Africa and Asia. [Ca testis part 1 p2]
- Geographically, incidence is highest in Scandinavia and Germany. [Ca testis part 1 p2]
- Environmental risk factors include organochlorines and polyvinyl chloride. [Ca testis part 1 p2]
- Infertility carries a 3-fold increased risk of testicular cancer. [Ca testis part 1 p2]
- Prenatal exposure to hormone therapy (diethylstilbestrol) or radiation is a risk factor for testicular tumour. [Ca testis part 1 p2]

Age Groups and Epidemiology

- Testicular tumours have three age peaks: infancy (yolk sac tumour), 30-34 years (seminoma) and about 60 years (malignant lymphoma). [Ca testis part 1 p2]
- In men aged 15-19 years, testicular cancer is the second commonest malignancy after leukaemia. [Ca testis part 1 p2]
- The annual incidence is 5.4 cases per 100,000 (1 lakh) men. [Ca testis part 1 p2]
- Testicular tumours are more common on the right side than the left. [Ca testis part 1 p2]

Genetics and Associated Syndromes

- Overexpression of the CCND2 gene (cyclin D2) is seen in most germ cell tumours. [Ca testis part 1 p3]
- Down syndrome, persistent Mullerian duct syndrome and cutaneous ichthyosis are listed as syndromes associated with testicular germ cell tumour. [Ca testis part 1 p3]
- Germ cell transformation runs from primordial germ cell to gonocyte (fetal), to ITGCN (before puberty), and after puberty to either seminoma or embryonal carcinoma; embryonal carcinoma can then give rise to choriocarcinoma, yolk sac tumour and teratoma. [Ca testis part 1 p3]
- Seminoma is characterised by DNA hypomethylation and hyperdiploidy (hypertriploidy?), whereas the embryonal carcinoma lineage (choriocarcinoma, yolk sac tumour, teratoma) shows extraembryonic differentiation and hypermethylation. [Ca testis part 1 p3]

ITGCN (Intratubular Germ Cell Neoplasia)

- ITGCN is also called carcinoma in situ (CIS) of the testis. [Ca testis part 1 p4]
- The incidence of ITGCN is 0.8%. [Ca testis part 1 p4]
- In the fetal gonocyte model, primordial gonocytes persist abnormally beyond the neonatal period without differentiating into spermatozoa, and invasive growth is then driven by postnatal and pubertal gonadotropin stimulation. [Ca testis part 1 p4]
- In the second model, zygotene-pachytene spermatocytes transform to a precursor germ cell phenotype. [Ca testis part 1 p4]
- Overexpression of CCND2 and an extra copy of 12p activate CDK4 and CDK6, which protect against apoptosis and lead to genomic instability. [Ca testis part 1 p4]
- ITGCN is diagnosed on testicular biopsy done for infertility, on contralateral biopsy in GCT patients, or in the affected testis during testis-sparing surgery. [Ca testis part 1 p5]
- Treatment of ITGCN also depends on the patient's wish to avoid testosterone replacement therapy. [Ca testis part 1 p5]
- Low-dose radiotherapy for ITGCN gives local control rates similar to radical orchidectomy, and radical orchidectomy is the most definitive treatment. [Ca testis part 1 p6]
- Leydig cell function is relatively preserved after radiotherapy because Leydig cells are radioresistant compared with germ cells. [Ca testis part 1 p6]
- ITGCN in an orchidectomy specimen of a man with testis cancer has no prognostic implication for the risk of relapse. [Ca testis part 1 p6]

Pathology: Extragonadal GCT

- Retroperitoneal GCTs behave like testicular GCTs in biology and prognosis. [Ca testis part 1 p7]
- Mediastinal seminoma has the same prognosis as testicular seminoma. [Ca testis part 1 p7]

- Mediastinal NSGCT is less chemosensitive than testicular or retroperitoneal NSGCT, with a 5-year overall survival of 45%. [Ca testis part 1 p7]
- Mediastinal NSGCT is more likely to contain a yolk sac tumour component, so AFP is raised. [Ca testis part 1 p7]
- Mediastinal NSGCT is associated with haematological malignancies that carry extra 12p. [Ca testis part 1 p7]
- Mediastinal mature teratomas have low malignant potential and are generally cured surgically. [Ca testis part 1 p7]

Seminoma

- Grossly, seminoma is a soft, tan to white, diffuse or multinodular mass, with or without focal necrosis; necrosis is more prominent in other GCTs. [Ca testis part 1 p8]
- Seminoma cells have polygonal nuclei. [Ca testis part 1 p8]
- Immunohistochemistry has a limited role in diagnosing GCTs. [Ca testis part 1 p8]
- Syncytiotrophoblastic cells in seminoma have no clear prognostic significance. [Ca testis part 1 p8]

Spermatocytic Seminoma

- Spermatocytic seminoma is not bilateral. [Ca testis part 1 p9]
- Spermatocytic seminoma does not stain for glycogen. [Ca testis part 1 p9]
- Its three cell types are small lymphocyte-like cells, medium-sized cells with eosinophilic cytoplasm and round nuclei, and large mononucleated or multinucleated cells. [Ca testis part 1 p9]

NSGCT: Embryonal Carcinoma

- Embryonal carcinoma is the most undifferentiated GCT type. [Ca testis part 1 p10]
- Grossly, embryonal carcinoma is tan to yellow with haemorrhage and necrosis. [Ca testis part 1 p10]
- Embryonal carcinoma has variable microscopy: solid sheets, papillary, tubular or glandular-alveolar patterns. [Ca testis part 1 p10]
- Embryonal carcinoma is PLAP positive on immunohistochemistry. [Ca testis part 1 p10]
- Embryonal carcinoma causes a modest rise in hCG when syncytiotrophoblastic cells are present. [Ca testis part 1 p10]
- LDH concentration is a prognostic factor in metastatic embryonal carcinoma. [Ca testis part 1 p10]

Yolk Sac Tumour and Teratoma

- Yolk sac tumour is also called endodermal sinus tumour, orchidoblastoma, and juvenile or infantile embryonal carcinoma. [Ca testis part 1 p11]
- Yolk sac tumour is the commonest infantile testicular tumour and the second commonest prepubertal testicular tumour after teratoma. [Ca testis part 1 p11]
- Schiller-Duval bodies represent endodermal sinuses. [Ca testis part 1 p11]
- Cytoplasmic and extracellular eosinophilic hyaline globules are seen in 84% of yolk sac tumours. [Ca testis part 1 p11]
- Immature teratoma is incompletely differentiated, while mature teratoma is well differentiated and may contain hair, teeth, bone and cartilage. [Ca testis part 1 p11]

Clinical Features

- A localised testicular tumour may cause heaviness or a pressure-like sensation. [Ca testis part 1 p13]
- Acute pain from haemorrhage, or from rapid expansion with infarction, is much more common in NSGCT than in seminoma. [Ca testis part 1 p13]
- hCG-secreting tumours can cause hyperprolactinaemia. [Ca testis part 1 p13]
- Bulky retroperitoneal metastases can present as a palpable mass, abdominal pain, ureteric obstruction, lower-limb oedema or GI symptoms (from retroduodenal metastasis); back pain comes from psoas or nerve-root involvement. [Ca testis part 1 p13]
- Pulmonary metastases cause cough, dyspnoea and haemoptysis; bone metastases cause skeletal pain; brain metastases cause neurological symptoms (choriocarcinoma); hepatic metastases cause jaundice, ascites and nausea. [Ca testis part 1 p14]

Diagnosis: Scrotal Ultrasound

- Scrotal ultrasound for testicular tumour uses a 5-10 MHz probe. [Ca testis part 1 p15]
- On ultrasound, NSGCT shows haemorrhage, calcification and necrosis as well as heterogeneous echotexture. [Ca testis part 1 p15]
- Spontaneous regression (burned-out primary) is common in GCTs, most commonly in seminoma. [Ca testis part 1 p15]

Tumour Markers

- AFP is raised in 100% of yolk sac tumours. [Ca testis part 1 p16]
- Non-GCT malignancies that raise AFP are hepatocellular carcinoma and cancers of the biliary tract, pancreas, lung and stomach. [Ca testis part 1 p16]
- Alcohol abuse and ataxia telangiectasia are non-malignant causes of raised AFP. [Ca testis part 1 p16]
- Normal serum AFP is below 10 ng/mL. [Ca testis part 1 p16]
- The beta subunit is unique to hCG, which is normally produced by the placenta. [Ca testis part 1 p17]
- hCG is raised in 40-60% of embryonal carcinomas. [Ca testis part 1 p17]
- Non-GCT cancers that raise hCG include gastric, hepatic, pancreatic, lung, breast, biliary, kidney and bladder cancers. [Ca testis part 1 p17]
- Pregnancy raises hCG. [Ca testis part 1 p17]
- LDH has five isoenzymes, and LDH-1 is the one mostly raised in GCT. [Ca testis part 1 p17]
- LDH is non-specific because it also comes from cardiac, smooth and skeletal muscle. [Ca testis part 1 p17]
- LDH is not a good indicator of recurrence. [Ca testis part 1 p17]
- Serum PLAP (placental alkaline phosphatase) is raised in advanced seminoma. [Ca testis part 1 p18]
- Gamma-glutamyl transpeptidase (GGT) is raised in 30% of seminomas. [Ca testis part 1 p18]

Suspected Extragonadal GCT

- Extragonadal GCTs arise more often in the mediastinum than in the retroperitoneum, and seminoma is the commonest type. [Ca testis part 1 p20]
- An extragonadal mass with raised AFP and/or hCG and a normal testis examination is enough to start chemotherapy. [Ca testis part 1 p20]
- A suspected extragonadal GCT with normal tumour markers and a normal testis examination needs a biopsy of the mass to confirm the diagnosis. [Ca testis part 1 p21]

Staging (AJCC 8th edition)

- In AJCC 8th edition, pT1 pure seminoma is subdivided into pT1a (below 3 cm) and pT1b (3 cm or more). [Ca testis part 1 p22]
- Stage 0 is pTis, N0, M0, S0. [Ca testis part 1 p23]

Staging Imaging

- Staging CT is done with oral and IV contrast. [Ca testis part 1 p24]
- A retroperitoneal node size cut-off above 10 mm is commonly used on CT, though size criteria are not clearly defined. [Ca testis part 1 p24]
- Nodes of 5-9 mm in the primary landing zone are especially suspicious when they lie anterior to the great vessels. [Ca testis part 1 p24]
- MRI is an alternative to CT for staging but takes longer and costs more. [Ca testis part 1 p24]
- In some European countries, laparoscopic RPLND is done in CS I and IIA NSGCT as a staging procedure without curative intent, to identify patients who need chemotherapy. [Ca testis part 1 p24]
- All patients should have a routine chest X-ray before management decisions are made. [Ca testis part 1 p25]
- Mediastinal or hilar lymphadenopathy without retroperitoneal disease suggests a non-GCT cause such as lymphoma or sarcoidosis, and needs a biopsy. [Ca testis part 1 p25]

Sperm Cryopreservation

- At diagnosis, 52% of GCT patients are oligospermic and 10% are azoospermic. [Ca testis part 1 p26]

- Recovery of spermatogenesis after radiotherapy takes 2-3 years longer than after chemotherapy. [Ca testis part 1 p26]

Treatment of Seminoma: Stage I

- Carboplatin causes less ototoxicity, neurotoxicity and nephrotoxicity than cisplatin. [Testis CA Rx p2]
- After adjuvant carboplatin for stage I seminoma, follow-up still needs CT scans. [Testis CA Rx p2]
- After prophylactic radiotherapy for stage I seminoma, in-field recurrence is below 1%, so routine follow-up CT of the treated field is not required. [Testis CA Rx p2]

Seminoma Stage II

- For stage IIA seminoma, the 30 Gy dog-leg dose is given as 15 fractions of 2 Gy to the para-aortic and ipsilateral iliac nodes. [Testis CA Rx p3]
- For stage IIB seminoma, the boost to the enlarged nodes is 2 Gy x 3 = 6 Gy on top of 30 Gy to the para-aortic and ipsilateral iliac field. [Testis CA Rx p3]
- Disease-free survival for stage II seminoma treated with radiotherapy or chemotherapy is about 90%. [Testis CA Rx p3]

Advanced Seminoma (Stage IIC and III) and Residual Mass after Chemotherapy

- In advanced seminoma, induction chemotherapy gives a response rate of 70-90%, with relapse in 15-20%. [Testis CA Rx p4]
- 4 cycles of VIP is an alternative to 4 cycles of BEP for intermediate-prognosis seminoma. [Testis CA Rx p4]
- Radiotherapy has no role for a residual seminoma mass after chemotherapy. [Testis CA Rx p4]
- After first-line chemotherapy, 60-80% of seminoma patients have a radiologically detectable residual lesion, and 50-60% of these resolve spontaneously within 13-18 months. [Testis CA Rx p4]
- A residual seminoma mass smaller than 3 cm harbours malignancy in 0-4%, and one larger than 3 cm in 27-38%. [Testis CA Rx p4]
- If post-chemotherapy surgery for a PET-positive seminoma residual shows viable malignancy, give 2 further cycles of chemotherapy; if it shows necrosis, follow up. [Testis CA Rx p4]

Relapsed Seminoma

- A patient who relapses after single-agent carboplatin is considered chemotherapy-naive and should receive first-line cisplatin-based chemotherapy. [Testis CA Rx p5]
- Early relapse of seminoma after chemotherapy carries a poor prognosis, with long-term survival of 30-50%. [Testis CA Rx p5]
- In early relapse after chemotherapy, biopsy the relapse site before second-line VeIP, because it may contain teratoma. [Testis CA Rx p5]
- Pure seminoma accounts for less than 80% (as written; possibly a different figure) of late relapses after chemotherapy. [Testis CA Rx p5]

Treatment of NSGCT: Clinical Stage I

- Low-risk CS I NSGCT (no EC predominance and no LVI) may have adjuvant BEP (1 or 2 cycles) if the patient is unwilling to undergo surveillance, or nerve-sparing RPLND if he is unwilling to have either. [Testis CA Rx p6]
- High-risk CS I NSGCT patients unwilling to have chemotherapy can choose nerve-sparing RPLND or surveillance. [Testis CA Rx p6]
- Most relapses of CS I NSGCT present with a lymph-node mass larger than 3 cm, distant metastases or raised markers, and are treated with 3 or 4 cycles of BEP or VIP followed by resection of any residual tumour. [Testis CA Rx p6]

NSGCT Clinical Stage IIA

- Marker-positive (S1) CS IIA NSGCT gets 3 cycles of BEP, followed by resection of any residual tumour. [Testis CA Rx p7]
- When marker-negative CS IIA NSGCT is followed up and reimaged at 6 weeks, progressive disease with positive markers gets 3 cycles of BEP with or without residual tumour resection, and

progressive disease with negative markers gets nerve-sparing RPLND or chemotherapy. [Testis CA Rx p7]

- On the 6-week reimaging of marker-negative CS IIA NSGCT, an unchanged mass gets nerve-sparing RPLND, and a regressing mass gets follow-up. [Testis CA Rx p7]
- With chemotherapy or RPLND, survival in CS IIA/IIB NSGCT exceeds 90%. [Testis CA Rx p7]

NSGCT Clinical Stage IIB and Chemotherapy Regimens

- After primary chemotherapy for CS IIB NSGCT, persistently elevated tumour markers call for second-line chemotherapy. [Testis CA Rx p8]
- VIP stands for etoposide (VP-16), ifosfamide and cisplatin. [Testis CA Rx p8]

NSGCT Stage IIC/III and Response to Chemotherapy

- After induction chemotherapy, elevated markers (partial remission) or progressive disease are treated with second-line chemotherapy. [Testis CA Rx p9]
- Early relapse of NSGCT within 2 years can also be treated with high-dose chemotherapy with bone marrow (stem cell) transplant. [Testis CA Rx p9]
- Late relapse of NSGCT after 2 years carries poor survival; the resected tissue may show viable tumour, teratoma or adenocarcinoma transformation. [Testis CA Rx p9]

Desperation Surgery and Choriocarcinoma Syndrome

- Progressive disease after first- and second-line chemotherapy has a dismal prognosis, and the evidence for desperation surgery comes only from limited literature and small studies. [Testis CA Rx p10]
- The single resectable site in desperation surgery is usually the retroperitoneum. [Testis CA Rx p10]
- Choriocarcinoma syndrome is very high beta-hCG (above 50,000 IU/L) with a mediastinal, abdominal or testicular mass, gynaecomastia and hyperthyroidism, and it can cause tumour haemorrhage. [Testis CA Rx p10]
- Choriocarcinoma syndrome is a medical emergency that needs immediate chemotherapy. [Testis CA Rx p10]

Choriocarcinoma Syndrome (continued) and Brain Metastasis

- In choriocarcinoma syndrome, the first course is EP for 3 days with beta-blockers, followed by the BEP or VIP regimen, and radical orchidectomy is done between cycles. [Testis CA Rx p11]
- Brain metastases from GCT are highly vascular and tend to bleed during chemotherapy, with a 4-10% death rate from intracranial bleeding. [Testis CA Rx p11]
- Five-year survival with brain metastases is 33% in disseminated NSGCT and 57% in seminoma. [Testis CA Rx p11]
- Treatment options for brain metastases are chemotherapy, surgical resection, whole-brain radiation, stereotactic radiosurgery and multimodality treatment, and most patients receive multimodality treatment. [Testis CA Rx p11]
- Relapse in the brain after first-line chemotherapy gets second-line chemotherapy followed by radiotherapy or surgery. [Testis CA Rx p11]

CH 82 · TUMOURS OF THE PENIS

Carcinoma in Situ (Erythroplasia of Queyrat, Bowen Disease)

- Erythroplasia of Queyrat is well marginated and may ulcerate or discharge. [Carcinoma Penis p1]
- Bowen disease has the same histology as erythroplasia of Queyrat, forms well-marginated plaques of scaly erythema and can ulcerate. [Carcinoma Penis p1]
- The differential diagnosis of Bowen disease includes bowenoid papulosis and superficial basal cell carcinoma, as well as nummular eczema and psoriasis. [Carcinoma Penis p1]
- Metastasis from penile carcinoma in situ is extremely rare. [Carcinoma Penis p1]
- Penile carcinoma in situ has no association with internal malignant tumours. [Carcinoma Penis p1]
- Treatment of penile CIS starts only after histopathological confirmation. [Carcinoma Penis p1]
- KTP laser is an alternative to Nd:YAG laser for penile CIS. [Carcinoma Penis p1]

- Radiotherapy is used for penile CIS that resists topical therapy in men who are not surgical candidates. [Carcinoma Penis p1]

Aetiology

- The incidence of penile cancer is decreasing. [Carcinoma Penis p2]
- Penile cancer peaks in the 6th decade, although it is also reported in younger patients. [Carcinoma Penis p2]
- African American patients have a higher risk of penile cancer and poorer survival. [Carcinoma Penis p2]
- A higher number of lifetime sexual partners is a risk factor for penile cancer. [Carcinoma Penis p2]

Evaluation of the Primary Lesion

- Penile cancer usually presents with a mass or discharge; bleeding, pain and urinary retention are rare, and systemic symptoms may occur. [Carcinoma Penis p3]
- DRE and bimanual examination assess perineal body involvement and any pelvic mass. [Carcinoma Penis p3]
- Primary sites are glans 48%, prepuce 20%, glans and prepuce 10%, coronal sulcus 6% and shaft less than 2%. [Carcinoma Penis p3]
- The growth patterns are superficial spreading, flat, and ulcerated with vertical growth. [Carcinoma Penis p3]
- Biopsy may be followed by definitive surgery later, or frozen section may be followed by surgery at the same sitting. [Carcinoma Penis p3]
- Typical penile SCC shows keratinisation, epithelial pearl formation and varying mitotic activity. [Carcinoma Penis p3]

Histology and Grading

- Broders classification grades penile SCC as well differentiated (grade I), moderately differentiated (II), poorly differentiated (III) and anaplastic (IV). [Carcinoma Penis p4]
- Basaloid carcinoma is HPV-related in about 80%. [Carcinoma Penis p4]

Laboratory Evaluation

- Laboratory findings in penile cancer include anaemia, leukocytosis and hypoalbuminaemia. [Carcinoma Penis p4]
- Hypercalcaemia in penile cancer is treated with aggressive saline hydration to promote sodium and calcium excretion, diuretics if there is volume overload, bisphosphonates and calcitonin. [Carcinoma Penis p4]

Radiographic Studies

- Ultrasound of the primary uses a 7.5 MHz linear probe and shows a hypoechoic lesion, but it cannot reliably tell subepithelial glans invasion from corpus spongiosum involvement. [Carcinoma Penis p5]
- Artificial erection for MRI of the primary is induced with PGE1. [Carcinoma Penis p5]

Differential Diagnosis

- The differential diagnosis of penile carcinoma includes condyloma acuminatum, Buschke–Löwenstein tumour, lichen sclerosus (BXO) and infectious lesions: chancre, chancroid, herpes, granuloma inguinale and tuberculosis. [Carcinoma Penis p5]

Importance of Lymph Nodes

- A palpable inguinal node that appears during follow-up after treatment of the primary is metastatic in essentially 100% of cases. [Carcinoma Penis p8]

Investigations for Metastasis

- Metastatic work-up is done only when nodes are metastatic, using CT chest and abdomen or chest X-ray, liver function tests and bone scan, or PET-CT. [Carcinoma Penis p11]
- Stage 0 penile cancer is Tis or Ta. [Carcinoma Penis p11]

Organ Preservation and Penile Preservation

- Mohs surgery is listed among organ-preserving options for the primary, but it is not commonly done. [Carcinoma Penis p12]
- The goals of penile preservation are to keep glans sensation where possible and to maximise penile shaft length. [Carcinoma Penis p12]
- Local recurrence after limited excision with frozen section is under 10%. [Carcinoma Penis p12]
- Recurrence after glans resurfacing is about 20% after total stripping and about 50% after partial stripping. [Carcinoma Penis p12]
- Glans resurfacing preserves sensation. [Carcinoma Penis p12]
- Argon laser is another laser option for the penile primary, along with CO2, Nd:YAG and KTP lasers. [Carcinoma Penis p12]
- About 75% of men resume sexual activity, with satisfaction, after penile preservation. [Carcinoma Penis p12]
- In selected cases, recurrence after laser treatment can be managed by repeat ablation. [Carcinoma Penis p12]

Partial Amputation

- Partial penectomy is done when excision with a clear margin leaves at least 4 cm of unstretched penile stump, which allows standing micturition and possibly vaginal penetration. [Carcinoma Penis p15]

Radiotherapy as Organ Preservation

- Circumcision before radiotherapy also allows any infection to resolve. [Carcinoma Penis p16]
- Penile preservation at 5 years after EBRT is about 50%. [Carcinoma Penis p16]
- EBRT is widely available and easy to give, but needs a 45-day course; brachytherapy needs expertise and is not widely available. [Carcinoma Penis p16]
- For radiotherapy the penis is positioned either with the patient supine and a wax or Perspex block around the penis, or with the patient prone and the penis suspended in a small water-bath container. [Carcinoma Penis p16]
- Problems after penile radiotherapy include moist desquamation, urethral fistula, urethral stricture, penile necrosis, pain and oedema. [Carcinoma Penis p16]

Metastatic Disease

- Cisplatin plus 5-FU is an alternative combination to TIP for metastatic penile cancer. [Carcinoma Penis p20]
- Bleomycin regimens are avoided because of lung toxicity, which can be fatal. [Carcinoma Penis p20]
- In metastatic disease that responds to chemotherapy, surgery for resectable disease improves survival, to about 32 months. [Carcinoma Penis p20]

Non-Squamous Malignancies

- Penile BCC grows slowly and is diagnosed by biopsy; local excision is curative. [Carcinoma Penis p21]
- Penile melanoma presents as a blue-black or reddish-brown pigmented papule or ulcer on the glans. [Carcinoma Penis p21]
- Biopsy of penile melanoma shows atypical junctional cell activity with pigmented cells displaced into the dermis. [Carcinoma Penis p21]
- Treatment of penile melanoma depends on site: circumcision for the prepuce, glansectomy or partial penectomy for the glans, and partial or total penectomy for the shaft. [Carcinoma Penis p21]
- Benign penile sarcomas occur more often distally. [Carcinoma Penis p21]
- After vascular tumours, the commonest malignant penile sarcomas are neural, then myogenic, then fibrous. [Carcinoma Penis p21]
- Penile sarcoma presents with a mass, pain, penile enlargement, priapism or urinary retention. [Carcinoma Penis p21]
- Extramammary Paget disease presents with pruritus, discharge and discomfort. [Carcinoma Penis p21]

- Paget cells are large, oval or round, clear-staining hydropic cells with hypochromatic nuclei. [Carcinoma Penis p21]
- Extramammary Paget disease grows slowly but can become invasive. [Carcinoma Penis p21]
- In extramammary Paget disease, positive margins need further excision, and invasive tumours need more extensive resection with reconstruction. [Carcinoma Penis p21]
- If extramammary Paget disease has palpable lymphadenopathy, radical node dissection is done, but the prognosis is poor; neoadjuvant chemotherapy such as docetaxel may be added. [Carcinoma Penis p21]
- Penile adenosquamous carcinoma presents as a large firm mass, is CEA positive and is probably best treated by local excision and surveillance. [Carcinoma Penis p21]
- Primary penile lymphoreticular malignancy is rare; penile involvement may be metastatic, and leukaemia can cause priapism. [Carcinoma Penis p21]
- Low-dose radiotherapy has also been reported to treat primary penile lymphoma successfully. [Carcinoma Penis p21]
- Penile metastases arise rarely from the gastrointestinal tract, testis or kidney. [Carcinoma Penis p21]
- Tumour reaches the penis by direct extension, retrograde venous spread, lymphatic spread or arterial embolism. [Carcinoma Penis p21]
- Penile metastases present with penile swelling, nodularity or ulceration, as well as priapism. [Carcinoma Penis p21]
- The commonest histological finding in penile metastasis is replacement of the corpora cavernosa by the metastatic lesion. [Carcinoma Penis p21]
- If the primary of a penile metastasis can be excised or controlled, penectomy may be considered. [Carcinoma Penis p21]

CH 84 · INGUINAL NODE DISSECTION

Sentinel Lymph Node Biopsy and Dynamic SLNB

- Dynamic sentinel node biopsy is limited to larger centres with experienced surgeons. [Carcinoma Penis p9]
- Sentinel nodes in DSNB are assessed on permanent sections, not frozen section. [Carcinoma Penis p9]
- For DSNB, 0.3–0.4 mL of technetium-99m sulphur colloid is injected around the tumour 4 hours or 1 day before surgery. [Carcinoma Penis p9]
- Under anaesthesia, 1 mL of blue dye is injected intradermally around the circumference of the penile shaft. [Carcinoma Penis p9]
- A small inguinal incision is made where the gamma camera directs, and the nodes found by gamma signal and blue dye are removed; groin radioactivity should fall sharply after removal. [Carcinoma Penis p9]
- The pathologist should embed the entire sentinel node, cut it at 2 mm intervals and analyse it with staining and immunohistochemistry. [Carcinoma Penis p9]
- If the sentinel node is positive, inguinal lymph node dissection is done within 4 weeks. [Carcinoma Penis p9]

Lymph Node Management

- Before planning groin management, the primary should be completely resected, its histopathology should be available and the groins should be examined. [Carcinoma Penis p17]
- A mobile node mass of 4 cm or more carries a risk of extranodal extension and is managed as metastatic nodal disease. [Carcinoma Penis p17]
- If nodal disease progresses on induction chemotherapy, the options are a clinical trial, palliative chemotherapy or radiotherapy to the affected side, then hospice referral on further progression. [Carcinoma Penis p17]

Grafts in Substitution Urethroplasty: Flaps

- The groin flap, based on the superficial circumflex iliac artery, is an example of an axial flap. [Urethra Grafts p1]
- Flaps are also classified by elevation technique: a peninsular flap keeps both vascular and cutaneous continuity at its base; an island flap keeps vascular continuity but divides cutaneous continuity; a free flap divides both, and vascular continuity is re-established at the recipient site. [Urethra Grafts p1]

Grafts

- Graft take has three phases: imbibition, inosculation and revascularisation. (The corpus gives only imbibition and inosculation.) [Urethra Grafts p2]
- Appendix is listed as a substitution material for the urethra. (The corpus mentions appendix only for ureteric repair, Book 11.) [Urethra Grafts p2]

Buccal Mucosal Grafts

- Anatomically, buccal mucosa extends from the anterior tonsillar pillar (posteriorly) to the outer commissure of the lips (anteriorly). [Urethra Grafts p3]
- Superiorly and inferiorly, buccal mucosa is limited by the maxillary and mandibular vestibular folds. [Urethra Grafts p3]
- Buccal mucosa is supplied by branches of the maxillary and mandibular nerves. [Urethra Grafts p3]
- Its arterial supply is the buccal artery, a branch of the maxillary artery. [Urethra Grafts p3]
- The buccal donor site is concealed, which is a cosmetic advantage. [Urethra Grafts p3]
- Buccal mucosa has a high regenerative power. [Urethra Grafts p3]
- Buccal mucosa cannot provide enough tissue to be tubularised into long tubes. [Urethra Grafts p3]

Lingual Mucosa

- The lingual graft is the non-keratinised stratified squamous epithelium of the ventral tongue; the dorsum of the tongue is keratinised. [Urethra Grafts p4]
- The lingual graft is taken from the posterolateral aspect of the posterior tongue, and a graft 7–8 cm long can be obtained. [Urethra Grafts p4]
- Overzealous lingual harvest can injure the submandibular duct at the base of the tongue. [Urethra Grafts p4]
- Lingual mucosa is especially useful in patients with oral submucous fibrosis or unhealthy buccal mucosa. [Urethra Grafts p4]

Other Grafts

- Labial (lip) mucosa offers only limited tissue and may cause lip disfigurement. [Urethra Grafts p5]
- Bladder mucosa can be harvested under regional anaesthesia and gives long grafts. [Urethra Grafts p5]
- Bladder mucosa grafts are too invasive to harvest, carry the complications of bladder surgery, and are prone to encrustation. [Urethra Grafts p5]
- Tunica vaginalis is easily and universally available, needs no general anaesthesia, and requires no extensive surgery because it is a local tissue transfer. [Urethra Grafts p5]
- Tunica vaginalis is less reliable than oral mucosal grafts and may be unavailable after previous scrotal (hydrocele) surgery. [Urethra Grafts p5]

Histology of Oral Mucosa

- Oral mucosa has four layers: oral epithelium, superficial lamina propria, deep lamina propria, and muscularis containing minor salivary glands. [Urethra Grafts p6]
- The buccal mucosal graft consists of the oral epithelium and the superficial lamina propria. [Urethra Grafts p6]

CH 86 · URETHRAL STRICTURE DISEASE

BMG: Indications and Technique

- In dorsal onlay BMG urethroplasty, the dorsal urethrotomy opens the whole length of the stricture plus 0.5 cm on either side. [Urethra Grafts p7]
- A throat pack is placed before buccal harvest. [Urethra Grafts p7]
- The donor site is infiltrated submucosally with 2% lignocaine (Xylocaine) with adrenaline. [Urethra Grafts p8]
- The buccal graft is outlined by two parallel incisions about 1.5 cm apart, running from the angle of the mouth to the anterior tonsillar pillar. [Urethra Grafts p8]
- The harvested graft is kept in saline, with or without gentamicin, before defatting. [Urethra Grafts p8]
- In the dorsal onlay sequence, the graft is laid on the corpora cavernosa with the mucosa facing the lumen; one edge is sutured to one cut edge of the urethra, quilting sutures are placed, and the other edge is sutured to the other urethral margin over a catheter. [Urethra Grafts p8]

Long-Term Follow-Up

- In a report of 50 cases, Barbagli et al placed the graft ventrally, dorsally or laterally and found no significant difference in success. [Urethra Grafts p9]
- The restricture rate after BMG urethroplasty is around 5%, and other perineal complications are rare. [Urethra Grafts p9]
- About 15% of patients have long-term oral numbness and mouth tightness after buccal harvest. [Urethra Grafts p9]

RGU (Retrograde Urethrogram): Indications

- A suspected foreign body in the urethra is an indication for RGU. [RGU p1]

RGU: Patient Preparation

- Examine the genitalia beforehand to exclude a tight phimosis or local infection, which may make RGU difficult or impossible. [RGU p1]
- Ask the patient to pass urine before the procedure. [RGU p1]

RGU: Method

- Retract the foreskin and clean the glans and meatus with povidone-iodine (Betadine) or another antiseptic solution. [RGU p2]
- Instil a small amount of 2% lidocaine jelly into the urethra before contrast injection. [RGU p2]
- Take a plain (scout) film in the supine position before injecting contrast. [RGU p2]
- Contrast can be injected through a feeding tube placed just inside the tip of the penis (meatus), as an alternative to a catheter balloon in the fossa navicularis. [RGU p2]
- Take films in 30-degree left anterior oblique and 30-degree right anterior oblique positions during contrast injection. [RGU p2]
- The oblique position is described as the patient turned about 45 degrees, with the lower (dependent) leg flexed 90 degrees at the knee and the upper leg kept straight. [RGU p2]

RGU: Complications

- Complications of RGU are contrast extravasation or urethral rupture, contrast reactions, urethral trauma and urinary tract infection. [RGU p2]

Diagnosis & Evaluation of Urethral Stricture

- Stricture depth and density are hard to measure objectively; they are judged from physical examination, the appearance of the urethra on radiography, and USG or MRI. [Ureth Inj Stric p8]

Internal Urethrotomy

- For a bulbar stricture under 1.5 cm with minimal deep spongiofibrosis, OIU gives about 75% moderately long-term success. [Ureth Inj Stric p10]
- Erectile dysfunction after OIU is caused by cavernosal veno-occlusive dysfunction. [Ureth Inj Stric p10]

- Colchicine is used against recurrence because it blocks tubulin and so inhibits wound contraction. [Ureth Inj Stric p10]
- Mitomycin C injected submucosally works through antifibroblast and anticollagen activity. [Ureth Inj Stric p10]

Open Procedures

- Hockey-stick penile skin flaps and Q-flaps are flap options for anterior urethral stricture. [Ureth Inj Stric p11]
- Two-stage urethroplasty is indicated when the urethral plate is too poor or absent, or when the bulbar urethra is necrosed or stenosed. [Ureth Inj Stric p11]
- In stage I the whole scarred urethral plate is excised, and the graft is placed on dartos fascia, tunica albuginea or the corpora cavernosa. [Ureth Inj Stric p11]

Pelvic Fracture Urethral Injuries

- The transpubic approach is very rarely needed; its indications include marked prostatic displacement and retropubic fixation of the prostate, as well as fistulas. [Ureth Inj Stric p12]
- An abdominoperineal approach may also be needed when there is severe fibrosis. [Ureth Inj Stric p13]
- Alternatives to the voiding cystourethrogram at catheter removal (21 or 28 days) are flexible endoscopy and a pericatheter study. [Ureth Inj Stric p13]
- The suprapubic catheter is removed once the patient has voided well for 5–7 days. [Ureth Inj Stric p13]
- Follow-up after PFUI urethroplasty includes flexible endoscopy at 6 months and again at 1 year. [Ureth Inj Stric p13]

OTHER TOPICS IN THIS BOOK

Role of PET-CT in carcinoma testis

- In the post-chemotherapy seminoma residual mass algorithm, a raised beta-hCG level leads directly to salvage chemotherapy, without PET. [PET CT PSMA p2]

PET in other systems

- PET has no role in assessing the primary lesion in penile cancer. [PET CT PSMA p12]

11 Stones, Obstruction & Trauma

CH 93 · MANAGEMENT OF UPPER URINARY TRACT OBSTRUCTION

Stone composition

- Hounsfield unit values correlate with stone composition, but there is overlap between many stone types. [Compilation 05-2 p1]
- On CT, uric acid stones can be distinguished from calcium stones, but calcium stones usually cannot be distinguished from struvite stones. [Compilation 05-2 p1]
- Matrix stones can also be treated by ureteroscopy, although PCNL is the best choice. [Compilation 05-2 p1]

Ammonium acid urate stones

- Further associations of ammonium acid urate stones are inflammatory bowel disease, morbid obesity and recurrent uric acid stones. [Compilation 05-2 p72]
- Conditions leading to metabolic acidosis are the prime risk factor for ammonium acid urate stones. [Compilation 05-2 p72]
- Treatment by cause: bowel disease with standard fluid intake, oral calcium, alkalinisation and oxalate reduction; a history of uric acid calculi with more fluid, protein and salt restriction, potassium citrate alkalinisation and allopurinol; laxative abuse with a healthier bowel regimen; chronic infection as for struvite calculi. [Compilation 05-2 p72]

Pre-ureteral Vena Cava (Retrocaval Ureter): Treatment

- The repair is done over a double-J stent, which is removed after 4 to 6 weeks. [RCU p3]

Retroperitoneal Fibrosis: Epidemiology

- Although RPF typically affects patients aged 40–60 years, cases in patients younger than 18 years have also been reported. [Retroperitoneal Fibrosis p1]

Retroperitoneal Fibrosis: Clinical Presentation

- The plaque centred on the distal aorta at L4–L5 may extend distally to involve the iliac vessels. [Retroperitoneal Fibrosis p3]

Retroperitoneal Fibrosis: Medical Management

- Progesterone itself is listed alongside medroxyprogesterone acetate and tamoxifen as a hormonal agent for RPF. [Retroperitoneal Fibrosis p6]

Retroperitoneal Fibrosis: Surgery

- Open ureterolysis is performed through a midline transperitoneal incision. [Retroperitoneal Fibrosis p7]
- The deep biopsies of the mass are sent for both frozen section and permanent (paraffin) section. [Retroperitoneal Fibrosis p7]
- Laparoscopic ureterolysis uses a four-port transperitoneal approach and has the same success and complication rates as open ureterolysis. [Retroperitoneal Fibrosis p7]

Dietary hyperoxaluria

- Potatoes are an oxalate-rich food that contributes to dietary hyperoxaluria. [Stone Pathogenesis p14]
- Prolonged antibiotic use reduces or eliminates Oxalobacter formigenes, so less oxalate is degraded in the gut, more is absorbed, and hyperoxaluria results. [Stone Pathogenesis p14]

Idiopathic hyperoxaluria

- In the Arabian Peninsula population, hyperoxaluria with almost no hypercalciuria leads to idiopathic calcium oxalate stones. [Stone Pathogenesis p14]
- Abnormalities in oxalate transport are a proposed mechanism of idiopathic hyperoxaluria. [Stone Pathogenesis p14]

Hypocitraturia

- Citrate enhances the inhibitory effect of Tamm-Horsfall protein on crystallisation. [Stone Pathogenesis p16]
- Metabolic acidosis lowers urinary citrate both by enhanced tubular reabsorption and by decreased citrate synthesis in renal tubular cells. [Stone Pathogenesis p16]
- Strenuous exercise causes hypocitraturia through lactic acidosis. [Stone Pathogenesis p16]

Hypomagnesiuria

- Low urinary magnesium promotes stones also by decreasing urinary citrate, in addition to reducing magnesium complexation of calcium and oxalate salts. [Stone Pathogenesis p17]
- Poor dietary magnesium intake is a cause of hypomagnesiuria. [Stone Pathogenesis p17]

Hyperuricosuria (uric acid stones)

- The most common cause of hyperuricosuria is dietary purine intake. [Stone Pathogenesis p21]
- Multiple myeloma and polycythaemia are the myeloproliferative examples causing hyperuricosuria. [Stone Pathogenesis p21]
- Haemoglobinopathies, including thalassaemia, cause hyperuricosuria. [Stone Pathogenesis p21]

Pathogenesis of infectious (struvite) stones

- In infection stones, calcium, phosphate and carbonate concentrations allow precipitation of hydroxyapatite as well as carbonate apatite. [Stone Pathogenesis p23]

Bacteriology of infection stones

- The majority of urea-splitting organisms belong to the Enterobacteriaceae family. [Stone Pathogenesis p24]
- Some Mycoplasma species and some yeasts are also urease producers. [Stone Pathogenesis p24]

- Bacterial infection (E. coli, Proteus) alters the activity of urokinase and sialidase, which promotes stone formation. [Stone Pathogenesis p24]

Pathogenesis of cystine stones

- Cystine is a dimer of two cysteine molecules joined by a disulfide bond. [Stone Pathogenesis p25]

Matrix stones

- Matrix is the non-crystalline component of stones, and predominantly matrix stones are rare. [Stone Pathogenesis p26]
- Matrix stones are typically radiolucent and may be mistaken for a tumour or a uric acid stone on imaging. [Stone Pathogenesis p26]
- Pure matrix stones may be composed of up to 65% protein. [Stone Pathogenesis p26]
- Protein components of stone matrix include osteopontin and calprotectin. [Stone Pathogenesis p26]
- Beta-2-microglobulin (microfibrillar protein) matrix stones occur in chronic renal failure patients on dialysis. [Stone Pathogenesis p26]

Drug-related stones

- Phosphate-binding antacids indirectly promote stone formation. [Stone Pathogenesis p26]

CH 94 · UPPER URINARY TRACT TRAUMA

Controversies in fluid and dietary therapy for stones

- Caffeine increases stone risk by raising urinary calcium/creatinine and magnesium/creatinine and urine flow, but not the oxalate/creatinine ratio. [Compilation 05-2 p71]
- Lemonade, citrus juice, grapefruit juice, apple juice, tea, beer and wine were listed as possibly decreasing stone risk (marked with a question mark). [Compilation 05-2 p71]
- Overall, the evidence suggests that it is not the type of fluid but the amount taken per day that matters. [Compilation 05-2 p71]
- Low-carbohydrate, high-protein weight-loss diets (Atkins, South Beach, Sugar Busters) deliver a marked acid load to the kidney and increase stone risk. [Compilation 05-2 p71]
- Calcium restriction is an old concept; a moderate calcium supplement, preferably calcium citrate, is advised. [Compilation 05-2 p71]
- Vitamin D repletion does not change urinary calcium, but urinary calcium should be monitored. [Compilation 05-2 p71]
- Bisphosphonates with thiazides may reduce hypercalciuria and protect bone. [Compilation 05-2 p71]
- Vitamin C intake should be limited to less than 2 g/day. [Compilation 05-2 p71]

Renal Trauma: Mechanism and Presentation

- Of penetrating renal injuries, 86% are gunshot injuries and 14% are stab injuries. [Renal Trauma p1]
- Penetrating trauma at the anterior axillary line more often injures the renal hilum and pedicle, whereas trauma at the posterior axillary line more often causes parenchymal injury. [Renal Trauma p1]
- A rib fracture carries a threefold increase in the incidence of renal trauma. [Renal Trauma p1]

Indications for Imaging

- All paediatric patients with more than 5 red blood cells per high-power field should be imaged, without waiting for shock. [Renal Trauma p3]

AAST Grading (Revised 2011)

- Only 5% of renal injuries are grade II to grade V. [Renal Trauma p5]
- Only 5% of renal injury cases are grade II to grade V. [Upper Tract Trauma Stricture p5]

Management and Indications for Surgery

- In the acute setting, angioembolisation can stop significant renal bleeding and avoid nephrectomy, though the evidence for this is from case reports only. [Renal Trauma p6]

- Delayed diagnosis of an arterial injury, leading to delayed nephrectomy, is a relative indication for surgery. [Renal Trauma p7]
- Delayed bleeding occurs in 9% of renal injuries. [Renal Trauma p7]
- Delayed bleeding can be managed with bed rest and hydration, or with angioembolisation. [Renal Trauma p7]
- Post-traumatic hypertension from vascular stenosis, parenchymal compression or post-traumatic arteriovenous fistula is mediated by the renin-angiotensin-aldosterone system. [Renal Trauma p7]

Renal Trauma

- Causes of blunt renal trauma include road traffic accidents, falls from height and assault. [Upper Tract Trauma Stricture p1]
- Among penetrating renal injuries, gunshot wounds account for 86% and stab wounds for 14%. [Upper Tract Trauma Stricture p1]
- Penetrating trauma at or anterior to the anterior axillary line more often injures the renal hilum and pedicle, whereas trauma at the posterior axillary line more often causes parenchymal injury. [Upper Tract Trauma Stricture p1]
- A rib fracture carries a 3-fold increase in the incidence of renal trauma. [Upper Tract Trauma Stricture p1]

Hematuria

- Dipstick testing or urine microscopy of the first aliquot is used to detect microscopic haematuria; any gross haematuria is significant. [Upper Tract Trauma Stricture p2]

Indications of Imaging

- All paediatric trauma patients with more than 5 RBCs per high-power field should be imaged, without waiting for shock. [Upper Tract Trauma Stricture p3]

Indications for Surgery and Complications of Renal Trauma

- Delayed diagnosis of a renal arterial injury is a relative indication for surgery, in the form of delayed nephrectomy. [Upper Tract Trauma Stricture p7]
- Delayed bleeding occurs in 9% of renal injuries and may be managed by bed rest and hydration, or by angioembolisation. [Upper Tract Trauma Stricture p7]

Ureteral Injuries — Causes

- Ureteric injury occurs in less than 4% of penetrating trauma and less than 1% of blunt trauma. [Upper Tract Trauma Stricture p9]
- Gunshot ureteric injuries are associated with bladder injury in 5% of cases. [Upper Tract Trauma Stricture p9]
- After hysterectomy, the commonest surgical causes of ureteric injury in descending order are colorectal surgery, ovarian surgery and vascular surgery. [Upper Tract Trauma Stricture p9]
- Of ureteroscopic ureteric perforations, 0.2% require open surgery. [Upper Tract Trauma Stricture p9]
- The risk of ureteric perforation by intracorporeal lithotripter is highest with electrohydraulic lithotripsy, then Nd:YAG laser, then holmium:YAG laser. [Upper Tract Trauma Stricture p9]

Ureteral Injuries — CT Scan

- CT may miss an upper ureteric injury because the urinary extravasation can be contained by Gerota's fascia, obscuring the diagnosis. [Upper Tract Trauma Stricture p10]
- Circumferential contrast extravasation is a CT feature of ureteric injury. [Upper Tract Trauma Stricture p10]

Intra-operative Recognition of Ureteric Injury

- The wound trajectory guides intra-operative recognition of ureteric injury. [Upper Tract Trauma Stricture p11]
- Injecting 1–2 mL of methylene blue into the renal pelvis and looking for leakage helps identify a ureteric injury at operation. [Upper Tract Trauma Stricture p11]

Ureteric Injury — Damage Control

- A damage-control option for ureteric injury is to do nothing to the ureter at the first operation and plan re-operation at 24 hours. [Upper Tract Trauma Stricture p12]

Management of Ureteric Injury Recognised Intraoperatively

- Intra-operative recognition of ureteric injury can be as low as 0% during laparoscopic surgery. [Upper Tract Trauma Stricture p14]
- If the repair fails after stenting, endoscopic options include balloon dilation, laser incision and Acucise (cutting-balloon) endoureterotomy. [Upper Tract Trauma Stricture p14]

Ureteric injuries recognised in the post-op period

- If stenting and nephrostomy fail after a delayed-recognised ureteric injury, endoscopic methods to consider include laser, balloon and Acucise (cutting-balloon) incision before open reconstruction. [Upper Tract Trauma Stricture p15]

General principles of ureteral repair

- The ureteric anastomosis is made with an absorbable monofilament suture such as 5-0 PDS. [Upper Tract Trauma Stricture p16]

CH 95 · URINARY LITHIASIS: ETIOLOGY, EPIDEMIOLOGY AND PATHOGENESIS

PUJO and stone

- UPJ obstruction is associated with stones in 20–30% of patients. [Compilation 05-2 p2]
- Two clinical situations occur: a primary PUJO with secondary calculi, or a renal pelvic/UPJ stone causing oedema at the PUJ that gives a PUJO-like appearance. [Compilation 05-2 p2]
- When it is unclear whether there is a primary PUJO or oedema due to the stone, treat the stones first and repeat CT or ultrasound after 4–6 weeks; if hydronephrosis persists, do a DTPA scan, operating if it confirms PUJO and following up if it does not. [Compilation 05-2 p3]

Types of Stones

- Brushite is calcium hydrogen phosphate dihydrate ($CaHPO4·2H2O$). [Compilation 05 p2]

Renal Tubular Acidosis

- In distal RTA, the hypercalciuria that drives stone formation results from metabolic acidosis with a secondary rise in PTH. [Compilation 05 p18]

Fast and Calcium Load Test

- The patient follows a restricted diet for at least 7 days before the test. [Compilation 05 p45]
- For hydration, the patient drinks 300 ml of water at 12 hours and again at 9 hours before calcium loading. [Compilation 05 p45]
- Two hours before calcium loading, the patient empties the bladder and drinks 600 ml of water. [Compilation 05 p45]
- Urine is then collected for 2 hours as the fasting urine. [Compilation 05 p45]
- A 1 g oral calcium load is given as 250 ml of a liquid synthetic diet (Calcitest). [Compilation 05 p45]
- Urine is collected for the next 4 hours as the post-load urine. [Compilation 05 p45]
- Calcium and creatinine are measured in both the fasting and the post-load urine. [Compilation 05 p45]
- Normal post-load urinary calcium is less than 0.2 mg per mg creatinine. [Compilation 05 p45]

Classification of Nephrolithiasis (Metabolic Evaluation Parameters)

- "Unclassified calcium nephrolithiasis" is a category in the metabolic classification of nephrolithiasis. [Compilation 05 p47]

Role of CT (DECT)

- DECT distinguishes uric acid from non-uric-acid stones with an accuracy of 92-100%. [Compilation 05 p52]

Clinical Relevance of Stone Analysis: Mineral Names

- The mineral name of tricalcium phosphate stones is whitlockite. [Compilation 05 p53]

State of Saturation

- The concentration product is illustrated with sodium chloride, where CP equals the product of the sodium and chloride ion concentrations ([Na+] x [Cl-]). [General Stone Formation p1]

Nucleation

- Nuclei are defined as the earliest crystals that will not dissolve; small nuclei are unstable and dissolve, whereas large nuclei persist. [General Stone Formation p2]

Crystal Growth, Aggregation and Retention

- One mechanism of crystal retention is oxalate-induced renal tubular epithelial cell injury, which promotes adherence of calcium oxalate crystals. [General Stone Formation p3]
- A second mechanism of crystal retention is oxidative stress to renal tubular epithelial cells caused by calcium oxalate monohydrate. [General Stone Formation p3]
- Crystals retained in the tubular lumen without extrusion by epithelial erosion produce nephrocalcinosis, whereas retention followed by subepithelial location and epithelial erosion produces urolithiasis. [General Stone Formation p4]

Inhibitors of Crystal Formation

- Citrate reduces crystal growth, and this effect is much greater for calcium phosphate than for calcium oxalate. [General Stone Formation p5]
- Osteopontin is also known as uropontin. [General Stone Formation p6]
- Urinary prothrombin fragment 1 (written "urinary prothrombin factor 1") is a urinary inhibitor of crystal formation. [General Stone Formation p6]
- Polyanionic macromolecule inhibitors include the glycosaminoglycans chondroitin sulfate and heparan sulfate, acid mucopolysaccharide, and RNA. [General Stone Formation p6]

Promoters of Crystal Formation

- Oxidative stress is listed as a promoter of crystal formation. [General Stone Formation p6]

Calcium Metabolism

- Of the 30-40% of dietary calcium that is absorbed, 90% is absorbed in the small intestine and 10% in the colon. [General Stone Formation p7]
- Calcium is absorbed from the intestine in its ionic state. [General Stone Formation p7]
- Phosphate, citrate and sulfate (as well as oxalate and fatty acids) form complexes with calcium in the gut and reduce its absorption. [General Stone Formation p7]
- About 98% of filtered calcium is reabsorbed by the kidney. [General Stone Formation p7]

Oxalate and Magnesium Metabolism

- Absorbed dietary oxalate is taken up 50% in the small intestine and 50% in the colon. [General Stone Formation p8]
- Intestinal magnesium absorption is mostly by passive diffusion, with a component of active transport. [General Stone Formation p8]
- Magnesium is absorbed in the small intestine (mostly the ileum) and in the large intestine. [General Stone Formation p8]
- Intestinal magnesium absorption is hormonally regulated by vitamin D. [General Stone Formation p8]

Phosphorus Metabolism

- About 60% of dietary phosphorus is absorbed in the intestine. [General Stone Formation p9]
- Intestinal phosphate absorption is active, transcellular at low concentrations and paracellular at high concentrations. [General Stone Formation p9]
- Vitamin D increases intestinal phosphate absorption. [General Stone Formation p9]
- Low luminal pH decreases, and high luminal pH increases, intestinal phosphate transport. [General Stone Formation p9]
- About 80-90% of filtered phosphate is reabsorbed in the renal tubule. [General Stone Formation p9]

Epidemiology of Urolithiasis

- Stone episodes are more frequent in the summer months, as well as in hot, dry climates. [Stone Pathogenesis p1]
- Occupations with heat exposure, such as cooks, cause dehydration and so predispose to stones. [Stone Pathogenesis p1]
- The stone risk from obesity, diabetes and metabolic syndrome is especially marked in women. [Stone Pathogenesis p1]
- Obesity increases urinary excretion of oxalate, uric acid, sodium and phosphate. It predisposes to calcium oxalate stones (through increased promoters) as well as to uric acid stones (through low urine pH). [Stone Pathogenesis p1]
- Whether hard water or soft water affects stone risk is controversial. [Stone Pathogenesis p1]
- In the United States, White people form more stones than Hispanic, Asian and Asian-American people. [Stone Pathogenesis p1]

Hypercalciuria: definitions and renal calcium handling

- An alternative definition of hypercalciuria is urinary calcium excretion above 7 mmol/day in men or above 6 mmol/day in women. [Stone Pathogenesis p3]
- The normal kidney filters about 270 mmol of calcium daily and reabsorbs all but about 4 mmol of it (about 98%). [Stone Pathogenesis p3]

Absorptive Hypercalciuria

- Absorptive hypercalciuria is defined as increased urinary calcium excretion (above 0.2 mg calcium per mg creatinine) after an oral calcium load. [Stone Pathogenesis p4]
- Causes of absorptive hypercalciuria include vitamin D-dependent and vitamin D-independent processes, among them upregulation of the vitamin D receptor. [Stone Pathogenesis p4]
- Hypersensitivity to vitamin D is a further cause of absorptive hypercalciuria. [Stone Pathogenesis p4]
- Renal phosphate wasting (renal phosphate leak) is classified as absorptive hypercalciuria type 3. [Stone Pathogenesis p4]

Resorptive Hypercalciuria

- The thiazide challenge works because a thiazide increases renal calcium reabsorption even when serum calcium is already high, so in primary hyperparathyroidism it makes the hypercalcaemia worse. [Stone Pathogenesis p7]

Hypercalciuria due to other diseases

- Glucocorticoids inhibit intestinal calcium absorption. This is why they help treat the hypercalciuria of sarcoidosis, even though glucocorticoid excess itself causes resorptive hypercalciuria. [Stone Pathogenesis p8]
- Tuberculosis and thyrotoxicosis are further causes of hypercalciuria. [Stone Pathogenesis p8]

Oxalate metabolism in the liver

- Hydroxyproline is metabolised to 4-hydroxy-2-oxoglutarate, which HOG aldolase (HOGA, deficient in PH3) splits into glyoxylate and pyruvate. [Stone Pathogenesis p9]
- Glyoxylate can be reduced to glycolate by GRHPR, transaminated to glycine by AGT (alanine donates the amino group and becomes pyruvate, with vitamin B6 as cofactor; deficient in PH1), or oxidised to oxalate by lactate dehydrogenase (LDH). [Stone Pathogenesis p9]
- Hydroxypyruvate is converted to D-glycerate by GRHPR (deficient in PH2). When GRHPR is deficient, LDH converts it to L-glycerate instead, which is why L-glycerate is the PH2 urinary marker. [Stone Pathogenesis p9]

Primary Hyperoxaluria

- In primary hyperoxaluria, interstitial nephrocalcinosis causes interstitial fibrosis, and obstructing stones cause renal parenchymal inflammation. Together these produce progressive renal damage. [Stone Pathogenesis p10]
- Thirty mutations of the GRHPR gene have been identified in PH2. [Stone Pathogenesis p10]

- In PH3 the mechanism of the hyperoxaluria is unknown. [Stone Pathogenesis p10]
- PH3 hyperoxaluria is associated with severe hypercalciuria and recurrent calcium oxalate stones, and stone formation may stop later in life. [Stone Pathogenesis p10]
- PH3 has no reported cases of ESRD, and kidney transplantation has not been reported in this group. [Stone Pathogenesis p10]
- Untreated PH1 leads to ESRD by 15 years of age in 50% of children, and overall 30% of patients die. [Stone Pathogenesis p10]

CH 97 · SURGICAL MANAGEMENT OF URETERAL CALCULI

Ureteral calculi: natural history and management

- Alpha-blockers increase the chance of passage of proximal/mid ureteric stones under 5 mm by 29%. [Compilation 05-2 p7]
- Associated ureteral pathology is an indication for PCNL (antegrade approach) for a proximal/mid ureteric stone. [Compilation 05-2 p7]

Ureteric stone in megaureter

- A stone in a non-obstructed megaureter can be treated with MET, URS or ESWL. [Compilation 05-2 p8]
- A stone in an obstructed megaureter is treated by retropulsion to the kidney plus PCNL plus ureteroneocystostomy, or laparoscopic/robotic ureteroneocystostomy with retrieval of the calculus, or URS with endoureterotomy (for a segment under 3 cm), or open ureteroneocystostomy with retrieval of the calculus. [Compilation 05-2 p8]

Ureteral stricture with stone

- A pathological stricture below a stone is managed by considering its cause: endoureterotomy with double-J stent and URS at the same or a second sitting, or open/laparoscopic/robotic ureterotomy with definitive repair. [Compilation 05-2 p9]

Stones and renal function

- Staghorn calculus, recurrent pyelonephritis or abscess, XGPN and chronic obstruction from stone threaten renal function; ultrasound or CT showing cortical atrophy or thinned parenchyma should prompt a diuretic renogram before deciding between nephrectomy, drainage with reassessment, or stone surgery. [Compilation 05-2 p10]

Solitary kidney and urinary diversion

- In an obstructed solitary kidney, either drain first (PCN or double-J stent) with later definitive surgery, or do primary URSL with a double-J stent, depending on clinical stability, UTI and electrolyte abnormality. [Compilation 05-2 p11]
- In urinary diversion with possible retrograde access, a large stone burden is treated by PCNL and a small one by flexible URS; without retrograde access, PCNL or antegrade URS is used. [Compilation 05-2 p11]
- The ureteric orifice in a diversion is located by loopogram, pouchogram or intravenous indigo carmine. [Compilation 05-2 p11]

Transplant kidney stone

- Transplant stones present with vague symptoms, graft-site discomfort, fever, oliguria and raised creatinine. [Compilation 05-2 p12]
- A transplant ureteric stone under 1.5 cm is treated with SWL or antegrade/retrograde URS; one over 1.5 cm, or after failed URS/SWL, is treated by mini-PCNL. [Compilation 05-2 p12]
- Tract dilatation in a transplant kidney may be tough because of the fibrotic capsule. [Compilation 05-2 p12]

EHL vs pneumatic vs laser lithotripsy

- With EHL held more than 1 mm from tissue, energy under 500 mJ causes no injury, whereas over 1000 mJ causes ureteric perforation. [Compilation 05-2 p35]

- EHL can reach the lower pole (flexible probe). [Compilation 05-2 p35]
- The tissue effect of pneumatic lithotripsy is focal erosion and haemorrhage. [Compilation 05-2 p35]
- Pneumatic lithotripsy is the least efficient of the three, although easily available. [Compilation 05-2 p35]
- Pneumatic probe sizes are 2.4, 3, 4.8, 6 and 10.5 Fr. [Compilation 05-2 p35]
- Ultrasonic lithotripsy probes range from 2.5 Fr to 12 Fr. [Compilation 05-2 p70]
- Disadvantages of ultrasonic lithotripsy include risk of ureteral injury and blocking of the hollow suction probe by stone. [Compilation 05-2 p70]

Ureteral avulsion

- Transureteroureterostomy is contraindicated in stone formers. [Compilation 05-2 p43]

Steinstrasse

- Steinstrasse lies in the distal ureter in about 64% (65%), the proximal ureter in about 29% (30%) and the mid ureter in about 8% (5%). [Compilation 05-2 p67]
- Bladder irritation is a presentation of steinstrasse. [Compilation 05-2 p67]
- Meatal stenosis and ureteric stricture are causes of steinstrasse. [Compilation 05-2 p67]

Clinically insignificant residual fragments after SWL

- After SWL, residual fragments are all fragments remaining in the kidney 3 months after the last SWL session. [Compilation 05-2 p68]
- About 24–36% (about 30%) of patients have residual fragments at 3 months after SWL. [Compilation 05-2 p68]
- Multiple stones and congenital anomalies are risk factors for residual fragments after SWL. [Compilation 05-2 p68]
- Plain KUB overestimates stone-free rates by about 35%. [Compilation 05-2 p68]
- Ultrasound assessment of residual fragments suffers from interobserver variability. [Compilation 05-2 p68]
- Whether CIRF need treatment is controversial, because residual fragments carry a significant risk of stone recurrence. [Compilation 05-2 p69]
- Alkali citrate after ESWL decreases stone growth and stone recurrence from residual fragments. [Compilation 05-2 p69]

Lithotripters: electrohydraulic lithotripsy (EHL)

- During ureteroscopic EHL, the fibre tip is projected 2–5 mm beyond the tip of the ureteroscope. [Lithotripters ESWL p2]
- The EHL probe is placed about 1 mm from the stone surface. [Lithotripters ESWL p2]
- EHL causes retropulsion of the stone. [Lithotripters ESWL p2]
- EHL produces larger fragments than other intracorporeal sources. [Lithotripters ESWL p2]
- The success rate of EHL is about 90%. [Lithotripters ESWL p2]

Lasers

- The ruby laser is a continuous-wave laser that produces heat and is not appropriate for clinical lithotripsy. [Lithotripters ESWL p3]
- The pulsed dye laser does not fragment hard stones, and the eye protection it requires hampers visualisation of the stone; it is no longer used. [Lithotripters ESWL p3]
- Holmium laser fibres come in 200, 365, 550 and 1000 µm sizes; the 200 and 365 µm fibres are the ones used for lithotripsy. [Lithotripters ESWL p4]
- When more holmium power is needed, increase the frequency in preference to the pulse energy, because higher pulse energy narrows the safety margin. [Lithotripters ESWL p4]
- Holmium laser lithotripsy gives stone-free rates of 85–100%. [Lithotripters ESWL p4]
- Disadvantages of the holmium laser include a high initial cost and the cost of the laser fibres. [Lithotripters ESWL p4]
- The holmium laser can cut guidewires and baskets. [Lithotripters ESWL p4]

Ballistic lithotripsy

- The first ballistic lithotrite was the Swiss Lithoclast. [Lithotripters ESWL p5]
- Ballistic lithotripsy causes no thermal injury. [Lithotripters ESWL p5]
- Ballistic lithotripters need little maintenance and their probes have a long life. [Lithotripters ESWL p5]
- The success rate of ballistic lithotripsy is 73–100%. [Lithotripters ESWL p5]
- The goal of ballistic fragmentation is to generate fragments smaller than 2 mm. [Lithotripters ESWL p5]

Ultrasonic lithotripter

- The ultrasonic lithotripter's piezoceramic plate resonates to produce ultrasonic waves at 23,000–25,000 Hz, which make the stone resonate and break; the unit has a generator, a foot pedal and suction. [Lithotripters ESWL p6]
- Ultrasonic lithotripsy produces minimal heat, so soft tissue damage is minimal. [Lithotripters ESWL p6]
- Ultrasonic lithotripsy is particularly helpful for large stones and leads to less steinstrasse. [Lithotripters ESWL p6]
- Ultrasonic lithotripsy needs a rigid scope with a large working channel to take the 4.5 Fr hollow probe. [Lithotripters ESWL p6]
- The success rate of ultrasonic lithotripsy is 70–100%. [Lithotripters ESWL p6]

ESWL: types of generators

- In the electrohydraulic generator, the high-voltage spark causes explosive vaporisation of water, producing a spherically expanding shock wave. [Lithotripters ESWL p7]
- The electromagnetic generator produces plane or cylindrical shock waves, which are focused by an acoustic lens or a reflector. [Lithotripters ESWL p8]
- Electromagnetic generators cause less pain because the energy enters over a larger skin area. [Lithotripters ESWL p8]
- The piezoceramic elements of a piezoelectric generator are made of barium titanate. [Lithotripters ESWL p9]
- A piezoelectric generator needs 300–3000 piezoelectric crystals to generate a sufficiently large shock pressure. [Lithotripters ESWL p9]
- Piezoelectric generators focus accurately. [Lithotripters ESWL p9]

ESWL: imaging systems

- Modern lithotripters use a rotatable C-arm for fluoroscopy, in place of the HM3's two fixed X-ray units at 90°. [Lithotripters ESWL p10]
- Fluoroscopic targeting is familiar to most urologists, shows radio-opaque stones, and shows radiolucent stones as filling defects when contrast is used. [Lithotripters ESWL p10]
- Fluoroscopic targeting involves radiation exposure, has difficulty with radiolucent stones and needs high maintenance. [Lithotripters ESWL p10]
- Ultrasound targeting is inexpensive, uses no radiation and detects non-opaque stones. [Lithotripters ESWL p10]
- Ultrasound targeting needs a highly trained operator and makes it difficult to see stones in the middle third of the ureter or when a stent is in situ. [Lithotripters ESWL p10]

Mechanism of stone comminution

- Cavitation acts indirectly: asymmetrical bubble collapse produces microjets, and symmetrical bubble collapse produces a spherical wave front. The direct shock wave acts as a plane wave front. Together they cause compression-induced tensile cracks and spalling. [Lithotripters ESWL p11]
- Modern lithotripters have small focal zones, so they make little use of the squeezing-splitting (circumferential compression) mechanism. [Lithotripters ESWL p12]
- When there is no boundary, the bubble collapses symmetrically and gives a spherical wave front; when there is a boundary such as the stone surface, collapse forms cavitation microjets that apply a local compressive force progressing spherically into the stone. [Lithotripters ESWL p14]

Bioeffects of ESWL

- Acute vascular effects of SWL are damage and rupture of veins and small arteries, rupture of glomerular and peritubular capillaries, venous thrombi and intraparenchymal haemorrhage. [Lithotripters ESWL p15]
- Acute tubular effects of SWL are mild tubular dilatation with cast formation and mild tubular necrosis. [Lithotripters ESWL p15]
- Acute glomerular effects of SWL are breaks in Bowman's capsule and damage to mesangial cells and podocytes. [Lithotripters ESWL p15]
- Chronic renal changes after SWL are nephron loss, dilated veins, streaky fibrosis, diffuse interstitial fibrosis, calcium and haemosiderin deposits, and hyalinised acellular scars running from cortex to medulla. [Lithotripters ESWL p15]
- Proposed chronic effects of SWL include an increased rate of stone recurrence and the induction of brushite stone disease. [Lithotripters ESWL p15]
- First-generation lithotripters cause more renal injury than second- and third-generation devices. [Lithotripters ESWL p16]
- Kidney size matters for renal injury: a juvenile kidney is more vulnerable than an adult kidney. [Lithotripters ESWL p16]
- Pre-existing renal impairment aggravates SWL renal injury. [Lithotripters ESWL p16]
- Ramping starts with 100–500 shocks at low energy before raising the power to the standard treatment energy. [Lithotripters ESWL p16]

Factors affecting stone fragmentation

- Matrix stones are resistant to SWL. [Lithotripters ESWL p17]
- Apatite stones break less well than calcium oxalate dihydrate stones. [Lithotripters ESWL p17]
- Stones in the upper ureter or the iliac ureter have lower stone-free rates after SWL. [Lithotripters ESWL p18]
- Stones in the middle calyx and the pelvic ureter have among the best stone-free rates after SWL, together with the renal pelvis and upper calyx. [Lithotripters ESWL p18]
- Obesity makes SWL imaging and focusing difficult, in addition to increasing the skin-to-stone distance. [Lithotripters ESWL p19]
- A BMI of 20–28 gives the best SWL outcome. [Lithotripters ESWL p19]
- An air pocket covering just 2% of the coupling area reduces stone breakage by 20–40%. [Lithotripters ESWL p19]
- Higher power settings narrow the focal zone, so respiratory movement carries the stone in and out of it and breakage falls. [Lithotripters ESWL p19]
- Successful SWL is defined as being stone-free 3 months after treatment. [Lithotripters ESWL p20]
- Residual fragments larger than 5 mm are generally considered a failure of the SWL session. [Lithotripters ESWL p20]

Clavien-Dindo classification of PCNL complications

- After PCNL, 77% of patients have no complication (Clavien 0), 11% have grade I, 7% grade II, 4.1% grade III and 0.6% grade IV complications. [PCNL Complications p1]
- Grade I allows antiemetics, antipyretics, analgesics, diuretics, electrolytes and physiotherapy. [PCNL Complications p1]
- Total parenteral nutrition, like blood transfusion, counts as grade II. [PCNL Complications p1]
- Grade IIIa is an intervention not under general anaesthesia, and grade IIIb is an intervention under general anaesthesia. [PCNL Complications p1]
- Grade IVa is single-organ dysfunction (including dialysis), and grade IVb is multi-organ dysfunction. [PCNL Complications p1]

Acute haemorrhage

- Further factors in acute bleeding are patient characteristics, supracostal access, tract dilatation, renal pelvic perforation, excessive torque and misuse of instruments. [PCNL Complications p2]

- Parenchymal tract bleeding is mostly not significant, and the access sheath provides haemostasis during the operation. [PCNL Complications p2]
- If a vein or artery within the collecting system is injured, stop the procedure if vision is lost. [PCNL Complications p2]
- Venous bleeding is managed with a nephrostomy tube or a Councill catheter, with or without a gelatin matrix. [PCNL Complications p2]
- Arterial bleeding within the collecting system can be fulgurated under direct vision. [PCNL Complications p2]

Delayed haemorrhage

- Delayed haemorrhage occurs in about 1% of patients. [PCNL Complications p3]
- Pseudoaneurysm is far commoner than arteriovenous fistula, and an arteriovenous fistula usually bleeds continuously rather than intermittently. [PCNL Complications p3]
- Any history of bright red blood in the urine after PCNL warrants hospital admission and angiography. [PCNL Complications p3]
- Alternatives to embolisation are an endovascular stent, or ultrasound-guided puncture of the pseudoaneurysm with injection of thrombin or fibrin tissue adhesive. [PCNL Complications p3]

Collecting system injury

- Small infundibular tears are common and are not a problem if there is no haemorrhage. [PCNL Complications p4]
- Ureteral injury is rare and heals with a stent. [PCNL Complications p4]
- Renal pelvic injury occurs during initial access or dilatation, or from pushing too hard on a renal pelvic stone during lithotripsy. [PCNL Complications p4]
- A perforation is usually recognised during the operation by collapse of the previously distended collecting system. [PCNL Complications p4]
- After a recognised perforation, abort the procedure, or finish quickly with low irrigation pressure if it is near completion. [PCNL Complications p4]
- Drainage is by a nephroureteral stent, or a double-J stent plus nephrostomy, with a nephrostogram and tube removal at 2 to 7 days. [PCNL Complications p4]
- A missed perforation presents with postoperative abdominal pain and ileus, with or without fever. [PCNL Complications p4]
- A urinoma from a missed perforation may need percutaneous drainage if a stent and nephrostomy are already in place; if there are no drainage tubes, attempt a double-J stent or nephrostomy, with or without a drain. [PCNL Complications p4]

Colon injury

- Colon injury occurs in less than 1%, the same as the incidence of retrorenal colon. [PCNL Complications p5]
- Left colon injuries outnumber right colon injuries 2:1. [PCNL Complications p5]
- Risk factors for colon injury include advanced age, dilated colon, previous surgery and thin body habitus. [PCNL Complications p5]
- A missed colon injury presents with unexplained fever, prolonged ileus, unexplained leucocytosis, rectal bleeding, peritoneal inflammation, faecaluria or pneumaturia. [PCNL Complications p5]
- Conservative management adds broad-spectrum antibiotics and nil by mouth for a few days, then clear liquids and a gradually increased diet. [PCNL Complications p5]
- The colostomy tube is removed after contrast confirms there is no communication. [PCNL Complications p5]
- Intraperitoneal colon perforation needs open surgical repair. [PCNL Complications p5]

Small bowel and other visceral injuries

- Small bowel injury is rare and is detected by signs of peritonitis or by a nephroenteric fistula on the nephrostogram. [PCNL Complications p6]
- Duodenal injury can be managed conservatively with a percutaneous intraduodenal catheter or nasoduodenal drainage, combined with upper urinary tract drainage, fasting and parenteral feeding; otherwise open surgery is needed. [PCNL Complications p6]

- Splenic and liver injuries are rare unless the kidney is accessed above the 10th rib, and very rare with access between the 10th and 11th ribs. [PCNL Complications p6]
- Gallbladder or biliary tree injury can cause biliary peritonitis. [PCNL Complications p6]
- Postoperative pancreatitis can occur without evidence of pancreatic trauma. [PCNL Complications p6]

Pleural injury

- Pleural injury occurs in 24% of accesses above the 11th rib and 4.6% of accesses above the 12th rib. [PCNL Complications p7]
- Urinothorax is a form of pleural complication. [PCNL Complications p7]
- Every supracostal access should have formal chest radiography, during or after the operation. [PCNL Complications p7]
- A hydrothorax detected during the operation is treated with an intercostal tube drain; a minimal one found after the operation is managed conservatively, and a large one or one causing breathing problems needs an intercostal drain. [PCNL Complications p7]
- In a urinoma, rule out distal obstruction. [PCNL Complications p7]

Metabolic and physiological complications

- Saline extravasation can cause congestive cardiac failure and respiratory distress, but is rare in uncomplicated cases. [PCNL Complications p8]
- Venous gas embolism presents with hypoxaemia, pulmonary oedema, increased airway pressure, hypotension, jugular venous distension, facial plethora, dysrhythmias, and right heart strain with wide QRS on ECG. [PCNL Complications p8]

Venous thromboembolism

- The risk of deep vein thrombosis after PCNL is low, and thromboprophylaxis is not mentioned in the AUA best practice statements. [PCNL Complications p8]
- The best preventive measure is early ambulation, and stockings or intermittent compression pumps can be used during the operation. [PCNL Complications p8]

Fever and sepsis

- Additional risk factors for fever and sepsis are paraplegia, indwelling stents, previous PCNL and hydronephrosis. [PCNL Complications p9]
- If appropriate prophylaxis was given, most patients with fever after PCNL do not have an infection. [PCNL Complications p9]
- If bacteriological cure is not possible before PCNL, the bacterial count should be suppressed as much as possible. [PCNL Complications p9]
- A negative preoperative urine culture does not mean that sepsis will not occur. [PCNL Complications p9]

Musculoskeletal and positioning problems

- Positioning injuries include facial nerve injury, nose injury, stroke, peripheral nerve injury and rhabdomyolysis. [PCNL Complications p9]

Tube dislodgement

- Nephrostomy dislodgement is prevented by proper skin fixation. [PCNL Complications p10]
- The strength of nephrostomy retention is, in order, balloon, then Cope loop, then Malecot. [PCNL Complications p10]
- Tissue ingrowth can make tube removal difficult, especially with a Malecot. [PCNL Complications p10]

Collecting system obstruction

- Transient obstruction from clot or oedema is managed conservatively, and late strictures are mostly managed endoscopically. [PCNL Complications p10]
- Infundibular stenosis occurs in 2% after PCNL. [PCNL Complications p10]

Loss of renal function and death

- Renal function falls slightly immediately after PCNL, reaching a nadir at 48 hours, but long-term loss is negligible. [PCNL Complications p10]

- When function is reduced by stone obstruction, PCNL may improve renal function. [PCNL Complications p10]
- Death after PCNL is extremely rare and is usually due to underlying cardiovascular disease. [PCNL Complications p10]

CH 98 · SURGICAL MANAGEMENT OF UPPER URINARY TRACT CALCULI

Ectopic (pelvic) kidney with stone

- Problems for stone treatment in an ectopic kidney include high ureteral insertion, UPJO and malrotation. [Compilation 05-2 p3]
- Laparoscopic pyelolithotomy, with or without pyeloplasty, is an option for stones in an ectopic kidney. [Compilation 05-2 p3]
- SWL for a pelvic kidney needs the prone position, and impaired drainage should be excluded first (end of line partly cut off). [Compilation 05-2 p3]

Evaluation of outcome: residual fragments

- Plain KUB X-ray has a sensitivity of about 60% for residual fragments and can detect radio-opaque stones as small as 2 mm. [Compilation 05-2 p4]
- Nephrotomography is superior to the plain X-ray for residual fragments but is obsolete in many centres. [Compilation 05-2 p4]
- Ultrasound has a sensitivity of 50–70% for residual fragments and can also detect hydronephrosis. [Compilation 05-2 p4]
- CT is the preferred method to detect residual stone after PCNL (100% sensitivity), at the cost of radiation exposure and high cost. [Compilation 05-2 p4]
- Residual stones may pass spontaneously, cause symptoms if they impact and obstruct, act as a nidus for further stone formation, or be a source of persistent infection. [Compilation 05-2 p4]
- Clinically insignificant residual fragments (CIRF) are defined as fragments of 4 mm or less that are non-obstructive and non-infectious, with sterile urine, in an asymptomatic patient. [Compilation 05-2 p5]
- Outcomes of intervention are reported as stone-free rate, success rate (stone-free rate plus CIRF) and the effectiveness coefficient. [Compilation 05-2 p5]
- The effectiveness coefficient, used to compare treatment modalities, is % stone free divided by (100% + % retreatment + % auxiliary procedures), multiplied by 100. [Compilation 05-2 p5]

Combined ultrasonic–pneumatic devices

- On the combined device, the "Small Stone" pedal gives finer control and activates only the ultrasonic action of the inner probe, while the "Large Stone" pedal activates both modalities. [Compilation 05-2 p37]
- Both modalities appear safe, but contact time with dual-modality settings should be minimised. [Compilation 05-2 p37]
- The Lithoclast Ultra combines a pneumatic probe and an ultrasonic probe as two separate handpieces joined together. [Compilation 05-2 p69]
- The CyberWand uses a dual ultrasonic probe design. [Compilation 05-2 p69]

Colon injury in PCNL

- A colon injury detected during the operation is managed easily by separate drainage of the colon and the urinary tract. [Compilation 05-2 p53]
- Most colon perforations are extraperitoneal and can be managed conservatively, by pulling the nephrostomy tube back into the colon to act as a colostomy and draining the kidney with a double-J stent or another nephrostomy. [Compilation 05-2 p53]

Strategies for Surgical Management of Stones: Natural History

- Staghorn stones are named because they look like the antlers of a deer or stag on imaging. [Compilation 05 p75]

- Without treatment, complete loss of renal function can occur in 50% of patients with staghorn stones after 2 years. [Compilation 05 p75]
- Only about 15% of non-staghorn renal stones pass spontaneously, mainly those of 5 mm or less. [Compilation 05 p75]
- 10-20% (about 15%) of initially asymptomatic renal stones will require surgery at 3-4 years. [Compilation 05 p75]

Stone Burden (Non-Lower-Pole Algorithm and Non-Staghorn Stones)

- For non-staghorn stones of 1 cm or less, failed URS and SWL is an indication for PCNL. [Compilation 05 p77]
- For stones over 2 cm, URS is the alternative to PCNL in patients who are too frail for PCNL or who refuse blood transfusion (as well as in coagulopathy). [Compilation 05 p78]
- Super-mini PCNL uses 10F, 12F or 14F sheaths with active suction. [Compilation 05 p78]
- Micro-PCNL uses a 4.85 Fr instrument. [Compilation 05 p78]

Staghorn Stone: Other Options

- "Sandwich therapy" (PCNL combined with ESWL) is an option for staghorn stones. [Compilation 05 p79]

Stone Location: Lower Pole

- For an asymptomatic lower pole stone under 1 cm, observation is also an option. [Compilation 05 p80]
- A long infundibular length (more than 3 cm) is unfavourable lower pole anatomy. [Compilation 05 p80]
- A narrow infundibular width (less than 5 mm) is unfavourable lower pole anatomy. [Compilation 05 p80]

CH 99 · PREVENTION OF URINARY STONE DISEASE

Metabolic Evaluation: Indications

- Osteoporosis is an indication for metabolic evaluation in a stone former. [Compilation 05 p41]
- A pathological fracture is an indication for metabolic evaluation in a stone former. [Compilation 05 p41]
- Renal insufficiency is an indication for metabolic evaluation in a stone former. [Compilation 05 p41]
- Infirm health, meaning inability to tolerate repeated stone episodes, is an indication for metabolic evaluation. [Compilation 05 p41]

Metabolic Evaluation: Low-Risk Group (Abbreviated Protocol)

- In a low-risk single stone former, further evaluation is optional and should be discussed with the patient. [Compilation 05 p42]
- The multichannel blood screen of the abbreviated protocol includes a basic metabolic panel (sodium, potassium, chloride, CO2, BUN and creatinine), calcium, uric acid and intact PTH. [Compilation 05 p42]
- Digital tomosynthesis is a newer radiographic technique for documenting stones: a plain KUB film plus a single tomographic sweep of the X-ray emitter, reconstructed by digital software. [Compilation 05 p43]
- NCCT KUB is usually used for the acute stone episode rather than for metabolic evaluation, where plain KUB and IVU document current stones and their density. [Compilation 05 p43]

Metabolic Evaluation: High-Risk or Recurrent Formers

- The extensive diagnostic evaluation is time-consuming and difficult to perform, requires adherence to a restricted diet (older protocols), needs two 24-hour urine collections and has a high diagnostic yield. [Compilation 05 p44]
- The simplified diagnostic evaluation omits the fast and calcium load test and may not require strict dietary adherence. [Compilation 05 p44]
- A single 24-hour urine specimen may lead to misdiagnosis. [Compilation 05 p44]
- 24-hour urine is collected on ice or with an added preservative. [Compilation 05 p44]
- Blood tests in the extensive evaluation include a complete blood count and a comprehensive metabolic panel (14 tests covering kidney and liver function, glucose, parathyroid, electrolytes and

fluid balance), plus calcium, PTH, uric acid and creatinine. [Compilation 05 p44]
- Urinary sulfate is one of the 24-hour urine parameters. [Compilation 05 p45]

Medical Management: Fluids

- Fluid helps by mechanical diuresis, which prevents stasis, and by preventing supersaturation of stone components. The amount of fluid matters more than the type. [Compilation 05 p57]
- Water hardness is a minor concern. Drinking hard water raises urinary calcium, magnesium and citrate, with no significant change in urinary pH, oxalate or uric acid. [Compilation 05 p57]
- Carbonated beverages may give some protection by raising urinary citrate, and lemonade or soda flavoured with citric acid is helpful. [Compilation 05 p57]
- Soda flavoured with phosphoric acid increases the risk of stones. [Compilation 05 p57]

Medical Management: Diet

- Recommended protein restriction is 0.8-1 g/kg/day. [Compilation 05 p58]
- Controversy: some studies suggest that patients with hypocitraturic calcium oxalate stones may benefit from sodium supplementation. [Compilation 05 p58]
- Very high doses of vitamin C should be avoided by limiting daily intake to less than 2 g, because vitamin C may increase urinary oxalate excretion. [Compilation 05 p58]
- The daily calcium requirement is 1000 mg; moderate dietary calcium from dairy products, or calcium supplements taken with meals, lowers stone risk. [Compilation 05 p59]
- Calcium citrate is a stone-friendly supplement because of the inhibitory action of citrate. [Compilation 05 p59]
- Vitamin D supplementation is controversial because it may increase stones. Vitamin D repletion is likely safe, but 24-hour urinary calcium should be monitored. [Compilation 05 p59]
- Bisphosphonates, alone or with thiazide diuretics, improve bone density and decrease hypercalciuria, which reduces stone formation. [Compilation 05 p59]
- The Atkins diet (high protein, low carbohydrate) increases stone formation. [Compilation 05 p60]

Treatment of Absorptive Hypercalciuria

- Thiazides do not decrease intestinal calcium absorption. [Compilation 05 p61]
- Thiazide side effects include muscle cramps, lassitude, sleepiness and decreased libido. [Compilation 05 p61]
- Thiazides become less effective with long-term therapy, so a drug holiday may be needed. [Compilation 05 p61]
- Sodium cellulose phosphate is a non-absorbable resin that binds calcium and inhibits its absorption. It is no longer used. [Compilation 05 p61]
- Orthophosphate (a sodium or potassium salt of phosphorus) may have a role when other treatments are ineffective. [Compilation 05 p62]
- Orthophosphate works by inhibiting vitamin D synthesis and by directly altering renal tubular calcium reabsorption. [Compilation 05 p62]
- Problems with orthophosphate include parathyroid stimulation, soft-tissue calcification, a marked rise in urinary phosphorus, increased urinary saturation of brushite, GI distress, hypomagnesaemia and hyperoxaluria (written "hypo-oxaluria?" in the notes). [Compilation 05 p62]
- Absorptive hypercalciuria type II may need no specific drug treatment because the defect is not severe. It is managed with high fluid intake, moderate calcium intake and avoidance of a high-sodium diet. [Compilation 05 p62]

Treatment of Renal and Resorptive Hypercalciuria

- In renal hypercalciuria, thiazides correct the renal calcium leak, which normalises the secondary hyperparathyroidism, vitamin D and calcium levels. [Compilation 05 p63]
- Resorptive hypercalciuria has no established drug therapy. Orthophosphate may be used in mild to moderate cases. [Compilation 05 p63]
- Oestrogen may be used for resorptive hypercalciuria in postmenopausal women when surgery is not possible. [Compilation 05 p63]

Treatment of Cystine Stones

- Urine alkalinisation for cystinuria is difficult because the target pH is often unrealistic to achieve. [Compilation 05 p64]
- Mental status changes are a side effect of alpha-mercaptopropionylglycine (tiopronin). [Compilation 05 p64]
- Captopril works in cystinuria through its free -SH (sulfhydryl) group but is not often used. Side effects include hypotension, hyperkalaemia, chronic cough and bad taste. [Compilation 05 p65]

Treatment of Ammonium Acid Urate Stones

- Ammonium acid urate stones are treated with alkalinisation and the common therapies: fluids, moderate calcium intake and a low-oxalate diet. [Compilation 05 p65]
- The underlying cause must be corrected. Laxative abuse is managed by encouraging healthy bowel habits, and bowel disease or chronic diarrhoea is rectified if possible. [Compilation 05 p65]

Rx of Infectious Stones

- Besides DVT, side effects of acetohydroxamic acid include tremor, headache, palpitations, GI upset, oedema, rash, alopecia, anaemia, abdominal pain and loss of taste, and these side effects prevent its widespread use. [Compilation 05 p66]
- Renacidin (hemiacidrin) irrigation is associated with significant side effects and even death. [Compilation 05 p66]

Rx: Enteric Hyperoxaluria

- Calcium citrate is a suitable calcium preparation: the calcium binds oxalate in the intestine and reduces its absorption, and the citrate also helps prevent stones. [Compilation 05 p67]
- Magnesium gluconate is better tolerated than magnesium oxide or magnesium hydroxide for magnesium supplementation in enteric hyperoxaluria. [Compilation 05 p67]
- Cholestyramine works by binding bile salts in the intestine, which leaves calcium free to bind oxalate as calcium oxalate and so reduces oxalate absorption. [Compilation 05 p67]
- Probiotics that degrade oxalate (Oxalobacter formigenes or lactic acid bacteria) are a treatment option for enteric hyperoxaluria. [Compilation 05 p67]
- Pyridoxine is an alternative for patients with refractory enteric hyperoxaluria. [Compilation 05 p67]

Rx of Calcium Nephrolithiasis: Hypomagnesuric Calcium Stones

- Hypomagnesiuric calcium stone formers typically have low urinary citrate and low urine volume as well as low urinary magnesium. [Compilation 05 p68]
- The drugs for hypomagnesiuric calcium stones are magnesium oxide and magnesium hydroxide. [Compilation 05 p68]
- Potassium-magnesium citrate is a newer drug for hypomagnesiuric calcium stones. [Compilation 05 p68]
- Magnesium therapy increases urinary citrate, decreases urinary oxalate and gives a favourable calcium-to-magnesium ratio in urine. [Compilation 05 p68]

Hyperuricosuric Calcium Stones

- With allopurinol for hyperuricosuric calcium stones, moderate sodium restriction (150 mEq/day) is advisable. [Compilation 05 p69]
- Reduced dietary purine intake is a therapy for hyperuricosuric calcium stones. [Compilation 05 p69]
- In hyperuricosuric calcium stone disease, potassium citrate (30-60 mEq/day in divided doses) with high fluid intake decreases the supersaturation of both uric acid and calcium oxalate. [Compilation 05 p69]
- The evidence that higher-dose potassium citrate causes calcium phosphate stones by over-alkalinisation is conflicting. [Compilation 05 p69]

Hypocitraturic Calcium Oxalate Stones

- In chronic diarrhoeal states, slow-release potassium citrate tablets are poorly absorbed because of rapid intestinal transit, so a liquid preparation or a more frequent dosing schedule may be needed. [Compilation 05 p70]

- In distal RTA, potassium citrate corrects the metabolic acidosis and hypokalaemia as well as the urinary citrate. [Compilation 05 p70]
- In distal RTA, the overall rise in urine pH with potassium citrate is small because these patients already have a raised urine pH. [Compilation 05 p70]

CH 125 · GENITAL AND LOWER URINARY TRACT TRAUMA

Penile fracture: etiology and pathophysiology

- The tunica albuginea is composed of collagen and elastin. [Penile Fracture p1]
- The outer longitudinal layer determines the strength and thickness of the tunica albuginea, which vary at different locations along the shaft. [Penile Fracture p1]
- The tunica albuginea resists rupture until intracavernous pressure rises above 1500 mmHg. [Penile Fracture p1]
- Although rupture can occur anywhere along the shaft, most penile fractures are distal to the suspensory ligament. [Penile Fracture p2]
- The tunical tear is usually transverse and 1-2 cm long. [Penile Fracture p2]
- In vigorous intercourse, the rigid penis slips out of the vagina and strikes the perineum or pubic bone, which produces a buckling injury. [Penile Fracture p2]
- In the US and Europe, intercourse-related fractures happen more often in stressful situations such as extramarital sex. [Penile Fracture p2]

Diagnosis and investigations

- Because fear and embarrassment are common, patients may present very late. [Penile Fracture p3]
- MRI is useful but expensive, of limited availability and time-consuming. [Penile Fracture p4]
- Cavernosography is discouraged because it is time-consuming and unfamiliar to most urologists. [Penile Fracture p4]
- Retrograde urethrography is time-consuming and can be inaccurate, so intraoperative flexible cystoscopy is now often done routinely just before catheter placement. [Penile Fracture p4]

Management

- A distal circumcision (degloving) incision gives exposure to all three penile compartments. [Penile Fracture p4]
- In uncircumcised men, the distal circumcision incision puts the distal prepuce at risk of ischaemia, so limited circumcision at the end of the repair should be strongly considered. [Penile Fracture p5]

Outcome and complications

- Surgical reconstruction gives faster recovery, lower morbidity, fewer complications and long-term penile curvature in fewer than 5%. [Penile Fracture p5]
- Conservative management gives penile curvature in more than 10% of patients, abscess or debilitating plaques in 25-30%, and longer hospital stay and recovery time. [Penile Fracture p5]

Anterior Urethral Injury

- Constricting bands are a recognised cause of anterior urethral injury (listed with blunt trauma, sexual intercourse, penetrating trauma and iatrogenic injury). [Ureth Inj Stric p1]
- In a complete anterior urethral disruption managed with a suprapubic catheter, delayed reconstruction is usually done after 3 months, preceded by MCU/RGU. [Ureth Inj Stric p1]
- Causes of anterior urethral injury are blunt trauma, sexual intercourse, penetrating trauma, constricting bands and iatrogenic injury. Sexual intercourse and constricting bands are not listed as causes in the corpus. [Urethral Trauma p1]

Posterior Urethral Injury

- Posterior urethral injury is usually part of multisystem trauma caused by road traffic accidents, falls or industrial accidents. [Ureth Inj Stric p2]
- In children the injury commonly extends to the bladder neck because they have a rudimentary prostate. (The corpus states the extension but not this reason.) [Ureth Inj Stric p2]

- Posterior urethral injury is usually part of multisystem trauma from road traffic accidents, falls or industrial accidents. [Urethral Trauma p2]
- In children the tear extends to the bladder neck more often, because the prostate is still rudimentary. The corpus states that the tear extends but does not give this reason. [Urethral Trauma p2]

Posterior Urethral Injury: Management

- At open perineal posterior urethroplasty, the end point is that a dilator of at least 28 Fr passes into the bladder without resistance. (The corpus gives only spatulation to 30 Fr or more.) [Ureth Inj Stric p3]

About Other Options

- Some authors advocate immediate anastomotic reconstruction when there is a high-riding bladder, in addition to bladder neck injury. [Ureth Inj Stric p4]
- Female urethral disruption needs immediate primary repair, or at least realignment over a catheter, to avoid a later urethrovaginal fistula or urethral obliteration. [Ureth Inj Stric p4]
- In female urethral injury, vaginal lacerations should also be sutured to prevent vaginal stenosis. [Ureth Inj Stric p4]
- Delayed reconstruction in the female is very difficult because the small urethra is embedded in scar and is hard to mobilise. [Ureth Inj Stric p4]
- Primary realignment technique: a Coudé catheter is passed antegrade (through the bladder) out to the urethral meatus, tied to a second catheter, and the second catheter is drawn back into the bladder. [Ureth Inj Stric p4]
- Prolonged realignment attempts should be avoided because they risk infection of the pelvic haematoma. [Ureth Inj Stric p4]

Realignment and Delayed Endoscopic Treatment

- Realignment may not prevent symptomatic stenosis, but it may allow the resulting stricture to be managed endoscopically, and it eases open posterior urethroplasty by aligning the prostate and urethra and shortening the stricture. [Ureth Inj Stric p5]

Main Complications

- Anejaculation occurs in less than 4% after posterior urethral injury and repair. (The corpus gives <4% for incontinence only.) [Ureth Inj Stric p6]
- Men who are impotent are more likely to restenose after urethroplasty, because the bulbar urethra is ischaemic. [Urethral Trauma p6]
- Anejaculation after posterior urethroplasty occurs in under 4%, the same rate as incontinence. The corpus lists ejaculatory dysfunction but gives no rate. [Urethral Trauma p6]

Management: Delayed Posterior Urethroplasty

- The scar is fully excised once a dilator of at least 28 Fr passes into the bladder without resistance. A margin note says to dilate up to 32 Fr. The corpus gives only "spatulate to 30 Fr". [Urethral Trauma p3]

Other Options: Immediate Repair, Female Injury, Primary Realignment

- Some authors advise immediate anastomotic reconstruction when the bladder is high-riding, as well as for bladder neck injury. [Urethral Trauma p4]
- A female urethral disruption needs immediate primary repair, or at least realignment over a catheter, to prevent a later urethrovaginal fistula or urethral obliteration. [Urethral Trauma p4]
- Suture any vaginal lacerations with a female urethral injury, to prevent vaginal stenosis. [Urethral Trauma p4]
- Delayed reconstruction is very difficult in women, because the short urethra is embedded in scar and hard to mobilise. [Urethral Trauma p4]
- One primary-realignment technique passes a Coude catheter antegrade to the meatus, ties it to a second catheter and draws that catheter into the bladder. [Urethral Trauma p4]
- Avoid prolonged realignment attempts, because they risk infecting the pelvic haematoma. The corpus says to avoid them but does not give this reason. [Urethral Trauma p4]

Delayed Endoscopic Treatment

- A stricture that recurs after optical internal urethrotomy (OIU) or cut-to-the-light needs a 3-month period of urethral rest before definitive surgery. The corpus gives urethral rest as 6–8 weeks of suprapubic drainage, in the context of stricture assessment. [Urethral Trauma p5]

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Ureteral Injuries: Causes

- Ureteral injury occurs in under 4% of penetrating trauma and under 1% of blunt trauma. [Ureteral Injury p1]
- Gunshot wounds of the ureter come with bladder injury in about 5%. (The corpus gives only small bowel, colon, vessels and kidney 10–28%.) [Ureteral Injury p1]
- Hysterectomy is the commonest surgical cause, at about 54% of iatrogenic injuries (the corpus says "about half"). Colorectal surgery comes next, then ovarian surgery, then vascular surgery. Ovarian surgery is not named as a cause in the corpus. [Ureteral Injury p1]
- About 0.2% of ureteroscopic injuries need open surgery. [Ureteral Injury p1]
- Ureteral injury risk by intracorporeal lithotripsy energy runs electrohydraulic (EHL) > Nd:YAG laser > Ho:YAG laser. The corpus notes the EHL risk but does not give this ranking. [Ureteral Injury p1]

Diagnosis: CT, RGP, IVP

- Helical CT needs delayed images at 5–15 minutes. The corpus gives a single 10-minute delay. [Ureteral Injury p2]
- CT can miss an upper ureteric injury, because Gerota's fascia can contain the urinary extravasation and hide the diagnosis. [Ureteral Injury p2]
- Circumferential contrast extravasation is a CT sign of ureteral injury. [Ureteral Injury p2]
- Some centres use retrograde pyelography as the primary diagnostic test for ureteral injury. [Ureteral Injury p2]
- Use IVP intraoperatively when ureteral injury is suspected but preoperative imaging was negative. [Ureteral Injury p2]

Intra-operative Recognition

- The wound trajectory guides the search for a ureteral injury during the operation. [Ureteral Injury p3]
- Inject 1–2 mL of methylene blue and watch the pelvis for leakage. The corpus mentions IV methylene blue but gives no volume and does not describe this technique. [Ureteral Injury p3]
- If the intraoperative single-shot IVP is negative but injury is still suspected, watch for signs of delayed presentation: fever, leucocytosis and peritoneal irritation. [Ureteral Injury p3]

Management: Damage Control and Other Options

- One damage-control option is to do nothing to the ureter at the first operation and plan a re-operation at 24 hours. [Ureteral Injury p4]
- Autotransplantation is indicated when several attempts at ureteral repair have failed. [Ureteral Injury p4]

Ureteral Injury after External Trauma (algorithm)

- In an unstable blunt-trauma patient, a normal intraoperative one-shot IVP should prompt a search for other sources of haematuria (bladder, kidney, urethra). [Ureteral Injury p5]
- In an unstable penetrating-trauma patient with haematuria, do an intraoperative one-shot IVP. If it is normal, or if there is no haematuria, explore the ureter when the bullet or knife track passes near it. [Ureteral Injury p5]

Management: Recognised Intraoperatively

- At laparoscopy, the intraoperative recognition rate can be as low as 0% (compared with about one-third at open surgery). [Ureteral Injury p6]
- If a stented repair fails, the endoscopic options include Acucise endoureterotomy, balloon dilation and laser incision. Acucise is not named in the corpus. [Ureteral Injury p6]

Injuries Recognised in the Post-op Period

- When the retrograde or antegrade stent fails for a major injury found late, place a PCN and wait about 6 weeks before definitive repair. (Paper 3 has the 6-week delay; the notes add the PCN step.) [Ureteral Injury p7]

General Principles of Ureteral Repair

- Debride minimally in general, but more judiciously (wider) in gunshot injuries. [Ureteral Injury p8]
- Use absorbable monofilament suture such as 5-0 PDS for the anastomosis. The corpus says only "5-0 absorbable" or "fine absorbable". [Ureteral Injury p8]

Ureteral strictures: etiology

- Malignant causes of ureteral stricture include transitional cell carcinoma and cervical cancer. [Upper Tract Trauma Stricture p17]
- Periureteral fibrosis causing stricture may be secondary to an aortic aneurysm or endometriosis, and some strictures remain idiopathic. [Upper Tract Trauma Stricture p17]

Ureteral strictures: indications for intervention

- Recurrent pyelonephritis is an indication for intervention in ureteral stricture, alongside the need to exclude malignancy and ongoing obstruction or pain. [Upper Tract Trauma Stricture p18]

Endourological options: ureteral stent and balloon dilatation

- Ureteral stenting is effective acutely for most ureteral strictures, especially intrinsic ones, with about 88% success. [Upper Tract Trauma Stricture p19]
- Balloon dilatation is most commonly done retrogradely; the antegrade approach is used when retrograde access is not possible. [Upper Tract Trauma Stricture p19]
- A high-pressure balloon 4 cm long and 5-8 mm in diameter is used (margin note: "Endoburst"). [Upper Tract Trauma Stricture p19]
- The balloon is kept inflated as a tamponade for about 10 minutes, after which the waist usually disappears. [Upper Tract Trauma Stricture p19]

Endourological options: balloon results and endoureterotomy

- Balloon dilatation is done under fluoroscopy and followed by diuretic renography at 1, 6 and 12 months, continuing to 2 years. [Upper Tract Trauma Stricture p20]
- Alternatively, a low-profile balloon can be passed through the ureteroscope and the stricture dilated under direct vision. [Upper Tract Trauma Stricture p20]
- The overall success rate of balloon dilatation is 50-75%. [Upper Tract Trauma Stricture p20]
- Endoureterotomy can be performed with a cold knife, a cutting electrode or a holmium laser. [Upper Tract Trauma Stricture p20]
- Holmium laser endoureterotomy has a success rate of 65-80%. [Upper Tract Trauma Stricture p20]
- After endoureterotomy a larger double-J stent (8-12 Fr) is left for 6 weeks, with diuretic renogram follow-up at 1, 6, 12 and 24 months. [Upper Tract Trauma Stricture p21]

Surgical repair: psoas hitch

- The psoas hitch may be fixed to the psoas major muscle as an alternative to the psoas minor tendon. [Upper Tract Trauma Stricture p23]

Surgical repair: Boari flap technique

- Incisions for a Boari flap are Pfannenstiel, midline or Gibson. [Upper Tract Trauma Stricture p25]

Surgical repair: Davis intubated ureterotomy

- Davis intubated ureterotomy was used for very long strictures, and a modern modification adds a buccal mucosa graft. [Upper Tract Trauma Stricture p26]

Surgical repair: ileal ureteral substitution

- A variation of ileal ureter is ileocalycostomy, used when the renal pelvis is scarred or intrarenal. [Upper Tract Trauma Stricture p27]

Management of ureteroenteric anastomotic stricture

- Length and renal function of a ureteroenteric stricture are assessed with CT urography, IVU, loopogram, antegrade nephrostogram and DTPA renogram. [Upper Tract Trauma Stricture p28]

OTHER TOPICS IN THIS BOOK

Treatment of stones in calyceal diverticula

- PCNL should be considered first-line for most diverticular stones, with a stone-free rate of 70–100%. [Anomalies p3]
- Diverticular ablation rates after PCNL are over 70%. [Anomalies p4]
- A resectoscope with a roller-ball electrode allows easy fulguration of the diverticular lining, and dilatation and trauma can also cause ablation. [Anomalies p4]
- Finding a flattened renal papilla means an obstructed calyx rather than a diverticulum. [Anomalies p4]
- In an anteriorly located diverticulum, dilating or incising the neck is difficult because of unfavourable angles. [Anomalies p4]
- Puncture is difficult in small and upper-pole diverticula. [Anomalies p4]
- Passing a guidewire from the diverticulum into the pelvis is almost impossible. [Anomalies p4]
- When direct puncture fails, a neighbouring calyx can be punctured and the diverticulum entered indirectly, by perforating its wall or retrogradely through the diverticular neck. [Anomalies p4]
- Retrograde contrast may fail to fill the diverticulum because of a narrow infundibulum, so ultrasound-guided puncture or CT-guided direct contrast injection into the diverticulum can be used to target it. [Anomalies p4]
- URS is first-line for diverticular stones under 2 cm in the upper or middle calyx when the diverticular neck is short and identifiable. [Anomalies p5]
- URS gives a stone-free rate of 50–90%, but diverticular obliteration only about 20%, and a staged procedure may be needed. [Anomalies p5]
- At URS the diverticular ostium can be identified in only 75% of cases. [Anomalies p5]
- SWL for diverticular stones gives a stone-free rate of 4–58%, and it neither ablates the lining nor dilates the neck. [Anomalies p5]
- Laparoscopic or robotic treatment of diverticular stones gives a stone-free rate of about 100%. [Anomalies p5]

CO2 laser

- The CO2 laser is an infrared gas laser that emits a continuous wave. [LASER p4]
- Indications for the CO2 laser also include skin resurfacing and vocal cord lesions. [LASER p4]
- The CO2 laser is delivered through a rigid articulated-arm device rather than a fibre. [LASER p4]
- The tissue effects of the CO2 laser are vaporisation and coagulation. [LASER p4]

Lasers in lithotripsy: holmium versus erbium:YAG

- Erbium:YAG is the second laser (with holmium) used for lithotripsy. [LASER p5]
- The Er:YAG wavelength is 2940 nm. [LASER p5]
- Er:YAG acts by optical (photomechanical) energy rather than photothermal vaporisation. [LASER p5]
- Shock-wave production is weak with holmium but strong with Er:YAG. [LASER p5]
- Holmium produces a pear-shaped vapour bubble (with more heat loss), whereas Er:YAG produces a torpedo-shaped bubble. [LASER p5]
- Er:YAG has a smaller, safer soft-tissue penetration depth than holmium. [LASER p5]
- Holmium fragmentation is slow with fine fragments, whereas Er:YAG fragmentation is fast with large fragments. [LASER p5]
- Er:YAG energy cannot be carried by standard silica quartz fibres, which are compatible only with holmium. [LASER p5]
- Er:YAG has a shorter pulse duration than holmium. [LASER p5]

Problems with Er:YAG technology

- The main barrier to Er:YAG in endourology is the lack of fibres with adequate mechanical, chemical and thermal reliability. [LASER p6]
- Sapphire fibres are too brittle and too thick to use in endourology. [LASER p6]
- ZBLAN fluoride glass fibres have a low melting point and inadequate durability. [LASER p6]
- Hybrid fibres (fluoride core, sapphire tip and circumferential cladding) are under evaluation for Er:YAG. [LASER p6]

General points: laser physics

- Laser emission begins when an atom stimulated by an external energy source has its electrons excited; they release the excess energy as photons, which form highly concentrated energy. [LASER p10]

What are Nanobacteria?

- Nanobacteria were proposed as cell-walled microorganisms smaller than the generally accepted lower size limit for life, which is about 200 nm for bacteria such as mycoplasma. [Nano T B p6]
- Listed properties of nanobacteria include being bacteria-like, pleomorphic and infectious. [Nano T B p6]
- Nanobacteria are said to form a biofilm. [Nano T B p6]

Role in Urolithiasis: Formation of Calcification

- Proposed sequence: Nanobacterium sanguineum is engulfed by phagocytosis, provokes only a weak inflammatory response because of low virulence, and produces a biofilm that host defences cannot penetrate. [Nano T B p7]
- Within the biofilm, calcium and phosphate fixation leads to calcium carbonate and hydroxyapatite formation, giving calcification and stones. [Nano T B p7]
- Nanobacteria have been shown to form calcium stones in modified Loeffler's medium. [Nano T B p7]
- In vivo, stone formation in rats inoculated with nanobacteria was dose dependent. [Nano T B p7]

Role of Fetuin

- Fetuin is an inhibitor of mineralisation; it is a plasma protein secreted by the liver that binds calcium and phosphorus and so inhibits biomineralisation. [Nano T B p8]
- When calcium and phosphorus reach saturation levels, fetuin's inhibitory role is overcome and it precipitates as mineraloprotein complexes that grow and form a nidus for stone formation. [Nano T B p8]
- Factors that overcome fetuin's inhibitory role are fetuin gene polymorphism, saturation levels of calcium and phosphorus, and an altered microenvironment such as pH. [Nano T B p8]

Supine PCNL

- Supine PCNL carries less risk of hypothermia and sepsis than prone PCNL. [Tubless Supine PCNL p1]
- Supine PCNL reduces surgeon fatigue because the surgeon can operate sitting. [Tubless Supine PCNL p1]
- In the supine position, access to the posterior calyces is not always possible. [Tubless Supine PCNL p2]
- Supine PCNL is less familiar to most urologists. [Tubless Supine PCNL p2]
- In supine positioning, the patient is placed at the table edge with a support such as a 3 L saline bag under the ipsilateral flank. [Tubless Supine PCNL p2]

Tubeless PCNL

- Tubeless PCNL allows a faster return to work. [Tubless Supine PCNL p3]
- A ureteric stent may not drain the kidney as well as a nephrostomy tube. [Tubless Supine PCNL p4]
- Low-volume stone disease (less than 3 cm) is an indication for tubeless PCNL. [Tubless Supine PCNL p4]

Renal Tubular Acidosis: comparison and type 1

- Plasma bicarbonate in type 1 RTA is very low, below 12 mEq/L. [Acid Base RTA p5]
- Plasma bicarbonate in type 4 RTA is only mildly low, above 17 mEq/L. [Acid Base RTA p5]
- Autoimmune thyroiditis is associated with distal (type 1) RTA. [Acid Base RTA p5]
- Toxic nephropathy is associated with distal (type 1) RTA. [Acid Base RTA p5]

- Lithium is a drug cause of distal (type 1) RTA. [Acid Base RTA p5]
- Renal transplantation is a secondary cause of distal (type 1) RTA. [Acid Base RTA p5]
- Sickle cell anaemia is a secondary cause of distal (type 1) RTA. [Acid Base RTA p5]
- In distal RTA, sodium bicarbonate alkalinises the urine, lowers aldosterone and so helps correct sodium and potassium. [Acid Base RTA p5]

Type 2 (proximal) RTA

- Primary proximal RTA may be idiopathic or familial. [Acid Base RTA p7]
- Amphotericin B is listed as an acquired cause of proximal (type 2) RTA. [Acid Base RTA p7]
- 6-mercaptopurine is an acquired cause of proximal (type 2) RTA. [Acid Base RTA p7]

Type 4 RTA

- Patients with type 4 RTA often have associated azotaemia and hypertension. [Acid Base RTA p8]

Stone Analysis: Methods

- MR spectrometry is listed as a method of stone analysis. [Stone Analysis p1]
- Chromatography is listed as a method of stone analysis. [Stone Analysis p1]
- Ultraviolet-visible spectroscopy is listed as a method of stone analysis. [Stone Analysis p1]
- Photomicroscopy is listed as a method of stone analysis. [Stone Analysis p1]
- Fluorescence is listed as a method of stone analysis. [Stone Analysis p1]

Fourier Transform Infrared Spectroscopy (FTIR)

- FTIR obtains an infrared spectrum of a solid, liquid or gas by measuring the coherence of a radiation source (written as "radioactive source" in the notes). [Stone Analysis p2]
- The Fourier transform is a mathematical process that converts the raw data (the interferogram) into the actual spectrum. [Stone Analysis p2]
- The FTIR pathway is: infrared source, then interferometer, then stone powder on the surface of the spectrophotometer, then interferogram, then Fourier-transform computer, then spectrum, then result. [Stone Analysis p2]
- FTIR is fast and needs less than 2 mg of stone fragments. [Stone Analysis p2]
- FTIR gives information about the core and the surface of a stone separately. [Stone Analysis p2]
- The ATR (attenuated total reflectance) technique increases the sensitivity of FTIR. [Stone Analysis p2]

Stone analysis procedure (FTIR)

- The stone is received in a sterile container, then cleaned, dried and stored in an air-conditioned environment. [Stone Analysis p3]
- Large stones are fractured so that the core and the surface can be analysed separately. [Stone Analysis p3]
- The stone is pulverised with a pestle and mortar to a homogeneous powder. [Stone Analysis p3]
- About 2 mg of the powder is placed on the flat surface of the spectrophotometer. [Stone Analysis p3]
- The spectra are collected with software and compared with synthetic kidney-stone spectra in a reference library. [Stone Analysis p3]
- A report is issued after the library comparison. [Stone Analysis p3]

Pregnancy & Stones: epidemiology

- Stones in pregnancy are more common in multiparous women. [Stone in Pregnancy p1]
- After adjusting for age, the incidence is the same in multiparous and primiparous women. [Stone in Pregnancy p1]

Aetiology and evaluation

- The increased renal blood flow and GFR of pregnancy cause hyperuricosuria as well as hypercalciuria. [Stone in Pregnancy p2]
- The rise in urine pH is most marked in the second trimester. [Stone in Pregnancy p2]
- Diagnosis can be difficult because the gravid uterus may mask signs and symptoms. [Stone in Pregnancy p2]

- The commonest presentation is flank pain, with microscopic or macroscopic haematuria in most cases and UTI in some; nausea and vomiting also occur. [Stone in Pregnancy p2]

Ultrasound modifications and other imaging

- Colour Doppler can tell a dilated ureter from the iliac artery. [Stone in Pregnancy p3]
- A dilated ureter below the iliac vessels (infra-iliac) suggests a distal stone. [Stone in Pregnancy p3]
- The resistive index does not change in normal pregnancy, so a raised RI points to obstruction. [Stone in Pregnancy p3]
- A limited IVP in pregnancy consists of a scout film, a 30-second film and a 20-minute film. [Stone in Pregnancy p3]
- Nuclear renography uses limited radiation and can diagnose obstruction, but the bladder acts as a reservoir that exposes the fetus; this is reduced by extra water intake, frequent voiding and possibly a catheter. [Stone in Pregnancy p3]
- The HASTE MRI protocol (half-Fourier single-shot turbo spin echo) may improve the diagnosis of stones in pregnancy. [Stone in Pregnancy p3]

Radiation in pregnancy

- The fetus is most sensitive to radiation in the first trimester, during organogenesis. [Stone in Pregnancy p4]
- The fetal effects of radiation are teratogenesis, carcinogenesis and mutagenesis, and they depend on the dose and the stage of pregnancy. [Stone in Pregnancy p4]
- In the second and third trimesters the teratogenic risk falls, but a risk of childhood malignancy remains. [Stone in Pregnancy p4]
- No threshold radiation dose has been defined below which harmful effects cannot occur. [Stone in Pregnancy p4]
- The reliability of low-dose CT for diagnosing calculi in pregnancy is controversial. [Stone in Pregnancy p4]

Treatment

- A DJ stent in pregnancy can be placed under local anaesthesia. [Stone in Pregnancy p5]
- During ureteroscopy in pregnancy, fetal X-ray exposure is reduced by using a below-table X-ray source as well as a lead apron under the patient. [Stone in Pregnancy p5]

Nephrostomy and first-trimester upper ureteric stone

- The drawbacks of nephrostomy in pregnancy are the morbidity of carrying a tube, tube blockage, the need for flushing and replacement, and the tube acting as a source of infection. [Stone in Pregnancy p6]
- For an upper ureteric stone in the first trimester, a nephrostomy or DJ stent followed by definitive surgery after delivery is not a good option, because the drain must stay for a long time and needs many changes; ureteroscopy with laser is the best option. [Stone in Pregnancy p6]

Complications of URS

- The overall complication rate of ureteroscopy is 3.5%. [urs p1]
- Fever is the most frequent complication of ureteroscopy, at 1.8% (corpus gives only sepsis ~5%). [urs p1]
- Ureteral perforation rates have decreased over time because of improved technique and instruments. [urs p1]
- Causes of perforation include splitting of the ureter after balloon dilatation. [urs p1]
- Causes of perforation include basket and grasper injury. [urs p1]
- Causes of perforation include pressurised irrigation. [urs p1]
- After perforation, give antibiotics in addition to stopping and stenting. [urs p1]
- After the stent is removed at 4 weeks following perforation, imaging should be done. [urs p1]

Stricture

- An impacted stone is defined as one past which a guidewire cannot be passed (the alternative definition, not moved in 2 months, is in the corpus). [urs p2]

- Previous ureteral surgery is a risk factor for stricture after ureteroscopy. [urs p2]

Avulsion

- Ureteral avulsion is the most catastrophic complication of ureteroscopy. [urs p3]
- Avulsion can occur when a large instrument that was forcefully advanced is withdrawn. [urs p3]
- The commonest presentation of avulsion is a portion of ureter being withdrawn from the patient along with the instrument. [urs p3]
- Although delayed repair of avulsion is preferred, immediate repair is also an option. [urs p3]
- After avulsion, any urinoma should be drained percutaneously. [urs p3]

Submucosal stone and lost stone

- Stone extrusion can occur in 2% of ureteroscopic procedures. [urs p4]
- A stone that migrates submucosally is difficult to remove; treatment is laser excision with double-J stenting, and late stricture formation is a complication. [urs p4]
- A lost stone (complete extrusion outside the ureter) can be left in place, because attempted retrieval increases the risk of injury and extravasation. [urs p4]
- For a lost stone, stop the procedure, place a stent and give antibiotics to prevent urinoma or abscess; late stricture formation can follow. [urs p4]

Other complications

- Fluid extravasation during ureteroscopy can cause congestive cardiac failure and pulmonary oedema. [urs p4]
- Deep vein thrombosis is a complication of ureteroscopy. [urs p4]
- Musculoskeletal injuries are a complication of ureteroscopy. [urs p4]

Ureteral access sheath

- Use of the ureteral access sheath was first reported by Takayasu and Aso in 1974. [urs p5]
- The ureteral access sheath was popularised two decades later. [urs p5]
- The hydrophilic coating creates a low-friction surface, and the sheath is strengthened to resist kinking. [urs p5]
- The access sheath decreases operative time. [urs p5]
- With an access sheath there is no need to drain the bladder separately. [urs p5]
- Higher stone-free rates have been reported with the access sheath. [urs p5]
- The access sheath facilitates active extraction of stone fragments with minimal ureteric trauma. [urs p5]
- Access sheath manufacturers include Cook, Boston Scientific, Rocamed, Coloplast and Olympus, and sheaths are available in different lengths and diameters. [urs p6]

12 Bladder Cancer & Diversion

CH 129 · TUMOURS OF THE URINARY BLADDER

Inverted Papilloma

- Inverted papilloma is associated with chronic inflammation and bladder outlet obstruction. [Compilation 01 p1]
- Inverted papilloma is composed of anastomosing islands of normal urothelial cells that invaginate from the surface into the lamina propria but not into the muscularis propria. [Compilation 01 p1]
- Although commonest at the trigone, inverted papilloma can occur anywhere in the bladder. [Compilation 01 p1]
- The notes divide inverted papilloma into Type I (benign) and Type II (malignant). [Compilation 01 p1]

Papilloma

- About 75% of urothelial papillomas carry FGFR3 mutations. [Compilation 01 p1]

Nephrogenic Adenoma

- Causes of the chronic irritation behind nephrogenic adenoma include trauma, previous surgery, renal transplantation, intravesical chemotherapy, stones, catheters and infection. [Compilation 01 p1-2]
- The glandular tubules of nephrogenic adenoma resemble renal tubules and involve the mucosa and submucosa. [Compilation 01 p1-2]
- Nephrogenic adenoma is a vascular lesion and presents with gross haematuria in most cases. [Compilation 01 p1-2]
- Two theories of origin are proposed: metaplasia from chronic irritation, or nests of displaced mesonephric tissue in the urothelium that are activated by mucosal injury. [Compilation 01 p1-2]
- Treatment is transurethral resection together with elimination of the source of chronic irritation. [Compilation 01 p1-2]

Leiomyoma

- Leiomyoma is the commonest non-epithelial benign tumour of the bladder. [Compilation 01 p2]
- On cystoscopy a leiomyoma shows a smooth indentation of the bladder covered by normal mucosa. [Compilation 01 p2]

Bladder Cancer: Epidemiology

- Bladder cancer is commoner in whites than in African Americans, the opposite of most other tumours. [Compilation 01 p3]
- In adolescents and young adults, bladder tumours are usually well differentiated and non-invasive, the opposite of most other tumours. [Compilation 01 p3]

Etiology: Genetic Factors

- The slow NAT-2 polymorphism carries an odds ratio of 1.4 for bladder cancer compared with the fast polymorphism. [Compilation 01 p3-4]
- GSTM1 (glutathione-S-transferase M1) conjugates reactive chemicals such as arylamines and nitrosamines, and the null GSTM1 polymorphism increases bladder cancer risk 1.5 times. [Compilation 01 p3-4]
- Slow NAT-2 combined with null GSTM1 leads to high levels of 3-aminobiphenyl (as written) and a higher risk of bladder cancer. [Compilation 01 p3-4]

Etiology: External Risk Factors

- Aromatic amine exposure includes diesel exhaust and paint substances, in addition to benzidine, beta-naphthylamine and polycyclic aromatic hydrocarbons. [Compilation 01 p4-6]
- The latent period after aromatic amine exposure is 10-20 years. [Compilation 01 p4-6]
- Aromatic amines enter by inhalation or through the skin. [Compilation 01 p4-6]
- Bladder cancer risk is linearly related to the duration and intensity of smoking. [Compilation 01 p4-6]
- Smoking cessation for 1-3 years still leaves a relative risk of 2.6, falling to 1.1 after 15 years. [Compilation 01 p4-6]
- HPV codes two oncoproteins, E6 and E7; E6 interacts with TP53, which has a role in bladder cancer formation and progression. [Compilation 01 p4-6]
- BK virus causes severe haemorrhagic cystitis in bone marrow transplant patients, but no consistent link between BK virus infection and bladder carcinoma has been shown. [Compilation 01 p4-6]
- Chronic UTI carries a relative risk of 1.4-1.6 for bladder cancer, possibly through production of carcinogens such as nitrosamines by bacteria. [Compilation 01 p4-6]
- Evidence for radiation as a cause of bladder cancer includes atomic bomb survivors. [Compilation 01 p4-6]
- Green tea is protective because of antioxidant polyphenolic compounds, especially epigallocatechin-3-gallate. [Compilation 01 p4-6]

TNM Classification of Bladder Cancer

- All lymph nodes above the aortic bifurcation are classed as distant lymph nodes. [Compilation 01 p8]
- The true-pelvis regional nodes for N1/N2 are the hypogastric, obturator, external iliac and presacral nodes. [Compilation 01 p8]

Staging and Pathology

- Five-year survival by stage is over 85% for stage 0a/0is, 65-75% for stage I, 55% for stage II, about 30% for stage III and 15-25% for stage IV, with 5-20% for M1 disease. [Compilation 01 p9]
- Urothelial carcinoma is the second commonest cause of death among genitourinary tumours. [Compilation 01 p9]

Grade Distribution and Precursor Lesions

- NMIBC grade distribution is about 25% PUNLMP, 50% low grade and 25% high grade. [Compilation 01 p10]
- The membrane of umbrella cells is rigid and made of uroplakin proteins, and umbrella cells form part of the urine-bladder barrier that stops urinary toxins from transforming urothelial cells. [Compilation 01 p10]
- Hyperplasia may show loss of parts of chromosome 9, the most frequent deletion in hyperplasia. [Compilation 01 p10]
- Dysplasia shows allelic loss of chromosome 9 and occasional TP53 abnormalities. [Compilation 01 p10]

Origin, Recurrence, Dissemination

- Deletion of part or all of chromosome 9 is the earliest mutation seen in low malignant potential NMIBC; the chromosome 9 tumour suppressor gene (9q21 site, as written) acts as a negative regulator of RB. [Compilation 01 p11-12]
- Multifocality and rapidly recurring tumours are strong predictors of finding histologically abnormal urothelium in a normal-appearing bladder. [Compilation 01 p11-12]
- Angiolymphatic invasion is a poor prognostic sign with a 40% risk of nodal disease and is an independent predictor of overall and cancer-specific survival. [Compilation 01 p11-12]
- Immunohistochemistry with CD31 and CD34 monoclonal antibodies should be used to identify blood vessels accurately when assessing angiolymphatic invasion. [Compilation 01 p11-12]
- Pagetoid spread is spread of cancer cells underneath a layer of normal-appearing surface urothelium, seen mainly in urothelial CIS. [Compilation 01 p11-12]
- Pagetoid spread is seen in 15% of bladders with CIS and in 11% of high-grade papillary NMIBC. [Compilation 01 p11-12]
- Pagetoid spread can extend into the prostatic urethra and distal ureters. [Compilation 01 p11-12]

Prognostic Factors and Molecular Pathways

- Proliferation markers such as PCNA and MIB-1 are found in high-grade tumours and indicate poor prognosis. [Compilation 01 p14-15]
- Three pathways are proposed; the third runs from normal urothelium through hyperplasia or dysplasia to high-grade papillary carcinoma and then muscle-invasive disease. [Compilation 01 p14-15]
- PUNLMP has the same proliferation rate as low-grade urothelial carcinoma. [Compilation 01 p14-15]

Primary CIS and Detection of Urothelial Cancer

- Primary CIS is CIS not associated with a previous or concurrent papillary or invasive lesion, and with no previous CIS. [Compilation 01 p16]
- Microscopic haematuria is present in all patients with bladder cancer. [Compilation 01 p16]
- Haematuria can be intermittent, so any episode should be evaluated even if a later urinalysis is negative. [Compilation 01 p16]
- Of patients with gross haematuria, 50% have a demonstrable cause and 12% have a bladder tumour. [Compilation 01 p16]
- Microscopic haematuria carries a 4.1% risk of bladder malignancy. [Compilation 01 p16]
- Irritative voiding symptoms with microscopic haematuria double the risk of bladder cancer, especially CIS. [Compilation 01 p16]
- At presentation NMIBC is about 70% Ta, 20% T1 and 10% CIS. [Compilation 01 p16]

Cystoscopy, Blue Light and NBI

- Cysview solution (under 2 ounces) is delivered to the bladder 1 hour before cystoscopy, and normal white-light cystoscopy is followed by blue-light cystoscopy. [Compilation 01 p17]
- Side effects of Cysview include allergic reaction, bladder pain, cystitis, dysuria and haematuria. [Compilation 01 p17]
- In NBI the depth of light penetration into the bladder wall increases with increasing wavelength. [Compilation 01 p17]
- In NBI, vascular structures appear green or dark brown against a pink or white mucosal background. [Compilation 01 p17]
- White-light cystoscopy has a sensitivity of 87% and specificity of 85%, whereas NBI has a sensitivity of 100% and specificity of 82%. [Compilation 01 p17]

Random Bladder Biopsies and Urine Cytology

- Random bladder biopsies have an overall detection rate of 2.5% in specimens with known or suspected bladder tumour and 23% in specimens with a concurrent bladder tumour. [Compilation 01 p18]
- Random biopsies are reasonable in high-risk patients who have had intravesical therapy. [Compilation 01 p18]
- The performance of urine cytology depends on the cytopathologist and the number of samples, as well as on tumour stage and grade. [Compilation 01 p18]
- Instrumented (bladder wash) cytology improves sensitivity and specificity but needs an invasive procedure. [Compilation 01 p18]

Urinary Molecular Markers

- The FDA-approved urinary markers listed are NMP-22, FISH (UroVysion), BTA stat, BTA TRAK and ImmunoCyt/uCyt+. [Compilation 01 p19-24]
- The NMP-22 cut-off is 10 units/mL. [Compilation 01 p19-24]
- NMP-22 concentration in urine is 20 times higher in bladder cancer patients than in non-cancer patients. [Compilation 01 p19-24]
- NMP-22 gives false positives with active UTI or significant haematuria. [Compilation 01 p19-24]
- NMP-22 results are available in 30 minutes. [Compilation 01 p19-24]
- Lowering the NMP-22 cut-off to 5 improves sensitivity but significantly worsens specificity. [Compilation 01 p19-24]
- NMP-22 sensitivity rises with stage: 35% for Ta, 65% for T1 and 88% for T2. [Compilation 01 p19-24]
- Combining cystoscopy with NMP-22 gives a higher detection rate. [Compilation 01 p19-24]
- NMP-22 (BladderChek test) costs less than cytology. [Compilation 01 p19-24]
- FISH is useful for high-grade disease and may be anticipatory, turning positive before a new tumour forms. [Compilation 01 p19-24]
- In BTA, the complement factor H and factor H-related proteins are present in bladder cancer cells and inhibit the complement cascade to prevent cell lysis. [Compilation 01 p19-24]
- BTA false positives also occur with urolithiasis, inflammation, recent instrumentation, other genitourinary malignancies and intravesical BCG therapy. [Compilation 01 p19-24]
- ImmunoCyt uses three fluorescent monoclonal antibodies: LDQ10 and M344 (fluorescein-labelled, against mucin) and 19A211 (Texas Red, against a CEA antigen). [Compilation 01 p19-24]
- ImmunoCyt requires 500 negative cells on the slide to give a negative report. [Compilation 01 p19-24]
- ImmunoCyt false positives occur with BPH, infection and urolithiasis. [Compilation 01 p19-24]
- FGFR-3 (CertNDx) is high in low-grade pTa tumours and is a marker of favourable disease with low recurrence and improved survival. [Compilation 01 p19-24]
- FGFR-3 (CertNDx) is not affected by the degree of haematuria or other urinary tract diseases. [Compilation 01 p19-24]
- Microsatellite analysis uses PCR amplification to detect tumour-associated loss of heterozygosity, with sensitivity 72-97% and specificity 80-100%, and was described as the most promising marker. [Compilation 01 p19-24]

- DNA methylation analysis of CpG islands has a sensitivity of 75%, but methylated CpG islands can be found in normal urothelial cells of older patients. [Compilation 01 p19-24]
- Lewis blood group antigen X has no commercially available test and is usually absent from adult urothelial cells except occasional umbrella cells. [Compilation 01 p19-24]
- CK20 and CYFRA 21-1 are cytoskeletal proteins detectable in urine by protein or mRNA assays, and CYFRA 21-1 is not a useful marker in its current form. [Compilation 01 p19-24]
- Survivin is an anti-apoptotic protein highly expressed in urothelial cancer cells; it may predict response to intravesical therapy but is relatively poor at detecting advanced-stage or high-grade tumours. [Compilation 01 p19-24]
- Urinary hyaluronic acid has a role in cell proliferation and its amount correlates with disease stage. [Compilation 01 p19-24]
- Telomerase activity is detected in 80% of urine samples from bladder cancer patients. [Compilation 01 p19-24]
- BLCA-4 is a nuclear transcription factor marker with high specificity and sensitivity. [Compilation 01 p19-24]

Optical imaging techniques in high-grade bladder cancer

- White-light cystoscopy cannot distinguish low-grade from high-grade tumour and cannot distinguish CIS from inflammation. [Compilation 01 p35]
- For blue-light cystoscopy (photodynamic diagnosis), 5-ALA or HAL is instilled into the bladder 1 hour beforehand and blue light of 375–440 nm is used. [Compilation 01 p35]
- Blue-light cystoscopy does not distinguish high-grade from low-grade tumour, and gives false positives after BCG, after a previous biopsy or TURBT, and with inflammation. [Compilation 01 p35]
- Narrow-band imaging does not distinguish high-grade from low-grade tumour. [Compilation 01 p35]
- Confocal laser endomicroscopy illuminates tissue with a low-power laser and detects the fluorescent light emitted from it, with topical or intravenous fluorescein, to a depth of about 120 µm. [Compilation 01 p35]
- Confocal laser endomicroscopy can distinguish high-grade from low-grade bladder tumours, but it has a restricted view, needs contrast and cannot detect muscle invasion. [Compilation 01 p35]
- Raman shift spectroscopy, named after the Indian physicist C.V. Raman, measures the energy that photons gain or lose from intramolecular bonds in tissue; the emitted wavelengths are studied as wave numbers, and the Raman shift is the change in photon energy. [Compilation 01 p36]
- Raman spectroscopy can assess invasion and distinguish low-grade from high-grade bladder tumour, but it is still in trials. [Compilation 01 p36]
- UV autofluorescence uses ultraviolet excitation of tissue and can detect invasion and distinguish low-grade from high-grade tumours, but its drawback is toxicity. [Compilation 01 p36]
- Multiphoton microscopy relies on absorption of two infrared photons (fluorescence imaging) to a depth of about 0.5 mm; it distinguishes benign from malignant tissue and allows cytological grading, but it does not detect muscle invasion. [Compilation 01 p37]
- Scanning fibre endoscopy uses a small flexible endoscope with laser-based imaging to give a panoramic view of the urothelium, and is still in trials. [Compilation 01 p37]
- Molecular imaging visualises and characterises biological processes at the molecular and cellular level. [Compilation 01 p37]
- In bladder molecular imaging, fluorescently labelled anti-CD47 antibody is instilled, the bladder is emptied, and optical imaging with confocal laser endomicroscopy follows; because anti-CD47 binds only tumour cells, the fluorescent tissue is biopsied. [Compilation 01 p37]

CH 130 · MANAGEMENT STRATEGIES FOR NON–MUSCLE-INVASIVE BLADDER CANCER

Carcinoma in situ: natural history and treatment

- In patients undergoing cystectomy, the presence of CIS is associated with upstaging in 55%, compared with 5% when CIS is absent. [Compilation 01 p27]

- Response to BCG is an important prognostic factor in CIS: progression to muscle invasion occurs in 10–20% of complete responders but in 66% of non-responders. [Compilation 01 p28]
- In isolated CIS, intravesical chemotherapy gives a complete response in about 50% and reduces recurrence only, whereas BCG reduces both recurrence and progression. [Compilation 01 p29]
- CIS concurrent with low-grade Ta/T1 tumour is treated with BCG or intravesical chemotherapy; CIS concurrent with high-grade Ta/T1 tumour needs re-TURBT first, followed by BCG or chemotherapy. [Compilation 01 p29]
- Response to intravesical therapy for CIS is assessed at 3 months; responders continue a maintenance regimen, while non-responders have TURBT plus a second course of intravesical therapy (BCG preferred), and those who still do not respond go on to partial or radical cystectomy. [Compilation 01 p29]
- CIS of the upper tract is treated according to UTUC guidelines. [Compilation 01 p30]

Repeat TURBT

- Reasons for incomplete removal at the first TURBT are extensive tumour volume, anatomical inaccessibility, medical instability requiring premature termination, and risk of perforation. [Compilation 01 p32]

Laser and office-based management

- The most efficient laser probe for bladder tumour is an end-fire non-contact fibre with a 5–15 degree angle of divergence, which allows a depth of penetration up to 5 mm. [Compilation 01 p34]
- A white appearance of laser-treated tissue means protein denaturation, and the procedure should then be stopped. [Compilation 01 p34]
- The drawbacks of laser treatment of bladder tumours are higher cost and no tissue for biopsy. [Compilation 01 p34]
- For office-based fulguration, xylocaine (lidocaine) jelly with a dwell time of 15–30 minutes gives mucosal analgesia. [Compilation 01 p34]

Perioperative (single immediate) intravesical therapy

- Primary bladder tumours most often arise on the floor and side walls, whereas early recurrences after resection appear at the dome because floating tumour cells implant there. [Compilation 01 p38]
- Maintenance therapy after a single immediate instillation does not reduce the risk of recurrence further. [Compilation 01 p38]
- Epirubicin is the agent used for single immediate instillation in Europe. [Compilation 01 p38]
- The problems of perioperative instillation are cost and complexity. [Compilation 01 p38]
- Perioperative instillation has no effect on high-grade tumours. [Compilation 01 p39]
- Perioperative instillation can make subsequent cystectomy more difficult. [Compilation 01 p39]
- Perioperative instillation decreases recurrence by up to 12%, and by 50% if therapy is optimised. [Compilation 01 p39]

Intravesical BCG

- Morales's original BCG regimen used the percutaneous route as well, and Brosman showed similar success with an intravesical-only regimen. [Compilation 01 p39]
- The cytokines expressed after intravesical BCG include GM-CSF, IL-1 and IL-6, in addition to TNF-α, IFN-γ, IL-2, IL-8 and IL-12. [Compilation 01 p39]
- BCG stimulates an anti-angiogenic response as well as a cell-based immune response. [Compilation 01 p40]
- Restricting diuretics and caffeine, as well as fluids, before BCG instillation limits dilution. [Compilation 01 p41]
- After BCG instillation, the patient should clean the toilet with bleach. [Compilation 01 p41]
- There are no data on BCG in patients with prosthetic materials (theoretical risk of more infections) or with vesicoureteric reflux. [Compilation 01 p41]
- Over 10 years, about 30% of patients with CIS treated with BCG remain free of recurrence or progression. [Compilation 01 p42]

- In CIS treated with BCG, progression at 5 years is 19% in initial responders and 95% in non-responders. [Compilation 01 p42]
- BCG can effectively treat residual papillary tumour, but it should not be used as a substitute for surgical resection. [Compilation 01 p42]
- BCG-refractory disease is disease that does not improve or worsens during BCG. [Compilation 01 p43]
- Declaring BCG failure may take up to 6 months, because the response rate of high-grade bladder cancer treated with BCG rose from 57% to 80% between 3 and 6 months after therapy. [Compilation 01 p44]

BCG toxicity

- Antituberculous monotherapy should not be used for BCG toxicity. [Compilation 01 p45]
- Cycloserine should not be used for BCG infection because it causes severe psychiatric symptoms. [Compilation 01 p46]
- Grade 1 BCG toxicity can be managed with topical antispasmodics such as phenazopyridine, in addition to anticholinergics, analgesics and anti-inflammatory drugs. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p21]
- In grade 2 BCG toxicity, consult immediately with a physician experienced in managing mycobacterial infections and their complications. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p22]
- In grade 2 BCG toxicity, give oral isoniazid 300 mg/day and rifampin 600 mg/day until the symptoms resolve. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p22]
- For grade 3 allergic reactions to BCG (joint pain, rash), give isoniazid 300 mg/day and rifampin 600 mg/day for 3–6 months depending on response. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p23]
- For grade 3 BCG solid-organ involvement, give isoniazid 300 mg/day, rifampin 600 mg/day and ethambutol 15 mg/kg/day as a single daily dose for 3–6 months. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p23]
- Organs that can be involved in grade 3 BCG toxicity include the bone and joints, in addition to the epididymis, liver, lung, kidney and prostate. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p23]
- Consider prednisone 40 mg/day when the response is inadequate or for septic shock, never without effective antimycobacterial therapy. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p23]

Interferon and other immunotherapy

- IFN-α is the most extensively studied intravesical interferon, but as a single agent it is more expensive and less effective than BCG or intravesical chemotherapy. [Compilation 01 p46]
- Other investigational intravesical immunotherapy agents are IL-2 and a cell wall DNA extract (written "mitochondrial"; probably mycobacterial cell wall extract). [Compilation 01 p47]

Intravesical chemotherapy

- Mitomycin C acts by inhibiting DNA synthesis. [Compilation 01 p47]
- Doxorubicin is an anthracycline antibiotic that inhibits topoisomerase II and inhibits protein synthesis. [Compilation 01 p47]
- Valrubicin is a semisynthetic analogue of doxorubicin. [Compilation 01 p48]
- Thiotepa causes serious systemic side effects, especially myelosuppression, because of its low molecular weight. [Compilation 01 p48]
- A SWOG comparison of doxorubicin and BCG (Lamm et al, 1991) showed progression in 15% of BCG patients compared with 37% of chemotherapy patients. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p24]
- Because intravesical chemotherapy carries no risk of BCG infectious complications, it is the favoured adjuvant approach in many European countries. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p24]

Refractory high-grade disease

- For high-grade recurrence after previous intravesical chemotherapy, BCG gives better disease-free survival. [Compilation 01 p48]

- Salvage chemotherapy for BCG-intolerant high-grade disease carries a high rate of progression. [Compilation 01 p48]
- More than 2 BCG courses are not recommended because they fail 80% of the time. [Compilation 01 p48]

Alternative options: photodynamic therapy and radiation

- For bladder photodynamic therapy, porfimer sodium can be given systemically as an alternative to intravesical HAL. [Compilation 01 p49]
- Red-light laser treatment (630 nm, for 12–20 minutes) is given 2–3 days after the photosensitiser, once it has cleared from normal tissue. [Compilation 01 p49]
- Emerging photosensitisers for bladder photodynamic therapy are hypericin and radachlorin. [Compilation 01 p49]
- TUR plus radiation gives a complete response in 50–75% of NMIBC patients. [Compilation 01 p49]

Risk categorization of NMIBC

- For intermediate-risk NMIBC treated with intravesical chemotherapy, treatment is given for a maximum of 1 year, and the optimal chemotherapy schedule is not known. [Compilation 01 p52]
- Multiple large T1 high-grade tumours belong to the highest-risk subgroup of NMIBC, for which radical cystectomy is advised (with full-dose BCG for 1–3 years in those who refuse). [Compilation 01 p52]
- T1 high-grade tumour with CIS in the prostatic urethra is listed in the highest-risk subgroup of NMIBC, for which radical cystectomy is advised. [Compilation 01 p52]

CIS: Classification

- Secondary CIS is defined as CIS detected during follow-up of a patient whose previous tumour was not CIS, as distinct from concurrent CIS, which is CIS found in the presence of another urothelial tumour in the bladder. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p4]
- CIS cannot be treated by endoscopic procedures alone. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p4]

TUR (BT)

- Detrusor muscle in the specimen is the surrogate marker of resection quality and is required in every TURBT except for TaG1/low-grade tumours. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p7]
- Perform TURBT as a systematic stepwise procedure in which the resectoscope is inserted under vision with inspection of the whole urethra. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p8]
- The final steps of a systematic TURBT are recording the findings in the operation record and giving a precise description of the specimen for pathology evaluation. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p8]

Additional bladder biopsies during TURBT

- Random biopsies of normal-looking mucosa are taken from the trigone, bladder dome, and right, left, anterior and posterior bladder walls. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p9]
- Random biopsies of normal-looking mucosa are also recommended when a high-risk exophytic tumour of non-papillary appearance is expected. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p9]

Prostatic urethral biopsy

- Visible abnormalities of the prostatic urethra are an indication for prostatic urethral biopsy. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p10]
- If prostatic urethral biopsy is not performed at the initial TURBT, it should be completed at the time of the second resection. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p10]

Second TURBT

- Absence of muscle in the initial specimen is an indication for second TURBT except in Ta low-grade tumours and primary CIS. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p11]
- BCG appears more effective at preventing progression in high-grade papillary tumours and CIS when a restaging TURBT has been performed before BCG. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p12]

Intravesical BCG: procedure

- The BCG dose given in the slides is 120 mg in 50 mL, reduced if side effects develop. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p17]

Photodynamic therapy

- Photodynamic therapy is an alternative option for refractory NMIBC. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p26]
- In bladder photodynamic therapy, a photosensitising agent such as hexaminolevulinate (HAL) is instilled and the affected areas are treated with red laser light in the presence of molecular oxygen, causing free-radical cytotoxic injury. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p26]
- Use of photodynamic therapy is limited by side effects: bladder contracture or irritability in 50% and dermal photosensitivity in 19%. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p26]

Radiation therapy

- Radiation therapy is an option for NMIBC patients who refuse radical cystectomy after failure of intravesical therapy or who are unfit for major surgery. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p27]
- Radiation therapy has no significant effect on CIS. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p27]

EAU 2018 risk stratification and management of NMIBC

- In the EAU 2018 scheme, intermediate risk is a residual category: all tumours not defined as low risk or high risk. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p30]
- For EAU intermediate risk, one-year full-dose BCG consists of induction plus three weekly instillations at 3, 6 and 12 months. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p30]
- In the EAU 2018 scheme, TaG1G2/low-grade tumours are high risk only when they are multiple, recurrent and large (>3 cm), with all three features present. [TREATMENT OF NON-MUSCLE INVASIVE BLADDER CANCER p31]

WHO Grading System (slide 5)

- WHO 1973 Grade 1 carcinomas were reassigned in the WHO 2004 classification to either PUNLMP or low-grade carcinoma. [nmibcimg]
- WHO 1973 Grade 2 carcinomas were reassigned in the WHO 2004 classification to either low-grade or high-grade carcinoma. [nmibcimg]
- All WHO 1973 Grade 3 carcinomas were reassigned to high-grade carcinoma in the WHO 2004 classification. [nmibcimg]

Need for Additional Therapies (slide 13)

- Urothelial papilloma has a recurrence rate of 0-8%, grade progression of 2%, stage progression of 0% and survival of 100%. [nmibcimg]
- PUNLMP has a grade progression rate of 11% and survival of 93-100%. [nmibcimg]
- Low-grade papillary carcinoma has a grade progression rate of 7% and survival of 82-96%. [nmibcimg]
- High-grade carcinoma (papillary and CIS) has a recurrence rate of 55-58% and survival of 74-90%. [nmibcimg]

BCG Treatment Schedule (slide 19)

- The BCG induction dose is given as 120 mg weekly for 6 weeks. [nmibcimg]
- An alternative BCG maintenance phase is 1 dose per month for 10 months. [nmibcimg]

Unsuccessful Treatment with BCG (slide 20)

- Detection of muscle-invasive bladder cancer at any time during follow-up counts as BCG failure. [nmibcimg]
- CIS without concomitant papillary tumour that is present at both 3 and 6 months counts as BCG-refractory. [nmibcimg]
- A high-grade tumour that appears during BCG therapy counts as BCG-refractory. [nmibcimg]
- Low-grade recurrence during or after BCG treatment is not considered BCG failure. [nmibcimg]

Various Intravesical Therapies (slide 25)

- Doxorubicin (Adriamycin) can be used perioperatively, is used in low- to intermediate-risk disease, causes cystitis in 20-40%, and its other toxicities are fever, allergy and contracted bladder in 5%. [nmibcimg]
- Epirubicin is used in low- to intermediate-risk disease, causes cystitis in 10-30%, and contracted bladder is rare. [nmibcimg]
- Thiotepa is used in low- to intermediate-risk disease, causes cystitis in 10-30%, and causes myelosuppression in 8-19%. [nmibcimg]
- Mitomycin causes rash in 8-19% of patients. [nmibcimg]
- BCG causes cystitis in 60-80% of patients. [nmibcimg]
- Intravesical interferon is not used perioperatively, is a salvage agent, causes cystitis in under 5%, and causes flu-like symptoms in 20%. [nmibcimg]
- Intravesical gemcitabine is classed as a salvage agent whose other toxicity is occasional nausea. [nmibcimg]

Treatment options for BCG failure and recurrences after BCG (slide 32)

- For non-high-grade recurrence after BCG given for a primary intermediate-risk tumour, the options are a repeat BCG course or intravesical chemotherapy, and then radical cystectomy. [nmibcimg]

Follow up after therapy (slide 33)

- In intermediate-risk NMIBC, consider upper tract imaging especially at recurrence, and image the upper tracts if haematuria occurs. [nmibcimg]

CH 131 · MANAGEMENT OF MUSCLE-INVASIVE AND METASTATIC BLADDER CANCER

Pelvic lymph node dissection

- The inferior (distal) boundary of standard PLND is given as Cooper's ligament or the circumflex iliac vein. [Compilation 01 p60]

Muscle-invasive and metastatic bladder tumour: introduction and staging

- If left untreated, 85% of patients with MIBC die within 2 years. [MIBC p1]

Prostatic urethral biopsy, bimanual examination and imaging

- The prostatic urethral loop biopsy may start at the bladder neck in shorter prostates and runs to the mid-to-distal verumontanum; the 5 and 7 o'clock area beside the verumontanum has the highest concentration of prostatic ducts, so CIS is most likely found there. [MIBC p2]
- Bimanual examination technique: the dominant hand is placed on the mass suprapubically and one or two fingers of the non-dominant hand are placed in the rectum or vagina; the bladder is kept empty but no Foley catheter is left inside. [MIBC p2]
- The bimanual clinical staging scheme (T2a non-palpable to T4b fixed to sidewall) is attributed to Marshall. [MIBC p2]
- If imaging is done after TUR, it should be delayed for about 7 days, because inflammatory artefact may be mistaken for T3 disease. [MIBC p2]
- For local staging, CT has a sensitivity of about 90% and specificity of about 95%. [MIBC p2]
- For nodal metastasis, CT specificity is 70–100%. [MIBC p2]

Imaging (MRI) and radical cystectomy

- MRI may have an advantage over CT in predicting tumour grade and chemosensitivity, but it is poorer than CT at detecting lymph node metastasis. [MIBC p3]
- Radical cystectomy with bilateral PLND gives excellent local control, with only about 4% local recurrence in node-negative disease. [MIBC p3]

Organ preservation and lymph node zones

- 15–75% of patients have prostate involvement on final cystoprostatectomy specimens; if prostate preservation is considered, TUR sampling of the bladder neck and prostatic urethra is advisable. [MIBC p4]
- Prostate-sparing techniques are apical sparing, posterior sparing (posterior prostate and seminal vesicles) and total prostate sparing; all give better preservation of continence and erection. [MIBC p4]
- The primary lymphatic zone of the bladder ("HOPE") comprises hypogastric, obturator, external iliac and presacral nodes. [MIBC p4]
- The secondary lymphatic zones are the common iliac, paracaval, para-aortic and interaortocaval nodes. [MIBC p4]

Pelvic lymph node dissection templates

- Bladder cancer has no skip metastases, so sentinel lymph node biopsy has no role. [MIBC p5]
- Standard PLND is called level I (up to the common iliac bifurcation) and extended PLND level II (up to the aortic bifurcation). [MIBC p5]
- Lymph node density (positive nodes as a percentage of nodes removed) below 20% is associated with better survival, although this is controversial. [MIBC p5]

Intraoperative decision making

- Grossly positive nodes outside the pelvis on pre-operative imaging should be confirmed by CT-guided biopsy; if positive, give neoadjuvant chemotherapy and then radical cystectomy. [MIBC p6]
- Extensive periureteral disease is a further reason not to proceed with cystectomy. [MIBC p6]
- Risk factors for upper-tract recurrence after cystectomy are bladder CIS, tumour in the distal ureter and high-grade pTa–T1 disease. [MIBC p6]
- Ureteric frozen section is controversial because it is debatable whether a negative CIS margin reduces recurrence. [MIBC p6]
- Non-muscle-invasive disease on final pathology is listed as a risk factor for male urethral recurrence. [MIBC p6]
- Diffuse CIS of the prostatic urethra or ducts is an indication to consider urethrectomy. [MIBC p6]
- Because pre-operative prostatic urethral biopsy has only moderate sensitivity and specificity, some suggest urethrectomy only if the apical urethral margin of the RC specimen is positive. [MIBC p6]

Managing the female urethra

- Inguinal lymphadenopathy increases the risk of urethral tumour in women. [MIBC p8]
- Complete urethrectomy for bladder-neck tumours is controversial, because 60% of women with tumour at this site have no urethral involvement on final pathology. [MIBC p8]

Neoadjuvant and adjuvant chemotherapy

- Neoadjuvant chemotherapy also lets the clinician assess each individual's response to therapy. [MIBC p9]
- NCCN says clinicians should strongly consider cisplatin-based NAC for cT2 N0 M0, and recommends NAC for cT3–T4a N0 M0. [MIBC p9]
- Adjuvant chemotherapy allows immediate cystectomy without delay. [MIBC p9]
- Systemic chemotherapy after RC is often difficult because of surgical deconditioning and deteriorating renal function, as well as peri-operative complications. [MIBC p9]

Bladder preservation

- Obvious T3 disease on imaging and multifocal tumours make a patient a suboptimal candidate for bladder preservation. [MIBC p11]
- TUR monotherapy should be chosen only with a negative restaging TUR, no hydronephrosis, no adenopathy, tumour <3 cm and no multifocal disease, and the patient must be counselled about recurrence. [MIBC p11]
- After partial cystectomy, recurrence is NMIBC in 12–50%, MIBC in 20–60% and metastatic in 15–50%. [MIBC p11]

- Partial cystectomy may be combined with neoadjuvant chemoradiation or with adjuvant chemotherapy (small studies). [MIBC p11]
- Radiation monotherapy uses 1.5–2 Gy per fraction after debulking TUR, and local failure occurs in more than 30% of patients. [MIBC p11]

Metastatic bladder cancer

- Ultimate 5-year overall survival in metastatic bladder cancer is 5–20%. [MIBC p15]
- Carboplatin improves tolerability but has lower efficacy than cisplatin. [MIBC p15]
- PGC (paclitaxel, gemcitabine, cisplatin) is another first-line option for cisplatin-fit patients. [MIBC p15]
- Nivolumab, an antibody blocking the PD-1 receptor, is listed as a second-line option. [MIBC p15]
- Second-line treatment is chosen regardless of the time to progression after first line; with performance status ≥2, options are immunotherapy, clinical trials or best supportive care. [MIBC p15]
- Patients with performance status ≥2 and GFR <60 mL/min have no combination chemotherapy data and receive pembrolizumab or atezolizumab monotherapy or best supportive care. [MIBC p15]

Other drugs tried for metastatic bladder cancer

- Vinflunine, an antitubulin vinca alkaloid, is approved in Europe as a second-line agent. [MIBC p17]
- Taxanes (paclitaxel, docetaxel, larotaxel) have been tried in metastatic bladder cancer. [MIBC p17]
- Antifolates pemetrexed and piritrexim (synthetic) have been tried. [MIBC p17]
- VEGF inhibitors bevacizumab, sunitinib and sorafenib have been tried alone or in combination. [MIBC p17]
- EGFR-directed therapy includes cetuximab with or without paclitaxel or GC, gefitinib, lapatinib and erlotinib. [MIBC p17]
- Other agents tried are trastuzumab (anti-HER2), ipilimumab (CTLA-4 inhibitor) and cabozantinib (anti-angiogenic). [MIBC p17]

Radical Cystectomy: Patient Preparation and Positioning

- Pre-operative counselling by an enterostomal therapist is part of patient preparation, alongside stoma site marking. [Radical Cystectomy p1]
- Radical cystectomy can be done under general or spinal anaesthesia. [Radical Cystectomy p1]
- Men are placed supine with the table flexed 15 degrees at the level of the anterior superior iliac spine (ASIS). [Radical Cystectomy p1]
- Women are placed supine with spreader bars to give access to the vagina, or in low lithotomy. [Radical Cystectomy p1]
- The operative field is painted and draped from the xiphoid to the upper thighs. [Radical Cystectomy p1]
- Povidone-iodine, not chlorhexidine, is used on the genitals. [Radical Cystectomy p1]
- The open incision runs from 2 cm above the umbilicus to just above the pubis. [Radical Cystectomy p1]

Open Radical Cystectomy: Technique

- The peritoneum is opened lateral to each medial umbilical ligament down to the internal inguinal rings, and the vas deferens or round ligament is divided there. [Radical Cystectomy p2]
- On the right, the white line is incised and carried around the caecum to mobilise the root of the small bowel mesentery. [Radical Cystectomy p2]
- Dividing the superior vesical artery on each side allows maximal ureteric length, after which the ureter is divided between ligatures. [Radical Cystectomy p2]
- Pelvic lymphadenectomy can be done at this stage, or after the bladder has been removed. [Radical Cystectomy p2]
- Historically, women had the bladder, urethra, anterior vagina, uterus and cervix removed because women present with more advanced disease than men. [Radical Cystectomy p3]
- In women, the urethral meatus is incised either from the pelvis or from the vaginal introitus. [Radical Cystectomy p3]

Robotic Radical Cystectomy: Position and Ports

- Theatre layout: the anaesthetist is at the head, with a monitor on each side of the head end; a right and a left assistant stand at each side; the scrub nurse with a third monitor is at the lower left; the da Vinci system is docked between the legs; the surgeon sits at the console away from the table. [Radical Cystectomy p4]
- The patient is fixed to the table in dorsal lithotomy with a 30 degree head-down tilt. [Radical Cystectomy p4]
- Port layout: a 12 mm camera port 5 cm above the umbilicus; 8 mm robotic arm ports at about umbilical level, including the fourth robotic arm laterally on one side; a 12 mm assistant port laterally on the other side, plus an additional 12 mm assistant port. [Radical Cystectomy p4]
- A 15 mm lower midline port is placed for the Endo GIA stapler if intracorporeal diversion is planned. [Radical Cystectomy p4]

Robotic Radical Cystectomy: Procedure

- The superior vesical artery (anterior pedicle) is found at the most posterior part of the plane developed between the external iliac vessels and the lateral bladder wall. [Radical Cystectomy p5]
- The bedside assistant divides the posterior pedicles with LigaSure or staplers; the endopelvic fascia on each side and the prostatic pedicles are then divided. [Radical Cystectomy p5]
- When the bladder is dropped anteriorly, the urachus is divided near the umbilicus and wide peritoneal wings are taken down before the Retzius space is fully dissected. [Radical Cystectomy p5]
- The whole specimen is placed in a 15 cm Endo Catch bag. [Radical Cystectomy p6]
- Robotic extended pelvic lymph node dissection takes around 45 to 60 minutes and uses a split-and-roll technique. [Radical Cystectomy p6]
- The specimen is extracted through a Pfannenstiel incision with intracorporeal diversion, or through a vertical midline infraumbilical incision with extracorporeal diversion. [Radical Cystectomy p6]
- In robotic cystectomy in women, the uterine vessels are ligated as well as the ovarian vessels. [Radical Cystectomy p6]
- In women, the round ligaments are transected and the uterus, tubes and ovaries are sutured to the obliterated umbilical ligaments to help visualisation for the posterior dissection. [Radical Cystectomy p6]

CH 132 · SURGICAL MANAGEMENT OF BLADDER CANCER: TRANSURETHRAL, OPEN AND ROBOTIC

Endoscopic management of NMIBC (TURBT)

- Resection is done with a 12–30 degree lens, because this deflection lets the surgeon see the loop where it is placed. [Compilation 01 p30]
- Video TUR offers magnification, resident teaching, documentation and recording, and less exposure of the surgeon to body fluids. [Compilation 01 p31]
- Obturator nerve block is one way to avoid the obturator reflex during TURBT. [Compilation 01 p31]
- Uncontrolled haematuria or bladder perforation complicates TURBT in fewer than 5%, while minor perforation is common. [Compilation 01 p31]
- If perforation during TURBT is intraperitoneal, the risk of tumour seeding is low. [Compilation 01 p31]
- Intravenous indigo carmine or methylene blue, or a retrograde pyelogram, can show whether the ureteric orifice is obstructed after resection near it. [Compilation 01 p31]
- If ureteric orifice obstruction develops after TURBT, options are balloon dilatation, endoscopic incision or ureteric reimplantation. [Compilation 01 p31]

CH 133 · USE OF INTESTINAL SEGMENTS IN URINARY DIVERSION

General classification of urinary diversions

- Variations of ureterosigmoidostomy (rectal-bladder urinary diversion) include ileocaecal sigmoidostomy, the rectal bladder, and the sigmoid hemi-Kock with proximal colonic intussusception, in which urine is passed by evacuation. [Cutaneous Diversion p1]

Rectal-bladder urinary diversion

- Contraindications to rectal-bladder diversion are dilated ureters, extensive pelvic radiation, existing renal insufficiency, an incompetent anal sphincter and pre-existing colorectal disease. [Cutaneous Diversion p3]
- Anal sphincter competence is checked by instilling 400–500 ml of a thin mixture of oatmeal and water into the rectum and asking the patient to retain it for 1 hour in the upright position. [Cutaneous Diversion p3]
- The types of rectal-bladder diversion are ureterosigmoidostomy, the folded rectosigmoid bladder, the augmented valved rectum, the hemi-Kock and T-Kock procedures, and the sigma-rectum pouch (Mainz II). [Cutaneous Diversion p3]
- In the folded rectosigmoid bladder, the ureters are implanted through serosa-lined tunnels. [Cutaneous Diversion p3]
- In the augmented valved rectum, a proximal intussusception of the sigmoid confines urine to a smaller surface area. [Cutaneous Diversion p3]
- In the hemi-Kock and T-Kock procedures, the ileal patch is anastomosed to the rectum. [Cutaneous Diversion p3]
- The Mainz II (sigma-rectum) pouch is a simple, reproducible operation that creates a low-pressure reservoir with increased capacity by partial reconfiguration of the rectosigmoid junction. [Cutaneous Diversion p3]

Continent catheterising pouches: stoma sites and principles of continence

- The alternative stoma site is the lower abdominal quadrant, through the rectus bulge and below the bikini line. [Cutaneous Diversion p4]
- The intussuscepted nipple valve (or the flap valve that avoids intussusception) is the most technically demanding continence mechanism. [Cutaneous Diversion p4]
- A fourth continence principle is a hydraulic valve, as in the Benchekroun nipple, which has been largely abandoned. [Cutaneous Diversion p4]

Types of continent cutaneous pouch

- The Kock pouch (continent ileal reservoir) uses 15 cm of terminal ileum with an intussuscepted nipple valve, preserves the ileocaecal valve, and has a high complication rate. [Cutaneous Diversion p5]
- The double T-pouch uses 70 cm of terminal ileum in a W-shaped configuration and preserves the ileocaecal valve. [Cutaneous Diversion p5]
- The double T-pouch has fewer complications than the Kock pouch. [Cutaneous Diversion p5]
- The Mainz I pouch has a nipple valve made by intussusception of the ileum and ileocaecal valve, and a high complication rate (31%). [Cutaneous Diversion p5]
- In the Mainz I pouch, the bowel is laid out as segments of 10–15 cm, 10–15 cm, 10–15 cm and 20–25 cm. [Cutaneous Diversion p5]
- In the Indiana pouch, the entire right colon is reconfigured in a Heineke–Mikulicz fashion and an appendicectomy is done. [Cutaneous Diversion p5]
- In the Indiana pouch, buttressing of the ileocaecal valve alone is probably enough, and the remaining ileal neourethra is tapered to a 12–14 Fr lumen and brought out. [Cutaneous Diversion p5]
- The Indiana pouch is one of the most reliable reservoirs, is easy to construct and has a low complication rate. [Cutaneous Diversion p5]
- The Penn pouch is an ileocaecal pouch using 10–12 cm of ileum, and it was the first diversion to employ the Mitrofanoff principle. [Cutaneous Diversion p6]
- In the Penn pouch, the appendix is excised with a button of caecum and reversed on itself before tunnelled reimplantation. [Cutaneous Diversion p6]
- The Penn pouch is easy to construct and suits patients without a previous appendicectomy. [Cutaneous Diversion p6]
- A gastric pouch uses a wedge-shaped segment of stomach from the greater curvature based on the right gastroepiploic artery. [Cutaneous Diversion p6]

- In a gastric pouch, a contralateral distal ureter tunnelled into the reservoir (similar to the appendix), with a transureteroureterostomy, can serve as the continence mechanism; alternatively a Mitrofanoff procedure is used. [Cutaneous Diversion p6]
- Possible advantages of gastric pouches are fewer infections and fewer metabolic complications, and use when the bowel is irradiated or short bowel syndrome is anticipated. [Cutaneous Diversion p6]
- Gastric pouches can be especially useful in paediatric patients, but experience is limited at present. [Cutaneous Diversion p6]

Complications

- Internal hernia is a complication related to the bowel anastomosis. [Cutaneous Diversion p7]
- Pouchitis needs at least 10 days of antibiotics. [Cutaneous Diversion p7]
- Retention in a continent cutaneous pouch occurs most commonly with a nipple valve. [Cutaneous Diversion p7]

Urinary Diversion: indications and types

- Before renal transplantation, urinary diversion is indicated in patients whose bladder cannot receive the transplant ureter. [Compilation 01-2 p26]

Complications of urinary diversion

- Complications of urinary diversion are grouped as intra-operative (such as haemorrhage, trauma and ureteric stripping) and post-operative (metabolic, intestinal-anastomotic, conduit or pouch and stoma related, and general complications such as chest and wound problems). [Compilation 01-2 p27]
- Listed conduit complications include conduit infarction, volvulus, conduit–enteric fistula, excessive conduit length and conduit stenosis, as well as GI bleeding. [Compilation 01-2 p27]
- Cutaneous ureterostomy is an incontinent form of urinary diversion, alongside the ileal and other conduits. [Compilation 01-2 p27]

Principles of Different Bowel Segment Use: stomach

- The right gastroepiploic artery is much larger than the left, and a gastric antral segment is mobilised on the right gastroepiploic artery. [Compilation 01-2 p29]

Principles of Different Bowel Segment Use: small intestine and colon

- Compared with ileum, jejunum has a thin mesentery and large vessels in its arcade, whereas ileum has a thick mesentery with multiple arcades of small vessels. [Compilation 01-2 p30]
- The bowel should be divided so that the isolated segment's arcade is supplied by a palpable artery arising from the base of the mesentery. [Compilation 01-2 p30]
- After pelvic radiation, the last 5 cm of terminal ileum and the bowel lying in the pelvis (written as "5 feet bowel 6 feet distal to ligament of Treitz") may show radiation effect and should not be used. [Compilation 01-2 p30]
- The ileocolic artery is the direct continuation of the SMA and also supplies the terminal 15 cm of ileum, which is usually left behind for B12 absorption, so the mesentery must be dissected accordingly. [Compilation 01-2 p30]
- The midpoint between the right colic and middle colic arteries is a further weak point of the colonic blood supply, where anastomosis should be avoided. [Compilation 01-2 p30]

ERAS: aim and goal

- The aim of ERAS is to standardise and improve perioperative management; its goal is faster and more efficient recovery using evidence-based practices. [ERAS p1]

Pre-op ERAS elements

- Patients should stop smoking 8 weeks before surgery. [ERAS p1]
- Excessive alcohol intake should be avoided before surgery. [ERAS p1]
- Preoperative optimisation includes nutritional screening with a nutritional risk score. [ERAS p1]
- Immunonutrition uses arginine, fish oil and nucleotides. [ERAS p1]

- Preoperative carbohydrate loading can be given orally or intravenously, is safe in diabetics, and can be given up to 2 hours before surgery. [ERAS p1]
- There is a concern about cardiovascular events with alvimopan. [ERAS p1]
- For premedication, short-acting benzodiazepines are preferred to reduce anxiety; long-acting benzodiazepines are associated with cognitive impairment. [ERAS p2]
- The EAU advises continuing the prophylactic antibiotic for up to 24 hours, extended to 72 hours if there are specific infection risk factors or the operation is prolonged (more than 3 hours). [ERAS p2]
- There are no prospective studies of a specific anaesthetic protocol for urological surgery. [ERAS p2]

Intra-op ERAS elements

- Excess fluid and hypervolaemia cause splanchnic hypoperfusion, which leads to ileus. [ERAS p3]
- The optimal duration of urinary catheter drainage is not known, but early removal appears beneficial. [ERAS p3]

Post-op ERAS elements

- PONV is prevented by multimodal antiemetic prophylaxis (ondansetron) and dexamethasone. [ERAS p3]
- PONV prevention also includes intraoperative fluid optimisation and stenting of the ureteroileal anastomoses. [ERAS p3]
- PONV prevention improves patient satisfaction and reduces hospital stay and readmission. [ERAS p3]
- Prophylactic oral laxatives are used for ileus prophylaxis. [ERAS p3]
- TPN is not given routinely unless enteral nutrition is delayed for more than 5–7 days. [ERAS p4]
- Opioid-sparing multimodal analgesia should be combined with regional or local anaesthesia, using oral or IV paracetamol with or without NSAIDs. [ERAS p4]
- Good analgesia enables early mobilisation, which reduces DVT, ileus and chest infection. [ERAS p4]
- Discharge criteria are resumed oral intake, normal bowel function and effective oral pain control. [ERAS p4]
- Audit is needed to ensure successful implementation of ERAS and to assess its effect on clinical and financial outcomes. [ERAS p4]
- ERAS outcomes include cost-effectiveness and improved quality of life, as well as faster recovery and shorter stay. [ERAS p4]

Ileal conduit

- The ureteroileal anastomoses can also be done after the ileoileostomy has been made. [Ileal Conduit p1]
- Keep the base of the conduit mesentery as wide as possible. [Ileal Conduit p1]
- The isolated conduit segment lies posterior and caudal to the ileoileostomy. [Ileal Conduit p1]
- Mobilising the caecum, small bowel and colon cephalad makes it easier to bring both ureters into the right lower quadrant. [Ileal Conduit p1]
- In an ideal Bricker anastomosis, the spatulated ureters pass under the conduit without any tension or kink. [Ileal Conduit p1]
- The Bricker anastomosis is a single-layer, full-thickness, interrupted anastomosis using 4-0 or 5-0 Vicryl. [Ileal Conduit p2]
- For the Bricker anastomosis, the serosa and muscularis of the conduit are incised, a small disc of mucosa and submucosa is excised, and the spatulated ureter is then sewn to the opening. [Ileal Conduit p2]
- The ureteric stents are brought out through the conduit so that their distal ends lie outside the stoma. [Ileal Conduit p2]
- The conduit may also be sutured to the lateral abdominal wall to prevent an internal hernia. [Ileal Conduit p2]

Ileostomy (rosebud stoma)

- To make the stoma, a circular plug of skin is excised at the marked site. [Ileal Conduit p2]

- A Kelly clamp is spread to make a hiatus through the rectus muscle and the posterior rectus sheath. [Ileal Conduit p2]
- The anterior rectus sheath is sutured to the serosa of the conduit. [Ileal Conduit p2]
- The rosebud stoma is everted with four three-point sutures, each passing through the mucosal edge of the bowel, then through the bowel serosa and muscularis, then out through the dermis (not the epidermis), which everts the edges so the stoma pouts. [Ileal Conduit p2]

Complications of intestinal segments: classification

- Complications of intestinal segments in urinary diversion are classified as metabolic, neuromechanical and technical-surgical. [Intestines in Uro p1]

Stomach: hypochloraemic metabolic alkalosis

- Normally the kidney compensates for gastric H+ loss by excreting bicarbonate in the urine, so alkalosis becomes a problem in renal failure because bicarbonate excretion is impaired. [Intestines in Uro p1]
- Severe metabolic alkalosis from a gastric segment can cause respiratory insufficiency, as well as lethargy, seizures and ventricular arrhythmias. [Intestines in Uro p1]
- Failure to empty a gastric reservoir properly distends it and releases gastrin, which increases HCl production and worsens hypochloraemic alkalosis. [Intestines in Uro p1]
- In the gastric alkalosis cycle, renal compensation (H+ retention with K+ loss) causes hypokalaemia; vomiting then causes water and sodium loss, dehydration with raised aldosterone, acute renal failure and possibly death. [Intestines in Uro p1]

Jejunum: hyponatraemic, hypochloraemic, hyperkalaemic acidosis

- The jejunal conduit syndrome also causes azotaemia. [Intestines in Uro p3]
- Clinical features of the jejunal syndrome include nausea, muscle weakness and elevated temperature, besides vomiting and dehydration. [Intestines in Uro p3]
- Hyperalimentation solutions worsen the electrolyte abnormalities of the jejunal syndrome. [Intestines in Uro p3]
- About 25% of patients with a jejunal conduit may develop the electrolyte abnormalities. [Intestines in Uro p3]

Ileum and colon: hyperchloraemic metabolic acidosis

- Hyperchloraemic acidosis after ileal or colonic diversion is usually minor, and severe disturbances are much less common. [Intestines in Uro p4]
- Symptoms of hyperchloraemic acidosis are easy fatigability, anorexia, lethargy, weight loss and polydipsia, with worsening diarrhoea in ureterosigmoidostomy. [Intestines in Uro p4]
- Oral sodium bicarbonate produces considerable intestinal gas, which may not be tolerated, and has a bad taste. [Intestines in Uro p4]
- Sodium citrate, potassium citrate and citric acid solutions (for example Bicitra or tricitrate preparations) are alternatives when the sodium load is a problem, as in cardiac or renal patients. [Intestines in Uro p4]
- In persistent acidosis, chlorpromazine or nicotinic acid is combined with low-dose alkalinisation. [Intestines in Uro p4]
- The side effect of chlorpromazine in this use is tardive dyskinesia. [Intestines in Uro p4]
- Nicotinic acid can worsen liver dysfunction or peptic ulcer disease and can cause headache, double vision, flushing and dermatitis. [Intestines in Uro p4]

Hypokalaemia in urinary diversion

- Hypokalaemia after diversion is caused by renal potassium wasting from renal damage and osmotic diuresis, with gut loss playing only a minor role. [Intestines in Uro p6]
- An ileal segment absorbs some potassium from urine whereas a colonic segment absorbs none, which is why colonic diversions and ureterosigmoidostomy are more likely to cause potassium loss. [Intestines in Uro p6]

Renal function assessment and altered sensorium

- Histological changes develop over time in intestinal segments exposed to urine, and they are less marked in colon. [Intestines in Uro p7]
- Magnesium deficiency causing altered sensorium after diversion results from nutritional depletion and increased renal loss. [Intestines in Uro p7]
- For ammoniagenic encephalopathy, the reservoir is emptied continuously with a Foley catheter or a rectal tube. [Intestines in Uro p7]
- Lactulose works by combining with ammonium in the gut and reducing its absorption. [Intestines in Uro p7]
- Arginine glutamate is given as 50 g in 1 L of 5% dextrose, and it complexes with ammonia to form glutamine. [Intestines in Uro p7]

Abnormal drug absorption and osteomalacia

- Drugs at risk are those absorbed from the gut and excreted unchanged by the kidney, because they are re-exposed to the intestinal segment and reabsorbed to toxic levels. [Intestines in Uro p8]
- Chemotherapy is usually well tolerated in patients with intestinal diversions, though methotrexate toxicity has been reported. [Intestines in Uro p8]
- In osteomalacia after diversion, mineralised bone is reduced and the osteoid component increases. [Intestines in Uro p8]
- In persistent acidosis, bone hydroxyapatite buffers H+ and releases calcium, which is lost in the urine. [Intestines in Uro p8]
- An altered calcium to PTH ratio contributes to osteomalacia after diversion. [Intestines in Uro p8]
- Symptoms of osteomalacia after diversion are lethargy, joint pain and proximal myopathy. [Intestines in Uro p8]
- Investigations in osteomalacia after diversion show low serum calcium and low phosphate, as well as raised alkaline phosphatase. [Intestines in Uro p8]

Growth and development

- Children with intestinal diversions are susceptible to fractures and to complications after orthopaedic procedures. [Intestines in Uro p9]
- Growth problems occur in those whose intestinal urinary diversions were made in childhood and kept for more than 10 years. [Intestines in Uro p9]

Stones

- Pre-existing pyelonephritis is a risk factor for stones after intestinal diversion. [Intestines in Uro p9]

Infection

- Intestinal diversions have an increased incidence of bacteraemia and septic episodes as well as bacteriuria. [Intestines in Uro p10]
- Proposed reasons for infection are that intestine cannot inhibit bacterial proliferation, makes urine less bacteriostatic, allows bacterial translocation when distended, and has an altered immunological mucosal barrier. [Intestines in Uro p10]
- About 13% of patients with intestinal diversion have septic or major infectious complications. [Intestines in Uro p10]
- Continent diversions have a high incidence of bacteriuria and septic episodes. [Intestines in Uro p10]

Intestinal motility, short bowel and nutritional problems

- When the ileocaecal valve is lost, fat reaching the colon causes fatty diarrhoea and deficiency of the fat-soluble vitamins A, D, E and K. [Intestines in Uro p11]
- Osteoporosis after bowel loss results from vitamin D deficiency and reduced calcium absorption, because calcium binds fat to form soaps. [Intestines in Uro p11]
- Significant jejunal loss, rare in urology, causes malabsorption of fat, calcium and folic acid. [Intestines in Uro p11]

- Colonic loss reduces fluid, electrolyte and bicarbonate reabsorption and causes dehydration. [Intestines in Uro p11]
- Any major bowel loss can cause nocturnal bowel movements, faecal urgency, faecal incontinence, diarrhoea and nutritional deficiencies. [Intestines in Uro p11]

Cancer

- In ileal segments, the tumours reported are adenomatous polyps and anaplastic carcinoma. [Intestines in Uro p12]
- Exposure of gut epithelium to urine alone is probably not the cause of secondary cancer. [Intestines in Uro p12]

Bowel preparation

- After mechanical preparation, the concentration of organisms per gram of faeces is unchanged, so any spillage gives the same inoculum as unprepared bowel. [Intestines in Uro p13]
- Conventional bowel preparation used a 3-day diet, whole-gut irrigation, an elemental diet and 10% mannitol, and left the patient exhausted and nutritionally depleted. [Intestines in Uro p13]
- Whole-gut irrigation uses Ringer lactate or normal saline through a nasogastric tube over several hours. [Intestines in Uro p13]
- Whole-gut irrigation is contraindicated in unstable cardiovascular disease, congestive cardiac failure, cirrhosis, severe renal disease (fluid overload) and bowel obstruction. [Intestines in Uro p13]
- The advantages of whole-gut irrigation are dietary freedom, a short preparation time and no enema. [Intestines in Uro p13]
- The PEG–electrolyte regimen is a low-residue diet on day 2 and clear liquids on day 1 before surgery, then PEG solution at 1–1.5 L per hour for 3 hours, up to 4 L or until the rectal effluent is clear. [Intestines in Uro p13]
- In children, PEG solution is given at 20–40 mL/kg per hour. [Intestines in Uro p13]
- Metoclopramide 10 mg may be given for nausea during PEG preparation. [Intestines in Uro p13]
- The oral cathartic doses are sodium phosphate 45 mL or magnesium citrate 148 mL; they are typically well tolerated and need smaller volumes. [Intestines in Uro p13]
- Sodium phosphate can cause potassium, calcium and phosphate derangements in frail patients and needs caution in renal failure. [Intestines in Uro p13]
- Kanamycin is noted as the best oral antibiotic preparation: 1 g orally every hour for 4 doses then 4 times a day on the first day of the regimen, then 1 g orally 4 times a day on each of the next 2 days. [Intestines in Uro p13]

Bowel preparation: antibiotics

- A third-generation cephalosporin is suggested for intravenous prophylaxis before intestinal surgery. [Intestines in Uro p16]

Diarrhoea and pseudomembranous enterocolitis

- Pseudomembranous enterocolitis presents with abdominal pain and diarrhoea, usually without fever or chills, though systemic toxicity may supervene. [Intestines in Uro p16]
- Pseudomembranous enterocolitis may progress to toxic megacolon, which carries a mortality of 15–20%. [Intestines in Uro p16]
- Antibiotics suppress normal bowel flora, allowing Clostridium difficile to grow and release toxin, which causes diffuse inflammation with cream-coloured plaques, erythema, oedema, polymorphonuclear infiltrate and mucosal sloughing. [Intestines in Uro p16]
- The disease can be a pseudomembranous colitis, an enteritis, or both. [Intestines in Uro p16]
- Diagnosis of pseudomembranous enterocolitis is by symptoms and endoscopy, confirmed by culture or by toxin identification, which gives a quick result. [Intestines in Uro p16]
- Treatment is vancomycin or metronidazole with withdrawal of the offending antibiotic, and subtotal colectomy if toxic megacolon develops. [Intestines in Uro p16]

Intestinal anastomosis: types

- Sutured intestinal anastomoses include the two-layer enteroenterostomy, the single-layer enteroenterostomy and the end-to-side sutured ileocolic anastomosis, which can be used with or without discrepant bowel sizes. [Intestines in Uro p19]
- The theoretical benefits of stapled anastomosis are better blood supply to the healing margin, less tissue handling, minimal oedema from uniformly placed staples, a wider lumen, ease and speed, and less postoperative ileus. [Intestines in Uro p19]
- The staplers used are the linear stapler, the anastomotic stapler and the circular stapler. [Intestines in Uro p19]
- In laparoscopic and robotic surgery, bowel anastomosis uses endoscopic stapling devices, while the ureteroileal anastomosis is sutured. [Intestines in Uro p19]
- A purely laparoscopic intestinal anastomosis at radical cystectomy has a much higher complication rate from intestinal and ureteric anastomotic leaks than an exteriorised approach through a mini-laparotomy (extracorporeal surgery). [Intestines in Uro p19]
- Compression anastomosis with the biofragmentable ring has proved safe, effective and time-efficient. [Intestines in Uro p19]

Postoperative care

- Small-bowel activity returns within a few hours of surgery, and stomach activity as early as 24 hours. [Intestines in Uro p21]
- Parenteral nutrition is indicated when preoperative nutrition is impaired, when a postoperative complication delays oral feeding, or when paralytic ileus persists on postoperative day 5. [Intestines in Uro p21]
- Some surgeons recommend jejunal feeding tubes for early enteral feeding. [Intestines in Uro p21]
- Gastrostomy tube decompression helps patients with severe pulmonary disease, because it eases pulmonary toilet and improves comfort. [Intestines in Uro p21]
- In patients with significant comorbidity, an H2 blocker or antacid is given through the stomach tube every 2 hours to keep the gastric pH above 5 and prevent stress ulceration. [Intestines in Uro p21]

Complications of intestinal anastomosis

- Complications of intestinal anastomosis include faecal or urinary fistula with sepsis, wound infection, pelvic abscess and wound dehiscence. [Intestines in Uro p22]
- Recurrent cancer is the second most common cause of bowel obstruction after adhesions. [Intestines in Uro p22]
- Haemorrhage from the anastomosis is commonest when stomach is used. [Intestines in Uro p22]
- Intestinal stenosis is caused by oedema when immediate and by ischaemia when delayed. [Intestines in Uro p22]
- Pseudo-obstruction usually occurs within the first 3 days in patients with multiple comorbidities. [Intestines in Uro p22]
- A stricture of the isolated intestinal segment forms through lymphoid depletion, bacterial infection, then submucosal oedema with fibrosis. [Intestines in Uro p22]
- Elongation of the isolated intestinal segment is usually due to distal obstruction. [Intestines in Uro p22]

Abdominal stomas

- Early stomal complications include bowel necrosis, bleeding, prolapse, obstruction and stomal retraction, as well as parastomal hernia and stenosis. [Intestines in Uro p23]
- Peristomal dermatitis may be irritative, pseudoverrucous, or erythematous with erosive lesions. [Intestines in Uro p23]

Ureteroileal anastomosis: principles and renal outcome

- A refluxing anastomosis is an advantage in patients whose remaining urothelium is prone to malignant change, because the upper tracts can be imaged by instilling contrast into the conduit. [Intestines in Uro p24]

- Overall, about 12.5% of patients develop renal failure after ureteroileal diversion, from lack of ureteric motility, infection, stones and obstruction at the anastomosis. [Intestines in Uro p24]
- The ureter is cleaned of its adventitia for only 2–3 mm at the distal end where the anastomosis is made. [Intestines in Uro p24]
- To create a submucosal tunnel for a non-refluxing anastomosis, saline is injected submucosally with a 25-gauge needle to raise the mucosa and ease dissection. [Intestines in Uro p24]
- Ureterointestinal stricture is prevented by soft silastic stents and better suture material. [Intestines in Uro p24]

Ureterocolic anastomosis

- The Leadbetter–Clarke combined technique is similar to the extravesical technique used in transplantation. [Intestines in Uro p25]
- The Strickler ureterointestinal anastomosis is a non-refluxing ureterocolic tunnel in which the ureter enters from the lateral side. [Intestines in Uro p25]
- The Pagano technique brings both ureters through a single incision into two submucosal tunnels, and it has the lowest stricture rate among ureterocolonic anastomoses. [Intestines in Uro p25]
- The Cordonnier and Nesbit techniques are refluxing ureterocolic anastomoses and are not preferred for ureterosigmoidostomy. [Intestines in Uro p25]

Ureteroileal anastomosis: techniques

- The Wallace anastomosis is not recommended in patients with extensive CIS or a high likelihood of recurrent tumour, because a single tumour can block both ureters. [Intestines in Uro p26]
- A tunnelled small-bowel antireflux anastomosis is not widely used. [Intestines in Uro p26]
- In the split-nipple technique (Griffiths), the spatulated ureter is turned back on itself and the nipple is implanted into the bowel. [Intestines in Uro p26]
- In the Le Duc technique, the ureter is laid onto the inner bowel wall and becomes a submucosal tunnel when the mucosa re-epithelialises over it. [Intestines in Uro p26]
- In the hammock anastomosis, the conjoined ureters are sutured to the intestinal mucosa, with longitudinal seromuscular incisions cross-hatched by multiple incisions. [Intestines in Uro p26]
- In the ureteral dipping technique, the ureter is dipped into the bowel lumen and secured to the seromuscular layer without a mucosal anastomosis. [Intestines in Uro p26]
- The U-SB anastomosis uses serosal compression of an extramural ureter as the antireflux mechanism. [Intestines in Uro p26]
- Intussuscepted ileocaecal valves and intussuscepted ileal valves are further intestinal antireflux valves, besides the nipple valve. [Intestines in Uro p26]

Complications of ureteroileal anastomosis

- Urinary fistula from the ureteroileal anastomosis usually occurs within 7–10 days. [Intestines in Uro p27]
- Endoscopic treatments for ureterointestinal stricture are balloon dilatation, cold-knife incision and laser incision. [Intestines in Uro p27]
- Pyelonephritis after diversion, early or long-term, causes considerable morbidity and significant mortality. [Intestines in Uro p27]

Renal deterioration

- Renal deterioration after diversion does not depend much on the type of urinary diversion. [Intestines in Uro p30]

CH 134 · CONTINENT URINARY DIVERSION

Principles of Creating Reservoir

- Splitting the bowel on its antimesenteric wall interrupts the coordinated activity fronts, but this activity is regained over a period of time. [Compilation 01-2 p32]

Orthotopic Urinary Diversion: Principles

- The three principles of orthotopic diversion can be remembered as the "3 Cs": capacity, compliance and continence. [Orthotopic diversion p1]

Pre-requisites / Patient Selection: Tumour-related Factors

- The patient must agree to indefinite follow-up before an orthotopic neobladder is offered. [Orthotopic diversion p2]
- In men with prostatic mucosal, ductal or stromal invasion, urethrectomy is an option when neoadjuvant chemotherapy is not possible. [Orthotopic diversion p2]

Pre-requisites / Patient Selection: Patient-related Factors

- Obesity is not a contraindication to a neobladder; in obese patients stoma creation itself can be difficult. [Orthotopic diversion p3]

Surgical Techniques: Choice of Bowel Segment

- An advantage of a gastric segment is that its acid secretion reduces bacterial contamination of the urine. [Orthotopic diversion p5]

Different Methods of Orthotopic Reservoir

- Orthotopic reservoirs can be grouped as ileal (Camey II, Hemi-Kock, Ghoneim serous-lined extramural tunnel, Hautmann, Studer, T pouch), ileocolic (Mainz, Le Bag) and colonic (right colon pouch, sigmoid pouch). [Orthotopic diversion p6]
- The Le Bag pouch is an ileocolic neobladder made from ileum, caecum and colon. [Orthotopic diversion p6]
- Camey II detubularises the ileum and folds it in a transverse U orientation. [Orthotopic diversion p7]
- The Barre modification of the Camey pouch places the ileum in a Z configuration. [Orthotopic diversion p7]
- In the Ghoneim (serous-lined extramural tunnel) neobladder, the ureters are passed into the 40 cm W-shaped ileal reservoir through extramural serous-lined tunnels made with continuous seromuscular sutures over the ileum, which form a backing for the ureters. [Orthotopic diversion p7]

Ileal Neobladder (Hautmann Pouch)

- The Hautmann pouch originally used a Le Duc antireflux ureteric anastomosis, later replaced by a freely refluxing end-to-side anastomosis. [Orthotopic diversion p8]
- In the Hautmann pouch, the ureters are implanted from inside the neobladder over stents, the limbs are joined with running absorbable sutures, and the rest of the anterior wall is then closed. [Orthotopic diversion p8]
- A modification of the Hautmann pouch uses only 40 cm of ileum. [Orthotopic diversion p8]
- For short ureters, one or both ends of the Hautmann W can be left long (as chimneys) for the ureteroileal anastomosis. [Orthotopic diversion p8]

Studer Pouch

- In the Studer pouch, the two 22 cm limbs are opened from the apex towards the opening of the afferent limb. [Orthotopic diversion p9]
- When the Studer reservoir is folded and closed, the distal one fingerbreadth is left open for the urethral anastomosis. [Orthotopic diversion p9]
- The Studer afferent limb can be combined with a flap-valve or modified T-pouch antireflux technique for the ureteroileal anastomosis. [Orthotopic diversion p9]

T-pouch Modification

- The T pouch is a modification of Ghoneim's serous-lined extramural tunnel, designed to prevent its long-term complications while keeping an antireflux mechanism; it can also be used with a Studer neobladder. [Orthotopic diversion p10]
- In the T pouch, the antireflux mechanism is made by anchoring the distal 3–4 cm of the 15 cm ileal segment into the serosa-lined ileal trough formed by the bases of the two adjacent 22 cm ileal segments. [Orthotopic diversion p10]

Mainz Ileocolonic Reservoir and Le Bag Reservoir

- The Mainz ileocolonic neobladder uses a W-shaped bowel segment that is opened fully; the posterior wall is made by joining the three opposing limbs and the reservoir is closed side to side. [Orthotopic diversion p11]
- In the Mainz neobladder, a button-hole incision in the base of the caecum is used for the urethral anastomosis. [Orthotopic diversion p11]
- The Le Bag reservoir uses about 20 cm of colon and 20 cm of ileum, opened on the antimesenteric border and sutured to form the posterior wall. [Orthotopic diversion p11]
- In the Le Bag reservoir, the urethra is joined end-to-side to the caecum, the ureters are anastomosed, and the neobladder is closed anteriorly. [Orthotopic diversion p11]

Right Colon Pouch and Sigmoid Pouch (Reddy)

- The orthotopic right colon pouch uses the entire right colon and caecum; the ileal stump is closed. [Orthotopic diversion p12]
- The right colon is opened along the anterior taenia, leaving the proximal 2–3 inches of caecum intact. [Orthotopic diversion p12]
- In the right colon pouch, the ureters are joined by an antirefluxing ureterocolonic anastomosis, a urethrocolonic anastomosis is made, and the colon is folded in a Heineke–Mikulicz manner and closed. [Orthotopic diversion p12]
- The sigmoid (Reddy) pouch is based on the inferior mesenteric artery. [Orthotopic diversion p12]
- In the sigmoid pouch, the medial taenia is incised down to a point just short of the site of the urethral anastomosis, before the posterior wall is sutured. [Orthotopic diversion p12]

Disadvantages, Advantages and Outcome of Neobladder

- An advantage listed for the neobladder is that the kidneys are protected from infections. [Orthotopic diversion p13]
- Mortality after orthotopic neobladder is 0–6%. [Orthotopic diversion p13]
- Early complications occur in 5–30% and late complications in 10–35% of neobladder patients. [Orthotopic diversion p13]

Complications

- Complications are divided into those not directly related to the diversion (early: bleeding, thrombotic events, infection, cardiovascular and pulmonary complications; late: bowel obstruction, ventral hernia, thrombotic and cardiovascular events) and those directly related to it. [Orthotopic diversion p14]
- Afferent limb obstruction, together with ureteroileal obstruction, is a diversion-related complication. [Orthotopic diversion p14]

Pouch Perforation, Continence and Retention

- Neobladder perforation is rare because outlet resistance is low. [Orthotopic diversion p15]
- Decreased sphincter tone at night is a reason for night-time incontinence after neobladder. [Orthotopic diversion p15]
- Reversal of the normal night-time ADH effect is a reason for night-time incontinence after neobladder. [Orthotopic diversion p15]
- Use of an excessive length of intestine (more than 60 cm) is a risk factor for retention after neobladder. [Orthotopic diversion p16]
- A nerve-sparing or prostate-sparing procedure is a risk factor for retention after neobladder. [Orthotopic diversion p16]
- Biofeedback and pelvic floor muscle training are options for neobladder retention, alongside clean intermittent catheterisation. [Orthotopic diversion p16]

Follow-up

- Physical examination at follow-up includes pelvic and rectal examination. [Orthotopic diversion p17]
- Yearly PSA is checked if prostate cancer was found on cystectomy pathology. [Orthotopic diversion p17]

- Imaging depends on cystectomy pathology: for pT2 or lower, CT and chest radiograph at 4 months, 12 months, then yearly to 5 years, then every 2 years. [Orthotopic diversion p17]
- For pT2 or lower disease, ultrasound may replace CT after 3 years. [Orthotopic diversion p17]
- For pT3, node-positive or higher disease, CT of abdomen and pelvis with chest CT or chest radiograph is done at each visit. [Orthotopic diversion p17]

CH 135 · MINIMALLY INVASIVE URINARY DIVERSION

Contraindications

- Bladder pathology that would preclude use of the native bladder is a contraindication to augmentation. [Augmentation Cystoplasty p2]

Gastrocystoplasty

- Gastrocystoplasty is rarely done nowadays. [Augmentation Cystoplasty p3]
- An advantage of gastrocystoplasty is less bacterial colonisation. [Augmentation Cystoplasty p3]
- Haematuria–dysuria syndrome occurs in about 70% after gastrocystoplasty. [Augmentation Cystoplasty p3]
- Peptic ulceration and raised gastrin are disadvantages of gastrocystoplasty. [Augmentation Cystoplasty p3]

Operative technique: incision and native bladder

- Intestinal cystoplasty is done through a midline or a lower transverse incision. [Augmentation Cystoplasty p4]
- Formerly the supratrigonal bladder was removed, but now the native bladder is usually preserved as long as it is widely opened. [Augmentation Cystoplasty p4]
- The bladder is generally bivalved by a sagittal incision from several centimetres above the bladder neck anteriorly to just above the trigone posteriorly, which makes anastomosis of the bowel easy and keeps the native bladder's contribution to overall capacity. [Augmentation Cystoplasty p4]
- A greater circumference can be achieved by making transverse incisions into the two bivalved halves. [Augmentation Cystoplasty p4]

Ileocystoplasty and caecocystoplasty

- The length of bowel used depends on the native bladder volume, the patient's urine output and the desired final capacity. [Augmentation Cystoplasty p5]
- The isolated ileal segment is irrigated with 0.25% neomycin solution before being opened on its antimesenteric border. [Augmentation Cystoplasty p5]
- The ileum may be reconfigured into an S shape as well as U or W, using full-thickness sutures with inversion of the mucosa. [Augmentation Cystoplasty p5]
- The ileum-to-bladder anastomosis is started posteriorly, which is easier, and then completed anteriorly, with absorbable sutures in one or two layers. [Augmentation Cystoplasty p5]
- Caecocystoplasty alone is no longer done. [Augmentation Cystoplasty p5]

Ileocaecocystoplasty (Mainz)

- In ileocaecocystoplasty, the ileum can alternatively be opened and used as a patch on the caecal segment. [Augmentation Cystoplasty p6]
- The right colon is mobilised up to the hepatic flexure. [Augmentation Cystoplasty p6]
- In a typical ileocaecal augmentation the colonic segment equals the ileal segment in length, whereas in the Mainz ileocaecocystoplasty the ileum is twice the length of the caecum. [Augmentation Cystoplasty p6]
- In the Mainz ileocaecocystoplasty the suprapubic tube is placed through the native bladder. [Augmentation Cystoplasty p6]
- If the appendix in the ileocaecal segment is not to be used, appendicectomy is done. [Augmentation Cystoplasty p7]
- For a continent stoma the appendix can be kept in situ with its base tunnelled beneath a taenia. [Augmentation Cystoplasty p7]
- Ileum with an imbricated ileocaecal valve can be used to bridge short ureters, the valve acting as an antireflux mechanism. [Augmentation Cystoplasty p7]

Sigmoid cystoplasty and gastrocystoplasty technique

- The sigmoid can be detubularised either by opening it along the antimesenteric border and closing the two ends, or by U- or S-shaped reconfiguration. [Augmentation Cystoplasty p8]
- A large suprapubic tube is kept after sigmoid cystoplasty. [Augmentation Cystoplasty p8]
- Gastrocystoplasty can use the gastric body or the antrum. The body technique takes a 10–20 cm segment, which has more acid-producing cells, on the right gastroepiploic artery, and a bladder drainage tube is kept. [Augmentation Cystoplasty p8]

Postoperative management

- A nasogastric tube is kept until bowel function returns, although some surgeons do not use one. [Augmentation Cystoplasty p9]
- Extravesical drains are removed after several days if there is no drainage. [Augmentation Cystoplasty p9]
- At 3 weeks the suprapubic catheter is clamped after the cystogram, and CIC is also done once or twice at night. [Augmentation Cystoplasty p9]
- The post-void residual is checked, and CIC continues if it is high. [Augmentation Cystoplasty p9]

Complications

- Gastrointestinal complications include bowel obstruction and early satiety after use of stomach. [Augmentation Cystoplasty p10]
- Persistent rhythmic bowel contractions may cause persistent incontinence, VUR, hydronephrosis or delayed perforation. [Augmentation Cystoplasty p10]
- Patients with delayed spontaneous perforation are very sick. [Augmentation Cystoplasty p11]
- Delayed spontaneous perforation is caused by catheter-induced injury or transmural infection. [Augmentation Cystoplasty p11]
- Complications related to the bowel anastomosis include stricture and internal hernias, as well as leak, peritonitis and obstruction. [Continent Pouches p4]
- Pouchitis is a complication of the reservoir and needs at least 10 days of antibiotics. [Continent Pouches p4]
- Retention in a continent pouch occurs most commonly with a nipple valve. [Continent Pouches p4]

Pregnancy after augmentation

- The expanding uterus is probably not detrimental to the pedicle of the reconstruction. [Augmentation Cystoplasty p11]
- If caesarean section is done, care must be taken of the augmentation pedicle and any stoma. [Augmentation Cystoplasty p11]

Continent Catheterizing Pouches: sites of stoma

- The alternative stoma site to the umbilicus is the lower quadrant of the abdomen, brought out through the rectus bulge and below the bikini line. [Continent Pouches p1]

Four general principles for continence

- For right colon pouches, appendiceal tunnelling procedures are the simplest of all continence mechanisms to perform. [Continent Pouches p1]
- The intussuscepted nipple valve is the most technically demanding continence mechanism (the flap valve avoids the need for intussusception). [Continent Pouches p1]
- The fourth principle is provision of a hydraulic valve, as in the Benchekroun nipple, which is a largely abandoned procedure. [Continent Pouches p1]

Various types / procedures

- The Kock pouch (continent ileal reservoir) is noted as using 15 cm of terminal ileum, and it preserves the ileocaecal valve. [Continent Pouches p2]
- The Kock pouch has a high complication rate and has been abandoned. [Continent Pouches p2]
- The double T-pouch uses 70 cm of terminal ileum arranged in a W-shaped configuration. [Continent Pouches p2]

- In the double T-pouch, an antiperistaltic 12–15 cm segment of distal ileum provides continence and the proximal 10 cm serves as the antireflux mechanism. [Continent Pouches p2]
- The double T-pouch preserves the ileocaecal valve and has lower complication rates than the Kock pouch. [Continent Pouches p2]
- The Mainz I pouch is built in a W configuration from limbs of about 10–15 cm, 10–15 cm, 10–15 cm and 20–25 cm (caecum/ascending colon and ileum), with the nipple valve made by intussusception of ileum through the ileocaecal valve. [Continent Pouches p2]
- The Mainz I pouch has a high complication rate (31%). [Continent Pouches p2]
- An appendicectomy is done in the Indiana pouch. [Continent Pouches p2]
- In the Indiana pouch, buttressing of only the ileocaecal valve is probably needed, while the remaining ileum (neourethra) can be tapered and brought out. [Continent Pouches p2]
- Variations of the Indiana pouch are the Florida pouch and the Miami pouch. [Continent Pouches p2]
- The Indiana pouch is one of the most reliable of all reservoirs, is easy to construct and has a low complication rate. [Continent Pouches p2]
- The Penn pouch is an ileocaecal pouch with 10–12 cm of ileum and was the first diversion to employ the Mitrofanoff principle. [Continent Pouches p3]
- In the Penn pouch, the appendix is excised with a button of caecum and reversed on itself before a tunnelled reimplant (the alternative is burying the appendix in the taenia with the tip brought out as the stoma). [Continent Pouches p3]
- The Penn pouch is easy to construct and is suitable for patients without a history of prior appendicectomy. [Continent Pouches p3]
- Gastric pouches use a wedge-shaped segment of stomach from the greater curvature based on the left gastroepiploic artery. [Continent Pouches p3]
- In a gastric pouch, the contralateral distal ureter (after transureteroureterostomy) can be tunnelled into the gastric reservoir, like an appendix, as the continence mechanism. [Continent Pouches p3]
- Alternatively, a gastric pouch can use a Mitrofanoff procedure, with a composite gastric reservoir and ileum segment and an appendix channel. [Continent Pouches p3]
- Possible advantages of gastric pouches are fewer infections and fewer metabolic complications. [Continent Pouches p3]
- Gastric pouches may be especially useful in paediatric patients, but experience is limited at present. [Continent Pouches p3]

Recent Concepts: alternatives to GI cystoplasty

- The alternatives to GI cystoplasty are ureterocystoplasty, autoaugmentation and synthetic material, and none of them carries a tumour risk. [Compilation 01-2 p33]
- Ureterocystoplasty can achieve a bladder capacity of up to 470 mL, but it is applicable only in limited cases. [Compilation 01-2 p33]
- In autoaugmentation, the detrusor over the bladder dome is split or excised and its edges are sutured to the psoas muscle, so the mucosa bulges out. [Compilation 01-2 p33]
- Autoaugmentation can increase bladder capacity by up to 40%. [Compilation 01-2 p33]
- Autoaugmentation is not popular because collagen deposition over the mucosa leads to contraction and about a 50% reduction in capacity, so long-term results are poor when capacity is the concern. [Compilation 01-2 p33]
- Autoaugmentation is indicated only in patients with reasonable capacity but poor compliance due to uninhibited contractions. [Compilation 01-2 p33]
- Seromuscular enterocystoplasty gives good early results but is technically more difficult, and its long-term results are not known. [Compilation 01-2 p33]
- Problems with the bioengineered bladder are that harvesting the patient's own bladder cells is invasive and carries anaesthetic risk; in urine cell culture, cells isolated from the urine are cultured in vitro instead. [Compilation 01-2 p33]
- Urinary "buttons" are an alternative to a conduit or suprapubic catheter for urinary drainage; they have an internal part like a Foley tip with a balloon and an external flat button that sits on the skin.

[Compilation 01-2 p33]

- The urinary button has a one-way valve that maintains continence, and no stoma bag is needed. [Compilation 01-2 p33]

OTHER TOPICS IN THIS BOOK

Management

- Endoscopic management is transurethral resection of the diverticular neck with fulguration of the entire urothelial lining of the sac. [Bladder Diverticulum p5]
- Endoscopic management is indicated for patients who are not candidates for open repair, or for patients undergoing TURP who have an associated poorly draining diverticulum. [Bladder Diverticulum p5]
- Open diverticulectomy can be transvesical (the commonest, including a transvesical submucosal technique), extravesical, or combined transvesical and extravesical. [Bladder Diverticulum p5]
- The combined transvesical and extravesical approach is used for large diverticula and/or considerable peridiverticular inflammation. [Bladder Diverticulum p5]
- Laparoscopic and robotic diverticulectomy give a smaller incision, less pain, shorter hospital stay and shorter convalescence. [Bladder Diverticulum p5]

Complications of diverticulectomy

- Ureteric injury is the most troublesome complication of diverticulectomy. [Bladder Diverticulum p6]
- A partial ureteric injury recognised intraoperatively is repaired and stented, while a complete transection is reimplanted. [Bladder Diverticulum p6]
- Other complications of diverticulectomy include bleeding, vascular injury and rectal injury. [Bladder Diverticulum p6]

Cystitis Glandularis

- Cystitis glandularis is usually seen near the trigone and has a cobblestone appearance at cystoscopy. [Compilation 04-3 p9]
- Typical cystitis glandularis is superficial, whereas the intestinal type is less common and resembles neoplasia. [Compilation 04-3 p9]
- Cystitis glandularis can rarely cause ureteric obstruction with hydronephrosis. [Compilation 04-3 p9]
- Eliminating chronic irritation in cystitis glandularis includes replacing an indwelling (per-urethral) catheter with clean intermittent catheterisation. [Compilation 04-3 p9]
- Distal obstruction such as prostatic enlargement or urethral stricture should be managed in cystitis glandularis. [Compilation 04-3 p9]
- Bladder augmentation or cystectomy has rarely been reported for intractable cystitis glandularis or for bladder contracture. [Compilation 04-3 p9]

Late Complications of Neobladder

- Late complications are influenced by seven factors: tumour recurrence, adjuvant or salvage chemotherapy, radiation, the type of bowel segment, the technique of diversion, the patient's general condition (nutrition) and associated comorbidities. [UTUC notes 2 p21]
- Late complications are classified as either not directly related to the diversion or directly related to it. [UTUC notes 2 p21]
- Late complications directly related to the diversion are incontinence, perforation/rupture, retention, afferent limb obstruction, urethral stricture, vaginal fistula and stones. Urethral stricture is the item not listed in the corpus. [UTUC notes 2 p21]
- Afferent limb obstruction risk is highest with the antireflux valve in the hemi-Kock pouch and the T-pouch. [UTUC notes 2 p22]
- Pouch stones are rare in Studer and Hautmann neobladders. [UTUC notes 2 p22]
- To prevent a pouch–vaginal fistula, close the vaginal cuff carefully and watertight, as well as placing an omental flap. [UTUC notes 2 p22]

13 Kidney & UTUC

CH 136 · BENIGN RENAL TUMOURS

Oncocytoma

- Oncocytoma makes up 3 to 7% of kidney tumours, with a female-to-male ratio of 2:1 and a peak in the 7th decade. [Compilation 06 p11-13]
- The commonest genetic abnormality in oncocytoma is loss of heterozygosity of chromosome 1 and/or chromosome 14. [Compilation 06 p11-13]
- Grossly, oncocytoma is mahogany or tan, homogeneous, well circumscribed and has a pseudocapsule. [Compilation 06 p11-13]
- Microscopically, oncocytoma consists of large, uniform, round or polygonal, highly eosinophilic cells arranged in a nested pattern. [Compilation 06 p11-13]
- The differential diagnosis of oncocytoma includes clear cell RCC with eosinophilic characteristics, as well as chromophobe RCC. [Compilation 06 p11-13]
- On angiography oncocytoma shows a spoke-wheel pattern, though this has low predictive value. [Compilation 06 p11-13]
- Oncocytoma shows its peak HU during the nephrographic phase, whereas RCC peaks during the corticomedullary phase. [Compilation 06 p11-13]
- An optimal immunohistochemical panel for separating RCC from oncocytoma on core biopsy uses 3 markers (vimentin, ECAM and glutathione-S-transferase alpha), with 100% sensitivity and 100% specificity. [Compilation 06 p11-13]
- Intraoperative frozen section is not sensitive for diagnosing oncocytoma. [Compilation 06 p11-13]

Angiomyolipoma

- Angiomyolipoma makes up less than 10% of renal tumours. [Compilation 06 p14-17]
- Angiomyolipoma strongly expresses oestrogen receptor beta, progesterone receptor and androgen receptor. [Compilation 06 p14-17]
- About 70–80% of angiomyolipomas are sporadic and 20–30% are genetic (tuberous sclerosis). [Compilation 06 p14-17]
- On CT, more than 20 pixels with HU below −20, or more than 5 pixels with HU below −30, gives a 100% positive predictive value for angiomyolipoma. [Compilation 06 p14-17]
- Indentation of the renal parenchyma on CT is characteristic of angiomyolipoma. [Compilation 06 p14-17]
- Aneurysmal dilatations are seen on CT angiography in 50% of angiomyolipomas. [Compilation 06 p14-17]
- Angiomyolipoma may be confused with liposarcoma; HMB-45 staining, positive in angiomyolipoma, confirms the diagnosis. [Compilation 06 p14-17]
- Selective embolisation of angiomyolipoma often needs repeat procedures. [Compilation 06 p14-17]

Renal Cysts

- Renal cysts are the commonest benign renal lesions, making up 70% of asymptomatic renal masses. [Compilation 06 p19-20]
- Risk factors for renal cysts are increasing age, male gender, hypertension and renal insufficiency. [Compilation 06 p19-20]
- The Bosniak classification can be used with MRI, but MRI tends to exaggerate some findings. [Compilation 06 p19-20]
- For a parapelvic cyst, a laparoscopic approach is better than a percutaneous one because of nearby vital structures. [Compilation 06 p19-20]
- Complications of renal cysts are pain from haemorrhage, infection and rupture. [Compilation 06 p19-20]

MEST and Cystic Nephroma

- Female predilection, a history of hormonal ablation therapy in affected males, and frequent ER/PR expression suggest that sex hormones have a role in the pathogenesis of MEST and cystic

nephroma. [Compilation 06 p21-22]

- On imaging, cystic nephroma is a solitary, centrally located lesion with curvilinear calcification and septal enhancement, and cannot be differentiated from cystic RCC. [Compilation 06 p21-22]
- Histologically, cystic nephroma has a fibrous pseudocapsule and cysts lined by flattened, cuboidal or hobnail epithelium, with a stroma that resembles ovarian stroma. [Compilation 06 p21-22]
- In cystic nephroma the epithelial component stains for cytokeratin, and the stromal component for ER, PR, CD10 and calretinin. [Compilation 06 p21-22]
- MEST peaks in the 5th decade, usually in women with a history of oestrogen replacement therapy. [Compilation 06 p21-22]
- MEST appears as a Bosniak III/IV cyst indistinguishable from cystic RCC; consider the diagnosis pre-operatively in a woman on HRT, and treat by partial nephrectomy if feasible, otherwise radical nephrectomy. [Compilation 06 p21-22]

Metanephric Adenoma and Papillary Adenoma

- Metanephric adenoma is radiologically indistinguishable from RCC and shares some features with Wilms tumour. [Compilation 06 p23]
- About 10% of patients with metanephric adenoma have polycythaemia. [Compilation 06 p23]
- Metanephric adenoma shows peripheral or central calcification and enhances less than the surrounding cortex and medulla. [Compilation 06 p23]
- Metanephric adenoma is usually confirmed after excision by positivity for WT1 and S100 and by testing for BRAF V600E mutation. [Compilation 06 p23]
- Papillary adenoma may be a premalignant lesion for papillary RCC, and many believe all solid epithelial renal masses are potentially malignant and need treatment. [Compilation 06 p23]
- The incidence of papillary adenoma rises with age, male sex and acquired renal cystic disease. [Compilation 06 p23]

Classification of Renal Tumours

- Leukaemia is listed among the malignant renal masses. [Compilation 06 p45]
- Infected renal cyst and rheumatic granuloma are listed among the inflammatory renal masses. [Compilation 06 p45]

Tuberous Sclerosis — genetics and pathogenesis

- Tuberous sclerosis was classically called epiloia, the triad of epilepsy, low intelligence and adenoma sebaceum. [Tuberous Sclerosis p1]
- TSC1 and TSC2 are tumour suppressor genes that regulate cell proliferation and differentiation. [Tuberous Sclerosis p1]
- Pathogenesis: loss of TSC1/TSC2 leads to accumulation of Rheb, then accumulation of FKBP38, which activates mTORC1 (mTOR complex 1), the step blocked by mTOR inhibitors. [Tuberous Sclerosis p1]

Urological manifestations

- Renal disease is the second commonest cause of morbidity and mortality in TSC, after neurological disease. [Tuberous Sclerosis p2]
- Renal cysts are the second most common renal manifestation after angiomyolipoma. [Tuberous Sclerosis p2]
- Oncocytoma and perirenal cysts are also renal manifestations of TSC. [Tuberous Sclerosis p2]

Diagnostic criteria

- Major criteria (besides renal AML, facial angiofibromas and cortical tubers) include ungual or periungual fibroma, 3 or more hypomelanotic macules, shagreen patch, multiple retinal hamartomas, subependymal nodule, subependymal (giant cell) astrocytoma, cardiac rhabdomyoma and lymphangioleiomyomatosis. [Tuberous Sclerosis p2]
- Minor criteria are multiple renal cysts, non-renal hamartoma, hamartomatous rectal polyps, retinal achromic patch, bone cysts, gingival fibromas, confetti skin lesions, dental enamel pits and cerebral white matter radial migration lines. [Tuberous Sclerosis p2]

Management of AML in TSC

- Before any surgery for TSC-associated AML, the neurological status must be fully evaluated and the patient optimised because of seizure risk; involve a neurophysician. [Tuberous Sclerosis p3]
- Sirolimus, as well as everolimus, has been used (in trials) for TSC-associated AML. [Tuberous Sclerosis p3]
- Embolisation is preferred in acute haemorrhage because emergency surgery almost always ends in nephrectomy. [Tuberous Sclerosis p4]
- Ablative therapies may be considered for comorbid patients or those with multiple bilateral tumours. [Tuberous Sclerosis p4]

CH 137 · MALIGNANT RENAL TUMOURS

RCC – General Aspects

- The incidence of RCC is about 12 per lakh (100,000) per year. [Compilation 06 p1]
- Wilms tumour is commoner in younger children, but RCC is as common as Wilms tumour in the second decade of life. [Compilation 06 p1]
- Childhood RCC is more likely than adult RCC to be symptomatic, locally advanced, high grade and of unfavourable histology. [Compilation 06 p1]

Etiology

- Papillary and clear cell RCC arise from proximal convoluted tubule cells, whereas medullary carcinoma, collecting duct carcinoma and chromophobe RCC arise from collecting duct cells. [Compilation 06 p2]
- All forms of tobacco are implicated in RCC, and the risk begins to fall after smoking stops. [Compilation 06 p2]
- For obesity, the relative risk of RCC is 1.07 for each additional unit of BMI. [Compilation 06 p2]
- Proposed mechanisms linking obesity to RCC are raised IGF-1, raised oestrogen, increased arteriolar nephrosclerosis and local inflammation. [Compilation 06 p2]
- Hypertension is thought to promote RCC by causing renal injury and inflammation, and metabolic and functional changes in the renal tubules. [Compilation 06 p2]

Clinical Presentation

- IVC obstruction by RCC can also present with dilated superficial anterior abdominal wall veins, proteinuria or deteriorating ipsilateral renal function, pulmonary embolic episodes and a right-sided cardiac mass. [Compilation 06 p3]
- Symptoms of systemic (metastatic) RCC also include cervical lymphadenopathy and jaundice (common metastatic sites being lung, liver, lymph nodes and bone). [Compilation 06 p3]

Paraneoplastic Syndromes and Hypercalcaemia

- Other paraneoplastic manifestations of RCC listed are anaemia, pyrexia, cachexia with weight loss, and amyloidosis. [Compilation 06 p4]
- Focused radiation therapy to metastatic bone lesions is part of the management of hypercalcaemia in RCC. [Compilation 06 p4]

Stauffer's Syndrome

- Clinical features of Stauffer syndrome are fever, weight loss and signs of liver dysfunction such as oedema and bleeding tendencies. [Compilation 06 p5]
- Laboratory abnormalities in Stauffer syndrome include raised prothrombin time, low albumin, raised SGOT/SGPT, low platelets and neutropenia, in addition to raised ALP. [Compilation 06 p5]
- Liver biopsy in Stauffer syndrome shows a non-specific hepatitis with a prominent lymphocytic infiltrate. [Compilation 06 p5]

Changes in TNM Staging in Recent Times

- T1 was divided into T1a and T1b in 2002, reflecting the different prognosis of the two groups. [Compilation 06 p7-8]

- T2 was subsequently subdivided (T2a/T2b), supporting the prognostic relevance of the 10 cm break-point. [Compilation 06 p7-8]
- In the 2009 revision, isolated renal vein thrombosis was downgraded from T3b to T3a, reflecting its more favourable prognosis. [Compilation 06 p7-8]
- Nodal disease was previously subdivided by the number of nodes involved; this was abolished because it showed no prognostic significance. [Compilation 06 p7-8]

Robson Staging

- Robson stage I is tumour within the renal capsule. [Compilation 06 p8]
- Robson stage II is tumour invading perinephric fat but confined within Gerota's fascia. [Compilation 06 p8]
- Robson stage III is involvement of regional lymph nodes and/or the renal vein or IVC. [Compilation 06 p8]
- Robson stage IV is invasion of adjacent organs or distant metastasis. [Compilation 06 p8]

Pathological Subtypes of RCC

- Fuhrman grading by nuclear size and features: grade I is about 10 µm, round and uniform with absent or inconspicuous nucleoli; grade II is about 15 µm, irregular, with small nucleoli visible at 400x; grade III is about 20 µm, irregular, with prominent nucleoli; grade IV is 20 µm or more, bizarre and multilobed, with prominent nucleoli and heavy chromatin clumps. [Compilation 06 p27-29]
- Clear cell RCC can result from VHL gene hypermethylation as well as mutation. [Compilation 06 p27-29]
- Clear cell RCC shows loss of chromosomes 8p and 14q in addition to 9p. [Compilation 06 p27-29]
- Grossly, clear cell RCC is a well circumscribed, lobulated, golden-yellow tumour that arises from the proximal convoluted tubule. [Compilation 06 p27-29]
- Chromophobe RCC has a well circumscribed, homogeneous, tan or light-brown cut surface and stains positive with Hale's colloidal iron. [Compilation 06 p27-29]
- Papillary RCC is a fleshy tumour with a fibrous pseudocapsule in which necrosis and haemorrhage are common, and it arises from the proximal convoluted tubule. [Compilation 06 p27-29]
- Papillary RCC type I has basophilic cells with low-grade nuclei and type II has eosinophilic cells with high-grade nuclei; type I has a better prognosis than type II. [Compilation 06 p27-29]
- Collecting duct carcinoma makes up less than 1% of RCC and is firm, with high-grade nuclei and frequent mitoses. [Compilation 06 p27-29]

Pathological Subtypes of RCC (contd): Medullary RCC and rare subtypes

- Many cases of renal medullary carcinoma are already locally advanced or metastatic when diagnosed. [Compilation 06 p30]
- Renal medullary carcinoma shares its histology with collecting duct carcinoma. [Compilation 06 p30]
- Post-neuroblastoma RCC is listed as a separate pathological subtype of RCC. [Compilation 06 p30]

Familial RCC Syndromes

- In succinate dehydrogenase (SDH)-deficient RCC, the SDHB gene is on chromosome 1 and the SDHD gene on chromosome 11. [Compilation 06 p32-33]
- The renal tumours seen in SDH-RCC syndrome include chromophobe RCC, clear cell RCC, type 2 papillary RCC and oncocytoma. [Compilation 06 p32-33]
- Papillary thyroid carcinoma is an extrarenal association of SDH-RCC syndrome, along with paraganglioma. [Compilation 06 p32-33]
- Birt–Hogg–Dubé syndrome can also give clear cell RCC and papillary RCC occasionally, in addition to chromophobe and hybrid oncocytic tumours. [Compilation 06 p32-33]

VHL Syndrome

- The VHL protein has an alpha-domain, which binds the elongin B/elongin C–cullin-2–RBX1 complex, and a beta-domain; mutation of the alpha-domain disrupts this VHL complex. [Compilation 06 p34-35]
- By haemangioblastoma risk, VHL type 1, type 2A and type 2B all have a high incidence of haemangioblastoma, whereas type 2C has neither RCC nor haemangioblastoma. [Compilation 06 p34-35]

NSS for VHL and Other Familial RCC

- Familial RCC tumours are generally of low stage, even though they are multiple and bilateral and recur more often. [Compilation 06 p40]
- SDH-RCC, like HLRCC, is an exception to the 3 cm rule because it is more aggressive. [Compilation 06 p40]

Screening for RCC in Target Populations

- Relatives of VHL patients who test negative on genetic analysis need only less stringent follow-up. [Compilation 06 p41]
- Patients with tuberous sclerosis should have periodic RCC screening with ultrasound or CT. [Compilation 06 p41]

Diagnostic Techniques in Evaluation of RCC

- IVP is no longer used for renal masses because it lacks sensitivity and specificity, and it misses small anterior or posterior lesions that do not distort the collecting system or the renal contour. [Compilation 06 p46-48]
- Features on IVP that suggest malignancy are calcification within the mass, increased tissue density, an irregular margin and distortion of the collecting system. [Compilation 06 p46-48]
- About 10–20% of renal masses have indeterminate CT features, and renal mass biopsy should be considered for them. [Compilation 06 p46-48]
- A renal pseudotumour shows on CT as an enhancing renal segment that is isodense with the rest of the kidney; biopsy is considered when it is uncertain. [Compilation 06 p46-48]
- Strategies to reduce the risk of nephrogenic systemic fibrosis are to avoid gadolinium in CKD patients, to dialyse after the procedure, and to use periprocedural hydration with CT instead. [Compilation 06 p46-48]
- Renal arteriography now has a limited role. Its indications are a renal mass with concomitant renal artery disease, equivocal cases where the presence or absence of neovascularity may help diagnosis, and showing the vascularity of an IVC tumour thrombus, which is seen in 35–40% of cases. [Compilation 06 p46-48]

Renal Mass Biopsy

- Pneumothorax after renal mass biopsy occurs in less than 1%. [Compilation 06 p49-52]
- Needle-tract seeding after renal mass biopsy is a concern mainly for centrally located infiltrative renal masses and for poorly differentiated urothelial cancer. [Compilation 06 p49-52]
- Renal mass biopsy is done under sedation with an 18 G, 16 cm long needle, which is considered ideal. [Compilation 06 p49-52]
- A co-axial technique, taking multiple biopsies through one co-axial cannula, avoids tumour seeding. [Compilation 06 p49-52]
- Biopsy cores should be taken away from necrotic areas of the tumour. [Compilation 06 p49-52]

Metastatic Evaluation of Renal Mass

- Liver function tests should be part of the metastatic evaluation in every case of renal mass, along with chest X-ray and abdominopelvic CT or MRI. [Compilation 06 p51]
- FDG PET-CT has good specificity but suboptimal sensitivity in RCC, and its best role is in patients with equivocal findings on conventional imaging. [Compilation 06 p51]

Prognostic Factors for RCC

- Adverse clinical prognostic factors also include symptomatic (rather than incidental) presentation, systemic symptoms, raised ESR, raised CRP and raised alkaline phosphatase. [Compilation 06 p53]
- Adrenal involvement and metastatic burden are anatomical prognostic factors. [Compilation 06 p53]
- Microvascular invasion and surgical margin status are histological prognostic factors. [Compilation 06 p53]

Molecular Prognostic Factors for RCC

- CA-IX also serves as a marker of response to systemic therapy, so it is valuable in advanced disease. [Compilation 06 p54]
- B7-H1 (PD-L1), a T-cell co-regulatory molecule, is a strong independent predictor of disease progression in RCC, even after accounting for other molecular factors and established clinical and pathological predictors. [Compilation 06 p54]
- Ki-67 correlates with reduced survival in clear cell RCC, is an independent prognostic factor, and has been incorporated into predictive algorithms. [Compilation 06 p54]

Management of Localised RCC: Partial Nephrectomy

- Treatment options for local recurrence after partial nephrectomy are repeat partial nephrectomy, radical nephrectomy, thermal ablation or active surveillance. [Compilation 06 p59]

Radical Nephrectomy

- Extended lymphadenectomy from the crus of the diaphragm to the aortic bifurcation is controversial; enlarged hilar and immediately adjacent para-aortic or paracaval nodes are removed. [Compilation 06 p61]
- A large tumour of more than 10 cm is an indication to remove the adrenal at radical nephrectomy, in addition to direct involvement, locally advanced disease, upper-pole location and diffuse involvement. [Compilation 06 p61]

Active Surveillance

- Masses suited to active surveillance are small, solid, enhancing and homogeneous; masses with poorly defined margins or heterogeneous masses are not appropriate. [Compilation 06 p63]

Surveillance after PN / RN

- In the notes' surveillance scheme, low-risk patients are pT1 N0 Mx, and moderate-to-high-risk patients are pT2–4 N0 Mx or pT any N1 M0. [Compilation 06 p64]
- For low-risk patients after partial nephrectomy, CT or MRI is done at 3–6 months, followed by ultrasound, CT or MRI yearly for 3 years. [Compilation 06 p64]
- For low-risk patients after radical nephrectomy, ultrasound, CT or MRI is done at 3–6 months, and imaging after 12 months is at the clinician's discretion. [Compilation 06 p64]
- In low-risk patients, a chest X-ray is done annually for 3 years and after that as clinically indicated. [Compilation 06 p64]
- For moderate-to-high-risk patients, CT or MRI at 3–6 months is followed by ultrasound, CT or MRI every 6 months for 3 years, and then annually to year 5. [Compilation 06 p64]
- In moderate-to-high-risk patients, site-specific imaging is performed as warranted, imaging beyond 5 years is at the clinician's discretion, and chest X-ray follows the same protocol as for low risk. [Compilation 06 p64]
- General surveillance measures after nephrectomy include history and examination and laboratory testing: renal function tests, urinalysis (CUE), eGFR, complete blood count, LDH, liver function tests, alkaline phosphatase and calcium. [Compilation 06 p64]

Zonal NePhRO scoring system

- The Zonal NePhRO score has four components (Nearness, Physical zones, Radius, Organisation), each scored 1, 2 or 3 points according to zone 1, 2 or 3. [Compilation 06 p66]
- Nearness scores 1 if the mass touches the cortex, 2 if it touches the medulla, and 3 if it touches the collecting system or crosses into the renal sinus. [Compilation 06 p66]
- Physical zone scores 1 for a lower-pole mass below the collecting system, 2 for a mass lateral to but not touching the pelvicalyceal system, and 3 for an upper-pole mass or one touching the pelvicalyceal system. [Compilation 06 p66]
- Radius scores 1 for a diameter under 2.5 cm, 2 for 2.5–4 cm, and 3 for more than 4 cm. [Compilation 06 p66]

- Organisation scores 1 for a mass more than 50% exophytic, 2 for 50–75% endophytic, and 3 for more than 75% endophytic. [Compilation 06 p66]
- A Zonal NePhRO total of 4–6 is low risk, 7–9 intermediate and 10–12 high risk. [Compilation 06 p66]

IVC involvement in locally advanced RCC

- Besides leg oedema and a non-reducing right varicocele, IVC tumour thrombus should be suspected with dilated abdominal veins, proteinuria, pulmonary embolism, a right atrial mass or a nonfunctioning ipsilateral kidney. [Compilation 06 p71-73]
- Inferior venacavography can be used to assess IVC thrombus but is rarely done. [Compilation 06 p71-73]
- If the IVC is completely occluded, grafting and reconstruction are not needed after resection, because collateral flow has developed. [Compilation 06 p71-73]

Locally invasive RCC

- The usual presentation of locally invasive RCC is pain, due to invasion of the posterior abdominal wall, nerve roots and paraspinal muscles. [Compilation 06 p73-74]
- Neoadjuvant options for locally invasive RCC are systemic therapy (no defined protocols) and preoperative embolisation. [Compilation 06 p73-74]

Lymph node dissection in RCC

- For a right-sided tumour, the lymphadenectomy removes paracaval, precaval, retrocaval and interaortocaval nodes from the right crus of the diaphragm to the bifurcation of the IVC. [Compilation 06 p74]
- For a left-sided tumour, the lymphadenectomy runs from the superior mesenteric artery to the aortic bifurcation and removes nodes anterior and lateral to the aorta and retroaortic nodes; interaortocaval nodes are removed only if palpable or radiologically enlarged. [Compilation 06 p74]

Local recurrence after RN / NSS

- Complete resection of a local recurrence may be associated with significant morbidity. [Compilation 06 p75]

CH 138 · UROTHELIAL TUMOURS OF THE UPPER URINARY TRACT AND URETER

UTUC: Pathology Statistics

- In the renal pelvis, papillary tumours are invasive in about 50% and sessile tumours in about 80%, a pattern similar to the bladder. [Compilation 01-2 p80]

Prognostic Factors (UTUC)

- Broders classification of UTUC grade: Grade I is papilloma, Grade II well differentiated, Grade III moderately differentiated and Grade IV anaplastic or poorly differentiated. [UTUC notes 2 p1]
- Hydronephrosis is an independent predictor of both advanced disease stage and survival. The corpus has it only as a predictor of invasion and a high-risk feature. [UTUC notes 2 p1]
- A tumour size above 3–4 cm may be associated with worse survival and a higher risk of bladder recurrence. The corpus gives only 2 cm as the best cut-off for at least pT2. [UTUC notes 2 p1-2]
- Advanced age predicts recurrence-free, disease-specific and overall survival in UTUC. [UTUC notes 2 p2]
- By race, Black non-Hispanic patients with UTUC have higher mortality than White non-Hispanic patients. [UTUC notes 2 p2]
- Multifocality is defined as tumour at two or more sites within the urothelium, and it independently predicts poor clinical outcome. [UTUC notes 2 p2]
- Tumour necrosis correlates with aggressive clinicopathological features, including stage, grade, lymphovascular invasion, presence of CIS and lymph node metastasis. [UTUC notes 2 p2]
- A lymph node density of 30% or more predicts poor outcome in UTUC. [UTUC notes 2 p2]
- Lymphovascular invasion is an independent prognostic factor for disease-specific survival, and it correlates with higher tumour stage, grade and disease recurrence. [UTUC notes 2 p2-3]

Molecular Biology (Chromosomal Abnormalities) and Markers

- No molecular prognostic marker for UTUC is yet validated for clinical use. [UTUC notes 2 p3]
- Cell-cycle marker: TP53 mutation is a prognostic marker in UTUC. [UTUC notes 2 p3]
- CDKN1B (formerly called p27) is a cell-cycle prognostic marker in UTUC. [UTUC notes 2 p3]
- Apoptosis markers Bcl-2 and survivin are associated with advanced UTUC. [UTUC notes 2 p3]
- Cell migration and invasion markers, E-cadherin and MMP (matrix metalloproteinase) expression, indicate poor prognosis. [UTUC notes 2 p3]
- Angiogenesis marker HIF-1alpha expression indicates poor prognosis. [UTUC notes 2 p3]
- Cell proliferation markers include overexpression of Ki-67, EGFR, NF-kB and HER-2. [UTUC notes 2 p3]
- Cell differentiation marker uroplakin III expression is associated with lower stage. [UTUC notes 2 p3]
- Mitosis marker Aurora-A expression indicates poor prognosis. [UTUC notes 2 p3]
- cMET and RON belong to the MET proto-oncogene family. The notes record cMET as linked to worse prognosis and RON (with cMET) as having no correlation with outcome. [UTUC notes 2 p3]
- COX-2 expression indicates poor prognosis in UTUC. [UTUC notes 2 p3]

Risk Stratification (Treatment)

- The notes give an older size cut-off of 1 cm (low risk below 1 cm, high risk above 1 cm) alongside the current 2 cm cut-off. The corpus gives only 2 cm (EAU) and 1.5 cm (AUA/SUO). [UTUC notes 2 p4]

Metastatic Disease (UTUC)

- Metastatic UTUC is treated with platinum-based chemotherapy, with or without surgical resection. [UTUC notes 2 p11]
- Other regimens for advanced UTUC are PCG (paclitaxel, carboplatin, gemcitabine), CG (carboplatin–gemcitabine) and cabozantinib, an inhibitor of the MET and VEGF pathways. [UTUC notes 2 p11]
- Radiotherapy has no role in metastatic UTUC, either as sole treatment or as an adjunct to chemotherapy. [UTUC notes 2 p11]
- RNU gives no survival benefit in metastatic UTUC and may be used only for palliation. [UTUC notes 2 p11]
- No chemotherapy regimen is defined as standard for UTUC. [UTUC notes 2 p11]

Urothelial Tumors of Upper Urinary Tract: Epidemiology

- The Balkan countries with the highest UTUC incidence are Albania, Bulgaria, Bosnia and Herzegovina, Croatia, Greece and Romania (South Eastern Europe). [utuc p1]
- UTUC incidence in western countries is about 2 per 100,000. [utuc p1]
- Synchronous bilateral UTUC occurs in 1.6%. [utuc p1]
- Metachronous contralateral UTUC occurs in 2–6% after a UTUC. [utuc p1]
- The notes record metachronous occurrence of 80% "after bladder cancer". This is written as stated; it does not match the 2–4% figure for UTUC after bladder cancer on p4 of the notes and in Book 13 p.13, so check the source. [utuc p1]
- Worse survival is associated with increasing age, male gender, black non-Hispanic race and advanced stage. [utuc p1]
- Overall 5-year disease-specific survival for UTUC is 75%. [utuc p1]
- The sex ratio is reversed in Taiwan, with a male:female ratio of 1:2. [utuc p1]
- UTUC is commoner in whites than in African-Americans. [utuc p1]

Etiology: Genetic (HNPCC / Lynch syndrome)

- Lynch syndrome cancers listed in the notes include pancreas, breast, ovary and stomach cancers as well as colon and uterine cancers. [utuc p1-2]
- Lynch-associated UTUC presents at a younger age, with a mean age of 55. [utuc p1-2]
- Lynch-associated UTUC is more likely in females. [utuc p1-2]

Etiology: Aristolochic acid nephropathy

- The source plants of aristolochic acid are Aristolochia fangchi and Aristolochia clematitis. [utuc p2]
- Aristolochic acid mutates codon 139 of the p53 gene, and this mutation is seen in both Balkan endemic nephropathy and Chinese herb nephropathy. [utuc p2]
- Balkan endemic nephropathy is a degenerative interstitial nephropathy that gives a much higher incidence of UTUC but not of bladder cancer. [utuc p2]
- UTUC in Balkan endemic nephropathy is usually low grade, multiple and bilateral. [utuc p2]
- Chinese herb nephropathy is caused by a Chinese herb containing A. fangchi and leads to renal failure and UTUC. [utuc p2]
- The very high incidence of UTUC in Taiwan and China is attributed to the use of Aristolochia in herbal medicine. [utuc p2]

Etiology: Smoking, coffee, analgesics

- The smoking-related risk of UTUC is dose related, and stopping smoking decreases the risk. [utuc p2-3]
- Smoking causes more ureteral cancers than renal pelvic cancers. [utuc p2-3]
- Coffee has no proven association with UTUC. [utuc p2-3]
- Phenacetin was used in combinations with caffeine, codeine, acetaminophen and aspirin. [utuc p2-3]
- Phenacetin has been replaced by paracetamol, which is non-toxigenic. [utuc p2-3]
- Analgesic (phenacetin) UTUC has a long latent period of 20 years. [utuc p2-3]

Etiology: Arsenic and occupation

- In Taiwan, arsenic in well water causes blackfoot disease, dry gangrene and increased UTUC. [utuc p3-4]
- In arsenic-related UTUC, ureteric tumours outnumber renal pelvic tumours 2:1. [utuc p3-4]
- In arsenic-related UTUC, the male:female ratio is reversed at 1:2. [utuc p3-4]
- Occupational risk includes aromatic hydrocarbons in the chemical, petrol and plastic industries. [utuc p3-4]
- Exposure to coal, asphalt and tar is an occupational risk factor. [utuc p3-4]
- Chlorinated solvents used in printing are a risk factor, through both skin contact and vapour inhalation. [utuc p3-4]

Etiology: Chronic inflammation and iatrogenic causes

- Ifosfamide, as well as cyclophosphamide, increases UTUC risk through its metabolite acrolein. [utuc p4]
- Habitual use of anthranoid and chemical laxatives for more than 1 year increases UTUC risk 9-fold. [utuc p4]

Natural history: UTUC after bladder cancer

- UTUC develops 17–170 months after bladder cancer, and rates as high as 25% have been reported. [utuc p4]
- The risk of UTUC after bladder cancer is increased with NMIBC, low grade, multiple and multifocal bladder tumours, a history of previous UTUC, a positive ureteral margin, N0 status, involvement of the male prostatic urethra or the female urethra, vesicoureteral reflux, and bladder cancers close to the ureteral orifice. [utuc p4]

Natural history: Bladder recurrence after UTUC; origin and pattern

- Besides downstream seeding, bladder recurrence after UTUC is explained by longer exposure to carcinogens and by the greater number of urothelial cells subjected to carcinogenic events. [utuc p5]
- 20% of UTUC are metastatic at initial presentation. [utuc p5]
- CIS in the upper tract carries more chance of bilateral and multifocal disease. [utuc p5]
- With CIS, bilateral disease occurs in about 25%, compared with 3–5% normally. [utuc p5]
- CIS carries a higher risk of subsequent pan-urothelial disease. [utuc p5]

- Pan-urothelial disease means both upper tracts in a female, and in a male either both upper tracts or one upper tract plus the prostatic urethra. [utuc p5]

Diagnosis

- CTU with 3D reconstruction can recognise small filling defects under 5 mm. [utuc p7-9]
- A drawback of CTU is its higher radiation dose. [utuc p7-9]
- The differential diagnosis of a radiolucent filling defect is UTUC, blood clot, stone, sloughed papilla, overlying bowel gas, fungus ball and external compression. [utuc p7-9]
- The attenuation of UTUC ranges from 10 to 70 HU (average 46 HU). [utuc p7-9]
- Ureteroscopic sampling can be done by forceps or by brushings. [utuc p7-9]
- The risks of ureteroscopic biopsy include pyelovenous and pyelolymphatic migration of tumour cells, extravasation and dissemination. [utuc p7-9]
- Antegrade endoscopy risks tumour cell implantation in the retroperitoneum as well as along the nephrostomy tract. [utuc p7-9]
- Cytology sensitivity for grade II tumours is 45%. [utuc p7-9]
- Ureteral washings are more accurate because irrigation releases loosely adherent cells. [utuc p7-9]
- Brush biopsy through a retrograde catheter or ureteroscope has a sensitivity and specificity of about 90%. [utuc p7-9]
- Complications of brush biopsy are massive haemorrhage, perforation and extravasation. [utuc p7-9]
- Ionic high-osmolar contrast material, as used in retrograde pyelography, worsens cytological abnormality, so samples should be obtained before these agents are used. [utuc p7-9]

Treatment: Risk stratification and site-based algorithm

- An earlier tumour-size cut-off of 1 cm is recorded for low-risk versus high-risk UTUC, alongside the current 2 cm. [utuc p11-12]
- For a low-risk calyceal tumour, the options are URS first and RNU second. [utuc p11-12]
- For a high-risk distal ureteric tumour, distal ureterectomy with or without LND is an alternative to RNU. [utuc p11-12]

CH 139 · SURGICAL MANAGEMENT OF UPPER URINARY TRACT UROTHELIAL TUMOURS

Kidney-Sparing Surgery (Open)

- Open kidney-sparing options for renal pelvic UTUC are pyelotomy with tumour excision and fulguration of the base, and partial nephrectomy guided by intraoperative ultrasound. [UTUC notes 2 p6]
- Reported recurrence after open kidney-sparing surgery for UTUC is 7–60%, with inferior oncological outcomes and the drawback of an open procedure. [UTUC notes 2 p6]

Endoscopic Ablation: Lasers and URS

- Holmium:YAG settings for UTUC ablation are 0.6–1.2 J at 8–10 Hz, giving efficient ablation and precise cutting. [UTUC notes 2 p7]
- Neodymium:YAG is used at 20–30 W, ablates tumour by coagulative necrosis, gives excellent haemostasis and leaves only a narrow safety margin in the ureter. [UTUC notes 2 p7]
- Ureteroscopic management of UTUC can be done as an outpatient procedure. [UTUC notes 2 p7]

Radical Nephroureterectomy and Bladder Cuff

- A non-refluxing ureteric reimplant after distal ureterectomy also limits ascending infection. [UTUC notes 2 p9]
- The bladder cuff excised with the distal ureter at RNU is about 1 cm. [UTUC notes 2 p9]
- Five distal ureter and cuff techniques are described: (i) traditional open intravesical, extravesical or combined; (ii) transvesical ligation and detachment using a Collins knife and endoloop; (iii) transurethral resection of the ureteric orifice (pluck); (iv) intussusception (stripping); (v) total laparoscopic. [UTUC notes 2 p10]

- The pluck technique was abandoned because of concern about tumour seeding into the extravesical space. [UTUC notes 2 p10]
- The intussusception (stripping) technique is contraindicated when ureteric tumours are present. [UTUC notes 2 p10]
- The total laparoscopic distal ureter technique is contraindicated by distal ureteric tumours. [UTUC notes 2 p10]
- The notes record a strong EAU 2018 recommendation for adjuvant intravesical instillation (for example mitomycin C) after RNU to prevent bladder recurrence. The corpus attributes the strong recommendation to AUA/SUO only. [UTUC notes 2 p10]

Lymphadenectomy in UTUC

- The EAU 2018 guideline gives template lymphadenectomy at RNU only a weak recommendation, and more studies are needed. [UTUC notes 2 p11]

Radical nephroureterectomy with bladder cuff

- The bladder cuff excised with the distal ureter is about 1 cm. [utuc p13-14]
- RNU is indicated for low-grade, non-invasive renal pelvic and upper ureteric tumours that are large, multifocal and rapidly recurring. [utuc p13-14]
- Distal ureter management can be done by five techniques: traditional open (intravesical, extravesical or combined); transvesical ligation and detachment (Collins knife or endoloop); transurethral resection of the ureteral orifice ("pluck"); intussusception ("stripping"); and a totally laparoscopic technique. [utuc p13-14]
- The pluck technique was abandoned because of tumour seeding of the extravesical space. [utuc p13-14]
- The intussusception (stripping) technique is contraindicated when there are ureteral tumours. [utuc p13-14]
- A totally laparoscopic distal ureter technique is contraindicated by distal ureteric tumours. [utuc p13-14]
- Limited (regional) lymphadenectomy done with RNU has limited morbidity. [utuc p13-14]

Lymphadenectomy

- EAU 2018 gives only a weak recommendation for extended lymphadenectomy in UTUC. [utuc p15-16]
- Nodal involvement in UTUC increases with stage and grade. [utuc p15-16]
- The problems with lymphadenectomy in UTUC are a lack of uniform templates in the literature and a less consistent pattern of spread than in bladder cancer. [utuc p15-16]

Metastatic disease

- Other regimens for metastatic UTUC include PCG (paclitaxel, carboplatin, gemcitabine), carboplatin–gemcitabine and cabozantinib, an inhibitor of the MET and VEGF pathways. [utuc p15]
- Radiotherapy has no role in UTUC, either as the sole treatment or as an adjunct to chemotherapy. [utuc p15]
- RNU gives no survival benefit in metastatic UTUC and may be done only for palliation. [utuc p15]

Follow-up

- After open surgery, or once endoscopy has made the patient tumour free, physical examination, cystoscopy and urine cytology (for high grade only) are done 3-monthly for 1 year, 6-monthly in the 2nd and 3rd years, and yearly thereafter. [utuc p17]
- The contralateral upper tract is imaged yearly by IVU or RGP. [utuc p17]
- After organ-sparing therapy, ipsilateral endoscopy is done 6-monthly for the first several years and then yearly. [utuc p17]
- For metastatic evaluation in patients at significant risk of progression, physical examination, chest X-ray and a comprehensive metabolic panel with liver enzymes are done 3-monthly in the 1st year, 6-monthly in the 2nd and 3rd years and yearly in the 4th and 5th years. After 5 years only the urothelium is evaluated. [utuc p17]

- CT or MRI of the abdomen and pelvis is done 6-monthly in the 1st and 2nd years and yearly in years 3–5. [utuc p17]
- A bone scan is done only if alkaline phosphatase is raised or there is bone pain. The corpus gives this rule for RCC only. [utuc p17]

CH 141 · CONTEMPORARY OPEN SURGERY OF THE KIDNEY

Laparoscopic Renal Surgery: Approaches and Port Placement

- Five approaches are described: transperitoneal, retroperitoneal, hand-assisted, robotic, and LESS/NOTES. [Compilation 06-2 p21]
- In transperitoneal access the umbilical port is for the camera; the anterior axillary line port, placed at or below the level of the umbilicus, is for instruments and passage of sutures, bulldog clamps or staplers to secure and divide the hilar vessels. [Compilation 06-2 p21]
- The third (instrument) port is placed 2 cm below the xiphoid; on the right side an additional port is placed for liver retraction. [Compilation 06-2 p21]
- All ports are shifted laterally in obese patients. [Compilation 06-2 p21]
- In the retroperitoneal approach, a 5 mm trocar is placed just off the tip of the 12th rib and a 12 mm trocar posterior and superior to the camera port; the first port is a blunt-tip trocar placed after balloon dilatation through a 15 mm transverse incision in the posterior axillary line. [Compilation 06-2 p22]
- In hand-assisted surgery the surgeon's left hand goes into the hand port. [Compilation 06-2 p22]
- The robotic 3-arm configuration uses 4-5 ports (8 mm robotic ports, a 12 mm camera port and a 12 mm assistant trocar, with an additional assistant trocar); the 4-arm configuration uses 5-6 ports in the same general layout, with robotic trocars shifted to avoid clashing of the robotic arms. [Compilation 06-2 p23]

LESS and NOTES

- In LESS all ports lie within a single incision, often concealed within the umbilicus, and virtually all renal operations have been performed via LESS. [Compilation 06-2 p24]
- LESS is done in modified or full flank position using multiple low-profile trocars within one small incision, or purpose-specific access devices, and flexible laparoscopes and instruments may be used. [Compilation 06-2 p24]
- LESS is the most difficult of the minimally invasive techniques. [Compilation 06-2 p24]
- NOTES (natural orifice transluminal endoscopic surgery) has limited experience in humans; hybrid NOTES with standard laparoscopic assistance, using the vagina as access and extraction site during nephrectomy, has been described. [Compilation 06-2 p24]
- Transvesical nephrectomy has recently been done in animals. [Compilation 06-2 p24]

Simple and Radical Laparoscopic Nephrectomy

- The ureter is identified but not divided early, and the lower pole may be dissected. [Compilation 06-2 p25]
- The renal artery may be divided using multiple titanium clips or a stapler. [Compilation 06-2 p25]
- Intact removal can be done through an extended trocar site or a Pfannenstiel incision. [Compilation 06-2 p25]

Partial nephrectomy: hypothermia, warm ischaemia and hilar control

- Tumour complexity is an indication to consider hypothermia (cold ischaemia) during partial nephrectomy. [Compilation 06-2 p28]
- Although the historical safe limit of warm ischaemia is 30 minutes, some data suggest that even 90 minutes may be reasonable. [Compilation 06-2 p28]
- Selective renal artery clamping needs extensive hilar dissection and has longer operative times and higher perioperative transfusion rates. [Compilation 06-2 p28]
- Clamping the renal artery alone (rather than artery and vein) has the theoretical advantage of allowing retrograde venous blood flow and possibly partial oxygen supply to the kidney, but this is controversial. [Compilation 06-2 p29]

- Parenchymal compression and clamping is not feasible for central, hilar or large tumours, or in pure laparoscopic surgery without specialised instruments. [Compilation 06-2 p29]
- Long-term overall survival is similar after laparoscopic and open partial nephrectomy. [Compilation 06-2 p29]
- Five-year cancer-specific survival after laparoscopic partial nephrectomy has been reported to be up to 100%. [Compilation 06-2 p29]

Anterior approaches: Midline

- The midline incision extends from the xiphisternum to the symphysis pubis. [Kidney Approaches p1]
- Layers in the midline incision are skin, subcutaneous tissue, linea alba, preperitoneal fat and peritoneum. [Kidney Approaches p1]
- Renovascular operations are listed as an indication for the midline approach (the corpus lists renovascular surgery under chevron only). [Kidney Approaches p1]
- An advantage of the midline approach is that coexisting intraperitoneal pathology can be dealt with at the same time. [Kidney Approaches p1]

Anterior approaches: Anterior subcostal

- The anterior subcostal approach is done in the supine position, on the right or left side. [Kidney Approaches p1]
- The anterior subcostal incision is placed two fingerbreadths below the costal margin and runs from the midline to the tip of the 11th rib. [Kidney Approaches p1]
- Layers in the anterior subcostal incision are skin, subcutaneous fascia, external oblique, internal oblique and transversus abdominis muscles, transversalis fascia, then the retroperitoneum. [Kidney Approaches p1]
- The anterior subcostal approach provides good exposure of the renal hilum. [Kidney Approaches p1]

Anterior approaches: Chevron

- The chevron incision starts at the 11th rib on one side, crosses the midline and ends at the 11th rib on the other side. [Kidney Approaches p1]

Flank approaches: Subcostal flank

- The subcostal flank incision is placed one fingerbreadth below and parallel to the 12th rib. [Kidney Approaches p2]
- The subcostal flank incision runs from the lateral border of the sacrospinalis to the lateral border of the rectus muscle. [Kidney Approaches p2]
- The subcostal flank approach is also a poor choice for radical nephrectomy of a large mass because hilar access is poor. [Kidney Approaches p2]
- Layers in the subcostal flank incision are skin, subcutaneous tissue, latissimus dorsi and external oblique, serratus posterior inferior, lumbodorsal fascia, internal oblique and transversus abdominis, then peritoneum. [Kidney Approaches p2]
- In the flank approach, sweeping the peritoneum medially gives a retroperitoneal approach, while incising it gives a transperitoneal approach. [Kidney Approaches p2]

Flank approaches: Supracostal

- The supracostal skin incision uses the same landmarks as the subcostal flank incision. [Kidney Approaches p2]
- In the supracostal approach, latissimus dorsi, serratus posterior inferior and the intercostal muscles are divided posteriorly, the external and internal obliques anteriorly, then the lumbodorsal fascia and transversus abdominis, and the peritoneum is swept medially. [Kidney Approaches p2]
- The supracostal approach is extrapleural and extraperitoneal. [Kidney Approaches p2]
- The supracostal approach has less morbidity than the transcostal approach. [Kidney Approaches p2]

Dorsal lumbotomy

- Dorsal lumbotomy is also suited to thin adults who need bilateral renal surgery. [Kidney Approaches p3]

- Disadvantages of dorsal lumbotomy are poor exposure, particularly of the renal hilum, and the need to turn the patient prone after induction of anaesthesia, with its related complications. [Kidney Approaches p3]
- Dorsal lumbotomy is poorly tolerated by older adults, patients with cardiopulmonary problems and obese patients. [Kidney Approaches p3]
- The dorsal lumbotomy incision runs vertically from the inferior border of the 12th rib to the iliac crest, along the lateral line of the sacrospinalis. [Kidney Approaches p3]
- Dorsal lumbotomy steps are: incise the subcutaneous tissue; incise the posterior layer of the lumbodorsal fascia; retract the sacrospinalis medially; divide the costovertebral ligament; divide the fused middle and anterior layers of the lumbodorsal fascia; retract quadratus lumborum medially; incise the transversalis fascia; and enter the paranephric space. [Kidney Approaches p3]

Thoracoabdominal

- The thoracoabdominal approach is ideal for large renal masses, upper pole masses and locally advanced tumours, as well as renal tumours with IVC thrombus. [Kidney Approaches p3]
- For the thoracoabdominal approach the patient is placed semioblique with a bean bag supporting the flank. [Kidney Approaches p3]
- The thoracoabdominal approach uses the 8th, 9th, 10th or 11th intercostal space depending on the tumour. [Kidney Approaches p3]
- The thoracoabdominal skin incision starts at the lateral aspect of the sacrospinalis, runs through the intercostal space and can extend as far as the contralateral rectus abdominis. [Kidney Approaches p3]
- Disadvantages of the thoracoabdominal approach are the morbidity of opening the pleural cavity, which needs a chest tube, and the risk of phrenic nerve injury. [Kidney Approaches p3]

CH 142 · LAPAROSCOPIC AND ROBOTIC SURGERY OF THE KIDNEY

Laparoscopic Partial Nephrectomy and Renal Hypothermia

- Robot-assisted laparoscopic PN reduces the difficulty of tumour resection and renorrhaphy and is advantageous for surgeons who lack advanced laparoscopic skills. [Compilation 06-2 p26]
- In robotic PN the bedside assistant clamps the renal hilum, provides suction and retraction, delivers and cuts sutures, and places clips. [Compilation 06-2 p26]
- Alternative haemostatic methods in laparoscopic PN include lasers, microwave, ultrasonic and radiofrequency energy. [Compilation 06-2 p27]
- Collecting system repair is associated with longer operating time. [Compilation 06-2 p27]
- Renal hypothermia is rarely used in laparoscopic PN because of its complexity and because most tumours do not need very long warm ischaemia for resection. [Compilation 06-2 p27]

Complications of laparoscopic renal surgery

- Tumour rupture is an intraoperative complication of laparoscopic renal surgery. [Compilation 06-2 p30]
- Postoperative complications of laparoscopic renal surgery also include seroma, wound dehiscence, internal hernia, pulmonary complications, urinary retention or UTI, and renal insufficiency. [Compilation 06-2 p30]
- Testicular infarction or ischaemia and epididymitis are postoperative complications of laparoscopic renal surgery. [Compilation 06-2 p30]
- In patients with multiple previous surgeries, trocars should be placed away from sites of suspected adhesions, and a retroperitoneal approach may be needed. [Compilation 06-2 p31]
- Imaging should be reviewed just before surgery and kept available in the operating room. [Compilation 06-2 p31]
- A bowel injury recognised at surgery is debrided and closed in two layers; consider a drain and keep the patient nil by mouth until bowel function returns. [Compilation 06-2 p32]
- Splenic injury is managed with topical haemostatic agents or argon beam coagulation; splenectomy is rarely needed. [Compilation 06-2 p33]

- Liver injury is managed with haemostatic agents, and gallbladder injury is best treated by cholecystectomy. [Compilation 06-2 p33]
- Pulmonary complications are prevented by stopping smoking and spirometry, and treated with spirometry, deep breathing exercises and early ambulation. [Compilation 06-2 p33]
- Wound infection is treated with wound swab culture, appropriate antibiotics, drainage of any abscess and wound dressings. [Compilation 06-2 p34]
- An incisional hernia under 2 cm may be closed anatomically, whereas one over 2 cm may need mesh. [Compilation 06-2 p34]

CH 143 · FOCAL THERAPY FOR RENAL TUMOURS

Thermal Ablation

- Indications for thermal ablation include local recurrence after nephron-sparing surgery, and hereditary renal cancer in which many multifocal lesions make repeated partial nephrectomies cumbersome. [Compilation 06 p62]
- Local recurrence-free outcome after thermal ablation is 80–90%. [Compilation 06 p62]
- After ablation, repeat biopsy is indicated for new enhancement, progressive increase in size, new nodules, or satellite or port-site lesions. [Compilation 06 p62]

Focal therapy for renal tumours: indications, advantages, therapies

- Compared with nephron-sparing surgery, focal therapy is technically easier, less time-consuming and can be delivered by open, laparoscopic or percutaneous approaches. [Compilation 06-2 p37]
- Laser interstitial thermal therapy and targeted embolisation with ablation are listed among future renal ablative therapies needing further evaluation. [Compilation 06-2 p38]

Cryoablation

- Cooper and Lee developed the first modern cryoprobe, using pressurised liquid nitrogen (−196 °C); its disadvantage was the inability to monitor the extent of the cryolesion. [Compilation 06-2 p39]
- Argon-gas cryoprobes can generate temperatures down to −185.7 °C. [Compilation 06-2 p39]

Radiofrequency ablation

- The typical RFA current frequency is 460 kHz. [Compilation 06-2 p41]
- RFA probes include a single-electrode monopolar probe, a Christmas-tree-shaped device, and the LeVeen insulated monopolar probe with 12 deployable tines (RF antennas). [Compilation 06-2 p42]
- Bipolar RFA probes produce higher temperatures than monopolar devices. [Compilation 06-2 p42]
- Impedance-based RFA systems typically start at 40–80 W and increase by 10 W per minute to a maximum of 130–200 W until an impedance of 200–500 ohms is reached. [Compilation 06-2 p43]
- Selective embolisation can help prevent the heat-sink effect. [Compilation 06-2 p43]

Complications of renal ablation

- Post-ablation syndrome consists of low-grade fever, pain, malaise and myalgias; it is usually self-limiting or settles with NSAIDs. [Compilation 06-2 p45]
- Mild haematuria or clot colic can follow renal ablation. [Compilation 06-2 p45]
- Pain and paraesthesia at the percutaneous probe insertion site is the most common complication of ablation (8%). [Compilation 06-2 p45]
- Thermal injury to the genitofemoral nerve causes chronic groin pain, to the psoas muscle causes impaired hip flexion, and to the adrenal gland causes a hypertensive crisis. [Compilation 06-2 p45]

Choice of approach and oncological outcomes of ablation

- Laparoscopic ablation is preferred for tumours that are difficult percutaneously: anterior tumours or those adjacent to bowel, tumours close to the collecting system, and tumours without suitable imaging access. [Compilation 06-2 p46]
- Percutaneous ablation is ideal for a posterior tumour at least 0.5 cm away from the renal pelvis or pelvicalyceal system. [Compilation 06-2 p46]

- Local recurrence-free outcome is 80–86% after cryoablation and 88–92% after RFA. [Compilation 06-2 p46]

HIFU

- Renal HIFU is delivered by an extracorporeal route, using ultrasound waves from 20 kHz to 20 MHz. [Compilation 06-2 p47]
- Medical uses of HIFU include destruction of solid tumours of bone, kidney, prostate, liver, pancreas and brain, palliation of bone metastases and ablation of breast fibroadenoma. [Compilation 06-2 p47]
- HIFU is also used for neuropathic pain, Parkinson disease, kidney stones and prostate cancer (Sonablate and Ablatherm devices). [Compilation 06-2 p48]
- Disadvantages of HIFU are skin burns and poor tissue targeting, the latter improved by laparoscopic HIFU. [Compilation 06-2 p48]
- HIFU minimises or eliminates the risk of tumour seeding, haemorrhage and urinary extravasation; precision is its primary advantage. [Compilation 06-2 p48]
- The HIFU focal point measures 3 × 3 × 11 mm and usually lies 3–4 cm from the transducer. [Compilation 06-2 p49]
- HIFU uses parabolic transducers, which can be non-invasive or minimally invasive, and focusing is done either geometrically or electronically. [Compilation 06-2 p49]

Irreversible electroporation

- Pathology after IRE shows absence of cellular viability immediately, diffuse cellular necrosis by day 7 and chronic inflammation or fibrosis by day 14. [Compilation 06-2 p50]
- IRE electrodes (single or multiple) are placed laparoscopically or percutaneously under CT or ultrasound guidance. [Compilation 06-2 p50]

Microwave ablation

- On the electromagnetic spectrum, ultrasound is about 10^6 Hz, microwave about 10^9 Hz and laser about 3 × 10^14 Hz. [Compilation 06-2 p51]
- Microwave energy is transmitted by an insulated coaxial cable and delivered through an antenna or emitter. [Compilation 06-2 p51]
- Further advantages of MWA are a more uniform heating zone, less dependence on tissue properties, and no need for tissue contact. [Compilation 06-2 p52]

CH 144 · TREATMENT OF ADVANCED RENAL CELL CARCINOMA

Cytoreductive nephrectomy in mRCC

- A third proposed explanation for the benefit of cytoreductive nephrectomy is that systemic agents, particularly cytokines, fail to induce a meaningful response in the primary renal tumour in most patients. [Compilation 06 p80]

Cytoreductive Nephrectomy and Palliative Surgery in mRCC

- The role of cytoreductive nephrectomy is unclear in non-clear cell histology. [Compilation 06-2 p1]
- Palliative resection of metastases may be needed for metastases in weight-bearing joints as well as bones, and for vertebral metastases with impending spinal cord compression; solitary brain metastases are also listed as an indication for palliative resection. [Compilation 06-2 p1]
- Cytoreductive nephrectomy done for palliation of local symptoms does not always result in clinical benefit. [Compilation 06-2 p1]

Immunotherapy for mRCC: Interferon and Interleukin-2

- IFN-alpha was the earliest cytokine to be evaluated in mRCC. [Compilation 06-2 p3]
- IL-2 can be given intravenously or subcutaneously. [Compilation 06-2 p4]
- High-dose IL-2 is approved by the US FDA for mRCC. [Compilation 06-2 p4]
- Additional IL-2 side effects are third-space fluid retention and respiratory compromise. [Compilation 06-2 p4]
- The combination of IFN-alpha with IL-2 has high toxicity with no survival advantage. [Compilation 06-2 p4]

Allogeneic Haematopoietic Stem Cell Transplant

- Allogeneic haematopoietic stem cell transplant replaces the recipient's immune and haematopoietic systems with those of a healthy HLA-compatible donor, and acts through a graft-versus-tumour (allogeneic antitumour) effect. [Compilation 06-2 p5]

Targeted Therapy: Molecular Mechanism

- The VHL complex (cullin-2, elongin B, elongin C, Rbx1 and an E2 enzyme) binds the beta-domain of the VHL protein; a mutant alpha-domain disrupts the complex, so HIF accumulates. [Compilation 06-2 p7]
- HIF-driven TGF-alpha acts on the EGF receptor (EGFR). [Compilation 06-2 p7]

Bevacizumab

- Bevacizumab is a humanised antibody (more than 90% human) that blocks angiogenesis. [Compilation 06-2 p8]
- Further bevacizumab side effects are fatigue, infection, bowel perforation, headache, nose bleed, allergy and nasal septal perforation; bleeding is the main side effect. [Compilation 06-2 p8]
- Other uses of bevacizumab are eye diseases and cancers of the lung, colon and rectum, breast, brain and ovary. [Compilation 06-2 p8]
- Bevacizumab is given intravenously every 14 days. [Compilation 06-2 p8]
- In AVOREN, no benefit of bevacizumab plus IFN-alpha was seen in MSKCC poor-risk patients. [Compilation 06-2 p8]
- Bevacizumab plus IFN-alpha can be used in patients who failed standard first-line therapy, and in good- and intermediate-risk patients only. [Compilation 06-2 p8]

Sorafenib and Sunitinib

- Sorafenib side effects also include hypertension, fatigue and diarrhoea. [Compilation 06-2 p9]
- Sunitinib works by inhibiting vascular endothelial development and proliferation and by inhibiting vascular pericyte function. [Compilation 06-2 p9]
- Sunitinib side effects also include diarrhoea, rash, asthenia and bone marrow suppression. [Compilation 06-2 p9]
- An alternative sunitinib schedule is 2 weeks on and 1 week off. [Compilation 06-2 p10]
- The notes cite the "EFFECT" and "STAR" trials for sunitinib advantages: better quality of life than IFN-alpha, mostly mild-to-moderate side effects manageable with close monitoring and proactive management, and superior PFS and OS compared with IFN-alpha. [Compilation 06-2 p10]
- Sunitinib as second-line monotherapy after cytokines gives a 30-40% partial response rate. [Compilation 06-2 p10]

Pazopanib, Axitinib and Other TKIs

- The pazopanib dose is 800 mg once daily (oral). [Compilation 06-2 p11]
- Axitinib is a second-generation inhibitor; the dose is 5 mg twice daily orally, increased to 7 mg twice daily. [Compilation 06-2 p11]
- Nintedanib and dovitinib are listed among other VEGF-pathway agents. [Compilation 06-2 p11]

mTOR Pathway Inhibition

- The mTOR protein is encoded by the FRAP-1 gene and regulates the translation and stability of HIF-1alpha, driving cell growth, multiplication and angiogenesis. [Compilation 06-2 p12]
- Temsirolimus side effects also include fatigue, rash and hypophosphataemia. [Compilation 06-2 p12]

Sequential Therapy, Chemotherapy and Non-clear Cell RCC

- After disease progression on mTOR inhibition, VEGF-targeted therapy is given. [Compilation 06-2 p13]
- After both VEGF and mTOR inhibition, sorafenib is listed as treatment. [Compilation 06-2 p13]
- Gemcitabine plus doxorubicin is used for sarcomatoid and rapidly progressive mRCC. [Compilation 06-2 p14]
- In papillary tumours, sunitinib is better than everolimus. [Compilation 06-2 p14]

CARMENA Trial

- CARMENA was a phase III, prospective, multicentre, open-label randomised trial with overall survival as primary end point. [Compilation 06-2 p17]
- 226 patients were assigned to CN plus sunitinib and 224 to sunitinib alone; 40 in the CN arm and 11 in the sunitinib arm did not receive sunitinib. [Compilation 06-2 p17]
- The safety populations were 186 (CN arm) and 213 (sunitinib arm); 165 and 161 patients died respectively. [Compilation 06-2 p17]
- Follow-up period was 50.9 months. [Compilation 06-2 p17]
- In CARMENA there was no significant difference in response rate or progression-free survival between the arms. [Compilation 06-2 p18]

OTHER TOPICS IN THIS BOOK

Autosomal dominant polycystic kidney disease: clinical features

- GI symptoms in ADPKD arise from renomegaly or colonic diverticula. [Cystic Disease p6]
- Anaemia is unusual in the chronic renal failure of ADPKD because erythropoietin levels are raised. [Cystic Disease p6]
- Risk factors for earlier ESRD in ADPKD also include a first episode of haematuria before 30 years, hypertension before 35 years, hyperlipidaemia and sickle cell trait. [Cystic Disease p6]
- In ADPKD the kidneys are bilaterally enlarged, keep a reniform shape, and are diffusely cystic throughout cortex and medulla. [Cystic Disease p7]
- Although RCC is no commoner in ADPKD, the incidence of renal adenoma is high. [Cystic Disease p7]
- Hepatic cysts in ADPKD are usually asymptomatic and are more common in females (oestrogen effect). [Cystic Disease p7]
- Of ADPKD patients who have a subarachnoid haemorrhage from a berry aneurysm, 9% die. [Cystic Disease p7]

ADPKD: evaluation

- A family history covering 3 generations is a must in evaluating ADPKD. [Cystic Disease p7]
- The ultrasound criteria are 100% sensitive for at-risk individuals aged 30 years or more and for younger individuals with PKD1 mutations, but only 67% sensitive for PKD2 individuals under 30 years. [Cystic Disease p8]
- CT and MRI are superior to ultrasound for diagnosing ADPKD. [Cystic Disease p8]
- Routine MRA or CTA screening for asymptomatic berry aneurysms is not recommended. [Cystic Disease p8]
- Aneurysm screening is indicated, besides a family history of aneurysm or SAH, after a previous aneurysm rupture, in preparation for major surgery, in a high-risk occupation (for example an airline pilot), and when the patient is very anxious about a possible intracranial aneurysm. [Cystic Disease p8]
- Without a supporting family history, ADPKD is a presumptive diagnosis when there are bilateral renal cysts plus at least 2 of: bilateral renal enlargement, 3 or more hepatic cysts, a cerebral artery aneurysm, or a solitary cyst of the arachnoid, pineal gland or spleen. [Cystic Disease p8]

ADPKD: treatment and emerging therapy

- ADPKD has no cure; the goals are to lessen its complications and delay the onset of ESRD. [Cystic Disease p9]
- UTI in ADPKD, especially in women, must be treated effectively. [Cystic Disease p9]
- Chloramphenicol is another lipophilic antibiotic that penetrates the cyst wall. [Cystic Disease p9]
- Somatostatin, like tolvaptan, is used to reduce vasopressin-induced cAMP levels in ADPKD. [Cystic Disease p10]
- mTOR inhibitors reduce kidney cyst burden in ADPKD. [Cystic Disease p10]
- Emerging ADPKD therapies include epidermal growth factor receptor (ErbB1 and ErbB2) inhibitors, Src kinase inhibitors and CDK inhibitors. [Cystic Disease p10]

Role of PET-CT in RCC

- In RCC, PET-CT is indicated in patients at high risk of metastasis who have equivocal findings on CT or MRI. [PET CT PSMA p1]
- FDG-PET has good specificity but suboptimal sensitivity for staging RCC. [PET CT PSMA p1]
- Immuno-PET/CT with iodine-124-labelled cG250 (124I-cG250; cG250 is the antibody that recognises CA-IX, and iodine-124 is a positron-emitting radionuclide) may have a promising role in the diagnosis and staging of clear cell RCC. [PET CT PSMA p1]
- In the notes, 124I-cG250 PET/CT for clear cell RCC has a sensitivity of 96% and a specificity of 100%. The injected complex binds CA-IX on clear cell RCC and is detected by PET/CT. [PET CT PSMA p1]

Management of Positive Upper Tract Urine Cytology or CIS

- If the bladder has been adequately treated and the cytology stays positive for more than 1 year, the patient needs extravesical evaluation. [UTUC notes 2 p12]
- Upper-tract CIS is often multifocal, so segmental resection does not work for it. Nephroureterectomy is kept for radiographic or endoscopic evidence of more than surface disease, and frequent re-evaluation is needed. [UTUC notes 2 p12]

Adjuvant Therapy (UTUC)

- One adjuvant option after percutaneous organ-sparing therapy is brachytherapy of the percutaneous nephrostomy (PCN) tract. [UTUC notes 2 p13]
- PCN-tract brachytherapy uses iridium wire. Its problem is cutaneous fistula formation, which then needs radical nephroureterectomy. [UTUC notes 2 p13]
- After complete excision, adjuvant radiotherapy is used to reduce local relapse after radical surgery for T3, T4 or N+ disease. [UTUC notes 2 p13]
- Adjuvant radiotherapy given without chemotherapy does not protect against distant failure. [UTUC notes 2 p13]

Follow-up (UTUC)

- Contralateral upper-tract imaging is done yearly by IVU or retrograde pyelogram (RGP). [UTUC notes 2 p14]
- Metastatic evaluation is for patients at significant risk of progression. It comprises physical examination, chest X-ray and a comprehensive metabolic panel with liver enzymes: 3-monthly in year 1, 6-monthly in years 2 and 3, and yearly in years 4 and 5. [UTUC notes 2 p14]
- After 5 years, follow-up is evaluation of the urothelium only. [UTUC notes 2 p14]
- Cross-sectional CT/MRI of the abdomen and pelvis is done 6-monthly in years 1 and 2, then yearly from year 3 to 5. [UTUC notes 2 p14]

Extended LND (UTUC)

- The rate of lymph node involvement in UTUC (12–25%) rises with stage and grade. [UTUC notes 2 p15]
- There are two problems with UTUC lymphadenectomy. Templates are not uniform across the literature, and the pattern of spread is less consistent than in bladder cancer. [UTUC notes 2 p15]

14 Adrenal & Retroperitoneum

CH 145 · SURGICAL AND RADIOGRAPHIC ANATOMY OF THE ADRENALS

Adrenal glands: gross anatomy and relations

- Grossly, the adrenal medulla is brown, in contrast to the yellow-orange cortex. [Adrenal Grneral p1]
- The greater curvature of the stomach is an anterior relation of the left adrenal gland. [Adrenal Grneral p1]
- The adrenal arterial branches penetrate the capsule to form a subcapsular plexus. [Adrenal Grneral p1]

Vasculature, lymphatics and nerve supply

- The adrenal medulla has a dual blood supply: arterial blood directly from medullary arterioles and venous blood from the cortical sinusoidal capillaries. [Adrenal Grneral p2]
- The overlapping arterial and venous anatomy of the adrenal affects (complicates) partial adrenalectomy. [Adrenal Grneral p2]
- Lymph from the right adrenal drains to the paracaval nodes and from the left adrenal to the para-aortic nodes. [Adrenal Grneral p2]
- Preganglionic sympathetic fibres form a nerve plexus at the adrenal capsule and traverse the cortex to reach the medulla. [Adrenal Grneral p2]
- Catecholamines released from the medulla reach the circulation through fenestrated capillaries. [Adrenal Grneral p2]
- Postganglionic fibres from the splanchnic ganglia supply the adrenal cortex. [Adrenal Grneral p3]
- Parasympathetic innervation of the adrenal cortex and medulla is not well defined. [Adrenal Grneral p3]

Histology

- Zona glomerulosa cells are small polyhedral cells with a vacuolated appearance (less cytoplasm, more lipid), arranged in ovoid clusters. [Adrenal Grneral p3]
- Aldosterone is synthesised in the mitochondria and smooth endoplasmic reticulum, at 100–150 µg/day. [Adrenal Grneral p3]
- Zona fasciculata cells are large polyhedral cells arranged in cords. [Adrenal Grneral p3]
- Zona reticularis cells are smaller cells arranged in anastomosing cords, and produce more than 20 mg/day of DHEA, DHEA sulphate and androstenedione. [Adrenal Grneral p3]

Embryology

- Over the 2 weeks after condensation, the coelomic epithelial cells proliferate and migrate to the cranial end of the mesonephros to form the adrenogenital ridge (intermediate mesoderm), whose cephalic portion forms the adrenal cortex and caudal portion forms the gonads. [Adrenal Grneral p4]
- Neural crest cells migrate into the medial aspect of the fetal adrenal cortex by the 9th week and reach a central location by the 18th week to form the medulla. [Adrenal Grneral p4]
- At 8–10 weeks the cortex differentiates morphologically into a definitive zone, which replaces the fetal zone after birth, and a fetal zone. [Adrenal Grneral p4]
- Although twice the adult weight at birth, the neonatal adrenal is not completely developed. [Adrenal Grneral p5]
- The zona glomerulosa and zona fasciculata develop within the definitive cortex over the first year under the influence of ACTH. [Adrenal Grneral p5]
- The zona reticularis reaches adult volume only at 6–8 years of age, which explains the relatively late importance of adrenal sex steroid production. [Adrenal Grneral p5]

Anomalies

- Unilateral adrenal agenesis is rare. [Adrenal Grneral p5]
- The surgical importance of adrenal rests is compensatory hypertrophy after adrenalectomy, inadvertent excision during unrelated surgery, and neoplasia. [Adrenal Grneral p5]

Radiological points

- CT is the most widely used adrenal imaging, and perinephric fat allows clear visualisation of the gland with outstanding resolution. [Adrenal Grneral p6]
- The limbs of the normal adrenal are usually thinner than the adjacent diaphragmatic crura. [Adrenal Grneral p6]
- MRI is slightly inferior to CT in spatial resolution. [Adrenal Grneral p6]
- For adrenal MRI the transverse plane is most commonly used, with a slice thickness of 3 mm or less. [Adrenal Grneral p6]
- On T1-weighted MRI the normal adrenal has a uniform intermediate signal slightly lower than liver and renal cortex. [Adrenal Grneral p6]

- On ultrasound the normal adrenal is hypoechoic, but it is difficult to distinguish from surrounding structures because of retroperitoneal fat, and right-sided lesions are visualised better. [Adrenal Grneral p6]

CH 146 · PATHOPHYSIOLOGY, EVALUATION AND MEDICAL MANAGEMENT OF ADRENAL DISORDERS

Adrenal insufficiency (Addison disease): prevalence and causes

- The estimated prevalence of adrenal insufficiency is 39–60 per million. [Addison Disease p1]
- Autoimmune adrenalitis is the commonest cause of primary adrenal insufficiency in the Western world, whereas tuberculosis is the commonest cause in developing countries. [Addison Disease p1]
- Autoimmune polyendocrine syndrome is a cause of primary adrenal insufficiency. [Addison Disease p1]
- Cytomegalovirus (CMV) adrenalitis in HIV patients is a cause of primary adrenal insufficiency. [Addison Disease p1]
- Fungal infection is a cause of primary adrenal insufficiency. [Addison Disease p1]
- The infiltrative causes of primary adrenal insufficiency include amyloidosis, sarcoidosis and haemochromatosis. [Addison Disease p1]
- Secondary adrenal insufficiency can also result from pituitary or hypothalamic radiation, autoimmune disease and trauma. [Addison Disease p1]
- The clinical features of adrenal insufficiency are mostly non-specific and constitutional. [Addison Disease p1]
- Skin pigmentation in primary adrenal insufficiency is due to melanocortin receptor stimulation by the high serum ACTH. [Addison Disease p1]

Symptoms and signs

- Symptoms of adrenal insufficiency include dizziness, fatigue and lack of energy, anorexia and weight loss, gastric pain, nausea and vomiting, joint pain and myalgias, dry itchy skin, and loss of libido in women. [Addison Disease p2]
- Signs of adrenal insufficiency include fever, low blood pressure and postural hypotension. [Addison Disease p2]
- Laboratory signs of adrenal insufficiency include hypoglycaemia, hypercalcaemia, anaemia, lymphocytosis and eosinophilia. [Addison Disease p2]
- Women with adrenal insufficiency may lose axillary or pubic hair, and children may show absent adrenarche or pubarche. [Addison Disease p2]

Adrenal crisis (acute adrenal insufficiency)

- Adrenal crisis is life-threatening and is often preceded by hypotension unresponsive to fluids. [Addison Disease p2]
- Adrenal crisis presents with abdominal pain, nausea, vomiting, fever and hypotension. [Addison Disease p2]
- Children in adrenal crisis can have hypoglycaemic seizures. [Addison Disease p2]

Diagnostic tests

- The diagnosis of adrenal insufficiency may be made on clinical grounds and is then secured by tests. [Addison Disease p3]
- Abnormal aldosterone and renin levels are a diagnostic feature of primary adrenal insufficiency only. [Addison Disease p3]
- The corticotropin (ACTH stimulation) test is the confirmatory test. [Addison Disease p3]
- Secondary causes need detailed testing of the hypothalamic–pituitary–adrenal axis by an endocrinologist. [Addison Disease p3]

Treatment

- Cortisone acetate is an alternative to hydrocortisone for glucocorticoid replacement, with the remainder of the daily dose given in the evening to follow the circadian rhythm. [Addison Disease p3]

- Adrenal androgen replacement is usually limited to patients who have constitutional symptoms despite glucocorticoid and mineralocorticoid replacement. [Addison Disease p3]
- Predictors of persistent hypertension after adrenalectomy also include use of 2 or more antihypertensive drugs before surgery, a first-degree relative with hypertension, and the degree of cardiovascular remodelling and renal insufficiency. [Primary Aldosteronism p12]
- Spironolactone can cause menstrual disturbances. [Primary Aldosteronism p12]
- Even on mineralocorticoid receptor antagonists, patients will still need other oral antihypertensives, regular exercise with weight loss, and a low-sodium diet. [Primary Aldosteronism p12]
- In FH type I, a mineralocorticoid receptor antagonist is added if BP is not controlled with glucocorticoid alone or if iatrogenic Cushing syndrome develops. [Primary Aldosteronism p12]

Peri-operative steroid dosing

- The patient's usual steroid dose is sufficient peri-operatively, but although this is controversial, surgeons continue to give stress-dose steroids. [Addison Disease p4]
- For major surgery, the usual glucocorticoid dose is given within 2 hours before surgery. [Addison Disease p4]

Prognosis

- The impact of primary adrenal insufficiency on life expectancy is not established, but it has a significant impact on quality of life. [Addison Disease p4]
- Secondary adrenal insufficiency is an established cause of premature death. [Addison Disease p4]

Catecholamine metabolism

- Normetanephrine and metanephrine are further conjugated to sulphated normetanephrine and metanephrine sulphate, which together make up total metanephrines. [Adrenal Grneral p7]
- PNMT is unique to the adrenal medulla, although the brain and the organ of Zuckerkandl also express it. [Adrenal Grneral p7]
- COMT is present in the adrenal medulla and also in the liver and kidney; the notes record 90% (metanephrine) and 20% (normetanephrine) against medullary COMT. [Adrenal Grneral p7]

Steroid hormone synthesis and actions

- Aldosterone synthase converts corticosterone to 18-hydroxycorticosterone, which 18-oxidase converts to aldosterone. [Adrenal Grneral p8]
- Androstenedione is converted to testosterone (and then DHT) and to oestrone, and testosterone is converted to oestradiol. [Adrenal Grneral p8]
- In the distal nephron aldosterone causes sodium and chloride reabsorption, hydrogen and potassium excretion, and free water reabsorption, so sodium concentration stays the same while total body water volume changes. [Adrenal Grneral p8]
- Atrial natriuretic peptide is the main inhibitory regulator of aldosterone. [Adrenal Grneral p9]
- Glucocorticoids enhance skeletal and cardiac muscle contraction, are anti-inflammatory and immunosuppressive, inhibit bone formation and collagen synthesis, cause protein catabolism, maintain normal GFR, and increase vascular contractility while decreasing vascular permeability. [Adrenal Grneral p9]

Primary Aldosteronism: history and RAAS

- Primary aldosteronism was first described by Dr Jerome Conn in 1955. [Primary Aldosteronism p1]
- Primary aldosteronism is the most common form of secondary hypertension. [Primary Aldosteronism p1]
- Besides angiotensin II and potassium, ACTH is an additional regulator that stimulates adrenal aldosterone secretion. [Primary Aldosteronism p1]

Mineralocorticoid escape and subtypes of Conn syndrome

- Mineralocorticoid escape is the escape from the sodium-retaining effects of excess aldosterone in primary hyperaldosteronism; it is mediated by pressure natriuresis and atrial natriuretic peptide (ANP) and limits volume expansion to 1.5 kg or less. [Primary Aldosteronism p2]

- Bilateral (idiopathic) adrenal hyperplasia in primary aldosteronism is mostly micronodular. [Primary Aldosteronism p2]
- Hypertension and hypokalaemia are more severe with an aldosterone-producing adenoma than with bilateral hyperplasia. [Primary Aldosteronism p2]

Types of familial hyperaldosteronism

- Familial hyperaldosteronism type I is inherited as autosomal dominant. [Primary Aldosteronism p2]
- In FH type I, aldosterone shows a circadian rhythm governed by ACTH instead of by the RAAS. [Primary Aldosteronism p2]
- FH type I presents with a family history of early-onset hypertension or cerebrovascular accident (stroke). [Primary Aldosteronism p2]
- Familial hyperaldosteronism type II is autosomal dominant, and its adenoma or hyperplasia is indistinguishable from sporadic cases. [Primary Aldosteronism p2]
- Familial hyperaldosteronism type III causes bilateral adrenal hyperplasia with refractory hypertension, severe hypokalaemia and overproduction of hybrid steroids. [Primary Aldosteronism p2]

Clinical features

- Primary aldosteronism is usually diagnosed in the third to sixth decade of life in patients being evaluated for refractory hypertension. [Primary Aldosteronism p3]
- Only a minority of patients have symptoms of hypokalaemia, such as headache, palpitations and weakness. [Primary Aldosteronism p3]
- Cardiac damage in primary aldosteronism includes left ventricular hypertrophy (LVH) and a prolonged QT interval. [Primary Aldosteronism p3]
- Patients with primary aldosteronism have a higher risk of myocardial infarction, atrial fibrillation and stroke, and appropriate treatment decreases this risk. [Primary Aldosteronism p3]
- Renal damage in primary aldosteronism shows as proteinuria. [Primary Aldosteronism p3]
- Primary aldosteronism is associated with a higher rate of type 2 diabetes mellitus. [Primary Aldosteronism p3]
- Decreased libido is in the 90–100% group of features, and menstrual disturbance is in the 70–90% group. [Cushing Syndrome p3]
- Features seen in 50–70% are muscle weakness, buffalo hump, easy bruising, diabetes or glucose intolerance, osteoporosis and depression. [Cushing Syndrome p3]
- Features seen in 20–50% are headache, backache, limb oedema, recurrent infections and nephrolithiasis. [Cushing Syndrome p3]
- Acne and alopecia occur in 0–20%. [Cushing Syndrome p3]
- Definitive treatment of Cushing syndrome lowers the stone risk but does not bring it back to that of the general population. [Cushing Syndrome p3]
- Other names for subclinical Cushing syndrome are pre-Cushing syndrome, non-Cushingoid Cushing syndrome and preclinical Cushing syndrome. [Cushing Syndrome p3]
- Virilisation alone is due to excess 17-ketosteroids and DHEA; it causes male-pattern baldness, hirsutism and dysmenorrhoea. [Malignant Adrenal Lesions p2]
- Feminisation in men presents with testicular atrophy and gynaecomastia, from peripheral conversion of testosterone to oestrogen. [Malignant Adrenal Lesions p2]
- Hyperaldosteronism is the rarest functional presentation and causes hypertension with hypokalaemia. [Malignant Adrenal Lesions p2]
- Non-functional (mass-effect or systemic) features include back pain, nausea and vomiting, besides abdominal fullness and constitutional symptoms. [Malignant Adrenal Lesions p2]
- A single ACC commonly secretes several hormones, and its functional status and hormone output may change as it grows. [Malignant Adrenal Lesions p2]

Screening

- Screening for primary aldosteronism is also indicated when other secondary causes of hypertension (such as phaeochromocytoma or renovascular hypertension) are being considered.

[Primary Aldosteronism p4]

- Unexplained hypokalaemia, whether spontaneous or diuretic-induced, is an indication to screen for primary aldosteronism. [Primary Aldosteronism p4]
- The aldosterone-to-renin ratio (ARR) has a sensitivity of 66–100% and a specificity of 61–96%. [Primary Aldosteronism p4]
- Measuring plasma renin activity is time-consuming, and plasma renin concentration has been proposed as an alternative. [Primary Aldosteronism p4]

Confirmatory testing

- Confirmatory testing is mandatory because aldosterone and plasma renin activity levels vary with day-to-day oscillation, posture, diet and antihypertensive drugs. [Primary Aldosteronism p5]
- Only 50–87% of screen-positive patients have a positive confirmatory test. [Primary Aldosteronism p5]
- The fludrocortisone suppression test (once considered the gold standard) gives fludrocortisone 0.1 mg 6-hourly with NaCl 2 g 8-hourly for 4 days; PAC is then measured in the upright position, and PAC >6 ng/dL is diagnostic of primary aldosteronism. [Primary Aldosteronism p5]
- In the oral sodium loading test, supplemental NaCl can be given to ensure an intake of at least 12.8 g of sodium per day, with 24-hour urine aldosterone, sodium and creatinine measured. [Primary Aldosteronism p5]
- Sodium loading is contraindicated in cardiac and renal disease. [Primary Aldosteronism p5]
- The IV saline infusion test is done in the morning after an overnight fast. [Primary Aldosteronism p5]
- The furosemide upright test and the ACTH stimulation test are not widely used as confirmatory tests. [Primary Aldosteronism p5]

Subtype differentiation: CT and genetic screening

- CT alone is acceptable below age 40 because incidental (non-functioning) adrenal adenomas are rare in this population. [Primary Aldosteronism p7]
- Genetic screening is not done for all patients because familial hyperaldosteronism is rare; it is considered when there is a family history of primary aldosteronism, onset before 20 years, or a family history of stroke at a young age. [Primary Aldosteronism p7]
- On CT an aldosterone-producing adenoma is a unilateral, low-density, non-enhancing lesion (<10 HU) with an average size of 1.6–1.8 cm and a normal-appearing opposite gland. [Primary Aldosteronism p7]

Adrenal venous sampling

- If AVS is non-selective, the data are discarded and repeat AVS is considered, with ACTH stimulation if it was not used initially. [Primary Aldosteronism p8]
- AVS is preceded by 1 hour of recumbency and is performed in the morning after an overnight fast. [Primary Aldosteronism p8]
- AVS uses percutaneous access through the femoral vein, with catheter tips positioned to sample the right adrenal vein, left adrenal vein and IVC. [Primary Aldosteronism p8]
- The lower selectivity cut-off (adrenal vein to IVC cortisol >1.1:1) and the higher one (>5:1), and likewise the lateralisation cut-off of 2:1 to 4:1, depend on whether cosyntropin (ACTH) stimulation is used. [Primary Aldosteronism p8]

Inconclusive AVS: alternatives

- When AVS is inconclusive, the options are nuclear scintigraphy (which gives both functional and anatomical information), postural stimulation testing and measurement of cortisol metabolites. [Primary Aldosteronism p10]
- NP-59 (I-131 6β-iodomethyl-norcholesterol) is a cholesterol-based cortisol analogue tracer that labels adrenocortical cells. [Primary Aldosteronism p10]
- Before NP-59 injection, saturated potassium iodide (Lugol's solution) is given to protect the thyroid from uptake of free I-131. [Primary Aldosteronism p10]
- ACTH is suppressed for NP-59 scanning with dexamethasone 1 mg 6-hourly for 7 days. [Primary Aldosteronism p10]

- NP-59 imaging begins on day 4 after injection and may continue through day 10. [Primary Aldosteronism p10]
- Unilateral NP-59 uptake within 5 days of injection suggests an adenoma, whereas bilateral uptake suggests idiopathic hyperplasia. [Primary Aldosteronism p10]
- NP-59 detection is proportional to the size and function of the adenoma, and NP-59 SPECT-CT improves sensitivity for smaller adenomas. [Primary Aldosteronism p10]
- The posture stimulation test measures aldosterone after overnight recumbency and again after 4 hours upright; normally, the upright position raises angiotensin and aldosterone 2–4 times above the recumbent level. [Primary Aldosteronism p10]
- In idiopathic hyperplasia, aldosterone rises by at least 33% over baseline on standing, whereas an aldosterone-producing adenoma or FH type I shows no change with posture. [Primary Aldosteronism p10]
- The posture test gives no information on location, and 30–50% of adenomas show a rise in PAC while 20% of idiopathic hyperplasias show no rise with posture. [Primary Aldosteronism p10]
- 18-hydroxycorticosterone (18-OHB), the immediate precursor of aldosterone, above 100 ng/dL suggests adenoma and below 100 ng/dL suggests hyperplasia; its accuracy is below 80% and it gives no information on side (right versus left). [Primary Aldosteronism p10]

Adrenal Adenoma

- Adrenal adenomas are yellow on gross inspection because of their lipid content. [Benign Adrenal p1-3]
- Histologically it can be difficult to tell an adenoma from a malignancy; the Weiss criteria or immunohistochemistry are used. [Benign Adrenal p1-3]
- About 90% of adrenal adenomas are benign and about 90% are non-functional. [Benign Adrenal p1-3]
- On follow-up, only about 2% of non-functioning adenomas become metabolically active. [Benign Adrenal p1-3]
- Adrenal adenomas are usually smooth, homogeneous tumours smaller than 4 cm. [Benign Adrenal p1-3]
- Adrenal adenomas do not show significant FDG uptake on PET. [Benign Adrenal p1-3]
- Normalising serum cortisol after treatment of a cortisol-secreting adenoma reduces, but does not eliminate, cardiovascular risk. [Benign Adrenal p1-3]
- After removal of an aldosterone-secreting adenoma, aldosterone usually normalises within 3 months, although baseline essential hypertension may still need medication. [Benign Adrenal p1-3]

Adrenal Oncocytoma

- Adrenal oncocytoma is extremely rare, with only about 50 cases reported. [Benign Adrenal p4]
- Oncocytomas also arise in the kidney, the salivary glands and other endocrine organs. [Benign Adrenal p4]
- Adrenal oncocytoma has a female-to-male ratio of 2.5:1 and a left-to-right ratio of 3.5:1. [Benign Adrenal p4]
- The roughly 10% of adrenal oncocytomas that are functional secrete cortisol or sex steroids. [Benign Adrenal p4]
- About 23% of reported adrenal oncocytomas are of indeterminate malignant potential. [Benign Adrenal p4]
- Adrenal oncocytomas can grow up to 20 cm. [Benign Adrenal p4]
- Two cases of adrenal oncocytoma have been reported in pregnancy. [Benign Adrenal p4]
- Adrenal oncocytoma is nearly always diagnosed on histopathology after surgical resection; evaluation is the same as for other adrenal masses. [Benign Adrenal p4]
- Unlike renal oncocytoma, a central stellate scar is not regularly reported in adrenal oncocytoma. [Benign Adrenal p4]
- The prognosis of adrenal oncocytoma is uncertain but probably benign, with little evidence. [Benign Adrenal p4]

Myelolipoma

- Myelolipoma is rare, with an incidence below 0.1%. [Benign Adrenal p5-6]
- About 15% of myelolipomas are extra-adrenal (85% are adrenal); extra-adrenal sites include presacral, thoracic, retroperitoneal, pelvic, renal, hepatic and gastric locations. [Benign Adrenal p5-6]

- Myelolipoma results from clonal stem-cell proliferation. [Benign Adrenal p5-6]
- The haematopoietic elements in a myelolipoma are relatively normal trilineage cells, but they do not contribute to circulating blood cells because the tumour has no capillary venous sinuses. [Benign Adrenal p5-6]
- One hypothesis is that myelolipoma arises as a result of an adrenal stress response; other theories are unproven. [Benign Adrenal p5-6]
- Myelolipoma affects men and women equally. [Benign Adrenal p5-6]
- Rupture of a myelolipoma has been reported in lesions larger than 10 cm. [Benign Adrenal p5-6]
- Myelolipoma is sometimes associated with a functional adenoma, most commonly cortisol-secreting. [Benign Adrenal p5-6]
- On CT, the higher-density myeloid components of a myelolipoma enhance with contrast. [Benign Adrenal p5-6]
- About a quarter of myelolipomas contain calcification, and many show haemorrhage. [Benign Adrenal p5-6]
- The macroscopic fat of a myelolipoma can also be identified on MRI. [Benign Adrenal p5-6]
- Adrenal myelolipoma must be distinguished from an upper-pole renal angiomyolipoma and from liposarcoma. [Benign Adrenal p5-6]
- Some experts perform a biopsy of a suspected myelolipoma if the diagnosis is in doubt. [Benign Adrenal p5-6]
- Growth rates of myelolipoma vary, and some actually decrease in size over time. [Benign Adrenal p5-6]
- Spontaneous haemorrhage of a myelolipoma is extremely rare. [Benign Adrenal p5-6]

Ganglioneuroma

- Ganglioneuroma is a benign neuroectodermal neoplasm containing ganglion cells and Schwann cells. [Benign Adrenal p7]
- Malignant transformation of ganglioneuroma is rare. [Benign Adrenal p7]
- Ganglioneuromas can grow extremely large. [Benign Adrenal p7]
- VIP secretion with diarrhoea has been reported in ganglioneuroma. [Benign Adrenal p7]
- On both plain and contrast CT, a ganglioneuroma has lower attenuation than muscle. [Benign Adrenal p7]
- A metabolically silent adrenal mass with attenuation below 40 HU on non-contrast CT and stippled calcification suggests ganglioneuroma. [Benign Adrenal p7]
- The natural history of unresected ganglioneuromas is not known. [Benign Adrenal p7]

Adrenal Cysts

- Adrenal cysts account for 1–22% of incidentally detected adrenal lesions. [Benign Adrenal p8-10]
- Adrenal cysts can be diagnosed at any age, even in the prenatal period. [Benign Adrenal p8-10]
- Adrenal cysts are bilateral in 8–10% of cases. [Benign Adrenal p8-10]
- Adrenal cysts are commoner in women, with a peak between the 3rd and 6th decades. [Benign Adrenal p8-10]
- Adrenal cysts are associated with polycystic kidney disease, Beckwith–Wiedemann syndrome and Klippel–Trenaunay–Weber syndrome. [Benign Adrenal p8-10]
- Adrenal pseudocysts have no cellular lining and result from previous haemorrhage or infarction. [Benign Adrenal p8-10]
- Endothelial adrenal cysts lack a proliferating epithelium and are lymphangiomatous or angiomatous. [Benign Adrenal p8-10]
- Epithelial adrenal cysts have a true epithelial lining and include glandular cysts, embryonal cysts and cystic adenomas. [Benign Adrenal p8-10]
- Parasitic adrenal cysts are rare and are caused by Echinococcus. [Benign Adrenal p8-10]
- It is difficult to distinguish a benign adrenal cyst from a cystic adrenal neoplasm. [Benign Adrenal p8-10]
- On CT, an adrenal cyst is a well-defined, sharply marginated mass of fluid attenuation without enhancement; intracystic debris and rim-like calcification, as well as haemorrhage, raise its attenuation. [Benign Adrenal p8-10]

- On MRI, an adrenal cyst is hypointense on T1 and hyperintense on T2. [Benign Adrenal p8-10]
- Because imaging cannot rule out malignancy, a cyst may be excised or aspirated; aspiration can be diagnostic or therapeutic. [Benign Adrenal p8-10]
- Adrenal cyst aspirate is sent for Gram stain, cytology, cortisol and amylase (amylase for left-sided lesions). [Benign Adrenal p8-10]
- Active surveillance of an adrenal cyst should be done with caution because a chance of malignancy exists. [Benign Adrenal p8-10]
- Small, asymptomatic, thin-walled cysts with homogeneous near-water attenuation can be observed. [Benign Adrenal p8-10]
- The prognosis of an adrenal cyst depends on its histology. [Benign Adrenal p8-10]

Cushing syndrome: definition and HPA axis

- Cushing syndrome is rare, occurring in 2–5 per million people per year. [Cushing Syndrome p1]
- Stress and circadian regulation drive the hypothalamus, which releases CRH; oxytocin and vasopressin also stimulate the pituitary but play a lesser role. [Cushing Syndrome p1]
- The pituitary makes ACTH (corticotropin) by cleaving POMC, and cortisol feeds back negatively on both the pituitary and the hypothalamus. [Cushing Syndrome p1]
- McCune–Albright syndrome comprises AIMAH, fibrous dysplasia, skin problems and other endocrine abnormalities. [Cushing Syndrome p1]
- Carney complex is autosomal dominant and comprises PPNAD, spotty skin and mucous membrane lesions, and multiple tumours, including Sertoli cell tumours. [Cushing Syndrome p1]

Causes of Cushing syndrome

- Cushing disease is named after Harvey Cushing, a neurosurgeon, who described it in 1932. [Cushing Syndrome p2]
- The commonest pituitary cause is an ACTH-producing microadenoma; rare pituitary causes are macroadenoma, pituitary carcinoma and hyperplasia of ACTH-producing cells. [Cushing Syndrome p2]
- Two thirds of patients with Cushing disease are female. [Cushing Syndrome p2]
- Ectopic ACTH-producing tumours are mostly malignant, and they include islet cell tumours, thymic tumours and medullary carcinoma of the thyroid, as well as bronchial carcinoid, small cell lung cancer and phaeochromocytoma. [Cushing Syndrome p2]
- Ectopic CRH production is a rare ACTH-dependent cause (less than 1%), most often from bronchial carcinoma. [Cushing Syndrome p2]
- AIMAH accounts for less than 1% of Cushing syndrome. [Cushing Syndrome p2]

Establishing the diagnosis and identifying the cause

- Second-line tests are the 2-day low-dose DST, which is less practical, and midnight plasma cortisol testing. [Cushing Syndrome p4]
- Tests no longer recommended for diagnosis are random serum cortisol, plasma ACTH level, urinary 17-ketosteroids, the insulin tolerance test and the loperamide test. [Cushing Syndrome p4]
- When ACTH is low but abdominal imaging is normal, consider PPNAD or exogenous steroids. [Cushing Syndrome p4]
- When ACTH is low and adrenal lesions are bilateral, adrenal vein sampling is used to lateralise. [Cushing Syndrome p4]
- In ACTH-dependent disease, modern imaging has difficulty localising pituitary microadenomas and ectopic ACTH-producing tumours, which is why IPSS is needed. [Cushing Syndrome p4]

Cortisol excess without Cushing syndrome

- Other causes of cortisol excess without Cushing syndrome are pregnancy, glucocorticoid resistance, poorly controlled diabetes, malnutrition, anorexia and intense chronic exercise. [Cushing Syndrome p5]

Treatment: exogenous and Cushing disease

- Exogenous Cushing syndrome is managed by tapering the steroid gradually when possible. [Cushing Syndrome p6]
- In steroid withdrawal syndrome, the patient cannot tolerate a steroid dose reduction even though HPA axis tests appear normal. [Cushing Syndrome p6]
- After trans-sphenoidal surgery for Cushing disease, relapse occurs in about 25%, so long-term follow-up is needed. [Cushing Syndrome p6]
- With a pituitary macroadenoma, 15% or fewer patients are cured by surgery. [Cushing Syndrome p6]
- After trans-sphenoidal surgery a severe Addisonian state is common, and glucocorticoid replacement is needed for the year after surgery. [Cushing Syndrome p6]
- Hypopituitarism follows trans-sphenoidal surgery in under 5% to 50% of cases, a wide range. [Cushing Syndrome p6]
- Re-operation after failed pituitary surgery is challenging, its success rates vary, and it carries a higher risk of hypopituitarism. [Cushing Syndrome p6]
- Salvage pituitary radiation is an option after failure, but ACTH falls slowly and hypopituitarism may follow. [Cushing Syndrome p6]

Bilateral adrenalectomy and ectopic ACTH syndrome

- Advantages of bilateral adrenalectomy are no risk of hypopituitarism and a very high success rate with rapid resolution of hypercortisolism. [Cushing Syndrome p7]
- Nelson syndrome can rarely cause raised intracranial pressure. [Cushing Syndrome p7]
- Prophylactic pituitary radiotherapy after bilateral adrenalectomy to prevent Nelson syndrome is controversial. [Cushing Syndrome p7]
- Residual functioning adrenal tissue rarely remains after bilateral adrenalectomy. [Cushing Syndrome p7]
- Resection of the tumour is the single best treatment for ectopic ACTH syndrome, but it is possible in only about 10%. [Cushing Syndrome p7]
- In about 35% of ectopic ACTH syndrome the primary tumour is unresectable or cannot be found, and bilateral adrenalectomy with lifelong replacement is then the best option. [Cushing Syndrome p7]

ACTH-independent disease, medical treatment and prognosis

- A cortisol-secreting adrenal mass may be treated by adrenalectomy or partial adrenalectomy. [Cushing Syndrome p8]
- Medical treatment is used to bridge the patient to surgery or when surgery is not possible, and it is best left to the endocrinologist. [Cushing Syndrome p8]
- Aminoglutethimide is another steroid synthesis blocker that can be used in Cushing syndrome. [Cushing Syndrome p8]
- Abiraterone, which inhibits 17-alpha-hydroxylase, is a possible future medical treatment for Cushing syndrome. [Cushing Syndrome p8]
- Life expectancy in Cushing syndrome is shorter than normal, and treatment reduces this risk but does not remove it. [Cushing Syndrome p8]

Adrenal Incidentaloma: definition, frequency and algorithm

- The frequency of adrenal incidentaloma rises with age, from less than 0.5% in people in their 20s to up to 7% in people in their 70s. [Incidentloma p1]
- The median diameter of an adrenal incidentaloma is 3 cm. [Incidentloma p1]
- A non-functional incidentaloma on follow-up is re-imaged at 6, 12 and 24 months, and the functional work-up is repeated yearly for 4 years. [Incidentloma p1]
- During follow-up, growth of the mass by more than 1 cm or a positive repeat functional work-up is an indication for adrenalectomy. [Incidentloma p1]

Imaging: ultrasonography and CT

- Ultrasonography is suboptimal for detecting and characterising adrenal masses. [Incidentloma p2]

- On the right side, the IVC and liver give ultrasound a better window to the adrenal gland. [Incidentloma p2]
- On the left side, ultrasound is less sensitive, and adrenal lesions are overlooked or mistaken for splenic, pancreatic, para-aortic lymphatic or gastric anatomy. [Incidentloma p2]
- An unenhanced CT attenuation below 10 HU has a sensitivity of 71% and a specificity of 98% for adrenal adenoma. [Incidentloma p2]
- Enhanced CT is taken about 1 minute after the contrast bolus with 5 mm images, and an enhancing rim (with irregular margins) suggests malignancy. [Incidentloma p2]
- Rapid washout of enhancement on the 15-minute delayed CT is seen in lipid-poor adenomas and also in lipid-rich adenomas. [Incidentloma p2]

Imaging: MRI

- Chemical-shift MRI uses the interference between signals from fat and water: signal loss on out-of-phase (opposed-phase) sequences compared with in-phase images shows intracellular lipid and so identifies an adenoma. [Incidentloma p3]
- Gadolinium-enhanced MRI washout studies are not used for adrenal mass characterisation. [Incidentloma p3]

Functional imaging

- Functional imaging is not used routinely for incidentalomas, because cross-sectional imaging plus biochemical evaluation already gives high diagnostic accuracy. [Incidentloma p4]
- NP-59, a radiolabelled cholesterol analogue, is taken up by adrenocortical cells and can distinguish adenomas from other adrenal lesions, but it is time-consuming and not currently approved. [Incidentloma p4]
- 11C-metomidate inhibits 11β-hydroxylase and aldosterone synthase and is taken up preferentially by adrenocortical cells, but it cannot differentiate benign adrenal lesions from adrenocortical carcinoma. [Incidentloma p4]
- 131I-MIBG has only a limited role in the routine evaluation of an incidentaloma. [Incidentloma p4]
- Alternatives to MIBG include 18F-FDG PET, 18F-dopamine PET and 111In-octreotide scanning; the routine use of dopamine PET and octreotide scanning has not been evaluated. [Incidentloma p4]
- 18F-FDG PET can differentiate benign from malignant adrenal masses with a sensitivity of 93–100% and a specificity of 80–100%, and it is reserved for cases where imaging and clinical data are inconclusive. [Incidentloma p4]

Cortisol hypersecretion testing: LD-DST

- The three first-line tests for cortisol excess (1-mg DST, late-night salivary cortisol and 24-h UFC) have relatively equivalent accuracy. [Incidentloma p6]
- A post-dexamethasone cortisol cut-off of below 1.8 µg/dL has a specificity of about 80%. [Incidentloma p6]
- Further CYP3A4 inducers that accelerate dexamethasone metabolism and cause false-positive DST results are phenobarbitone, primidone, ethosuximide, pioglitazone and rifapentine. [Incidentloma p6]
- Further CYP3A4 inhibitors that slow dexamethasone metabolism are cimetidine, fluoxetine, aprepitant and fosaprepitant. [Incidentloma p6]
- Mitotane, like oestrogen, raises cortisol-binding globulin and total cortisol and so can give a false-positive DST. [Incidentloma p6]
- Carbamazepine, fenofibrate and carbenoxolone falsely increase urinary free cortisol results. [Incidentloma p6]

Late-night salivary cortisol

- Late-night salivary cortisol tests the loss of normal diurnal variation, that is, the inability to suppress cortisol at night, and is abnormal even in mild Cushing syndrome. [Incidentloma p7]
- Salivary cortisol reflects serum cortisol levels in near real time. [Incidentloma p7]
- A normal late-night salivary cortisol is below 145 ng/dL when measured by liquid chromatography–tandem mass spectrometry. [Incidentloma p7]

- Late-night salivary cortisol has a sensitivity and specificity above 90%. [Incidentloma p7]
- Late-night salivary cortisol has high false-positive rates in depression, altered sleep patterns and chronic illness. [Incidentloma p7]
- Patients should avoid tobacco on the day of salivary cortisol testing and should not eat or brush their teeth within 30 minutes of sample collection. [Incidentloma p7]

24-hour urinary free cortisol

- The Endocrine Society practice guideline (2008) states that the low-dose DST or late-night salivary cortisol has supplanted 24-h UFC for initial screening. [Incidentloma p7]
- When UFC is used in the second and third trimesters of pregnancy, cut-off values 2 to 3 times above the normal laboratory threshold are needed to allow for the physiological changes of pregnancy. [Incidentloma p7]
- The usual normal cut-off for 24-h urinary free cortisol is 50–100 µg per 24 hours. [Incidentloma p7]

Aldosterone hypersecretion testing

- The fludrocortisone suppression test is a confirmatory test for primary aldosteronism, though it has fallen out of favour. [Incidentloma p9]

Catecholamine hypersecretion testing

- The upper limits of normal for plasma free metanephrines are 61 ng/L for metanephrine and 112 ng/L for normetanephrine. [Incidentloma p10]
- A positive 24-h urinary fractionated metanephrine result is metanephrine above 1531 nmol/day, normetanephrine above 4001 nmol/day, or total metanephrines above 1563 nmol/day. [Incidentloma p10]
- 24-h urinary fractionated metanephrines have a specificity of about 91%. [Incidentloma p10]

Adrenal biopsy

- Bleeding is the most common complication of adrenal biopsy, and it can complicate or prevent later laparoscopic resection. [Incidentloma p11]
- Adrenal biopsy can cause pneumothorax and haemothorax. [Incidentloma p11]

Malignant adrenal lesions: adrenal carcinoma, epidemiology

- Malignant adrenal lesions comprise adrenocortical carcinoma, metastasis, malignant phaeochromocytoma and neuroblastoma. [Malignant Adrenal Lesions p1]
- Sporadic ACC is usually unilateral, whereas 2–6% are bilateral and hereditary. [Malignant Adrenal Lesions p1]
- McCune–Albright syndrome is also listed among the syndromic associations of ACC. [Malignant Adrenal Lesions p1]

Pathophysiology of ACC

- TP53 loss (as in Li–Fraumeni syndrome) is found in 20–33% of sporadic ACC. [Malignant Adrenal Lesions p1]
- IGF (IGF-2) overexpression, as in Beckwith–Wiedemann syndrome, is present in about 90% of ACC. [Malignant Adrenal Lesions p1]
- Constitutive activation of the Wnt/β-catenin pathway occurs in about 30% of ACC. [Malignant Adrenal Lesions p1]

Imaging

- CT features of ACC also include irregular enhancement and necrotic areas with cystic degeneration. [Malignant Adrenal Lesions p4]
- Local tumour extension and lymphadenopathy are ominous features suggesting locally advanced or metastatic disease. [Malignant Adrenal Lesions p4]
- The commonest sites of ACC metastasis are lung and liver. [Malignant Adrenal Lesions p4]
- On MRI, ACC is isointense to liver and spleen on T1, intermediate to increased in intensity on T2, and shows marked gadolinium uptake. [Malignant Adrenal Lesions p4]
- ACC is 18F-FDG PET avid, but PET is rarely needed because cross-sectional imaging is usually sufficient. [Malignant Adrenal Lesions p4]

Pathological evaluation (Weiss criteria)

- The high nuclear grade criterion in the Weiss system corresponds to Fuhrman grade 3 or 4. [Malignant Adrenal Lesions p6]
- Weiss score of 3 or more has a sensitivity of 100% and specificity of 96% for malignancy. [Malignant Adrenal Lesions p6]
- Weiss criteria should be used with caution, with alternative criteria, in paediatric ACC and in adrenal tumours with oncocytic features. [Malignant Adrenal Lesions p6]

Management algorithm

- In unresectable stage IV disease the diagnosis is confirmed by needle biopsy and mitotane monotherapy is started, with follow-up every 2 months; stable or regressing disease continues mitotane, and progression moves to EDP chemotherapy. [Malignant Adrenal Lesions p7]
- Resectable (oligometastatic) stage IV disease is treated by debulking surgery or metastasectomy plus mitotane. [Malignant Adrenal Lesions p7]

Surgery and follow-up

- Local and systemic adjuvant therapy is often given despite lack of clear evidence that it improves survival. [Malignant Adrenal Lesions p8]
- Cytoreduction or metastasectomy may not improve survival but reduces tumour-related (hormonal) symptoms and facilitates additional therapy. [Malignant Adrenal Lesions p8]
- PET has no established role in follow-up and surveillance; CT is used. [Malignant Adrenal Lesions p8]
- Radiotherapy has a limited role for the primary ACC. [Malignant Adrenal Lesions p8]

Adjuvant radiotherapy and medical therapy

- Adjuvant tumour-bed radiotherapy reduced local recurrence to 14% compared with 80% without radiotherapy, with no significant difference in disease-free or overall survival. [Malignant Adrenal Lesions p9]
- Adjuvant radiotherapy is best for patients with positive or indeterminate margins without metastasis. [Malignant Adrenal Lesions p9]
- Mitotane acts by direct cytotoxicity to the adrenal cortex, oxidative damage through free-radical generation, and inhibition of steroid-synthesis enzymes. [Malignant Adrenal Lesions p9]
- Further side effects of mitotane include lethargy, depression, feminisation, rash, transaminitis, dyslipidaemia and thrombocytopenia. [Malignant Adrenal Lesions p9]
- The response rate to mitotane alone is 15–35%. [Malignant Adrenal Lesions p9]

Symptom control and prognosis

- Mitotane controls hormonal symptoms only after a delay, so adrenostatic drugs are used to lower cortisol in symptomatic patients; aminoglutethimide is also listed with ketoconazole, metyrapone and etomidate. [Malignant Adrenal Lesions p10]
- Tumour necrosis and atypical mitotic figures are additional adverse prognostic factors. [Malignant Adrenal Lesions p10]

Adrenal metastasis

- Adrenal metastases most commonly come from melanoma and lung cancer; other primaries include RCC, breast, MCT (as written), contralateral ACC, gastrointestinal, prostate, cervical, basal cell, pancreatic, cholangiocarcinoma, urothelial, squamous cell carcinoma, seminoma, thymoma and CML. [Malignant Adrenal Lesions p12]
- Bilateral adrenal metastases larger than 4 cm can cause adrenal insufficiency. [Malignant Adrenal Lesions p12]
- On CT, adrenal metastases are well circumscribed, homogeneous lesions with HU >10 on non-contrast CT that fail to show significant washout. [Malignant Adrenal Lesions p12]

Pheochromocytoma: epidemiology and paraganglia

- Pheochromocytoma accounts for only about 0.5% of all cases of hypertension. [Pheochromocytoma p1]
- Paraganglia are chromaffin-producing neural crest tissue that anatomically parallels the sympathetic and parasympathetic ganglia, found in the head, neck, thorax, abdomen and pelvis.

[Pheochromocytoma p1]

Pathophysiology

- Pheochromocytomas differ in their enzymatic composition and in their ability to metabolise their own catecholamines; this produces great variability in the amount and ratio of catecholamines secreted, which explains the heterogeneous clinical behaviour of these tumours. [Pheochromocytoma p1]

Pathophysiology of malignant pheochromocytoma

- Pathological appearance, local invasion and immunohistochemistry (including Ki-67) are of limited value in defining malignancy; only clinical metastasis defines it. [Pheochromocytoma p2]

Clinical characteristics

- The classical "rule of 10s" also includes 10% bilateral and 10% paediatric tumours. [Pheochromocytoma p3]
- Overall malignancy in adrenal pheochromocytoma is under 5%, whereas about one-third of extra-adrenal tumours are malignant. [Pheochromocytoma p3]
- Familial (hereditary) tumours present at a younger age; sporadic tumours are usually diagnosed in the 4th to 5th decade of life (30s and 40s). [Pheochromocytoma p3]
- Tumours arising from the right adrenal are larger, more common and recur more frequently. [Pheochromocytoma p3]
- Further reported symptoms include tiredness, nausea, flushing, weight loss, anxiety and panic attacks, and orthostatic hypotension in addition to sustained or paroxysmal hypertension. [Pheochromocytoma p3]
- Episodic hypertension, the classic presenting sign, occurs in only 30–50% of patients. [Pheochromocytoma p3]

Clinical characteristics of hereditary and malignant pheochromocytoma

- The commonest sites of metastasis are bone, lung, liver and lymph nodes. [Pheochromocytoma p4]
- Metastases may be present at diagnosis or appear during surveillance after excision of the primary; most are discovered within 5 years. [Pheochromocytoma p4]

Diagnostic tests: cross-sectional imaging

- On CT a pheochromocytoma appears well circumscribed; its high attenuation reflects low lipid content and rich vascularity. [Pheochromocytoma p5]
- On MRI pheochromocytoma shows no signal drop-out on out-of-phase (chemical-shift) sequences, which separates it from a lipid-rich adenoma. [Pheochromocytoma p5]

Diagnostic tests: functional imaging

- FDG-PET avidity is independent of the tumour's biochemical activity or phenotype. [Pheochromocytoma p6]
- FDG-PET is more accurate than MIBG for metastatic disease in nearly all patients except those with MEN-2 mutations, in whom FDG-PET sensitivity is below 50%. [Pheochromocytoma p6]
- MIBG scintigraphy for pheochromocytoma has a sensitivity of 83–100% and a specificity of 95–100%. [Pheochromocytoma p6]
- Routine functional imaging in every mass with positive biochemistry is controversial. [Pheochromocytoma p6]

Biochemical evaluation

- Catecholamines and their metabolites circulate in two forms: free, and conjugated with a sulfate moiety. [Pheochromocytoma p7]
- Catecholamine measurement (urine or blood) has a sensitivity of about 85% and a specificity of about 85%, because release is often paroxysmal. [Pheochromocytoma p7]
- Urinary catecholamine measurement is still recommended (traditionally obtained) together with urinary fractionated metanephrine testing. [Pheochromocytoma p7]
- Whether plasma free metanephrines or 24-hour urinary fractionated metanephrines should be the initial screening test is controversial. [Pheochromocytoma p8]

- Vanillylmandelic acid (VMA), the primary end metabolite of catecholamines, has a sensitivity below 65% and a specificity of 99% (especially in non-familial cases). [Pheochromocytoma p8]
- Clonidine, an alpha-2 agonist, suppresses catecholamine (especially norepinephrine) production by the sympathetic nervous system but not by a pheochromocytoma; normetanephrine is compared before and after clonidine. [Pheochromocytoma p8]
- The main problem with the clonidine suppression test is significant hypotension. [Pheochromocytoma p8]
- Chromogranin A belongs to the granins, found in secretory vesicles of neuroendocrine and nervous tissue; its sensitivity for pheochromocytoma is about 85%. [Pheochromocytoma p9]
- Chromogranin A is used when free plasma metanephrines are mildly to moderately raised (less than 4-fold). [Pheochromocytoma p9]
- Chromogranin A is cleared by the kidney, so its specificity falls when GFR is below 80 mL/min. [Pheochromocytoma p9]

Screening for hereditary pheochromocytoma

- Gene testing is chosen by phenotype after informed consent: family history or age under 50 years, test VHL, RET, SDHB and SDHD; multiple tumours, VHL, SDHB and SDHD; malignant tumour, VHL and SDHB; bilateral tumours, VHL, RET and SDHD. [Pheochromocytoma p9]
- A positive result is followed by screening of first-degree relatives. [Pheochromocytoma p9]

Treatment: preoperative preparation

- Phenoxybenzamine is started at 10 mg orally twice daily and increased by 10–20 mg every 2–3 days. [Pheochromocytoma p10]
- Evidence on the efficacy of selective alpha-1 blockers is contradictory. [Pheochromocytoma p10]
- Beta-1-blocker doses are atenolol 12.5–25 mg twice or three times daily, or metoprolol 25–50 mg orally three or four times daily. [Pheochromocytoma p10]
- Use of metyrosine (catecholamine synthesis blockade) depends on institutional preference; its dose is increased by 250–500 mg every 2–3 days. [Pheochromocytoma p10]
- Metyrosine needs about 3 days to reach its full clinical effect, and 500 mg orally is given the night before surgery. [Pheochromocytoma p11]
- Further CNS side effects of metyrosine are galactorrhoea and mood depression. [Pheochromocytoma p11]
- Calcium channel blocker doses, added if BP stays poorly controlled, are amlodipine 10–20 mg daily, nicardipine 60–90 mg daily, nifedipine 30–90 mg daily and verapamil 180–540 mg daily (verapamil avoids reflex tachycardia). [Pheochromocytoma p11]
- Patients are admitted 1–2 days before surgery for BP and heart-rate monitoring and intravascular volume restoration. [Pheochromocytoma p11]
- An alternative regimen (Option 2) gives a calcium channel blocker followed by a catecholamine synthesis inhibitor as an outpatient for 7–14 days, then 1–2 days as an inpatient for BP and heart-rate monitoring and IV fluid restoration. [Pheochromocytoma p12]

Treatment of malignant pheochromocytoma

- Treatment of malignant pheochromocytoma is largely palliative; surgical metastasectomy of resectable disease is standard of care, but its survival benefit is uncertain. [Pheochromocytoma p13]
- Ablation techniques and embolisation have been described for local control of malignant tumour. [Pheochromocytoma p13]
- With 131I-MIBG therapy, catecholamine levels fall in 40% of patients and tumours shrink in 30%. [Pheochromocytoma p14]
- High-dose MIBG therapy is controversial because of significant toxicity. [Pheochromocytoma p14]
- CVD chemotherapy gives a short-lived response (under 2 years), with radiographic tumour shrinkage in 50% and a biochemical response in 75%. [Pheochromocytoma p14]
- CVD is given mainly when MIBG therapy has failed or when there is no MIBG uptake on initial studies. [Pheochromocytoma p14]

Follow-up and prognosis

- Metabolic testing is repeated 2 weeks after adrenalectomy. [Pheochromocytoma p14]

- Post-operative cross-sectional imaging is reasonable to document tumour resection and healing of the resection bed; later imaging is guided by biochemical results. [Pheochromocytoma p15]
- Risk factors for recurrence are extra-adrenal disease, hereditary disease, bilateral disease and a large tumour. [Pheochromocytoma p15]
- Up to 50% of recurrences are malignant. [Pheochromocytoma p15]
- Bone metastases appear to carry the most benign prognosis. [Pheochromocytoma p15]

Hereditary syndromes of pheochromocytoma (table)

- MEN-2A also features cutaneous lichen amyloidosis. [Pheochromocytoma p16]
- SDHB (PGL-4) tumours include head and neck paragangliomas (carotid body tumour, vagal, jugular and tympanic paragangliomas) as well as abdominal and thoracic paragangliomas. [Pheochromocytoma p16]

CH 147 · SURGERY OF THE ADRENAL GLANDS

Adrenalectomy: laparoscopic contraindications

- Severe cardiopulmonary disease is an absolute contraindication to laparoscopic adrenalectomy. [Adrenalectomy p1]
- Local recurrence of a previously resected adrenal mass is an absolute contraindication to laparoscopic adrenalectomy. [Adrenalectomy p1]
- Malignant phaeochromocytoma is a relative contraindication to laparoscopic adrenalectomy. [Adrenalectomy p1]
- Morbid obesity is a relative contraindication to laparoscopic adrenalectomy. [Adrenalectomy p1]
- Significant adhesions are a relative contraindication to laparoscopic adrenalectomy. [Adrenalectomy p1]
- A history of recurrent pyelonephritis is a relative contraindication to laparoscopic adrenalectomy. [Adrenalectomy p1]
- A virilising tumour (which should be presumed to be ACC) is a relative contraindication to laparoscopic adrenalectomy. [Adrenalectomy p1]

Preparation of patient for adrenal surgery

- In Conn syndrome, magnesium should be repleted along with potassium before surgery. [Adrenalectomy p2]
- In Conn syndrome, intravascular fluid status should be normalised before surgery. [Adrenalectomy p2]
- The notes list a stress dose of cortisol in the preoperative preparation for Conn syndrome. [Adrenalectomy p2]
- Patients with an incidentaloma should receive the same anaesthetic precautions as for phaeochromocytoma, because about 5% of phaeochromocytomas have normal diagnostic studies. [Adrenalectomy p2]
- For ACC, consent should be taken for removal of adjacent organs. [Adrenalectomy p2]
- For ACC, failure to identify vena cava involvement before surgery is a recognised pitfall, so caval involvement must be looked for preoperatively. [Adrenalectomy p2]
- In Cushing syndrome, diabetes should be controlled before surgery. [Adrenalectomy p2]
- In Cushing syndrome, a cardiopulmonary work-up is part of preoperative preparation. [Adrenalectomy p2]
- In Cushing syndrome, preoperative antibiotics are given. [Adrenalectomy p2]

Surgical approach

- The transperitoneal approach gives better access to the renal hilum and great vessels. [Adrenalectomy p3]
- The retroperitoneal approach gives a shorter hospital stay. [Adrenalectomy p3]
- The retroperitoneal approach is better for morbidly obese patients. [Adrenalectomy p3]
- The disadvantage of the retroperitoneal approach is a smaller operative field. [Adrenalectomy p3]

Oncological principles of resection for ACC

- The intact anterior peritoneum should be preserved if there is no evidence of invasion through the peritoneal layer. [Adrenalectomy p3]
- Barriers such as mops should be used to exclude the remainder of the peritoneal cavity. [Adrenalectomy p3]
- Bleeding and fluid spillage into the peritoneal cavity should be minimised. [Adrenalectomy p3]
- Gloves, gowns and instruments should be changed after tumour removal and before closure. [Adrenalectomy p3]

Complications of adrenalectomy

- Access-related intraoperative complications include abdominal wall bleeding. [Adrenalectomy p4]
- Access-related intraoperative complications include cutaneous nerve injury. [Adrenalectomy p4]
- Intraoperative haemorrhage can arise from a lumbar vein. [Adrenalectomy p4]
- Intraoperative haemorrhage can arise from a hepatic vein. [Adrenalectomy p4]
- Haemorrhage can arise from the remnant adrenal gland after partial adrenalectomy. [Adrenalectomy p4]
- Ischaemic complications include inadvertent ligation of the renal artery or renal vein. [Adrenalectomy p4]
- Ischaemic complications include inadvertent ligation of the superior mesenteric artery or superior mesenteric vein. [Adrenalectomy p4]
- After surgery for primary hyperaldosteronism, hypokalaemia can occur from continued potassium loss in the immediate postoperative period. [Adrenalectomy p4]
- Postoperative hyperkalaemia after surgery for primary hyperaldosteronism is secondary to failure (suppression) of the contralateral adrenal. [Adrenalectomy p4]
- After surgery for Cushing syndrome, hyperglycaemia is a complication. [Adrenalectomy p4]
- After surgery for Cushing syndrome, poor wound healing is a complication. [Adrenalectomy p4]
- After surgery for Cushing syndrome, there is an increased risk of infection. [Adrenalectomy p4]
- After surgery for Cushing syndrome, fractures secondary to osteoporosis can occur. [Adrenalectomy p4]
- Pancreatitis or pancreatic fistula is a general postoperative complication. [Adrenalectomy p4]
- Pneumonia is a general postoperative complication. [Adrenalectomy p4]
- Prolonged ileus is a general postoperative complication. [Adrenalectomy p4]
- Postoperative collections are a general complication. [Adrenalectomy p4]

Postoperative management

- Post-excision hypoglycaemia occurs because high catecholamine levels inhibit insulin release through alpha-2 receptor stimulation; once the tumour is removed, rebound hyperinsulinaemia follows. [Pheochromocytoma p12]

CH 140 · RETROPERITONEAL TUMOURS

Retroperitoneal Sarcoma: Management Algorithm

- For an unresectable or metastatic RP sarcoma that stays unresectable after neoadjuvant therapy, the palliative options are chemotherapy, radiotherapy, surgery for symptom control, supportive care, and observation if the patient is asymptomatic. [Compilation 01-2 p69]
- After resection, follow-up imaging may be CT or MRI, and chest imaging should be considered. [Compilation 01-2 p69]

Retroperitoneal Sarcoma: Treatment Options

- Pre-operative radiotherapy allows partial regression of tumour size and reduces tumour cell implantation. [Compilation 01-2 p70]
- The pre-operative radiotherapy dose is 45–50 Gy in total, at 1.8 Gy per fraction. [Compilation 01-2 p70]
- Adjuvant radiotherapy is considered for high-grade tumours, positive margins, or tumours larger than 5 cm. [Compilation 01-2 p70]
- Adjuvant radiotherapy should not be regarded as compensation for incomplete or poorly conducted surgery. [Compilation 01-2 p70]

- The MAID chemotherapy regimen for sarcoma consists of mesna, doxorubicin (Adriamycin), ifosfamide and dacarbazine. [Compilation 01-2 p70]
- Lymph node dissection in RP sarcoma is done only for bulky or grossly involved lymph nodes. [Compilation 01-2 p70]

Retroperitoneal Tumours

- Retroperitoneal sarcomas arise de novo. Benign retroperitoneal tumours are much less common and do not transform to malignant. [UTUC notes 2 p18]
- In the classification by tissue of origin, adipose tissue gives angiolipoma, lipoblastoma and lipomatosis on the benign side (as well as lipoma). [UTUC notes 2 p18]
- Fibrous tissue gives fibroma, angiofibroma and elastofibroma (benign) and fibrosarcoma (malignant). [UTUC notes 2 p18]
- Fibrohistiocytic tissue gives fibrous histiocytoma and xanthoma (benign) and malignant fibrous histiocytoma (malignant). [UTUC notes 2 p18]
- Muscle gives angiomyoma and rhabdomyoma (benign, alongside leiomyoma) and angiomyosarcoma and rhabdomyosarcoma (malignant, alongside leiomyosarcoma). [UTUC notes 2 p18]
- Neural tissue lists phaeochromocytoma as benign and malignant schwannoma and malignant phaeochromocytoma as malignant, alongside schwannoma, neurofibroma, ganglioneuroma, MPNST and neuroblastoma. [UTUC notes 2 p18]
- Synovial tissue gives synovioma (benign) and synovial carcinoma/sarcoma (malignant). [UTUC notes 2 p18]
- Vascular tissue gives haemangioma (benign) and angiosarcoma and Kaposi sarcoma (malignant). [UTUC notes 2 p18]
- Lymphatic tissue gives lymphangioma (benign) and lymphangiosarcoma (malignant). [UTUC notes 2 p18]

Staging (RP sarcoma)

- The commonest site of metastasis from retroperitoneal sarcoma is the lung, followed by liver, bone and brain. Spread is haematogenous. [UTUC notes 2 p19]
- T staging by size is T1 under 5 cm, T2 5 to 10 cm, T3 10 to 15 cm and T4 over 15 cm. [UTUC notes 2 p19]
- N1 must be histologically proven (pN1). M is staged M0 or M1. [UTUC notes 2 p19]
- Residual tumour classification: R0 means entirely resected with no microscopic or macroscopic residual tumour; R1 means microscopic residual tumour; R2 means macroscopic residual tumour. [UTUC notes 2 p19]
- R3 means tumour spillage and dissemination during resection. [UTUC notes 2 p19]

Clinical Presentation & Work-Up (RP sarcoma)

- Presenting features include an abdominal lump, pain, constitutional symptoms, early satiety, fatigue and leg oedema. [UTUC notes 2 p20]
- Further indications for image-guided core biopsy before treatment are suspected metastasis from a primary elsewhere, a surgically unresectable mass, and a suspected sarcoma with metastases on imaging. [UTUC notes 2 p20]
- On CT, a mass of higher density without a fat component favours biopsy, because it suggests lymphoma, metastasis or germ cell tumour rather than liposarcoma. [UTUC notes 2 p20]

15 Renal, Nephrology & Transplant

CH 88 · SURGICAL, RADIOLOGIC AND ENDOSCOPIC ANATOMY OF THE KIDNEY AND URETER

Lumbodorsal fascia

- The lumbodorsal fascia is a deep investing membrane covering most of the posterior thorax and abdomen, but is only a thin fibrous lamina in the thorax. [Kidney Approaches p4]

- The anterior and middle layers of the lumbodorsal fascia insert onto the transverse processes of the vertebrae, and the posterior layer inserts onto the spinous processes. [Kidney Approaches p4]
- Laterally, the lumbodorsal fascia bridges the aponeuroses of internal oblique and transversus abdominis, the costal angles and the iliac crest to the medially placed vertebral column and sacrum. [Kidney Approaches p4]
- Relations of the lumbodorsal fascia layers: serratus posterior inferior and latissimus dorsi lie superficial to the posterior layer; erector spinae lies between the posterior and middle layers; quadratus lumborum lies between the middle and anterior layers; and psoas major lies anterior to the anterior layer. [Kidney Approaches p4]

CH 90 · FUNDAMENTALS OF NEPHROLOGY

Haemodialysis venous catheters

- The right internal jugular vein is preferred for a dialysis catheter because it gives a short, straight path to the right atrium. [Dialysis AVF p1]
- The left internal jugular vein is used if the right internal jugular and right subclavian veins are contraindicated. [Dialysis AVF p1]
- Cuffed catheters migrate less than uncuffed ones. [Dialysis AVF p1]
- Dialysis catheters come in double-D or side-by-side port designs. [Dialysis AVF p1]
- Catheters may be impregnated with antiseptic or silver-based coatings to reduce bacterial growth. [Dialysis AVF p1]

AV fistula: pre-operative evaluation

- In the vein dilatation test, the internal diameter of the vein should rise by more than 50% on proximal occlusion. [Dialysis AVF p2]
- In the arteriolar dilatation test, the normal triphasic pulse contour becomes biphasic after the fist is clenched for 2 minutes and then opened. [Dialysis AVF p3]
- Arteriography is indicated when the pulse is absent or reduced at the desired site, or when the mean arterial pressure differs by more than 20 mmHg between the two arms. [Dialysis AVF p3]

AV fistula: locations

- The snuffbox fistula, the most distal site, lies between the tendons of extensor pollicis longus and extensor pollicis brevis and is a variant of the radiocephalic fistula. [Dialysis AVF p3]
- An ulnar artery to forearm basilic vein fistula is an uncommon conventional option. [Dialysis AVF p3]
- Unconventional upper-limb fistulas include forearm basilic vein to radial artery at the wrist, and loop configurations of the forearm basilic or cephalic vein to the brachial artery. [Dialysis AVF p4]
- The Konner modification of the Gracz fistula joins the perforating veins of the proximal forearm to the proximal radial artery. [Dialysis AVF p4]
- An elbow fistula is indicated when the radial or ulnar artery at the wrist is under 2 mm in diameter, calcified or segmentally stenosed, or when the cephalic vein is under 2.5 mm. [Dialysis AVF p4]
- Leg fistulas include saphenous vein to popliteal artery and superficial femoral artery to femoral vein. [Dialysis AVF p4]

AV fistula: procedure, maturation and complications

- The end-to-side method avoids venous hypertension in the hand. [Dialysis AVF p5]
- The side-to-side method sometimes causes "red hand syndrome". [Dialysis AVF p5]
- The modified "piggyback" Slot technique reduces anastomotic stenosis and torsion of the anastomotic vein. [Dialysis AVF p5]
- Immediate fistula blood flow should be 200–300 mL/min; a flow under 120 mL/min predicts failure or closure. [Dialysis AVF p5]
- Hand exercises are advised after fistula creation. [Dialysis AVF p5]
- A mature fistula also needs a straight cannulation segment of at least 6 cm. [Dialysis AVF p6]
- Red hand syndrome is a complication of AV fistula. [Dialysis AVF p6]

- In steal, arterial flow diverts into the fistula, and with distal small-vessel arteriopathy this causes hand ischaemia. [Dialysis AVF p6]
- Thrombosis is the commonest complication of AV fistula. [Dialysis AVF p6]

AV grafts

- AV grafts may be straight, looped or curved. [Dialysis AVF p6]
- Graft sites include brachial artery to axillary vein, brachial artery to basilic vein, radial artery to basilic vein at the wrist, and axillary artery to axillary vein (same side, or the opposite-side artery as a "necklace graft"). [Dialysis AVF p6]
- Advantages of AV grafts are a large surface area for needle placement and easy cannulation and surgical handling. [Dialysis AVF p7]

Haemodialysis prescription

- Typical dialysate bicarbonate is 25 mmol/L and calcium 1.5 mmol/L. [Dialysis AVF p8]
- Dialysate flow rate is 500 mL/min. [Dialysis AVF p8]
- Dialysate temperature is 35–36 °C. [Dialysis AVF p8]
- Typical fluid removal per session is 2.2 L. [Dialysis AVF p8]

Peritoneal dialysis

- Automated PD includes continuous cycling PD (CCPD), intermittent PD (IPD) and nocturnal intermittent PD, which are more suitable for children on PD. [Dialysis AVF p9]
- PD fluid uses lactate as the buffer, in a balanced salt solution devoid of potassium. [Dialysis AVF p10]
- The three-pore model of the peritoneal membrane has ultrasmall pores (water channels), small pores (the major pathway, for water and small solutes) and large pores (for large molecules). [Dialysis AVF p11]
- Further absolute contraindications to PD are a large diaphragmatic defect and acute ischaemic or infectious bowel disease. [Dialysis AVF p11]
- PD catheter types are the straight single-cuff, straight double-cuff, double-cuff coiled, swan-neck and Toronto Western catheters. [Dialysis AVF p12]
- In each PD exchange the peritoneal cavity is drained over 10–15 minutes and 2–2.5 L is instilled over 10–15 minutes. [Dialysis AVF p12]
- Adequacy of PD also includes no oedema or hypertension, serum potassium below 5 mmol/L, serum bicarbonate above 24 mEq/L, BMI 20–30 kg/m², albumin above 3.5, and a stable mid-arm circumference. [Dialysis AVF p13]
- Large-solute clearance is judged by albumin clearance below 0.15 mL/min. [Dialysis AVF p13]
- Risk of death is lower on PD than on HD in the first 3 years. [Dialysis AVF p13]
- PD patients eventually need HD or a transplant. [Dialysis AVF p13]
- Inflow failure of a PD catheter is due to a fibrin plug, omental wrapping or catheter kinking. [Dialysis AVF p14]
- Thickening of the peritoneal membrane and metabolic acidosis are complications of PD. [Dialysis AVF p14]

Hyperfiltration Injury: Definition

- Hyperfiltration is variably defined as an absolute rise in whole-kidney GFR, a rise in filtration fraction, or a rise in single-nephron GFR (SNGFR). [Hyperfiltration Injury p1]
- Even after loss of 90% of renal function, most kidneys keep excretory and secretory function at a level compatible with life because of compensatory physiological and morphological changes. [Hyperfiltration Injury p1]
- Of the increase in kidney size after nephron loss, 10–25% is due to hyperplasia and the rest to hypertrophy. [Hyperfiltration Injury p1]

Etiopathogenesis

- A high-protein meal is a physiological cause of hyperfiltration (as is pregnancy). [Hyperfiltration Injury p2]
- FSGS and ADPKD are listed as causes of hyperfiltration. [Hyperfiltration Injury p2]

- Sickle cell disease, thalassaemia and cancer are other causes of hyperfiltration (with obesity). [Hyperfiltration Injury p2]

Pathogenesis

- Raised glomerular pressure activates the renin–angiotensin–aldosterone system, which causes hypertension and also drives tubulointerstitial inflammation. [Hyperfiltration Injury p3]
- Increased glomerular permeability leaks protein, which injures tubular cells and leads through cytokine-mediated tubulointerstitial inflammation to tubulointerstitial fibrosis, alongside the podocyte-injury route to glomerulosclerosis. [Hyperfiltration Injury p3]

Treatment

- ACE inhibitors and ARBs can promote regression of glomerulosclerosis and so retard the progression of diabetic nephropathy. [Hyperfiltration Injury p4]
- Pro-insulin C-peptide is a potential renoprotective agent; in diabetics it constricts the afferent arterioles and so prevents hyperfiltration. [Hyperfiltration Injury p4]
- Rosiglitazone improves endothelial dysfunction, reduces glomerular hyperfiltration, and lowers microalbuminuria and progression to ESRD. [Hyperfiltration Injury p4]
- Obesity should be avoided as a further factor that may cause renal damage (as well as nephrotoxic drugs). [Hyperfiltration Injury p4]

Causes of proteinuria

- The causes of proteinuria can be grouped as pre-renal (haemoglobinuria, myoglobinuria, multiple myeloma), renal (functional or pathological) and post-renal (surgical). [Proteinuria p1]
- Epilepsy is a cause of functional proteinuria, which is mild and transient, along with fever and exercise. [Proteinuria p1]
- Glomerular causes of pathological proteinuria include SLE, IgA nephropathy, Alport syndrome and post-streptococcal glomerulonephritis, as well as diabetes and amyloidosis. The corpus names only diabetes, glomerulonephritis, FSGS and amyloid. [Proteinuria p1]
- Sarcoidosis and Balkan nephropathy are tubular causes of proteinuria. [Proteinuria p1]
- Post-renal (surgical) proteinuria is protein that enters the urine after it reaches the renal pelvis. [Proteinuria p1]
- Urinary post-renal causes are UTI, neoplasm and calculus; extraurinary post-renal causes are genital infections such as prostatitis. [Proteinuria p1]

Evaluation: albumin dipstick

- The albumin dipstick is a qualitative and semi-quantitative test. [Proteinuria p2-3]
- The dipstick works by a change in the colour of the indicator caused by a protein-induced pH change; the darker the green, the higher the protein concentration. [Proteinuria p2-3]
- Alkaline urine causes a false-positive dipstick result for protein. [Proteinuria p2-3]
- The dipstick threshold of 20–30 mg/dL is too high to detect the early phases of kidney disease. [Proteinuria p2-3]

Evaluation: quantitative and specific tests

- The 24-hour urinary protein excretion is the gold-standard test; it quantifies total protein using chemical assay or dye-binding techniques. [Proteinuria p3]
- Protein electrophoresis helps distinguish glomerular from tubular proteinuria. In tubular proteinuria only 10–20% is albumin, and immunoglobulin predominates. [Proteinuria p3]
- Immunoassay identifies specific proteins. [Proteinuria p3]
- Microalbuminuria indicates a higher risk of diabetic nephropathy and CKD. Besides 24-hour urine collection, it can be detected by immunoturbidimetry. [Proteinuria p3]

Algorithm for a positive dipstick

- Proteinuria is classified by timing as transient, persistent or intermittent. Intermittent proteinuria that is not related to posture is quantified like persistent proteinuria, while posture-related proteinuria is orthostatic. [Proteinuria p4]

- In glomerular proteinuria (more than 2 g/24 h, mainly albumin), haematuria with dysmorphic red cells and red cell casts calls for a glomerular haematuria work-up. With no or little haematuria, look for specific diseases such as diabetes and amyloidosis. [Proteinuria p4]
- When proteinuria is 300–2000 mg/24 h and mainly globulin, normal proteins indicate tubular proteinuria (evaluate for the cause). Abnormal proteins indicate overflow proteinuria, which calls for evaluation of the specific protein. [Proteinuria p4]

Renal Artery Stenosis: Manifestations and Causes

- RAS may be asymptomatic and found incidentally. [Renal Artery Stenosis p1]
- Accelerated cardiovascular and cerebrovascular disease is listed as one of the four manifestations of RAS. [Renal Artery Stenosis p1]
- Renovascular hypertension may arise with 50% stenosis when post-stenotic dilatation is present, as well as with 70% stenosis of one or both arteries. [Renal Artery Stenosis p1]
- Arterial thrombosis is a cause of renovascular hypertension and ischaemic nephropathy. [Renal Artery Stenosis p1]
- Arteriovenous malformation is a cause of renovascular hypertension and ischaemic nephropathy. [Renal Artery Stenosis p1]
- Endovascular or iatrogenic injury, for example after aortic aneurysm repair, is a cause of renovascular hypertension and ischaemic nephropathy. [Renal Artery Stenosis p1]

Pathology: Fibromuscular Dysplasia

- FMD is usually bilateral. [Renal Artery Stenosis p2]
- FMD is a non-atherosclerotic, non-inflammatory vascular disease that causes abnormal growth within the arterial wall, most commonly affecting the renal and carotid arteries. [Renal Artery Stenosis p2]
- FMD has four histological types: intimal fibroplasia, medial fibroplasia, perimedial fibroplasia and medial hyperplasia. [Renal Artery Stenosis p2]
- Intimal fibroplasia accounts for 10% of FMD, affects children and younger adults, and shows collagen deposition within the intima. [Renal Artery Stenosis p2]
- Intimal fibroplasia may be complicated by dissection or arterial wall haematoma, usually involves the proximal renal artery, and progresses with loss of renal function if not treated. [Renal Artery Stenosis p2]
- Medial fibroplasia accounts for 70-80% of FMD and affects women aged 25-50 years. [Renal Artery Stenosis p2]
- In medial fibroplasia, fibrotic tissue in the stenotic segments alternates with thinned-out media in the aneurysmal segments. [Renal Artery Stenosis p2]
- Perimedial fibroplasia accounts for 10-20% of FMD, affects younger women (5-15 years) and involves the mid renal artery. [Renal Artery Stenosis p2]
- Perimedial fibroplasia shows widespread collagen deposition over the outer half of the media, also gives a string-of-beads appearance, and progresses with loss of renal function unless treated. [Renal Artery Stenosis p2]
- Medial hyperplasia is a rare type of FMD with no fibrosis. [Renal Artery Stenosis p2]

Pathology: Atherosclerotic RAS

- Atherosclerotic RAS is bilateral in 20-40% of cases. [Renal Artery Stenosis p3]
- 30% of atherosclerotic RAS is non-ostial, usually 1-3 cm distal to the ostium. [Renal Artery Stenosis p3]

Pathogenesis of Ischaemic Nephropathy

- Ischaemic nephropathy progresses through four phases: cellular activation and injury, fibrogenic signalling, fibrogenic, and destructive. [Renal Artery Stenosis p5]
- Reduced renal blood flow causes secretion of pro-inflammatory mediators and attraction of inflammatory cells, which leads to fibrosis. [Renal Artery Stenosis p5]
- Angiotensin II is a profibrogenic peptide as well as a powerful vasoconstrictor, and acts through NF-kB and TGF-beta. [Renal Artery Stenosis p5]

- NF-kB drives fibroblast proliferation and differentiation, and makes tubular cells secrete profibrotic cytokines (producing matrix) and chemoattractant adhesion proteins (producing inflammation); all of these converge on fibrosis. [Renal Artery Stenosis p5]

Clinical Features

- Severe or refractory hypertension with grade III-IV hypertensive retinopathy should raise suspicion of renovascular disease. [Renal Artery Stenosis p6]
- Paradoxical worsening of hypertension with diuretic use should raise suspicion of renovascular disease. [Renal Artery Stenosis p6]
- The abdominal bruit of renovascular disease is systolic-diastolic and radiates to both flanks. [Renal Artery Stenosis p6]
- Diffuse vascular disease and/or evidence of cholesterol embolisation should raise suspicion of renovascular disease. [Renal Artery Stenosis p6]

Investigations

- Intra-arterial DSA (digital subtraction angiography), as well as conventional renal angiography, is a gold standard for diagnosis. [Renal Artery Stenosis p7]
- MRA involves no radiation and can also assess renal blood flow and GFR. [Renal Artery Stenosis p7]
- MRA has a sensitivity of 100% and a specificity of 70-95%. [Renal Artery Stenosis p7]
- MRA may miss lesions in distal, intrarenal and accessory arteries. [Renal Artery Stenosis p7]
- Gadolinium MRA carries a risk of nephrogenic systemic fibrosis (NSF) when GFR is below 30 mL/min. [Renal Artery Stenosis p7]
- Spiral CT angiography has a sensitivity of 98% and a specificity of 94%. [Renal Artery Stenosis p7]
- The sensitivity and specificity of spiral CT decline when serum creatinine is above 1.7 mg/dL. [Renal Artery Stenosis p7]
- Duplex Doppler is accurate in patients with renal failure. [Renal Artery Stenosis p7]
- Duplex Doppler is time-consuming, and intravascular or even main-artery lesions may be poorly seen in obesity or with bowel gas. [Renal Artery Stenosis p7]

Angiography

- Angiography is indicated when clinical suspicion is high but the screening test is negative, because screening tests can miss distal or intrarenal disease. [Renal Artery Stenosis p8]

Medical Therapy

- Medical therapy controls BP in more than 90% of patients, usually needing several antihypertensive drugs. [Renal Artery Stenosis p10]
- During medical therapy, renal size and cortical blood flow should be monitored as well as serum creatinine and GFR or creatinine clearance. [Renal Artery Stenosis p10]
- ACE inhibitors in bilateral RAS cause acute reversible renal failure in 10-20% of patients, particularly when there is volume contraction. [Renal Artery Stenosis p10]

Angioplasty with or without Stenting

- Intolerance of medical therapy is an indication for angioplasty with or without stenting. [Renal Artery Stenosis p11-12]
- Haemodynamically significant RAS with accelerated or malignant hypertension is an indication for angioplasty with or without stenting. [Renal Artery Stenosis p11-12]
- The procedural success rate of renal angioplasty is 85-100%. [Renal Artery Stenosis p11-12]
- Hypertension is cured or improved in about 90% of cases after angioplasty. [Renal Artery Stenosis p11-12]
- Restenosis after PTRA alone is 30% for non-ostial lesions and 50% for ostial lesions, and 15-25% after PTRA with a stent. [Renal Artery Stenosis p11-12]
- After angioplasty, renal function improves in 40% and stabilises in another 40%. [Renal Artery Stenosis p11-12]
- The STAR trial, along with ASTRAL and CORAL, showed no significant difference in hypertension between medically treated patients and those treated with angioplasty. [Renal Artery Stenosis p11-12]

- The complication rate of renal angioplasty is 5-15%; major complications occur in less than 5%. [Renal Artery Stenosis p11-12]
- Renal artery spasm or renal infarction is a minor complication of angioplasty, reversed with intra-arterial nitroglycerin. [Renal Artery Stenosis p11-12]
- Malpositioning or dislocation of the stent is a complication of renal stenting. [Renal Artery Stenosis p11-12]
- Improvement or stabilisation of renal function after revascularisation is most likely when a pyelogram is seen on IVP, kidney length is more than 9 cm, collaterals fill the distal vasculature beyond a total occlusion on angiography, or renal biopsy shows viable glomeruli with minimal arteriolar glomerulosclerosis. [Renal Artery Stenosis p11-12]

Surgery

- Surgery is indicated for malignant or accelerated hypertension that does not respond to, or is not tolerated on, medical therapy when angioplasty is technically impossible. [Renal Artery Stenosis p13]
- Iliorenal bypass is an option for renal artery reconstruction when the aorta is diseased. [Renal Artery Stenosis p13]
- The supraceliac or lower thoracic aorta can be used as inflow for bypass when the infrarenal aorta is diseased, because it is less involved by atherosclerosis. [Renal Artery Stenosis p13]
- Ablative surgery for RAS can be done by laparoscopic or open nephrectomy. [Renal Artery Stenosis p13]

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Lap donor nephrectomy: advantages

- Laparoscopic donor nephrectomy does not increase complications compared with open donor nephrectomy. [LDN p1]
- Laparoscopic donor nephrectomy reduces wound-related problems and incisional hernias. [LDN p1]

Lap donor nephrectomy: risks and complications

- Non-life-threatening postoperative complications of laparoscopic donor nephrectomy include ileus, port-site infection, deep vein thrombosis and lymphatic leak. [LDN p1]
- Intraoperative complications of laparoscopic donor nephrectomy include bleeding, visceral injury and ureteric stripping or a short ureter. [LDN p1]
- The life-threatening complications that cause donor death are haemorrhage and pulmonary embolism. [LDN p1]

Complications specific to donor nephrectomy

- To protect the remaining kidney from CKD, the donor should avoid nephrotoxic drugs after donation. [LDN p2]
- Hypotension should be avoided during donor nephrectomy to prevent graft dysfunction. [LDN p2]

Disadvantages of LDN

- Bowel injury is a specific risk of laparoscopic donor nephrectomy. [LDN p3]
- Internal hernia is a risk of laparoscopic donor nephrectomy. [LDN p3]

Hand-assisted LDN

- Hand-assisted laparoscopic donor nephrectomy needs less traction on the kidney. [LDN p3]
- Hand-assisted laparoscopic donor nephrectomy allows faster kidney removal and less warm ischaemia. [LDN p3]
- Hand assistance shortens the learning curve for laparoscopic donor nephrectomy. [LDN p3]
- Disadvantages of hand-assisted laparoscopic donor nephrectomy are higher cost, an awkward working position for the surgeon, and a higher risk of traumatic injury from manual manipulation. [LDN p3]

Renal Preservation Prior to Transplant

- Preservation allows time for cross-matching, sharing of organs between centres and preparation of the recipient when deceased-donor organs are used. [Transplant p1]

Harmful Effects: Hypothermia and Reperfusion Injury

- Ischaemia and hypothermia activate lysosomal enzymes in the renal cell. [Transplant p2]
- Ischaemia and hypothermia cause loss of cellular potassium and magnesium, which (with the rise in cellular calcium) leads to ATN and cell swelling. [Transplant p2]

Renal Preservation Solutions

- The original Collins solution is listed as a preservation solution in its own right, separate from Euro-Collins. [Transplant p3]

UW Solution: Contents and Disadvantages

- Dexamethasone in UW solution acts, together with MgSO4, as a membrane-stabilising agent. [Transplant p4]
- KH2PO4 in UW solution serves as a source of potassium. [Transplant p4]
- A disadvantage of UW solution is that particles become caught in the capillary bed and prevent reperfusion. [Transplant p4]

Methods of Preservation

- In cold storage the kidney is kept in a bag so that the crushed ice is not in direct contact with the kidney surface. [Transplant p5]
- Machine-driven pulsatile perfusion maintains kidney viability for up to 72 hours. [Transplant p5]

Rejection (Hyperacute, Acute, Chronic)

- Acute rejection can occur in all transplants unless immunosuppression is achieved, the exception being transplants between (identical) twins. [Transplant p6]

Acute Rejection: Presentation and Investigations

- Urinalysis in acute rejection may show proteinuria, pyuria or haematuria. [Transplant p7]
- Imaging in acute rejection may show decreased cortical blood flow and decreased tubular function. [Transplant p7]
- An adequate transplant biopsy contains at least 10 glomeruli and 2 small arteries, and is stained with H&E, PAS or silver, and trichrome. [Transplant p7]
- Two separate biopsy cores should be obtained. [Transplant p7]

Pathogenesis of Rejection

- Antibody produced by plasma cells damages the graft by antibody-dependent cell-mediated cytotoxicity as well as by complement-mediated damage. [Transplant p8]
- Cytokines from CD4+ T-helper cells activate macrophages and NK cells, producing a delayed-type hypersensitivity reaction. [Transplant p8]

Treatment of Cell-Mediated Rejection

- Methylprednisolone for cell-mediated rejection is given at 3–5 mg/kg (250–500 mg/day) for 3–5 days. [Transplant p10]
- Muromonab (OKT3) is given at 5 mg IV for 7–14 days. [Transplant p10]
- ATG is given at 1–1.5 mg/kg IV. [Transplant p10]
- Anti-T-cell antibodies are more effective than corticosteroids in restoring kidney function and preventing graft loss, but have more adverse effects. [Transplant p10]
- Alemtuzumab (anti-CD52) is given as 30–60 mg on day 0 and day 2. [Transplant p10]

Treatment of Antibody-Mediated Rejection

- Plasma exchange is given as 2–5 treatments, daily or on alternate days, combined with IVIG. [Transplant p11]
- IVIG is given at 100–200 mg/kg after plasma exchange. [Transplant p11]
- Rituximab is given at 375 mg/m2 with plasma exchange and IVIG. [Transplant p11]
- Bortezomib is given at 1.3 mg/m2 for 4 doses over 1–2 weeks, usually combined with plasma exchange and IVIG. [Transplant p11]

- Eculizumab, used for prevention in cross-match-positive transplant, is given at 600–1200 mg weekly for 4 doses, then every two weeks until successful antibody reduction. [Transplant p11]

Difference Between ATN and Acute Rejection

- Post-transplant ATN commonly occurs within the first week after transplant. [Transplant p12]
- Post-transplant ATN is commoner with a cadaveric donor (25%) than with a live donor (5%). [Transplant p12]
- Delayed graft function from ATN usually recovers within 10–14 days; treatment is conservative, and dialysis may be required. [Transplant p12]
- On Doppler ultrasound the resistive index is about 0.9 in ATN and about 1.0 in acute rejection. [Transplant p12]
- Hypertension is a presenting feature of acute rejection. [Transplant p12]

Banff Classification of Rejection

- Banff T-cell-mediated rejection includes a chronic category (chronic T-cell-mediated rejection) besides acute and borderline. [Transplant p13]

Immunosuppressive Therapy: Classification

- Muromonab (OKT3) is a monoclonal anti-CD3 antibody. [Transplant p14]
- Daclizumab, like basiliximab, is a monoclonal anti-CD25 (IL-2 receptor) antibody. [Transplant p14]

Immunosuppressive drugs: others

- Leflunomide is listed as an immunosuppressant that inhibits pyrimidine synthesis. [Transplant p15]

Corticosteroids

- Corticosteroids act on T cells, B cells and macrophages; their mechanism is complex and not fully understood. [Transplant p15]
- Corticosteroids suppress production of TNF-alpha, chemokines and MHC class II, as well as IL-2. [Transplant p15]
- At higher doses, corticosteroids also have a receptor-independent effect. [Transplant p15]
- Corticosteroids inhibit leukotrienes and prostaglandins and regulate NF-kB. [Transplant p15]
- Mechanism pathway: the steroid binds the cytoplasmic steroid receptor/heat shock protein complex; the heat shock protein is released, and the steroid–receptor complex enters the nucleus, binds the promoter region of target genes, and induces or suppresses cytokine genes. [Transplant p16]
- Renal side effects of corticosteroids are hypokalaemia and diuretic resistance. [Transplant p16]
- Gastrointestinal side effects of corticosteroids are peptic ulcer, pancreatitis and colonic perforation. [Transplant p16]
- Haematological side effects of corticosteroids are leucocytosis and polycythaemia. [Transplant p16]
- Neuromuscular and psychiatric side effects of corticosteroids are myopathy, psychosis and insomnia. [Transplant p16]
- Cosmetic side effects of corticosteroids are cushingoid features, hirsutism, acne and buffalo hump. [Transplant p16]
- In pregnancy, corticosteroids can cause fetal adrenal suppression. [Transplant p16]

Calcineurin inhibitors

- Calcineurin inhibitors increase TGF-beta, which adds to immunosuppression but also causes tissue fibrosis. [Transplant p17]
- Calcineurin inhibitors cause hypophosphataemia. [Transplant p18]
- Calcineurin inhibitors cause type 4 renal tubular acidosis. [Transplant p18]
- Myocardial hypertrophy is more common with tacrolimus than with ciclosporin. [Transplant p18]
- Coarse facial features and oedema are more common with ciclosporin than with tacrolimus. [Transplant p18]

Mycophenolate mofetil

- Mycophenolate causes diarrhoea by inhibiting gastrointestinal epithelial cell replication, which disrupts fluid absorption. [Transplant p19]
- If pregnancy is desired, wait 1 year after the graft has stabilised and switch mycophenolate to azathioprine. [Transplant p19]

Azathioprine

- Azathioprine is converted to 6-mercaptopurine, which inhibits purine synthesis. [Transplant p20]
- Azathioprine inhibits T-cell proliferation much more than B-cell proliferation, so it mainly inhibits cell-mediated immunity. [Transplant p20]
- Azathioprine side effects include hepatotoxicity and alopecia. [Transplant p20]

mTOR inhibitors

- mTOR inhibitors cause leucopenia, thrombocytopenia and diarrhoea. [Transplant p22]
- Hyperlipidaemia is more common with sirolimus than with everolimus. [Transplant p22]

Biological agents: antithymocyte globulin (ATG)

- ATG is available in two forms, equine ATG and rabbit ATG. [Transplant p22]
- ATG is also used to treat aplastic anaemia. [Transplant p22]
- ATG binds multiple epitopes on the T-cell surface. It causes rapid lymphocytopenia through phagocytosis, apoptosis and complement-dependent cytolysis. [Transplant p22]
- ATG is a potent immunosuppressant that depresses T and B cells for up to 24 hours after administration. [Transplant p23]
- ATG side effects include thrombocytopenia, a serum sickness-like syndrome and, rarely, ARDS. [Transplant p23]

Rituximab

- Rituximab depletes B cells through antibody-dependent cell-mediated cytotoxicity, complement-mediated lysis and apoptosis. [Transplant p23]
- Rituximab infusion reactions (rash, itching, low blood pressure, shortness of breath) occur within 2 hours of injection. Headache, fever, chills and dizziness also occur. [Transplant p24]
- Rituximab can reactivate hepatitis B. [Transplant p24]
- Rituximab can cause progressive multifocal leukoencephalopathy. [Transplant p24]
- Rituximab can cause toxic epidermal necrolysis. [Transplant p24]

Intravenous immunoglobulin (IVIG)

- The IVIG mechanism is not fully understood. It inhibits antibody production, increases catabolism of circulating antibodies, inhibits cytokines and inhibits complement-mediated injury. [Transplant p24]
- IVIG side effects are thrombotic events, flushing, chills, headache, myalgia, anaphylactic reaction and meningitis. [Transplant p25]

Drugs used for induction

- The induction dose of methylprednisolone is 500 mg IV intraoperatively, followed by a taper over 1–5 days. [Transplant p26]
- The induction dose of rATG is 1–1.5 mg/kg IV for 4–14 days. [Transplant p26]
- OKT3 is an induction agent, given as 5 mg IV for 7–14 days. [Transplant p26]
- Daclizumab, like basiliximab, is an anti-CD25 induction agent. [Transplant p26]

Immunosuppressive regimens

- Regimens are classified by the number of drugs. Monotherapy is ciclosporin, tacrolimus or an mTOR inhibitor alone. Dual therapy is one of these plus a steroid, azathioprine or MMF. Triple therapy is steroid plus an antiproliferative plus ciclosporin, tacrolimus or an mTOR inhibitor. Quadruple therapy is ATG induction plus triple immunosuppression. [Transplant p27]

ABO-Incompatible Transplantation: Problems and Solutions

- Approximate recipient blood-type frequencies are A 45%, O 40%, B 10% and AB 5%. [Transplant p28]
- Approximately 35% of medically suitable donor-recipient pairs are ABO incompatible. [Transplant p28]
- A problem of ABO-incompatible transplantation is a significantly higher rate of infection. [Transplant p28]
- A problem of ABO-incompatible transplantation is adverse events related to antibody depletion therapy. [Transplant p28]
- ABO-incompatible transplantation requires intensified immunosuppression. [Transplant p28]
- ABO-incompatible transplantation carries a high cost. [Transplant p28]
- A further problem is lack of experience with ABO-incompatible transplantation in many centres of developing countries. [Transplant p29]
- Because there are no RCTs comparing the different desensitisation protocols, no single ideal protocol can be selected for all purposes. [Transplant p29]
- Anti-A/anti-B immunoglobulin titres are measured by the tube technique, the gel technique or flow cytometry. [Transplant p29]
- A high pre-operative anti-A/B antibody titre is associated with poor graft survival. [Transplant p29]
- High post-operative anti-A/B titres are associated with an increased risk of antibody-mediated rejection. [Transplant p29]
- Monitoring in ABO-incompatible transplantation covers the anti-ABO IgG titre and the B-cell count. [Transplant p29]
- The rituximab dose for B-cell depletion is 375 mg/m2. [Transplant p30]
- In plasmapheresis, the removed plasma is replaced by human albumin, colloid solution and/or fresh frozen plasma. [Transplant p30]
- Disadvantages of plasmapheresis are removal of coagulation factors and loss of antiviral and antibacterial antibodies, which increases infections. [Transplant p30]
- Double filtration plasmapheresis selectively removes the desired immunoglobulin from plasma and needs less fluid substitution than plasmapheresis. [Transplant p30]
- Immunoadsorption is a safe and more effective selective removal of anti-A or anti-B antibodies, and volume replacement is not necessary. [Transplant p31]
- IVIG acts by complement downregulation, interaction with Fc receptors, inhibition of B- and T-cell proliferation (reducing antibody production), inhibition of CD8 T-cell toxicity and increased B-cell apoptosis. [Transplant p31]
- High-dose IVIG can cause haemolysis and an increased risk of antibody-mediated rejection in ABO-incompatible transplantation. [Transplant p31]
- Serious side effects of IVIG are neutropenia, thrombotic events and neurotoxicity. [Transplant p31]
- Johns Hopkins ABO-incompatible protocol (the author marked it "please check yourself"): plasma exchange and rituximab (500 mg) daily or on alternate days starting 2 weeks before transplant, tacrolimus and mycophenolate mofetil from 1 week before, and ATG with methylprednisolone on the day before transplant. [Transplant p32]

Swap Transplant

- In a domino paired exchange, a non-directed (altruistic, Good Samaritan) donor gives a kidney to the first recipient, whose incompatible donor then gives to the next recipient, completing the exchange. [Transplant p33]
- An ongoing donor chain started by a non-directed donor is either continued or ended by offering the last living donor's kidney to a patient on the deceased-donor waiting list. [Transplant p34]
- ABO-incompatible kidney transplantation reduces the deceased-donor waiting list and waiting time. [Transplant p35]

Post Transplant Complications: Infections

- Any live vaccine should be given at least 4 weeks before transplant. [Transplant p36]
- Vaccines to give before transplant include HPV, hepatitis B and pneumococcus. [Transplant p36]

- Live vaccines barred after transplant include polio and MMR. [Transplant p36]
- Vaccines safe after transplant include hepatitis A and B, diphtheria and tetanus. [Transplant p36]
- Bacterial infections in the first month after transplant are UTI, wound infection, aspiration pneumonia, intra-abdominal infections and catheter and drain-site infections. [Transplant p37]
- Multidrug-resistant bacteria are an important cause of mortality and morbidity in the first month after transplant. [Transplant p37]
- The main viral infection in the first month after transplant is HSV; other viruses are uncommon then. [Transplant p37]
- Fungal infection in the first month after transplant is Candida, including azole-resistant Candida. [Transplant p37]
- From 1 to 6 months after transplant, opportunistic viral infections (secondary to immunosuppression) include CMV, HHV-6, HSV, VZV, EBV, HBV, HCV and BK virus. [Transplant p37]
- Opportunistic fungal infections from 1 to 6 months include Aspergillus, Cryptococcus and mucormycosis. [Transplant p37]
- Opportunistic bacterial infections from 1 to 6 months include Nocardia, TB and Legionella. [Transplant p37]
- Opportunistic parasitic infections from 1 to 6 months include Pneumocystis jirovecii, Toxoplasma and Leishmaniasis. [Transplant p37]
- After 6 months, recipients with good graft function, no chronic viral infection and low-dose immunosuppression mainly get community-acquired infections. [Transplant p38]
- After 6 months, recipients with chronic viral infection develop progressive liver disease (HBV/HCV), BK nephropathy or PTLD (EBV). [Transplant p38]
- After 6 months, recipients with multiple episodes of rejection get opportunistic bacterial, viral and fungal infections. [Transplant p38]
- UTI prophylaxis after kidney transplant is TMP-SMX for 3–6 months, or nitrofurantoin. [Transplant p38]
- Pneumocystis pneumonia prophylaxis is TMP-SMX, or inhaled pentamidine. [Transplant p38]
- Oral candidiasis prophylaxis is nystatin suspension for 1–3 months, or clotrimazole lozenges (written as "cotrimoxazole"; clotrimazole is the likely intended drug). [Transplant p38]
- Vaginal candidiasis prophylaxis is intravaginal nystatin or clotrimazole (written as "cotrimoxazole"). [Transplant p38]
- HSV/CMV prophylaxis is acyclovir or valacyclovir for 3 months. [Transplant p38]
- Peptic ulcer prophylaxis after transplant is an H2-receptor blocker, plus a second agent if symptomatic. [Transplant p39]
- CMV disease occurs mainly 1–6 months after transplant, as primary infection (recipient negative, donor positive), reactivation of latent virus (recipient positive, donor positive or negative) or superinfection with a new virus in a seropositive recipient (both positive). [Transplant p39]
- CMV can affect any organ (for example hepatitis and retinitis), lowers the platelet count and total leucocyte count, and increases the risk of opportunistic infection and PTLD. [Transplant p39]
- CMV prevention is with acyclovir or valganciclovir. [Transplant p39]
- CMV is diagnosed by a CMV PCR assay. [Transplant p39]
- CMV disease is treated with IV ganciclovir. [Transplant p39]
- Ganciclovir-resistant CMV is treated with cidofovir or foscarnet. [Transplant p39]

OTHER TOPICS IN THIS BOOK

Urinalysis: proteinuria

- Dipstick protein grades are trace 5–20 mg/dL, 1+ 30 mg/dL, 2+ 100 mg/dL, 3+ 300 mg/dL and 4+ >2000 mg/dL. [CUE Campbell p4]
- On the tetrabromophenol blue dipstick, protein is seen as shades of green. [CUE Campbell p4]
- Alkaline urine is listed as a cause of a false-negative dipstick protein result. [CUE Campbell p4]
- Precipitation with 3% sulfosalicylic acid detects protein from 15 mg/dL. [CUE Campbell p4]
- Orthostatic proteinuria is attributed to pressure on the renal vein while standing. [CUE Campbell p4]

Nanobacteria in Other Urological Diseases

- In ADPKD, nanobacteria are proposed to target the tubule cells that are the primary source of cysts. [Nano T B p9]
- Higher levels of nanobacterial antigens have been detected in the urine of ADPKD patients. [Nano T B p9]
- Prostatic calcifications in chronic prostatitis/CPPS are typically calcium apatite, said to be the hallmark of nanobacterial action. [Nano T B p9]
- Higher nanobacterial antigen levels have been found in the serum of CPPS patients using ELISA. [Nano T B p9]

Acid-Base Disorders: regulation

- Acid-base balance in the body is maintained by three systems: the kidney, the lungs and the buffer systems. [Acid Base RTA p1]
- Distal H+ secretion alone would lower urine pH below 2.5, which would be toxic to renal tubular cells, so the secreted acid is buffered with ammonium in the collecting duct cells. [Acid Base RTA p1]
- The lungs exhale CO2, which is acidic; a rise in blood CO2 drives hyperventilation, which removes CO2. [Acid Base RTA p1]
- Buffer systems are the body's own weak acids and weak bases, which guard against sudden shifts in pH. [Acid Base RTA p1]
- The most important buffer system in blood is carbonic acid and bicarbonate ions. [Acid Base RTA p1]

Normal values and acid-base disorders

- Normal arterial values are pH 7.35-7.45, pO2 80-100 mmHg, pCO2 35-45 mmHg and HCO3 22-28 mEq/L. [Acid Base RTA p2]
- The acid-base disorders are metabolic acidosis, renal tubular acidosis, metabolic alkalosis, respiratory acidosis and respiratory alkalosis. [Acid Base RTA p2]
- In the proximal tubule, luminal H+ (secreted in exchange for Na+) combines with filtered HCO3 to form H2CO3, which carbonic anhydrase converts to CO2 and water; these enter the cell, where carbonic anhydrase regenerates H+ and HCO3, and the HCO3 passes to the blood. [Acid Base RTA p2]
- In metabolic acidosis the primary change is a fall in HCO3 with a compensatory fall in pCO2, and in metabolic alkalosis a rise in HCO3 with a compensatory rise in pCO2. [Acid Base RTA p2]
- In respiratory acidosis the primary change is a rise in pCO2 with a compensatory rise in HCO3, and in respiratory alkalosis a fall in pCO2 with a compensatory fall in HCO3. [Acid Base RTA p2]

Metabolic Acidosis

- Lactic acidosis and diabetic ketoacidosis are causes of increased-anion-gap metabolic acidosis from an increased acid load. [Acid Base RTA p3]
- Ethylene glycol, methanol and salicylate intoxication are causes of increased-anion-gap metabolic acidosis. [Acid Base RTA p3]
- The sodium bicarbonate dose for metabolic acidosis is 0.4 x body weight x bicarbonate deficit. [Acid Base RTA p3]
- Alkali (sodium bicarbonate) treatment is given when pH is below 7.2 or HCO3 is below 10. [Acid Base RTA p3]

Metabolic Alkalosis

- Metabolic alkalosis is defined by HCO3 above 26 mEq/L. [Acid Base RTA p4]
- Causes of metabolic alkalosis include excessive vomiting, certain diuretics and severe dehydration. [Acid Base RTA p4]
- Causes of metabolic alkalosis include excessive use of alkali drugs, heavy ingestion of antacids and endocrine disorders. [Acid Base RTA p4]
- The mainstay of treatment of metabolic alkalosis is volume replacement, electrolyte replacement and treatment of the underlying cause. [Acid Base RTA p4]
- In severe metabolic alkalosis, dilute acid in the form of intravenous ammonium chloride solution is occasionally given. [Acid Base RTA p4]

16 BPH

OTHER TOPICS IN THIS BOOK

Lasers in BPH: advantages, disadvantages and indications

- High cost is a disadvantage of the holmium laser. [LASER p2]
- The thulium laser gives a smoother incision than holmium. [LASER p2]
- The thulium laser causes less tissue scarring because there is less scattering of thermal energy. [LASER p2]
- The (continuous-wave) thulium laser is still investigational for stone disease. [LASER p2]
- KTP/LBO photoselective vaporisation has a simple (short) learning curve. [LASER p2]

Lasers in BPH: chromophore, emission and historical techniques

- The KTP/LBO laser can deliver energy in continuous and pulsed modes. [LASER p3]
- The Nd:YAG laser can be continuous or pulsed (Q-switched). [LASER p3]
- Historical Nd:YAG prostate techniques were laser ablation and interstitial laser coagulation. [LASER p3]
- The holmium fibre can be end-firing or side-firing and is flexible. [LASER p3]
- Thulium laser generators are 70 W or 120 W, and thulium uses rigid 550 µm or 880 µm fibres. [LASER p3]
- KTP and LBO use rigid side-firing fibres. [LASER p3]

Green light laser and mode of action

- In the green light generator, a laser diode pumps the Nd:YAG through a coupling lens, and the beam passes an input monitor, the Nd:YAG rod, the KTP crystal and an output monitor before emerging as green light. [LASER p7]
- Photoselective vaporisation creates a cavity in the prostate. [LASER p7]

PVP technique

- The PVP fibre is swept from inside out in a painting motion. [LASER p8]
- If the PVP fibre is too close, contact vaporisation occurs and damages the fibre. [LASER p8]
- If the PVP fibre is too far away, vaporisation is inefficient and more coagulation occurs. [LASER p8]
- Absence of large bubbles during PVP means more coagulation, a longer operation and more postoperative dysuria and LUTS. [LASER p8]

Irrigation Fluids in TURP

- The uses of irrigation fluid in TURP are better visualisation, distension of the bladder and prostatic urethra, and flushing out of blood and tissue debris. [TUR Fluids p1]
- Further properties of an ideal irrigation fluid are that it should be non-metabolisable, easy to sterilise, rapidly excretable and non-combustible. [TUR Fluids p1]
- Glycine is also available as 1.2% and 2.2% solutions besides the standard 1.5%. [TUR Fluids p1]
- Distilled water irrigation causes haemoglobinuria (in addition to haemolysis). [TUR Fluids p1]
- Ringer's lactate is an iso-osmolar irrigant with less risk of TUR syndrome, but it is ionised and so cannot be used for monopolar resection. [TUR Fluids p1]
- Mannitol 5% irrigant has the disadvantages of high cost and acute intravascular volume expansion (in addition to osmotic diuresis). [TUR Fluids p2]
- Sorbitol irrigant can also cause osmotic diuresis (besides lactic acidosis and hyperglycaemia). [TUR Fluids p2]
- The sorbitol plus mannitol mixture is marketed as Cytal, and its main disadvantage is expense. [TUR Fluids p2]

Prostatic Urethral Lift (UroLift)

- UroLift is an option for patients with multiple comorbidities who are not fit for regional or general anaesthesia (Book 16 lists only long-term catheter or prostatic artery embolisation for unfit men). [urolift p1]
- UroLift has a short learning curve. [urolift p1]
- Inclusion criterion: IPSS 12 or more. [urolift p2]
- Inclusion criterion: Qmax 15 mL/s or less. [urolift p2]
- Inclusion criterion: post-void residual less than 350 mL. [urolift p2]
- UroLift is not suitable for a high bladder neck (in addition to a large median lobe). [urolift p2]
- Cystoscopy is done with a 2.9 mm 0 degree telescope through a 20 Fr sheath. [urolift p2]
- The cystoscope bridge is replaced with the UroLift system headpiece, and the telescope is passed through the headpiece. [urolift p2]
- An inappropriately placed implant can be removed with a grasper or biopsy forceps. [urolift p3]

17 Prostate Cancer

CH 153 · EPIDEMIOLOGY, AETIOLOGY AND PREVENTION OF PROSTATE CANCER

Epidemiology

- Lung cancer is the commonest cancer worldwide, and prostate cancer ranks second. [CaP Etio Preven p1]
- Prostate cancer is the most common non-cutaneous malignancy in US men. [CaP Etio Preven p1]
- From 1989 to 1992, incidence rose sharply because of PSA screening. From 1992 to 1995 it fell steeply, which is called the "cull effect". [CaP Etio Preven p1]
- Jamaican men of African descent share the highest risk with African Americans. [CaP Etio Preven p1]
- Risk follows a gradient: White Americans have the highest, then men of Asian descent living in the US, then similar men living in Asia. [CaP Etio Preven p1]
- Only 2% of all prostate cancers are diagnosed in men younger than 50 years. [CaP Etio Preven p1]
- The average age at death from prostate cancer is 77 years. [CaP Etio Preven p1]
- Prostate cancer is the 6th leading cause of cancer deaths worldwide. [Compilation 02 p44]
- About 15% of all prostate cancer is caused by germline mutation. [Compilation 02 p44]

Stage at diagnosis; genetic risk factors

- Pathological stage at diagnosis is migrating downward, partly because of PSA screening, and about 80% of newly diagnosed men have localised disease. [CaP Etio Preven p2]
- Prostate cancer susceptibility mutations have been described on chromosomes 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 17, 19, 22 and X. [CaP Etio Preven p2]
- An affected second-degree relative gives a relative risk of 1.68. [CaP Etio Preven p2]
- HOXB13 is located on chromosome 17. [CaP Etio Preven p2]

Inflammation and infection; metabolic syndrome

- Infection, oestrogens and cooked meat are proposed triggers of epithelial damage. The damage leads to inflammatory infiltrates and proliferative inflammatory atrophy, then oxidant stress, then DNA damage, and finally cancer. [CaP Etio Preven p3]
- Propionibacterium acnes is the predominant bacterial species in both benign and malignant prostatic tissue and is associated with prostatic inflammation. [CaP Etio Preven p3]
- No infectious agent has been proven to cause prostate cancer. [CaP Etio Preven p3]
- Among the single components of the metabolic syndrome, increased waist circumference (as well as hypertension) is associated with a significantly greater risk of prostate cancer. [CaP Etio Preven p3]

Drugs, obesity and diet

- Toremifene citrate, a breast cancer drug, did not reduce prostate cancer in studies. [CaP Etio Preven p4]

- Statin use has no association with overall prostate cancer incidence but a protective association with advanced prostate cancer, which suggests an effect in the late stages of carcinogenesis (repeated on p10). [CaP Etio Preven p4]
- Obesity is associated with higher rates of biochemical failure. [CaP Etio Preven p4]
- Fried foods increase the risk of prostate cancer. [CaP Etio Preven p4]
- Phytoestrogen intake is associated with a decreased risk of prostate cancer (repeated on p5). [CaP Etio Preven p4]
- Dietary fat intake shows no relationship with prostate cancer risk. [CaP Etio Preven p4]

Hormonally active medication, smoking

- Hypogonadal men who receive testosterone do not have an increased risk of prostate cancer. [CaP Etio Preven p5]
- Oestrogens have direct and indirect effects on prostatic growth and development and probably play a role in the initiation and progression of prostate cancer. [CaP Etio Preven p5]
- Proposed mechanisms by which smoking raises prostate cancer risk are cadmium exposure, raised circulating androgen levels and cellular oxidative stress. [CaP Etio Preven p5]
- Current smokers have a higher risk of biochemical failure and metastasis, as well as of prostate cancer-specific mortality. [CaP Etio Preven p5]

Vitamin D, ejaculation, STIs and other factors

- Low vitamin D levels are associated with a higher risk of prostate cancer. [CaP Etio Preven p6]
- Frequent ejaculation has a protective effect against prostate cancer. [CaP Etio Preven p6]
- No single sexually transmitted agent is directly implicated. Infections may raise risk through inflammation. [CaP Etio Preven p6]
- Night-shift workers have a higher risk of prostate cancer. [CaP Etio Preven p6]
- Circumcision is protective against prostate cancer. [CaP Etio Preven p6]

Chemoprevention: definition, goal and levels

- Chemoprevention is the use of natural or synthetic agents that reverse, suppress or prevent the carcinogenic process, so that clinically evident cancer does not develop. [CaP Etio Preven p7]
- The goal of chemoprevention is to lower the incidence of a given cancer while also reducing treatment-related side effects and morbidity. [CaP Etio Preven p7]
- Primary prevention targets the general population of healthy individuals at risk of prostate cancer. [CaP Etio Preven p7]
- Secondary prevention targets individuals with a premalignant lesion such as HGPIN, to prevent progression to frank cancer. [CaP Etio Preven p7]
- Tertiary prevention aims to prevent a second primary cancer in individuals who already have one. [CaP Etio Preven p7]
- The clinical rationale for chemoprevention is that the established risk factors (age, ethnicity and family history) cannot be modified. [CaP Etio Preven p7]

Biologic rationale, challenges; PCPT

- The biologic rationale is that premalignant changes appear as long as 20–30 years before carcinoma. This gives time to intervene with lifestyle change (diet, smoking cessation, exercise) or with chemoprevention. [CaP Etio Preven p8]
- The challenge is twofold: to find an effective intervention with acceptable toxicity, and to identify a population at high enough risk for chemoprevention to be appropriate and cost-effective. [CaP Etio Preven p8]
- PCPT was the first large-scale chemoprevention trial for prostate cancer. [CaP Etio Preven p8]
- PCPT enrolled men aged 55 years or older with a normal DRE (in addition to PSA ≤3 ng/ml). [CaP Etio Preven p8]

REDUCE trial

- REDUCE required a negative prior biopsy (6–12 cores) within the previous 6 months. [CaP Etio Preven p9]

- REDUCE also required a prostate volume of 80 ml or less. [CaP Etio Preven p9]
- In REDUCE, the excess of Grade Group 8–10 cancers with dutasteride appeared during years 3 and 4 of the study. [CaP Etio Preven p9]

FDA reassessment; toremifene; SELECT

- In its reassessment, the FDA estimated that for every 150–200 men treated with a 5ARI, one additional man would be diagnosed with a high-grade cancer in order to avert 3–4 low-grade cancers. [CaP Etio Preven p10]
- Toremifene citrate is an oestrogen receptor modulator, FDA-approved for breast cancer, and its RCT failed to show a reduction in prostate cancer. [CaP Etio Preven p10]
- SELECT was planned to run for 12 years. [CaP Etio Preven p10]

Green tea catechins; prevention framework

- Green tea catechins induce apoptosis and inhibit prostate cancer in vitro. [CaP Etio Preven p11]
- In a small RCT of 60 men with HGPIN, green tea catechins were associated with a lower risk of prostate cancer than placebo, and confirmatory trials are needed. [CaP Etio Preven p11]
- In the prostate cancer prevention framework, the intensity of intervention rises with cancer risk (set by age, family history, ethnicity, baseline PSA and genetic susceptibility). It steps up from lifestyle and dietary change, to vitamins and micronutrients, to pharmacological agents, and finally to prophylactic surgery. [CaP Etio Preven p11]

Screening trials (ERSPC and PLCO)

- The ERSPC core age group was 55–69 years. [Compilation 02 p8]
- ERSPC was about twice as large as PLCO, with less contamination. [Compilation 02 p8]
- In ERSPC, DRE was added to the PSA cut-off of 3 ng/ml during the first and second rounds of screening. [Compilation 02 p8]
- PLCO used annual screening for 6 years with both DRE and PSA, at a PSA cut-off of 4 ng/ml. [Compilation 02 p8]

CH 154 · PROSTATE CANCER BIOMARKERS

Prostate Cancer Tumour Markers: Biomarker Development

- The Early Detection Research Network (EDRN) was set up to identify, develop and validate promising biomarkers and technologies for the early detection of cancer. [CaP Markers p1]
- Biomarker development has 5 phases; phase 1 is biomarker discovery in biomarker development laboratories. [CaP Markers p1]
- Phase 2 measures the sensitivity and specificity of the new biomarker and its ability to tell cases from controls. [CaP Markers p1]
- Phase 3 tests the ability of the new biomarker to detect preclinical disease and measures how much lead time it gives before clinical presentation. [CaP Markers p1]
- In phase 4 the indications for applying the new biomarker are clearly defined. [CaP Markers p1]
- Phase 5 is cancer control studies, which test whether the new biomarker reduces the burden of disease; a reduction in mortality is the successful outcome. [CaP Markers p1]

Assessment of Biomarker Performance

- Sensitivity is the proportion of people with the disease who have a positive test: sensitivity = TP / (TP + FN) × 100. [CaP Markers p2]
- Specificity is the proportion of people without the disease who have a negative test: specificity = TN / (TN + FP) × 100. [CaP Markers p2]
- Accuracy = (TP + TN) / total population. [CaP Markers p2]
- The area under the ROC curve (AUC) measures accuracy; an AUC of 1 indicates an ideal test and 0.5 a poorly performing biomarker. The curve plots sensitivity on the vertical axis against 1 − specificity on the horizontal axis. [CaP Markers p2]

- Positive predictive value is the probability that a positive test indicates the disease: PPV = TP / (TP + FP). [CaP Markers p2]
- Negative predictive value is the probability that a negative test indicates absence of disease: NPV = TN / (TN + FN). [CaP Markers p2]
- NPV and PPV are more relevant to clinical practice than sensitivity and specificity. [CaP Markers p2]
- The likelihood ratio expresses how much a new biomarker raises or lowers the probability of disease before biopsy. [CaP Markers p2]

FDA-Approved Biomarkers and CLIA

- The FDA-approved prostate cancer biomarkers are PSA, the free-to-total PSA ratio, PHI and PCA3. [CaP Markers p4]
- A CLIA-certified test (Clinical Laboratory Improvement Amendments) can be offered commercially, but certification does not guarantee that the biomarker has undergone rigorous clinical evaluation. [CaP Markers p4]

Prostate-Specific Antigen (PSA): Biology

- PSA is a glycoprotein secreted by prostatic epithelial cells in zymogen form. [CaP Markers p4]
- The molecular weight of PSA is 33 kD. [CaP Markers p4]
- PSA liquefies semen by acting on semenogelin and fibronectin. [CaP Markers p4]
- The PSA precursor sequence is pre-pro PSA, then inactive proPSA, then active PSA after hK2 cleaves the 7-amino-acid leader sequence in the lumen. [CaP Markers p4]
- proPSA occurs as −7, −4 and −2 proPSA isoforms; partial cleavage of −7 proPSA gives −4 and −2 proPSA. [CaP Markers p4]
- The three free PSA subforms (proPSA, BPSA and other free PSA such as intact PSA) circulate in roughly equal concentrations. [CaP Markers p4]
- Free PSA exposes all its epitopes (A to E); α1-antichymotrypsin (ACT) blocks the E epitope only, so PSA–ACT remains measurable. [CaP Markers p4]
- In prostate cancer both proPSA and bound PSA rise in serum. [CaP Markers p4]
- The normal PSA level in seminal fluid is 0.5–5.0 mg/mL. [CaP Markers p4]
- PSA expression peaks twice, at 0–6 months of age and after about 10 years of age, in line with testosterone levels. [CaP Markers p4]
- The fall in serum PSA with raised BMI may be due to haemodilution. [CaP Markers p4]

Factors Affecting PSA

- PSA becomes detectable at puberty, when LH and testosterone rise. [CaP Markers p7]
- The assay used can cause pseudo-acceleration or pseudo-deceleration of PSA. [CaP Markers p7]
- Genetic factors may influence PSA levels. [CaP Markers p7]
- The PSA change caused by DRE falls within the error of the assay. [CaP Markers p7]
- Long-distance cycling raises PSA by about 10% after bicycle rides of more than 55 km. [CaP Markers p7]

Free PSA

- Whether percentage free PSA gives prognostic information is debatable. [CaP Markers p8]
- The use of percentage free PSA is unclear when total PSA is below 4 ng/mL. [CaP Markers p8]

Free PSA Isoforms and PSA Derivatives

- PHI is approved for men aged 50 years or older with PSA 4–10 ng/mL and a negative DRE. [CaP Markers p9]
- BPSA is generally limited to the transition zone, is highly expressed in BPH, correlates with prostate volume and is unlikely to distinguish cancer from BPH. [CaP Markers p9]
- Volume-based PSA derivatives also include complexed PSA density and PSA transition zone density. [CaP Markers p9]
- The usefulness of PSA density has not been confirmed in all studies. [CaP Markers p9]
- Some studies use a PSA velocity cut-off of 0.4 ng/mL/year, and apply velocity to men with PSA below 4 ng/mL as well as 4–10 ng/mL. [CaP Markers p9]

- A PSA velocity above 2 ng/mL/year is associated with increased mortality after radiotherapy or radical prostatectomy. [CaP Markers p9]

Prostate-Specific Membrane Antigen (PSMA)

- PSMA is a type II transmembrane glycoprotein in the cell membrane of prostatic epithelial cells, with an extracellular C-terminus that binds glutamate and glutamate-like structures. [CaP Markers p10]
- The gene for PSMA is on chromosome 11p. [CaP Markers p10]
- PSMA is also expressed in the central nervous system and the intestine. [CaP Markers p10]
- Raised PSMA in prostate cancer is used for diagnosis, prognosis and imaging; its greatest current use is targeted imaging and therapeutics. [CaP Markers p10]
- Raised PSMA probably indicates a worse prognosis. [CaP Markers p10]

Human Kallikrein 2 (hK2)

- hK2 shares 80% amino acid homology with PSA, has similar specificity for prostatic tissue and is hormonally regulated by androgens. [CaP Markers p11]
- The concentration of hK2 in prostatic tissue, seminal fluid and serum is less than 2% that of total PSA. [CaP Markers p11]
- In serum, hK2 exists in bound and free forms, mostly free. [CaP Markers p11]
- hK2 expression varies independently of PSA: it is less intense than PSA in benign tissue and more intense than PSA in cancer. [CaP Markers p11]
- hK2 correlates with more aggressive pathological features. [CaP Markers p11]
- Other kallikrein tumour markers, KLK4, KLK14 and KLK15, are found mainly in tissue rather than serum. [CaP Markers p11]

Endoglin

- Endoglin (CD105) is a transmembrane glycoprotein and cell-surface coreceptor for TGF-β1 and TGF-β3; it sits on vascular endothelial cells and has a key role in angiogenesis. [CaP Markers p12]
- Pre-operative plasma endoglin is associated with node-positive disease and predicts biochemical recurrence. [CaP Markers p12]
- Urinary endoglin levels correlate with tumour volume. [CaP Markers p12]
- Endoglin is used alone or with other markers, and its clinical utility needs further study. [CaP Markers p12]

Circulating Tumour Cells (CTCs)

- CTCs are defined as CD45-negative cells that are positive for EpCAM and/or cytokeratin. [CaP Markers p12]
- There is no single gold-standard method for identifying CTCs. [CaP Markers p12]
- CellSearch is the only FDA-approved method for identifying CTCs; it uses antibodies against EpCAM, CD45 (negative) and cytokeratins 8, 18 and 19 (positive). [CaP Markers p12]
- A CTC count of 5 or more per 7.5 mL of blood indicates poor prognosis in prostate, colorectal and breast cancer. [CaP Markers p12]
- The CTC-chip, developed at Harvard Medical School, has the highest sensitivity and can detect a single EpCAM-positive cell among 1 billion blood cells. [CaP Markers p12]

Autoimmune Responses

- The body's autoimmune response to cancer (autoimmune signatures) can be a target for biomarker development. [CaP Markers p13]

Urine-Based Biomarkers: PCA3

- The PCA3 gene is on chromosome 9q. [CaP Markers p13]
- PCA3 is not expressed outside the prostate, and its levels are far higher in malignant than in benign tissue. [CaP Markers p13]
- The commercial PCA3 assay is Progensa (Hologic). [CaP Markers p13]

- The attentive DRE for PCA3 uses three firm strokes on each lobe towards the median sulcus, and the first 20–30 mL of urine is collected within 1 hour. [CaP Markers p13]
- A PCA3 cut-off of 10 has also been proposed, alongside 25 and 35, with different sensitivity and specificity. [CaP Markers p13]
- Use of PCA3 is reported to lead to better health outcomes. [CaP Markers p13]

Urine-Based Biomarkers: Gene Fusions and Metabolomics

- The Mi-Prostate Score gives both the likelihood of detecting prostate cancer on biopsy and the probability of detecting high-grade cancer. [CaP Markers p14]
- Metabolomics exploits differences in metabolic processes between cancer cells and benign cells. [CaP Markers p14]
- The most prominent metabolite is sarcosine, a metabolite of glycine found in the urine of patients with prostate cancer. [CaP Markers p14]
- The Prostarix assay is a commercial metabolomics panel run on a post-DRE sample; it is not FDA approved. [CaP Markers p14]

Annexin A3 and MicroRNA

- Annexin A3 is a calcium- and phospholipid-binding protein whose level is inversely related to the presence of prostate cancer; it is used alone or with other markers. [CaP Markers p15]
- MicroRNAs are single-stranded RNAs that regulate mRNA and have diagnostic and prognostic roles; miR-141 is a marker for prostate cancer. [CaP Markers p15]

Tissue-Based Biomarkers: AMACR

- The AMACR (α-methylacyl coenzyme A racemase) gene is on chromosome 5 and is upregulated in prostate cancer. [CaP Markers p15]
- AMACR is responsible for β-oxidation of fatty acids. [CaP Markers p15]
- AMACR expression in biopsy tissue has 97% sensitivity and 100% specificity for prostate cancer. [CaP Markers p15]

Epigenetic Modifications

- Epigenetic change in prostate cancer includes histone acetylation as well as DNA methylation. [CaP Markers p16]
- Retinoic acid receptor β2 (RARβ2) is one of the key hypermethylated genes in prostate cancer, along with GSTP1, APC and RASSF1A. [CaP Markers p16]

Circulating tumour cells in prostate cancer

- In prostate cancer, CTCs are defined as CD45-negative and EpCAM- and/or cytokeratin-positive cells. [CTC p2]
- A count of 5 or more CTCs per 7.5 mL of blood indicates poor prognosis. [CTC p2]
- CellSearch is the FDA-approved CTC method, and the CTC-chip was developed at Harvard Medical School. [CTC p2]

Prostate Cancer Tumour Markers: Factors affecting PSA

- Urinary retention causes a variable degree of rise in PSA levels. [Compilation 02 p52]

CH 155 · PROSTATE BIOPSY: TECHNIQUES AND IMAGING

Prostate Biopsy: Ultrasonographic Anatomy

- Corpora amylacea are age-related calcifications along the surgical capsule that highlight the plane between the peripheral and transition zones. [Compilation 02 p70]
- The sagittal plane on TRUS shows the prostatic urethra along its entire course. [Compilation 02 p70]
- The internal sphincter extends from the bladder neck around the urethra to the level of the verumontanum. [Compilation 02 p70]
- On TRUS the internal sphincter appears as a hypoechoic ring around the upper prostatic urethra, giving it a funnelled appearance proximally as it arises from the bladder neck. [Compilation 02 p70]

Seminal Vesicles, Vasa and TRUS Uses

- On TRUS the seminal vesicles have a smooth saccular appearance and should be symmetrical. [Compilation 02 p73]
- In the transverse plane the vasa deferentia course just above the seminal vesicles before diving caudally towards the prostate near the midline. [Compilation 02 p73]
- TRUS is also used for treatment planning of radiofrequency ablation and microwave therapy. [Compilation 02 p73]
- The TRUS probe most commonly used is 7 MHz (range 6–10 MHz). [Compilation 02 p73]
- The 6–10 MHz probe gives a high-resolution image with a focal range of 1 to 4 cm from the transducer, which is best for the peripheral zone where most cancers arise. [Compilation 02 p73]

TRUS Probes, Coupling Medium and Machine Settings

- Lower-frequency probes give lower resolution but increase the accuracy of volume measurement and improve anterior delineation of large glands. [Compilation 02 p74]
- A water-density coupling medium is used because the acoustic properties of soft tissue resemble water and clinically useful ultrasound energy does not propagate through air. [Compilation 02 p74]
- End-fire and side-fire probes show no significant difference in positive biopsy rate. [Compilation 02 p74]
- Side-fire probes may give better patient tolerance (marked with a question mark in the notes). [Compilation 02 p74]
- Magnification should be low for prostate measurements and high during biopsy. [Compilation 02 p74]
- The optimal brightness setting gives a medium-grey image of the normal peripheral zone, which is the reference point for judging echogenicity. [Compilation 02 p74]
- Relative to normal peripheral zone, hypoechoic is darker, isoechoic is the same, hyperechoic is lighter and anechoic is completely black. [Compilation 02 p74]

TRUS Technique and Probe Manipulation

- For transverse imaging, moving the probe cephalad into the rectum images the prostate base, seminal vesicles and bladder neck. [Compilation 02 p75]
- For transverse imaging, moving the probe caudally towards the anal sphincter images the prostate apex and proximal urethra. [Compilation 02 p75]
- For sagittal imaging, clockwise rotation of the probe shows the left side and counterclockwise rotation shows the right side. [Compilation 02 p75]
- For sagittal imaging, angling the probe towards the floor shows the right side and angling towards the ceiling shows the left side. [Compilation 02 p75]

Prostate Cancer and Treated Prostate on TRUS

- On TRUS, 60% of prostate cancers are hypoechoic and 1% are hyperechoic (the remaining 39% isoechoic). [Compilation 02 p76]
- Other hypoechoic prostatic lesions include granulomatous prostatitis and lymphoma, as well as prostatic infarct. [Compilation 02 p76]
- All hypoechoic lesions should be included in the biopsy specimen, although they are not pathognomonic of cancer. [Compilation 02 p76]
- After EBRT the prostate decreases in volume by 6 months. [Compilation 02 p76]
- Brachytherapy seeds are hyperechoic on TRUS and cast posterior shadowing. [Compilation 02 p76]
- After radical prostatectomy, TRUS is considered normal if there is smooth tapering of the bladder neck to the urethra. [Compilation 02 p76]
- After radical prostatectomy, a nodule anterior to the anastomosis on TRUS represents the ligated dorsal vein complex. [Compilation 02 p76]

Indications for Repeat Biopsy

- PIN at multiple biopsy sites carries a 20–30% cancer risk and is an indication for repeat biopsy. [Compilation 02 p77]

Preparing the Patient for Biopsy

- A small amount of urine in the bladder is useful because it demonstrates the prostate–bladder junction. [Compilation 02 p78]
- Oral TMP-SMX is one of the antibiotic choices for biopsy prophylaxis. [Compilation 02 p78]
- In patients at risk of endocarditis, preoperative IV ampicillin plus gentamicin is followed by 2–3 days of oral fluoroquinolone. [Compilation 02 p78]
- A cleansing enema decreases faeces in the rectum and provides a superior acoustic window. [Compilation 02 p78]

Analgesia and Positioning

- The periprostatic lidocaine is injected at the hyperechoic fat pad that marks the junction of the seminal vesicles and the prostate, near the bladder base. [Compilation 02 p79]
- Direct intraprostatic injection can augment the benefit of periprostatic injection. [Compilation 02 p79]
- In the left lateral decubitus position the knees and hips are flexed 90 degrees, a pillow between the knees helps maintain the position, and the buttocks should be flush with the end of the table. [Compilation 02 p79]
- The right lateral decubitus position can also be used for TRUS biopsy. [Compilation 02 p79]
- Lithotomy is preferred for placement of fiducial gold markers and brachytherapy treatment planning, as well as transperineal biopsy. [Compilation 02 p79]

Biopsy Technique and Templates

- The biopsy gun advances the needle 0.5 cm and samples the next 1.5 cm of tissue, with the tip extending 0.5 cm beyond the area sampled. [Compilation 02 p80]
- Prostate biopsy cores are fixed in 10% formalin. [Compilation 02 p80]
- Sextant biopsy takes one core from base, mid-gland and apex on each side in the parasagittal plane. [Compilation 02 p80]
- Some recommend seminal vesicle biopsy when PSA is above 30. [Compilation 02 p80]

Repeat biopsy

- A third or fourth biopsy is also justified by poor prognostic factors on the first or second biopsy, not only by high suspicion of cancer. [Prostate notes (02 dragged) p1]
- At repeat biopsy, attention should be directed towards the apex. [Prostate notes (02 dragged) p1]

Transperineal and transurethral biopsy

- Transperineal biopsy is done in dorsal lithotomy with an end-fire transducer. [Prostate notes (02 dragged) p1]
- On the transperineal route, the seminal vesicles and focal hypoechoic lesions in the peripheral zone are hard to see. [Prostate notes (02 dragged) p1]
- A transperineal biopsy takes a minimum of 6 cores, 3 from each side. [Prostate notes (02 dragged) p1]
- Transurethral biopsy is not done, because TRUS biopsy samples the transition zone adequately. [Prostate notes (02 dragged) p1]
- Solitary transition-zone cancers without a peripheral-zone tumour are rare, under 5% of patients. [Prostate notes (02 dragged) p1]

Complications of biopsy

- Further risk factors for post-biopsy infection are non-white race, more comorbidities, prostatomegaly and the number of previous biopsies. [Prostate notes (02 dragged) p2]
- Prostatitis and epididymitis are listed as biopsy complications. [Prostate notes (02 dragged) p2]
- Erectile dysfunction after prostate biopsy is incompletely characterised. [Prostate notes (02 dragged) p2]

Advanced and investigational techniques

- Colour and power Doppler TRUS aim to detect the neovascularity of prostate cancer. [Prostate notes (02 dragged) p3]
- Unenhanced Doppler cannot show the microvessels of prostate cancer, which are 10–15 µm in diameter. [Prostate notes (02 dragged) p3]

- Microbubble contrast agents for Doppler are IV bubbles of air or a high-molecular-weight gas, encapsulated so that they last longer. [Prostate notes (02 dragged) p3]
- Microbubbles measure 1–10 µm. [Prostate notes (02 dragged) p3]
- The microbubble capsule is an albumin or polymer hard shell, or a lipid or surfactant coat. [Prostate notes (02 dragged) p3]
- Elastography may prove better than colour Doppler at finding malignant areas: it measures tissue stiffness, and reduced elasticity suggests malignancy. [Prostate notes (02 dragged) p3]
- Prostate HistoScanning is an evolving technology that differentiates, characterises and displays prostate tissue by analysing back-scattered ultrasound. [Prostate notes (02 dragged) p3]
- Computerised machine templates turn 2D ultrasound data into a 3D model for targeted imaging (TargetScan, Artemis). [Prostate notes (02 dragged) p3]

mpMRI-targeted biopsy

- Commercial MRI/TRUS fusion platforms include UroNav and Artemis (with ProFuse software). [Prostate notes (02 dragged) p4]

CH 156 · PATHOLOGY OF PROSTATIC NEOPLASIA

Prostatic intraepithelial neoplasia

- As evidence that HGPIN is a precursor, more HGPIN is linked to a greater number of multifocal carcinomas. [Prostate notes (02 dragged) p6]
- HGPIN and cancer share the same biomarker and molecular changes. [Prostate notes (02 dragged) p6]
- The risk of cancer on a further biopsy within a year of HGPIN is 26.4%. [Prostate notes (02 dragged) p6]
- PIN on its own does not raise PSA. [Prostate notes (02 dragged) p6]

HGPIN on biopsy and its mimics

- For unifocal HGPIN, repeat biopsy after 3 years. [Prostate notes (02 dragged) p7]
- For PINATYP, repeat biopsy within 6 months. [Prostate notes (02 dragged) p7]

Gleason system

- Gleason grading is done at relatively low magnification, 10x. [Prostate notes (02 dragged) p8]
- The original Gleason score ranges from 2 to 10. [Prostate notes (02 dragged) p8]
- Scores 2–4 are not given on needle biopsy for three reasons: experts grade most of these tumours higher on review, reproducibility is poor, and they do not reliably mean favourable findings at RP. [Prostate notes (02 dragged) p8]

Gleason patterns

- Pattern 1 is a well-circumscribed nodule of closely packed, medium-sized, round to oval glands. [Prostate notes (02 dragged) p9]
- Pattern 2 is like pattern 1 and fairly circumscribed, but the nodule edge may show minimal infiltration, and the glands are looser and less uniform. [Prostate notes (02 dragged) p9]
- Pattern 3 glands are usually smaller than in patterns 1 and 2, vary markedly in shape and size, and infiltrate in and among non-neoplastic acini. [Prostate notes (02 dragged) p9]
- Pattern 4 includes hypernephromatoid glands. [Prostate notes (02 dragged) p9]
- In pattern 5, comedocarcinoma has central necrosis surrounded by papillary, cribriform or solid masses. [Prostate notes (02 dragged) p9]

Clinical significance of Gleason grade

- Gleason score 6 tumours are more homogeneous, and RP is predicted to cure them. [Prostate notes (02 dragged) p10-11]
- Grade group 4 or 5 in more than 5% of the prostatic volume at RP carries a high chance of biochemical recurrence. [Prostate notes (02 dragged) p10-11]
- Gleason score predicts side-specific extraprostatic extension and extension into the neurovascular bundle. [Prostate notes (02 dragged) p10-11]

- Gleason score helps set the intervention criteria during active surveillance. [Prostate notes (02 dragged) p10-11]
- Gleason score predicts prognosis after cryotherapy and after HIFU. [Prostate notes (02 dragged) p10-11]

Gleason grading of variants

- Signet-ring cell carcinoma is graded as pattern 5. [Prostate notes (02 dragged) p11]
- Sarcomatoid carcinoma is graded as pattern 5. [Prostate notes (02 dragged) p11]
- Xanthomatous carcinoma is graded as pattern 3 or 4. [Prostate notes (02 dragged) p11]
- Atrophic carcinoma is graded as pattern 3 (written "3/2"). [Prostate notes (02 dragged) p11]
- Pseudohyperplastic carcinoma is graded as pattern 2 or 3. [Prostate notes (02 dragged) p11]

Spread of prostate cancer

- The rectum is rarely involved, and rectal involvement can be hard to tell from a rectal primary. [Prostate notes (02 dragged) p12]
- Lung metastases are commonly found at autopsy. [Prostate notes (02 dragged) p12]
- After nodes and bone, the next commonest metastatic sites are bladder, liver and adrenal gland. [Prostate notes (02 dragged) p12]

CH 157 · DIAGNOSIS AND STAGING OF PROSTATE CANCER

Screening for prostate cancer

- An advantage of screening is that it lowers the rate of advanced disease at the time of diagnosis. [Compilation 02 p2]
- EAU (2018, weak recommendation) advises a risk-adapted strategy based on the initial PSA, with 2-yearly follow-up for men initially at risk (PSA >1 ng/ml at 40 years or PSA >2 ng/ml at 60 years) and follow-up postponed to 8 years for men not at risk; the other EAU early-detection statements are strong recommendations. [Compilation 02 p2]

Diagnosis: DRE, PSA and PSA derivatives

- Only about 18% of prostate cancers are detected by DRE alone. [Compilation 02 p3]
- DRE detects a tumour only when its volume is 0.2 ml or more, so it misses a substantial portion of early cancers. [Compilation 02 p3]
- DRE can either overestimate or underestimate the extent of the tumour. [Compilation 02 p3]
- PSA is a better predictor of cancer than DRE alone or TRUS alone. [Compilation 02 p3]
- Risk of prostate cancer by PSA level (PCPT): 17% at PSA 1.1–2.0, 24% at 2.1–3.0 and 27% at 3.1–4.0 ng/ml. [Compilation 02 p3]
- Complexed PSA is one of the PSA derivatives (related tests) used alongside free PSA, PSA isoforms, hK2, PHI and the 4K panel. [Compilation 02 p3]

Staging systems

- Whitmore–Jewett staging: stage A is a tumour found incidentally and not detectable clinically; stage B is found on physical examination but has not spread outside the capsule; stage C has extended through the capsule; stage D has spread to other organs. [Compilation 02 p5]

CH 158 · ACTIVE MANAGEMENT STRATEGIES FOR LOCALISED PROSTATE CANCER

Evaluation before treatment of localised cancer

- Men planning surgery should have a fuller work-up (bone scan, CT or MRI, fitness tests), as higher-risk men do. [Prostate notes (02 dragged) p16]
- Monoclonal antibody scans and MR spectroscopy are not widely used but may prove useful in future. [Prostate notes (02 dragged) p16]

CH 159 · ACTIVE SURVEILLANCE OF PROSTATE CANCER

Active Surveillance

- In men with low-grade disease, prostate cancer progression and mortality increase slowly up to 15 years but do not increase rapidly beyond that time. [Compilation 02 p34]
- Overdiagnosis and overtreatment place a financial burden on the patient and the healthcare system. [Compilation 02 p34]
- Problems of active surveillance include the risk of losing the window of opportunity for cure of early localised disease, the need for the patient to accept a strict follow-up protocol, and biopsy-related infections and erectile dysfunction; nerve-sparing surgery may be difficult after repeated biopsies. [Compilation 02 p34]
- Active surveillance is a less established strategy in patients with a long life expectancy. [Compilation 02 p36]
- Candidates should understand that they carry a risk of extraprostatic extension and lymph node metastasis. [Compilation 02 p36]
- Tumour extension on DRE is a trigger for intervention during active surveillance. [Compilation 02 p38]
- Disease-specific and all-cause survival on active surveillance compare favourably with long-term outcomes after radical prostatectomy or radiotherapy in patients of similar risk. [Compilation 02 p38]
- 25–50% of men on active surveillance develop objective evidence of tumour progression. [Compilation 02 p38]

Watchful Waiting

- Watchful waiting monitoring comprises history, physical examination with DRE, PSA and creatinine every 6 months, and a bone scan yearly. [Compilation 02 p39]
- Palliative treatment in watchful waiting, usually ADT, is started on symptomatic progression, evidence of upper tract obstruction or evidence of metastatic disease. [Compilation 02 p39]

CH 160 · OPEN RADICAL PROSTATECTOMY

Radical Prostatectomy: Open Procedure, Preoperative Preparation and Operative Steps

- Bowel preparation for open RP is a clear liquid diet the day before surgery, magnesium citrate in the evening and an enema on the morning of surgery. [Compilation 02-3 p26]
- Prophylactic antibiotics are given 1 hour before surgery and continued for a total of 24 hours. [Compilation 02-3 p26]
- General anaesthesia is given with systolic blood pressure kept below 100 mmHg. [Compilation 02-3 p26]
- The patient is supine, with or without flexion of the table at the level of the anterior superior iliac spine, and a 16 Fr Foley catheter is placed and connected to drainage. [Compilation 02-3 p26]
- The endopelvic fascia is incised where it is transparent and shows the underlying levator ani. [Compilation 02-3 p27]
- The superficial branch of the dorsal vein is divided together with the puboprostatic ligaments. [Compilation 02-3 p27]
- The apical dissection is the most complex part of the operation. [Compilation 02-3 p27]
- The lateral pedicles are thick, so they are divided lateral to the seminal vesicles in superficial, middle and deep layers. [Compilation 02-3 p28]
- The commonest sites of positive margins after RP are, in order, the apex, then the posterior surface, then the posterolateral surface. [Compilation 02-3 p28]

Complications of Open RP

- Causes of bladder neck contracture after RP are inadequate mucosal approximation, haematoma and urinary extravasation. [Compilation 02-3 p29]
- For a recalcitrant bladder neck contracture, triamcinolone 200 mg in 5 mL is injected into the bladder neck after cold-knife incision. [Compilation 02-3 p29]

- Causes of incontinence after RP include intrinsic sphincter deficiency (usually in older men), injury during ligation and division of the DVC, damage to the urethral smooth muscle, and preoperative overactive bladder due to bladder outlet obstruction. [Compilation 02-3 p30]

Surgical Modifications of Classical RP: Nerve-sparing RP

- Before one NVB is excised, the contralateral NVB should first be freed from the prostate, starting at the apex, to avoid traction injury to it. [Compilation 02-3 p32]
- The drawback of high anterior release of the NVB is that there is less soft tissue at the apex, so positive surgical margins could increase. [Compilation 02-3 p33]
- In high anterior release, the levator fascia over the anterior apex is incised and the incision is carried distally along the lateral edge of the apex; the NVB is exposed and released before the urethra is transected, taking care not to enter the underlying prostatic fascia. [Compilation 02-3 p33]

Bladder Neck Sparing RP

- Bladder neck sparing RP aims to hasten the return of urinary control. [Compilation 02-3 p34]
- Bladder neck sparing makes little difference to positive margin rates. [Compilation 02-3 p34]
- Bladder neck sparing gives a clinically significant continence advantage at 6 months, which largely disappears by 1 year. [Compilation 02-3 p34]
- Bladder neck contracture rates may be high after bladder neck sparing, so it is not done routinely. [Compilation 02-3 p34]
- The Heidelberg technique preserves the circular fibres of the bladder neck, with a urethro-urethral anastomosis. [Compilation 02-3 p34]

Seminal Vesicle Sparing RP

- Most men with low-risk disease can undergo seminal vesicle sparing RP. [Compilation 02-3 p34]
- The non-oncological risks and benefits of seminal vesicle sparing are uncertain and compromised cancer control remains a concern; randomised trials are needed, so it is not done routinely. [Compilation 02-3 p34]

Management of Localised Prostate Cancer: Evaluation

- Monoclonal antibody scans, PET scans, MR spectroscopy and lymphotropic MRI were listed as not widely used for staging localised prostate cancer but possibly more useful in future (monoclonal antibody scans and MR spectroscopy are not in the corpus). [Localised CaP p1]

Radical Prostatectomy: Nerve Sparing and Surgical Approaches

- Patient factors that make nerve sparing questionable are poor preoperative erections or medical conditions that produce ED (diabetes, hypertension, neurological disorders) and the lack of a current or future sexual relationship. [Localised CaP p2]
- Perineal RP has a shorter operative time than retropubic RP. [Localised CaP p2]
- Perineal RP carries a higher risk of rectal injury, with faecal incontinence. [Localised CaP p2]

Retropubic, Laparoscopic and Robotic RP

- Advantages of the retropubic approach are familiar anatomy, low risk of rectal injury, wide exposure with access to the pelvic nodes, preservation of the neurovascular bundles and a lower risk of cancer at the surgical margins. [Localised CaP p3]
- Problems listed for laparoscopic RP are a higher risk of severe complications, difficult haemostasis with injury to the neurovascular bundle and cavernous nerves, bowel injury and urinary ascites, and uncertain cancer control (no long-term studies, higher positive margins in early studies). [Localised CaP p3]

Selection of Patients

- The general upper age limit for radical prostatectomy is 76 years. [Localised CaP p4]
- Neoadjuvant hormones make nerve-sparing surgery difficult. [Localised CaP p4]
- Neoadjuvant chemotherapy before RP rarely produces a complete pathological response. [Localised CaP p4]

- The general upper age limit for RP is 76 years. [Prostate notes (02 dragged) p19]
- The ideal RP candidate has a tumour that is biologically significant and completely resectable. [Prostate notes (02 dragged) p19]
- Men should be told that nerve sparing is not always possible and that they may need PDE5 inhibitors, intraurethral or intracorporeal vasodilators, vacuum erection devices, venous flow constrictors or a penile prosthesis. [Prostate notes (02 dragged) p19]
- Neoadjuvant hormones make nerve-sparing surgery more difficult. [Prostate notes (02 dragged) p19]

Post-operative Care and Cancer Control after RP

- After RP the patient ambulates with assistance on the evening or afternoon of the day of surgery. [Localised CaP p6]
- Catheter removal before 7 days after RP gives a 15-20% retention rate. [Localised CaP p6]
- A protective pad is used until complete urinary control is achieved. [Localised CaP p6]
- The first serum PSA is checked 1 month after RP. [Localised CaP p6]
- Lymphovascular invasion is listed among the adverse pathological factors after RP (with non-organ-confined disease, extracapsular extension, positive margins, seminal vesicle invasion and nodal metastases). [Localised CaP p6]

Recurrence, Continence and Potency after RP

- In recurrent neuroendocrine tumours, PSA is not much elevated, so DRE has a role in detecting palpable recurrent tumour. [Localised CaP p7]
- Cancer-specific recurrence after RP continues to increase for at least 15 years, and the risk of cancer mortality may increase for 25 years or more, so long follow-up is needed. [Localised CaP p7]
- Potency after RP is defined as the ability to maintain erections sufficiently rigid for penetration and sexual intercourse, with or without the help of a PDE5 inhibitor. [Localised CaP p7]

Erectile Function and Complications of RP

- Era of surgery is listed among the determinants of return of erections after RP (with age, preoperative potency and extent of nerve sparing). [Localised CaP p8]
- About 95% of men in their 40s and 50% of men in their 70s can achieve satisfactory erections after RP. [Localised CaP p8]
- Return of erections begins as partial erections 3-6 months after surgery. [Localised CaP p8]
- Early complications of RP occur in under 10% in experienced hands. [Localised CaP p8]

Obturator Nerve Injury, Thromboprophylaxis and Late Complications

- Options for obturator nerve injury include nerve grafting with a cutaneous or genitofemoral nerve, or conservative management without repair with physical therapy (in addition to tension-free primary repair). [Localised CaP p9]
- Perioperative heparin injections given as thromboprophylaxis may predispose to lymphoceles. [Localised CaP p9]

Radical prostatectomy: advantages and nerve-sparing selection

- One advantage of RP is that treatment failure is easier to detect. [Prostate notes (02 dragged) p17]
- Nerve sparing is of doubtful value with poor preoperative erections, with conditions that cause ED (diabetes, hypertension, neurological disorders), or when the man has no current or expected sexual relationship. [Prostate notes (02 dragged) p17]

Surgical approaches

- Perineal RP has a shorter operating time than retropubic RP. [Prostate notes (02 dragged) p17-18]
- Retropubic RP offers familiar anatomy, a low risk of rectal injury, wide exposure with access to pelvic nodes, and a lower risk of cancer at the surgical margins. [Prostate notes (02 dragged) p17-18]
- Laparoscopic RP carries a higher risk of severe complications. [Prostate notes (02 dragged) p17-18]
- In laparoscopic RP, difficult haemostasis can injure the neurovascular bundle and damage the cavernous nerves. [Prostate notes (02 dragged) p17-18]

- Bowel injury and urinary ascites are listed as laparoscopic problems. [Prostate notes (02 dragged) p17-18]
- Cancer control after laparoscopic RP is uncertain: there are no long-term studies, and early studies showed more positive margins. [Prostate notes (02 dragged) p17-18]

Surgical technique

- The ampullary portion of the vas is dissected with the seminal vesicles. [Prostate notes (02 dragged) p20-21]
- The neurovascular bundle is usually found at the apex, but dissection can also start at the base. [Prostate notes (02 dragged) p20-21]

Postoperative care

- Patients walk with assistance on the evening or afternoon of surgery. [Prostate notes (02 dragged) p21]
- The catheter is removed 3–21 days after surgery, depending on the integrity of the anastomosis. [Prostate notes (02 dragged) p21]
- Removing the catheter before 7 days causes retention in 15–20%. [Prostate notes (02 dragged) p21]
- Serum PSA is checked at 1 month. [Prostate notes (02 dragged) p21]

Cancer control after RP

- Preoperative PSA and the treatment given are listed as independent prognostic factors, along with stage and Gleason score. [Prostate notes (02 dragged) p21-22]
- Lymphovascular invasion is an adverse factor after RP. [Prostate notes (02 dragged) p21-22]
- After a PSA rise, only 34% go on to metastasis or cancer death. [Prostate notes (02 dragged) p21-22]
- Neuroendocrine tumours are an exception to PSA follow-up, because they raise PSA little, so DRE is used to find a palpable recurrence. [Prostate notes (02 dragged) p21-22]
- Cancer-specific recurrence keeps rising for at least 15 years after RP, and mortality risk may rise for 25 years or more, so follow-up must be long. [Prostate notes (02 dragged) p21-22]

Erectile function after RP

- Potency after RP means erections rigid enough for penetration and intercourse, with or without a PDE5 inhibitor. [Prostate notes (02 dragged) p22-23]
- Return of erections also depends on the era of surgery. [Prostate notes (02 dragged) p22-23]
- Erections are usually not as good as before surgery. [Prostate notes (02 dragged) p22-23]
- Satisfactory erections return in 95% of men in their 40s and 50% in their 70s. [Prostate notes (02 dragged) p22-23]
- Erections return first as partial erections, 3–6 months after surgery. [Prostate notes (02 dragged) p22-23]

Complications of RP

- Early complications occur in under 10% in experienced hands. [Prostate notes (02 dragged) p23-24]
- Obturator nerve injury can be treated with a nerve graft (cutaneous or genitofemoral nerve). [Prostate notes (02 dragged) p23-24]
- Obturator nerve injury can also be managed without repair, with physical therapy. [Prostate notes (02 dragged) p23-24]
- Perioperative heparin injections may predispose to lymphoceles. [Prostate notes (02 dragged) p23-24]

CH 161 · LAPAROSCOPIC AND ROBOTIC-ASSISTED LAPAROSCOPIC RADICAL PROSTATECTOMY AND PELVIC LYMPHADENECTOMY

Laparoscopic / Robotic Prostatectomy

- Bowel preparation before laparoscopic or robotic RP is magnesium citrate or bowel prep the day before, or only a Fleet enema on the morning of surgery. [Compilation 02-3 p36]
- In steep Trendelenburg, the legs are spread slightly apart to allow access to the rectum. [Compilation 02-3 p36]
- RALP port sizes: a 12 mm camera port, a 12 mm and a 5 mm assistant port, and three 8 mm robotic ports. [Compilation 02-3 p37]
- Laparoscopic RP port layout: two 10–12 mm ports and two 5 mm ports. [Compilation 02-3 p37]

- In the transperitoneal approach any urine leak is intraperitoneal, whereas in the extraperitoneal approach a urine leak is contained. [Compilation 02-3 p37]
- Further ways to identify the bladder neck robotically are retracting the bladder dome in a cephalad direction and bimanual palpation (pinching) of the bladder neck with the two robotic or laparoscopic instruments. [Compilation 02-3 p38]
- PLND is generally recommended with intermediate- or high-risk features such as PSA above 10, a large palpable nodule or Gleason score of 7 or more. [Compilation 02-3 p40]
- Optic neuropathy is a position-related complication of robotic RP, along with raised intraocular pressure. [Compilation 02-3 p41]
- Open conversion during laparoscopic or robotic RP occurs in under 2%. [Compilation 02-3 p41]
- A disadvantage of present robotic systems is that they are very bulky. [Compilation 02-3 p42]

CH 162 · RADIATION THERAPY FOR PROSTATE CANCER

Radiation for localised disease: prognostic factors

- The 5-year PSA-failure-free survival with radiation therapy is about 85% in low-risk, 50% in intermediate-risk and 33% in high-risk localised disease. [Compilation 02 p11]
- Splitting intermediate risk into favourable and unfavourable also uses the pretreatment PSA velocity and mpMRI findings (besides percentage of positive cores, number of unfavourable factors and primary Gleason pattern). [Compilation 02 p11]

Post-treatment PSA follow-up, bounce and biochemical failure after RT

- After radiotherapy, PSA is monitored at 6-monthly intervals until it reaches its nadir. [Compilation 02 p12]
- Besides radiation prostatitis, PSA bounce can be caused by instrumentation and by testosterone recovery after ADT. [Compilation 02 p12]
- For a rising PSA within 30 months of brachytherapy: if PSA is 10 ng/ml or more, look for metastasis; if below 10 ng/ml but beyond 30 months, do a biopsy; otherwise continue careful follow-up. [Compilation 02 p12]
- Both the level and the timing of the PSA nadir correlate with distant metastasis and cause-specific survival, and some studies use 18 months as the cut-off for time to nadir. [Compilation 02 p12]
- The nadir PSA comes from residual benign prostatic epithelium, residual local cancer cells or subclinical disseminated micrometastases. [Compilation 02 p12]
- The nadir PSA is a significant predictor of outcome, but no absolute nadir threshold can or should be used to define cure; the lower the nadir, the higher the chance of success. [Compilation 02 p12]
- In most studies after conventional-dose radiotherapy, a median PSA nadir of 0.4–0.5 ng/ml is seen with no evidence of disease, above 1 ng/ml with local failure and above 2 ng/ml with distant failure. [Compilation 02 p12]
- A post-nadir PSA doubling time under 3–6 months suggests distant failure, while a doubling time of 1 year or more suggests local failure. [Compilation 02 p12]
- If neoadjuvant hormonal therapy is given, patients often start radiotherapy with an already undetectable PSA, so the true nadir and time to nadir cannot be determined. [Compilation 02 p12]

Post-radiation therapy biopsy

- Problems of post-RT biopsy are its timing after treatment, interpretation of indeterminate biopsies and sampling error. [Compilation 02 p15]
- Radiotherapy markedly alters gland architecture, the basis of Gleason scoring, so post-RT biopsies are often falsely reported as high grade. [Compilation 02 p15]
- Basal cells labelled by an anti-cytokeratin monoclonal antibody distinguish benign tissue from malignant tissue, where basal cells are absent. [Compilation 02 p15]
- Immunohistochemical proliferation markers such as Ki-67 and proliferating cell nuclear antigen (PCNA) can be used to detect residual tumour in post-RT biopsies. [Compilation 02 p15]

External beam radiotherapy: techniques

- Conventional (pre-CT) radiotherapy set its portals by skeletal anatomy (pubic symphysis and femoral heads), inferring prostate position from the Foley balloon, bladder contrast and a rectal marker. [Compilation 02 p16]
- CT-based 3D tumour visualisation (3D conformal radiation therapy) began in the 1990s, increasing precision and lowering dose to surrounding organs. [Compilation 02 p16]
- With IMRT inverse planning the prostate can receive about 1.8 Gy daily while most of the adjacent bladder receives less than half that dose. [Compilation 02 p16]
- Image guidance is used to reduce interfraction and intrafraction motion; previously oncologists enlarged the beam to accommodate prostate motion. [Compilation 02 p16]
- For interfraction motion, daily ultrasound localisation of the prostate immediately before treatment is an option, as are fiducial markers with software to locate the prostate each day. [Compilation 02 p16]
- For intrafraction motion, the prostate can be immobilised with an intrarectal balloon that pins it against the pubic bone, or tracked in real time with the Calypso system. [Compilation 02 p16]
- The current EBRT dose is 76–80 Gy; doses above 80 Gy are not beneficial, and rectal toxicity limits the dose. [Compilation 02 p16]

EBRT side effects

- Radiation side effects result from injury to the microvasculature of the bladder, rectum, striated sphincter muscle and urethra. [Compilation 02 p18]
- About one-third of patients develop acute symptoms of proctitis and cystitis, usually when the dose exceeds 50 Gy; most settle after radiotherapy ends, but 5–10% have permanent symptoms. [Compilation 02 p18]
- Erectile dysfunction occurs in about 50% of patients after EBRT, from injury to the vasculature of the cavernous nerves and corpora cavernosa, and usually begins about 1 year after treatment. [Compilation 02 p18]

Heavy particle beams and SBRT

- Heavy particle beam therapy includes neutron as well as proton beams, is of limited availability and expensive, and is perceived to have higher complication rates without much gain in tumour control. [Compilation 02 p19]
- The Bragg peak is the peak on the Bragg curve, which plots the energy loss of ionising radiation during its travel through matter; the proton beam is widened (spread out) so that a larger tumour volume is treated. [Compilation 02 p19]
- SBRT (CyberKnife) uses a linear accelerator mounted on a robotic arm to give large doses in a small number of fractions. [Compilation 02 p19]
- Extreme hypofractionation is often given on fewer than 5 days a week to allow normal tissue recovery, so the entire course can be completed within 2 weeks. [Compilation 02 p19]

Brachytherapy (LDR and HDR) and its toxicity

- Early brachytherapy used free-hand implantation, with lower success rates than today's image-guided implantation. [Compilation 02 p20]
- In HDR brachytherapy the iridium-192 source (1–10 curie) is applied for variable periods of seconds to minutes. [Compilation 02 p20]
- Using ADT to shrink large glands before brachytherapy delays any PSA rise that would show tumour persistence or recurrence. [Compilation 02 p20]
- Brachytherapy toxicity depends on implant technique and quality; age, comorbidities (diabetes, peripheral vascular disease) and tobacco use predispose to more toxicity. [Compilation 02 p20]
- About 10% of patients need a TURP after brachytherapy, and only a channel TURP should be done. [Compilation 02 p20]
- Causes of impotence after brachytherapy are underlying small vessel disease, cavernous nerve damage and damage to the vascular tissue around the penile bulb. [Compilation 02 p20]

Molecular Therapies & Radiation

- About 50% of men with locally advanced prostate cancer ultimately develop biochemical failure within 10 years of treatment. [Compilation 02 p28]
- Molecular-based therapies target vulnerable aspects of cancer growth or of the cancer's ability to repair injury, and they do so in a targeted way to minimise effects on normal tissue. [Compilation 02 p28]
- The problem with siRNA radiosensitisation is that it also sensitises non-cancerous cells, and the challenging aspect is delivery to specific cell types. [Compilation 02 p28]
- Defects in MHC class I expression are seen in 85% of primary prostate cancers and in all metastatic tumours, which means evasion of host immunity may be a critical factor in prostate cancer development. [Compilation 02 p29]
- Radio-gene therapy is based on enzyme/prodrug systems, oncolytic viruses and cytotoxins, which cause cytolytic or apoptotic cell death, and combining them with radiotherapy may give superior cell killing. [Compilation 02 p29]
- Enzyme/prodrug gene therapy converts an inactive prodrug using an enzyme vectored only to tumour cells, so the active drug stays away from normal cells. [Compilation 02 p30]
- HSV thymidine kinase plus ganciclovir suicide gene therapy, combined with a replication-competent adenovirus and radiation, has recently been tested in prostate cancer and appears safe. [Compilation 02 p30]
- Oncolytic viruses have a lytic phase in their normal life cycle that is lethal to the host cell; candidates for therapy include adenovirus, vaccinia, Newcastle disease virus, HSV, influenza virus and mumps virus. [Compilation 02 p30]
- Adenovirus suits prostate cancer therapy because it has low pathogenicity for normal tissue, is not tumourigenic, and causes only mild to moderate symptoms when infection occurs. [Compilation 02 p30]

Radiobiology of Hypofractionation and Alpha-to-Beta Ratio

- The alpha-to-beta ratio is defined as the dose at which cell killing from the linear component equals cell killing from the quadratic component. [Compilation 02 p31]
- Early-responding tissues typically have a high alpha-to-beta ratio, while late-responding tissues have a low one. [Compilation 02 p31]
- The alpha-to-beta ratio of head and neck cancers is 13–16 Gy. [Compilation 02 p31]
- The alpha-to-beta ratio of the bladder is 5 Gy. [Compilation 02 p31]

BCR after definitive RT: definitions and PSA bounce

- For the old ASTRO definition, the three consecutive PSA rises were measured 6 months apart (as written in the notes). [Prostate notes (02 dragged) p45]
- With the Phoenix definition, it often takes considerably longer for PSA to rise by 2 ng/ml, so failure is called later. [Prostate notes (02 dragged) p45]
- PSA bounce after RT results from radiation prostatitis or from delivery of a sublethal dose of radiation. [Prostate notes (02 dragged) p45]

CH 163 · FOCAL THERAPY OF PROSTATE CANCER

Focal Therapy for Prostate Cancer: Localisation of Disease

- Prostate HistoScanning is listed as an ultrasound-based localisation option alongside TRUS, contrast-enhanced TRUS and elastography. [Focal Therapy Prostate p1]
- These ultrasound-based modalities (TRUS, CE-TRUS, elastography, HistoScanning) are poor at localising disease, assessing tumour burden and detecting small tumour foci. [Focal Therapy Prostate p1]
- Saturation biopsies give only a small additional benefit unless they are carried out with a template mapping technique. [Focal Therapy Prostate p2]
- On T2-weighted MRI, water is bright and fat is dark. [Focal Therapy Prostate p2]
- On MR spectroscopy (MRS), prostate cancer shows an increased ratio of choline plus creatine to citrate. [Focal Therapy Prostate p2]

Cryotherapy

- Intracellular dehydration during freezing is accompanied by pH changes that contribute to cell death. [Focal Therapy Prostate p2]
- Injury during warming results from osmotic cellular swelling and vascular hyperpermeability. [Focal Therapy Prostate p2]
- Recommended freezing is rapid freezing down to −40 °C, held for a minimum of 3 minutes, as part of a double freeze–thaw cycle. [Focal Therapy Prostate p3]
- The later repair phase after cryotherapy shows fibrosis, calcification and hyalinisation. [Focal Therapy Prostate p3]
- Multiprobe systems with smaller individual probes allow more precise freezing and reduce toxicity. [Focal Therapy Prostate p3]
- Third-generation cryotherapy devices use gas-based systems to provide rapid freeze–thaw cycles. [Focal Therapy Prostate p3]
- Cryoneedles are inserted through the perineum either through a brachytherapy grid or freehand. [Focal Therapy Prostate p4]
- Thermocouples are placed in the treatment area, in Denonvilliers' fascia and in other key areas such as the rhabdosphincter and the neurovascular bundles. [Focal Therapy Prostate p4]
- Cystoscopy is done at the start of the procedure to verify needle position. [Focal Therapy Prostate p4]
- COLD (Cryo On-Line Database) registry: 3-year BCR-free survival was 75.7%, 41.9% of patients had new erectile dysfunction and incontinence was 1.6%. [Focal Therapy Prostate p4]

High-Intensity Focused Ultrasound (HIFU)

- Ultrasound is mechanical vibration above the threshold of human hearing (16 kHz). [Focal Therapy Prostate p4]
- An alternating voltage makes piezoelectric crystals (lead zirconate titanate) oscillate at the same frequency as the current, producing ultrasound waves that propagate as alternating cycles of compression and rarefaction. [Focal Therapy Prostate p4]
- Diagnostic ultrasound uses 1–20 MHz, whereas therapeutic HIFU uses 0.8–3.5 MHz. [Focal Therapy Prostate p5]
- Low-intensity ultrasound (0.125–3 W/cm^2) can stimulate normal physiological processes in response to injury and can accelerate the transport of drugs across the skin. [Focal Therapy Prostate p5]
- High-intensity ultrasound causes tissue damage during the high-pressure (compression) phase. [Focal Therapy Prostate p5]
- A temperature of 56 °C held for 1 second causes irreversible cell death by coagulative necrosis. [Focal Therapy Prostate p5]
- Histology after HIFU: coagulative necrosis is followed by an inflammatory response, then granulation tissue (immature fibroblasts and new capillaries) at the periphery of the necrotic area at about 1 week, and proliferative repair tissue at the periphery by 2 weeks. [Focal Therapy Prostate p6]
- Patient movement is a factor that prevents successful HIFU treatment. [Focal Therapy Prostate p6]
- HIFU is done under spinal or general anaesthesia with a suprapubic or per-urethral catheter placed, and is delivered under TRUS (commonest), TRUS–MR fusion or in-gantry MRI guidance. [Focal Therapy Prostate p6]

Photodynamic Therapy

- In photodynamic therapy the photosensitising drug is activated by light of a specific wavelength after a defined drug–light interval. [Focal Therapy Prostate p7]
- Vasculature-activated drugs have a short drug–light interval (minutes), allow single-session treatment and need no light restrictions after a few hours. [Focal Therapy Prostate p7]
- Tissue-activated drugs have long drug–light intervals (hours to days), so drug and light are given in separate sessions; they are cleared slowly and accumulate in the skin, so patients must avoid sunlight for a few weeks to prevent sunburn. [Focal Therapy Prostate p7]
- Some photosensitisers, such as aminolevulinic acid (ALA), accumulate preferentially in tumour tissue. [Focal Therapy Prostate p7]

- Transperineal light delivery uses hollow plastic needles placed under TRUS, through which optical fibres carry low-power laser light to the treatment site. [Focal Therapy Prostate p8]
- Other light-delivery routes are transurethral delivery and open insertion of fibres at laparotomy. [Focal Therapy Prostate p8]

Focal Photothermal (Laser) Therapy

- Unlike photodynamic therapy, focal laser photothermal therapy uses neither oxygen nor photosensitisers. [Focal Therapy Prostate p9]
- MRI is used for treatment planning and TRUS for fibre placement; transperineally an open-ended catheter is introduced into the target lesion and the optical laser fibre is inserted through it. [Focal Therapy Prostate p9]
- Focal laser therapy is done under sedation or general anaesthesia. [Focal Therapy Prostate p9]

Focal Irreversible Electroporation

- IRE needle electrodes are inserted at the boundaries of the lesion, not within the treatment area. [Focal Therapy Prostate p10]
- Once the needles are positioned, IRE treatment usually takes less than 5 minutes. [Focal Therapy Prostate p10]

Radiofrequency Ablation

- RFA coagulative necrosis is characterised by cell membrane disruption, protein denaturation and vascular thrombosis. [Focal Therapy Prostate p11]
- RFA may be monopolar or bipolar; bipolar RFA has less risk of accidental burns. [Focal Therapy Prostate p11]
- Wet RFA uses constant saline infusion during ablation to reduce charring, in contrast to dry RFA. [Focal Therapy Prostate p11]

Complications and Cancer Control of Focal Therapy

- Rectal toxicity after focal therapy includes rectal bleeding and diarrhoea (besides perineal pain and rectourethral fistula). [Focal Therapy Prostate p12]
- The trifecta outcome after focal therapy is achieved in 50–89%. [Focal Therapy Prostate p12]

Follow-up after Focal Therapy

- For PSA density after focal therapy, BPH contributes about 0.3 ng/mL per gram and cancer about 3.5 ng/mL per cm^3; this may be used in future. [Focal Therapy Prostate p13]
- The role of PSA kinetics (PSA velocity and PSA doubling time) after focal therapy needs future trials. [Focal Therapy Prostate p13]

BCR after HIFU

- HIFU is considered safe for men with clinically localised (T1–T2) prostate cancer, low PSA (less than 15 ng/mL) and low prostate volume. [Prostate notes (02 dragged) p51]
- The Stuttgart definition of biochemical failure after HIFU is PSA nadir plus 1.2 ng/mL. [Prostate notes (02 dragged) p51]
- There is no clear-cut guideline for managing HIFU failure; salvage radiotherapy and salvage radical prostatectomy are the options. [Prostate notes (02 dragged) p51]

CH 164 · TREATMENT OF LOCALLY ADVANCED PROSTATE CANCER

Locally advanced prostate cancer: EAU guideline statements

- EAU advises against intraoperative frozen section of pelvic nodes to decide whether or not to complete the radical prostatectomy; the RP should simply be completed (also repeated on p3). [L.A. CaP p1]

Imaging modalities and novel markers

- TRUS understages prostate cancer and has high interobserver variability for local staging. [L.A. CaP p2]

- Colour and power Doppler TRUS for local staging of prostate cancer are still under investigation. [L.A. CaP p2]
- Traditional clinical and pathological parameters are limited in their ability to predict the local extent of the tumour accurately, which drives the search for novel markers. [L.A. CaP p2]
- TMPRSS2–ETS gene fusion is associated with a high pathological stage. [L.A. CaP p2]
- Sarcosine is a marker associated with metastatic prostate cancer. [L.A. CaP p2]

Staging work-up and radical prostatectomy

- Pelvic CT or MRI is indicated for cT3 or cT4 disease or when the calculated probability of lymph node invasion exceeds 10%. [L.A. CaP p3]
- After radical prostatectomy for locally advanced disease, cancer-specific survival is 60% at 15 years and overall survival is 75% at 10 years. [L.A. CaP p3]

Neoadjuvant ADT before RP

- Neoadjuvant ADT before RP does not alter the rate of lymph node metastases. [L.A. CaP p4]

Adjuvant radiotherapy after RP

- Adjuvant RT is given to the prostatic fossa after recovery of urinary function, that is at about 3 months after surgery. [L.A. CaP p4]

Adjuvant ADT and adjuvant RT in pN1 disease

- Observation in pN1 disease additionally requires microscopic-only nodal involvement and no extranodal extension (alongside 2 or fewer nodes and PSA below 0.1 ng/mL). [L.A. CaP p5]
- Men with low-volume nodal disease (fewer than 3 positive nodes) who have Gleason 7–10 with pT3 or pT4 disease or R1 margins (and possibly those with 3–4 nodes) are the ones most likely to benefit from adjuvant RT after surgery. [L.A. CaP p5]
- There is no recommendation on the extent (field) of adjuvant RT in pN1 disease, although whole-pelvis RT appears beneficial. [L.A. CaP p5]
- Men with pN1 disease most likely to benefit from RT after surgery are those with low-volume nodal disease (fewer than 3 nodes) plus Gleason 7–10, pT3–pT4 or R1 margins, and possibly those with 3–4 positive nodes. [Prostate notes (02 dragged) p34]
- There is no recommendation on the field extent of adjuvant RT in pN1 disease. [Prostate notes (02 dragged) p34]

Adjuvant chemotherapy and primary radiotherapy

- Adjuvant chemotherapy after RP should be used in clinical trials only. [L.A. CaP p6]
- When ADT is combined with RT, neoadjuvant ADT starts 2–3 months before RT, while adjuvant ADT starts at the onset of RT. [L.A. CaP p6]
- Brachytherapy (as primary treatment) is less studied in high-risk and locally advanced cases. [L.A. CaP p6]

Focal ablative therapy and ADT

- Focal ablative therapy may have a role in locally advanced disease when combined with ADT or other systemic therapy. [L.A. CaP p7]
- Intermittent androgen deprivation protocol: give total androgen deprivation until PSA falls below 4.0 ng/mL (usually within 24–32 weeks), then withhold treatment, monitor PSA, and restart when PSA returns to the pretreatment level or rises above 10–20 ng/mL. [L.A. CaP p7]

Management of Post-operative Biochemical Recurrence

- Of men with biochemical recurrence after RP, 50% recur within 3 years and 99% within 10 years; recurrence is rare after 15 years. [Localised CaP p10]
- Salvage RT should be started before PSA reaches 0.5 ng/ml. [Localised CaP p10]
- Salvage RT response is more favourable with PSA recurrence long after surgery, slowly rising PSA, low-grade tumour, no seminal vesicle or nodal involvement, and possibly positive margins. [Localised CaP p10]

Pre-op ADT, Salvage RP and Adjuvant RT

- Preoperative ADT might be beneficial in high-risk disease by giving a longer time to biochemical recurrence. [Localised CaP p11]
- Adjuvant RT is typically given 4-6 months after RP, when PSA is still undetectable and urinary continence has been regained. [Localised CaP p11]

Surveillance after RP and pN1

- Criteria favouring surveillance or possibly delayed RT for BCR include pathological stage below pT3a and time to recurrence over 3 years (with PSA doubling time over 12 months and Gleason score 7 or less). [Localised CaP p12]
- Observation of pN1 after extended PLND requires absence of extranodal extension (in addition to 2 or fewer nodes with microscopic involvement and PSA below 0.1). [Localised CaP p12]

Pre-op ADT, salvage RP and adjuvant RT after RP

- Nerve sparing may be difficult after pre-operative androgen deprivation therapy. [Prostate notes (02 dragged) p26]
- Extensive extracapsular extension is listed as an indication for adjuvant RT after RP. [Prostate notes (02 dragged) p26]
- Adjuvant RT is typically given 4–6 months after RP, while PSA is still undetectable and once the patient has regained urinary continence. [Prostate notes (02 dragged) p26]

Surveillance after RP / delayed RT; pN1 after eLND

- Surveillance with possible delayed RT for BCR suits men with pathological stage below pT3a and a time to recurrence of more than 3 years, together with PSADT >12 months and Gleason ≤7. [Prostate notes (02 dragged) p27]
- In pN1 disease after eLND, observation (EAU) also requires the absence of extranodal extension, in addition to ≤2 microscopically involved nodes and PSA <0.1. [Prostate notes (02 dragged) p27]

Locally advanced prostate cancer: EAU guidelines

- Do not perform frozen section of the pelvic nodes during RP to decide whether or not to complete the prostatectomy; complete the RP regardless (also repeated on p32). [Prostate notes (02 dragged) p30]

Definition, risk assessment and imaging of locally advanced disease

- TRUS understages prostate cancer and has interobserver variability in local staging. [Prostate notes (02 dragged) p31]
- TMPRSS2–ETS fusion is associated with a high pathological stage. [Prostate notes (02 dragged) p31]
- Sarcosine is a novel marker associated with metastatic prostate cancer. [Prostate notes (02 dragged) p31]

Incidence, staging work-up and radical prostatectomy in locally advanced disease

- About 10% of men with newly diagnosed prostate cancer have locally advanced disease. [Prostate notes (02 dragged) p32]
- Pelvic CT or MRI is indicated for cT3 or cT4 disease, or when the calculated probability of lymph node invasion is above 10%. [Prostate notes (02 dragged) p32]
- Outcomes of RP in locally advanced disease are about 60% cancer-specific survival at 15 years and about 75% overall survival at 10 years. [Prostate notes (02 dragged) p32]

Neoadjuvant ADT before RP; adjuvant RT

- Adjuvant RT to the prostatic fossa is given after urinary function has recovered, that is at about 3 months after RP. [Prostate notes (02 dragged) p33]

Adjuvant chemotherapy; RT as primary treatment of locally advanced disease

- Adjuvant chemotherapy after RP should be given only within clinical trials. [Prostate notes (02 dragged) p35]
- When ADT is given with RT, neoadjuvant ADT is started 2–3 months before RT, while adjuvant ADT is started at the onset of RT. [Prostate notes (02 dragged) p35]

Focal ablation and androgen deprivation therapy in locally advanced disease

- Intermittent androgen deprivation protocol: treat with total androgen deprivation until PSA falls below 4.0 ng/ml (usually within 24–32 weeks), then withhold treatment and monitor PSA, restarting when PSA returns to the pretreatment level or rises above 10–20 ng/ml. [Prostate notes (02 dragged) p36]

CH 165 · MANAGEMENT OF RECURRENT AND NEWLY METASTATIC PROSTATE CANCER

Salvage Therapy after Radiorecurrence

- Men with recurrent prostate cancer usually fail after the age of 65 years and have additional comorbidities, so options range from watchful waiting and delayed systemic ADT to local salvage therapies. [Focal Therapy Prostate p15]
- In over 90% of men with recurrent prostate cancer, the strategy used is watchful waiting with delayed systemic ADT. [Focal Therapy Prostate p15]

Endocrine / Hormonal Therapy — classification

- Cetrorelix is listed as an LHRH antagonist, together with degarelix and abarelix. [Harmone CaP p1]
- Diethylstilbestrol (DES) is an oestrogen that suppresses testosterone; it is about a thousand-fold more potent than testosterone at suppressing LH and FSH. [Harmone CaP p1]
- DES is not associated with bone loss, but it causes severe thromboembolic complications even at low doses, together with cardiovascular toxicity, so it is not considered standard first-line treatment. [Harmone CaP p1]
- Aminoglutethimide is an inhibitor of androgen synthesis, listed with ketoconazole and abiraterone. [Harmone CaP p1]

LHRH Agonists

- The flare is seen with all LHRH agonist preparations and can cause a severe, life-threatening exacerbation of symptoms. [Harmone CaP p2]
- Some authorities advise starting the antiandrogen 1 week before the LHRH agonist. [Harmone CaP p2]
- Leuprolide is marketed as Eligard (subcutaneous) and Lupron Depot (intramuscular). [Harmone CaP p2]
- Leuprolide doses are 3.75 mg or 7.5 mg monthly, 11.25 mg or 22.5 mg 3-monthly, 30 mg 4-monthly and 45 mg 6-monthly. [Harmone CaP p2]
- Monitoring on an LHRH agonist is PSA in the first few weeks of therapy, then LH, FSH and serum testosterone after 4 weeks of therapy. [Harmone CaP p2]
- LHRH agonist adverse events seen in more than 10% are hot flushes, cardiovascular changes or ischaemia, fatigue, pain and peripheral oedema. [Harmone CaP p2]
- LHRH agonist adverse events seen in 1–10% are asthenia, gynaecomastia, headache, testicular atrophy, anorexia, anaemia, nausea, vomiting, constipation and urinary frequency; less often reported are impotence and haematuria. [Harmone CaP p2]

LHRH Antagonists

- The severe allergic reaction seen with the first-generation antagonist abarelix is not seen with third- and fourth-generation antagonists; patients given abarelix must be observed for 30 minutes after the injection. [Harmone CaP p3]
- Degarelix is marketed as Firmagon and is supplied as 80 mg and 120 mg powder for injection. [Harmone CaP p3]
- Degarelix adverse effects seen in more than 10% are hot flushes, injection-site erythema and pain, weight gain and raised liver enzymes. [Harmone CaP p3]
- Degarelix adverse effects seen in 1–10% are arthralgia, back pain, fatigue, hypertension and injection-site swelling or induration. [Harmone CaP p3]
- Degarelix does not cause the systemic allergic reactions seen with abarelix. [Harmone CaP p3]

Steroidal Antiandrogens

- Megestrol acetate and medroxyprogesterone acetate are steroidal antiandrogens, like cyproterone acetate, and all have poor overall efficacy. [Harmone CaP p7]
- Cyproterone acetate acts in two ways: it blocks the androgen receptor directly, and its progestational action crosses the blood–brain barrier to inhibit LH centrally, rapidly lowering testosterone by 70–80%. [Harmone CaP p7]
- Cyproterone side effects reflect the hypogonadal state: loss of libido, erectile dysfunction, lassitude, gynaecomastia (under 20%) and rare fulminant hepatotoxicity. [Harmone CaP p7]

Non-steroidal Antiandrogens

- Because non-steroidal antiandrogens do not cause hypogonadism, potency is preserved, with long-term preservation in about 20%. [Harmone CaP p8]
- Flutamide was the first pure antiandrogen. [Harmone CaP p8]
- Flutamide has a short half-life (6 hours); its active metabolite is 2-hydroxyflutamide, which is excreted by the kidneys. [Harmone CaP p8]
- Flutamide is not associated with fluid retention or thromboembolism. [Harmone CaP p8]
- Overall survival with flutamide is significantly shorter than with DES, and bicalutamide is likewise inferior to DES. [Harmone CaP p8]
- Bicalutamide is not affected by moderate hepatic impairment (as well as age and renal insufficiency). [Harmone CaP p8]
- Bicalutamide 50 mg/day as monotherapy is inferior to orchidectomy in metastatic or locally advanced disease. [Harmone CaP p8]
- Nilutamide has a half-life of 56 hours and is cleared by the liver. [Harmone CaP p8]
- Nilutamide also causes alcohol intolerance and nausea, and its pneumonitis can progress to interstitial fibrosis. [Harmone CaP p8]

Combined Androgen Blockade

- The fact that serum testosterone does not fall to zero after surgical or medical castration is evidence of additional (adrenal) androgen sources. [Harmone CaP p11]
- The largest CAB trial, in 1286 patients, found no difference between surgical castration with or without flutamide. [Harmone CaP p11]

Intermittent Androgen Deprivation Therapy

- IADT rests on two ideas: in preclinical animal models, intermittent rather than continuous androgen deprivation lengthened the time to emergence of refractory cancer growth; and patients and doctors increasingly questioned the benefit of continuous ADT given its profound side effects. [Harmone CaP p12]
- Possible benefits of IADT are off-treatment periods that relieve side effects, improving quality of life and compliance; delayed progression to CRPC; and efficacy similar to continuous ADT, especially in men with lower disease burden. [Harmone CaP p12]
- Possible IADT candidates are men who reach PSA below 4 during a 5–7 month induction period, tolerate continuous ADT poorly, have non-metastatic disease or metastases limited to lymph nodes, have locally advanced disease, are older, have Gleason score 7 or more, and have a longer PSA doubling time. [Harmone CaP p12]

Bilateral Orchidectomy and Complications of Androgen Ablation

- Subcapsular orchidectomy does not improve cancer control compared with simple orchidectomy. [Harmone CaP p14]
- The longer a man stays on ADT the greater the risk of osteopenia, osteoporosis and fracture; 4 years of ADT will place an average man in the osteopenia range. [Harmone CaP p14]
- Bone mineral density should be measured at the hip by DEXA. [Harmone CaP p14]
- Transdermal estradiol increases bone mineral density in men with prostate cancer. [Harmone CaP p14]

Hot Flashes

- A hot flash is a subjective feeling of warmth in the upper torso and head followed by objective perspiration. [Harmone CaP p15]
- Hot flashes decrease in intensity and frequency over time but may persist in some men; only men who find them bothersome are treated. [Harmone CaP p15]
- Megestrol acetate for hot flashes is given at 20 mg twice daily; its problem is appetite stimulation, and the dose can be reduced to 5 mg twice daily. [Harmone CaP p15]
- Oestrogenic treatments for hot flashes are low-dose DES and transdermal estradiol; up to 90% get partial or complete relief, at the cost of painful gynaecomastia and thromboembolic events. [Harmone CaP p15]
- Venlafaxine (an antidepressant) for hot flashes is given at 12.5 mg twice daily and reduces hot flashes in more than 50% of patients. [Harmone CaP p15]
- The proposed mechanism of hot flashes is a rise in hypothalamic adrenergic concentrations with altered beta-endorphins and calcitonin gene-related peptide acting on the hypothalamic thermoregulatory centre. [Harmone CaP p15]

Sexual Dysfunction

- On ADT, 10–17% can maintain erections adequate for intercourse and up to 20% maintain some sexual activity. [Harmone CaP p16]
- Libido is more severely compromised than erection, and the loss is related to the duration of ADT; the libido loss is extremely difficult to treat. [Harmone CaP p16]
- Erectile dysfunction on ADT is treated with oral PDE5 inhibitors or intracavernosal alprostadil, although these may not be used long term. [Harmone CaP p16]

Cognitive Function

- ADT reduces verbal fluency, which improves with testosterone (the other cognitive symptoms do not). [Harmone CaP p16]
- ADT also causes memory loss, anxiety, psychological distress and a decline in tasks that need complex information processing. [Harmone CaP p16]

Body Habitus, Diabetes and Cardiovascular Effects

- After 1 year of ADT, weight rises by 1.8–3.8%. [Harmone CaP p17]
- ADT increases appetite, and low testosterone is associated with raised insulin levels and increased abdominal girth. [Harmone CaP p17]
- Diabetes or metabolic syndrome is present in more than 50% of men on long-term ADT. [Harmone CaP p17]
- ADT causes subcutaneous fat accumulation and a rise in HDL. [Harmone CaP p17]
- Cardiovascular morbidity and mortality from ADT are more pronounced in men with lower-risk prostate cancer, and arise from its effects on lipids, glucose metabolism and body habitus. [Harmone CaP p17]

Gynaecomastia and Anaemia

- Prophylactic breast radiotherapy gives no benefit once gynaecomastia has begun. [Harmone CaP p17]
- Established gynaecomastia is treated by liposuction and subcutaneous mastectomy. [Harmone CaP p17]
- ADT-related anaemia is very common and is thought to be due to loss of testosterone stimulation of erythropoietin production. [Harmone CaP p17]

Management of post-operative biochemical recurrence

- About 50% of biochemical recurrences after radical prostatectomy appear within 3 years of surgery. [Prostate notes (02 dragged) p25]
- About 99% of biochemical recurrences after radical prostatectomy appear within 10 years. [Prostate notes (02 dragged) p25]
- Biochemical recurrence is rare more than 15 years after radical prostatectomy. [Prostate notes (02 dragged) p25]

- Absence of seminal vesicle and lymph node involvement predicts a favourable response to salvage RT. [Prostate notes (02 dragged) p25]
- Positive surgical margins (marked with a query in the notes) predict a favourable response to salvage RT. [Prostate notes (02 dragged) p25]

Management of BCR after definitive therapy: BCR after RP

- Besides PSA doubling time, pathological Gleason score and time from surgery to BCR, the course from BCR to metastasis and death also depends on age and advanced tumour stage. [Prostate notes (02 dragged) p40]
- Ultrasensitive PSA (uPSA) assays are those that measure PSA at concentrations below 0.1 ng/ml. [Prostate notes (02 dragged) p40]

Ultrasensitive PSA; imaging in BCR after RP

- Ultrasensitive PSA may create anxiety in patients with clinically meaningless PSA rises and may lead to overtreatment. [Prostate notes (02 dragged) p41]
- When PSA is below 10 ng/ml after RP, only about 4% of patients have a positive bone scan. [Prostate notes (02 dragged) p41]
- TRUS and TRUS-guided biopsy of the prostatic fossa have reasonable sensitivity for local recurrence at low PSA levels after RP. [Prostate notes (02 dragged) p41]
- Indium-111 capromab pendetide (ProstaScint) scanning uses an IgG monoclonal antibody that binds the intracellular domain of PSMA, which is exposed only in apoptotic or necrotic cells; this limited its clinical adoption. [Prostate notes (02 dragged) p41]
- Newer anti-PSMA antibodies such as J591 bind the external (extracellular) domain of PSMA. [Prostate notes (02 dragged) p41]
- The sensitivity and specificity of these antibody scans are only about 60–70%. [Prostate notes (02 dragged) p41]

PET/CT and MRI in BCR; salvage RT after RP

- 18F-FDG PET also has low sensitivity in recurrent prostate cancer because benign tissue and post-operative scar show similar uptake. [Prostate notes (02 dragged) p42]
- Acetate is one of the better PET tracers for recurrent prostate cancer, alongside choline and fluorocholine. [Prostate notes (02 dragged) p42]
- Choline PET is US FDA approved for detecting prostate cancer recurrence. [Prostate notes (02 dragged) p42]
- Almost 60% of men with BCR after RP still receive ADT, even though it is not curative. [Prostate notes (02 dragged) p42]

Salvage RT: outcomes, failure and dose

- Salvage RT given at PSA ≤0.5 ng/ml gives a progression-free survival of 48%. [Prostate notes (02 dragged) p43]
- Salvage RT also improves all-cause mortality. [Prostate notes (02 dragged) p43]
- Causes of failure after salvage RT are persistent local disease, recurrence of local disease, persistence of metastasis and development of metastatic disease. [Prostate notes (02 dragged) p43]
- Salvage RT may be delayed in men with stage pT3a or lower, in addition to PSADT >12 months and Gleason ≤7. [Prostate notes (02 dragged) p43]

Concurrent ADT with salvage RT; ADT for biochemical failure after RP

- One theoretical benefit of adding ADT to salvage RT is that it sterilises micrometastatic disease. [Prostate notes (02 dragged) p44]

Imaging and salvage radical prostatectomy after RT failure

- Imaging after RT is difficult because of fibrosis and shrinkage of the prostate. [Prostate notes (02 dragged) p46]
- TRUS has a limited ability to identify suspicious lesions in the irradiated prostate. [Prostate notes (02 dragged) p46]

- T2-weighted MRI for local recurrence after RT has a sensitivity of only 27%, though with high specificity. [Prostate notes (02 dragged) p46]
- Salvage RP is the most definitive salvage treatment, with 70–80% 10-year cancer-specific survival. [Prostate notes (02 dragged) p46]
- In older series, about 50% of men had major complications after salvage RP. [Prostate notes (02 dragged) p46]
- Better surgical planning, meticulous attention to anatomy and improved radiation delivery have reduced the morbidity of salvage RP. [Prostate notes (02 dragged) p46]

Salvage cryotherapy

- Urinary incontinence after salvage cryotherapy was as high as 72% in earlier series (now 4–6.5%). [Prostate notes (02 dragged) p47]
- Further complications of salvage cryotherapy are rectal pain, scrotal pain or oedema, and haematuria. [Prostate notes (02 dragged) p47]

Salvage brachytherapy, salvage HIFU and ADT after RT failure

- HDR brachytherapy has gained attention as salvage because it can implant the seminal vesicles and treat extracapsular extension while conformal delivery limits the dose to surrounding organs. [Prostate notes (02 dragged) p48]
- Complications of salvage HIFU also include urinary retention, urinary tract infection and urethrocutaneous fistula. [Prostate notes (02 dragged) p48]
- Overall, about 93.5% of men receive ADT as second-line therapy after radiation failure. [Prostate notes (02 dragged) p48]

BCR after definitive cryotherapy

- There is no single accepted definition of biochemical recurrence after primary cryotherapy. [Prostate notes (02 dragged) p50]
- Definitions that have been used for BCR after cryotherapy include PSA above 0.5 ng/mL, PSA above 1 ng/mL, the ASTRO definition, and nadir plus 2 ng/mL. [Prostate notes (02 dragged) p50]
- Management strategies for cryotherapy failure are not well established; options are repeat cryotherapy, salvage radiotherapy (the feasible choice) and radical prostatectomy. [Prostate notes (02 dragged) p50]

Bone problems in carcinoma prostate on ADT

- Skeletal complications in men on ADT include nerve entrapment and hypercalcaemia, besides fracture, uncontrolled pain and spinal cord compression. [Prostate notes (02 dragged) p73]
- Risk factors for bone loss on ADT are older age, family history, Asian or Hispanic ethnicity, corticosteroid use, pre-existing bone loss and longer duration of ADT. [Prostate notes (02 dragged) p73]
- Bone-specific ALP is a useful blood test in the bone evaluation of men on ADT. [Prostate notes (02 dragged) p73]
- Anaemia in these men reflects ineffective erythropoiesis. [Prostate notes (02 dragged) p73]
- Urine N-telopeptide (NTX) is a biomarker of bone turnover used in evaluation. [Prostate notes (02 dragged) p73]
- The TRAP (tartrate-resistant acid phosphatase) test is used as a marker of bone resorption. [Prostate notes (02 dragged) p74]
- Bone scintigraphy has the best sensitivity but low specificity for bone metastases. [Prostate notes (02 dragged) p74]
- Plain X-ray has a sensitivity of only 40–50% for bone metastases; X-ray, CT/MRI and PET or PSMA PET are needed in equivocal cases. [Prostate notes (02 dragged) p74]
- Bone biopsy gives an accurate diagnosis but is usually unnecessary. [Prostate notes (02 dragged) p74]

STAMPEDE Trial

- STAMPEDE stands for "Systemic Therapy in Advancing or Metastatic Prostate Cancer: Evaluation of Drug Efficacy". [Prostate Trials p1]

- STAMPEDE is a multi-arm, multi-stage (MAMS) randomised controlled trial; the corpus says only "multi-arm". [Prostate Trials p1]
- The participants were men starting long-term hormone therapy, and the trial was run in a tertiary-care setting. [Prostate Trials p1]
- The study hypothesis was that the research interventions would improve survival over standard of care alone. [Prostate Trials p1]
- The definitive primary outcome was overall survival, and the intermediate primary outcome was failure-free survival. [Prostate Trials p1]
- The secondary outcomes were toxicity, quality of life, cost-effectiveness and symptomatic skeletal events. [Prostate Trials p1]

STAMPEDE: interventions compared and outcomes

- Control arm A (standard of care) was ADT, with or without radiotherapy and with or without docetaxel. [Prostate Trials p2]
- Arm B (standard of care plus zoledronic acid) gave no increase in overall survival. [Prostate Trials p2]
- Arm C (standard of care plus docetaxel) increased overall survival by 10 months but caused more side effects. [Prostate Trials p2]
- Arm D (standard of care plus celecoxib) gave no change in overall survival. [Prostate Trials p2]
- Arm E (standard of care plus zoledronic acid plus docetaxel) increased overall survival. [Prostate Trials p2]
- Arm F (standard of care plus zoledronic acid plus celecoxib) gave no change in overall survival. [Prostate Trials p2]
- Arm G (standard of care plus abiraterone acetate plus prednisolone) also increased progression-free survival and time to radiographic progression, in addition to overall survival. [Prostate Trials p2]
- Arm J was standard of care plus abiraterone plus enzalutamide; the notes record it as in follow-up. [Prostate Trials p2]
- Arm K was standard of care plus metformin; the notes record it as recruiting. [Prostate Trials p2]
- Arm L was transdermal oestradiol, with or without radiotherapy and with or without docetaxel; the notes record it as recruiting. [Prostate Trials p2]

CHAARTED Trial

- CHAARTED was a phase III randomised trial. [Prostate Trials p3]
- CHAARTED enrolled 790 patients between July 2006 and November 2012. [Prostate Trials p3]
- Patients were randomised 1:1 to ADT alone or ADT plus docetaxel. [Prostate Trials p3]
- Docetaxel had to be started within 4 months of starting ADT. [Prostate Trials p3]
- The primary end point of CHAARTED was overall survival. [Prostate Trials p3]

LATITUDE Trial

- LATITUDE was a phase III, double-blind, randomised trial. [Prostate Trials p5-6]
- Patients were randomised 1:1 to ADT plus abiraterone 1000 mg once daily plus prednisolone 5 mg daily (n = 597) or ADT plus placebo (n = 602). [Prostate Trials p5-6]
- Overall survival and (radiographic) progression-free survival were co-primary end points. [Prostate Trials p5-6]
- The secondary end points were time to pain progression, time to PSA progression, time to the next symptomatic skeletal event, time to chemotherapy and time to subsequent prostate cancer therapy. [Prostate Trials p5-6]
- Abiraterone plus prednisolone significantly improved all secondary end points, as well as progression-free survival. [Prostate Trials p5-6]
- Overall safety was consistent with earlier studies in metastatic castration-sensitive prostate cancer. [Prostate Trials p5-6]

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Sipuleucel-T (Provenge)

- Sipuleucel-T is a personalised vaccine made from the patient's own CD54-positive dendritic cells, which are the major class of antigen-presenting cells collected at leukapheresis. [CRPC p11]
- Sipuleucel-T should not be used in men who need narcotic analgesia for cancer-related pain. [Prostate notes (02 dragged) p66]
- Reported sipuleucel-T side effects, besides mild infusion fevers and rigors, include bladder pain, swelling of the face, arms or feet, bloody or cloudy urine, body aches, chest pain, confusion, cough, diarrhoea and sleeplessness. [Prostate notes (02 dragged) p66]
- Trials are evaluating sipuleucel-T in combination with other AR-directed therapies. [Prostate notes (02 dragged) p66]

Zoledronate

- Osteonecrosis of the jaw on zoledronate is an unusual complication, and its cause is not well understood. [CRPC p22]

Denosumab (Xgeva)

- Denosumab is a fully human monoclonal antibody against RANKL. [CRPC p23]
- Denosumab 120 mg is marketed as Xgeva and was FDA approved in November 2010. [Prostate notes (02 dragged) p78]
- Denosumab side effects include muscle and joint pain, skin rash, nausea, diarrhoea, constipation, breathlessness, flu-like symptoms, back pain, fatigue, eye changes and rare allergy. [Prostate notes (02 dragged) p78]
- Other denosumab indications are bone metastases from breast, kidney and lung cancer, osteoporosis unresponsive to other drugs, and giant cell tumour of bone that cannot be treated surgically. [Prostate notes (02 dragged) p78]

Treatment of mCRPC (Extra Questions)

- The choice between abiraterone and chemotherapy in mCRPC depends on patient preference, cost, availability and specific side effects. [Compilation 02-3 p44]
- Hypertension, diabetes and hypokalaemia favour docetaxel over abiraterone, whereas neuropathy or neutropenia favours abiraterone. [Compilation 02-3 p44]
- Docetaxel is a finite course of injections at lower cost, whereas abiraterone is prolonged oral therapy at higher cost. [Compilation 02-3 p44]

Radiation for palliation: bone metastases

- Bone metastases cause localised pain that is frequently continuous and unrelenting regardless of site; diagnosis is by examination, plain X-ray and bone scan, with CT or MRI if bone involvement is suspected but X-ray and bone scan are negative. [Compilation 02 p24]
- Prophylactic surgical fixation (orthopaedic referral) is warranted for an intramedullary lytic lesion of 50% or more of the bone's cross-sectional diameter, or a lytic lesion involving a length of cortex equal to or greater than the cross-sectional diameter or more than 2.5 cm in axial length. [Compilation 02 p24]
- Pathological fracture is relatively infrequent with blastic prostate cancer metastases. [Compilation 02 p24]

Spinal cord compression

- Spinal cord compression can cause paraplegia and autonomic dysfunction; pain is the predominant symptom (95% of patients) and precedes diagnosis by about 4 months. [Compilation 02 p26]
- Once paraplegia occurs, return of function is infrequent, so early diagnosis and therapy are critical. [Compilation 02 p26]
- Surgery before radiotherapy is also indicated for compression of the cord by bone and when the tissue diagnosis is unknown. [Compilation 02 p26]

Systemic radionuclide therapy

- Phosphorus-32 and rhenium-186 are further bone-seeking radionuclides (with strontium-89 and samarium-153) used historically in CRPC with multiple painful bone metastases; their problem is bone marrow suppression. [Compilation 02 p27]
- Radium-223 was approved by the US FDA in 2013. [Compilation 02 p27]

Inhibition of Androgen Synthesis: Aminoglutethimide

- Aminoglutethimide blocks the side-chain cleavage enzyme, so cholesterol is not converted to pregnenolone. [Harmone CaP p5]
- Aminoglutethimide causes a medical adrenalectomy because it also blocks aldosterone and cortisol synthesis, so fludrocortisone and cortisone replacement is necessary. [Harmone CaP p5]
- Aminoglutethimide side effects are anorexia, skin rash, lethargy, vertigo, nystagmus and hypothyroidism. [Harmone CaP p5]
- The aminoglutethimide dose is 1000 mg/day, the same as abiraterone. [Harmone CaP p5]

Ketoconazole

- Ketoconazole is an orally active, broad-spectrum azole antifungal that interferes with two cytochrome P450-dependent pathways: lanosterol to cholesterol (by inhibiting 14-methylation) and C21 to C19 steroids (by blocking 17,20-desmolase). [Harmone CaP p5]
- Ketoconazole stops both adrenal steroid synthesis and Leydig-cell testosterone synthesis, so hydrocortisone 20 mg twice daily is given with it. [Harmone CaP p5]
- The effects of ketoconazole are immediately reversible, so continuous dosing is needed (400 mg three times daily). [Harmone CaP p5]
- In hormonally intact men on ketoconazole, testosterone begins to rise again and can reach the low-normal range within 5 months of therapy. [Harmone CaP p5]
- Ketoconazole is now used in CRPC as a second-line hormonal agent (in non-metastatic CRPC). [Harmone CaP p5]
- Ketoconazole side effects are gynaecomastia, weakness and lethargy, hepatic dysfunction, nausea and visual disturbance. [Harmone CaP p5]

Antiandrogen Withdrawal Phenomenon

- After antiandrogen withdrawal, the PSA falls within 4 weeks after stopping flutamide and within 6 weeks after stopping bicalutamide or nilutamide. [Harmone CaP p10]
- Objective response after antiandrogen withdrawal can also improve, but a measurable response is less common than a PSA fall. [Harmone CaP p10]
- Why some men respond is not clearly understood, but men who had rapid PSA responses to androgen ablation have higher rates of the withdrawal phenomenon. [Harmone CaP p10]

Mechanism of development of castration resistance

- The classic models of castration resistance are the hypersensitive AR pathway, AR promiscuity, the outlaw AR pathway, the bypass AR pathway and the lurker cell model. [Prostate notes (02 dragged) p54]
- In the hypersensitive pathway the AR becomes activated by even lower levels of androgen, and approximately one third of castration-resistant tumours show AR gene amplification. [Prostate notes (02 dragged) p54]
- In the outlaw AR model, growth factor peptides such as EGF and IGF-1 increase AR transcriptional activity in the absence of androgen, so the cancer progresses in the castrate state. [Prostate notes (02 dragged) p54]
- In the lurker cell model, a small pre-existing population of possible epithelial stem cells in the prostate grows out under androgen deprivation (an inherent rather than acquired alteration). [Prostate notes (02 dragged) p54]

Definition and states of CRPC

- Symptomatic progression alone is not sufficient to diagnose CRPC (EAU definition). [Prostate notes (02 dragged) p55]

- In non-metastatic CRPC, about one third of men develop metastases within 2 years. [Prostate notes (02 dragged) p55]
- Bone metastasis in mCRPC begins with arrest of circulating tumour cells in cortical and medullary bone spaces, which adhere to bone surfaces and expand, causing pain and fracture. [Prostate notes (02 dragged) p55]

CRPC management: life-prolonging and targeted therapy

- Apalutamide was developed as ARN-509. [Prostate notes (02 dragged) p56]
- Sipuleucel-T is marketed as Provenge. [Prostate notes (02 dragged) p56]
- PI3K/Akt/mTOR inhibitors rapamycin, everolimus and temsirolimus failed as single agents in CRPC; newer agents under study include BEZ235, MK2206, BKM120 and PX866. [Prostate notes (02 dragged) p56]
- Anti-angiogenic agents tried in CRPC include bevacizumab, aflibercept and tasquinimod. [Prostate notes (02 dragged) p56]
- Cabozantinib targets the MET signalling pathway in CRPC. [Prostate notes (02 dragged) p56]
- Custirsen targets the apoptosis pathway in CRPC. [Prostate notes (02 dragged) p56]

Mitoxantrone

- Mitoxantrone is a semi-synthetic anthracycline and topoisomerase II inhibitor that disrupts DNA synthesis and repair by intercalating between DNA bases. [Prostate notes (02 dragged) p57]
- Mitoxantrone 12 mg/m2 is given 3-weekly as a palliative third-line agent. [Prostate notes (02 dragged) p57]
- Mitoxantrone remains useful for docetaxel- or cabazitaxel-refractory disease or when those agents are not tolerated. [Prostate notes (02 dragged) p57]
- Side effects of mitoxantrone are nausea and vomiting, hair loss, immunosuppression and irreversible cardiomyopathy. [Prostate notes (02 dragged) p57]

Docetaxel

- Docetaxel also blocks anti-apoptotic signalling by suppressing BCL-XL. [Prostate notes (02 dragged) p58]
- In mCRPC docetaxel 75 mg/m2 3-weekly with prednisolone 5 mg twice daily is given for up to 10 doses. [Prostate notes (02 dragged) p58]
- Other uses of docetaxel include breast, lung, stomach, ovarian and head and neck cancers. [Prostate notes (02 dragged) p58]
- Additional docetaxel toxicities are anaemia and thrombocytopenia (as part of myelosuppression), nausea, vomiting, diarrhoea and stomatitis. [Prostate notes (02 dragged) p58]

Cabazitaxel

- Cabazitaxel causes fatigue and peripheral neuropathy, the neuropathy being less than with docetaxel. [Prostate notes (02 dragged) p59]

Abiraterone (Zytiga)

- Abiraterone is a steroidal anti-androgen and an irreversible inhibitor of CYP17, the key regulator of extragonadal androgen synthesis. [Prostate notes (02 dragged) p60]
- On abiraterone, deoxycorticosterone rises about 10-fold, corticosterone about 40-fold, 17-OH progesterone about 3-fold and 11-deoxycortisol about 4-fold, while cortisol falls about 2-fold, aldosterone about 1.5-fold and DHEA about 3-fold. [Prostate notes (02 dragged) p60]
- On abiraterone, androstenedione falls below 2 ng/dL and testosterone below 1 ng/dL. [Prostate notes (02 dragged) p60]
- The mineralocorticoid excess caused by abiraterone suppresses renin, which lowers aldosterone. [Prostate notes (02 dragged) p60]

Abiraterone (continued)

- Abiraterone also acts as a partial antagonist of the androgen receptor. [Prostate notes (02 dragged) p61]
- Besides CYP17, abiraterone inhibits 3-beta-hydroxysteroid dehydrogenase (3βHSD), CYP11B1 (steroid 11-beta-hydroxylase) and other CYP450 enzymes. [Prostate notes (02 dragged) p61]
- Abiraterone acetate is the prodrug metabolised to abiraterone, and its more potent metabolite is delta-4-abiraterone (D4A), which can lower testosterone below 1 ng/dL. [Prostate notes (02 dragged) p61]

- First-line abiraterone with prednisolone in mCRPC is for men with ECOG performance status 0 or 1 and asymptomatic or mildly symptomatic disease. [Prostate notes (02 dragged) p61]
- Very common (more than 10%) abiraterone adverse events are hypokalaemia, oedema, diarrhoea, hypertension and urinary tract infection. [Prostate notes (02 dragged) p61]
- Common (1–10%) abiraterone adverse events are cardiac failure, angina, arrhythmias including atrial fibrillation and tachycardia, hypertriglyceridaemia, raised transaminases, dyspepsia, fractures, haematuria and sepsis. [Prostate notes (02 dragged) p62]
- Uncommon (0.1–1%) abiraterone adverse events are adrenal insufficiency, myopathy, rhabdomyolysis and allergic alveolitis. [Prostate notes (02 dragged) p62]

Enzalutamide (Xtandi)

- Enzalutamide is marketed as Xtandi. [Prostate notes (02 dragged) p62]
- Enzalutamide remains a strong AR antagonist in the castration-resistant state even when the AR is overexpressed, and it also impairs binding of the AR to the transcriptional complex. [Prostate notes (02 dragged) p62]
- Enzalutamide 160 mg is taken as four capsules once daily, swallowed whole and not chewed. [Prostate notes (02 dragged) p62]
- Enzalutamide side effects include asthenia, back pain, arthralgia, hypertension, diarrhoea, hot flushes and neutropenia. [Prostate notes (02 dragged) p62]
- CNS side effects of enzalutamide include visual hallucinations, cognitive disorder, memory impairment, vertigo, insomnia and seizures, because the drug crosses the blood–brain barrier and inhibits the GABA-A receptor. [Prostate notes (02 dragged) p62]

Apalutamide (Erleada, ARN-509)

- Apalutamide (Erleada), approved in February 2018 on the SPARTAN trial, was the first drug specifically approved for non-metastatic CRPC. [Prostate notes (02 dragged) p65]
- Apalutamide is a non-steroidal pure AR antagonist that also inhibits AR nuclear translocation and DNA binding. [Prostate notes (02 dragged) p65]
- Apalutamide 240 mg is taken as four 60 mg tablets once daily. [Prostate notes (02 dragged) p65]
- Apalutamide side effects include fatigue, nausea, abdominal pain, diarrhoea, breast tenderness and seizures. [Prostate notes (02 dragged) p65]
- Apalutamide is contraindicated in pregnancy and in men with a history of seizures. [Prostate notes (02 dragged) p65]

PROSTVAC-VF

- The three TRICOM costimulatory molecules in PROSTVAC-VF are LFA-3, ICAM-1 and B7.1. [Prostate notes (02 dragged) p68]
- PROSTVAC-VF is given subcutaneously as 1 priming dose followed by 6 boosting doses, generating PSA-specific T cells that kill PSA-expressing cells. [Prostate notes (02 dragged) p68]
- PROSTVAC-VF is relatively inexpensive. [Prostate notes (02 dragged) p68]
- The phase III trial of PROSTVAC-VF in asymptomatic or minimally symptomatic mCRPC was the PROSPECT trial. [Prostate notes (02 dragged) p68]

Immune checkpoint blockade (ipilimumab)

- Ipilimumab, an anti-CTLA-4 monoclonal antibody, showed promising activity in mCRPC but is not FDA approved for prostate cancer. [Prostate notes (02 dragged) p69]
- In mCRPC trials ipilimumab was given IV every 3 weeks for 12 weeks, then every 12 weeks. [Prostate notes (02 dragged) p69]
- Ipilimumab benefit in mCRPC was greatest in men without visceral metastases and with normal haemoglobin and normal ALP. [Prostate notes (02 dragged) p69]
- Ipilimumab side effects include fatigue, nausea, constipation and pruritus. [Prostate notes (02 dragged) p69]
- Immune-related adverse events of ipilimumab also include adrenal insufficiency, other endocrinopathies, dermatitis and vitiligo. [Prostate notes (02 dragged) p69]

Targeted treatments in CRPC

- The mTOR inhibitors rapamycin, everolimus and temsirolimus failed to show significant activity in mCRPC, both as single agents and combined with bicalutamide. [Prostate notes (02 dragged) p70]
- PI3K/Akt/mTOR pathway agents under investigation in CRPC include BEZ235, MK2206, BKM120 and PX866. [Prostate notes (02 dragged) p70]
- Bevacizumab did not improve overall survival in CRPC and caused serious toxicity: hypertension, GI perforation, haemorrhage, mucositis and pneumonitis. [Prostate notes (02 dragged) p70]
- Aflibercept is a VEGF trap, a decoy receptor that binds circulating VEGF and prevents it reaching cellular VEGF receptors; it gave no significant OS or PFS difference in CRPC. [Prostate notes (02 dragged) p71]
- Sunitinib in CRPC increased PFS but not OS and is not FDA approved for this use. [Prostate notes (02 dragged) p71]
- Tasquinimod inhibits angiogenesis driven by HIF-1-alpha upregulation and aberrant VEGF expression, and also inhibits S100A9, a protein involved in differentiation and cell-cycle progression; it reached phase III trials. [Prostate notes (02 dragged) p71]
- Tasquinimod side effects are fatigue, nausea, constipation, anorexia, raised lipase and amylase, anaemia and DVT. [Prostate notes (02 dragged) p71]

MET signalling and apoptosis pathway in CRPC

- c-MET is the cell-surface receptor for hepatocyte growth factor. [Prostate notes (02 dragged) p72]
- High c-MET expression is seen in approximately 50% of primary prostate cancers at diagnosis and is universal in bone metastases. [Prostate notes (02 dragged) p72]
- Cabozantinib (XL184) is an oral tyrosine kinase inhibitor of c-MET, VEGFR2 and RET. [Prostate notes (02 dragged) p72]
- Cabozantinib side effects are diarrhoea, fatigue, anorexia, rash, hand-foot syndrome, mucositis and raised liver enzymes. [Prostate notes (02 dragged) p72]
- Custirsen is an intravenous antisense oligonucleotide that inhibits clusterin, an anti-apoptotic protein, and may reverse resistance to taxane chemotherapy. [Prostate notes (02 dragged) p72]
- The custirsen trials in CRPC were SYNERGY and AFFINITY. [Prostate notes (02 dragged) p72]

Pain and epidural cord compression

- Epidural metastasis is a fairly common and potentially devastating complication needing early diagnosis and treatment. [Prostate notes (02 dragged) p75]
- Chemotherapy is rarely used for epidural cord compression. [Prostate notes (02 dragged) p75]

Bisphosphonates

- Bisphosphonates are anti-resorptive drugs that inhibit osteoclast activity and proliferation, maintaining or increasing bone density and strength. [Prostate notes (02 dragged) p76]
- Indications for bisphosphonates beyond prostate cancer include bone metastases from breast and lung cancer, postmenopausal osteoporosis, Paget disease of bone and multiple myeloma. [Prostate notes (02 dragged) p76]
- Other uses of bisphosphonates are fibrous dysplasia, Gaucher disease, paediatric osteogenesis imperfecta, giant cell tumour of the appendicular skeleton and elderly men with non-traumatic fracture. [Prostate notes (02 dragged) p76]
- Zoledronate was first approved for hypercalcaemia and for decreased bone density in postmenopausal women. [Prostate notes (02 dragged) p76]
- Zoledronate 4 mg is given IV in 100 mL normal saline every 4 weeks for several months (an Indian brand is Zobone, Sun Pharma). [Prostate notes (02 dragged) p76]
- Zoledronate can cause fatigue and anaemia. [Prostate notes (02 dragged) p76]
- Other bisphosphonates are pamidronate, etidronate and clodronate; risedronate and alendronate are oral daily or weekly tablets, and ibandronate is an oral monthly tablet. [Prostate notes (02 dragged) p78]

Radiopharmaceuticals

- Radiopharmaceutical therapy is unsealed-source or molecular radiotherapy. [Prostate notes (02 dragged) p80]

- Alpha particles are about 7000 times heavier than beta particles, so 1–2 hits kill a cell, whereas beta particles need 100–1000 hits. [Prostate notes (02 dragged) p80]
- Radium-223 has replaced the beta emitters strontium-89 and samarium-153. [Prostate notes (02 dragged) p80]
- Radium-223 is used in men ineligible for chemotherapy or with docetaxel-refractory disease. [Prostate notes (02 dragged) p80]

SPARTAN Trial

- SPARTAN was a multicentre, placebo-controlled, double-blind, phase 3 randomised controlled trial of apalutamide's safety and efficacy. [Prostate Trials p7-8]
- SPARTAN randomised 1027 patients 2:1 to apalutamide plus ADT or placebo plus ADT. [Prostate Trials p7-8]
- Apalutamide was given as 240 mg once daily by mouth, taken as 4 tablets. [Prostate Trials p7-8]
- The notes give the study duration as about 43 months. [Prostate Trials p7-8]
- The secondary outcomes were time to metastasis, progression-free survival, time to symptomatic progression, overall survival and time to starting chemotherapy. [Prostate Trials p7-8]
- Apalutamide also improved progression-free survival, time to symptomatic progression and time to PSA progression. [Prostate Trials p7-8]

OTHER TOPICS IN THIS BOOK

Various Problems in PSA Screening

- Lead-time bias can be compensated for in two ways: by incorporating the lead time into comparisons of outcomes between screened and unscreened cohorts, and by measuring disease-specific mortality rather than survival time. [Compilation 02-3 p46-49]
- Aggressive tumours have a short potential screening period (a short interval between possible detection by screening and the onset of symptoms), whereas slow-growing tumours have a longer potential screening period. [Compilation 02-3 p46-49]
- Because of this, aggressive tumours are more likely to become symptomatic in the intervals between screening rounds and to be detected clinically, so the screen-detected tumours appear to have better survival, which is wrongly credited to screening when it really reflects tumour biology; the effect is stronger when screening is done at regular intervals. [Compilation 02-3 p46-49]
- Overdiagnosis is the extreme form of length-time bias. [Compilation 02-3 p46-49]
- There is no level 1 evidence that PSA screening is cost-effective in reducing prostate cancer mortality. [Compilation 02-3 p46-49]

SPARTAN Trial (adverse effects)

- In SPARTAN, fracture (as well as hypothyroidism and rash) was more common in the apalutamide group. [Compilation 02-3 p57]

Multiparametric MRI (Prostate)

- MR spectroscopy is listed as the fourth component of multiparametric prostate MRI, with diffusion-weighted and dynamic contrast-enhanced imaging making up the physiological (functional) part and T2-weighted imaging the anatomical part. [MRI p3]
- Diffusion-weighted imaging allows both qualitative and quantitative assessment of prostate cancer aggressiveness. [MRI p3]
- Cancer shows early and strong enhancement on dynamic contrast-enhanced MRI, which is used for detection and for monitoring treatment response. [MRI p3]
- On MR spectroscopy, prostate cancer shows low citrate and high choline. [MRI p3]
- MR spectroscopy can be used for detection and monitoring of response to therapy, but its routine use is not suggested. [MRI p3]

Clinical Applications of Prostate MRI

- For MRI detection, a clinically significant cancer is defined as one of 0.5 mL or more. [MRI p4]

- MRI has poor sensitivity for low-volume Gleason 3+3 cancer, which can be argued to be an advantage because these indolent tumours are the ones that raise concern about overdiagnosis and overtreatment. [MRI p4]
- In-bore MRI-guided biopsy is time consuming, whereas MRI/TRUS fusion biopsy can be done at the bedside. [MRI p4]
- During active surveillance, MRI can act as a substitute for, or a guide to, repeat biopsy. [MRI p4]
- After a prostate biopsy, MRI should be delayed by an interval of 6 weeks, because haemorrhage and inflammation cause artefacts. [MRI p4]

Various PET-CTs in carcinoma prostate

- 18F-FDG PET-CT has a higher sensitivity than a bone scan in prostate cancer. [PET CT PSMA p3]
- In prostate cancer, 18F-FDG PET-CT does not detect soft tissue metastases. [PET CT PSMA p3]
- 11C-choline has low urinary excretion, so it gives high-quality pelvic imaging. [PET CT PSMA p3]
- 11C-acetate PET/CT can detect local recurrence, nodal metastasis and distant metastasis. [PET CT PSMA p4]
- The disadvantage of 11C-acetate PET/CT is a lower detection rate at low PSA levels. [PET CT PSMA p4]
- PSMA PET-CT has a short scanning time. [PET CT PSMA p4]
- PSMA PET-CT detects lesions against a low background signal. [PET CT PSMA p4]
- The main disadvantage of PSMA PET-CT is its lack of availability. [PET CT PSMA p4]

PSMA PET-CT scan: introduction and PSMA biology

- PSMA is overexpressed 100-1000 fold in prostate cancer. [PET CT PSMA p5]
- PSMA expression increases with androgen-refractory disease, metastatic disease and disease recurrence (as well as with tumour aggressiveness). [PET CT PSMA p5]
- PSMA is a type II transmembrane protein with three portions: intracellular, transmembrane and extracellular. [PET CT PSMA p5]
- The PSMA gene is located on the short arm of chromosome 11 (11p). [PET CT PSMA p5]
- In normal prostate, PSMA is located on the apical side of the epithelium. In neoplasia or dysplasia it translocates from the apical to the luminal surface, and expression is higher on less differentiated cells. [PET CT PSMA p5]
- PSMA-based PET imaging agents fall into three groups: new-generation antibodies (64Cu- or 89Zr-labelled), aptamers (64Cu-labelled), and low-molecular-weight PSMA inhibitors (11C-, 18F-, 68Ga-, 64Cu- or 86Y-labelled). [PET CT PSMA p5]
- Promising PSMA PET agents are 68Ga-HBED-CC conjugate (HBED-CC is the chelating agent), 18F-DCFBC and BAY 1075553. [PET CT PSMA p6]
- A potential indication for PSMA PET is as a marker of response to ADT. [PET CT PSMA p6]
- A potential indication for PSMA PET is local tumour detection in patients with a previous negative biopsy. [PET CT PSMA p6]
- PSMA is expressed in prostate cancer at all stages. [PET CT PSMA p6]

PSMA-based targeted therapy in CaP

- Targeted RNA therapy uses PSMA-targeted RNA aptamers. siRNA (small interfering RNA) or shRNA aptamers bind PSMA on cancer cells and are internalised. The RNA then interferes with protein synthesis and inhibits DNA repair, which increases radiation-induced cell death (radiosensitisation). The notes expand "Sh" as "short heparin"; the standard term is short hairpin RNA. [PET CT PSMA p7]
- In antisense siRNA therapy, the antisense RNA blocks translation of the mRNA, which inhibits protein synthesis and causes cell death. [PET CT PSMA p7]
- In one form of PSMA-based systemic radionuclide therapy, radioactive lutetium-177 is bound to the J591 monoclonal antibody, which targets PSMA-expressing cancer cells in bone and soft tissue. [PET CT PSMA p8]
- The destructive range of radium-223 is about 8 cells. [PET CT PSMA p8]
- The destructive range of lutetium-177 is 0.25 mm, or about 125 cells. [PET CT PSMA p8]

- Radium-223 (alpha) has a higher power to kill cells than lutetium-177 (beta). [PET CT PSMA p8]
- Radium-223 cannot be imaged, whereas lutetium-177 can be imaged with a gamma camera. [PET CT PSMA p8]
- Oedema is listed as a side effect of radium-223, along with GI effects. [PET CT PSMA p8]

Basic principles of PET

- One advantage of PET is that it gives details of the nervous system. [PET CT PSMA p10]
- PET disadvantages are ionising radiation, the need for radioactive substances, and poor resolution when PET is used alone. [PET CT PSMA p10]
- A cyclotron produces the 18F radionuclide, which is attached to the FDG molecule to make 18F-FDG. [PET CT PSMA p10]
- Inside the cell, hexokinase phosphorylates 18F-FDG to FDG-6-phosphate, which traps it inside the cell. [PET CT PSMA p10]

PET vs PET-CT

- In PET-CT, the CT provides the data for attenuation correction. [PET CT PSMA p11]
- PET-CT can tell structures with normally high metabolic activity apart from those with abnormally increased activity. [PET CT PSMA p11]
- PET-CT gives accurate localisation of small areas of increased radiotracer activity. Its spatial resolution is excellent, whereas PET alone has poor resolution. [PET CT PSMA p11]

Preparation for PET-CT

- Contraindications to PET-CT are pregnancy, use of caffeine, tobacco or alcohol in the past 24 hours, use of sedatives, and use of drugs that induce or inhibit metabolism. [PET CT PSMA p11]
- Before PET-CT the patient is nil by mouth for 4-6 hours and avoids caffeine and alcohol. [PET CT PSMA p11]
- A blood glucose below 150 mg/dl is desirable before FDG PET-CT. [PET CT PSMA p12]
- Strenuous exercise should be avoided before PET-CT. [PET CT PSMA p12]
- Oral water-soluble iodinated contrast is given for bowel opacification, except in head and neck studies. [PET CT PSMA p12]
- The radiopharmaceutical dose is 10 millicurie IV. [PET CT PSMA p12]
- Patient activity is limited for 20 minutes after injection. [PET CT PSMA p12]
- The study starts 60 minutes after injection. [PET CT PSMA p12]

? Please check: handwriting read with doubt

- **Basic Principles · Adverse reactions to intravascular iodinated contrast media:** margin note "0.1 mg/kg? < 1:1000 (IM), 1:10,000 (IV)" beside the epinephrine dose (p 14) [Contrast 2 p2]
- **Basic Principles · Key Points: Fibrin-based Sealants:** "Available as liquid form?" (note about Tisseel's presentation, possibly "frozen"/"liquid foam") (p 2) [Hemostatic agents p2]
- **Basic Principles · Surgical simulators available in urology:** "MyoSim (GreenLight / PVP)" — name of the GreenLight/PVP laser simulator (p2) [Simulation etc p2]
- **Oncology, DSD, Gender & Ageing Ch 24 · Activators of Angiogenesis:** "Angiogenin/Angiostatin?" listed among activators (angiostatin is normally an inhibitor and is also listed under inhibitors on the next page) (p 2) [Angiogenesis p2]
- **Infections & Inflammation · Haemorrhagic Cystitis:** "Cystopen (100)" (p 50) [Compilation 04-3 p50]
- **Infections & Inflammation · Piperacillin + Tazobactam:** "CrCl 20–40: 2.25 g IV 6-hourly?; <20: 2.25 g …?" (bottom line cut off) (p 1) [Compilation 04-4 p1]
- **Infections & Inflammation Ch 26 · Complicated Cystitis:** "Fever subsides?" — text cut off at the foot of the complicated cystitis table (p33) [Compilation 04-3 p33]
- **Infections & Inflammation Ch 26 · Imaging in UTI:** "Haematuria found in 40-?0% of cases of cystitis" (p 23; read as 40–60%) [Compilation 04 p27]

- **Paediatric Urology 1 Ch 41 · Medical treatment (watchful waiting):** "TMP-SMX, Nitrofurantoin, Prob..." — the third agent for children over 2 years is cut off at the page edge (p 22). [vur p22]
- **Paediatric Urology 1 Ch 50 · Multicystic dysplastic kidney:** "Ureteric bud theory ... gene mutations like GNF, WNT, PAX2" — "GNF" read as GDNF (p 16) [Cystic Disease p18]
- **Paediatric Urology 1 Ch 63 · Post-operative Chemotherapy (NWTSG/COG):** red vertical margin note beside diffuse anaplasia stage I, "Under trial?" (p 14) [Wilms tumor p14]
- **Paediatric Urology 1 Ch 63 · Tumour Markers:** red margin note beside microRNA, "GAAM?" (p 20) [Wilms tumor p20]
- **Paediatric Urology 2 Ch 43 · Surgical Options in the Newborn with Classic Exstrophy:** "Waswan approach" (p 16), read as Warsaw approach. [Exstrophy p16-18]
- **Paediatric Urology 2 Ch 43 · Surgical Options in the Newborn with Classic Exstrophy:** "(2-3)" written above the bladder neck and epispadias repair step of the Warsaw approach (p 16), probably 2–3 years of age. [Exstrophy p16-18]
- **Female Urology Ch 117 · Abdominal Sacrocolpopexy:** "Isolated uterine prolapse and/or [ente?]rocele" as an indication for sacrocolpopexy; the word before "-tocele" is illegible (p 10). [POP p10]
- **Voiding Dysfunction Ch 110 · Evaluation: OABSS questionnaire:** OABSS Q3 option 4 written "one/three? times a day", corrected in the notes; read as "once a day" (p 5) [OAB p5]
- **Voiding Dysfunction Ch 110 · Behavioural therapy and drugs:** annotation on bladder training "encourages inhibitory influences on bladder?" (p 6) [OAB p6]
- **Voiding Dysfunction Ch 111 · Pathophysiology and Aetiology:** "CIBS" in the neurogenic list, probably GBS (Guillain–Barré syndrome) (p 2) [Underactive Bladder p2]
- **Andrology & Sexual Medicine Ch 69 · Single-gene disorder: CBAVD:** "Isolated CBAVD: when some functional form of the gene that causes CF are completely absent?" (p 2; the sense of the sentence is unclear) [Infertility p2]
- **Andrology & Sexual Medicine Ch 73 · Etiology:** "neurotoxin (eretina?)" — name of the Phoneutria toxin (p 2) [Priapism p2]
- **Testis, Penis & Urethra Ch 82 · Non-Squamous Malignancies:** "Hemangioepithelioma?" as the commonest malignant penile sarcoma, probably haemangioendothelioma (p 22) [Carcinoma Penis p21]
- **Testis, Penis & Urethra Ch 82 · Non-Squamous Malignancies:** "docetaxel / c..." second neoadjuvant agent for extramammary Paget disease is cut off (p 23) [Carcinoma Penis p21]
- **Testis, Penis & Urethra Ch 85 · Other Grafts:** "Sxcoralion" listed as a disadvantage of bladder mucosa graft; best reading "excoriation?" (possibly meatal eversion/prolapse, which Paper 3 p.195 names for bladder mucosa) (p 5) [Urethra Grafts p5]
- **Testis, Penis & Urethra Ch 86 · RGU: Method:** "Usually the penile urethra is [word left blank / ? parallel] to the flexed femur" (p 2) [RGU p2]
- **Stones, Obstruction & Trauma · Lasers in lithotripsy: holmium versus erbium:YAG:** "0.79 µm?" as the Er:YAG soft-tissue penetration depth (p 5) [LASER p5]
- **Stones, Obstruction & Trauma · Stone analysis procedure (FTIR):** storage temperature "22–22°C?" (probably 20–22°C) (p 3) [Stone Analysis p3]
- **Stones, Obstruction & Trauma Ch 97 · Steinstrasse:** "Stone size <2.5 cm in kidney, <1.5 cm in ureter?" (faint prevention line) (p 67) [Compilation 05-2 p67]
- **Bladder Cancer & Diversion Ch 129 · Origin, Recurrence, Dissemination:** "25% (or 95%?) of invasive urothelial carcinomas have angiolymphatic system invasion" (p 12) [Compilation 01 p11-12]
- **Bladder Cancer & Diversion Ch 133 · Post-op ERAS elements:** "Combine use of NO & propofol" for PONV prevention (reading clear, but NO may mean nitrous oxide, to be avoided, rather than used) (p 3) [ERAS p3]
- **Bladder Cancer & Diversion Ch 133 · Ileum and colon: hyperchloraemic metabolic acidosis:** "Oral sodium bicarbonate (Sodosis-500?)" (p 5; brand name) [Intestines in Uro p4]

- **Bladder Cancer & Diversion Ch 133 · Bowel preparation:** "Phosphate neuropathy?" (p 15; may mean phosphate nephropathy) [Intestines in Uro p13]
- **Bladder Cancer & Diversion Ch 133 · Bowel preparation:** "Day 3 ... Day 2 ... Day 3" for the kanamycin regimen (p 15; the day labels are written as shown) [Intestines in Uro p13]
- **Bladder Cancer & Diversion Ch 135 · Various types / procedures:** "1/3rd" label on the Penn pouch diagram, probably the proximal third of the transverse colon included in the pouch? (p 3) [Continent Pouches p3]
- **Kidney & UTUC Ch 137 · Prognostic Factors for RCC:** "Venous involvement: 45–65%?" (5-year survival figure, line cut off at the page edge) (p 53) [Compilation 06 p53]
- **Kidney & UTUC Ch 138 · Etiology: Smoking, coffee, analgesics:** "Latency period approx. 22 years?" (could be "> 2 years") for analgesic nephropathy (p 3) [utuc p2-3]
- **Kidney & UTUC Ch 139 · Lymphadenectomy:** "Common iliac + HOPE" (best reading; probably a mnemonic for the distal ureter template) (p 16) [utuc p15-16]
- **Kidney & UTUC Ch 143 · Microwave ablation:** "Independent of wall pressure?" listed as an advantage of microwave ablation (p 52) [Compilation 06-2 p52]
- **Adrenal & Retroperitoneum Ch 146 · Adrenal Oncocytoma:** "cytokines?" listed with cortisol and sex steroids as products of functional oncocytomas (p 4) [Benign Adrenal p4]
- **Adrenal & Retroperitoneum Ch 146 · Adrenal Cysts:** "TG" also listed among the tests on cyst aspirate (possibly triglycerides?) (p 9) [Benign Adrenal p8-10]
- **Adrenal & Retroperitoneum Ch 146 · Adrenal Incidentaloma: definition, frequency and algorithm:** "85%? of incidentalomas are adenomas" (p 1; the figure looks like "05%") [Incidentloma p1]
- **Adrenal & Retroperitoneum Ch 146 · Catecholamine hypersecretion testing:** "normetanephrine >4001? nmol/day" (p 10; the figure is written as shown and may be a slip) [Incidentloma p10]
- **Renal, Nephrology & Transplant Ch 90 · Peritoneal dialysis:** "Smallest pore (water channel) <0.8 nm?; small pore (4 nm?); large pore (>15 nm?)" (p 11) [Dialysis AVF p11]
- **Renal, Nephrology & Transplant Ch 91 · Treatment of Cell-Mediated Rejection:** "ATG 1–1.5 mg/kg IV for 7–?" (duration cut off at page edge) (p 10) [Transplant p10]
- **Renal, Nephrology & Transplant Ch 91 · ABO-Incompatible Transplantation: Problems and Solutions:** margin note "? 1:80 titre level" (p 29); the corpus target titre is commonly ≤1:8 (Paper 2 p.318). [Transplant p29]
- **Renal, Nephrology & Transplant Ch 91 · Post Transplant Complications: Infections:** "H2 receptor blocker + An... (if symptomatic)", second agent possibly "antacid?" (p 39). [Transplant p39]
- **Prostate Cancer Ch 153 · REDUCE trial:** REDUCE age range "50–70 yrs?" (the trial's published range is 50–75) (p 9) [CaP Etio Preven p9]
- **Prostate Cancer Ch 166 · Spinal cord compression:** maintenance steroid dose written as "4 [overwritten] mg every 6 hourly" (p 26); best reading 4 mg. [Compilation 02 p26]
- **Prostate Cancer Ch 166 · Systemic radionuclide therapy:** margin note "β ... cells, SSP-rural²?" (p 27). [Compilation 02 p27]
- **Prostate Cancer Ch 166 · Abiraterone (Zytiga):** "Estradiol (<80 ng/dl?)" — likely a lower value; figure on abiraterone diagram (p 60) [Prostate notes (02 dragged) p60]